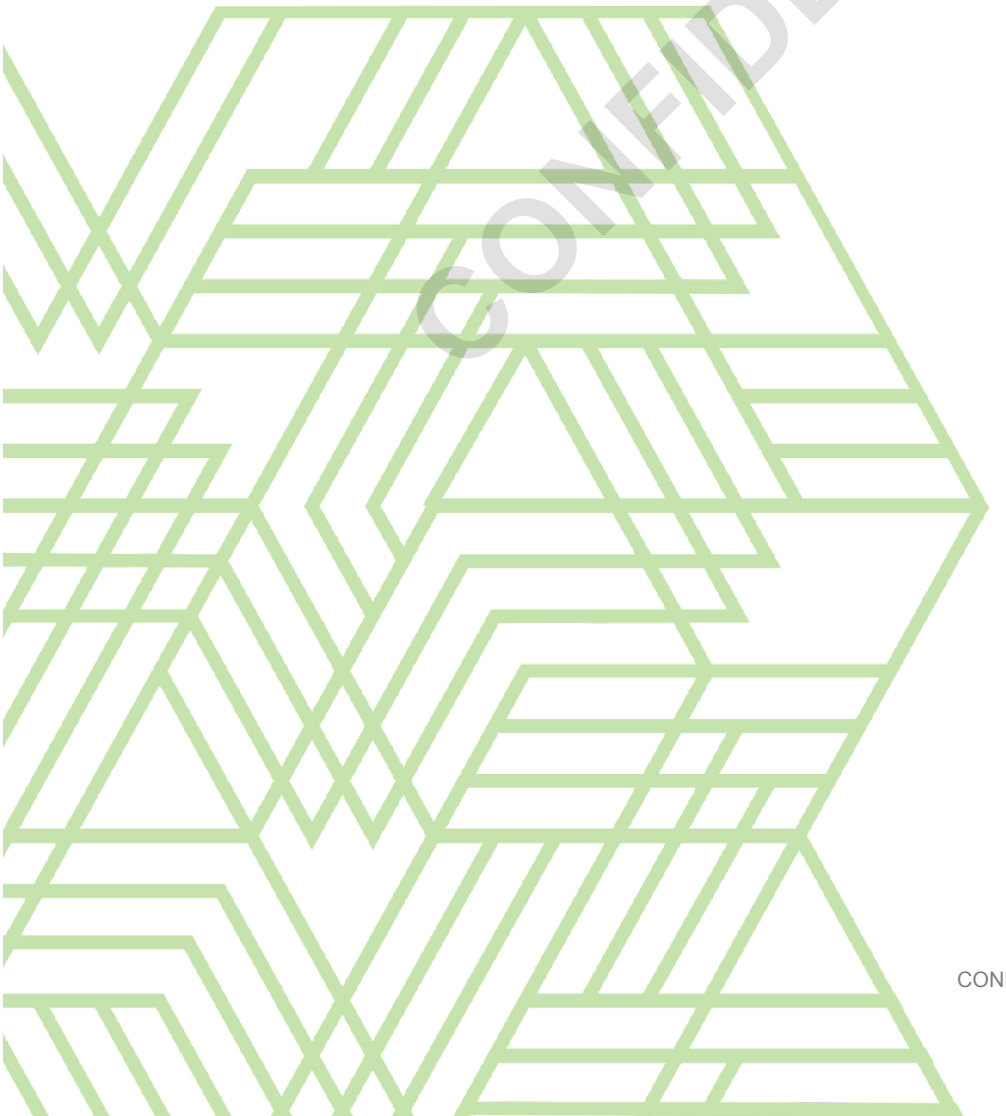
# ATM33/e (Rev. B)

## Register Specification

**SUMMARY:** This document provides register definitions and descriptions of the ATM33/e Rev. B Wireless SoC.

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- AT_PRRF_NONSECURE –
  - PATCH_TAG0
  - PATCH_TAG1
  - PATCH_TAG2
  - PATCH_TAG3
  - PATCH_TAG4
  - PATCH_TAG5
  - PATCH_TAG6
  - PATCH_TAG7
  - PATCH_TAG8
  - PATCH_TAG9
  - PATCH_TAG10
  - PATCH_TAG11
  - PATCH_TAG12
  - PATCH_TAG13
  - PATCH_TAG14
  - PATCH_TAG15
  - PATCH0
  - PATCH1
  - PATCH2
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  - PATCH33

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    - PATCH34
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    - PATCH41
    - PATCH42
    - PATCH43
    - PATCH44
    - PATCH45
    - PATCH46
    - PATCH47
    - RRAM_WRT_CONFIG_LO
    - RRAM_WRT_CONFIG_HI
    - RRAM_READ_CONFIG_LO
    - RRAM_READ_CONFIG_HI
    - RRAM_CACHE_CONFIG
    - RRAM_MEM_CONFIG
    - RRAM_STATUS
- AT_PRRF_SECURE –
    - PATCH_TAG0
    - PATCH_TAG1
    - PATCH_TAG2
    - PATCH_TAG3
    - PATCH_TAG4
    - PATCH_TAG5
    - PATCH_TAG6
    - PATCH_TAG7
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    - PATCH_TAG11
    - PATCH_TAG12
    - PATCH_TAG13
    - PATCH_TAG14
    - PATCH_TAG15
    - PATCH0
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    - PATCH42
    - PATCH43
    - PATCH44
    - PATCH45
    - PATCH46
    - PATCH47
    - RRAM_WRT_CONFIG_LO
    - RRAM_WRT_CONFIG_HI
    - RRAM_READ_CONFIG_LO
    - RRAM_READ_CONFIG_HI
    - RRAM_CACHE_CONFIG
    - RRAM_MEM_CONFIG
    - RRAM_STATUS
- GPIO0_NONSECURE –
    - DATA
    - DATAOUT
    - INENABLE_SET
    - INENABLE_CLR
    - OUTENSET
    - OUTENCLR
    - ALTFUNCSET
    - ALTFUNCCLR
    - INTENSET
    - INTENCLR
    - INTTYPESET
    - INTTYPECLR
    - INTPOLSET
    - INTPOLCLR
    - INTSTATUS_CLEAR
    - PULLUP_ENABLE_SET
    - PULLUP_ENABLE_CLR
    - PULLDOWN_ENABLE_SET
    - PULLDOWN_ENABLE_CLR
    - MASKLOWBYTE
    - MASKHIGHBYTE
    - PID4
    - PID5
    - PID6
    - PID7
    - PID0
    - PID1
    - PID2
    - PID3
    - CID0
    - CID1
    - CID2
    - CID3
- GPIO1_NONSECURE –
    - DATA
    - DATAOUT

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    - INENABLE_SET
    - INENABLE_CLR
    - OUTENSET
    - OUTENCLR
    - ALTFUNCSET
    - ALTFUNCCLR
    - INTENSET
    - INTENCLR
    - INTTYPESET
    - INTTYPECLR
    - INTPOLSET
    - INTPOLCLR
    - INTSTATUS_CLEAR
    - PULLUP_ENABLE_SET
    - PULLUP_ENABLE_CLR
    - PULLDOWN_ENABLE_SET
    - PULLDOWN_ENABLE_CLR
    - MASKLOWBYTE
    - MASKHIGHBYTE
    - PID4
    - PID5
    - PID6
    - PID7
    - PID0
    - PID1
    - PID2
    - PID3
    - CID0
    - CID1
    - CID2
    - CID3
- SHA2_NONSECURE –
    - CONTROL
    - SIDELOAD_CTRL
    - CMD
    - MESSAGE_LENGTH_LO
    - MESSAGE_LENGTH_HI
    - WIPE_V
    - FIFO_PUSH
    - DIGEST0
    - DIGEST1
    - DIGEST2
    - DIGEST3
    - DIGEST4
    - DIGEST5
    - DIGEST6
    - DIGEST7
    - KEY_0
    - KEY_1
    - KEY_2
    - KEY_3
    - KEY_4
    - KEY_5
    - KEY_6
    - KEY_7
    - STATUS
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - ERR_CODE
    - ID
- WRPR0_NONSECURE –
    - APB0_CTRL
    - APB1_CTRL

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- APB2_CTRL
- APB3_CTRL
- APB4_CTRL
- APB5_CTRL
- APB6_CTRL
- APB7_CTRL
- APB8_CTRL
- APB9_CTRL
- APB10_CTRL
- APB11_CTRL
- APB12_CTRL
- APB13_CTRL
- APB14_CTRL
- APB15_CTRL
- CLK_HPC_PIN_OUT_CTRL
- CLK_MPC_PIN_OUT_CTRL
- CLK_LPC_PIN_OUT_CTRL
- DBG_CTRL
- PSEL_A
- PSEL_B
- PSEL_C
- PSEL_D
- PSEL_E
- PSEL_F
- PSEL_G
- PSEL_H
- PSEL_I
- PUPD_OVRD
- PUPD_OVRD_VAL
- REMAP
- AHB_INTERPOSERS
- SCRATCHPAD_A
- SCRATCHPAD_B
- SCRATCHPAD_C
- SCRATCHPAD_D
- RRAM_WRITE_PROTECTION0
- RRAM_WRITE_PROTECTION1
- RRAM_WRITE_PROTECTION2
- RRAM_WRITE_PROTECTION3
- RRAM_WRITE_PROTECTION4
- RRAM_WRITE_PROTECTION5
- RRAM_WRITE_PROTECTION6
- RRAM_WRITE_PROTECTION7
- RRAM_STICKY_WRITE_PROTECTION0
- RRAM_STICKY_WRITE_PROTECTION1
- RRAM_STICKY_WRITE_PROTECTION2
- RRAM_STICKY_WRITE_PROTECTION3
- RRAM_STICKY_WRITE_PROTECTION4
- RRAM_STICKY_WRITE_PROTECTION5
- RRAM_STICKY_WRITE_PROTECTION6
- RRAM_STICKY_WRITE_PROTECTION7
- RRAM_STICKY_WRITE_PROTECTION_STATUS0
- RRAM_STICKY_WRITE_PROTECTION_STATUS1
- RRAM_STICKY_WRITE_PROTECTION_STATUS2
- RRAM_STICKY_WRITE_PROTECTION_STATUS3
- RRAM_STICKY_WRITE_PROTECTION_STATUS4
- RRAM_STICKY_WRITE_PROTECTION_STATUS5
- RRAM_STICKY_WRITE_PROTECTION_STATUS6
- RRAM_STICKY_WRITE_PROTECTION_STATUS7
- RRAM_STICKY_READ_PROTECTION0
- RRAM_STICKY_READ_PROTECTION_STATUS0
- PROT_BITS_SET0
- PROT_BITS_SET1
- PROT_BITS_STAT0
- PROT_BITS_STAT1

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    - QSPI_STICKY_WRITE_PROTECTION0
    - QSPI_STICKY_WRITE_PROTECTION1
    - QSPI_STICKY_WRITE_PROTECTION_STATUS0
    - QSPI_STICKY_WRITE_PROTECTION_STATUS1
    - PDSN
    - CALCOUNTRC
    - SECURE_DEBUG_CTRL
    - SECURE_DEBUG_STAT
    - CHIPID1
    - CHIPID2
    - CHIPREV
    - CORE_ID
- UART0_NONSECURE –
    - DATA
    - STATE
    - CTRL
    - INTSTATUS_CLEAR
    - BAUDDIV
    - RX_LWM
    - TX_LWM
    - RX_FIFO_SPACES
    - TX_FIFO_SPACES
    - HW_FLOW_OVRD
    - BAUD_RATE_DETECT
    - BAUD_DIV_CEIL
    - BRD_STAT
    - INTMASK
    - RX_FIFO_ENTRIES
    - INTSTATUS_SET
    - PID4
    - PID5
    - PID6
    - PID7
    - PID0
    - PID1
    - PID2
    - PID3
    - CID0
    - CID1
    - CID2
    - CID3
- UART1_NONSECURE –
    - DATA
    - STATE
    - CTRL
    - INTSTATUS_CLEAR
    - BAUDDIV
    - RX_LWM
    - TX_LWM
    - RX_FIFO_SPACES
    - TX_FIFO_SPACES
    - HW_FLOW_OVRD
    - BAUD_RATE_DETECT
    - BAUD_DIV_CEIL
    - BRD_STAT
    - INTMASK
    - RX_FIFO_ENTRIES
    - INTSTATUS_SET
    - PID4
    - PID5
    - PID6
    - PID7
    - PID0
    - PID1
    - PID2

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    - PID3
    - CID0
    - CID1
    - CID2
    - CID3
- PWM_NONSECURE –
    - PWM_CTRL
    - PWM0_CTRL
    - PWM0_DUR
    - PWM0_CFG
    - PWM1_CTRL
    - PWM1_DUR
    - PWM1_CFG
    - PWM2_CTRL
    - PWM2_DUR
    - PWM2_CFG
    - PWM3_CTRL
    - PWM3_DUR
    - PWM3_CFG
    - PWM4_CTRL
    - PWM4_DUR
    - PWM4_CFG
    - PWM5_CTRL
    - PWM5_DUR
    - PWM5_CFG
    - PWM6_CTRL
    - PWM6_DUR
    - PWM6_CFG
    - PWM7_CTRL
    - PWM7_DUR
    - PWM7_CFG
    - FSM_STATE
    - FIFO_CFG
    - FIFO_CARRIER1_DUR
    - FIFO_CARRIER2_DUR
    - FIFO_STAT
    - FIFO_STAT1
    - FIFO_SEL_ERR_SOURCE
    - INTERRUPTS
    - INTERRUPTS_MASK
    - INTERRUPTS_CLEAR
    - INTERRUPTS_SET
    - FIFO_DATA
    - CORE_ID
    - REV_HASH
    - REV_KEY
- SPI0_NONSECURE –
    - CTRL
    - TRANSACTION_SETUP
    - TRANSACTION_SETUP_DMA
    - TRANSACTION_STATUS
    - DATA_BYTES_LOWER
    - DATA_BYTES_UPPER
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CORE_ID
- KSM_NONSECURE –
    - CTRL0
    - TIME_PARAM0
    - TIME_PARAM1
    - MATRIX_SIZE
    - KEYBOARD_PACKET
    - LOW_POWER

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    - LPBACK_KSI
    - STATUS
    - INTERRUPTS
    - INTERRUPT_MASK
    - INTERRUPT_CLEAR
    - DEBUG
    - DEBUG2
    - CORE_ID
- SPI2_NONSECURE –
    - CTRL
    - TRANSACTION_SETUP
    - TRANSACTION_SETUP_DMA
    - TRANSACTION_STATUS
    - DATA_BYTES_LOWER
    - DATA_BYTES_UPPER
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CORE_ID
- QDEC_NONSECURE –
    - SETUP_FOR_AXIS_0
    - ACCUM_PURGE_STATUS_FOR_AXIS_0
    - ACCUMULATOR_VALUE_FOR_AXIS_0
    - ERR_PURGE_STATUS_FOR_AXIS_0
    - ERROR_COUNT_FOR_AXIS_0
    - SETUP_FOR_AXIS_1
    - ACCUM_PURGE_STATUS_FOR_AXIS_1
    - ACCUMULATOR_VALUE_FOR_AXIS_1
    - ERR_PURGE_STATUS_FOR_AXIS_1
    - ERROR_COUNT_FOR_AXIS_1
    - SETUP_FOR_AXIS_2
    - ACCUM_PURGE_STATUS_FOR_AXIS_2
    - ACCUMULATOR_VALUE_FOR_AXIS_2
    - ERR_PURGE_STATUS_FOR_AXIS_2
    - ERROR_COUNT_FOR_AXIS_2
    - CORE_ID
- SPI1_NONSECURE –
    - CTRL
    - TRANSACTION_SETUP
    - TRANSACTION_SETUP_DMA
    - TRANSACTION_STATUS
    - DATA_BYTES_LOWER
    - DATA_BYTES_UPPER
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CORE_ID
- SLWTIMER_NONSECURE –
    - CONTROL
    - INIT_LOW
    - INIT_HIGH
    - CNT_LOW
    - CNT_HIGH
    - THRES0_LOW
    - THRES0_HIGH
    - THRES1_LOW
    - THRES1_HIGH
    - THRES2_LOW
    - THRES2_HIGH
    - STATUS
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - CORE_ID

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- QSPI_NONSECURE –
  - TRANSACTION_SETUP
  - READ_DATA
  - OVERRIDE_DIN
  - MODE
  - REMOTE_AHB_SETUP
  - REMOTE_AHB_SETUP_2
  - REMOTE_AHB_SETUP_3
  - REMOTE_AHB_SETUP_4
  - REMOTE_AHB_SETUP_5
  - REMOTE_AHB_SETUP_6
  - REMOTE_AHB_SETUP_7
  - REMOTE_AHB_WLE_CNT
  - REMOTE_AHB_WIP_CNT
  - REMOTE_AHB_DBG0
  - REMOTE_AHB_DBG1
  - REMOTE_AHB_DBG2
  - CORE_ID
- I2C0_NONSECURE –
  - TRANSACTION_SETUP
  - CLOCK_CONTROL
  - OUTGOING_DATA
  - OUTGOING_DATA1
  - IO_CTRL
  - INCOMING_DATA
  - INCOMING_DATA1
  - TRANSACTION_STATUS
  - INTERRUPT_MASK
  - INTERRUPT_STATUS
  - SET_INTERRUPT
  - RESET_INTERRUPT
  - ID
- I2C1_NONSECURE –
  - TRANSACTION_SETUP
  - CLOCK_CONTROL
  - OUTGOING_DATA
  - OUTGOING_DATA1
  - IO_CTRL
  - INCOMING_DATA
  - INCOMING_DATA1
  - TRANSACTION_STATUS
  - INTERRUPT_MASK
  - INTERRUPT_STATUS
  - SET_INTERRUPT
  - RESET_INTERRUPT
  - ID
- NVM_NONSECURE –
  - OPMODE
  - CTRL
  - DATA0
  - DATA1
  - STATUS
  - T_SP_VQ
  - T_HP_VQ
  - T_SP_PG
  - T_HP_PG
  - T_HP_CK
  - T_S_PGM
  - T_H_PGM
  - T_CKHP
  - T_CKLP
  - T_SPS_CK
  - T_SR_CK
  - T_HR_CK
  - T_CKLR

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    - T_CKHR
    - T_SRS_CK
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - INTERRUPT_SET
    - INTERRUPT_RESET
    - CORE_ID
- WRPR1_NONSECURE –
    - APB0_CTRL
    - APB1_CTRL
    - APB2_CTRL
    - APB3_CTRL
    - APB4_CTRL
    - APB5_CTRL
    - APB6_CTRL
    - APB7_CTRL
    - APB8_CTRL
    - APB9_CTRL
    - APB10_CTRL
    - APB11_CTRL
    - APB12_CTRL
    - APB13_CTRL
    - APB14_CTRL
    - APB15_CTRL
    - CORE_ID
- RADIO_NONSECURE –
    - CTRL
    - TRANSACTION_SETUP
    - TRANSACTION_SETUP_DMA
    - TRANSACTION_STATUS
    - DATA_BYTES_LOWER
    - DATA_BYTES_UPPER
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CORE_ID
- PMU_NONSECURE –
    - CTRL
    - TRANSACTION_SETUP
    - TRANSACTION_SETUP_DMA
    - TRANSACTION_STATUS
    - DATA_BYTES_LOWER
    - DATA_BYTES_UPPER
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CORE_ID
- PSEQ_NONSECURE - WURX_CONFIG, WURX_CONFIG1 and WURX_CONFIG2 are for ATM3330e and ATM3330 Only
    - CTRL0
    - CTRL1
    - RETAIN_ALL_WAKE_MASK
    - HIB_WAKE_MASK
    - GPIO_WAKE_MASK
    - GPIO_WAKE_TYPE
    - GPIO_WAKE_POL
    - GPIO_WAKE_BOTH_EDGES
    - GPIO_WAKE_STATUS
    - WURX_CONFIG
    - WURX_CONFIG1
    - WURX_CONFIG2
    - GADC_CONFIG
    - XTAL_BITS0
    - XTAL_BITS1

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- OVERRIDES
- OVERRIDES2
- OVERRIDES3
- OVERRIDES4
- OVERRIDES5
- SYSRAM_OVERRIDES
- SYSRAM_OVERRIDES2
- SYSRAM_OVERRIDES3
- SYSRAM_OVERRIDES4
- SYSRAM_OVERRIDES5
- SYSRAM_OVERRIDES6
- SYSRAM_OVERRIDES7
- SYSRAM_OVERRIDES8
- EMRAM_OVERRIDES
- EMRAM_OVERRIDES2
- EMRAM_OVERRIDES3
- EMRAM_OVERRIDES4
- EMRAM_OVERRIDES5
- EMRAM_OVERRIDES6
- EMRAM_OVERRIDES7
- EMRAM_OVERRIDES8
- ICACHE_OVERRIDES
- ICACHE_OVERRIDES2
- ICACHE_OVERRIDES3
- ICACHE_OVERRIDES4
- ICACHE_OVERRIDES5
- ICACHE_OVERRIDES6
- ICACHE_OVERRIDES7
- ICACHE_OVERRIDES8
- RRAMCACHE_OVERRIDES
- RRAMCACHE_OVERRIDES2
- RRAMCACHE_OVERRIDES3
- RRAMCACHE_OVERRIDES4
- RRAMCACHE_OVERRIDES5
- RRAMCACHE_OVERRIDES6
- RRAMCACHE_OVERRIDES7
- RRAMCACHE_OVERRIDES8
- QSPICACHE_OVERRIDES
- COUNTER_CONTROL
- CURRENT_REAL_TIME
- CURRENT_COUNT_DOWN_TIME
- INIT_COUNT_DOWN
- INST_PENDING
- STATUS
- PERSISTENT0
- PERSISTENT1
- PERSISTENT2
- PERSISTENT3
- PERSISTENT4
- PERSISTENT5
- PERSISTENT6
- PERSISTENT7
- SENSOR_HUB_CONTROL
- KSMQDEC_CONTROL
- SPII2C_CONTROL
- FLASH_CONTROL
- FLASH_CONTROL2
- PMU_STATUS
- PMU_INTERRUPT
- SYSRAM_WRITE_BLOCK_CFG
- SYSRAM_SSRS_CFG
- INTERRUPT_STATUS
- INTERRUPT_MASK
- RESET_INTERRUPT
- FPGA_CFG

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    - PLL
    - RADIO_STATUS
    - RRAM_READ_CONFIG_LO
    - RRAM_READ_CONFIG_HI
    - SENSOR_HUB_IMM_BREAKOUT_MASK
    - FSM_TUNINGA
    - FSM_TUNINGB
    - WDOG_CONTROL
    - WDOG_TIMER
    - CORE_ID
- SWD_NONSECURE –
    - DAP_ADDR_CTRL
    - DAP_WDATA_CTRL
    - SWD_CLK_DIV_CTRL
    - PATTERN_UPPER_CTRL
    - PATTERN_LOWER_CTRL
    - TAIL_DUR
    - PATTERN_DUR
    - DAP_BRIDGE_STATUS
    - DAP_RESP_STATUS
    - RW_CTRL
    - APNDP_CTRL
    - TRIGGER_CTRL
    - DAP_RDATA
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - DTOP_BYPASS_CTRL
    - DTOP_BYPASS_STAT
    - DTOP_BYPASS_GPO
    - DTOP_BYPASS_RIF_MODE_CNTL
    - DTOP_BYPASS_RIF_CAL_CNTL
    - DTOP_BYPASS_RIF_RXDC7
    - CORE_ID
- GADC_NONSECURE –
    - CTRL
    - CTRL1
    - CH1_DATAPATH_CONFIG
    - CH2_DATAPATH_CONFIG
    - CH3_DATAPATH_CONFIG
    - CH4_DATAPATH_CONFIG
    - CH5_DATAPATH_CONFIG
    - CH6_DATAPATH_CONFIG
    - CH7_DATAPATH_CONFIG
    - CH8_DATAPATH_CONFIG
    - CH9_DATAPATH_CONFIG
    - CH10_DATAPATH_CONFIG
    - CH11_DATAPATH_CONFIG
    - CH12_DATAPATH_CONFIG
    - GAIN_CONFIG0
    - GAIN_CONFIG1
    - DATAPATH_OUTPUT
    - INTERRUPTS
    - INTERRUPT_MASK
    - INTERRUPT_CLEAR
    - CORE_ID
- TRNG_NONSECURE –
    - CONTROL
    - STATUS
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - TRNG

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    - CORE_ID
- RCOS_CAL_NONSECURE –
    - CAL_CTRL
    - SLOW_CLK_SET
    - SLOW_CLK_CNT
    - FAST_CLK_CNT
    - SOURCE_CLK_SEL
    - FORCE_RESET
    - STAT
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CORE_ID
- SHUB_NONSECURE –
    - SETUP0
    - SETUP1
    - PORT0_SETUP
    - PORT0_SENSOR0_SETUP
    - PORT0_SENSOR1_SETUP
    - PORT0_SENSOR2_SETUP
    - PORT0_SENSOR3_SETUP
    - PORT0_SENSOR4_SETUP
    - PORT0_SENSOR5_SETUP
    - PORT0_SENSOR6_SETUP
    - PORT0_SENSOR7_SETUP
    - PORT0_SPI_ENG0_CTRL
    - PORT0_I2C_IDW_0
    - PORT0_I2C_IDW_1
    - PORT0_I2C_IDW_2
    - PORT0_I2C_IDW_3
    - PORT0_I2C_IDW_4
    - PORT0_I2C_IDW_5
    - PORT0_I2C_IDW_6
    - PORT0_I2C_IDW_7
    - PORT0_I2C_ADDR_0
    - PORT0_I2C_ADDR_1
    - PORT0_I2C_ADDR_2
    - PORT0_I2C_ADDR_3
    - PORT0_I2C_ADDR_4
    - PORT0_I2C_ADDR_5
    - PORT0_I2C_ADDR_6
    - PORT0_I2C_ADDR_7
    - PORT0_I2C_IDR_0
    - PORT0_I2C_IDR_1
    - PORT0_I2C_IDR_2
    - PORT0_I2C_IDR_3
    - PORT0_I2C_IDR_4
    - PORT0_I2C_IDR_5
    - PORT0_I2C_IDR_6
    - PORT0_I2C_IDR_7
    - PORT0_I2C_DAT_CTRL
    - PORT0_I2C_DAT_LAST_CTRL
    - PORT0_ALM0_QUAN_CTRL
    - PORT0_ALM1_QUAN_CTRL
    - PORT0_ALM0_TRIG_CTRL
    - PORT0_ALM1_TRIG_CTRL
    - PORT0_ALM0_THRHLD_MAX_0
    - PORT0_ALM0_THRHLD_MAX_1
    - PORT0_ALM0_THRHLD_MAX_2
    - PORT0_ALM0_THRHLD_MIN_0
    - PORT0_ALM0_THRHLD_MIN_1
    - PORT0_ALM0_THRHLD_MIN_2
    - PORT0_ALM1_THRHLD_MAX_0
    - PORT0_ALM1_THRHLD_MAX_1

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    - PORT0_ALM1_THRHLD_MAX_2
    - PORT0_ALM1_THRHLD_MIN_0
    - PORT0_ALM1_THRHLD_MIN_1
    - PORT0_ALM1_THRHLD_MIN_2
    - PORT0_STATUS
    - PORT1_SETUP
    - PORT1_SENSOR0_SETUP
    - PORT1_SPI_ENG0_CTRL
    - PORT1_I2C_IDW_0
    - PORT1_I2C_ADDR_0
    - PORT1_I2C_IDR_0
    - PORT1_I2C_DAT_CTRL
    - PORT1_I2C_DAT_LAST_CTRL
    - PORT1_ALM0_QUAN_CTRL
    - PORT1_ALM0_TRIG_CTRL
    - PORT1_ALM0_THRHLD_MAX_0
    - PORT1_ALM0_THRHLD_MAX_1
    - PORT1_ALM0_THRHLD_MAX_2
    - PORT1_ALM0_THRHLD_MIN_0
    - PORT1_ALM0_THRHLD_MIN_1
    - PORT1_ALM0_THRHLD_MIN_2
    - PORT1_STATUS
    - FLASH_CTRL0
    - FLASH_CTRL1
    - FLASH_CTRL2
    - FLASH_CTRL3
    - FLASH_CTRL4
    - FLASH_STATUS0
    - FLASH_STATUS1
    - SHUB_STATUS0
    - PC_CTRL
    - PC_CTRL2
    - REG7
    - SHUB_SRAM_CLK_SEL
    - REGFILE_SEL
    - REGFILE_REGW
    - ID
- PDM0_NONSECURE –
    - CONTROL
    - FILTER_CONFIG
    - FILTER_CONFIG1
    - GAIN_CONTROL_CONFIG
    - INTERRUPTS
    - INTERRUPT_MASK
    - INTERRUPT_CLEAR
    - BUFFER_ACCESS_MODE
    - AUTO_COUNTER
    - BUFFER_DEPTH
    - CORE_ID
    - REV_HASH
    - REV_KEY
- PDM1_NONSECURE –
    - CONTROL
    - FILTER_CONFIG
    - FILTER_CONFIG1
    - GAIN_CONTROL_CONFIG
    - INTERRUPTS
    - INTERRUPT_MASK
    - INTERRUPT_CLEAR
    - BUFFER_ACCESS_MODE
    - AUTO_COUNTER
    - BUFFER_DEPTH
    - CORE_ID
    - REV_HASH
    - REV_KEY

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- WRPR2_NONSECURE –
  - APB0_CTRL
  - APB1_CTRL
  - APB2_CTRL
  - APB3_CTRL
  - APB4_CTRL
  - APB5_CTRL
  - APB6_CTRL
  - APB7_CTRL
  - APB8_CTRL
  - APB9_CTRL
  - APB10_CTRL
  - APB11_CTRL
  - APB12_CTRL
  - APB13_CTRL
  - APB14_CTRL
  - APB15_CTRL
  - CORE_ID
- I2S_NONSECURE –
  - I2S_CTRL0
  - I2S_CTRL1_TX
  - I2S_CTRL1_RX
  - I2S_CTRL2_TX
  - I2S_CTRL2_RX
  - I2S_CTRL3
  - I2S_CTRL4
  - I2S_CTRL5_TX
  - I2S_CTRL5_RX
  - I2S_INTRP_CTRL
  - I2S_PP1_RDATA
  - I2S_PP0_WDATA
  - I2S_PP1_CTRL
  - I2S_PP0_CTRL
  - I2S_IRQ0
  - I2S_IRQM0
  - I2S_IRQS0
  - I2S_IRQC0
  - I2S_IRQ1
  - I2S_IRQM1
  - I2S_IRQS1
  - I2S_IRQC1
  - BUFFER_DEPTH
  - PP_ST
  - I2S_SPARE
  - I2S_DBG
  - ID
- CLKRSTGEN_NONSECURE –
  - CLK_BP_CTRL
  - CLK_BP_CTRL_STAT
  - CLK_AUD_CTRL
  - PLL_CTRL
  - USER_RESETS
  - USER_CLK_DISABLES
  - USER_CLK_GATE_FORCE_ON
  - USER_CLK_GATE_CTRL
  - CONFIGURATION
  - CLKSYNC
  - CORE_ID
- AES_NONSECURE –
  - ALERT_TEST
  - KEY_SHARE0_0
  - KEY_SHARE0_1
  - KEY_SHARE0_2
  - KEY_SHARE0_3
  - KEY_SHARE0_4

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- KEY_SHARE0_5
- KEY_SHARE0_6
- KEY_SHARE0_7
- KEY_SHARE1_0
- KEY_SHARE1_1
- KEY_SHARE1_2
- KEY_SHARE1_3
- KEY_SHARE1_4
- KEY_SHARE1_5
- KEY_SHARE1_6
- KEY_SHARE1_7
- IV_0
- IV_1
- IV_2
- IV_3
- DATA_IN_0
- DATA_IN_1
- DATA_IN_2
- DATA_IN_3
- DATA_OUT_0
- DATA_OUT_1
- DATA_OUT_2
- DATA_OUT_3
- CTRL_SHADOWED
- TRIGGER
- STATUS
- STATUS1
- SIDELOAD_CTRL
- ID

- AT_DMA_NONSECURE –
  - OPMODE
  - CONST_WDATA
  - SRC_ADDR
  - TAR_ADDR
  - SIZE
  - SRC_CTRL
  - TAR_CTRL
  - FIFO_DPTH_ADDR
  - FIFO_PORT_SEL
  - SPI_PORT_SEL
  - ERR_STAT
  - STATUS
  - TOTAL_WRITE_REMAINDER
  - INTERRUPT_STATUS
  - INTERRUPT_MASK
  - SET_INTERRUPT
  - RESET_INTERRUPT
  - CFG_HNONSEC
  - CHAN1_OPMODE
  - CHAN1_CONST_WDATA
  - CHAN1_SRC_ADDR
  - CHAN1_TAR_ADDR
  - CHAN1_SIZE
  - CHAN1_SRC_CTRL
  - CHAN1_TAR_CTRL
  - CHAN1_FIFO_DPTH_ADDR
  - CHAN1_FIFO_PORT_SEL
  - CHAN1_SPI_PORT_SEL
  - CHAN1_ERR_STAT
  - CHAN1_STATUS
  - CHAN1_TOTAL_WRITE_REMAINDER
  - CHAN1_INTERRUPT_STATUS
  - CHAN1_INTERRUPT_MASK
  - CHAN1_SET_INTERRUPT
  - CHAN1_RESET_INTERRUPT

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    - CHAN1_CFG_HNONSEC
    - CHAN2_OPMODE
    - CHAN2_CONST_WDATA
    - CHAN2_SRC_ADDR
    - CHAN2_TAR_ADDR
    - CHAN2_SIZE
    - CHAN2_SRC_CTRL
    - CHAN2_TAR_CTRL
    - CHAN2_FIFO_DPTH_ADDR
    - CHAN2_FIFO_PORT_SEL
    - CHAN2_SPI_PORT_SEL
    - CHAN2_ERR_STAT
    - CHAN2_STATUS
    - CHAN2_TOTAL_WRITE_REMAINDER
    - CHAN2_INTERRUPT_STATUS
    - CHAN2_INTERRUPT_MASK
    - CHAN2_SET_INTERRUPT
    - CHAN2_RESET_INTERRUPT
    - CHAN2_CFG_HNONSEC
    - CHAN3_OPMODE
    - CHAN3_CONST_WDATA
    - CHAN3_SRC_ADDR
    - CHAN3_TAR_ADDR
    - CHAN3_SIZE
    - CHAN3_SRC_CTRL
    - CHAN3_TAR_CTRL
    - CHAN3_FIFO_DPTH_ADDR
    - CHAN3_FIFO_PORT_SEL
    - CHAN3_SPI_PORT_SEL
    - CHAN3_ERR_STAT
    - CHAN3_STATUS
    - CHAN3_TOTAL_WRITE_REMAINDER
    - CHAN3_INTERRUPT_STATUS
    - CHAN3_INTERRUPT_MASK
    - CHAN3_SET_INTERRUPT
    - CHAN3_RESET_INTERRUPT
    - CHAN3_CFG_HNONSEC
    - ERR_INTERRUPT_DLY
    - ARBITER_CTRL
    - ARBITER_GRANT_STAT
    - DIAGCNTL
    - ID
- GPIO0_SECURE –
    - DATA
    - DATAOUT
    - INENABLE_SET
    - INENABLE_CLR
    - OUTENSET
    - OUTENCLR
    - ALTFUNCSET
    - ALTFUNCCLR
    - INTENSET
    - INTENCLR
    - INTTYPESET
    - INTTYPECLR
    - INTPOLSET
    - INTPOLCLR
    - INTSTATUS_CLEAR
    - PULLUP_ENABLE_SET
    - PULLUP_ENABLE_CLR
    - PULLDOWN_ENABLE_SET
    - PULLDOWN_ENABLE_CLR
    - MASKLOWBYTE
    - MASKHIGHBYTE
    - PID4

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    - PID5
    - PID6
    - PID7
    - PID0
    - PID1
    - PID2
    - PID3
    - CID0
    - CID1
    - CID2
    - CID3
- GPIO1_SECURE –
    - DATA
    - DATAOUT
    - INENABLE_SET
    - INENABLE_CLR
    - OUTENSET
    - OUTENCLR
    - ALTFUNCSET
    - ALTFUNCCLR
    - INTENSET
    - INTENCLR
    - INTTYPESET
    - INTTYPECLR
    - INTPOLSET
    - INTPOLCLR
    - INTSTATUS_CLEAR
    - PULLUP_ENABLE_SET
    - PULLUP_ENABLE_CLR
    - PULLDOWN_ENABLE_SET
    - PULLDOWN_ENABLE_CLR
    - MASKLOWBYTE
    - MASKHIGHBYTE
    - PID4
    - PID5
    - PID6
    - PID7
    - PID0
    - PID1
    - PID2
    - PID3
    - CID0
    - CID1
    - CID2
    - CID3
- SHA2_SECURE –
    - CONTROL
    - SIDELOAD_CTRL
    - CMD
    - MESSAGE_LENGTH_LO
    - MESSAGE_LENGTH_HI
    - WIPE_V
    - FIFO_PUSH
    - DIGEST0
    - DIGEST1
    - DIGEST2
    - DIGEST3
    - DIGEST4
    - DIGEST5
    - DIGEST6
    - DIGEST7
    - KEY_0
    - KEY_1
    - KEY_2
    - KEY_3

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    - KEY_4
    - KEY_5
    - KEY_6
    - KEY_7
    - STATUS
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - ERR_CODE
    - ID
- WRPR0_SECURE –
    - APB0_CTRL
    - APB1_CTRL
    - APB2_CTRL
    - APB3_CTRL
    - APB4_CTRL
    - APB5_CTRL
    - APB6_CTRL
    - APB7_CTRL
    - APB8_CTRL
    - APB9_CTRL
    - APB10_CTRL
    - APB11_CTRL
    - APB12_CTRL
    - APB13_CTRL
    - APB14_CTRL
    - APB15_CTRL
    - CLK_HPC_PIN_OUT_CTRL
    - CLK_MPC_PIN_OUT_CTRL
    - CLK_LPC_PIN_OUT_CTRL
    - DBG_CTRL
    - PSEL_A
    - PSEL_B
    - PSEL_C
    - PSEL_D
    - PSEL_E
    - PSEL_F
    - PSEL_G
    - PSEL_H
    - PSEL_I
    - PUPD_OVRD
    - PUPD_OVRD_VAL
    - REMAP
    - AHB_INTERPOSERS
    - SCRATCHPAD_A
    - SCRATCHPAD_B
    - SCRATCHPAD_C
    - SCRATCHPAD_D
    - RRAM_WRITE_PROTECTION0
    - RRAM_WRITE_PROTECTION1
    - RRAM_WRITE_PROTECTION2
    - RRAM_WRITE_PROTECTION3
    - RRAM_WRITE_PROTECTION4
    - RRAM_WRITE_PROTECTION5
    - RRAM_WRITE_PROTECTION6
    - RRAM_WRITE_PROTECTION7
    - RRAM_STICKY_WRITE_PROTECTION0
    - RRAM_STICKY_WRITE_PROTECTION1
    - RRAM_STICKY_WRITE_PROTECTION2
    - RRAM_STICKY_WRITE_PROTECTION3
    - RRAM_STICKY_WRITE_PROTECTION4
    - RRAM_STICKY_WRITE_PROTECTION5
    - RRAM_STICKY_WRITE_PROTECTION6
    - RRAM_STICKY_WRITE_PROTECTION7

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- RRAM_STICKY_WRITE_PROTECTION_STATUS0
- RRAM_STICKY_WRITE_PROTECTION_STATUS1
- RRAM_STICKY_WRITE_PROTECTION_STATUS2
- RRAM_STICKY_WRITE_PROTECTION_STATUS3
- RRAM_STICKY_WRITE_PROTECTION_STATUS4
- RRAM_STICKY_WRITE_PROTECTION_STATUS5
- RRAM_STICKY_WRITE_PROTECTION_STATUS6
- RRAM_STICKY_WRITE_PROTECTION_STATUS7
- RRAM_STICKY_READ_PROTECTION0
- RRAM_STICKY_READ_PROTECTION_STATUS0
- PROT_BITS_SET0
- PROT_BITS_SET1
- PROT_BITS_STAT0
- PROT_BITS_STAT1
- QSPI_STICKY_WRITE_PROTECTION0
- QSPI_STICKY_WRITE_PROTECTION1
- QSPI_STICKY_WRITE_PROTECTION_STATUS0
- QSPI_STICKY_WRITE_PROTECTION_STATUS1
- PDSN
- CALCOUNTRC
- SECURE_DEBUG_CTRL
- SECURE_DEBUG_STAT
- CHIPID1
- CHIPID2
- CHIPREV
- CORE_ID

- UART0_SECURE –
  - DATA
  - STATE
  - CTRL
  - INTSTATUS_CLEAR
  - BAUDDIV
  - RX_LWM
  - TX_LWM
  - RX_FIFO_SPACES
  - TX_FIFO_SPACES
  - HW_FLOW_OVRD
  - BAUD_RATE_DETECT
  - BAUD_DIV_CEIL
  - BRD_STAT
  - INTMASK
  - RX_FIFO_ENTRIES
  - INTSTATUS_SET
  - PID4
  - PID5
  - PID6
  - PID7
  - PID0
  - PID1
  - PID2
  - PID3
  - CID0
  - CID1
  - CID2
  - CID3
- UART1_SECURE –
  - DATA
  - STATE
  - CTRL
  - INTSTATUS_CLEAR
  - BAUDDIV
  - RX_LWM
  - TX_LWM
  - RX_FIFO_SPACES
  - TX_FIFO_SPACES

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    - HW_FLOW_OVRD
    - BAUD_RATE_DETECT
    - BAUD_DIV_CEIL
    - BRD_STAT
    - INTMASK
    - RX_FIFO_ENTRIES
    - INTSTATUS_SET
    - PID4
    - PID5
    - PID6
    - PID7
    - PID0
    - PID1
    - PID2
    - PID3
    - CID0
    - CID1
    - CID2
    - CID3
- PWM_SECURE –
    - PWM_CTRL
    - PWM0_CTRL
    - PWM0_DUR
    - PWM0_CFG
    - PWM1_CTRL
    - PWM1_DUR
    - PWM1_CFG
    - PWM2_CTRL
    - PWM2_DUR
    - PWM2_CFG
    - PWM3_CTRL
    - PWM3_DUR
    - PWM3_CFG
    - PWM4_CTRL
    - PWM4_DUR
    - PWM4_CFG
    - PWM5_CTRL
    - PWM5_DUR
    - PWM5_CFG
    - PWM6_CTRL
    - PWM6_DUR
    - PWM6_CFG
    - PWM7_CTRL
    - PWM7_DUR
    - PWM7_CFG
    - FSM_STATE
    - FIFO_CFG
    - FIFO_CARRIER1_DUR
    - FIFO_CARRIER2_DUR
    - FIFO_STAT
    - FIFO_STAT1
    - FIFO_SEL_ERR_SOURCE
    - INTERRUPTS
    - INTERRUPTS_MASK
    - INTERRUPTS_CLEAR
    - INTERRUPTS_SET
    - FIFO_DATA
    - CORE_ID
    - REV_HASH
    - REV_KEY
- SPI0_SECURE –
    - CTRL
    - TRANSACTION_SETUP
    - TRANSACTION_SETUP_DMA
    - TRANSACTION_STATUS

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    - DATA_BYTES_LOWER
    - DATA_BYTES_UPPER
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CORE_ID
- KSM_SECURE –
    - CTRL0
    - TIME_PARAM0
    - TIME_PARAM1
    - MATRIX_SIZE
    - KEYBOARD_PACKET
    - LOW_POWER
    - LPBACK_KSI
    - STATUS
    - INTERRUPTS
    - INTERRUPT_MASK
    - INTERRUPT_CLEAR
    - DEBUG
    - DEBUG2
    - CORE_ID
- SPI2_SECURE –
    - CTRL
    - TRANSACTION_SETUP
    - TRANSACTION_SETUP_DMA
    - TRANSACTION_STATUS
    - DATA_BYTES_LOWER
    - DATA_BYTES_UPPER
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CORE_ID
- QDEC_SECURE –
    - SETUP_FOR_AXIS_0
    - ACCUM_PURGE_STATUS_FOR_AXIS_0
    - ACCUMULATOR_VALUE_FOR_AXIS_0
    - ERR_PURGE_STATUS_FOR_AXIS_0
    - ERROR_COUNT_FOR_AXIS_0
    - SETUP_FOR_AXIS_1
    - ACCUM_PURGE_STATUS_FOR_AXIS_1
    - ACCUMULATOR_VALUE_FOR_AXIS_1
    - ERR_PURGE_STATUS_FOR_AXIS_1
    - ERROR_COUNT_FOR_AXIS_1
    - SETUP_FOR_AXIS_2
    - ACCUM_PURGE_STATUS_FOR_AXIS_2
    - ACCUMULATOR_VALUE_FOR_AXIS_2
    - ERR_PURGE_STATUS_FOR_AXIS_2
    - ERROR_COUNT_FOR_AXIS_2
    - CORE_ID
- SPI1_SECURE –
    - CTRL
    - TRANSACTION_SETUP
    - TRANSACTION_SETUP_DMA
    - TRANSACTION_STATUS
    - DATA_BYTES_LOWER
    - DATA_BYTES_UPPER
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CORE_ID
- SLWTIMER_SECURE –
    - CONTROL

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    - INIT_LOW
    - INIT_HIGH
    - CNT_LOW
    - CNT_HIGH
    - THRES0_LOW
    - THRES0_HIGH
    - THRES1_LOW
    - THRES1_HIGH
    - THRES2_LOW
    - THRES2_HIGH
    - STATUS
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - CORE_ID
- QSPI_SECURE –
    - TRANSACTION_SETUP
    - READ_DATA
    - OVERRIDE_DIN
    - MODE
    - REMOTE_AHB_SETUP
    - REMOTE_AHB_SETUP_2
    - REMOTE_AHB_SETUP_3
    - REMOTE_AHB_SETUP_4
    - REMOTE_AHB_SETUP_5
    - REMOTE_AHB_SETUP_6
    - REMOTE_AHB_SETUP_7
    - REMOTE_AHB_WLE_CNT
    - REMOTE_AHB_WIP_CNT
    - REMOTE_AHB_DBG0
    - REMOTE_AHB_DBG1
    - REMOTE_AHB_DBG2
    - CORE_ID
- I2C0_SECURE –
    - TRANSACTION_SETUP
    - CLOCK_CONTROL
    - OUTGOING_DATA
    - OUTGOING_DATA1
    - IO_CTRL
    - INCOMING_DATA
    - INCOMING_DATA1
    - TRANSACTION_STATUS
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - ID
- I2C1_SECURE –
    - TRANSACTION_SETUP
    - CLOCK_CONTROL
    - OUTGOING_DATA
    - OUTGOING_DATA1
    - IO_CTRL
    - INCOMING_DATA
    - INCOMING_DATA1
    - TRANSACTION_STATUS
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - ID
- NVM_SECURE –
    - OPMODE
    - CTRL
    - DATA0
    - DATA1

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    - STATUS
    - T_SP_VQ
    - T_HP_VQ
    - T_SP_PG
    - T_HP_PG
    - T_HP_CK
    - T_S_PGM
    - T_H_PGM
    - T_CKHP
    - T_CKLP
    - T_SPS_CK
    - T_SR_CK
    - T_HR_CK
    - T_CKLR
    - T_CKHR
    - T_SRS_CK
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - INTERRUPT_SET
    - INTERRUPT_RESET
    - CORE_ID
- WRPR1_SECURE –
    - APB0_CTRL
    - APB1_CTRL
    - APB2_CTRL
    - APB3_CTRL
    - APB4_CTRL
    - APB5_CTRL
    - APB6_CTRL
    - APB7_CTRL
    - APB8_CTRL
    - APB9_CTRL
    - APB10_CTRL
    - APB11_CTRL
    - APB12_CTRL
    - APB13_CTRL
    - APB14_CTRL
    - APB15_CTRL
    - CORE_ID
- RADIO_SECURE –
    - CTRL
    - TRANSACTION_SETUP
    - TRANSACTION_SETUP_DMA
    - TRANSACTION_STATUS
    - DATA_BYTES_LOWER
    - DATA_BYTES_UPPER
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CORE_ID
- PMU_SECURE –
    - CTRL
    - TRANSACTION_SETUP
    - TRANSACTION_SETUP_DMA
    - TRANSACTION_STATUS
    - DATA_BYTES_LOWER
    - DATA_BYTES_UPPER
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CORE_ID
- PSEQ_SECURE – WURX_CONFIG, WURX_CONFIG1 and WURX_CONFIG2 are for ATM3330e and ATM3330 Only
    - CTRL0

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- CTRL1
- RETAIN_ALL_WAKE_MASK
- HIB_WAKE_MASK
- GPIO_WAKE_MASK
- GPIO_WAKE_TYPE
- GPIO_WAKE_POL
- GPIO_WAKE_BOTH_EDGES
- GPIO_WAKE_STATUS
- WURX_CONFIG
- WURX_CONFIG1
- WURX_CONFIG2
- GADC_CONFIG
- XTAL_BITS0
- XTAL_BITS1
- OVERRIDES
- OVERRIDES2
- OVERRIDES3
- OVERRIDES4
- OVERRIDES5
- SYSRAM_OVERRIDES
- SYSRAM_OVERRIDES2
- SYSRAM_OVERRIDES3
- SYSRAM_OVERRIDES4
- SYSRAM_OVERRIDES5
- SYSRAM_OVERRIDES6
- SYSRAM_OVERRIDES7
- SYSRAM_OVERRIDES8
- EMRAM_OVERRIDES
- EMRAM_OVERRIDES2
- EMRAM_OVERRIDES3
- EMRAM_OVERRIDES4
- EMRAM_OVERRIDES5
- EMRAM_OVERRIDES6
- EMRAM_OVERRIDES7
- EMRAM_OVERRIDES8
- ICACHE_OVERRIDES
- ICACHE_OVERRIDES2
- ICACHE_OVERRIDES3
- ICACHE_OVERRIDES4
- ICACHE_OVERRIDES5
- ICACHE_OVERRIDES6
- ICACHE_OVERRIDES7
- ICACHE_OVERRIDES8
- RRAMCACHE_OVERRIDES
- RRAMCACHE_OVERRIDES2
- RRAMCACHE_OVERRIDES3
- RRAMCACHE_OVERRIDES4
- RRAMCACHE_OVERRIDES5
- RRAMCACHE_OVERRIDES6
- RRAMCACHE_OVERRIDES7
- RRAMCACHE_OVERRIDES8
- QSPICACHE_OVERRIDES
- COUNTER_CONTROL
- CURRENT_REAL_TIME
- CURRENT_COUNT_DOWN_TIME
- INIT_COUNT_DOWN
- INST_PENDING
- STATUS
- PERSISTENT0
- PERSISTENT1
- PERSISTENT2
- PERSISTENT3
- PERSISTENT4
- PERSISTENT5
- PERSISTENT6

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    - PERSISTENT7
    - SENSOR_HUB_CONTROL
    - KSMQDEC_CONTROL
    - SPII2C_CONTROL
    - FLASH_CONTROL
    - FLASH_CONTROL2
    - PMU_STATUS
    - PMU_INTERRUPT
    - SYSRAM_WRITE_BLOCK_CFG
    - SYSRAM_SSRS_CFG
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - RESET_INTERRUPT
    - FPGA_CFG
    - PLL
    - RADIO_STATUS
    - RRAM_READ_CONFIG_LO
    - RRAM_READ_CONFIG_HI
    - SENSOR_HUB_IMM_BREAKOUT_MASK
    - FSM_TUNINGA
    - FSM_TUNINGB
    - WDOG_CONTROL
    - WDOG_TIMER
    - CORE_ID
- SWD_SECURE –
    - DAP_ADDR_CTRL
    - DAP_WDATA_CTRL
    - SWD_CLK_DIV_CTRL
    - PATTERN_UPPER_CTRL
    - PATTERN_LOWER_CTRL
    - TAIL_DUR
    - PATTERN_DUR
    - DAP_BRIDGE_STATUS
    - DAP_RESP_STATUS
    - RW_CTRL
    - APNDP_CTRL
    - TRIGGER_CTRL
    - DAP_RDATA
    - INTERRUPT_MASK
    - INTERRUPT_STATUS
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - DTOP_BYPASS_CTRL
    - DTOP_BYPASS_STAT
    - DTOP_BYPASS_GPO
    - DTOP_BYPASS_RIF_MODE_CNTL
    - DTOP_BYPASS_RIF_CAL_CNTL
    - DTOP_BYPASS_RIF_RXDC7
    - CORE_ID
- GADC_SECURE –
    - CTRL
    - CTRL1
    - CH1_DATAPATH_CONFIG
    - CH2_DATAPATH_CONFIG
    - CH3_DATAPATH_CONFIG
    - CH4_DATAPATH_CONFIG
    - CH5_DATAPATH_CONFIG
    - CH6_DATAPATH_CONFIG
    - CH7_DATAPATH_CONFIG
    - CH8_DATAPATH_CONFIG
    - CH9_DATAPATH_CONFIG
    - CH10_DATAPATH_CONFIG
    - CH11_DATAPATH_CONFIG
    - CH12_DATAPATH_CONFIG
    - GAIN_CONFIG0

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    - GAIN_CONFIG1
    - DATAPATH_OUTPUT
    - INTERRUPTS
    - INTERRUPT_MASK
    - INTERRUPT_CLEAR
    - CORE_ID
- TRNG_SECURE –
    - CONTROL
    - STATUS
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - TRNG
    - CORE_ID
- RCOS_CAL_SECURE –
    - CAL_CTRL
    - SLOW_CLK_SET
    - SLOW_CLK_CNT
    - FAST_CLK_CNT
    - SOURCE_CLK_SEL
    - FORCE_RESET
    - STAT
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CORE_ID
- SHUB_SECURE –
    - SETUP0
    - SETUP1
    - PORT0_SETUP
    - PORT0_SENSOR0_SETUP
    - PORT0_SENSOR1_SETUP
    - PORT0_SENSOR2_SETUP
    - PORT0_SENSOR3_SETUP
    - PORT0_SENSOR4_SETUP
    - PORT0_SENSOR5_SETUP
    - PORT0_SENSOR6_SETUP
    - PORT0_SENSOR7_SETUP
    - PORT0_SPI_ENG0_CTRL
    - PORT0_I2C_IDW_0
    - PORT0_I2C_IDW_1
    - PORT0_I2C_IDW_2
    - PORT0_I2C_IDW_3
    - PORT0_I2C_IDW_4
    - PORT0_I2C_IDW_5
    - PORT0_I2C_IDW_6
    - PORT0_I2C_IDW_7
    - PORT0_I2C_ADDR_0
    - PORT0_I2C_ADDR_1
    - PORT0_I2C_ADDR_2
    - PORT0_I2C_ADDR_3
    - PORT0_I2C_ADDR_4
    - PORT0_I2C_ADDR_5
    - PORT0_I2C_ADDR_6
    - PORT0_I2C_ADDR_7
    - PORT0_I2C_IDR_0
    - PORT0_I2C_IDR_1
    - PORT0_I2C_IDR_2
    - PORT0_I2C_IDR_3
    - PORT0_I2C_IDR_4
    - PORT0_I2C_IDR_5
    - PORT0_I2C_IDR_6
    - PORT0_I2C_IDR_7

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- PORT0_I2C_DAT_CTRL
- PORT0_I2C_DAT_LAST_CTRL
- PORT0_ALM0_QUAN_CTRL
- PORT0_ALM1_QUAN_CTRL
- PORT0_ALM0_TRIG_CTRL
- PORT0_ALM1_TRIG_CTRL
- PORT0_ALM0_THRHLD_MAX_0
- PORT0_ALM0_THRHLD_MAX_1
- PORT0_ALM0_THRHLD_MAX_2
- PORT0_ALM0_THRHLD_MIN_0
- PORT0_ALM0_THRHLD_MIN_1
- PORT0_ALM0_THRHLD_MIN_2
- PORT0_ALM1_THRHLD_MAX_0
- PORT0_ALM1_THRHLD_MAX_1
- PORT0_ALM1_THRHLD_MAX_2
- PORT0_ALM1_THRHLD_MIN_0
- PORT0_ALM1_THRHLD_MIN_1
- PORT0_ALM1_THRHLD_MIN_2
- PORT0_STATUS
- PORT1_SETUP
- PORT1_SENSOR0_SETUP
- PORT1_SPI_ENG0_CTRL
- PORT1_I2C_IDW_0
- PORT1_I2C_ADDR_0
- PORT1_I2C_IDR_0
- PORT1_I2C_DAT_CTRL
- PORT1_I2C_DAT_LAST_CTRL
- PORT1_ALM0_QUAN_CTRL
- PORT1_ALM0_TRIG_CTRL
- PORT1_ALM0_THRHLD_MAX_0
- PORT1_ALM0_THRHLD_MAX_1
- PORT1_ALM0_THRHLD_MAX_2
- PORT1_ALM0_THRHLD_MIN_0
- PORT1_ALM0_THRHLD_MIN_1
- PORT1_ALM0_THRHLD_MIN_2
- PORT1_STATUS
- FLASH_CTRL0
- FLASH_CTRL1
- FLASH_CTRL2
- FLASH_CTRL3
- FLASH_CTRL4
- FLASH_STATUS0
- FLASH_STATUS1
- SHUB_STATUS0
- PC_CTRL
- PC_CTRL2
- REG7
- SHUB_SRAM_CLK_SEL
- REGFILE_SEL
- REGFILE_REGW
- ID

- PDM0_SECURE –
  - CONTROL
  - FILTER_CONFIG
  - FILTER_CONFIG1
  - GAIN_CONTROL_CONFIG
  - INTERRUPTS
  - INTERRUPT_MASK
  - INTERRUPT_CLEAR
  - BUFFER_ACCESS_MODE
  - AUTO_COUNTER
  - BUFFER_DEPTH
  - CORE_ID
  - REV_HASH
  - REV_KEY

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- PDM1_SECURE –
  - CONTROL
  - FILTER_CONFIG
  - FILTER_CONFIG1
  - GAIN_CONTROL_CONFIG
  - INTERRUPTS
  - INTERRUPT_MASK
  - INTERRUPT_CLEAR
  - BUFFER_ACCESS_MODE
  - AUTO_COUNTER
  - BUFFER_DEPTH
  - CORE_ID
  - REV_HASH
  - REV_KEY
- WRPR2_SECURE –
  - APB0_CTRL
  - APB1_CTRL
  - APB2_CTRL
  - APB3_CTRL
  - APB4_CTRL
  - APB5_CTRL
  - APB6_CTRL
  - APB7_CTRL
  - APB8_CTRL
  - APB9_CTRL
  - APB10_CTRL
  - APB11_CTRL
  - APB12_CTRL
  - APB13_CTRL
  - APB14_CTRL
  - APB15_CTRL
  - CORE_ID
- I2S_SECURE –
  - I2S_CTRL0
  - I2S_CTRL1_TX
  - I2S_CTRL1_RX
  - I2S_CTRL2_TX
  - I2S_CTRL2_RX
  - I2S_CTRL3
  - I2S_CTRL4
  - I2S_CTRL5_TX
  - I2S_CTRL5_RX
  - I2S_INTRP_CTRL
  - I2S_PP1_RDATA
  - I2S_PP0_WDATA
  - I2S_PP1_CTRL
  - I2S_PP0_CTRL
  - I2S_IRQ0
  - I2S_IRQM0
  - I2S_IRQS0
  - I2S_IRQC0
  - I2S_IRQ1
  - I2S_IRQM1
  - I2S_IRQS1
  - I2S_IRQC1
  - BUFFER_DEPTH
  - PP_ST
  - I2S_SPARE
  - I2S_DBG
  - ID
- CLKRSTGEN_SECURE –
  - CLK_BP_CTRL
  - CLK_BP_CTRL_STAT
  - CLK_AUD_CTRL
  - PLL_CTRL

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    - USER_RESETS
    - USER_CLK_DISABLES
    - USER_CLK_GATE_FORCE_ON
    - USER_CLK_GATE_CTRL
    - CONFIGURATION
    - CLKSYNC
    - CORE_ID
- AES_SECURE –
    - ALERT_TEST
    - KEY_SHARE0_0
    - KEY_SHARE0_1
    - KEY_SHARE0_2
    - KEY_SHARE0_3
    - KEY_SHARE0_4
    - KEY_SHARE0_5
    - KEY_SHARE0_6
    - KEY_SHARE0_7
    - KEY_SHARE1_0
    - KEY_SHARE1_1
    - KEY_SHARE1_2
    - KEY_SHARE1_3
    - KEY_SHARE1_4
    - KEY_SHARE1_5
    - KEY_SHARE1_6
    - KEY_SHARE1_7
    - IV_0
    - IV_1
    - IV_2
    - IV_3
    - DATA_IN_0
    - DATA_IN_1
    - DATA_IN_2
    - DATA_IN_3
    - DATA_OUT_0
    - DATA_OUT_1
    - DATA_OUT_2
    - DATA_OUT_3
    - CTRL_SHADOWED
    - TRIGGER
    - STATUS
    - STATUS1
    - SIDELOAD_CTRL
    - ID
- AT_DMA_SECURE –
    - OPMODE
    - CONST_WDATA
    - SRC_ADDR
    - TAR_ADDR
    - SIZE
    - SRC_CTRL
    - TAR_CTRL
    - FIFO_DPTH_ADDR
    - FIFO_PORT_SEL
    - SPI_PORT_SEL
    - ERR_STAT
    - STATUS
    - TOTAL_WRITE_REMAINDER
    - INTERRUPT_STATUS
    - INTERRUPT_MASK
    - SET_INTERRUPT
    - RESET_INTERRUPT
    - CFG_HNONSEC
    - CHAN1_OPMODE
    - CHAN1_CONST_WDATA
    - CHAN1_SRC_ADDR

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- CHAN1_TAR_ADDR
- CHAN1_SIZE
- CHAN1_SRC_CTRL
- CHAN1_TAR_CTRL
- CHAN1_FIFO_DPTH_ADDR
- CHAN1_FIFO_PORT_SEL
- CHAN1_SPI_PORT_SEL
- CHAN1_ERR_STAT
- CHAN1_STATUS
- CHAN1_TOTAL_WRITE_REMAINDER
- CHAN1_INTERRUPT_STATUS
- CHAN1_INTERRUPT_MASK
- CHAN1_SET_INTERRUPT
- CHAN1_RESET_INTERRUPT
- CHAN1_CFG_HNONSEC
- CHAN2_OPMODE
- CHAN2_CONST_WDATA
- CHAN2_SRC_ADDR
- CHAN2_TAR_ADDR
- CHAN2_SIZE
- CHAN2_SRC_CTRL
- CHAN2_TAR_CTRL
- CHAN2_FIFO_DPTH_ADDR
- CHAN2_FIFO_PORT_SEL
- CHAN2_SPI_PORT_SEL
- CHAN2_ERR_STAT
- CHAN2_STATUS
- CHAN2_TOTAL_WRITE_REMAINDER
- CHAN2_INTERRUPT_STATUS
- CHAN2_INTERRUPT_MASK
- CHAN2_SET_INTERRUPT
- CHAN2_RESET_INTERRUPT
- CHAN2_CFG_HNONSEC
- CHAN3_OPMODE
- CHAN3_CONST_WDATA
- CHAN3_SRC_ADDR
- CHAN3_TAR_ADDR
- CHAN3_SIZE
- CHAN3_SRC_CTRL
- CHAN3_TAR_CTRL
- CHAN3_FIFO_DPTH_ADDR
- CHAN3_FIFO_PORT_SEL
- CHAN3_SPI_PORT_SEL
- CHAN3_ERR_STAT
- CHAN3_STATUS
- CHAN3_TOTAL_WRITE_REMAINDER
- CHAN3_INTERRUPT_STATUS
- CHAN3_INTERRUPT_MASK
- CHAN3_SET_INTERRUPT
- CHAN3_RESET_INTERRUPT
- CHAN3_CFG_HNONSEC
- ERR_INTERRUPT_DLY
- ARBITER_CTRL
- ARBITER_GRANT_STAT
- DIAGCNTL
- ID

- PMU_TOP –
  - PMU0
  - PMU1
  - PMU1A
  - PMU11
  - PMU15
  - PMU16
  - CHRG_CTRL
  - CHRG_CTRL2

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- PMU_RB_MPPT
- PMU_RB4
- PMU_RB5
- CORE_ID

- PMU_WURX – For ATM3330e and ATM3330 Only
  - WURX_GENERAL
  - WURX0_0
  - WURX0_1
  - WURX0_2
  - WURX0_3
  - WURX0_4
  - WURX1_0
  - WURX1_1
  - WURX1_2
  - WURX1_3
  - WURX1_4
  - WURX_SPARE


# AT_PRRF_NONSECURE

**Name** PATCH_TAG0
**Address** 0x1ffe00
**Description** Contains register fields associated with patch_tag0.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG1
**Address** 0x1ffe04
**Description** Contains register fields associated with patch_tag1.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG2
**Address** 0x1ffe08

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**Description** Contains register fields associated with patch_tag2.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22** | **21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG3
**Address** 0x1ffe0c
**Description** Contains register fields associated with patch_tag3.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22** | **21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG4
**Address** 0x1ffe10
**Description** Contains register fields associated with patch_tag4.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22** | **21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG5
**Address** 0x1ffe14
**Description** Contains register fields associated with patch_tag5.
**Reset** 0x00000000

| | nw | RS | ad |
|---|---|---|---|

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| Name | nword | RSVD | addr |
|---|---|---|---|
| Bits | 31 30 | 29 28 27 26 25 24 23 22 | 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG6
**Address** 0x1ffe18
**Description** Contains register fields associated with patch_tag6.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| Bits | 31 30 | 29 28 27 26 25 24 23 22 | 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG7
**Address** 0x1ffe1c
**Description** Contains register fields associated with patch_tag7.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| Bits | 31 30 | 29 28 27 26 25 24 23 22 | 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG8
**Address** 0x1ffe20
**Description** Contains register fields associated with patch_tag8.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| Bits | 31 30 | 29 28 27 26 25 24 23 22 | 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG9
**Address** 0x1ffe24
**Description** Contains register fields associated with patch_tag9.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| Bits | 31 30 | 29 28 27 26 25 24 23 22 | 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG10
**Address** 0x1ffe28
**Description** Contains register fields associated with patch_tag10.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| Bits | 31 30 | 29 28 27 26 25 24 23 22 | 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG11
**Address** 0x1ffe2c
**Description** Contains register fields associated with patch_tag11.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| Bits | 31 30 | 29 28 27 26 25 24 23 22 | 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG12
**Address** 0x1ffe30
**Description** Contains register fields associated with patch_tag12.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31 30** | **29 28 27 26 25 24 23 22** | **21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG13
**Address** 0x1ffe34
**Description** Contains register fields associated with patch_tag13.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31 30** | **29 28 27 26 25 24 23 22** | **21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG14
**Address** 0x1ffe38
**Description** Contains register fields associated with patch_tag14.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31 30** | **29 28 27 26 25 24 23 22** | **21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG15
**Address** 0x1ffe3c
**Description** Contains register fields associated with patch_tag15.
**Reset** 0x00000000

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| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31 30** | **29 28 27 26 25 24 23 22** | **21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH0
**Address** 0x1ffe40
**Description** Contains register fields associated with patch0.
**Reset** 0x00000000

| Name | patch0 |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch0 | (no description) | 32'b0 |

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**Name** PATCH1
**Address** 0x1ffe44
**Description** Contains register fields associated with patch1.
**Reset** 0x00000000

| Name | patch1 |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch1 | (no description) | 32'b0 |

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**Name** PATCH2
**Address** 0x1ffe48
**Description** Contains register fields associated with patch2.
**Reset** 0x00000000

| Name | patch2 |
|---|---|
| | |

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch2 | (no description) | 32'b0 |

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**Name** PATCH3
**Address** 0x1ffe4c
**Description** Contains register fields associated with patch3.
**Reset** 0x00000000

| Name | patch3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch3 | (no description) | 32'b0 |

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**Name** PATCH4
**Address** 0x1ffe50
**Description** Contains register fields associated with patch4.
**Reset** 0x00000000

| Name | patch4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch4 | (no description) | 32'b0 |

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**Name** PATCH5
**Address** 0x1ffe54
**Description** Contains register fields associated with patch5.
**Reset** 0x00000000

| Name | patch5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch5 | (no description) | 32'b0 |

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**Name** PATCH6
**Address** 0x1ffe58
**Description** Contains register fields associated with patch6.
**Reset** 0x00000000

| Name | patch6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch6 | (no description) | 32'b0 |

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**Name** PATCH7
**Address** 0x1ffe5c
**Description** Contains register fields associated with patch7.
**Reset** 0x00000000

| Name | patch7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch7 | (no description) | 32'b0 |

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**Name** PATCH8
**Address** 0x1ffe60
**Description** Contains register fields associated with patch8.
**Reset** 0x00000000

| Name | patch8 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch8 | (no description) | 32'b0 |

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**Name** PATCH9
**Address** 0x1ffe64
**Description** Contains register fields associated with patch9.
**Reset** 0x00000000

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| Name | patch9 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch9 | (no description) | 32'b0 |

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**Name** PATCH10
**Address** 0x1ffe68
**Description** Contains register fields associated with patch10.
**Reset** 0x00000000

| Name | patch10 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch10 | (no description) | 32'b0 |

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**Name** PATCH11
**Address** 0x1ffe6c
**Description** Contains register fields associated with patch11.
**Reset** 0x00000000

| Name | patch11 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch11 | (no description) | 32'b0 |

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**Name** PATCH12
**Address** 0x1ffe70
**Description** Contains register fields associated with patch12.
**Reset** 0x00000000

| Name | patch |
|---|---|

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| | 12 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch12 | (no description) | 32'b0 |

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**Name** PATCH13
**Address** 0x1ffe74
**Description** Contains register fields associated with patch13.
**Reset** 0x00000000

| Name | patch13 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch13 | (no description) | 32'b0 |

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**Name** PATCH14
**Address** 0x1ffe78
**Description** Contains register fields associated with patch14.
**Reset** 0x00000000

| Name | patch14 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch14 | (no description) | 32'b0 |

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**Name** PATCH15
**Address** 0x1ffe7c
**Description** Contains register fields associated with patch15.
**Reset** 0x00000000

| Name | patch15 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch15 | (no description) | 32'b0 |

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**Name** PATCH16
**Address** 0x1ffe80
**Description** Contains register fields associated with patch16.
**Reset** 0x00000000

| Name | patch16 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch16 | (no description) | 32'b0 |

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**Name** PATCH17
**Address** 0x1ffe84
**Description** Contains register fields associated with patch17.
**Reset** 0x00000000

| Name | patch17 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch17 | (no description) | 32'b0 |

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**Name** PATCH18
**Address** 0x1ffe88
**Description** Contains register fields associated with patch18.
**Reset** 0x00000000

| Name | patch18 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch18 | (no description) | 32'b0 |

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**Name** PATCH19
**Address** 0x1ffe8c
**Description** Contains register fields associated with patch19.
**Reset** 0x00000000

| Name | patch19 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch19 | (no description) | 32'b0 |

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**Name** PATCH20
**Address** 0x1ffe90
**Description** Contains register fields associated with patch20.
**Reset** 0x00000000

| Name | patch20 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch20 | (no description) | 32'b0 |

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**Name** PATCH21
**Address** 0x1ffe94
**Description** Contains register fields associated with patch21.
**Reset** 0x00000000

| Name | patch21 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch21 | (no description) | 32'b0 |

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**Name** PATCH22

CONFIDENTIAL

**Address** 0x1ffe98
**Description** Contains register fields associated with patch22.
**Reset** 0x00000000

| Name | patch22 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch22 | (no description) | 32'b0 |

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**Name** PATCH23
**Address** 0x1ffe9c
**Description** Contains register fields associated with patch23.
**Reset** 0x00000000

| Name | patch23 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch23 | (no description) | 32'b0 |

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**Name** PATCH24
**Address** 0x1ffea0
**Description** Contains register fields associated with patch24.
**Reset** 0x00000000

| Name | patch24 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch24 | (no description) | 32'b0 |

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**Name** PATCH25
**Address** 0x1ffea4
**Description** Contains register fields associated with patch25.
**Reset** 0x00000000

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| Name | patch25 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch25 | (no description) | 32'b0 |

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**Name** PATCH26
**Address** 0x1ffea8
**Description** Contains register fields associated with patch26.
**Reset** 0x00000000

| Name | patch26 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch26 | (no description) | 32'b0 |

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**Name** PATCH27
**Address** 0x1ffeac
**Description** Contains register fields associated with patch27.
**Reset** 0x00000000

| Name | patch27 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch27 | (no description) | 32'b0 |

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**Name** PATCH28
**Address** 0x1ffeb0
**Description** Contains register fields associated with patch28.
**Reset** 0x00000000

| Name | patch |
|---|---|

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| Name | patch28 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch28 | (no description) | 32'b0 |

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**Name** PATCH29
**Address** 0x1ffeb4
**Description** Contains register fields associated with patch29.
**Reset** 0x00000000

| Name | patch29 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch29 | (no description) | 32'b0 |

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**Name** PATCH30
**Address** 0x1ffeb8
**Description** Contains register fields associated with patch30.
**Reset** 0x00000000

| Name | patch30 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch30 | (no description) | 32'b0 |

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**Name** PATCH31
**Address** 0x1ffebc
**Description** Contains register fields associated with patch31.
**Reset** 0x00000000

| Name | patch31 |
|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch31 | (no description) | 32'b0 |

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---

**Name** PATCH32
**Address** 0x1ffec0
**Description** Contains register fields associated with patch32.
**Reset** 0x00000000

| Name | patch32 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch32 | (no description) | 32'b0 |

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---

**Name** PATCH33
**Address** 0x1ffec4
**Description** Contains register fields associated with patch33.
**Reset** 0x00000000

| Name | patch33 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch33 | (no description) | 32'b0 |

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---

**Name** PATCH34
**Address** 0x1ffec8
**Description** Contains register fields associated with patch34.
**Reset** 0x00000000

| Name | patch34 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| 31:0 | R/W | patch34 | (no description) | 32'b0 |
|---|---|---|---|---|

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**Name** PATCH35
**Address** 0x1ffecc
**Description** Contains register fields associated with patch35.
**Reset** 0x00000000

| Name | patch35 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch35 | (no description) | 32'b0 |

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**Name** PATCH36
**Address** 0x1ffed0
**Description** Contains register fields associated with patch36.
**Reset** 0x00000000

| Name | patch36 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch36 | (no description) | 32'b0 |

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**Name** PATCH37
**Address** 0x1ffed4
**Description** Contains register fields associated with patch37.
**Reset** 0x00000000

| Name | patch37 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch37 | (no description) | 32'b0 |

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**Name** PATCH38
**Address** 0x1ffed8
**Description** Contains register fields associated with patch38.
**Reset** 0x00000000

| Name | patch38 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch38 | (no description) | 32'b0 |

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---

**Name** PATCH39
**Address** 0x1ffedc
**Description** Contains register fields associated with patch39.
**Reset** 0x00000000

| Name | patch39 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch39 | (no description) | 32'b0 |

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---

**Name** PATCH40
**Address** 0x1ffee0
**Description** Contains register fields associated with patch40.
**Reset** 0x00000000

| Name | patch40 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch40 | (no description) | 32'b0 |

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**Name** PATCH41
**Address** 0x1ffee4
**Description** Contains register fields associated with patch41.

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**Reset** 0x00000000

| Name | patch41 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch41 | (no description) | 32'b0 |

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---

**Name** PATCH42
**Address** 0x1ffee8
**Description** Contains register fields associated with patch42.
**Reset** 0x00000000

| Name | patch42 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch42 | (no description) | 32'b0 |

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---

**Name** PATCH43
**Address** 0x1ffeec
**Description** Contains register fields associated with patch43.
**Reset** 0x00000000

| Name | patch43 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch43 | (no description) | 32'b0 |

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**Name** PATCH44
**Address** 0x1ffef0
**Description** Contains register fields associated with patch44.
**Reset** 0x00000000

| Name | patch44 |
|---|---|

CONFIDENTIAL

| Name | t c h 4 4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch44 | (no description) | 32'b0 |

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**Name** PATCH45
**Address** 0x1ffef4
**Description** Contains register fields associated with patch45.
**Reset** 0x00000000

| Name | p a t c h 4 5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch45 | (no description) | 32'b0 |

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**Name** PATCH46
**Address** 0x1ffef8
**Description** Contains register fields associated with patch46.
**Reset** 0x00000000

| Name | p a t c h 4 6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch46 | (no description) | 32'b0 |

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**Name** PATCH47
**Address** 0x1ffefc
**Description** Contains register fields associated with patch47.
**Reset** 0x00000000

| Name | p a t c h 4 |
|---|---|

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| | 7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch47 | (no description) | 32'b0 |

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**Name** RRAM_WRT_CONFIG_LO
**Address** 0x1fff00
**Description** Contains register fields associated with rram_wrt_config_lo.
**Reset** 0x00000000

| Name | d i n | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | din | (no description) | 32'b0 |

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**Name** RRAM_WRT_CONFIG_HI
**Address** 0x1fff04
**Description** Contains register fields associated with rram_wrt_config_hi.
**Reset** 0x00000000

| Name | d i n | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | din | (no description) | 32'b0 |

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**Name** RRAM_READ_CONFIG_LO
**Address** 0x1fff08
**Description** Contains register fields associated with rram_read_config_lo.
**Reset** 0x00000000

| Name | d o u t | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dout | (no description) | 32'b0 |

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**Name** RRAM_READ_CONFIG_HI
**Address** 0x1fff0c
**Description** Contains register fields associated with rram_read_config_hi.
**Reset** 0x00000000

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| Name | dout | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dout | (no description) | 32'b0 |

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**Name** RRAM_CACHE_CONFIG
**Address** 0x1fff10
**Description** Contains register fields associated with rram_cache_config.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | enable_fine_clock_gating | invalidate_cache | enable_cache | cache_mode |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | enable_fine_clock_gating | secondary clock gating control (enable_cache is primary). 1=rram_cache clock is enabled only when access falls in rram region. 0=always-on. | 1'b0 |
| 2 | R/W | invalidate_cache | (no description) | 1'b0 |
| 1 | R/W | enable_cache | (no description) | 1'b0 |
| 0 | R/W | cache_mode | (no description) | 1'b0 |

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**Name** RRAM_MEM_CONFIG
**Address** 0x1fff14
**Description** Contains register fields associated with rram_mem_config.
**Reset** 0x00000000

| | R | | | | | | | | | | | | | | | | | | | | | | | | | | | | en_f | tr | | | | | | | | | | | | | | trc | trc | rram_s |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Name | RSVD | en_fine_ckg | trc_tin | trc_lven | trc_info | rram_speedup_b |
|---|---|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 | 18 | 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 18 | R/W | en_fine_ckg | enable fine clock gating to the rram macro | 1'b0 |
| 17:3 | R/W | trc_tin | (no description) | 15'b0 |
| 2 | R/W | trc_lven | (no description) | 1'b0 |
| 1 | R/W | trc_info | (no description) | 1'b0 |
| 0 | R/W | rram_speedup_b | (no description) | 1'b0 |

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**Name** RRAM_STATUS
**Address** 0x1fff18
**Description** Contains register fields associated with rram_status.
**Reset** 0x00000000

| Name | RSVD | err | busy |
|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | err | (no description) | 1'b0 |
| 0 | R | busy | (no description) | 1'b0 |

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# AT_PRRF_SECURE

**Name** PATCH_TAG0
**Address** 0x101ffe00
**Description** Contains register fields associated with patch_tag0.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| Bits | 31 30 | 29 28 27 26 25 24 23 22 | 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG1

CONFIDENTIAL

**Address** 0x101ffe04
**Description** Contains register fields associated with patch_tag1.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31** **30** | **29** **28** **27** **26** **25** **24** **23** **22** | **21** **20** **19** **18** **17** **16** **15** **14** **13** **12** **11** **10** **9** **8** **7** **6** **5** **4** **3** **2** **1** **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG2
**Address** 0x101ffe08
**Description** Contains register fields associated with patch_tag2.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31** **30** | **29** **28** **27** **26** **25** **24** **23** **22** | **21** **20** **19** **18** **17** **16** **15** **14** **13** **12** **11** **10** **9** **8** **7** **6** **5** **4** **3** **2** **1** **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG3
**Address** 0x101ffe0c
**Description** Contains register fields associated with patch_tag3.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31** **30** | **29** **28** **27** **26** **25** **24** **23** **22** | **21** **20** **19** **18** **17** **16** **15** **14** **13** **12** **11** **10** **9** **8** **7** **6** **5** **4** **3** **2** **1** **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG4
**Address** 0x101ffe10
**Description** Contains register fields associated with patch_tag4.
**Reset** 0x00000000

| | n | R | a |
|---|---|---|---|

CONFIDENTIAL

| Name | nword | RSVD | addr |
|---|---|---|---|
| Bits | 31 30 | 29 28 27 26 25 24 23 22 | 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG5
**Address** 0x101ffe14
**Description** Contains register fields associated with patch_tag5.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| Bits | 31 30 | 29 28 27 26 25 24 23 22 | 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG6
**Address** 0x101ffe18
**Description** Contains register fields associated with patch_tag6.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| Bits | 31 30 | 29 28 27 26 25 24 23 22 | 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG7
**Address** 0x101ffe1c
**Description** Contains register fields associated with patch_tag7.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| Bits | 31 30 | 29 28 27 26 25 24 23 22 | 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG8
**Address** 0x101ffe20
**Description** Contains register fields associated with patch_tag8.
**Reset** 0x00000000

| Name | nword | RSVD | | | | | | | | addr | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG9
**Address** 0x101ffe24
**Description** Contains register fields associated with patch_tag9.
**Reset** 0x00000000

| Name | nword | RSVD | | | | | | | | addr | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG10
**Address** 0x101ffe28
**Description** Contains register fields associated with patch_tag10.
**Reset** 0x00000000

| Name | nword | RSVD | | | | | | | | addr | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG11
**Address** 0x101ffe2c
**Description** Contains register fields associated with patch_tag11.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31 30** | **29 28 27 26 25 24 23 22** | **21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG12
**Address** 0x101ffe30
**Description** Contains register fields associated with patch_tag12.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31 30** | **29 28 27 26 25 24 23 22** | **21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG13
**Address** 0x101ffe34
**Description** Contains register fields associated with patch_tag13.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31 30** | **29 28 27 26 25 24 23 22** | **21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG14
**Address** 0x101ffe38
**Description** Contains register fields associated with patch_tag14.

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**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31 30** | **29 28 27 26 25 24 23 22** | **21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH_TAG15
**Address** 0x101ffe3c
**Description** Contains register fields associated with patch_tag15.
**Reset** 0x00000000

| Name | nword | RSVD | addr |
|---|---|---|---|
| **Bits** | **31 30** | **29 28 27 26 25 24 23 22** | **21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | nword | (no description) | 2'b0 |
| 21:0 | R/W | addr | (no description) | 22'b0 |

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**Name** PATCH0
**Address** 0x101ffe40
**Description** Contains register fields associated with patch0.
**Reset** 0x00000000

| Name | patch0 |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch0 | (no description) | 32'b0 |

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**Name** PATCH1
**Address** 0x101ffe44
**Description** Contains register fields associated with patch1.
**Reset** 0x00000000

| Name | patc |
|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | h 1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch1 | (no description) | 32'b0 |

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**Name** PATCH2
**Address** 0x101ffe48
**Description** Contains register fields associated with patch2.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | p a t c h 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch2 | (no description) | 32'b0 |

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**Name** PATCH3
**Address** 0x101ffe4c
**Description** Contains register fields associated with patch3.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | p a t c h 3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch3 | (no description) | 32'b0 |

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**Name** PATCH4
**Address** 0x101ffe50
**Description** Contains register fields associated with patch4.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | p a t c h 4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch4 | (no description) | 32'b0 |

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**Name** PATCH5
**Address** 0x101ffe54
**Description** Contains register fields associated with patch5.
**Reset** 0x00000000

| Name | patch5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch5 | (no description) | 32'b0 |

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**Name** PATCH6
**Address** 0x101ffe58
**Description** Contains register fields associated with patch6.
**Reset** 0x00000000

| Name | patch6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch6 | (no description) | 32'b0 |

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**Name** PATCH7
**Address** 0x101ffe5c
**Description** Contains register fields associated with patch7.
**Reset** 0x00000000

| Name | patch7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch7 | (no description) | 32'b0 |

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**Name** PATCH8
**Address** 0x101ffe60
**Description** Contains register fields associated with patch8.

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**Reset** 0x00000000

| Name | patch8 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch8 | (no description) | 32'b0 |

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**Name** PATCH9
**Address** 0x101ffe64
**Description** Contains register fields associated with patch9.
**Reset** 0x00000000

| Name | patch9 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch9 | (no description) | 32'b0 |

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**Name** PATCH10
**Address** 0x101ffe68
**Description** Contains register fields associated with patch10.
**Reset** 0x00000000

| Name | patch10 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch10 | (no description) | 32'b0 |

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**Name** PATCH11
**Address** 0x101ffe6c
**Description** Contains register fields associated with patch11.
**Reset** 0x00000000

| Name | patc |
|---|---|

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| | h 1 1 |
|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch11 | (no description) | 32'b0 |

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**Name** PATCH12
**Address** 0x101ffe70
**Description** Contains register fields associated with patch12.
**Reset** 0x00000000

| Name | p a t c h 1 2 |
|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch12 | (no description) | 32'b0 |

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**Name** PATCH13
**Address** 0x101ffe74
**Description** Contains register fields associated with patch13.
**Reset** 0x00000000

| Name | p a t c h 1 3 |
|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch13 | (no description) | 32'b0 |

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**Name** PATCH14
**Address** 0x101ffe78
**Description** Contains register fields associated with patch14.
**Reset** 0x00000000

| Name | p a t c h 1 4 |
|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch14 | (no description) | 32'b0 |

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**Name** PATCH15
**Address** 0x101ffe7c
**Description** Contains register fields associated with patch15.
**Reset** 0x00000000

| Name | patch15 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch15 | (no description) | 32'b0 |

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**Name** PATCH16
**Address** 0x101ffe80
**Description** Contains register fields associated with patch16.
**Reset** 0x00000000

| Name | patch16 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch16 | (no description) | 32'b0 |

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**Name** PATCH17
**Address** 0x101ffe84
**Description** Contains register fields associated with patch17.
**Reset** 0x00000000

| Name | patch17 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch17 | (no description) | 32'b0 |

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**Name** PATCH18
**Address** 0x101ffe88
**Description** Contains register fields associated with patch18.
**Reset** 0x00000000

| Name | patch18 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch18 | (no description) | 32'b0 |

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**Name** PATCH19
**Address** 0x101ffe8c
**Description** Contains register fields associated with patch19.
**Reset** 0x00000000

| Name | patch19 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch19 | (no description) | 32'b0 |

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**Name** PATCH20
**Address** 0x101ffe90
**Description** Contains register fields associated with patch20.
**Reset** 0x00000000

| Name | patch20 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch20 | (no description) | 32'b0 |

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**Name** PATCH21

CONFIDENTIAL

**Address** 0x101ffe94
**Description** Contains register fields associated with patch21.
**Reset** 0x00000000

| Name | patch21 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch21 | (no description) | 32'b0 |

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**Name** PATCH22
**Address** 0x101ffe98
**Description** Contains register fields associated with patch22.
**Reset** 0x00000000

| Name | patch22 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch22 | (no description) | 32'b0 |

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**Name** PATCH23
**Address** 0x101ffe9c
**Description** Contains register fields associated with patch23.
**Reset** 0x00000000

| Name | patch23 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch23 | (no description) | 32'b0 |

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**Name** PATCH24
**Address** 0x101ffea0
**Description** Contains register fields associated with patch24.
**Reset** 0x00000000

CONFIDENTIAL

| Name | patch24 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch24 | (no description) | 32'b0 |

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---

**Name** PATCH25
**Address** 0x101ffea4
**Description** Contains register fields associated with patch25.
**Reset** 0x00000000

| Name | patch25 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch25 | (no description) | 32'b0 |

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---

**Name** PATCH26
**Address** 0x101ffea8
**Description** Contains register fields associated with patch26.
**Reset** 0x00000000

| Name | patch26 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch26 | (no description) | 32'b0 |

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**Name** PATCH27
**Address** 0x101ffeac
**Description** Contains register fields associated with patch27.
**Reset** 0x00000000

| Name | patch |
|---|---|

CONFIDENTIAL

| Name | patch27 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch27 | (no description) | 32'b0 |

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**Name** PATCH28
**Address** 0x101ffeb0
**Description** Contains register fields associated with patch28.
**Reset** 0x00000000

| Name | patch28 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch28 | (no description) | 32'b0 |

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---

**Name** PATCH29
**Address** 0x101ffeb4
**Description** Contains register fields associated with patch29.
**Reset** 0x00000000

| Name | patch29 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch29 | (no description) | 32'b0 |

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**Name** PATCH30
**Address** 0x101ffeb8
**Description** Contains register fields associated with patch30.
**Reset** 0x00000000

| Name | patch30 |
|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch30 | (no description) | 32'b0 |

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**Name** PATCH31
**Address** 0x101ffebc
**Description** Contains register fields associated with patch31.
**Reset** 0x00000000

| Name | patch31 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch31 | (no description) | 32'b0 |

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**Name** PATCH32
**Address** 0x101ffec0
**Description** Contains register fields associated with patch32.
**Reset** 0x00000000

| Name | patch32 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch32 | (no description) | 32'b0 |

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**Name** PATCH33
**Address** 0x101ffec4
**Description** Contains register fields associated with patch33.
**Reset** 0x00000000

| Name | patch33 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| 31:0 | R/W | patch33 | (no description) | 32'b0 |
|---|---|---|---|---|

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**Name** PATCH34
**Address** 0x101ffec8
**Description** Contains register fields associated with patch34.
**Reset** 0x00000000

| Name | patch34 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch34 | (no description) | 32'b0 |

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**Name** PATCH35
**Address** 0x101ffecc
**Description** Contains register fields associated with patch35.
**Reset** 0x00000000

| Name | patch35 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch35 | (no description) | 32'b0 |

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**Name** PATCH36
**Address** 0x101ffed0
**Description** Contains register fields associated with patch36.
**Reset** 0x00000000

| Name | patch36 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch36 | (no description) | 32'b0 |

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**Name** PATCH37
**Address** 0x101ffed4
**Description** Contains register fields associated with patch37.
**Reset** 0x00000000

| Name | patch37 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch37 | (no description) | 32'b0 |

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**Name** PATCH38
**Address** 0x101ffed8
**Description** Contains register fields associated with patch38.
**Reset** 0x00000000

| Name | patch38 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch38 | (no description) | 32'b0 |

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**Name** PATCH39
**Address** 0x101ffedc
**Description** Contains register fields associated with patch39.
**Reset** 0x00000000

| Name | patch39 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch39 | (no description) | 32'b0 |

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**Name** PATCH40
**Address** 0x101ffee0
**Description** Contains register fields associated with patch40.

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**Reset** 0x00000000

| Name | patch40 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch40 | (no description) | 32'b0 |

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**Name** PATCH41
**Address** 0x101ffee4
**Description** Contains register fields associated with patch41.
**Reset** 0x00000000

| Name | patch41 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch41 | (no description) | 32'b0 |

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**Name** PATCH42
**Address** 0x101ffee8
**Description** Contains register fields associated with patch42.
**Reset** 0x00000000

| Name | patch42 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch42 | (no description) | 32'b0 |

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**Name** PATCH43
**Address** 0x101ffeec
**Description** Contains register fields associated with patch43.
**Reset** 0x00000000

| | patch43 |
|---|---|

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| Name | t c h 4 3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch43 | (no description) | 32'b0 |

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**Name** PATCH44
**Address** 0x101ffef0
**Description** Contains register fields associated with patch44.
**Reset** 0x00000000

| Name | p a t c h 4 4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch44 | (no description) | 32'b0 |

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**Name** PATCH45
**Address** 0x101ffef4
**Description** Contains register fields associated with patch45.
**Reset** 0x00000000

| Name | p a t c h 4 5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch45 | (no description) | 32'b0 |

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**Name** PATCH46
**Address** 0x101ffef8
**Description** Contains register fields associated with patch46.
**Reset** 0x00000000

| Name | p a t c h 4 |
|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch46 | (no description) | 32'b0 |

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**Name** PATCH47
**Address** 0x101ffefc
**Description** Contains register fields associated with patch47.
**Reset** 0x00000000

| Name | patch47 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | patch47 | (no description) | 32'b0 |

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---

**Name** RRAM_WRT_CONFIG_LO
**Address** 0x101fff00
**Description** Contains register fields associated with rram_wrt_config_lo.
**Reset** 0x00000000

| Name | din | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | din | (no description) | 32'b0 |

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---

**Name** RRAM_WRT_CONFIG_HI
**Address** 0x101fff04
**Description** Contains register fields associated with rram_wrt_config_hi.
**Reset** 0x00000000

| Name | din | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | din | (no description) | 32'b0 |

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**Name** RRAM_READ_CONFIG_LO
**Address** 0x101fff08

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**Description** Contains register fields associated with rram_read_config_lo.
**Reset** 0x00000000

| Name | dout | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dout | (no description) | 32'b0 |

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---

**Name** RRAM_READ_CONFIG_HI
**Address** 0x101fff0c
**Description** Contains register fields associated with rram_read_config_hi.
**Reset** 0x00000000

| Name | dout | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dout | (no description) | 32'b0 |

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---

**Name** RRAM_CACHE_CONFIG
**Address** 0x101fff10
**Description** Contains register fields associated with rram_cache_config.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | enable_fine_clock_gating | invalidate_cache | enable_cache | cache_mode |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 3 | R/W | enable_fine_clock_gating | secondary clock gating control (enable_cache is primary). 1=rram_cache clock is enabled only when access falls in rram region. 0=always-on. | 1'b0 |
| 2 | R/W | invalidate_cache | (no description) | 1'b0 |
| 1 | R/W | enable_cache | (no description) | 1'b0 |
| 0 | R/W | cache_mode | (no description) | 1'b0 |

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**Name** RRAM_MEM_CONFIG
**Address** 0x101fff14
**Description** Contains register fields associated with rram_mem_config.
**Reset** 0x00000000

| Name | RSVD | | | |

**Address** 0x40100000
**Description** Data value[15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | Data | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | Data | Read Sampled at pin.<br>Write To data output register.<br>Read back value goes through double flip-flop<br>synchronization logic with a delay of two cycles. | 16'b0 |

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---

**Name** DATAOUT
**Address** 0x40100004
**Description** Data output Register value [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | Dataout | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | Dataout | Read Current value of data output register.<br>Write To data output register | 16'b0 |

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**Name** INENABLE_SET
**Address** 0x40100008
**Description** Input enable set [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | inenable_set | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inenable_set | Write 1 Sets the input enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the input signal is disabled.<br>Read back 1 Indicates the input signal is enabled. | 16'b0 |

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**Name** INENABLE_CLR
**Address** 0x4010000c
**Description** Input enable clear [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | inenable_clr | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inenable_clr | Write 1 Clears the input enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the input signal is disabled.<br>Read back 1 Indicates the input signal is enabled. | 16'b0 |

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**Name** OUTENSET
**Address** 0x40100010
**Description** Output enable set [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | outenset | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | outenset | Write 1 Sets the output enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the signal direction as input.<br>Read back 1 Indicates the signal direction as output. | 16'b0 |

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**Name** OUTENCLR
**Address** 0x40100014
**Description** Output enable clear [15:0]
**Reset** 0x00000000

| Name | RSV | | | | | | | | | | | | | | | outen | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | c l r | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | outenclr | Write 1 Clears the output enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the signal direction as input.<br>Read back 1 Indicates the signal direction as output. | 16'b0 |

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**Name** ALTFUNCSET
**Address** 0x40100018
**Description** Alternative function set [15:0]
**Reset** 0x00000000

| Name | R S V D | | | | | | | | | | | | | | | | a l t f u n c s e t | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | altfuncset | Write 1 Sets the ALTFUNC bit.<br>Write 0 No effect.<br>Read back 0 For I/O.<br>Read back 1 For an alternate function. | 16'b0 |

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**Name** ALTFUNCCLR
**Address** 0x4010001c
**Description** Alternative function clear [15:0]
**Reset** 0x00000000

| Name | R S V D | | | | | | | | | | | | | | | | a l t f u n c c l r | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | altfuncclr | Write 1 Clears the ALTFUNC bit.<br>Write 0 No effect.<br>Read back 0 For I/O.<br>Read back 1 For an alternate function. | 16'b0 |

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**Name** INTENSET
**Address** 0x40100020
**Description** Interrupt enable set [15:0]
**Reset** 0x00000000

| Name | RSVD | intenset |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intenset | Write 1 Sets the interrupt enable bit.<br>Write 0 No effect.<br>Read back 0 Interrupt disabled.<br>Read back 1 Interrupt enabled. | 16'b0 |

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**Name** INTENCLR
**Address** 0x40100024
**Description** Interrupt enable clear [15:0]
**Reset** 0x00000000

| Name | RSVD | intenclr |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intenclr | Write 1 Clears the interrupt enable bit.<br>Write 0 No effect.<br>Read back 0 Interrupt disabled.<br>Read back 1 Interrupt enabled. | 16'b0 |

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**Name** INTTYPESET
**Address** 0x40100028
**Description** Interrupt type set [15:0]
**Reset** 0x00000000

| Name | RSVD | inttypeset |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inttypeset | Write 1 Sets the interrupt type bit.<br>Write 0 No effect.<br>Read back 0 For LOW or HIGH level.<br>Read back 1 For falling edge or rising edge. | 16'b0 |

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**Name** INTTYPECLR
**Address** 0x4010002c
**Description** Interrupt type clear [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | inttypeclr | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inttypeclr | Write 1 Clears the interrupt enable bit.<br>Write 0 No effect.<br>Read back 0 For LOW or HIGH level.<br>Read back 1 For falling edge or rising edge. | 16'b0 |

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**Name** INTPOLSET
**Address** 0x40100030
**Description** Polarity-level, edge IRQ configuration [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | intpolset | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intpolset | Write 1 Sets the interrupt polarity bit.<br>Write 0 No effect.<br>Read back 0 For LOW level or falling edge.<br>Read back 1 For HIGH level or rising edge. | 16'b0 |

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**Name** INTPOLCLR
**Address** 0x40100034
**Description** Interrupt type clear [15:0]
**Reset** 0x00000000

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| Name | RSVD | intpolclr |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intpolclr | Write 1 Clears the interrupt polarity bit.<br>Write 0 No effect.<br>Read back 0 For LOW level or falling edge.<br>Read back 1 For HIGH level or rising edge. | 16'b0 |

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**Name** INTSTATUS_CLEAR
**Address** 0x40100038
**Description** Write one to clear interrupt request:
**Reset** 0x00000000

| Name | RSVD | intstatus_clear |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intstatus_clear | Write 1 clears the interrupt request.<br>Write 0 No effect.<br>Read back IRQ status Register. | 16'b0 |

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**Name** PULLUP_ENABLE_SET
**Address** 0x4010005c
**Description** pullup enable set [15:0]
**Reset** 0x00000000

| Name | RSV | pullup_ena |
|---|---|---|

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| | RSVD | pullup_enable_set |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pullup_enable_set | Write 1 Sets the pullup enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled up.<br>Read back 1 Indicates the pin is pulled up. | 16'b0 |

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**Name** PULLUP_ENABLE_CLR
**Address** 0x40100060
**Description** pullup enable clear [15:0]
**Reset** 0x00000000

| Name | RSVD | pullup_enable_clr |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pullup_enable_clr | Write 1 Clears the pullup enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled up.<br>Read back 1 Indicates the pin is pulled up. | 16'b0 |

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**Name** PULLDOWN_ENABLE_SET
**Address** 0x40100064
**Description** pulldown enable set [15:0]
**Reset** 0x00000000

| | RS | pulldown_ |
|---|---|---|

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| Name | VD | | | | | | | | | | | | | | | enable_set | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pulldown_enable_set | Write 1 Sets the pulldown enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled down.<br>Read back 1 Indicates the pin is pulled down. | 16'b0 |

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**Name** PULLDOWN_ENABLE_CLR
**Address** 0x40100068
**Description** pulldown enable clear [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | pulldown_enable_clr | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pulldown_enable_clr | Write 1 Clears the pulldown enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled down.<br>Read back 1 Indicates the pin is pulled down. | 16'b0 |

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**Name** MASKLOWBYTE
**Address** 0x40100400
**Description** Lower eight bits masked access. The register actually spans from 0x100 to 0x1FF
**Reset** 0x00000000

| | R | | | | | | | | | | | | | | | | | | | | | | | | | | | | | maskl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | SVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | owbyte | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | masklowbyte | Data for lower byte access, whith bits[9:2] of address value used as enable mask for each bit. | 8'b0 |

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**Name** MASKHIGHBYTE
**Address** 0x40100800
**Description** Higher eight bits masked access. The register actually spans from 0x200 to 0x2FF
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | maskhighbyte | | | | | | | | RSVD | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R/W | maskhighbyte | Data for higher byte access, whith bits[9:2] of address value used as enable mask for each bit. | 8'b0 |

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**Name** PID4
**Address** 0x40100fd0
**Description** PID4
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | block_count | | | | jep106_c_code | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | block_count | Block count | 4'b0 |
| 3:0 | R | jep106_c_code | jep106_c_code | 4'd4 |

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**Name** PID5
**Address** 0x40100fd4
**Description** PID5
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid5 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid5 | Peripheral ID Register 5. | 8'b0 |

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**Name** PID6
**Address** 0x40100fd8
**Description** PID6
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid6 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid6 | Peripheral ID Register 6. | 8'b0 |

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**Name** PID7
**Address** 0x40100fdc
**Description** PID7
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid7 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid7 | Peripheral ID Register 7. | 8'b0 |

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**Name** PID0
**Address** 0x40100fe0
**Description** PID0
**Reset** 0x00000021

| | RS | | | | | | | | | | | | | | | | | | | | | | part | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | VD | | | | | | | | | | | | | | | | | | | | | | part_num | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | part_num | Part number[7:0]. | 8'h21 |

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**Name** PID1
**Address** 0x40100fe4
**Description** PID1
**Reset** 0x000000b8

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | jep106_id_3_0 | | | | part_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | jep106_id_3_0 | jep106_id_3_0 | 4'hb |
| 3:0 | R | part_num | Part number[11:8]. | 4'h8 |

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**Name** PID2
**Address** 0x40100fe8
**Description** PID2
**Reset** 0x0000001b

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | rev | | | | jedec_used | jep106_id_6_4 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | rev | Revision | 4'h1 |
| 3 | R | jedec_used | jedec_used | 1'h1 |
| 2:0 | R | jep106_id_6_4 | jep106_id_6_4 | 3'h3 |

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**Name** PID3
**Address** 0x40100fec
**Description** PID3
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | ecorevnum | | | | mod_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | ecorevnum | ECO revision number. | 4'h0 |
| 3:0 | R | mod_num | modification number. | 4'h0 |

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**Name** CID0
**Address** 0x40100ff0
**Description** CID0
**Reset** 0x0000000d

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid0 | Component ID Register 0 | 8'h0d |

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**Name** CID1
**Address** 0x40100ff4
**Description** CID1
**Reset** 0x000000f0

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid1 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid1 | Component ID Register 1 | 8'hf0 |

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**Name** CID2
**Address** 0x40100ff8
**Description** CID2

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**Reset** 0x00000005

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid2 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid2 | Component ID Register 2 | 8'h05 |

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**Name** CID3
**Address** 0x40100ffc
**Description** CID3
**Reset** 0x000000b1

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid3 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid3 | Component ID Register 3 | 8'hb1 |

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# GPIO1_NONSECURE

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**Name** DATA
**Address** 0x40101000
**Description** Data value[15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | Data | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | Data | Read Sampled at pin.<br>Write To data output register.<br>Read back value goes through double flip-flop<br>synchronization logic with a delay of two cycles. | 16'b0 |

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**Name** DATAOUT
**Address** 0x40101004
**Description** Data output Register value [15:0]
**Reset** 0x00000000

| | R | | | | | | | | | | | | | | | | Dat | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | SVD | dataout |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | Dataout | Read Current value of data output register.<br>Write To data output register | 16'b0 |

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**Name** INENABLE_SET
**Address** 0x40101008
**Description** Input enable set [15:0]
**Reset** 0x00000000

| Name | RSVD | inenable_set |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inenable_set | Write 1 Sets the input enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the input signal is disabled.<br>Read back 1 Indicates the input signal is enabled. | 16'b0 |

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**Name** INENABLE_CLR
**Address** 0x4010100c
**Description** Input enable clear [15:0]
**Reset** 0x00000000

| Name | RSVD | inenable_clr |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inenable_clr | Write 1 Clears the input enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the input signal is disabled. | 16'b0 |

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| | | | | |
|---|---|---|---|---|
| | | | Read back 1 Indicates the input signal is enabled. | |

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**Name** OUTENSET
**Address** 0x40101010
**Description** Output enable set [15:0]
**Reset** 0x00000000

| Name | RSVD | outenset |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | outenset | Write 1 Sets the output enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the signal direction as input.<br>Read back 1 Indicates the signal direction as output. | 16'b0 |

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**Name** OUTENCLR
**Address** 0x40101014
**Description** Output enable clear [15:0]
**Reset** 0x00000000

| Name | RSVD | outenclr |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | outenclr | Write 1 Clears the output enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the signal direction as input.<br>Read back 1 Indicates the signal direction as output. | 16'b0 |

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**Name** ALTFUNCSET
**Address** 0x40101018
**Description** Alternative function set [15:0]
**Reset** 0x00000000

| Name | RSVD | altfunc |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | s e t | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | altfuncset | Write 1 Sets the ALTFUNC bit.<br>Write 0 No effect.<br>Read back 0 For I/O.<br>Read back 1 For an alternate function. | 16'b0 |

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**Name** ALTFUNCCLR
**Address** 0x4010101c
**Description** Alternative function clear [15:0]
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | R S V D | | | | | | | | | | | | | | | | a l t f u n c c l r | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | altfuncclr | Write 1 Clears the ALTFUNC bit.<br>Write 0 No effect.<br>Read back 0 For I/O.<br>Read back 1 For an alternate function. | 16'b0 |

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**Name** INTENSET
**Address** 0x40101020
**Description** Interrupt enable set [15:0]
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | R S V D | | | | | | | | | | | | | | | | i n t e n s e t | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intenset | Write 1 Sets the interrupt enable bit.<br>Write 0 No effect.<br>Read back 0 Interrupt disabled.<br>Read back 1 Interrupt enabled. | 16'b0 |

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**Name** INTENCLR

CONFIDENTIAL

**Address** 0x40101024
**Description** Interrupt enable clear [15:0]
**Reset** 0x00000000

| Name | RSVD | intenclr |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intenclr | Write 1 Clears the interrupt enable bit.<br>Write 0 No effect.<br>Read back 0 Interrupt disabled.<br>Read back 1 Interrupt enabled. | 16'b0 |

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**Name** INTTYPESET
**Address** 0x40101028
**Description** Interrupt type set [15:0]
**Reset** 0x00000000

| Name | RSVD | inttypeset |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inttypeset | Write 1 Sets the interrupt type bit.<br>Write 0 No effect.<br>Read back 0 For LOW or HIGH level.<br>Read back 1 For falling edge or rising edge. | 16'b0 |

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**Name** INTTYPECLR
**Address** 0x4010102c
**Description** Interrupt type clear [15:0]
**Reset** 0x00000000

| Name | RSVD | inttypeclr |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inttypeclr | Write 1 Clears the interrupt enable bit.<br>Write 0 No effect.<br>Read back 0 For LOW or HIGH level.<br>Read back 1 For falling edge or rising edge. | 16'b0 |

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**Name** INTPOLSET
**Address** 0x40101030
**Description** Polarity-level, edge IRQ configuration [15:0]
**Reset** 0x00000000

| Name | RSVD | intpolset |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intpolset | Write 1 Sets the interrupt polarity bit.<br>Write 0 No effect.<br>Read back 0 For LOW level or falling edge.<br>Read back 1 For HIGH level or rising edge. | 16'b0 |

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**Name** INTPOLCLR
**Address** 0x40101034
**Description** Interrupt type clear [15:0]
**Reset** 0x00000000

| Name | RSVD | intpolclr |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intpolclr | Write 1 Clears the interrupt polarity bit.<br>Write 0 No effect.<br>Read back 0 For LOW level or falling edge.<br>Read back 1 For HIGH level or rising edge. | 16'b0 |

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**Name** INTSTATUS_CLEAR
**Address** 0x40101038
**Description** Write one to clear interrupt request:
**Reset** 0x00000000

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| Name | RSVD | intstatus_clear |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intstatus_clear | Write 1 clears the interrupt request.<br>Write 0 No effect.<br>Read back IRQ status Register. | 16'b0 |

Back to top

**Name** PULLUP_ENABLE_SET
**Address** 0x4010105c
**Description** pullup enable set [15:0]
**Reset** 0x00000000

| Name | RSVD | pullup_enable_set |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pullup_enable_set | Write 1 Sets the pullup enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled up.<br>Read back 1 Indicates the pin is pulled up. | 16'b0 |

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**Name** PULLUP_ENABLE_CLR
**Address** 0x40101060
**Description** pullup enable clear [15:0]
**Reset** 0x00000000

| | | pul |
|---|---|---|

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| Name | RSVD | | | | | | | | | | | | | | | pullup_enable_clr | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pullup_enable_clr | Write 1 Clears the pullup enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled up.<br>Read back 1 Indicates the pin is pulled up. | 16'b0 |

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**Name** PULLDOWN_ENABLE_SET
**Address** 0x40101064
**Description** pulldown enable set [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | pulldown_enable_set | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pulldown_enable_set | Write 1 Sets the pulldown enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled down.<br>Read back 1 Indicates the pin is pulled down. | 16'b0 |

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---

**Name** PULLDOWN_ENABLE_CLR
**Address** 0x40101068
**Description** pulldown enable clear [15:0]
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | p | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

CONFIDENTIAL

| Name | RSVD | pulldown_enable_clr |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pulldown_enable_clr | Write 1 Clears the pulldown enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled down.<br>Read back 1 Indicates the pin is pulled down. | 16'b0 |

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**Name** MASKLOWBYTE
**Address** 0x40101400
**Description** Lower eight bits masked access. The register actually spans from 0x100 to 0x1FF
**Reset** 0x00000000

| Name | RSVD | masklowbyte |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | masklowbyte | Data for lower byte access, whith bits[9:2] of address value used as enable mask for each bit. | 8'b0 |

Back to top

**Name** MASKHIGHBYTE
**Address** 0x40101800
**Description** Higher eight bits masked access. The register actually spans from 0x200 to 0x2FF
**Reset** 0x00000000

| Name | RSV | maskhig | RSV |
|---|---|---|---|

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| | D | | | | | | | | | | | | | | | | h byte | | | | | | | | D | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R/W | maskhighbyte | Data for higher byte access, whith bits[9:2] of address value used as enable mask for each bit. | 8'b0 |

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**Name** PID4
**Address** 0x40101fd0
**Description** PID4
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | block_count | | | | jep106_c_code | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | block_count | Block count | 4'b0 |
| 3:0 | R | jep106_c_code | jep106_c_code | 4'd4 |

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**Name** PID5
**Address** 0x40101fd4
**Description** PID5
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid5 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid5 | Peripheral ID Register 5. | 8'b0 |

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**Name** PID6
**Address** 0x40101fd8
**Description** PID6
**Reset** 0x00000000

| | R | | | | | | | | | | | | | | | | | | | | | | p | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | SVD | | | | | | | | | | | | | | | | | | | | | | id6 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid6 | Peripheral ID Register 6. | 8'b0 |

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**Name** PID7
**Address** 0x40101fdc
**Description** PID7
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid7 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid7 | Peripheral ID Register 7. | 8'b0 |

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**Name** PID0
**Address** 0x40101fe0
**Description** PID0
**Reset** 0x00000021

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | part_num | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | part_num | Part number[7:0]. | 8'h21 |

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**Name** PID1
**Address** 0x40101fe4
**Description** PID1
**Reset** 0x000000b8

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | jep106_id_ | | | | part_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | 3_0 | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | jep106_id_3_0 | jep106_id_3_0 | 4'hb |
| 3:0 | R | part_num | Part number[11:8]. | 4'h8 |

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**Name** PID2
**Address** 0x40101fe8
**Description** PID2
**Reset** 0x0000001b

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | rev | | | | jedec_used | jep106_id_6_4 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | rev | Revision | 4'h1 |
| 3 | R | jedec_used | jedec_used | 1'h1 |
| 2:0 | R | jep106_id_6_4 | jep106_id_6_4 | 3'h3 |

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**Name** PID3
**Address** 0x40101fec
**Description** PID3
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | ecorevnum | | | | mod_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | ecorevnum | ECO revision number. | 4'h0 |
| 3:0 | R | mod_num | modification number. | 4'h0 |

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**Name** CID0

CONFIDENTIAL

**Address** 0x40101ff0
**Description** CID0
**Reset** 0x0000000d

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid0 | Component ID Register 0 | 8'h0d |

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**Name** CID1
**Address** 0x40101ff4
**Description** CID1
**Reset** 0x000000f0

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid1 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid1 | Component ID Register 1 | 8'hf0 |

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**Name** CID2
**Address** 0x40101ff8
**Description** CID2
**Reset** 0x00000005

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid2 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid2 | Component ID Register 2 | 8'h05 |

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**Name** CID3
**Address** 0x40101ffc
**Description** CID3
**Reset** 0x000000b1

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid3 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 7:0 | R | cid3 | Component ID Register 3 | 8'hb1 |

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# SHA2_NONSECURE

**Name** CONTROL
**Address** 0x40103000
**Description** Contains register fields associated with control.
**Reset** 0x00000000

| Name | RSVD | digest_swizzle | byte_swizzle | hmac_en | sha_en |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | digest_swizzle | digest word's byte swizzle | 1'b0 |
| 2 | R/W | byte_swizzle | message word byte swizzle | 1'b0 |
| 1 | R/W | hmac_en | HMAC datapath enable. If this bit is 1, HMAC operates when hash_start toggles | 1'b0 |
| 0 | R/W | sha_en | SHA256 enable. If 0, SHA engine won't initiate compression,<br>this is used to stop operation of the SHA engine until configuration has been done.<br>When the SHA engine is disabled the digest is cleared. | 1'b0 |

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**Name** SIDELOAD_CTRL
**Address** 0x40103004
**Description** Contains register fields associated with sideload_ctrl.
**Reset** 0x00000000

| Name | RSVD | sideload_update_en | sideload_ptr_clr |
|---|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | sideload_update_en | write 1 to enable sideload buffer push. | 1'b0 |
| 0 | R/W | sideload_ptr_clr | write 1 to clear the write pointer. Not Self-clearing. | 1'b0 |

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**Name** CMD
**Address** 0x40103008
**Description** Contains register fields associated with cmd.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | hash_process | hash_start |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | hash_process | If writes 1 into this field, SHA256 or HMAC calculates the digest or signing based on currently received message. | 1'b0 |
| 0 | R/W | hash_start | If writes 1 into this field, SHA256 or HMAC begins its operation. CPU should configure relative information first, such as message_length, secret_key. | 1'b0 |

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**Name** MESSAGE_LENGTH_LO
**Address** 0x4010300c
**Description** Contains register fields associated with message_length_lo.
**Reset** 0x00000000

| Name | length_lo | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | length_lo | Received Message Length calculated by the HMAC in bits [31:0] Message is byte granularity. lower 3bits [2:0] are ignored. | 32'b0 |

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**Name** MESSAGE_LENGTH_HI
**Address** 0x40103010
**Description** Contains register fields associated with message_length_hi.

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**Reset** 0x00000000

| Name | length_hi | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | length_hi | Received Message Length calculated by the HMAC in bits [63:32]<br>Message is byte granularity. lower 3bits [2:0] are ignored. | 32'b0 |

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**Name** WIPE_V
**Address** 0x40103014
**Description** Contains register fields associated with wipe_v.
**Reset** 0x00000000

| Name | wipe_v | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | wipe_v | a write to this register triggers a wipe_secret action. | 32'b0 |

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**Name** FIFO_PUSH
**Address** 0x40103018
**Description** Contains register fields associated with fifo_push.
**Reset** 0x00000000

| Name | wdata | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | wdata | Message FIFO input port. Byte writes are supported | 32'b0 |

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**Name** DIGEST0
**Address** 0x4010301c
**Description** Contains register fields associated with digest0.
**Reset** 0x00000000

| Name | di | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | digest0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest0 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST1
**Address** 0x40103020
**Description** Contains register fields associated with digest1.
**Reset** 0x00000000

| Name | digest1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest1 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST2
**Address** 0x40103024
**Description** Contains register fields associated with digest2.
**Reset** 0x00000000

| Name | digest2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest2 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST3
**Address** 0x40103028
**Description** Contains register fields associated with digest3.
**Reset** 0x00000000

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| Name | digest3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest3 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST4
**Address** 0x4010302c
**Description** Contains register fields associated with digest4.
**Reset** 0x00000000

| Name | digest4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest4 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST5
**Address** 0x40103030
**Description** Contains register fields associated with digest5.
**Reset** 0x00000000

| Name | digest5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest5 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST6
**Address** 0x40103034
**Description** Contains register fields associated with digest6.
**Reset** 0x00000000

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| Name | digest6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest6 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST7
**Address** 0x40103038
**Description** Contains register fields associated with digest7.
**Reset** 0x00000000

| Name | digest7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest7 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** KEY_0
**Address** 0x4010303c
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | key_0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_0 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_1
**Address** 0x40103040
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

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| Name | key_1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_1 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_2
**Address** 0x40103044
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | key_2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_2 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_3
**Address** 0x40103048
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | key_3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_3 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_4
**Address** 0x4010304c
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | key_4 |
|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_4 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_5
**Address** 0x40103050
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | key_5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_5 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_6
**Address** 0x40103054
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | key_6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_6 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_7
**Address** 0x40103058
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | key_7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_7 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** STATUS
**Address** 0x4010305c
**Description** Contains register fields associated with status.
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | fifo_depth | | | | | reserved | | fifo_full | fifo_empty |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8:4 | R | fifo_depth | FIFO entry count | 5'b0 |
| 3:2 | R | reserved | (no description) | 2'b0 |
| 1 | R | fifo_full | FIFO full. Data written to the FIFO whilst it is full will cause back-pressure on the interconnect | 1'b0 |
| 0 | R | fifo_empty | FIFO empty | 1'b1 |

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**Name** INTERRUPT_STATUS
**Address** 0x40103060
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | hmac_err | pop_when_empty | push_when_full | hmac_done | chunk_done | fifo_full | fifo_empty |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6 | R | hmac_err | HMAC error occurred. ERR_CODE register shows which error occurred | 1'b0 |
| 5 | R | pop_when_empty | pop when Message FIFO is full condition | 1'b0 |
| 4 | R | push_when_full | push when Message FIFO empty condition | 1'b0 |
| 3 | R | hmac_done | HMAC-256 completes a message with key | 1'b0 |
| 2 | R | chunk_done | one chunk of message done. (512 bits of message and padded bits) | 1'b0 |
| 1 | R | fifo_full | Message FIFO full condition | 1'b0 |
| 0 | R | fifo_empty | Message FIFO empty condition | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x40103064

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**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | hmac_err | pop_when_empty | push_when_full | hmac_done | chunk_done | fifo_full | fifo_empty |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6 | R/W | hmac_err | interrupt mask bit | 1'b0 |
| 5 | R/W | pop_when_empty | interrupt mask bit | 1'b0 |
| 4 | R/W | push_when_full | interrupt mask bit | 1'b0 |
| 3 | R/W | hmac_done | interrupt mask bit | 1'b0 |
| 2 | R/W | chunk_done | interrupt mask bit | 1'b0 |
| 1 | R/W | fifo_full | interrupt mask bit | 1'b0 |
| 0 | R/W | fifo_empty | interrupt mask bit | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x40103068
**Description** Contains register fields associated with set_interrupt.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | intrpt_set | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6:0 | R/W | intrpt_set | if nth bit set, the nth interrupt is set; | 7'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x4010306c
**Description** Contains register fields associated with reset_interrupt.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | intr | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | intrpt_reset |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 | 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6:0 | R/W | intrpt_reset | if nth bit set, the nth interrupt is reset; | 7'b0 |

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**Name** ERR_CODE
**Address** 0x40103070
**Description** Contains register fields associated with err_code.
**Reset** 0x00000000

| Name | err_code |
|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | err_code | NoError = 32'h 0000_0000,<br>SwPushMsgWhenShaDisabled = 32'h 0000_0001,<br>SwHashStartWhenShaDisabled = 32'h 0000_0002,<br>SwUpdateSecretKeyInProcess = 32'h 0000_0003,<br>SwHashStartWhenActive = 32'h 0000_0004,<br>SwPushMsgWhenDisallowed = 32'h 0000_0005 | 32'b0 |

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**Name** ID
**Address** 0x40103ffc
**Description** Contains register fields associated with id.
**Reset** 0x53484132

| Name | id |
|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | 'SHA2' | 32'h53484132 |

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# WRPR0_NONSECURE

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**Name** APB0_CTRL
**Address** 0x40143000
**Description** Contains register fields associated with apb0_ctrl.

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**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB1_CTRL
**Address** 0x40143004
**Description** Contains register fields associated with apb1_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB2_CTRL
**Address** 0x40143008
**Description** Contains register fields associated with apb2_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB3_CTRL
**Address** 0x4014300c
**Description** Contains register fields associated with apb3_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB4_CTRL
**Address** 0x40143010
**Description** Contains register fields associated with apb4_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB5_CTRL
**Address** 0x40143014
**Description** Contains register fields associated with apb5_ctrl.
**Reset** 0x00000002

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| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB6_CTRL
**Address** 0x40143018
**Description** Contains register fields associated with apb6_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB7_CTRL
**Address** 0x4014301c
**Description** Contains register fields associated with apb7_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB8_CTRL
**Address** 0x40143020
**Description** Contains register fields associated with apb8_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB9_CTRL
**Address** 0x40143024
**Description** Contains register fields associated with apb9_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB10_CTRL
**Address** 0x40143028
**Description** Contains register fields associated with apb10_ctrl.
**Reset** 0x00000002

| | | c | s | clk |
|---|---|---|---|---|

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| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB11_CTRL
**Address** 0x4014302c
**Description** Contains register fields associated with apb11_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB12_CTRL
**Address** 0x40143030
**Description** Contains register fields associated with apb12_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB13_CTRL
**Address** 0x40143034
**Description** Contains register fields associated with apb13_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB14_CTRL
**Address** 0x40143038
**Description** Contains register fields associated with apb14_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB15_CTRL
**Address** 0x4014303c
**Description** Contains register fields associated with apb15_ctrl.
**Reset** 0x00000002

| Name | RS | | | | | | | | | | | | | | | | | | | | | | | | | | | clk | sre | clk_e |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | VD | | | | | | | | | | | | | | | | | | | | | | | | | | | _sel | set | nable |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** CLK_HPC_PIN_OUT_CTRL
**Address** 0x40143040
**Description** Contains register fields associated with clk_hpc_pin_out_ctrl.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_hpc_sel | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | clk_hpc_sel | (no description) | 2'b0 |

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**Name** CLK_MPC_PIN_OUT_CTRL
**Address** 0x40143044
**Description** Contains register fields associated with clk_mpc_pin_out_ctrl.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_mpc_sel | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | clk_mpc_sel | (no description) | 2'b0 |

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**Name** CLK_LPC_PIN_OUT_CTRL

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**Address** 0x40143048
**Description** Contains register fields associated with clk_lpc_pin_out_ctrl.
**Reset** 0x00000000

| Name | RSVD | clk_lpc_sel |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | clk_lpc_sel | (no description) | 1'b0 |

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**Name** DBG_CTRL
**Address** 0x4014304c
**Description** Contains register fields associated with dbg_ctrl.
**Reset** 0x0006c000

| Name | RSVD | dbg_cg | dbg_skip_6x6_holes | trace_is_ddr | trigger_soc_rst | sse200_mstexp0_clk_enable | sse200_mstexp1_clk_enable | sse200_exp_nmi | sse200_expclkreq_in | sse200_expsysfclkqactive | sse200_exp_mpc_g_sel | sse200_dbgen_sel | sse200_niden_sel | sse200_certdisableext | em_block_addr | em_block_wdata | dbg_enable | dbg_shift | dbg_sel |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 22 | R/W | dbg_cg | (no description) | 1'b0 |
| 21 | R/W | dbg_skip_6x6_holes | (no description) | 1'b0 |
| 20 | R/W | trace_is_ddr | (no description) | 1'b0 |
| 19 | R/W | trigger_soc_rst | (no description) | 1'b0 |
| 18 | R/W | sse200_mstexp0_clk_enable | (no description) | 1'b1 |
| 17 | R/W | sse200_mstexp1_clk_enable | (no description) | 1'b1 |
| 16 | R/W | sse200_exp_nmi | (no description) | 1'b0 |
| 15 | R/W | sse200_expclkreq_in | (no description) | 1'b1 |

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| | | | | |
|---|---|---|---|---|
| 14 | R/W | sse200_sysfclkqactive | (no description) | 1'b1 |
| 13 | R/W | sse200_exp_mpc_g_sel | (no description) | 1'b0 |
| 12 | R/W | sse200_dbgen_sel | (no description) | 1'b0 |
| 11 | R/W | sse200_niden_sel | (no description) | 1'b0 |
| 10 | R/W | sse200_certdisableext | (no description) | 1'b0 |
| 9 | R/W | em_block_addr | not intended to be used; helps with cpf isolation issues | 1'b0 |
| 8 | R/W | em_block_wdata | not intended to be used; helps with cpf isolation issues | 1'b0 |
| 7 | R/W | dbg_enable | (no description) | 1'd0 |
| 6:4 | R/W | dbg_shft | (no description) | 3'd0 |
| 3:0 | R/W | dbg_sel | (no description) | 4'd0 |

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**Name** PSEL_A
**Address** 0x40143050
**Description** Contains register fields associated with psel_A.
**Reset** 0x00000201

| Name | p3_sel | | | | | | | | p2_sel | | | | | | | | p1_sel | | | | | | | | p0_sel | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p3_sel | (no description) | 8'b0 |
| 23:16 | R/W | p2_sel | (no description) | 8'b0 |
| 15:8 | R/W | p1_sel | by default P1 is SWDIO | 8'd2 |
| 7:0 | R/W | p0_sel | by default P0 is SWDCLK | 8'd1 |

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**Name** PSEL_B
**Address** 0x40143054
**Description** Contains register fields associated with psel_B.
**Reset** 0x00000000

| Name | p7_sel | | | | | | | | p6_sel | | | | | | | | p5_sel | | | | | | | | p4_sel | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p7_sel | (no description) | 8'b0 |
| 23:16 | R/W | p6_sel | (no description) | 8'b0 |
| 15:8 | R/W | p5_sel | (no description) | 8'b0 |
| 7:0 | R/W | p4_sel | (no description) | 8'b0 |

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**Name** PSEL_C
**Address** 0x40143058
**Description** Contains register fields associated with psel_C.

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**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | p11_sel | | | | | | | | p10_sel | | | | | | | | p9_sel | | | | | | | | p8_sel | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p11_sel | (no description) | 8'b0 |
| 23:16 | R/W | p10_sel | (no description) | 8'b0 |
| 15:8 | R/W | p9_sel | (no description) | 8'b0 |
| 7:0 | R/W | p8_sel | (no description) | 8'b0 |

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**Name** PSEL_D
**Address** 0x4014305c
**Description** Contains register fields associated with psel_D.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | p15_sel | | | | | | | | p14_sel | | | | | | | | p13_sel | | | | | | | | p12_sel | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p15_sel | (no description) | 8'b0 |
| 23:16 | R/W | p14_sel | (no description) | 8'b0 |
| 15:8 | R/W | p13_sel | (no description) | 8'b0 |
| 7:0 | R/W | p12_sel | (no description) | 8'b0 |

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**Name** PSEL_E
**Address** 0x40143060
**Description** Contains register fields associated with psel_E.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | p19_sel | | | | | | | | p18_sel | | | | | | | | p17_sel | | | | | | | | p16_sel | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p19_sel | (no description) | 8'b0 |
| 23:16 | R/W | p18_sel | (no description) | 8'b0 |
| 15:8 | R/W | p17_sel | (no description) | 8'b0 |
| 7:0 | R/W | p16_sel | (no description) | 8'b0 |

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**Name** PSEL_F
**Address** 0x40143064
**Description** Contains register fields associated with psel_F.
**Reset** 0x00000000

| Name | p23_sel | | | | | | | | p22_sel | | | | | | | | p21_sel | | | | | | | | p20_sel | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p23_sel | (no description) | 8'b0 |
| 23:16 | R/W | p22_sel | (no description) | 8'b0 |
| 15:8 | R/W | p21_sel | (no description) | 8'b0 |
| 7:0 | R/W | p20_sel | (no description) | 8'b0 |

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**Name** PSEL_G
**Address** 0x40143068
**Description** Contains register fields associated with psel_G.
**Reset** 0x00000000

| Name | p27_sel | | | | | | | | p26_sel | | | | | | | | p25_sel | | | | | | | | p24_sel | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p27_sel | (no description) | 8'b0 |
| 23:16 | R/W | p26_sel | (no description) | 8'b0 |
| 15:8 | R/W | p25_sel | (no description) | 8'b0 |
| 7:0 | R/W | p24_sel | (no description) | 8'b0 |

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**Name** PSEL_H
**Address** 0x4014306c
**Description** Contains register fields associated with psel_H.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | p30_sel | | | | | | | | p29_sel | | | | | | | | p28_sel | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:16 | R/W | p30_sel | (no description) | 8'b0 |
| 15:8 | R/W | p29_sel | (no description) | 8'b0 |
| 7:0 | R/W | p28_sel | (no description) | 8'b0 |

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**Name** PSEL_I
**Address** 0x40143070
**Description** Contains register fields associated with psel_I.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | jtag_interface | RSVD | | | | | | | | | | atb_enable | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16 | R/W | jtag_interface | (no description) | 1'b0 |
| 4:0 | R/W | atb_enable | (no description) | 5'b0 |

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**Name** PUPD_OVRD
**Address** 0x40143074
**Description** pin 0-31 pull up/down override control
**Reset** 0x00000000

| Name | RSVD | pin | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 30:0 | R/W | pin | when set, overrides pull-up/pull-down from periphery | 31'd0 |

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**Name** PUPD_OVRD_VAL
**Address** 0x40143078
**Description** pin 0-31 pull up/down override value
**Reset** 0x00000000

| Name | RSVD | pin |
|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 30:0 | R/W | pin | pull-up/pull_down override value<br>when pupd_ovrd is set:<br>1= pull-up override<br>0= pull-down override | 31'd0 |

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**Name** REMAP
**Address** 0x4014307c
**Description** Contains register fields associated with remap.
**Reset** 0x00000000

| Name | RSVD | qspi_drop |
|---|---|---|
| **Bits** | **31 … 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | qspi_drop | drops the at_ahb_qspi address start from 0x10000000 to 0x0. No change in range. | 1'b0 |

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**Name** AHB_INTERPOSERS
**Address** 0x40143080
**Description** Contains register fields associated with ahb_interposers.
**Reset** 0x00000001

| Name | RSVD | qspi_intp_low_ceiling | qspi_intp_enable |
|---|---|---|---|
| **Bits** | **31 … 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | qspi_intp_low_ceiling | (no description) | 1'b0 |
| 0 | R/W | qspi_intp_enable | (no description) | 1'b1 |

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**Name** SCRATCHPAD_A
**Address** 0x40143084
**Description** Contains register fields associated with scratchpad_A.
**Reset** 0x00000000

| Name | misc | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | misc | mostly likely will be used to inject random seed in sim | 32'b0 |

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**Name** SCRATCHPAD_B
**Address** 0x40143088
**Description** Contains register fields associated with scratchpad_B.
**Reset** 0x00000000

| Name | misc | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | misc | mostly likely will be used to inject random seed in sim | 32'b0 |

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**Name** SCRATCHPAD_C
**Address** 0x4014308c
**Description** Contains register fields associated with scratchpad_C.
**Reset** 0x00000000

| Name | misc | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | misc | mostly likely will be used to inject random seed in sim | 32'b0 |

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**Name** SCRATCHPAD_D
**Address** 0x40143090
**Description** Contains register fields associated with scratchpad_D.
**Reset** 0x00000000

| Name | misc |
|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | misc | mostly likely will be used to inject random seed in sim | 32'b0 |

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**Name** RRAM_WRITE_PROTECTION0
**Address** 0x40143094
**Description** Contains register fields associated with rram_write_protection0.
**Reset** 0xffffffff

| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | if a bit is set, invalidates any writes to corresponding to 2kB block | 32'hffffffff |

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**Name** RRAM_WRITE_PROTECTION1
**Address** 0x40143098
**Description** Contains register fields associated with rram_write_protection1.
**Reset** 0xffffffff

| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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**Name** RRAM_WRITE_PROTECTION2
**Address** 0x4014309c
**Description** Contains register fields associated with rram_write_protection2.
**Reset** 0xffffffff

| | d |
|---|---|

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| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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**Name** RRAM_WRITE_PROTECTION3
**Address** 0x401430a0
**Description** Contains register fields associated with rram_write_protection3.
**Reset** 0xffffffff

| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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**Name** RRAM_WRITE_PROTECTION4
**Address** 0x401430a4
**Description** Contains register fields associated with rram_write_protection4.
**Reset** 0xffffffff

| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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**Name** RRAM_WRITE_PROTECTION5
**Address** 0x401430a8
**Description** Contains register fields associated with rram_write_protection5.
**Reset** 0xffffffff

| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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**Name** RRAM_WRITE_PROTECTION6
**Address** 0x401430ac
**Description** Contains register fields associated with rram_write_protection6.
**Reset** 0xffffffff

| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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**Name** RRAM_WRITE_PROTECTION7
**Address** 0x401430b0
**Description** Contains register fields associated with rram_write_protection7.
**Reset** 0xffffffff

| Name | disable |
|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | _bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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**Name** RRAM_STICKY_WRITE_PROTECTION0
**Address** 0x401430b4
**Description** Contains register fields associated with rram_sticky_write_protection0.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | if a bit is set, invalidates any writes to the corresponding 2kB block, can only revert with cpu reset | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION1
**Address** 0x401430b8
**Description** Contains register fields associated with rram_sticky_write_protection1.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION2
**Address** 0x401430bc
**Description** Contains register fields associated with rram_sticky_write_protection2.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION3

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**Address** 0x401430c0
**Description** Contains register fields associated with rram_sticky_write_protection3.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION4
**Address** 0x401430c4
**Description** Contains register fields associated with rram_sticky_write_protection4.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION5
**Address** 0x401430c8
**Description** Contains register fields associated with rram_sticky_write_protection5.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION6
**Address** 0x401430cc
**Description** Contains register fields associated with rram_sticky_write_protection6.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION7
**Address** 0x401430d0
**Description** Contains register fields associated with rram_sticky_write_protection7.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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---

**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS0
**Address** 0x401430d4
**Description** Contains register fields associated with rram_sticky_write_protection_status0.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | if read back value is 1, then the sticky bit has been set | 32'h0 |

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---

**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS1
**Address** 0x401430d8
**Description** Contains register fields associated with rram_sticky_write_protection_status1.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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---

**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS2
**Address** 0x401430dc
**Description** Contains register fields associated with rram_sticky_write_protection_status2.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS3
**Address** 0x401430e0
**Description** Contains register fields associated with rram_sticky_write_protection_status3.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS4
**Address** 0x401430e4
**Description** Contains register fields associated with rram_sticky_write_protection_status4.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS5
**Address** 0x401430e8
**Description** Contains register fields associated with rram_sticky_write_protection_status5.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS6
**Address** 0x401430ec
**Description** Contains register fields associated with rram_sticky_write_protection_status6.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| 31:0 | R | val | (no description) | 32'h0 |
|---|---|---|---|---|

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**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS7
**Address** 0x401430f0
**Description** Contains register fields associated with rram_sticky_write_protection_status7.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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**Name** RRAM_STICKY_READ_PROTECTION0
**Address** 0x401430f4
**Description** Contains register fields associated with rram_sticky_read_protection0.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | if a bit is set, invalidates any reads to the corresponding 2kB block, can only revert with cpu reset | 32'h0 |

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**Name** RRAM_STICKY_READ_PROTECTION_STATUS0
**Address** 0x401430f8
**Description** Contains register fields associated with rram_sticky_read_protection_status0.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | if read back value is 1, then the sticky bit has been set | 32'h0 |

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**Name** PROT_BITS_SET0
**Address** 0x401430fc
**Description** Contains register fields associated with prot_bits_set0.
**Reset** 0x00000000

| | rom_ |
|---|---|

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| Name | rom_read_locks_set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | rom_read_locks_set | These bits go on to set sticky bits that lock up rom read. Each bit maps to one 2kB sector. Not self clearing | 32'h00000000 |

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**Name** PROT_BITS_SET1
**Address** 0x40143100
**Description** Not self-clearing
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | suppress_otp_writes | suppress_rom_patch | apb_shub_lock_set | apb_qspi_lock_set | uart1_lock_set | uart0_lock_set |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | suppress_otp_writes | sets sticky bit preventing OTP writes | 1'b0 |
| 4 | R/W | suppress_rom_patch | sets sticky bit preventing ROM patches from being applied | 1'b0 |
| 3 | R/W | apb_shub_lock_set | sets sticky bit disabling the update of write address range. | 1'b0 |
| 2 | R/W | apb_qspi_lock_set | sets sticky bit disabling APB cores accessing qspi. | 1'b0 |
| 1 | R/W | uart1_lock_set | sets sticky bit disabling uart1_tx. | 1'b0 |
| 0 | R/W | uart0_lock_set | sets sticky bit disabling uart0_tx. | 1'b0 |

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**Name** PROT_BITS_STAT0
**Address** 0x40143104
**Description** Contains register fields associated with prot_bits_stat0.
**Reset** 0x00000000

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| Name | rom_read_locks_stat | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | rom_read_locks_stat | sticky bits that lock up rom read. Each bit maps to one 2kB sector. Not self clearing | 32'h00000000 |

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**Name** PROT_BITS_STAT1
**Address** 0x40143108
**Description** Not self-clearing
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | suppress_otp_writes_stat | suppress_rom_patch_stat | apb_shub_lock_stat | apb_qspi_lock_stat | uart1_lock_stat | uart0_lock_stat |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R | suppress_otp_writes_stat | sticky bit preventing OTP writes | 1'b0 |
| 4 | R | suppress_rom_patch_stat | sticky bit preventing ROM patches from being applied | 1'b0 |
| 3 | R | apb_shub_lock_stat | sticky bit disabling shub write address range. | 1'b0 |
| 2 | R | apb_qspi_lock_stat | sticky bit disabling APB cores accessing qspi. | 1'b0 |
| 1 | R | uart1_lock_stat | sticky bit disabling uart1_tx. | 1'b0 |
| | | | | |

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| | | | | |
|---|---|---|---|---|
| 0 | R | uart0_lock_stat | sticky bit disabling uart0_tx. | 1'b0 |

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**Name** QSPI_STICKY_WRITE_PROTECTION0
**Address** 0x4014310c
**Description** Contains register fields associated with qspi_sticky_write_protection0.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | if a bit is set, invalidates any writes to the corresponding 2kB block, can only revert with cpu reset | 32'h0 |

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**Name** QSPI_STICKY_WRITE_PROTECTION1
**Address** 0x40143110
**Description** Contains register fields associated with qspi_sticky_write_protection1.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** QSPI_STICKY_WRITE_PROTECTION_STATUS0
**Address** 0x40143114
**Description** Contains register fields associated with qspi_sticky_write_protection_status0.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | if read back value is 1, then the sticky bit has been set | 32'h0 |

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**Name** QSPI_STICKY_WRITE_PROTECTION_STATUS1
**Address** 0x40143118
**Description** Contains register fields associated with qspi_sticky_write_protection_status1.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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**Name** PDSN
**Address** 0x4014311c
**Description** pin drive strength
**Reset** 0x00000000

| Name | RSVD | is_high | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 30:0 | R/W | is_high | 0=high drive stength/lower impedance<br>1=low drive strength/higher impedance | 31'd0 |

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**Name** CALCOUNTRC
**Address** 0x40143120
**Description** Contains register fields associated with calCountRC.
**Reset** 0x00000000

| Name | RSVD | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 22:0 | R | val | (no description) | 23'd0 |

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**Name** SECURE_DEBUG_CTRL
**Address** 0x40143124
**Description** Contains register fields associated with secure_debug_ctrl.
**Reset** 0x00000000

| Name | RSVD | sticky_disabled | soft_disabled |
|---|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | sticky_disabled | software writes 1 during bootload time to disable SWD interface.<br>Read value reflects the state of the secure debug FSM. | 1'd0 |
| 0 | R/W | soft_disabled | if sticky_disabled is 0, SW can write 0 to enable SWD interface.<br>Read value reflects the state of the secure debug FSM. | 1'd0 |

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**Name** SECURE_DEBUG_STAT
**Address** 0x40143128
**Description** Contains register fields associated with secure_debug_stat.
**Reset** 0x00000000

| Name | RSVD | sec_dbg_fsm |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 | 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2:0 | R | sec_dbg_fsm | state of secure debug FSM. | 3'd0 |

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**Name** CHIPID1
**Address** 0x40143ff0
**Description** Contains register fields associated with chipid1.
**Reset** 0x50617269

| Name | char0 | char1 | char2 | char3 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R | char0 | P | 8'h50 |
| 23:16 | R | char1 | a | 8'h61 |
| 15:8 | R | char2 | r | 8'h72 |
| 7:0 | R | char3 | i | 8'h69 |

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**Name** CHIPID2
**Address** 0x40143ff4
**Description** Contains register fields associated with chipid2.
**Reset** 0x73000000

| | char | char | char | char |
|---|---|---|---|---|

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| Name | char4 | | | | | | | | char5 | | | | | | | | char6 | | | | | | | | char7 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R | char4 | s | 8'h73 |
| 23:16 | R | char5 | nul | 8'h0 |
| 15:8 | R | char6 | nul | 8'h0 |
| 7:0 | R | char7 | nul | 8'h0 |

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**Name** CHIPREV
**Address** 0x40143ff8
**Description** Contains register fields associated with chiprev.
**Reset** 0x00000100

| Name | RSVD | | | | | | | | | | | | | | | running_off_32khz_xtal | major | | | | | | | minor | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16 | R | running_off_32khz_xtal | (no description) | 1'b0 |
| 15:8 | R | major | (no description) | 8'd1 |
| 7:0 | R | minor | (no description) | 8'd0 |

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**Name** CORE_ID
**Address** 0x40143ffc
**Description** Contains register fields associated with core_id.
**Reset** 0x57525052

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | WRPR in ASCII | 32'h57525052 |

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# UART0_NONSECURE

**Name** DATA
**Address** 0x40144000
**Description** Contains register fields associated with DATA.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | Data | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | Data | Read Received data.<br>Write Transmit data. | 8'b0 |

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**Name** STATE
**Address** 0x40144004
**Description** Contains register fields associated with STATE.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | tx_locked_stat | tx_idle | stop_bit_err | nUARTCTS | nUARTRTS | rx_overrun | tx_overrun | rx_buf_full | tx_buf_full |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8 | R/W | tx_locked_stat | 1=(Sticky) indicates whether tx is locked. Once locked, it remains locked until a system reset. (Read only) // (Atmosic) | 1'b0 |
| 7 | R/W | tx_idle | 1=tx state is idle (Read only) // (Atmosic) | 1'b0 |
| 6 | R/W | stop_bit_err | 1=stop_bit err (Read only) // (Atmosic) | 1'b0 |
| 5 | R/W | nUARTCTS | current status of nUARTCTS (Read only) // (Atmosic) | 1'b0 |
| 4 | R/W | nUARTRTS | current status of nUARTRTS (Read only) // (Atmosic) | 1'b0 |
| 3 | R/W | rx_overrun | RX buffer overrun, write 1 to clear. | 1'b0 |
| 2 | R/W | tx_overrun | TX buffer overrun, write 1 to clear. | 1'b0 |
| 1 | R/W | rx_buf_full | 1=RX buffer is not empty, 0= RX buffer is empty read-only. | 1'b0 |
| 0 | R/W | tx_buf_full | TX buffer full, read-only. | 1'b0 |

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**Name** CTRL
**Address** 0x40144008

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**Description** Contains register fields associated with CTRL.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | dma_mode_rx | dma_mode_tx | hs_testmode | rx_overrun_int_en | tx_overrun_int_en | RxIntEn | TxIntEn | RxEn | TxEn |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8 | R/W | dma_mode_rx | 1= dma can drain the rx fifo and the rx_intr is cleared when empty | 1'b0 |
| 7 | R/W | dma_mode_tx | 1= dma can fill the tx fifo and the tx_intr is cleared when full or done. | 1'b0 |
| 6 | R/W | hs_testmode | High-speed test mode for TX only. | 1'b0 |
| 5 | R/W | rx_overrun_int_en | RX overrun interrupt enable. | 1'b0 |
| 4 | R/W | tx_overrun_int_en | TX overrun interrupt enable. | 1'b0 |
| 3 | R/W | RxIntEn | RX interrupt enable. | 1'b0 |
| 2 | R/W | TxIntEn | TX interrupt enable. | 1'b0 |
| 1 | R/W | RxEn | RX enable. | 1'b0 |
| 0 | R/W | TxEn | TX enable. | 1'b0 |

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**Name** INTSTATUS_CLEAR
**Address** 0x4014400c
**Description** Contains register fields associated with INTSTATUS_CLEAR.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | rx_overflow_intr | tx_overflow_intr | rxintr | txintr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | rx_overflow_intr | RX overrun interrupt. Write 1 to clear. | 1'b0 |
| 2 | R/W | tx_overflow_intr | TX overrun interrupt. Write 1 to clear. | 1'b0 |
| 1 | R/W | rxintr | RX interrupt. Write 1 to clear. | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 0 | R/W | txintr | TX interrupt. Write 1 to clear. | 1'b0 |

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**Name** BAUDDIV
**Address** 0x40144010
**Description** Contains register fields associated with BAUDDIV.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | baud_div | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:0 | R/W | baud_div | Baud rate divider. The minimum number is 8. | 20'b0 |

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**Name** RX_LWM
**Address** 0x40144014
**Description** Contains register fields associated with RX_LWM.
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | Rx_LWM | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | Rx_LWM | low water mark of RX FIFO; a trigger is issued when (Atmosic)<br>RX FIFO has waited long enough to have received these many items, starting from<br>the first item.<br>Constraint: ( 1<= RX_LWM <= 14) | 5'd1 |

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**Name** TX_LWM
**Address** 0x40144018
**Description** Contains register fields associated with TX_LWM.
**Reset** 0x00000010

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | Tx_LWM | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| | | | low water mark of TX FIF0; a trigger is issued when (Atmosic) | |

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| | | | | |
|---|---|---|---|---|
| 4:0 | R/W | Tx_LWM | tX FIFO is drained to these many unoccupied slots.<br>Constraint (1<= TX_LWM <= 16) | 5'd16 |

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**Name** RX_FIFO_SPACES
**Address** 0x4014401c
**Description** Contains register fields associated with RX_FIFO_SPACES.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_fifo_spaces | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | rx_fifo_spaces | number of unoccupied spaces in RX_FIFO (Atmosic) | 5'b0 |

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**Name** TX_FIFO_SPACES
**Address** 0x40144020
**Description** Contains register fields associated with TX_FIFO_SPACES.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | tx_fifo_spaces | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | tx_fifo_spaces | number of unoccupied spaces in TX_FIFO (Atmosic) | 5'b0 |

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**Name** HW_FLOW_OVRD
**Address** 0x40144024
**Description** Contains register fields associated with HW_FLOW_OVRD.
**Reset** 0x0000002a

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| Name | RSVD | RTS_TRIG_LVL | nCTS_OVRD | nCTS_VAL | nRTS_OVRD | nRTS_VAL |
|---|---|---|---|---|---|---|
| **Bits** | **31:7** | **6:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6:4 | R/W | RTS_TRIG_LVL | threshold of rx fifo to de-assert nRTS. The value corresponds to the number of open spaces in RX fifo.<br>minimum programmable value is 2. | 3'd2 |
| 3 | R/W | nCTS_OVRD | 0= nUARTCTS override override disabled; 1(default)= nUARTCTS override enabled (Atmosic) | 1'b1 |
| 2 | R/W | nCTS_VAL | 0(default) = nCTS is asserted to enable TX; 1= nCTS is deasserted | 1'b0 |
| 1 | R/W | nRTS_OVRD | 0= nUARTRTS override override disabled; 1(default)= nUARTRTS override enabled ;(Atmosic) | 1'b1 |
| 0 | R/W | nRTS_VAL | 0(default) = nRTS is asserted to enable external TX; 1= nRTS is deasserted | 1'b0 |

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**Name** BAUD_RATE_DETECT
**Address** 0x40144028
**Description** Contains register fields associated with BAUD_RATE_DETECT.
**Reset** 0x00000000

| Name | RSVD | brd |
|---|---|---|
| **Bits** | **31:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | brd | 1=set by SW, HW treat the next frame/s as baud rate detection byte, cleared by HW once baud rate is detected. 0=normal mode | 1'b0 |

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**Name** BAUD_DIV_CEIL
**Address** 0x4014402c
**Description** Contains register fields associated with BAUD_DIV_CEIL.
**Reset** 0x000fffff

| Name | RSVD | baud_div_ceil |
|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:0 | R/W | bauddiv_ceil | sets max reference for baud rate in auto detection. Auto detection restarts if start bit is not detected within this window. | 20'hfffff |

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**Name** BRD_STAT
**Address** 0x40144030
**Description** Contains register fields associated with BRD_STAT.
**Reset** 0x00000000

| Name | RSVD | start_bit_timeout | on_going |
|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | start_bit_timeout | Previous synch bit too long. The calibration restarts after timeout. | 1'b0 |
| 0 | R | on_going | auto detection is on going | 1'b0 |

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**Name** INTMASK
**Address** 0x40144034
**Description** Contains register fields associated with INTMASK.
**Reset** 0x0000000f

| Name | RSVD | rx_overflow_int_en | tx_overflow_int_en | rxint_en | txint_en |
|---|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | rx_overflow_int_en | mask enable for RX overflow interrupt. | 1'b1 |
| 2 | R/W | tx_overflow_int_en | mask enable for TX overflow interrupt. | 1'b1 |
| 1 | R/W | rxint_en | mask enable for RX interrupt. | 1'b1 |
| 0 | R/W | txint_en | mask enable for TX interrupt. | 1'b1 |

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**Name** RX_FIFO_ENTRIES
**Address** 0x40144038
**Description** Contains register fields associated with RX_FIFO_ENTRIES.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_fifo_entries | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | rx_fifo_entries | number of occupied spaces in RX_FIFO. DMA reads this value to determine how many reads to do in each round. | 5'b0 |

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**Name** INTSTATUS_SET
**Address** 0x4014403c
**Description** Contains register fields associated with INTSTATUS_SET.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_overflow_intr_set | tx_overflow_intr_set | rxintr_set | txintr_set |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | rx_overflow_intr_set | RX overrun interrupt. Write 1 to set. (self-clearing) | 1'b0 |
| 2 | R/W | tx_overflow_intr_set | TX overrun interrupt. Write 1 to set. (self-clearing) | 1'b0 |
| 1 | R/W | rxintr_set | RX interrupt. Write 1 to set. (self-clearing) | 1'b0 |
| 0 | R/W | txintr_set | TX interrupt. Write 1 to set. (self-clearing) | 1'b0 |

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**Name** PID4
**Address** 0x40144fd0
**Description** Contains register fields associated with PID4.
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | block_count | | | | jep106_c_code | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | block_count | Block count | 4'b0 |
| 3:0 | R | jep106_c_code | jep106_c_code | 4'd4 |

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**Name** PID5
**Address** 0x40144fd4
**Description** Contains register fields associated with PID5.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | pid5 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid5 | Peripheral ID Register 5. | 8'b0 |

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**Name** PID6
**Address** 0x40144fd8
**Description** Contains register fields associated with PID6.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | pid6 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid6 | Peripheral ID Register 6. | 8'b0 |

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**Name** PID7
**Address** 0x40144fdc
**Description** Contains register fields associated with PID7.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid7 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid7 | Peripheral ID Register 7. | 8'b0 |

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**Name** PID0
**Address** 0x40144fe0
**Description** Contains register fields associated with PID0.
**Reset** 0x00000021

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | part_num | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | part_num | Part number[7:0]. | 8'h21 |

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**Name** PID1
**Address** 0x40144fe4
**Description** Contains register fields associated with PID1.
**Reset** 0x000000b8

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | jep106_id_3_0 | | | | part_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | jep106_id_3_0 | jep106_id_3_0 | 4'hb |
| 3:0 | R | part_num | Part number[11:8]. | 4'h8 |

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**Name** PID2
**Address** 0x40144fe8
**Description** Contains register fields associated with PID2.
**Reset** 0x0000001b

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | rev | | | | jedec_used | jep106_id_6_4 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | rev | Revision | 4'h1 |
| 3 | R | jedec_used | jedec_used | 1'h1 |
| 2:0 | R | jep106_id_6_4 | jep106_id_6_4 | 3'h3 |

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**Name** PID3
**Address** 0x40144fec
**Description** Contains register fields associated with PID3.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | ecorevnum | | | | mod_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | ecorevnum | ECO revision number. | 4'h0 |
| 3:0 | R | mod_num | modification number. | 4'h0 |

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**Name** CID0
**Address** 0x40144ff0
**Description** Contains register fields associated with CID0.
**Reset** 0x0000000d

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid0 | Component ID Register 0 | 8'h0d |

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**Name** CID1
**Address** 0x40144ff4
**Description** Contains register fields associated with CID1.
**Reset** 0x000000f0

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid1 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid1 | Component ID Register 1 | 8'hf0 |

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**Name** CID2
**Address** 0x40144ff8
**Description** Contains register fields associated with CID2.
**Reset** 0x00000005

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid2 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid2 | Component ID Register 2 | 8'h05 |

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**Name** CID3
**Address** 0x40144ffc
**Description** Contains register fields associated with CID3.
**Reset** 0x000000b1

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid3 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid3 | Component ID Register 3 | 8'hb1 |

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# UART1_NONSECURE

**Name** DATA
**Address** 0x40145000
**Description** Contains register fields associated with DATA.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | Data | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | Data | Read Received data.<br>Write Transmit data. | 8'b0 |

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**Name** STATE
**Address** 0x40145004
**Description** Contains register fields associated with STATE.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | tx_locked_stat | tx_idle | stop_bit_err | nUARTCTS | nUARTRTS | rx_overrun | tx_overrun | rx_buf_full | tx_buf_full |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8 | R/W | tx_locked_stat | 1=(Sticky) indicates whether tx is locked. Once locked, it remains locked until a system reset. (Read only) // (Atmosic) | 1'b0 |
| 7 | R/W | tx_idle | 1=tx state is idle (Read only) // (Atmosic) | 1'b0 |
| 6 | R/W | stop_bit_err | 1=stop_bit err (Read only) // (Atmosic) | 1'b0 |
| 5 | R/W | nUARTCTS | current status of nUARTCTS (Read only) // (Atmosic) | 1'b0 |
| 4 | R/W | nUARTRTS | current status of nUARTRTS (Read only) // (Atmosic) | 1'b0 |
| 3 | R/W | rx_overrun | RX buffer overrun, write 1 to clear. | 1'b0 |
| 2 | R/W | tx_overrun | TX buffer overrun, write 1 to clear. | 1'b0 |
| 1 | R/W | rx_buf_full | 1=RX buffer is not empty, 0= RX buffer is empty read-only. | 1'b0 |
| 0 | R/W | tx_buf_full | TX buffer full, read-only. | 1'b0 |

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**Name** CTRL
**Address** 0x40145008
**Description** Contains register fields associated with CTRL.
**Reset** 0x00000000

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| Name | RSVD | dma_mode_rx | dma_mode_tx | hs_testmode | rx_overrun_int_en | tx_overrun_int_en | RxIntEn | TxIntEn | RxEn | TxEn |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8 | R/W | dma_mode_rx | 1= dma can drain the rx fifo and the rx_intr is cleared when empty | 1'b0 |
| 7 | R/W | dma_mode_tx | 1= dma can fill the tx fifo and the tx_intr is cleared when full or done. | 1'b0 |
| 6 | R/W | hs_testmode | High-speed test mode for TX only. | 1'b0 |
| 5 | R/W | rx_overrun_int_en | RX overrun interrupt enable. | 1'b0 |
| 4 | R/W | tx_overrun_int_en | TX overrun interrupt enable. | 1'b0 |
| 3 | R/W | RxIntEn | RX interrupt enable. | 1'b0 |
| 2 | R/W | TxIntEn | TX interrupt enable. | 1'b0 |
| 1 | R/W | RxEn | RX enable. | 1'b0 |
| 0 | R/W | TxEn | TX enable. | 1'b0 |

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**Name** INTSTATUS_CLEAR
**Address** 0x4014500c
**Description** Contains register fields associated with INTSTATUS_CLEAR.
**Reset** 0x00000000

| Name | RSVD | rx_overflow_intr | tx_overflow_intr | rxintr | txintr |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | rx_overflow_intr | RX overrun interrupt. Write 1 to clear. | 1'b0 |
| 2 | R/W | tx_overflow_intr | TX overrun interrupt. Write 1 to clear. | 1'b0 |
| 1 | R/W | rxintr | RX interrupt. Write 1 to clear. | 1'b0 |
| 0 | R/W | txintr | TX interrupt. Write 1 to clear. | 1'b0 |

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**Name** BAUDDIV
**Address** 0x40145010
**Description** Contains register fields associated with BAUDDIV.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | baud_div | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:0 | R/W | baud_div | Baud rate divider. The minimum number is 8. | 20'b0 |

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**Name** RX_LWM
**Address** 0x40145014
**Description** Contains register fields associated with RX_LWM.
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | Rx_LWM | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | Rx_LWM | low water mark of RX FIFO; a trigger is issued when (Atmosic)<br>RX FIFO has waited long enough to have received these many items, starting from<br>the first item.<br>Constraint: ( 1<= RX_LWM <= 14) | 5'd1 |

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**Name** TX_LWM
**Address** 0x40145018
**Description** Contains register fields associated with TX_LWM.
**Reset** 0x00000010

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | Tx_LWM | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | Tx_LWM | low water mark of TX FIF0; a trigger is issued when (Atmosic)<br>tX FIFO is drained to these many unoccupied slots.<br>Constraint (1<= TX_LWM <= 16) | 5'd16 |

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**Name** RX_FIFO_SPACES
**Address** 0x4014501c
**Description** Contains register fields associated with RX_FIFO_SPACES.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_fifo_spaces | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | rx_fifo_spaces | number of unoccupied spaces in RX_FIFO (Atmosic) | 5'b0 |

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**Name** TX_FIFO_SPACES
**Address** 0x40145020
**Description** Contains register fields associated with TX_FIFO_SPACES.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | tx_fifo_spaces | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | tx_fifo_spaces | number of unoccupied spaces in TX_FIFO (Atmosic) | 5'b0 |

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**Name** HW_FLOW_OVRD
**Address** 0x40145024
**Description** Contains register fields associated with HW_FLOW_OVRD.
**Reset** 0x0000002a

| | R | | | | | | | | | | | | | | | | | | | | | | | | | | | RTS_T | nCTS | nCT | nRTS | nRT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Name | SVD | | | | | | | | | | | | | | | | | | | | | | | | RIG_LVL | | | _OVRD | S_VAL | _OVRD | S_VAL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6:4 | R/W | RTS_TRIG_LVL | ~~threshold of rx fifo to de-assert nRTS. The value corresponds~~ to the number of open spaces in RX fifo.<br>minimum programmable value is 2. | 3'd2 |
| 3 | R/W | nCTS_OVRD | 0= nUARTCTS override override disabled; 1(default)= nUARTCTS override enabled (Atmosic) | 1'b1 |
| 2 | R/W | nCTS_VAL | 0(default) = nCTS is asserted to enable TX; 1= nCTS is deasserted | 1'b0 |
| 1 | R/W | nRTS_OVRD | 0= nUARTRTS override override disabled; 1(default)= nUARTRTS override enabled ;(Atmosic) | 1'b1 |
| 0 | R/W | nRTS_VAL | 0(default) = nRTS is asserted to enable external TX; 1= nRTS is deasserted | 1'b0 |

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**Name** BAUD_RATE_DETECT
**Address** 0x40145028
**Description** Contains register fields associated with BAUD_RATE_DETECT.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | brd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | brd | 1=set by SW, HW treat the next frame/s as baud rate detection byte, cleared by HW once baud rate is detected. 0=normal mode | 1'b0 |

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**Name** BAUD_DIV_CEIL
**Address** 0x4014502c
**Description** Contains register fields associated with BAUD_DIV_CEIL.
**Reset** 0x000fffff

| Name | RSVD | | | | | | | | | | | baud_div_ceil | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 19:0 | R/W | bauddiv_ceil | sets max reference for baud rate in auto detection. Auto detection restarts if start bit is not detected within this window. | 20'hffffff |

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**Name** BRD_STAT
**Address** 0x40145030
**Description** Contains register fields associated with BRD_STAT.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | start_bit_timeout | on_going |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | start_bit_timeout | Previous synch bit too long. The calibration restarts after timeout. | 1'b0 |
| 0 | R | on_going | auto detection is on going | 1'b0 |

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**Name** INTMASK
**Address** 0x40145034
**Description** Contains register fields associated with INTMASK.
**Reset** 0x0000000f

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_overflow_int_en | tx_overflow_int_en | rxint_en | txint_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | rx_overflow_int_en | mask enable for RX overflow interrupt. | 1'b1 |
| 2 | R/W | tx_overflow_int_en | mask enable for TX overflow interrupt. | 1'b1 |

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| | | | | |
|---|---|---|---|---|
| 1 | R/W | rxint_en | mask enable for RX interrupt. | 1'b1 |
| 0 | R/W | txint_en | mask enable for TX interrupt. | 1'b1 |

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**Name** RX_FIFO_ENTRIES
**Address** 0x40145038
**Description** Contains register fields associated with RX_FIFO_ENTRIES.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_fifo_entries | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | rx_fifo_entries | number of occupied spaces in RX_FIFO. DMA reads this value to determine how many reads to do in each round. | 5'b0 |

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**Name** INTSTATUS_SET
**Address** 0x4014503c
**Description** Contains register fields associated with INTSTATUS_SET.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_overflow_intr_set | tx_overflow_intr_set | rxintr_set | txintr_set |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | rx_overflow_intr_set | RX overrun interrupt. Write 1 to set. (self-clearing) | 1'b0 |
| 2 | R/W | tx_overflow_intr_set | TX overrun interrupt. Write 1 to set. (self-clearing) | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 1 | R/W | rxintr_set | RX interrupt. Write 1 to set. (self-clearing) | 1'b0 |
| 0 | R/W | txintr_set | TX interrupt. Write 1 to set. (self-clearing) | 1'b0 |

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**Name** PID4
**Address** 0x40145fd0
**Description** Contains register fields associated with PID4.
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | block_count | | | | jep106_c_code | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | block_count | Block count | 4'b0 |
| 3:0 | R | jep106_c_code | jep106_c_code | 4'd4 |

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**Name** PID5
**Address** 0x40145fd4
**Description** Contains register fields associated with PID5.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid5 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid5 | Peripheral ID Register 5. | 8'b0 |

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**Name** PID6
**Address** 0x40145fd8
**Description** Contains register fields associated with PID6.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid6 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid6 | Peripheral ID Register 6. | 8'b0 |

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**Name** PID7
**Address** 0x40145fdc
**Description** Contains register fields associated with PID7.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid7 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid7 | Peripheral ID Register 7. | 8'b0 |

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**Name** PID0
**Address** 0x40145fe0
**Description** Contains register fields associated with PID0.
**Reset** 0x00000021

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | part_num | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | part_num | Part number[7:0]. | 8'h21 |

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**Name** PID1
**Address** 0x40145fe4
**Description** Contains register fields associated with PID1.
**Reset** 0x000000b8

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | jep106_id_3_0 | | | | part_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | jep106_id_3_0 | jep106_id_3_0 | 4'hb |

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| | | | | |
|---|---|---|---|---|
| 3:0 | R | part_num | Part number[11:8]. | 4'h8 |

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**Name** PID2
**Address** 0x40145fe8
**Description** Contains register fields associated with PID2.
**Reset** 0x0000001b

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | rev | | | | jedec_used | jep106_id_6_4 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | rev | Revision | 4'h1 |
| 3 | R | jedec_used | jedec_used | 1'h1 |
| 2:0 | R | jep106_id_6_4 | jep106_id_6_4 | 3'h3 |

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**Name** PID3
**Address** 0x40145fec
**Description** Contains register fields associated with PID3.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | ecorevnum | | | | mod_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | ecorevnum | ECO revision number. | 4'h0 |
| 3:0 | R | mod_num | modification number. | 4'h0 |

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**Name** CID0
**Address** 0x40145ff0
**Description** Contains register fields associated with CID0.
**Reset** 0x0000000d

| Name | RSV | | | | | | | | | | | | | | | | | | | | | | | cid | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | D | | | | | | | | | | | | | | | | | | | | | | 0 | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid0 | Component ID Register 0 | 8'h0d |

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**Name** CID1
**Address** 0x40145ff4
**Description** Contains register fields associated with CID1.
**Reset** 0x000000f0

| Name | R S V D | | | | | | | | | | | | | | | | | | | | | | c i d 1 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid1 | Component ID Register 1 | 8'hf0 |

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**Name** CID2
**Address** 0x40145ff8
**Description** Contains register fields associated with CID2.
**Reset** 0x00000005

| Name | R S V D | | | | | | | | | | | | | | | | | | | | | | c i d 2 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid2 | Component ID Register 2 | 8'h05 |

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**Name** CID3
**Address** 0x40145ffc
**Description** Contains register fields associated with CID3.
**Reset** 0x000000b1

| Name | R S V D | | | | | | | | | | | | | | | | | | | | | | c i d 3 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid3 | Component ID Register 3 | 8'hb1 |

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# PWM_NONSECURE

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**Name** PWM_CTRL
**Address** 0x40146000
**Description** control signals for all pwm channels
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | alt_ok_to_run | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | alt_ok_to_run | alternative mechanism to start pwm channels; useful if application requires all used pwm channels start together<br>[0] = start pwm0 channel<br>[1] = start pwm1 channel<br>[2] = start pwm2 channel<br>[3] = start pwm3 channel<br>[4] = start pwm4 channel<br>[5] = start pwm5 channel<br>[6] = start pwm6 channel<br>[7] = start pwm7 channel | 8'd0 |

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**Name** PWM0_CTRL
**Address** 0x40146004
**Description** control signals for pwm0 channel
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | idle_polarity | invert | mode | | ok_to_run | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group. | 2'd0 |

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| | | | | |
|---|---|---|---|---|
| | | | 10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run.<br>11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM0_DUR
**Address** 0x40146008
**Description** durations config for pwm0 channel
**Reset** 0x00000000

| Name | hi_dur | | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM0_CFG
**Address** 0x4014600c
**Description** configuration for pwm0 channel
**Reset** 0x00000000

| Name | update | RSVD | | | | | | | | | | | | | | | | frame_count | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** PWM1_CTRL
**Address** 0x40146010
**Description** control signals for pwm1 channel
**Reset** 0x00000000

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | idle_polarity | invert | mode | | ok_to_run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run.<br>11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | 2'd0 |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM1_DUR
**Address** 0x40146014
**Description** durations config for pwm1 channel
**Reset** 0x00000000

| Name | hi_dur | | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM1_CFG
**Address** 0x40146018
**Description** configuration for pwm1 channel
**Reset** 0x00000000

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| Name | update | RSVD | frame_count |
|---|---|---|---|
| Bits | 31 | 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 | 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** PWM2_CTRL
**Address** 0x4014601c
**Description** control signals for pwm2 channel
**Reset** 0x00000000

| Name | RSVD | idle_polarity | invert | mode | ok_to_run |
|---|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 | 4 | 3 | 2 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run.<br>11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | 2'd0 |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM2_DUR
**Address** 0x40146020
**Description** durations config for pwm2 channel

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**Reset** 0x00000000

| Name | hi_dur | | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM2_CFG
**Address** 0x40146024
**Description** configuration for pwm2 channel
**Reset** 0x00000000

| Name | update | RSVD | | | | | | | | | | | | | | | | frame_count | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** PWM3_CTRL
**Address** 0x40146028
**Description** control signals for pwm3 channel
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | idle_polarity | invert | mode | ok_to_run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run.<br>11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | 2'd0 |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM3_DUR
**Address** 0x4014602c
**Description** durations config for pwm3 channel
**Reset** 0x00000000

| Name | hi_dur | | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM3_CFG
**Address** 0x40146030
**Description** configuration for pwm3 channel
**Reset** 0x00000000

| Name | update | RSVD | | | | | | | | | | | | | | | | | frame_count | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** PWM4_CTRL
**Address** 0x40146034
**Description** control signals for pwm4 channel
**Reset** 0x00000000

| Name | RSVD | idle_polarity | invert | mode | ok_to_run |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4** | **3** | **2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run.<br>11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | 2'd0 |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM4_DUR
**Address** 0x40146038
**Description** durations config for pwm4 channel
**Reset** 0x00000000

| Name | hi_dur | lo_dur |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |

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| | | | | |
|---|---|---|---|---|
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM4_CFG
**Address** 0x4014603c
**Description** configuration for pwm4 channel
**Reset** 0x00000000

| Name | update | RSVD | | | | | | | | | | | | | | | | | | frame_count | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** PWM5_CTRL
**Address** 0x40146040
**Description** control signals for pwm5 channel
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | idle_polarity | invert | mode | | ok_to_run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run. | 2'd0 |

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| | | | | |
|---|---|---|---|---|
| | | | 11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM5_DUR
**Address** 0x40146044
**Description** durations config for pwm5 channel
**Reset** 0x00000000

| Name | hi_dur | | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM5_CFG
**Address** 0x40146048
**Description** configuration for pwm5 channel
**Reset** 0x00000000

| Name | update | RSVD | | | | | | | | | | | | | | | | frame_count | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** PWM6_CTRL
**Address** 0x4014604c
**Description** control signals for pwm6 channel
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | i | |

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | idle_polarity | invert | mode | | ok_to_run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run.<br>11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | 2'd0 |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM6_DUR
**Address** 0x40146050
**Description** durations config for pwm6 channel
**Reset** 0x00000000

| Name | hi_dur | | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM6_CFG
**Address** 0x40146054
**Description** configuration for pwm6 channel
**Reset** 0x00000000

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| Name | pdate | RSVD | | | | | | | | | | | | | | | | me_count | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** PWM7_CTRL
**Address** 0x40146058
**Description** control signals for pwm7 channel
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | idle_polarity | invert | mode | | ok_to_run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run.<br>11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | 2'd0 |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM7_DUR
**Address** 0x4014605c
**Description** durations config for pwm7 channel
**Reset** 0x00000000

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| Name | hi_dur | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM7_CFG
**Address** 0x40146060
**Description** configuration for pwm7 channel
**Reset** 0x00000000

| Name | update | RSVD | | | | | | | | | | | | | | | | frame_count | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** FSM_STATE
**Address** 0x40146064
**Description** fsm state
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | ch7 | ch6 | ch5 | ch4 | ch3 | ch2 | ch1 | ch0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:14 | R | ch7 | current state of pwm7 channel. | 2'd0 |
| 13:12 | R | ch6 | current state of pwm6 channel. | 2'd0 |
| 11:10 | R | ch5 | current state of pwm5 channel. | 2'd0 |
| 9:8 | R | ch4 | current state of pwm4 channel. | 2'd0 |
| 7:6 | R | ch3 | current state of pwm3 channel. | 2'd0 |
| 5:4 | R | ch2 | current state of pwm2 channel. | 2'd0 |

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| | | | | |
|---|---|---|---|---|
| 3:2 | R | ch1 | current state of pwm1 channel. | 2'd0 |
| 1:0 | R | ch0 | current state of pwm0 channel. | 2'd0 |

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**Name** FIFO_CFG
**Address** 0x40146068
**Description** fifo configuration
**Reset** 0x0000000f

| Name | RSVD | ch_stop | flush | RSVD | lwm |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8** | **7** | **6** | **5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7 | R/W | ch_stop | when set, stops channel running in IR_FIFO_MODE after last fifo command is done.<br>otherwise, the last command keeps repeating. | 1'b0 |
| 6 | R/W | flush | Signal's rising edge clears out fifo. Not self-clearing | 1'b0 |
| 4:0 | R/W | lwm | low water mark (lwm) of FIF0; a trigger is issued when FIFO has been drained to these many open slots.<br>Constraint (1<= LWM <= 16)<br>triggers when the number of open slots changes from (lwm - 1) to (lwm).<br>lwm = 15 means an interrupt is issued as the number of open slots changes from 14 to 15.<br>In other words, the number of entries in the fifo changes from 2 to 1. | 5'd15 |

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**Name** FIFO_CARRIER1_DUR
**Address** 0x4014606c
**Description** carrier1 frequency durations for IR_FIFO_MODE
**Reset** 0x00000000

| Name | hi_dur | lo_dur |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** FIFO_CARRIER2_DUR
**Address** 0x40146070
**Description** carrier2 frequency durations for IR_FIFO_MODE
**Reset** 0x00000000

| | h | l |
|---|---|---|

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| Name | hi_dur | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** FIFO_STAT
**Address** 0x40146074
**Description** fifo status
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | empty | full |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | empty | fifo is empty. | 1'b0 |
| 0 | R | full | fifo is full. APB is stalled if a write is requested. | 1'b0 |

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**Name** FIFO_STAT1
**Address** 0x40146078
**Description** fifo status
**Reset** 0x00000010

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | num_open_slots | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | num_open_slots | open slots in FIFO | 5'd16 |

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**Name** FIFO_SEL_ERR_SOURCE
**Address** 0x4014607c

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**Description** HW sets and SW write 0 to clear and 1 to set these bits. Set takes precedence.
**Reset** 0x00000000

| Name | RSVD | ch7_fifo_sel_err | ch6_fifo_sel_err | ch5_fifo_sel_err | ch4_fifo_sel_err | ch3_fifo_sel_err | ch2_fifo_sel_err | ch1_fifo_sel_err | ch0_fifo_sel_err |
|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7 | R/W | ch7_fifo_sel_err | pwm7 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 6 | R/W | ch6_fifo_sel_err | pwm6 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 5 | R/W | ch5_fifo_sel_err | pwm5 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 4 | R/W | ch4_fifo_sel_err | pwm4 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 3 | R/W | ch3_fifo_sel_err | pwm3 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 2 | R/W | ch2_fifo_sel_err | pwm2 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 1 | R/W | ch1_fifo_sel_err | pwm1 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 0 | R/W | ch0_fifo_sel_err | pwm0 channel caused fifo_sel_err_intrpt; | 1'b0 |

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**Name** INTERRUPTS
**Address** 0x40146080
**Description** Contains register fields associated with interrupts.
**Reset** 0x00000000

| Name | RSVD | fifo_cmd_done_intrpt | fifo_sel_err_intrpt | fifo_lwm_hit_intrpt | fifo_overflow_intrpt |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | fifo_cmd_done_intrpt | HW sets when a channel is in IR_FIFO_MODE and last cmmand in fifo is done. | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 2 | R | fifo_sel_err_intrpt | HW sets when a channel is in IR_FIFO_MODE but not given the FIFO; FIFO being used by another channel.<br>With the conflict condition as described, an error assertion happens when channel is in idle and ok_to_run is issued.<br>Consequently, the internal state registers of the channel revert to default values. | 1'b0 |
| 1 | R | fifo_lwm_hit_intrpt | HW sets when lwm condition is hit. | 1'b0 |
| 0 | R | fifo_ovrflow_intrpt | HW sets when a write to fifo happens when fifo is full. | 1'b0 |

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**Name** INTERRUPTS_MASK
**Address** 0x40146084
**Description** Contains register fields associated with interrupts_mask.
**Reset** 0x00000003

| Name | RSVD | mask_intrpt3 | mask_intrpt2 | mask_intrpt1 | mask_intrpt0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | mask_intrpt3 | allows fifo_cmd_done_intrpt to propagate to channel's combined interrupt. | 1'b0 |
| 2 | R/W | mask_intrpt2 | allows fifo_sel_err_intrpt to propagate to channel's combined interrupt. | 1'b0 |
| 1 | R/W | mask_intrpt1 | allows lwm_hit_intrpt to propogate to channel's combined interrupt. | 1'b1 |
| 0 | R/W | mask_intrpt0 | allows fifo_ovrflow_intrpt to propogate to channel's combined interrupt. | 1'b1 |

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**Name** INTERRUPTS_CLEAR
**Address** 0x40146088
**Description** Contains register fields associated with interrupts_clear.
**Reset** 0x00000000

| Name | RSVD | clear_intrpt3 | clear_intrpt2 | clear_intrpt1 | clear_intrpt0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | clear_intrpt3 | clears fifo_cmd_done_intrpt; not auto cleared. | 1'b0 |
| 2 | R/W | clear_intrpt2 | clears fifo_sel_error_intrpt; not auto cleared. | 1'b0 |
| 1 | R/W | clear_intrpt1 | clears lwm_hit_intrpt; not auto cleared. | 1'b0 |

| | | | | |
|---|---|---|---|---|
| 0 | R/W | clear_intrpt0 | clears fifo_overflow_intrpt; not auto cleared. | 1'b0 |

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**Name** INTERRUPTS_SET
**Address** 0x4014608c
**Description** Contains register fields associated with interrupts_set.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | set_intrpt3 | set_intrpt2 | set_intrpt1 | set_intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | set_intrpt3 | sets fifo_cmd_done_intrpt; not auto cleared. | 1'b0 |
| 2 | R/W | set_intrpt2 | sets fifo_sel_error_intrpt; not auto cleared. | 1'b0 |
| 1 | R/W | set_intrpt1 | sets lwm_hit_intrpt; not auto cleared. | 1'b0 |
| 0 | R/W | set_intrpt0 | sets fifo_overflow_intrpt; not auto cleared. | 1'b0 |

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**Name** FIFO_DATA
**Address** 0x40146ff0
**Description** write data to be stored in fifo
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | wdata | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | wdata | [15] selects carrier frequency<br>0: use carrier frequency defined by fifo_carrier1_dur<br>1: use carrier frequency defined by fifo_carrier2_dur<br>[14] selects between mark and space IR regions<br>0: output lo for the space region in IR.<br>space region frame: hi_dur=0, lo_dur= hi+lo dur of selected carrier frequency<br>max lo_dur is limited to 16bits.<br>1: output value defined by hi/lo dur for the mark region in IR<br>[13:0]: frame_count | 16'd0 |

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**Name** CORE_ID
**Address** 0x40146ff4
**Description** Contains register fields associated with core_id.
**Reset** 0x50574d20

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| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | PWM in ASCII | 32'h50574d20 |

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**Name** REV_HASH
**Address** 0x40146ff8
**Description** Contains register fields associated with rev_hash.
**Reset** 0x00000000

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | crc32 of this document; don't change value to anything other than 32'h00000000; scripts will fill it out in the verilog | 32'h00000000 |

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**Name** REV_KEY
**Address** 0x40146ffc
**Description** Contains register fields associated with rev_key.
**Reset** 0x52455620

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | REV in ASCII | 32'h52455620 |

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# SPI0_NONSECURE

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**Name** CTRL
**Address** 0x40147000
**Description** SPI Control Register
**Reset** 0x00010301

| Name | dma_mode | RSVD | spi3wire_en | s_wr_opcode | s_rd_opcode | cpol | cpha | mstr |
|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20** | **19** | **18 17 16 15 14 13 12 11** | **10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | dma_mode | 1 = dma controls spi core | 1'b0 |
| 19 | R/W | spi3wire_en | 1=enable 3-wire spi transactions, mosi pad becomes bi-directional<br>0=default 4-wire spi transactions | 1'b0 |
| 18:11 | R/W | s_wr_opcode | slave write opcode | 8'h20 |
| 10:3 | R/W | s_rd_opcode | slave read opcode | 8'h60 |
| 2 | R/W | cpol | spi clock polarity 0=spi clock low in idle; cs high 1=spi clock high in idle; cs high | 1'b0 |
| 1 | R/W | cpha | spi clock phase 0=leading edge sample, trailing edge toggle 1=leading edge toggle, trailing edge sample | 1'b0 |
| 0 | R/W | mstr | 1=master interface<br>0=slave interface | 1'b1 |

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**Name** TRANSACTION_SETUP
**Address** 0x40147004
**Description** transaction setup
**Reset** 0x00000000

| Name | csn_bubble | invert_mosi | dummy_cycles | | | csn_stays_low | loopback | opcode | | | | | | | clkdiv | | | | | | | | | rwb | num_data_bytes | | | | start |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | csn_bubble | introduce a delay between spi_csn falling and first bit on mosi | 1'b0 |
| 30 | R/W | invert_mosi | invert mosi; keep static during entire spi transaction | 1'b0 |
| 29:26 | R/W | dummy_cycles | number of extra shift cycles to help with turn-around | 4'd0 |
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 24 | R/W | loopback | if 1, mosi is internally connected to miso | 1'b0 |
| 23:16 | R/W | opcode | opcode | 8'd0 |
| 15:6 | R/W | clkdiv | freq_sclk = f_pclk/(2*(clkdiv+1)) | 10'd0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; not auto cleared | 1'b0 |

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**Name** TRANSACTION_SETUP_DMA
**Address** 0x40147008
**Description** transaction setup by DMA
**Reset** 0x00000000

| | | csn_ | | num_ |
|---|---|---|---|---|

| Name | RSVD | csn_stays_low | RSVD | rwb | data_bytes | start |
|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26** | **25** | **24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6** | **5** | **4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; auto cleared | 1'b0 |

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**Name** TRANSACTION_STATUS
**Address** 0x4014700c
**Description** transaction status
**Reset** 0x00000000

| Name | RSVD | opcode_status | RSVD | running | done |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R | opcode_status | master or slave opcode status | 8'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | done | 1=indicate master or slave read/write transaction is done | 1'b0 |

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**Name** DATA_BYTES_LOWER
**Address** 0x40147010
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores read/write address
**Reset** 0x00000000

| Name | byte3 | byte2 | byte1 | byte0 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:24 | R/W | byte3 | data byte3 | 8'b0 |
| 23:16 | R/W | byte2 | data byte2 | 8'b0 |
| 15:8 | R/W | byte1 | data byte1 | 8'b0 |
| 7:0 | R/W | byte0 | data byte0 | 8'b0 |

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**Name** DATA_BYTES_UPPER
**Address** 0x40147014
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores write data; preload read data
**Reset** 0x00000000

| Name | byte7 | | | | | | | | byte6 | | | | | | | | byte5 | | | | | | | | byte4 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte7 | data byte7 | 8'b0 |
| 23:16 | R/W | byte6 | data byte6 | 8'b0 |
| 15:8 | R/W | byte5 | data byte5 | 8'b0 |
| 7:0 | R/W | byte4 | data byte4 | 8'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x40147018
**Description** interrupt mask
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | passthru3 | passthru2 | passthru1 | passthru0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | passthru3 | 1=allow interrupt3 to be OR'ed into core interrupt | 1'b0 |
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x4014701c
**Description** interrupt status
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | in | in | in | in |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | terrupt3 | terrupt2 | terruptt1 | terruptt0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | interrupt3 | from slave, error read-data invalid; independent of mask | 1'b0 |
| 2 | R | interrupt2 | from slave, error unknown command; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, transaction done capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, transaction done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x40147020
**Description** interrupt set
**Reset** 0x00000000

| Name | RSVD | set_interrupt3 | set_interrupt2 | set_interrupt1 | set_interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | set_interrupt3 | set interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | set_interrupt2 | set interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | set_interrupt1 | set interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt0; not auto cleared | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x40147024
**Description** interrupt clear
**Reset** 0x00000000

| Name | RSVD | reset_inte | reset_inte | reset_inte | reset_inte |
|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | rrupt3 | rrupt2 | rruptl | rrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | reset_interrupt3 | clear interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | reset_interrupt2 | clear interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | reset_interrupt1 | clear interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | reset_interrupt0 | clear interrupt0; not auto cleared | 1'b0 |

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**Name** CORE_ID
**Address** 0x40147ffc
**Description** core ID
**Reset** 0x53504920

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | SPI in ASCII | 32'h53504920 |

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# KSM_NONSECURE

**Name** CTRL0
**Address** 0x4014a000
**Description** Contains register fields associated with ctrl0.
**Reset** 0x00000800

| Name | RSVD | ksm_not_empty_to_interrupt | consecscan | RSVD | flush | pop | loopback | go |
|---|---|---|---|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 12 | R/W | ksm_not_empty_to_intrpt | in addition to fifo_write, /not empty/ signal will also feed into setting intrpt[0] | 1'b0 |
| 11:8 | R/W | consecscan | how many consecutive scans are needed before key press or key release event? (0 is INVALID) | 4'd8 |
| 3 | R/W | flush | flushes the key packet fifo | 1'b0 |
| 2 | R/W | pop | read from keyboard event fifo; not self reseting | 1'b0 |
| 1 | R/W | loopback | enable lpback_ksi (see below) | 1'b0 |
| 0 | R/W | go | start the keyboard scanning; will scan while go is high; not self reseting | 1'b0 |

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**Name** TIME_PARAM0
**Address** 0x4014a004
**Description** Contains register fields associated with time_param0.
**Reset** 0x00100010

| Name | t1 | | | | | | | | | | | | | | | | t2 | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | t1 | from col falling to sampling first row | 16'd16 |
| 15:0 | R/W | t2 | from sampling last row to col rising | 16'd16 |

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**Name** TIME_PARAM1
**Address** 0x4014a008
**Description** Contains register fields associated with time_param1.
**Reset** 0x00080040

| Name | t3 | | | | | | | | | | | | | | | | t4 | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | t3 | gap between adjacent row samplings | 16'd8 |
| 15:0 | R/W | t4 | gap between col rising and next col falling | 16'd64 |

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**Name** MATRIX_SIZE
**Address** 0x4014a00c
**Description** Contains register fields associated with matrix_size.
**Reset** 0x00001307

| Name | RSVD | | | | | | | | | | | | | | | | | | | num_row | | | | | RSVD | | | num_col | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 12:8 | R/W | num_row | the actual number of rows is num_row + 1 | 5'd19 |
| 4:0 | R/W | num_col | the actual number of cols is num_col + 1 | 5'd7 |

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**Name** KEYBOARD_PACKET
**Address** 0x4014a040
**Description** Contains register fields associated with keyboard_packet.
**Reset** 0x00000000

| Name | RSVD | empty | pressed_released_n | col | row |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12** | **11** | **10** | **9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 11 | R | empty | if 1, ignore this pack et | 1'd0 |
| 10 | R | pressed_released_n | 1=pressed 0=released | 1'b0 |
| 9:5 | R | col | the column of the pressed or released key; 0-indexed | 5'd0 |
| 4:0 | R | row | the row of the pressed or released key; 0-indexed | 5'd0 |

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**Name** LOW_POWER
**Address** 0x4014a044
**Description** Contains register fields associated with low_power.
**Reset** 0x00000000

| Name | RSVD | key |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11** | **10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 10:0 | R | key | this is the key press or release that woke the device | 11'd0 |

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**Name** LPBACK_KSI
**Address** 0x4014a058
**Description** Contains register fields associated with lpback_ksi.
**Reset** 0x000fffff

| | | 1 |
|---|---|---|

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | lpback_ksi | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:0 | R/W | lpback_ksi | the values to be driven on each of the ksi lines (while in loopback) | 20'hfffff |

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**Name** STATUS
**Address** 0x4014a080
**Description** Contains register fields associated with status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | command_taken | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | command_taken | fsm in on 32kHz clock; it takes two 32kHz cycles for the command to be picked up | 1'b0 |

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**Name** INTERRUPTS
**Address** 0x4014a0c0
**Description** Contains register fields associated with interrupts.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | intrpt3 | intrpt2 | intrpt1 | intrpt0 | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | intrpt3 | flush read is done | 1'b0 |
| 2 | R | intrpt2 | pop read is done | 1'b0 |
| 1 | R | intrpt1 | keyboard event fifo overran | 1'b0 |
| 0 | R | intrpt0 | hardware wrote packet to keyboard event fifo | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x4014a0c4
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x0000000f

| Name | RSVD | mask_intrpt3 | mask_intrpt2 | mask_intrpt1 | mask_intrpt0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | mask_intrpt3 | allow intrpt3 to propogate to core's single output interrupt | 1'b1 |
| 2 | R/W | mask_intrpt2 | allow intrpt2 to propogate to core's single output interrupt | 1'b1 |
| 1 | R/W | mask_intrpt1 | allow intrpt1 to propogate to core's single output interrupt | 1'b1 |
| 0 | R/W | mask_intrpt0 | allow intrpt0 to propogate to core's single output interrupt | 1'b1 |

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**Name** INTERRUPT_CLEAR
**Address** 0x4014a0c8
**Description** Contains register fields associated with interrupt_clear.
**Reset** 0x00000000

| Name | RSVD | clear_intrpt3 | clear_intrpt2 | clear_intrpt1 | clear_intrpt0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | clear_intrpt3 | clear interrupt 3; not self reseting | 1'b0 |
| 2 | R/W | clear_intrpt2 | clear interrupt 2; not self reseting | 1'b0 |
| 1 | R/W | clear_intrpt1 | clear interrupt 1; not self reseting | 1'b0 |
| 0 | R/W | clear_intrpt0 | clear interrupt 0; not self reseting | 1'b0 |

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**Name** DEBUG
**Address** 0x4014aff4
**Description** Contains register fields associated with debug.
**Reset** 0x00000000

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| Name | fifo | | | | | | | | | | | | | | | | kb | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R | fifo | {overrun, underrun, full, empty, 4'd0, rdaddr(unsigned int; 4b), wraddr(unsigned int; 4b)} | 16'b0 |
| 15:0 | R | kb | {4'b0, debounce_available(array of booleans; 12b)} | 16'b0 |

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**Name** DEBUG2
**Address** 0x4014aff8
**Description** Contains register fields associated with debug2.
**Reset** 0x00000000

| Name | kp | | | | | | | | | | | | | | | | kr | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R | kp | how many keys have been pressed? | 16'b0 |
| 15:0 | R | kr | how many keys have been released? | 16'b0 |

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**Name** CORE_ID
**Address** 0x4014affc
**Description** Contains register fields associated with core_id.
**Reset** 0x4b534d20

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | KSM in ASCII | 32'h4b534d20 |

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# SPI2_NONSECURE

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**Name** CTRL
**Address** 0x4014b000
**Description** SPI Control Register
**Reset** 0x00010301

| Name | dma_mo | RSVD | spi3wire | s_wr_opc | s_rd_opc | cpol | cpha | mstr |
|---|---|---|---|---|---|---|---|---|

| | de | | | | | | | | | | | | _en | | | ode | | | | | | ode | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | dma_mode | 1 = dma controls spi core | 1'b0 |
| 19 | R/W | spi3wire_en | 1=enable 3-wire spi transactions, mosi pad becomes bi-directional<br>0=default 4-wire spi transactions | 1'b0 |
| 18:11 | R/W | s_wr_opcode | slave write opcode | 8'h20 |
| 10:3 | R/W | s_rd_opcode | slave read opcode | 8'h60 |
| 2 | R/W | cpol | spi clock polarity 0=spi clock low in idle; cs high 1=spi clock high in idle; cs high | 1'b0 |
| 1 | R/W | cpha | spi clock phase 0=leading edge sample, trailing edge toggle 1=leading edge toggle, trailing edge sample | 1'b0 |
| 0 | R/W | mstr | 1=master interface<br>0=slave interface | 1'b1 |

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**Name** TRANSACTION_SETUP
**Address** 0x4014b004
**Description** transaction setup
**Reset** 0x00000000

| Name | csn_bubble | invert_mosi | dummy_cycles | csn_stays_low | loopback | opcode | clkdiv | rwb | num_data_bytes | start |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29 28 27 26** | **25** | **24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6** | **5** | **4 3 2 1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | csn_bubble | introduce a delay between spi_csn falling and first bit on mosi | 1'b0 |
| 30 | R/W | invert_mosi | invert mosi; keep static during entire spi transaction | 1'b0 |
| 29:26 | R/W | dummy_cycles | number of extra shift cycles to help with turn-around | 4'd0 |
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 24 | R/W | loopback | if 1, mosi is internally connected to miso | 1'b0 |
| 23:16 | R/W | opcode | opcode | 8'd0 |
| 15:6 | R/W | clkdiv | freq_sclk = f_pclk/(2*(clkdiv+1)) | 10'd0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; not auto cleared | 1'b0 |

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**Name** TRANSACTION_SETUP_DMA
**Address** 0x4014b008
**Description** transaction setup by DMA
**Reset** 0x00000000

| Name | RSVD | csn_stays_low | RSVD | rwb | num_data_bytes | start |
|---|---|---|---|---|---|---|
| **Bits** | **31:26** | **25** | **24:6** | **5** | **4:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; auto cleared | 1'b0 |

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**Name** TRANSACTION_STATUS
**Address** 0x4014b00c
**Description** transaction status
**Reset** 0x00000000

| Name | RSVD | opcode_status | RSVD | running | done |
|---|---|---|---|---|---|
| **Bits** | **31:16** | **15:8** | **7:2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R | opcode_status | master or slave opcode status | 8'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | done | 1=indicate master or slave read/write transaction is done | 1'b0 |

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**Name** DATA_BYTES_LOWER
**Address** 0x4014b010
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores read/write address
**Reset** 0x00000000

| Name | byt | byt | byt | byt |
|---|---|---|---|---|

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| | e 3 | | | | | | | | e 2 | | | | | | | | e 1 | | | | | | | | e 0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte3 | data byte3 | 8'b0 |
| 23:16 | R/W | byte2 | data byte2 | 8'b0 |
| 15:8 | R/W | byte1 | data byte1 | 8'b0 |
| 7:0 | R/W | byte0 | data byte0 | 8'b0 |

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**Name** DATA_BYTES_UPPER
**Address** 0x4014b014
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores write data; preload read data
**Reset** 0x00000000

| Name | byte 7 | | | | | | | | byte 6 | | | | | | | | byte 5 | | | | | | | | byte 4 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte7 | data byte7 | 8'b0 |
| 23:16 | R/W | byte6 | data byte6 | 8'b0 |
| 15:8 | R/W | byte5 | data byte5 | 8'b0 |
| 7:0 | R/W | byte4 | data byte4 | 8'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x4014b018
**Description** interrupt mask
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | passthru3 | passthru2 | passthru1 | passthru0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | passthru3 | 1=allow interrupt3 to be OR'ed into core interrupt | 1'b0 |
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x4014b01c

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**Description** interrupt status
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | interrupt3 | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | interrupt3 | from slave, error read-data invalid; independent of mask | 1'b0 |
| 2 | R | interrupt2 | from slave, error unknown command; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, transaction done capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, transaction done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x4014b020
**Description** interrupt set
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | set_interrupt3 | set_interrupt2 | set_interrupt1 | set_interrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | set_interrupt3 | set interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | set_interrupt2 | set interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | set_interrupt1 | set interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt0; not auto cleared | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x4014b024
**Description** interrupt clear
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | reset | reset | reset | reset |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

CONFIDENTIAL

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | t_interrupt3 | t_interrupt2 | t_interrupt1 | t_interrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | reset_interrupt3 | clear interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | reset_interrupt2 | clear interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | reset_interrupt1 | clear interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | reset_interrupt0 | clear interrupt0; not auto cleared | 1'b0 |

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**Name** CORE_ID
**Address** 0x4014bffc
**Description** core ID
**Reset** 0x53504920

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | SPI in ASCII | 32'h53504920 |

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# QDEC_NONSECURE

**Name** SETUP_FOR_AXIS_0
**Address** 0x4014c000
**Description** Contains register fields associated with setup_for_axis_0.
**Reset** 0x00000030

| Name | RSVD | | | | | | | | | | | | | | | | | | |

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | a d | e a d | | a l | a l | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | purge_err_post_read | (no description) | 1'b1 |
| 4 | R/W | purge_accum_post_read | (no description) | 1'b1 |
| 3 | R/W | reverse_polarity | (no description) | 1'b0 |
| 2 | R/W | qsig_a_loopback_val | (no description) | 1'b0 |
| 1 | R/W | qsig_b_loopback_val | (no description) | 1'b0 |
| 0 | R/W | enable_loopback | (no description) | 1'b0 |

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**Name** ACCUM_PURGE_STATUS_FOR_AXIS_0
**Address** 0x4014c004
**Description** Contains register fields associated with accum_purge_status_for_axis_0.
**Reset** 0x00000000

| Name | RSVD | accum_purge_sts |
|---|---|---|
| **Bits** | **31 … 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | accum_purge_sts | (no description) | 1'b0 |

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**Name** ACCUMULATOR_VALUE_FOR_AXIS_0
**Address** 0x4014c008
**Description** Contains register fields associated with accumulator_value_for_axis_0.
**Reset** 0x00000000

| Name | accum_val |
|---|---|
| **Bits** | **31 … 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | accum_val | (no description) | 32'b0 |

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**Name** ERR_PURGE_STATUS_FOR_AXIS_0
**Address** 0x4014c00c
**Description** Contains register fields associated with err_purge_status_for_axis_0.
**Reset** 0x00000000

| Name | RSVD | err_purge_sts |
|---|---|---|
| **Bits** | **31:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | err_purge_sts | (no description) | 1'b0 |

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**Name** ERROR_COUNT_FOR_AXIS_0
**Address** 0x4014c010
**Description** Contains register fields associated with error_count_for_axis_0.
**Reset** 0x00000000

| Name | err_cnt |
|---|---|
| **Bits** | **31:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | err_cnt | (no description) | 32'b0 |

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**Name** SETUP_FOR_AXIS_1
**Address** 0x4014c014
**Description** Contains register fields associated with setup_for_axis_1.
**Reset** 0x00000030

| Name | RSV | purge_err_p | purge_accum_ | reverse_p | qsig_a_loop | qsig_b_loop | enable_lo |
|---|---|---|---|---|---|---|---|
| **Bits** | **31:6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | D | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| |

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | accum_val | (no description) | 32'b0 |

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**Name** ERR_PURGE_STATUS_FOR_AXIS_1
**Address** 0x4014c020
**Description** Contains register fields associated with err_purge_status_for_axis_1.
**Reset** 0x00000000

| Name | RSVD | err_purge_sts |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | err_purge_sts | (no description) | 1'b0 |

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**Name** ERROR_COUNT_FOR_AXIS_1
**Address** 0x4014c024
**Description** Contains register fields associated with error_count_for_axis_1.
**Reset** 0x00000000

| Name | err_cnt |
|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | err_cnt | (no description) | 32'b0 |

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**Name** SETUP_FOR_AXIS_2
**Address** 0x4014c028
**Description** Contains register fields associated with setup_for_axis_2.
**Reset** 0x00000030

| | purg | purge | re | qsig | qsig | en |
|---|---|---|---|---|---|---|

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| Name | RSVD | | |

| Name | accum_val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | accum_val | (no description) | 32'b0 |

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**Name** ERR_PURGE_STATUS_FOR_AXIS_2
**Address** 0x4014c034
**Description** Contains register fields associated with err_purge_status_for_axis_2.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | err_purge_sts |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | err_purge_sts | (no description) | 1'b0 |

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**Name** ERROR_COUNT_FOR_AXIS_2
**Address** 0x4014c038
**Description** Contains register fields associated with error_count_for_axis_2.
**Reset** 0x00000000

| Name | err_cnt | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | err_cnt | (no description) | 32'b0 |

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**Name** CORE_ID
**Address** 0x4014cffc
**Description** Contains register fields associated with core_id.
**Reset** 0x51444543

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| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | QDEC in ASCII | 32'h51444543 |

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# SPI1_NONSECURE

**Name** CTRL
**Address** 0x4014d000
**Description** SPI Control Register
**Reset** 0x00010301

| Name | dma_mode | RSVD | | | | | | | | | | spi3wire_en | s_wr_opcode | | | | | | | | s_rd_opcode | | | | | | | | cpol | cpha | mstr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | dma_mode | 1 = dma controls spi core | 1'b0 |
| 19 | R/W | spi3wire_en | 1=enable 3-wire spi transactions, mosi pad becomes bi-directional<br>0=default 4-wire spi transactions | 1'b0 |
| 18:11 | R/W | s_wr_opcode | slave write opcode | 8'h20 |
| 10:3 | R/W | s_rd_opcode | slave read opcode | 8'h60 |
| 2 | R/W | cpol | spi clock polarity 0=spi clock low in idle; cs high 1=spi clock high in idle; cs high | 1'b0 |
| 1 | R/W | cpha | spi clock phase 0=leading edge sample, trailing edge toggle 1=leading edge toggle, trailing edge sample | 1'b0 |
| 0 | R/W | mstr | 1=master interface<br>0=slave interface | 1'b1 |

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**Name** TRANSACTION_SETUP
**Address** 0x4014d004
**Description** transaction setup
**Reset** 0x00000000

| | csn_b | invert | dummy_ | csn_sta | loop | opc | clk | rw | num_dat | sta |
|---|---|---|---|---|---|---|---|---|---|---|

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| Name | csn_bubble | invert_mosi | dummy_cycles | csn_stays_low | loopback | opcode | clkdiv | rwb | num_data_bytes | start |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29:26** | **25** | **24** | **23:16** | **15:6** | **5** | **4:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | csn_bubble | introduce a delay between spi_csn falling and first bit on mosi | 1'b0 |
| 30 | R/W | invert_mosi | invert mosi; keep static during entire spi transaction | 1'b0 |
| 29:26 | R/W | dummy_cycles | number of extra shift cycles to help with turn-around | 4'd0 |
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 24 | R/W | loopback | if 1, mosi is internally connected to miso | 1'b0 |
| 23:16 | R/W | opcode | opcode | 8'd0 |
| 15:6 | R/W | clkdiv | freq_sclk = f_pclk/(2*(clkdiv+1)) | 10'd0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; not auto cleared | 1'b0 |

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**Name** TRANSACTION_SETUP_DMA
**Address** 0x4014d008
**Description** transaction setup by DMA
**Reset** 0x00000000

| Name | RSVD | csn_stays_low | RSVD | rwb | num_data_bytes | start |
|---|---|---|---|---|---|---|
| **Bits** | **31:26** | **25** | **24:6** | **5** | **4:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; auto cleared | 1'b0 |

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**Name** TRANSACTION_STATUS
**Address** 0x4014d00c
**Description** transaction status
**Reset** 0x00000000

| | | op | | |
|---|---|---|---|---|

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| Name | RSVD | | | | | | | | | | | | | | | | opcode_status | | | | | | | | RSVD | | | | | | running | done |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R | opcode_status | master or slave opcode status | 8'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | done | 1=indicate master or slave read/write transaction is done | 1'b0 |

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**Name** DATA_BYTES_LOWER
**Address** 0x4014d010
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores read/write address
**Reset** 0x00000000

| Name | byte3 | | | | | | | | byte2 | | | | | | | | byte1 | | | | | | | | byte0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte3 | data byte3 | 8'b0 |
| 23:16 | R/W | byte2 | data byte2 | 8'b0 |
| 15:8 | R/W | byte1 | data byte1 | 8'b0 |
| 7:0 | R/W | byte0 | data byte0 | 8'b0 |

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**Name** DATA_BYTES_UPPER
**Address** 0x4014d014
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores write data; preload read data
**Reset** 0x00000000

| Name | byte7 | | | | | | | | byte6 | | | | | | | | byte5 | | | | | | | | byte4 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte7 | data byte7 | 8'b0 |
| 23:16 | R/W | byte6 | data byte6 | 8'b0 |
| 15:8 | R/W | byte5 | data byte5 | 8'b0 |
| 7:0 | R/W | byte4 | data byte4 | 8'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x4014d018
**Description** interrupt mask
**Reset** 0x00000000

| Name | RSVD | passthru3 | passthru2 | passthru1 | passthru0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | passthru3 | 1=allow interrupt3 to be OR'ed into core interrupt | 1'b0 |
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x4014d01c
**Description** interrupt status
**Reset** 0x00000000

| Name | RSVD | interrupt3 | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | interrupt3 | from slave, error read-data invalid; independent of mask | 1'b0 |
| 2 | R | interrupt2 | from slave, error unknown command; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, transaction done capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, transaction done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x4014d020
**Description** interrupt set
**Reset** 0x00000000

| | | set_ | set_ | set_ | set_ |
|---|---|---|---|---|---|

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | interrupt3 | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | set_interrupt3 | set interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | set_interrupt2 | set interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | set_interrupt1 | set interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt0; not auto cleared | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x4014d024
**Description** interrupt clear
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | reset_interrupt3 | reset_interrupt2 | reset_interrupt1 | reset_interrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | reset_interrupt3 | clear interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | reset_interrupt2 | clear interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | reset_interrupt1 | clear interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | reset_interrupt0 | clear interrupt0; not auto cleared | 1'b0 |

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**Name** CORE_ID
**Address** 0x4014dffc
**Description** core ID
**Reset** 0x53504920

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:0 | R | id | SPI in ASCII | 32'h53504920 |

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# SLWTIMER_NONSECURE

**Name** CONTROL
**Address** 0x4014e000
**Description** control
**Reset** 0x00000000

| Name | RSVD | auto_reload | RSVD | reset_thres2 | reset_thres1 | reset_thres0 | reset_expired | load | clear |
|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10** | **9** | **8** | **7 6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8 | R/W | auto_reload | auto reload slow timer with initial value after expiring | 1'b0 |
| 5 | R/W | reset_thres2 | reset thres2 interrupt; not self clearing | 1'b0 |
| 4 | R/W | reset_thres1 | reset thres1 interrupt; not self clearing | 1'b0 |
| 3 | R/W | reset_thres0 | reset thres0 interrupt; not self clearing | 1'b0 |
| 2 | R/W | reset_expired | reset expired interrupt; not self clearing | 1'b0 |
| 1 | R/W | load | load slow timer | 1'b0 |
| 0 | R/W | clear | clear slow timer | 1'b0 |

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**Name** INIT_LOW
**Address** 0x4014e004
**Description** initial count value
**Reset** 0x00000000

| Name | init_low |
|---|---|
| **Bits** | **31:30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | init_low | lower 32b of the 40b initial count value | 32'b0 |

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**Name** INIT_HIGH
**Address** 0x4014e008

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**Description** initial count value
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | init_high | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | init_high | upper 8b of the 40b initial count value | 8'b0 |

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**Name** CNT_LOW
**Address** 0x4014e00c
**Description** current count value
**Reset** 0x00000000

| Name | cnt_low | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | cnt_low | lower 32b of the 40b current count value | 32'b0 |

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**Name** CNT_HIGH
**Address** 0x4014e010
**Description** current count value
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cnt_high | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cnt_high | upper 8b of the 40b current count value | 8'b0 |

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**Name** THRES0_LOW
**Address** 0x4014e014
**Description** threshold 0 value

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**Reset** 0x00000000

| Name | thres0_low | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | thres0_low | lower 32b of the 40b threshold 0 value; when counter decrements from one above to this value, an optional interrupt triggers | 32'b0 |

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**Name** THRES0_HIGH
**Address** 0x4014e018
**Description** threshold 0 value
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | thres0_high | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | thres0_high | upper 8b of the 40b theshold 0 value | 8'b0 |

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**Name** THRES1_LOW
**Address** 0x4014e01c
**Description** threshold 1 value
**Reset** 0x00000000

| Name | thres1_low | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | thres1_low | lower 32b of the 40b threshold 1 value; when counter decrements from | 32'b0 |

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| | | | | |
|---|---|---|---|---|
| | | | one above to this value, an optional interrupt triggers | |

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**Name** THRES1_HIGH
**Address** 0x4014e020
**Description** threshold 1 value
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | thres1_high | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | thres1_high | upper 8b of the 40b theshold 1 value | 8'b0 |

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**Name** THRES2_LOW
**Address** 0x4014e024
**Description** threshold 2 value
**Reset** 0x00000000

| Name | thres2_low | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | thres2_low | lower 32b of the 40b threshold 2 value; when counter decrements from one above to this value, an optional interrupt triggers | 32'b0 |

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**Name** THRES2_HIGH
**Address** 0x4014e028
**Description** threshold 2 value
**Reset** 0x00000000

| Name | RSV | | | | | | | | | | | | | | | | | | | | | | thres2_ | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | thres2_high |
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | thres2_high | upper 8b of the 40b theshold 2 value | 8'b0 |

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**Name** STATUS
**Address** 0x4014e02c
**Description** status information
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | op_running |
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | op_running | the operation (either load or clear) is still running; handoff to 32kHz domain takes time | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x4014e030
**Description** interrupt status
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | hit_thres2 | hit_thres1 | hit_thres0 | expired |
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | hit_thres2 | threshold 2 interrupt triggered | 1'b0 |
| 2 | R | hit_thres1 | threshold 1 interrupt triggered | 1'b0 |
| 1 | R | hit_thres0 | threshold 0 interrupt triggered | 1'b0 |
| 0 | R | expired | counter transitioned from 1 to 0 | 1'b0 |

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**Name** INTERRUPT_MASK

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**Address** 0x4014e034
**Description** interrupt mask
**Reset** 0x0000000f

| Name | RSVD | passthru_hit_thres2 | passthru_hit_thres1 | passthru_hit_thres0 | passthru_expired |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | passthru_hit_thres2 | 1=allow hit_thres2 to be OR'd into core interrupt | 1'b1 |
| 2 | R/W | passthru_hit_thres1 | 1=allow hit_thres1 to be OR'd into core interrupt | 1'b1 |
| 1 | R/W | passthru_hit_thres0 | 1=allow hit_thres0 to be OR'd into core interrupt | 1'b1 |
| 0 | R/W | passthru_expired | 1=allow counter expiring to be OR'd into core interrupt | 1'b1 |

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**Name** CORE_ID
**Address** 0x4014effc
**Description** core ID
**Reset** 0x53544d52

| Name | id |
|---|---|
| **Bits** | **31:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | 'STMR' in hex | 32'h53544d52 |

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# QSPI_NONSECURE

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**Name** TRANSACTION_SETUP
**Address** 0x4014f000
**Description** Contains register fields associated with transaction_setup.
**Reset** 0x22000000

| | rem | remote |
|---|---|---|

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| Name | RSVD | | remote_spi_has_control | remote_ahb_qspi_has_control | RSVD | | csn_val | clk_val | RSVD | | dout_3_ctrl | | RSVD | | dout_2_ctrl | | RSVD | | dout_1_ctrl | | RSVD | | dout_0_ctrl | | sample_din | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29 | R/W | remote_spi_has_control | (no description) | 1'b1 |
| 28 | R/W | remote_ahb_qspi_has_control | (no description) | 1'b0 |
| 25 | R/W | csn_val | (no description) | 1'b1 |
| 24 | R/W | clk_val | (no description) | 1'b0 |
| 21:20 | R/W | dout_3_ctrl | {oe val, drive val} | 2'h0 |
| 17:16 | R/W | dout_2_ctrl | {oe val, drive val} | 2'h0 |
| 13:12 | R/W | dout_1_ctrl | {oe val, drive val} | 2'h0 |
| 9:8 | R/W | dout_0_ctrl | {oe val, drive val} | 2'h0 |
| 7:0 | R/W | sample_din | (no description) | 8'h0 |

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**Name** READ_DATA
**Address** 0x4014f004
**Description** Contains register fields associated with read_data.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | rdata | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | rdata | (no description) | 8'b0 |

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**Name** OVERRIDE_DIN
**Address** 0x4014f008
**Description** Contains register fields associated with override_din.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | o | din |

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| Name | RSVD | override_din | din_override_val |
|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 | 4 | 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | override_din | (no description) | 1'b0 |
| 3:0 | R/W | din_override_val | (no description) | 4'b0 |

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**Name** MODE
**Address** 0x4014f00c
**Description** Contains register fields associated with mode.
**Reset** 0x00000002

| Name | RSVD | is_quad | is_dual |
|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | is_quad | high priority than is_dual | 1'b1 |
| 0 | R/W | is_dual | for debug | 1'b0 |

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**Name** REMOTE_AHB_SETUP
**Address** 0x4014f010
**Description** Contains register fields associated with remote_ahb_setup.
**Reset** 0x02030000

| Name | RSVD | slip_half_cyc | hyper | skew_csn_act_ | serialize_pp_ad | invalidate_entire | enable_fine_clock | enable_cach | enable_clock | wdata_word_sw | wdata_halfword_ | wdata_byte_sw | quad_overhea | rdata_byte_sw | clkdivsel | opcode | RSVD | is_opcode | mode | dummy_cycle |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 14 13 12 11 10 9 8 | 7 | 6 5 | 4 3 2 1 0 |

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | le | | wen | dress | _cache | _gating | e | s | ap | swap | ap | d | ap | | | | | | | | | | | | | | | s | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 30 | R/W | slip_half_cycle | (no description) | 1'b0 |
| 29 | R/W | hyper | core runs at 32MHz instead of 16MHz | 1'b0 |
| 28 | R/W | skew_csn_act_wen | in a write sequence, controller stalls the start of the sequence; set if div-by-1 | 1'b0 |
| 27 | R/W | serialize_pp_address | (no description) | 1'b0 |
| 26 | R/W | invalidate_entire_cache | (no description) | 1'b0 |
| 25 | R/W | enable_fine_clock_gating | (no description) | 1'b1 |
| 24 | R/W | enable_cache | (no description) | 1'b0 |
| 23 | R/W | enable_clocks | for the core clocks not the qspi interface clock | 1'b0 |
| 22 | R/W | wdata_word_swap | (no description) | 1'b0 |
| 21 | R/W | wdata_halfword_swap | (no description) | 1'b0 |
| 20 | R/W | wdata_byte_swap | (no description) | 1'b0 |
| 19 | R/W | quad_overhead | adds overhead cycles during quad I/O read mode<br>1 = opcode is fully serial (8-cycles), adds two cycles for performance enhance; (macronix, gigadev, winbond)<br>0 = opcode is parallelized (2-cycles); no performance enhance cycles needed (micron) | 1'b0 |
| 18 | R/W | rdata_byte_swap | does a byte swap before passing rdata to backplane | 1'b0 |
| 17:16 | R/W | clkdivsel | 0=1x 1=2x 2=4x 3=8x | 2'd3 |
| 15:8 | R/W | opcode | (no description) | 8'h0 |
| 6 | R/W | is_opcode | (no description) | 1'b0 |
| 5:4 | R/W | mode | 0,3=single 1=dual 2=quad | 2'd0 |
| 3:0 | R/W | dummy_cycles | (no description) | 4'd0 |

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**Name** REMOTE_AHB_SETUP_2
**Address** 0x4014f014
**Description** Contains register fields associated with remote_ahb_setup_2.
**Reset** 0x20050206

| Name | opcode_se | | | | | | | | opcode_wip | | | | | | | | opcode_pp | | | | | | | | opcode_we | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | opcode_se | (no description) | 8'h20 |
| 23:16 | R/W | opcode_wip | (no description) | 8'h05 |
| 15:8 | R/W | opcode_pp | (no description) | 8'h02 |
| 7:0 | R/W | opcode_we | (no description) | 8'h06 |

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**Name** REMOTE_AHB_SETUP_3
**Address** 0x4014f018
**Description** Contains register fields associated with remote_ahb_setup_3.
**Reset** 0x09968000

| Name | check_wle | wip_bit | | | wip_polarity | wle_bit | | | wle_polarity | enable_performance_mode | opcode_performance_mode | | | | | | | | RSVD | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | check_wle | (no description) | 1'b0 |
| 30:28 | R/W | wip_bit | (no description) | 3'd0 |
| 27 | R/W | wip_polarity | (no description) | 1'b1 |
| 26:24 | R/W | wle_bit | (no description) | 3'd1 |
| 23 | R/W | wle_polarity | (no description) | 1'b1 |
| 22 | R/W | enable_performance_mode | enable performace enhance mode for supported flash | 1'b0 |
| 21:14 | R/W | opcode_performance_mode | performance enhance mode opcode<br>macronix = 8'h5A<br>gigadev = 8'hAx<br>winbond = 8'h20 | 8'h5A |

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**Name** REMOTE_AHB_SETUP_4
**Address** 0x4014f01c
**Description** Contains register fields associated with remote_ahb_setup_4.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | invert_addr | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R/W | invert_addr | (no description) | 24'd0 |

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**Name** REMOTE_AHB_SETUP_5
**Address** 0x4014f020
**Description** Contains register fields associated with remote_ahb_setup_5.
**Reset** 0x02008200

| Name | RSVD | | stall_we2pp | | | | | | | | | stall_wle | | | | | | | | | pp_stall_wip | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29:20 | R/W | stall_we2pp | wen or wle to pp depending on check_wle setting | 10'd32 |
| 19:10 | R/W | stall_wle | (no description) | 10'd32 |
| 9:0 | R/W | pp_stall_wip | (no description) | 10'd512 |

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**Name** REMOTE_AHB_SETUP_6
**Address** 0x4014f024
**Description** Contains register fields associated with remote_ahb_setup_6.
**Reset** 0x00001fff

| Name | RSVD | | | | | | | | | | | | | | | | se_stall_wip | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | se_stall_wip | (no description) | 16'd8191 |

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**Name** REMOTE_AHB_SETUP_7
**Address** 0x4014f028
**Description** Contains register fields associated with remote_ahb_setup_7.
**Reset** 0x35000000

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| Name | RSVD | remap_legacy | remap_msb_loc_sub2 | remap_table |
|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28 27 26 25 24** | **23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29 | R/W | remap_legacy | (no description) | 1'b1 |
| 28:24 | R/W | remap_msb_loc_sub2 | (no description) | 5'd21 |
| 23:0 | R/W | remap_table | (no description) | 24'b0 |

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**Name** REMOTE_AHB_WLE_CNT
**Address** 0x4014f02c
**Description** Contains register fields associated with remote_ahb_wle_cnt.
**Reset** 0x00000000

| Name | RSVD | wle_cnt |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10** | **9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9:0 | R | wle_cnt | number of wle checks before it is true. | 10'd0 |

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**Name** REMOTE_AHB_WIP_CNT
**Address** 0x4014f030
**Description** Contains register fields associated with remote_ahb_wip_cnt.
**Reset** 0x00000000

| Name | RSVD | wip_cnt |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10** | **9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9:0 | R | wip_cnt | number of wip checks before it is over. | 10'd0 |

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**Name** REMOTE_AHB_DBG0
**Address** 0x4014f034
**Description** Contains register fields associated with remote_ahb_dbg0.
**Reset** 0x00000000

| Name | dbg0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dbg0 | (no description) | 32'd0 |

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**Name** REMOTE_AHB_DBG1
**Address** 0x4014f038
**Description** Contains register fields associated with remote_ahb_dbg1.
**Reset** 0x00000000

| Name | dbg1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dbg1 | (no description) | 32'd0 |

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**Name** REMOTE_AHB_DBG2
**Address** 0x4014f03c
**Description** Contains register fields associated with remote_ahb_dbg2.
**Reset** 0x00000000

| Name | dbg2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dbg2 | (no description) | 32'd0 |

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**Name** CORE_ID
**Address** 0x4014fffc
**Description** Contains register fields associated with core_id.
**Reset** 0x51535049

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | QSPI in ASCII | 32'h51535049 |

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# I2C0_NONSECURE

**Name** TRANSACTION_SETUP
**Address** 0x40150000
**Description** Contains register fields associated with transaction_setup.
**Reset** 0x00200000

| Name | RSVD | mstr | go | RSVD | ack_value_to_drive | master_drives_ack | RSVD | tail | RSVD | head |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22** | **21** | **20** | **19 18** | **17** | **16** | **15 14 13 12 11 10 9 8 7 6** | **5 4** | **3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 21 | R/W | mstr | - 1=master interface<br>- 0=slave interface | 1'b1 |
| 20 | R/W | go | transaction lauches on rising edge; not self reseting | 1'b0 |
| 17 | R/W | ack_value_to_drive | (no description) | 1'd0 |
| 16 | R/W | master_drives_ack | (no description) | 1'd0 |
| 5:4 | R/W | tail | - 0=stop<br>- 1=stall<br>- 2=restart | 2'd0 |
| 1:0 | R/W | head | - 0=start<br>- 1=stall | 2'd0 |

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**Name** CLOCK_CONTROL
**Address** 0x40150004
**Description** Contains register fields associated with clock_control.
**Reset** 0x00000007

| Name | RSVD | clk_pu | clkdiv |
|---|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 10 | R/W | clk_pu | scl pin pullup enable | 1'b0 |
| 9:0 | R/W | clkdiv | freq_sclk = f_pclk/(4*(clkdiv+1)) | 10'd7 |

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**Name** OUTGOING_DATA
**Address** 0x40150008
**Description** Contains register fields associated with outgoing_data.
**Reset** 0x0000ff00

| Name | RSVD | data_pu | data_oe | data_o |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 | 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16 | R/W | data_pu | sda pin pullup enable | 1'b0 |
| 15:8 | R/W | data_oe | (no description) | 8'hff |
| 7:0 | R/W | data_o | can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator | 8'b0 |

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**Name** OUTGOING_DATA1
**Address** 0x4015000c
**Description** stores outgoing read data
**Reset** 0x00000000

| Name | data1_o |
|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data1_o | slave: stores outgoing read data | 32'd0 |

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**Name** IO_CTRL
**Address** 0x40150010
**Description** Contains register fields associated with io_ctrl.
**Reset** 0x00000000

| Name | RS | data_i |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | V<br>D | | | | | | | | | | | | | | | | | | | | | | | | | | | | | e<br>_<br>o<br>v<br>r<br>d |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | data_ie_ovrd | - 1= ie is high<br>- 0= ie is ~oe | 1'b0 |

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**Name** INCOMING_DATA
**Address** 0x40150014
**Description** stores incoming write data
**Reset** 0x00000000

| Name | d<br>a<br>t<br>a<br>_<br>i | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data_i | master: only lsb byte used slave: all 4 bytes can be used | 32'd0 |

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**Name** INCOMING_DATA1
**Address** 0x40150018
**Description** stores read/write address
**Reset** 0x00000000

| Name | d<br>a<br>t<br>a<br>1<br>_<br>i | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data1_i | slave: stores read/write address | 32'd0 |

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**Name** TRANSACTION_STATUS
**Address** 0x4015001c
**Description** Contains register fields associated with transaction_status.
**Reset** 0x00000000

| | R<br>S | | | | | | | | | | | | | | | | | | | | | | | | | | | s<br>_ | r<br>u<br>n | a<br>c<br>k<br>_ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | VD | | | | | | | | | | | | | | | | | | | | | | | | | | | rwb | ning | value |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R | s_rwb | slave read/write status bit<br>- 0=write<br>- 1=read | 1'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | ack_value | useless if mast_drives_ack is set | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x40150020
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | passthru2 | passthru1 | passthru0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x40150024
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R | interrupt2 | from slave, rising edge on read data valid error; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, rising edge on done transaction capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, rising edge on done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x40150028
**Description** not auto cleared
**Reset** 0x00000000

| Name | RSVD | set_interrupt2 | set_interrupt1 | set_interrupt0 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | set_interrupt2 | set interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | set_interrupt1 | set interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt0; not auto cleared | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x4015002c
**Description** not auto cleared
**Reset** 0x00000000

| Name | RSVD | reset_interrupt2 | reset_interrupt1 | reset_interrupt0 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | reset_interrupt2 | clear interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | reset_interrupt1 | clear interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | reset_interrupt0 | clear interrupt0; not auto cleared | 1'b0 |

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**Name** ID

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**Address** 0x40150ffc
**Description** Contains register fields associated with id.
**Reset** 0x49324320

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | I2C in ASCII | 32'h49324320 |

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# I2C1_NONSECURE

**Name** TRANSACTION_SETUP
**Address** 0x40151000
**Description** Contains register fields associated with transaction_setup.
**Reset** 0x00200000

| Name | RSVD | | | | | | | | | | mstr | go | RSVD | | ack_value_to_drive | master_drives_ack | RSVD | | | | | | | | | | tail | | RSVD | | head | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 21 | R/W | mstr | - 1=master interface<br>- 0=slave interface | 1'b1 |
| 20 | R/W | go | transaction lauches on rising edge; not self reseting | 1'b0 |
| 17 | R/W | ack_value_to_drive | (no description) | 1'd0 |
| 16 | R/W | master_drives_ack | (no description) | 1'd0 |
| 5:4 | R/W | tail | - 0=stop<br>- 1=stall<br>- 2=restart | 2'd0 |
| 1:0 | R/W | head | - 0=start<br>- 1=stall | 2'd0 |

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**Name** CLOCK_CONTROL
**Address** 0x40151004
**Description** Contains register fields associated with clock_control.

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**Reset** 0x00000007

| Name | RSVD | clk_pu | clkdiv |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11** | **10** | **9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 10 | R/W | clk_pu | scl pin pullup enable | 1'b0 |
| 9:0 | R/W | clkdiv | freq_sclk = f_pclk/(4*(clkdiv+1)) | 10'd7 |

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**Name** OUTGOING_DATA
**Address** 0x40151008
**Description** Contains register fields associated with outgoing_data.
**Reset** 0x0000ff00

| Name | RSVD | data_pu | data_oe | data_o |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17** | **16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16 | R/W | data_pu | sda pin pullup enable | 1'b0 |
| 15:8 | R/W | data_oe | (no description) | 8'hff |
| 7:0 | R/W | data_o | can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator | 8'b0 |

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**Name** OUTGOING_DATA1
**Address** 0x4015100c
**Description** stores outgoing read data
**Reset** 0x00000000

| Name | data1_o |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data1_o | slave: stores outgoing read data | 32'd0 |

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**Name** IO_CTRL
**Address** 0x40151010
**Description** Contains register fields associated with io_ctrl.

CONFIDENTIAL

**Reset** 0x00000000

| Name | RSVD | data_ie_ovrd |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | data_ie_ovrd | - 1= ie is high<br>- 0= ie is ~oe | 1'b0 |

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**Name** INCOMING_DATA
**Address** 0x40151014
**Description** stores incoming write data
**Reset** 0x00000000

| Name | data_i |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data_i | master: only lsb byte used slave: all 4 bytes can be used | 32'd0 |

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**Name** INCOMING_DATA1
**Address** 0x40151018
**Description** stores read/write address
**Reset** 0x00000000

| Name | data1_i |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data1_i | slave: stores read/write address | 32'd0 |

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**Name** TRANSACTION_STATUS

CONFIDENTIAL

**Address** 0x4015101c
**Description** Contains register fields associated with transaction_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | s_rwb | running | ack_value |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R | s_rwb | slave read/write status bit<br>- 0=write<br>- 1=read | 1'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | ack_value | useless if mast_drives_ack is set | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x40151020
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | passthru2 | passthru1 | passthru0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x40151024
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | interrupt | interrupt | interrupt |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R | interrupt2 | from slave, rising edge on read data valid error; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, rising edge on done transaction capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, rising edge on done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x40151028
**Description** not auto cleared
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | set_interrupt2 | set_interrupt1 | set_interrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | set_interrupt2 | set interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | set_interrupt1 | set interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt0; not auto cleared | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x4015102c
**Description** not auto cleared
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | reset_interrupt2 | reset_interrupt1 | reset_interrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 2 | R/W | reset_interrupt2 | clear interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | reset_interrupt1 | clear interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | reset_interrupt0 | clear interrupt0; not auto cleared | 1'b0 |

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**Name** ID
**Address** 0x40151ffc
**Description** Contains register fields associated with id.
**Reset** 0x49324320

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | I2C in ASCII | 32'h49324320 |

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# NVM_NONSECURE

**Name** OPMODE
**Address** 0x40152000
**Description** Contains register fields associated with opmode.
**Reset** 0x00000000

| Name | go | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | program | read |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | go | opcode start when rising edge on 'go' is detected. | 1'b0 |
| 1 | R/W | program | program opcode | 1'b0 |
| 0 | R/W | read | read opcode | 1'b0 |

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**Name** CTRL
**Address** 0x40152004
**Description** Contains register fields associated with ctrl.
**Reset** 0x00000100

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | num_pgm_bit | burst_mode | ser_dout_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8:2 | R/W | num_pgm_bit | In single mode, specifies bit location to program; actual bit location is (programmed number - 1)<br>ex. to program bit 0; num_pgm_bit=1, to program bit 63; num_pgm_bit=64<br>In burst mode, program bits location from 0 up to (programmed number - 1). | 7'd64 |
| 1 | R/W | burst_mode | burst or single bit programming<br>1 = serial burst program bits starting at bit 0 up to specified bit number in num_pgm_bit register using specified input data from data0/data1 registers<br>0 = single program bit specified in num_pgm_bit register programming sets efuse bit to 1 | 1'b0 |
| 0 | R/W | ser_dout_en | enable serial data out and allow soft repair during read | 1'b0 |

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**Name** DATA0
**Address** 0x40152008
**Description** Contains register fields associated with data0.
**Reset** 0x00000000

| Name | data0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data0 | otp data word-0,<br>can be used for program or read data based on target operation | 32'd0 |

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**Name** DATA1
**Address** 0x4015200c
**Description** Contains register fields associated with data1.
**Reset** 0x00000000

| Name | data1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data1 | otp data word-1,<br>can be used for program or read data based on target operation | 32'd0 |

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**Name** STATUS
**Address** 0x40152010
**Description** Contains register fields associated with status.
**Reset** 0x00000000

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| Name | running | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | program_done | read_done |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R | running | read or program running | 1'b0 |
| 1 | R | program_done | programming done | 1'b0 |
| 0 | R | read_done | read done | 1'b0 |

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**Name** T_SP_VQ
**Address** 0x40152014
**Description** VDDQ to CSB setup time into program mode
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 10 nsec;<br>each lsb accounts for 16usec, max when all bits programmed is around 500usec<br>default 16usec based on 16MHz system clock | 5'd1 |

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**Name** T_HP_VQ
**Address** 0x40152018
**Description** VDDQ to CSB hold time out of program mode
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 10 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_SP_PG
**Address** 0x4015201c
**Description** PGM to CSB setup time into program mode
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 5 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_HP_PG
**Address** 0x40152020
**Description** PGM to CSB hold time into program mode
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 5 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_HP_CK
**Address** 0x40152024
**Description** SCLK to CSB hold time into program mode
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 31 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd1 |

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**Name** T_S_PGM

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**Address** 0x40152028
**Description** PGM to SCLK setup time to program bits
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:0 | R/W | cycles | want >= 8 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 6'd0 |

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**Name** T_H_PGM
**Address** 0x4015202c
**Description** PGM to SCLK hold time to program bits
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:0 | R/W | cycles | want >= 8 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 6'd0 |

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**Name** T_CKHP
**Address** 0x40152030
**Description** SCLK high period for program
**Reset** 0x000000a0

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | cycles | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 10:0 | R/W | cycles | want >= 9 usec and <= 11 usec; default for 10 usec<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 11'd160 |

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**Name** T_CKLP
**Address** 0x40152034

CONFIDENTIAL

**Description** SCLK low period for program
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 10 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_SPS_CK
**Address** 0x40152038
**Description** CSB to SCLK hold time out of program mode
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 31 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd1 |

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**Name** T_SR_CK
**Address** 0x4015203c
**Description** SCLK to CSB setup time into read mode
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 6 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_HR_CK
**Address** 0x40152040
**Description** SCLK to CSB hold time into read mode

CONFIDENTIAL

**Reset** 0x0000000e

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | cycles | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9:0 | R/W | cycles | want >= 940 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 10'd14 |

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**Name** T_CKLR
**Address** 0x40152044
**Description** SCLK low period for read
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 10 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_CKHR
**Address** 0x40152048
**Description** SCLK high period for read
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 10 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_SRS_CK
**Address** 0x4015204c
**Description** CSB to SCLK hold time out of read mode

CONFIDENTIAL

**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 20 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** INTERRUPT_STATUS
**Address** 0x40152050
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | intrpt0 | saw a rising edge on read or program done; independent of mask | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x40152054
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt0 | 1=allow intrpt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_SET
**Address** 0x40152058
**Description** not auto cleared
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | i |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

CONFIDENTIAL

| Name | RSVD | intrpt0 |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt0 | (no description) | 1'b0 |

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**Name** INTERRUPT_RESET
**Address** 0x4015205c
**Description** not auto cleared
**Reset** 0x00000000

| Name | RSVD | intrpt0 |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt0 | (no description) | 1'b0 |

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**Name** CORE_ID
**Address** 0x40152ffc
**Description** CORE ID
**Reset** 0x4e564d20

| Name | id |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | NVM in ASCII | 32'h4e564d20 |

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# WRPR1_NONSECURE

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**Name** APB0_CTRL
**Address** 0x40153000
**Description** Contains register fields associated with apb0_ctrl.
**Reset** 0x00000002

| Name | RSV | clk_ | sres | clk_en |
|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB1_CTRL
**Address** 0x40153004
**Description** Contains register fields associated with apb1_ctrl.
**Reset** 0x00000002

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB2_CTRL
**Address** 0x40153008
**Description** Contains register fields associated with apb2_ctrl.
**Reset** 0x00000002

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB3_CTRL
**Address** 0x4015300c
**Description** Contains register fields associated with apb3_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB4_CTRL
**Address** 0x40153010
**Description** Contains register fields associated with apb4_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB5_CTRL
**Address** 0x40153014
**Description** Contains register fields associated with apb5_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enabl |
|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | e |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB6_CTRL
**Address** 0x40153018
**Description** Contains register fields associated with apb6_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB7_CTRL
**Address** 0x4015301c
**Description** Contains register fields associated with apb7_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB8_CTRL
**Address** 0x40153020

CONFIDENTIAL

**Description** Contains register fields associated with apb8_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB9_CTRL
**Address** 0x40153024
**Description** Contains register fields associated with apb9_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB10_CTRL
**Address** 0x40153028
**Description** Contains register fields associated with apb10_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

CONFIDENTIAL

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB11_CTRL
**Address** 0x4015302c
**Description** Contains register fields associated with apb11_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB12_CTRL
**Address** 0x40153030
**Description** Contains register fields associated with apb12_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB13_CTRL
**Address** 0x40153034
**Description** Contains register fields associated with apb13_ctrl.
**Reset** 0x00000002

CONFIDENTIAL

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB14_CTRL
**Address** 0x40153038
**Description** Contains register fields associated with apb14_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB15_CTRL
**Address** 0x4015303c
**Description** Contains register fields associated with apb15_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** CORE_ID
**Address** 0x40153ffc
**Description** Contains register fields associated with core_id.
**Reset** 0x57525052

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | WRPR in ASCII | 32'h57525052 |

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# RADIO_NONSECURE

**Name** CTRL
**Address** 0x40155000
**Description** SPI Control Register
**Reset** 0x00010301

| Name | dma_mode | RSVD | | | | | | | | | | spi3wire_en | s_wr_opcode | | | | | | | | s_rd_opcode | | | | | | | | cpol | cpha | mstr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | dma_mode | 1 = dma controls spi core | 1'b0 |
| 19 | R/W | spi3wire_en | 1=enable 3-wire spi transactions, mosi pad becomes bi-directional<br>0=default 4-wire spi transactions | 1'b0 |
| 18:11 | R/W | s_wr_opcode | slave write opcode | 8'h20 |
| 10:3 | R/W | s_rd_opcode | slave read opcode | 8'h60 |
| 2 | R/W | cpol | spi clock polarity 0=spi clock low in idle; cs high 1=spi clock high in idle; cs high | 1'b0 |
| 1 | R/W | cpha | spi clock phase 0=leading edge sample, trailing edge toggle 1=leading edge toggle, trailing edge sample | 1'b0 |
| 0 | R/W | mstr | 1=master interface<br>0=slave interface | 1'b1 |

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**Name** TRANSACTION_SETUP
**Address** 0x40155004

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**Description** transaction setup
**Reset** 0x00000000

| Name | csn_bubble | invert_mosi | dummy_cycles | | | | csn_stays_low | loopback | opcode | | | | | | clkdiv | | | | | | | | | rwb | num_data_bytes | | | | start |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | csn_bubble | introduce a delay between spi_csn falling and first bit on mosi | 1'b0 |
| 30 | R/W | invert_mosi | invert mosi; keep static during entire spi transaction | 1'b0 |
| 29:26 | R/W | dummy_cycles | number of extra shift cycles to help with turn-around | 4'd0 |
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 24 | R/W | loopback | if 1, mosi is internally connected to miso | 1'b0 |
| 23:16 | R/W | opcode | opcode | 8'd0 |
| 15:6 | R/W | clkdiv | freq_sclk = f_pclk/(2*(clkdiv+1)) | 10'd0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; not auto cleared | 1'b0 |

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**Name** TRANSACTION_SETUP_DMA
**Address** 0x40155008
**Description** transaction setup by DMA
**Reset** 0x00000000

| Name | RSVD | | | | | csn_stays_low | RSVD | | | | | | | | | | | | | | | | | rwb | num_data_bytes | | | | start |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; auto cleared | 1'b0 |

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**Name** TRANSACTION_STATUS
**Address** 0x4015500c
**Description** transaction status
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | opcode_status | | | | | | | | RSVD | | | | | | running | done |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R | opcode_status | master or slave opcode status | 8'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | done | 1=indicate master or slave read/write transaction is done | 1'b0 |

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**Name** DATA_BYTES_LOWER
**Address** 0x40155010
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores read/write address
**Reset** 0x00000000

| Name | byte3 | | | | | | | | byte2 | | | | | | | | byte1 | | | | | | | | byte0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte3 | data byte3 | 8'b0 |
| 23:16 | R/W | byte2 | data byte2 | 8'b0 |
| 15:8 | R/W | byte1 | data byte1 | 8'b0 |
| 7:0 | R/W | byte0 | data byte0 | 8'b0 |

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**Name** DATA_BYTES_UPPER
**Address** 0x40155014
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores write data; preload read data
**Reset** 0x00000000

| Name | byte7 | | | | | | | | byte6 | | | | | | | | byte5 | | | | | | | | byte4 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte7 | data byte7 | 8'b0 |
| 23:16 | R/W | byte6 | data byte6 | 8'b0 |
| 15:8 | R/W | byte5 | data byte5 | 8'b0 |
| 7:0 | R/W | byte4 | data byte4 | 8'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x40155018
**Description** interrupt mask
**Reset** 0x00000000

| Name | RSVD | passthru3 | passthru2 | passthru1 | passthru0 |
|---|---|---|---|---|---|
| **Bits** | **31 … 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | passthru3 | 1=allow interrupt3 to be OR'ed into core interrupt | 1'b0 |
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x4015501c
**Description** interrupt status
**Reset** 0x00000000

| Name | RSVD | interrupt3 | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31 … 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | interrupt3 | from slave, error read-data invalid; independent of mask | 1'b0 |
| 2 | R | interrupt2 | from slave, error unknown command; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, transaction done capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, transaction done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT

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**Address** 0x40155020
**Description** interrupt set
**Reset** 0x00000000

| Name | RSVD | set_interrupt3 | set_interrupt2 | set_interrupt1 | set_interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | set_interrupt3 | set interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | set_interrupt2 | set interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | set_interrupt1 | set interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt0; not auto cleared | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x40155024
**Description** interrupt clear
**Reset** 0x00000000

| Name | RSVD | reset_interrupt3 | reset_interrupt2 | reset_interrupt1 | reset_interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | reset_interrupt3 | clear interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | reset_interrupt2 | clear interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | reset_interrupt1 | clear interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | reset_interrupt0 | clear interrupt0; not auto cleared | 1'b0 |

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**Name** CORE_ID
**Address** 0x40155ffc
**Description** core ID

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**Reset** 0x53504920

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | SPI in ASCII | 32'h53504920 |

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# PMU_NONSECURE

**Name** CTRL
**Address** 0x40157000
**Description** SPI Control Register
**Reset** 0x00010301

| Name | dma_mode | RSVD | | | | | | | | | | spi3wire_en | s_wr_opcode | | | | | | | | s_rd_opcode | | | | | | | | cpol | cpha | mstr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | dma_mode | 1 = dma controls spi core | 1'b0 |
| 19 | R/W | spi3wire_en | 1=enable 3-wire spi transactions, mosi pad becomes bi-directional<br>0=default 4-wire spi transactions | 1'b0 |
| 18:11 | R/W | s_wr_opcode | slave write opcode | 8'h20 |
| 10:3 | R/W | s_rd_opcode | slave read opcode | 8'h60 |
| 2 | R/W | cpol | spi clock polarity 0=spi clock low in idle; cs high 1=spi clock high in idle; cs high | 1'b0 |
| 1 | R/W | cpha | spi clock phase 0=leading edge sample, trailing edge toggle 1=leading edge toggle, trailing edge sample | 1'b0 |
| 0 | R/W | mstr | 1=master interface<br>0=slave interface | 1'b1 |

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**Name** TRANSACTION_SETUP
**Address** 0x40157004
**Description** transaction setup
**Reset** 0x00000000

| | csn_b | invert | dummy_ | csn_sta | loop | opc | clk | r | num_dat | sta |
|---|---|---|---|---|---|---|---|---|---|---|

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| Name | csn_bubble | invert_mosi | dummy_cycles | csn_stays_low | loopback | opcode | clkdiv | rwb | num_data_bytes | start |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29:26** | **25** | **24** | **23:16** | **15:6** | **5** | **4:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | csn_bubble | introduce a delay between spi_csn falling and first bit on mosi | 1'b0 |
| 30 | R/W | invert_mosi | invert mosi; keep static during entire spi transaction | 1'b0 |
| 29:26 | R/W | dummy_cycles | number of extra shift cycles to help with turn-around | 4'd0 |
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 24 | R/W | loopback | if 1, mosi is internally connected to miso | 1'b0 |
| 23:16 | R/W | opcode | opcode | 8'd0 |
| 15:6 | R/W | clkdiv | freq_sclk = f_pclk/(2*(clkdiv+1)) | 10'd0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; not auto cleared | 1'b0 |

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**Name** TRANSACTION_SETUP_DMA
**Address** 0x40157008
**Description** transaction setup by DMA
**Reset** 0x00000000

| Name | RSVD | csn_stays_low | RSVD | rwb | num_data_bytes | start |
|---|---|---|---|---|---|---|
| **Bits** | **31:26** | **25** | **24:6** | **5** | **4:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; auto cleared | 1'b0 |

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**Name** TRANSACTION_STATUS
**Address** 0x4015700c
**Description** transaction status
**Reset** 0x00000000

| | | opc | | |
|---|---|---|---|---|

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| Name | RSVD | | | | | | | | | | | | | | | | opcode_status | | | | | | | | RSVD | | | | | | running | done |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R | opcode_status | master or slave opcode status | 8'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | done | 1=indicate master or slave read/write transaction is done | 1'b0 |

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**Name** DATA_BYTES_LOWER
**Address** 0x40157010
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores read/write address
**Reset** 0x00000000

| Name | byte3 | | | | | | | | byte2 | | | | | | | | byte1 | | | | | | | | byte0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte3 | data byte3 | 8'b0 |
| 23:16 | R/W | byte2 | data byte2 | 8'b0 |
| 15:8 | R/W | byte1 | data byte1 | 8'b0 |
| 7:0 | R/W | byte0 | data byte0 | 8'b0 |

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**Name** DATA_BYTES_UPPER
**Address** 0x40157014
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores write data; preload read data
**Reset** 0x00000000

| Name | byte7 | | | | | | | | byte6 | | | | | | | | byte5 | | | | | | | | byte4 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte7 | data byte7 | 8'b0 |
| 23:16 | R/W | byte6 | data byte6 | 8'b0 |
| 15:8 | R/W | byte5 | data byte5 | 8'b0 |
| 7:0 | R/W | byte4 | data byte4 | 8'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x40157018
**Description** interrupt mask
**Reset** 0x00000000

| Name | RSVD | passthru3 | passthru2 | passthru1 | passthru0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | passthru3 | 1=allow interrupt3 to be OR'ed into core interrupt | 1'b0 |
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x4015701c
**Description** interrupt status
**Reset** 0x00000000

| Name | RSVD | interrupt3 | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | interrupt3 | from slave, error read-data invalid; independent of mask | 1'b0 |
| 2 | R | interrupt2 | from slave, error unknown command; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, transaction done capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, transaction done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x40157020
**Description** interrupt set
**Reset** 0x00000000

| | | set_i | set_i | set_i | set_i |
|---|---|---|---|---|---|

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| Name | RSVD | | |

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# PSEQ_NONSECURE

**Name** CTRL0
**Address** 0x40158000
**Description** signals to prepare for and to request low power state
**Reset** 0x00280000

| Name | req_shub_adv | spi_latch_open | i2c_latch_open | pinpu_latch_open | req_soc_off | gpio_latch_open | ksm_latch_open | pinsel_latch_open | RSVD | wurx_det1_sel | wurx_det0_sel | block_dbg_wake | RSVD | pwm_latch_open | RSVD | req_hibernate_state | req_retain_all_state | RSVD | manage_xtal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | req_shub_adv | If set, requests SOC to go to do CPU-less adv. | 1'b0 |
| 30 | R/W | spi_latch_open | If set, opens all latches on spi0 slave control signals. | 1'b0 |
| 29 | R/W | i2c_latch_open | If set, opens all latches on i2c 0/1 sck/sda pu signals. | 1'b0 |
| 28 | R/W | pinpu_latch_open | If set, opens all latches on all pin pu signals. | 1'b0 |
| 27 | R/W | req_soc_off | If set, requests SOC to go SOC Off state. | 1'b0 |
| 26 | R/W | gpio_latch_open | If set, opens all latches on gpio control signals. | 1'b0 |
| 25 | R/W | ksm_latch_open | If set, opens all latches on ksm control signals. | 1'b0 |
| 24 | R/W | pinsel_latch_open | If set, opens all latches all pin select signals. | 1'b0 |
| 22:21 | R/W | wurx_det1_sel | Is wurx1 detect level or edge?<br>0=level<br>1=rising<br>2,3=falling | 2'd1 |
| 20:19 | R/W | wurx_det0_sel | Is wurx0 detect level or edge?<br>0=level<br>1=rising<br>2,3=falling | 2'd1 |
| 18 | R/W | block_dbg_wake | If set, debugger cannot wake chip from low power state. | 1'b0 |
| 6 | R/W | pwm_latch_open | If set, opens all latches on pwm output prior to entering low power to hold pwm. | 1'b0 |
| 3 | R/W | req_hibernate_state | If set, requests SOC to go to Hibernate low power state. | 1'b0 |
| 2 | R/W | req_retain_all_state | If set, requests SOC to go to Retain All low power state. | 1'b0 |
| 0 | R/W | manage_xtal | If set, permits SOC to power down clk_mpc crystal circuit during low power mode. | 1'b0 |

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**Name** CTRL1

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**Address** 0x40158004
**Description** signals to prepare for and to request low power state
**Reset** 0x00000055

| Name | RSVD | reset_pmu_spi_regs | reset_radio_spi_regs | reset_wurx_spi_regs | brownout_sel | endoflife_sel | energy4TX_sel | energy4CPU_sel |
|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11** | **10** | **9** | **8** | **7 6** | **5 4** | **3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 10 | R/W | reset_pmu_spi_regs | (no description) | 1'b0 |
| 9 | R/W | reset_radio_spi_regs | (no description) | 1'b0 |
| 8 | R/W | reset_wurx_spi_regs | (no description) | 1'b0 |
| 7:6 | R/W | brownout_sel | Is brownout level or edge?<br>0=level<br>1=rising<br>2=falling<br>3=negate level | 2'd1 |
| 5:4 | R/W | endoflife_sel | Is endoflife level or edge?<br>0=level<br>1=rising<br>2=falling<br>3=negate level | 2'd1 |
| 3:2 | R/W | energy4TX_sel | Is energy4TX level or edge?<br>0=level<br>1=rising<br>2=falling<br>3=negate level | 2'd1 |
| 1:0 | R/W | energy4CPU_sel | Is energy4CPU level or edge?<br>0=level<br>1=rising<br>2=falling<br>3=negate level | 2'd1 |

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**Name** RETAIN_ALL_WAKE_MASK
**Address** 0x40158008
**Description** signals to select which events can generate a wake event while in Retain All
**Reset** 0x00000000

| | qualif | qualif | RSVD | watc | wa | wat | wat | watc | wat | watc | w | w | wa |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | qualify_on_energy4CPU | qualify_on_energy4TX | RSVD | watch_ble_osc_on | watch_wurx | watch_brownout | watch_endoflife | watch_energy4TX | watch_energy4CPU | watch_pmu_timer | watch_pmu_lpcomp | watch_i2c | watch_spi | watch_shub | watch_ksm | watch_qdec | watch_wurx1 | watch_wurx0 | watch_cntdown | watch_gpio |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29:17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | qualify_on_energy4CPU | (no description) | 1'b0 |
| 30 | R/W | qualify_on_energy4TX | (no description) | 1'b0 |
| 16 | R/W | watch_ble_osc_on | (no description) | 1'b0 |
| 15 | R/W | watch_wurx | valid to wake on multiple wurx detects in certain amount of time | 1'b0 |
| 14 | R/W | watch_brownout | valid to wake on pmu asserting brownout | 1'b0 |
| 13 | R/W | watch_endoflife | valid to wake on pmu asserting endoflife | 1'b0 |
| 12 | R/W | watch_energy4TX | valid to wake on pmu asserting energy4TX | 1'b0 |
| 11 | R/W | watch_energy4CPU | valid to wake on pmu asserting energy4CPU | 1'b0 |
| 10 | R/W | watch_pmu_timer | valid to wake on pmu timer expiring | 1'b0 |
| 9 | R/W | watch_pmu_lpcomp | valid to wake on pmu lpcomp threshold crossing | 1'b0 |
| 8 | R/W | watch_i2c | valid to wake on i2c slave activity | 1'b0 |
| 7 | R/W | watch_spi | valid to wake on spi slave activity | 1'b0 |
| 6 | R/W | watch_shub | valid to wake on sensor hub trigger | 1'b0 |
| 5 | R/W | watch_ksm | valid to wake on keyboard activity | 1'b0 |
| 4 | R/W | watch_qdec | valid to wake on qdec activity | 1'b0 |
| 3 | R/W | watch_wurx1 | valid to wake on wurx1 detect | 1'b0 |
| 2 | R/W | watch_wurx0 | valid to wake on wurx0 detect | 1'b0 |
| 1 | R/W | watch_cntdown | valid to wake on count down timer expiring | 1'b0 |
| 0 | R/W | watch_gpio | valid to wake on gpio activity | 1'b0 |

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**Name** HIB_WAKE_MASK
**Address** 0x4015800c
**Description** signals to select which events can generate a wake event while in Hiberbate
**Reset** 0x00000000

| Name | qualify_on_energy4CPU | qualify_on_energy4TX | RSVD | watch_ble_osc | watch_wurx | watch_brownout | watch_endoflife | watch_energy4TX | watch_energy4CPU | watch_pmu_timer | watch_pmu_lpcomp | watch_i2c | watch_spi | watch_shub | watch_ksm | watch_qdec | watch_wurx1 | watch_wurx0 | watch_cntdown | watch_gpio |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| | y4CPU | gy4TX | | | | | | | | | | | | | | _on | | ut | fe | TX | CPU | er | omp | | | | | | | | n | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | qualify_on_energy4CPU | (no description) | 1'b0 |
| 30 | R/W | qualify_on_energy4TX | (no description) | 1'b0 |
| 16 | R/W | watch_ble_osc_on | (no description) | 1'b0 |
| 15 | R/W | watch_wurx | valid to wake on multiple wurx detects in certain amount of time | 1'b0 |
| 14 | R/W | watch_brownout | valid to wake on pmu asserting brownout | 1'b0 |
| 13 | R/W | watch_endoflife | valid to wake on pmu asserting endoflife | 1'b0 |
| 12 | R/W | watch_energy4TX | valid to wake on pmu asserting energy4TX | 1'b0 |
| 11 | R/W | watch_energy4CPU | valid to wake on pmu asserting energy4CPU | 1'b0 |
| 10 | R/W | watch_pmu_timer | valid to wake on pmu timer expiring | 1'b0 |
| 9 | R/W | watch_pmu_lpcomp | valid to wake on pmu lpcomp threshold crossing | 1'b0 |
| 8 | R/W | watch_i2c | valid to wake on i2c slave activity | 1'b0 |
| 7 | R/W | watch_spi | valid to wake on spi slave activity | 1'b0 |
| 6 | R/W | watch_shub | valid to wake on sensor hub trigger | 1'b0 |
| 5 | R/W | watch_ksm | valid to wake on keyboard activity | 1'b0 |
| 4 | R/W | watch_qdec | valid to wake on qdec activity | 1'b0 |
| 3 | R/W | watch_wurx1 | valid to wake on wurx1 detect | 1'b0 |
| 2 | R/W | watch_wurx0 | valid to wake on wurx0 detect | 1'b0 |
| 1 | R/W | watch_cntdown | valid to wake on count down timer expiring | 1'b0 |
| 0 | R/W | watch_gpio | valid to wake on gpio activity | 1'b0 |

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**Name** GPIO_WAKE_MASK
**Address** 0x40158018
**Description** mask function to specify which gpio's can generate a gpio event
**Reset** 0x00000000

| Name | upper | | | | | | | | | | | | | | | | lower | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | upper | upper 16 gpio's | 16'h0 |
| 15:0 | R/W | lower | lower 16 gpio's | 16'h0 |

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**Name** GPIO_WAKE_TYPE
**Address** 0x4015801c
**Description** specifies how a gpio generates an event
**Reset** 0x00000000

| Name | upper | lower |
|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | upper | (per bit) 1 = edge sensitive , 0 = level | 16'h0 |
| 15:0 | R/W | lower | (per bit) 1 = edge sensitive , 0 = level | 16'h0 |

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**Name** GPIO_WAKE_POL
**Address** 0x40158020
**Description** specifies how a gpio generates an event
**Reset** 0xffffffff

| Name | upper | | | | | | | | | | | | | | | | lower | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | upper | (per bit) 1 = level high/rising edge , 0 = level low/falling edge | 16'hffff |
| 15:0 | R/W | lower | (per bit) 1 = level high/rising edge , 0 = level low/falling edge | 16'hffff |

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**Name** GPIO_WAKE_BOTH_EDGES
**Address** 0x40158024
**Description** specifies how a gpio generates an event
**Reset** 0x00000000

| Name | upper | | | | | | | | | | | | | | | | lower | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | upper | (per bit) 1 = either edge will wake; higher precedence than gpio_wake_type and gpio_wake_pol | 16'h0 |
| 15:0 | R/W | lower | (per bit) 1 = either edge will wake; higher precedence than gpio_wake_type and gpio_wake_pol | 16'h0 |

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**Name** GPIO_WAKE_STATUS
**Address** 0x40158028
**Description** specifies which gpio caused the wake
**Reset** 0x00000000

| Name | upper | | | | | | | | | | | | | | | | lower | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:16 | R | upper | (no description) | 16'h0 |
| 15:0 | R | lower | (no description) | 16'h0 |

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**Name** WURX_CONFIG
**Address** 0x4015802c
**Description** wurx configuration
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | wurx_cutvdd_b | wurx_rstb | wurx_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | wurx_cutvdd_b | if low, removes power from wurx | 1'b0 |
| 1 | R/W | wurx_rstb | if low, holds wurx in reset | 1'b0 |
| 0 | R/W | wurx_en | if low, disables wurx datapath | 1'b0 |

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**Name** WURX_CONFIG1
**Address** 0x40158030
**Description** wurx configuration
**Reset** 0x00000022

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | num_wurx_event | | | | wurx_src_sel | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:2 | R/W | num_wurx_event | Number of multiple wurx detect events to watch | 4'd8 |
| 1:0 | R/W | wurx_src_sel | which multiple wurx detect source to watch<br>0=wurx0<br>1=wurx1<br>2,3=wurx0 or wurx1 | 2'd2 |

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**Name** WURX_CONFIG2
**Address** 0x40158034
**Description** wurx configuration
**Reset** 0x80008000

| Name | num_wurx_time1_cyc | | | | | | | | | | | | | | | num_wurx_time0_cyc | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:17 | R/W | num_wurx_time1_cyc | Max time window between two consecutive wurx detect events in 32KHz cycles, default 500msec window | 15'd16384 |
| 16:0 | R/W | num_wurx_time0_cyc | Full time window during multiple wurx detect events watching in 32KHz cycles, default 1sec window | 17'd32768 |

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**Name** GADC_CONFIG
**Address** 0x40158038
**Description** gadc configuration
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | gadc_cutvdd_b | gadc_rstb | gadc_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | gadc_cutvdd_b | if low, removes power from gadc | 1'b0 |
| 1 | R/W | gadc_rstb | if low, holds gadc in reset | 1'b0 |
| 0 | R/W | gadc_en | if low, disables gadc datapath | 1'b0 |

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**Name** XTAL_BITS0
**Address** 0x4015803c
**Description** control signals for clk_mpc crystal circuit

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**Reset** 0x000b6766

| Name | RSVD | xobufp | xobufn | xocapout | xocapin | RSVD |
|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21** | **20 19** | **18 17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 20:19 | R/W | xobufp | (no description) | 2'b01 |
| 18:17 | R/W | xobufn | (no description) | 2'b01 |
| 16:9 | R/W | xocapout | 16MHz crystal tuning cap on XTALO_16M pin, typically set to same value as xocapin | 8'hb3 |
| 8:1 | R/W | xocapin | 16MHz crystal tuning cap on XTALI_16M pin, typically set to same value as xocapout | 8'hb3 |

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**Name** XTAL_BITS1
**Address** 0x40158040
**Description** control signals for clk_mpc crystal circuit
**Reset** 0x00164000

| Name | RSVD | extclk | clkmpc_en | clkhpc_en | doublerdcc_en | rcoscfreq | RSVD |
|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22** | **21** | **20** | **19** | **18** | **17 16 15 14** | **13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 21 | R/W | extclk | Set to 1 if supplying external clock source on XTALI_16M pin | 1'b0 |
| 20 | R/W | clkmpc_en | enable root clk_mpc buffer; chip will not be operational if cleared | 1'b1 |
| 19 | R/W | clkhpc_en | enable root clk_hpc buffer | 1'b0 |
| 18 | R/W | doublerdcc_en | enable duty cycle correction for clock doubler | 1'b1 |
| 17:14 | R/W | rcoscfreq | 16MHz RC oscillator frequency tuning, 0 - slowest, 15 - fastest | 4'h9 |

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**Name** OVERRIDES
**Address** 0x40158044
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | for | forc | forc |
|---|---|---|---|---|

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| Name | RSVD | force_osc_on_req | force_precision_val | force_precision_req |
|---|---|---|---|---|
| **Bits** | **31:3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | force_osc_on_req | force that clk_mpc oscillator request always be on | 1'b0 |
| 1 | R/W | force_precision_val | when type of precision for clk_mpc is forced, what should it be?<br>1=high/xtal<br>0=low/osc | 1'b0 |
| 0 | R/W | force_precision_req | force the type of precision when clk_mpc is requested | 1'b0 |

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**Name** OVERRIDES2
**Address** 0x40158048
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | block_efuse_vddcut | block_rram_vddcut | block_sysrom_vddcut |
|---|---|---|---|---|
| **Bits** | **31:3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | block_efuse_vddcut | if set, this core's FSM cannot power down the e-fuse macros | 1'b0 |
| 1 | R/W | block_rram_vddcut | if set, this core's FSM cannot power down the rram macros | 1'b0 |
| 0 | R/W | block_sysrom_vddcut | if set, this core's FSM cannot power down the ROM macros | 1'b0 |

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**Name** OVERRIDES3
**Address** 0x4015804c
**Description** override portion of pmu control signals
**Reset** 0x00000000

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| Name | RSVD | override_ram_iso | override_ram_iso_val | override_efuse_vddcut | override_efuse_vddcut_val | override_efuse_iso | override_efuse_iso_val | override_rram_vddcut | override_rram_vddcut_val | override_rram_rst | override_rram_rst_val | override_rram_poc | override_rram_poc_val | override_rram_nap | override_rram_nap_val | override_rram_iso | override_rram_iso_val | override_rram_clken | override_rram_clken_val | override_sysrom_ls | override_sysrom_ls_val | override_sysrom_iso | override_sysrom_iso_val | override_sysrom_clken | override_sysrom_clken_val |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23 | R/W | override_ram_iso | if set, software now has direct control of the pd_mem ISO pins | 1'b0 |
| 22 | R/W | override_ram_iso_val | if override is set, what value should be driven onto pd_mem ISO pins? | 1'b0 |
| 21 | R/W | override_efuse_vddcut | if set, head switch on e-fuse is now software controllable | 1'b0 |
| 20 | R/W | override_efuse_vddcut_val | if override is set, should head switch on e-fuse be open or close?<br>1=open<br>0=close | 1'b0 |
| 19 | R/W | override_efuse_iso | if set, software now has direct control of the e-fuse ISO pin | 1'b0 |
| 18 | R/W | override_efuse_iso_val | if override is set, what value should be driven onto e-fuse ISO pin? | 1'b0 |
| 17 | R/W | override_rram_vddcut | if set, head switch on romc (rom,rram) is now software controllable | 1'b0 |
| 16 | R/W | override_rram_vddcut_val | if override is set, should head switch on romc be open or close?<br>1=open<br>0=close | 1'b0 |
| 15 | R/W | override_rram_rst | if set, software now has direct control of the rram RST pin | 1'b0 |
| 14 | R/W | override_rram_rst_val | if override is set, what value should be driven onto rram RST pin? | 1'b0 |
| 13 | R/W | override_rram_poc | if set, software now has direct control of the rram POC_H pin | 1'b0 |
| 12 | R/W | override_rram_poc_val | if override is set, what value should be driven onto rram POC_H pin? | 1'b0 |
| 11 | R/W | override_rram_nap | if set, software now has direct control of the rram NAP pin | 1'b0 |
| 10 | R/W | override_rram_nap_val | if override is set, what value should be driven onto rram NAP pin? | 1'b0 |
| 9 | R/W | override_rram_iso | if set, software now has direct control of the rram ISO pin | 1'b0 |
| 8 | R/W | override_rram_iso_val | if override is set, what value should be driven onto rram ISO pin? | 1'b0 |
| 7 | R/W | override_rram_clken | if set, software now has direct control of the rram clock enable pin | 1'b0 |
| 6 | R/W | override_rram_clken_val | if override is set, what value should be driven onto rram clock enable pin? | 1'b0 |
| 5 | R/W | override_sysrom_ls | if set, software now has direct control of the rom LS pin | 1'b0 |
| 4 | R/W | override_sysrom_ls_val | if override is set, what value should be driven onto rom LS pin? | 1'b0 |
| 3 | R/W | override_sysrom_iso | if set, software now has direct control of the rom ISO pin | 1'b0 |
| 2 | R/W | override_sysrom_iso_val | if override is set, what value should be driven onto rom ISO pin? | 1'b0 |
| 1 | R/W | override_sysrom_clken | if set, software now has direct control of the rom clock enable pin | 1'b0 |
| 0 | R/W | override_sysrom_clken_val | if override is set, what value should be driven onto rom clock enable pin? | 1'b0 |

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**Name** OVERRIDES4

**Address** 0x40158050
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | override_pnr_iso1 | override_pnr_iso1_val | override_pnr_save | override_pnr_save_val | override_pnr_restore | override_pnr_restore_val | override_pnr_sleep | override_pnr_sleep_val | override_pnr_kill | override_pnr_kill_val | override_pnr_iso | override_pnr_iso_val | override_pnr_frst | override_pnr_frst_val | override_pnr_clken | override_pnr_clken_val | override_ble_nrestore_val | override_ble_nrestore | override_ble_sleep | override_ble_sleep_val | override_ble_save | override_ble_save_val | override_ble_kill | override_ble_kill_val | override_ble_iso | override_ble_iso_val | override_ble_frst | override_ble_frst_val | override_ble_clken | override_ble_clken_val |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29 | R/W | override_pnr_iso1 | if set, software now has direct control of the iso1 signal going to the cpu domain; for scan_outs only | 1'b0 |
| 28 | R/W | override_pnr_iso1_val | if override is set, what value should be driven onto the cpu domain's iso1 signal? | 1'b0 |
| 27 | R/W | override_pnr_save | if set, software now has direct control of the SAVE signal going to the cpu domain | 1'b0 |
| 26 | R/W | override_pnr_save_val | if override is set, what value should be driven onto the cpu domain's SAVE signal? | 1'b0 |
| 25 | R/W | override_pnr_restore | if set, software now has direct control of the RESTORE signal going to the cpu domain | 1'b0 |
| 24 | R/W | override_pnr_restore_val | if override is set, what value should be driven onto the cpu domain's RESTORE signal? | 1'b0 |
| 23 | R/W | override_pnr_sleep | if set, software now has direct control of the sleep signal going to the cpu domain | 1'b0 |
| 22 | R/W | override_pnr_sleep_val | if override is set, what value should be driven onto the cpu domain's sleep signal? | 1'b0 |
| 21 | R/W | override_pnr_kill | if set, software now has direct control of the kill signal going to the cpu domain | 1'b0 |
| 20 | R/W | override_pnr_kill_val | if override is set, what value should be driven onto the cpu domain's kill signal? | 1'b0 |
| 19 | R/W | override_pnr_iso | if set, software now has direct control of the iso signal going to the cpu domain | 1'b0 |
| 18 | R/W | override_pnr_iso_val | if override is set, what value should be driven onto the cpu domain's iso signal? | 1'b0 |
| 17 | R/W | override_pnr_frst | if set, software now has direct control of the forced reset signal going to the cpu domain | 1'b0 |
| 16 | R/W | override_pnr_frst_val | if override is set, what value should be driven onto the cpu domain's forced reset signal? | 1'b0 |
| 15 | R/W | override_pnr_clken | if set, software now has direct control of the clock enable signal going to the cpu domain | 1'b0 |
| | | | if override is set, what value should be driven onto the cpu domain's clock | |

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| | | | | |
|---|---|---|---|---|
| 14 | R/W | override_pnr_clken_val | enable signal? | 1'b0 |
| 13 | R/W | override_ble_save | if set, software now has direct control of the SAVE signal going to the ble domain | 1'b0 |
| 12 | R/W | override_ble_save_val | if override is set, what value should be driven onto the ble domain's SAVE signal? | 1'b0 |
| 11 | R/W | override_ble_nrestore | if set, software now has direct control of the NRESTORE signal going to the ble domain | 1'b0 |
| 10 | R/W | override_ble_nrestore_val | if override is set, what value should be driven onto the ble domain's NRESTORE signal? | 1'b0 |
| 9 | R/W | override_ble_sleep | if set, software now has direct control of the sleep signal going to the ble domain | 1'b0 |
| 8 | R/W | override_ble_sleep_val | if override is set, what value should be driven onto the ble domain's sleep signal? | 1'b0 |
| 7 | R/W | override_ble_kill | if set, software now has direct control of the kill signal going to the ble domain | 1'b0 |
| 6 | R/W | override_ble_kill_val | if override is set, what value should be driven onto the ble domain's kill signal? | 1'b0 |
| 5 | R/W | override_ble_iso | if set, software now has direct control of the iso signal going to the ble domain | 1'b0 |
| 4 | R/W | override_ble_iso_val | if override is set, what value should be driven onto the ble domain's iso signal? | 1'b0 |
| 3 | R/W | override_ble_frst | if set, software now has direct control of the forced reset signal going to the ble domain | 1'b0 |
| 2 | R/W | override_ble_frst_val | if override is set, what value should be driven onto the ble domain's forced reset signal? | 1'b0 |
| 1 | R/W | override_ble_clken | if set, software now has direct control of the clock enable signal going to the ble domain | 1'b0 |
| 0 | R/W | override_ble_clken_val | if override is set, what value should be driven onto the ble domain's clock enable signal? | 1'b0 |

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**Name** OVERRIDES5
**Address** 0x40158054
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | override_xtal | override_xtal_val | override_retv | override_retv_val | override_radio_en_o | override_radio_en_o_val | override_cutvdd_b | override_cutvdd_b_val | override_radio_frst_b | override_radio_frst_b_val | override_wreq | override_wreq_val | override_radio_iso | override_radio_iso_val | override_ble_frstdp | override_ble_frstdp_val | override_pnr_frstdp | override_pnr_frstdp_val | RSVD |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | override_xtal | If set, software now has direct control of clk_mpc's crystal circuit enable. | 1'b0 |
| 30 | R/W | override_xtal_val | If override is set, what value is clk_mpc's cystal circuit enable?<br>0=off<br>1=on | 1'b0 |
| 29 | R/W | override_retv | If set, software now has direct control of the memory retention voltage. | 1'b0 |
| 28 | R/W | override_retv_val | If override is set, what value is the memory retention voltage?<br>0=nominal<br>1=ultra low | 1'b0 |
| 27 | R/W | override_radio_en_o | If set, software now has direct control of radio enable. | 1'b0 |
| 26 | R/W | override_radio_en_o_val | If override is set, what value is the radio enable?<br>0=off<br>1=on | 1'b0 |
| 25 | R/W | override_cutvdd_b | If set, software now has direct control of radio head switches. | 1'b0 |
| 24 | R/W | override_cutvdd_b_val | If override is set, what value is the radio enable?<br>0=head switches are open<br>1=head switches are closed | 1'b0 |
| 23 | R/W | override_radio_frst_b | If set, software now has direct control of radio forced reset . | 1'b0 |
| 22 | R/W | override_radio_frst_b_val | If override is set, what value is the radio forced reset?<br>0=forced reset is enabled<br>1=forced reset is disabled | 1'b0 |
| 21 | R/W | override_wreq | If set, software now has direct control of wake up request. | 1'b0 |
| 20 | R/W | override_wreq_val | If override is set, what value is value of wake up request? Useful for debug/test of pseq interrupt.<br>0=no wake<br>1=wake | 1'b0 |
| 19 | R/W | override_radio_iso | If set, software now has direct control of radio isolation signal . | 1'b0 |
| 18 | R/W | override_radio_iso_val | If override is set, what value is the radio isolation signal? | 1'b0 |
| 17 | R/W | override_ble_frstdp | If set, software now has direct control of ble domain's datapath forced reset.<br>Not all state is retained even in Retain All state. | 1'b0 |
| 16 | R/W | override_ble_frstdp_val | If override is set, what value is the ble domain's datapath forced reset. | 1'b0 |
| 15 | R/W | override_pnr_frstdp | If set, software now has direct control of cpu domain's datapath forced reset.<br>Not all state is retained even in Retain All state. | 1'b0 |
| 14 | R/W | override_pnr_frstdp_val | If override is set, what value is the cpu domain's datapath forced reset. | 1'b0 |

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**Name** SYSRAM_OVERRIDES
**Address** 0x40158058
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_sysram | RSVD | override_sysram_v |
|---|---|---|---|---|

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | override_sysram_vddcut | | | | | | | | | | | | override_sysram_vddcut_val | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:12 | R/W | override_sysram_vddcut | If set, software now has direct control of sysram's head switches.<br>[23:20] reserved<br>[19] 16kB macro at high address space<br>[18] 16kB macro<br>[17] 16kB macro<br>[16] 16kB macro<br>[15] 16kB macro<br>[14] 16kB macro<br>[13] 16kB macro<br>[12] 16kB macro at lowest address space | 8'd0 |
| 7:0 | R/W | override_sysram_vddcut_val | If override is set, what is state of sysram's head switches?<br>[11:8] reserved<br>[7] 16kB macro at high address space; [1] head switch open [0] head switch close<br>[6] 16kB macro; [1] head switch open [0] head switch close<br>[5] 16kB macro; [1] head switch open [0] head switch close<br>[4] 16kB macro; [1] head switch open [0] head switch close<br>[3] 16kB macro; [1] head switch open [0] head switch close<br>[2] 16kB macro; [1] head switch open [0] head switch close<br>[1] 16kB macro; [1] head switch open [0] head switch close<br>[0] 16kB macro at lowest address space; [1] head switch open [0] head switch close | 8'd0 |

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**Name** SYSRAM_OVERRIDES2
**Address** 0x4015805c
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | override_sysram_clken | | | | | | | | RSVD | | | | override_sysram_clken_val | | | | | | | |

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:12 | R/W | override_sysram_clken | If set, software now has direct control of sysram's clock enable signals.<br>[23:20] reserved<br>[19] 16kB macro at high address space<br>[18] 16kB macro<br>[17] 16kB macro<br>[16] 16kB macro<br>[15] 16kB macro<br>[14] 16kB macro<br>[13] 16kB macro<br>[12] 16kB macro at lowest address space | 8'd0 |
| 7:0 | R/W | override_sysram_clken_val | If override is set, what is state of sysram's clock enable signals?<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** SYSRAM_OVERRIDES3
**Address** 0x40158060
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_sysram_iso | RSVD | override_sysram_iso_val |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 | 19 18 17 16 15 14 13 12 | 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:12 | R/W | override_sysram_iso | If set, software now has direct control of sysram's isolation signals.<br>[23:20] reserved<br>[19] 16kB macro at high address space<br>[18] 16kB macro<br>[17] 16kB macro<br>[16] 16kB macro<br>[15] 16kB macro<br>[14] 16kB macro<br>[13] 16kB macro | 8'd0 |

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| | | | | |
|---|---|---|---|---|
| | | | [12] 16kB macro at lowest address space | |
| 7:0 | R/W | override_sysram_iso_val | If override is set, what is state of sysram's isolation signals?<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** SYSRAM_OVERRIDES4
**Address** 0x40158064
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | override_sysram_ls | | | | | | | | RSVD | | | | override_sysram_ls_val | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:12 | R/W | override_sysram_ls | If set, software now has direct control of sysram's LS signals.<br>[23:20] reserved<br>[19] 16kB macro at high address space<br>[18] 16kB macro<br>[17] 16kB macro<br>[16] 16kB macro<br>[15] 16kB macro<br>[14] 16kB macro<br>[13] 16kB macro<br>[12] 16kB macro at lowest address space | 8'd0 |
| 7:0 | R/W | override_sysram_ls_val | If override is set, what is state of sysram's LS signals?<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** SYSRAM_OVERRIDES5
**Address** 0x40158068
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | override_sysram_ds | | | | | | | | RSVD | | | | override_sysram_ds_val | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:12 | R/W | override_sysram_ds | If set, software now has direct control of sysram's DS signals.<br>[23:20] reserved<br>[19] 16kB macro at high address space<br>[18] 16kB macro<br>[17] 16kB macro<br>[16] 16kB macro<br>[15] 16kB macro<br>[14] 16kB macro<br>[13] 16kB macro<br>[12] 16kB macro at lowest address space | 8'd0 |
| 7:0 | R/W | override_sysram_ds_val | If override is set, what is state of sysram's DS signals?<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** SYSRAM_OVERRIDES6
**Address** 0x4015806c
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | overr | | | | | | | | | | | | overrid | | | | | | | |

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| Name | RSVD | override_sysram_sd | RSVD | override_sysram_sd_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20** | **19 18 17 16 15 14 13 12** | **11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:12 | R/W | override_sysram_sd | If set, software now has direct control of sysram's SD signals.<br>[23:20] reserved<br>[19] 16kB macro at high address space<br>[18] 16kB macro<br>[17] 16kB macro<br>[16] 16kB macro<br>[15] 16kB macro<br>[14] 16kB macro<br>[13] 16kB macro<br>[12] 16kB macro at lowest address space | 8'd0 |
| 7:0 | R/W | override_sysram_sd_val | If override is set, what is state of sysram's SD signals?<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** SYSRAM_OVERRIDES7
**Address** 0x40158070
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | block_sysram_vddcut |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | block_sysram_vddcut | Block core's FSM from cutting the vdd supplies to the sysram.<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** SYSRAM_OVERRIDES8
**Address** 0x40158074
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | block_sysram_sd |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | block_sysram_sd | Block core's FSM from setting the SD signal to the sysram.<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** EMRAM_OVERRIDES
**Address** 0x40158078
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | override | | override_ |
|---|---|---|---|---|

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| Name | RSVD | override_emram_vddcut | RSVD | override_emram_vddcut_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13 12** | **11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13:12 | R/W | override_emram_vddcut | If set, software now has direct control of exchange memory vddcut signals. | 2'd0 |
| 1:0 | R/W | override_emram_vddcut_val | If override is set, what are the states of exchange memory vddcut signals? | 2'd0 |

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**Name** EMRAM_OVERRIDES2
**Address** 0x4015807c
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_emram_clken | RSVD | override_emram_clken_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13 12** | **11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13:12 | R/W | override_emram_clken | If set, software now has direct control of exchange memory clock enable signals. | 2'd0 |
| 1:0 | R/W | override_emram_clken_val | If override is set, what are the states of exchange memory clock enable signals? | 2'd0 |

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**Name** EMRAM_OVERRIDES3

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**Address** 0x40158080
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_emram_iso | RSVD | override_emram_iso_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13 12** | **11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13:12 | R/W | override_emram_iso | If set, software now has direct control of exchange memory isolation signals. | 2'd0 |
| 1:0 | R/W | override_emram_iso_val | If override is set, what are the states of exchange memory isolation signals? | 2'd0 |

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**Name** EMRAM_OVERRIDES4
**Address** 0x40158084
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_emram_ls | RSVD | override_emram_ls_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13 12** | **11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 13:12 | R/W | override_emram_ls | If set, software now has direct control of exchange memory LS signals. | 2'd0 |
| 1:0 | R/W | override_emram_ls_val | If override is set, what are the states of exchange memory LS signals? | 2'd0 |

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**Name** EMRAM_OVERRIDES5
**Address** 0x40158088
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_emram_ds | RSVD | override_emram_ds_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13 12** | **11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13:12 | R/W | override_emram_ds | If set, software now has direct control of exchange memory DS signals. | 2'd0 |
| 1:0 | R/W | override_emram_ds_val | If override is set, what are the states of exchange memory DS signals? | 2'd0 |

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**Name** EMRAM_OVERRIDES6
**Address** 0x4015808c
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_emram_sd | RSVD | override_emram_sd_val |
|---|---|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13:12 | R/W | override_emram_sd | If set, software now has direct control of exchange memory SD signals. | 2'd0 |
| 1:0 | R/W | override_emram_sd_val | If override is set, what are the states of exchange memory SD signals? | 2'd0 |

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**Name** EMRAM_OVERRIDES7
**Address** 0x40158090
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | block_emram_vddcut |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | block_emram_vddcut | If set, core's FSM cannot open the exchange memory head switches. | 2'd0 |

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**Name** EMRAM_OVERRIDES8
**Address** 0x40158094
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | block_emram_sd |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | block_emram_sd | If set, core's FSM cannot set the exchange memory SD signals. | 2'd0 |

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**Name** ICACHE_OVERRIDES
**Address** 0x40158098
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | override_icache_vddcut | | | | RSVD | | | | | | | | override_icache_vddcut_val | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:12 | R/W | override_icache_vddcut | If set, software now has direct control of cpu icache's head switches.<br>[15] data1 macro<br>[14] data0 macro<br>[13] tag1 macro<br>[12] tag0 macro | 4'd0 |
| 3:0 | R/W | override_icache_vddcut_val | If override is set, what is state of cpu icache's head switches?<br>[3] data1 macro; [1] head switch open [0] head switch close<br>[2] data0 macro; [1] head switch open [0] head switch close<br>[1] tag1 macro; [1] head switch open [0] head switch close<br>[0] tag0 macro; [1] head switch open [0] head switch close | 4'd0 |

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**Name** ICACHE_OVERRIDES2
**Address** 0x4015809c
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | R | override_ | R | override_ic |
|---|---|---|---|---|

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| Name | SVD | | | | | | | | | | | | | | | | icache_clken | | | | SVD | | | | | | | | cache_clken_val | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:12 | R/W | override_icache_clken | If set, software now has direct control of cpu icache's clock enable signals.<br>[15] data1 macro<br>[14] data0 macro<br>[13] tag1 macro<br>[12] tag0 macro | 4'd0 |
| 3:0 | R/W | override_icache_clken_val | If override is set, what is state of cpu icache's clock enable signals?<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** ICACHE_OVERRIDES3
**Address** 0x401580a0
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | override_icache_iso | | | | RSVD | | | | | | | | override_icache_iso_val | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:12 | R/W | override_icache_iso | If set, software now has direct control of cpu icache's isolation signals.<br>[15] data1 macro<br>[14] data0 macro<br>[13] tag1 macro | 4'd0 |

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| | | | | |
|---|---|---|---|---|
| | | | [12] tag0 macro | |
| 3:0 | R/W | override_icache_iso_val | If override is set, what is state of cpu icache's isolation signals?<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** ICACHE_OVERRIDES4
**Address** 0x401580a4
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | override_icache_ls | | | | RSVD | | | | | | override_icache_ls_val | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:12 | R/W | override_icache_ls | If set, software now has direct control of cpu icache's LS signals.<br>[15] data1 macro<br>[14] data0 macro<br>[13] tag1 macro<br>[12] tag0 macro | 4'd0 |
| 3:0 | R/W | override_icache_ls_val | If override is set, what is state of cpu icache's LS signals?<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** ICACHE_OVERRIDES5
**Address** 0x401580a8
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | | | override_i | | | | override_icache_ls_val |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | |

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| Name | RSVD | override_icache_ds | RSVD | override_icache_ds_val |
|---|---|---|---|---|
| **Bits** | **31:16** | **15:12** | **11:4** | **3:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:12 | R/W | override_icache_ds | If set, software now has direct control of cpu icache's DS signals.<br>[15] data1 macro<br>[14] data0 macro<br>[13] tag1 macro<br>[12] tag0 macro | 4'd0 |
| 3:0 | R/W | override_icache_ds_val | If override is set, what is state of cpu icache's DS signals?<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** ICACHE_OVERRIDES6
**Address** 0x401580ac
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_icache_sd | RSVD | override_icache_sd_val |
|---|---|---|---|---|
| **Bits** | **31:16** | **15:12** | **11:4** | **3:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:12 | R/W | override_icache_sd | If set, software now has direct control of cpu icache's SD signals.<br>[15] data1 macro<br>[14] data0 macro<br>[13] tag1 macro<br>[12] tag0 macro | 4'd0 |

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| | | | | |
|---|---|---|---|---|
| 3:0 | R/W | override_icache_sd_val | If override is set, what is state of cpu icache's SD signals?<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** ICACHE_OVERRIDES7
**Address** 0x401580b0
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | block_icache_vddcut |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3:0 | R/W | block_icache_vddcut | Block core's FSM from cutting the vdd supplies to the icache.<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** ICACHE_OVERRIDES8
**Address** 0x401580b4
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | block_icache_sd |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3 2 1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3:0 | R/W | block_icache_sd | Block core's FSM from setting the SD signal to the icache.<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** RRAMCACHE_OVERRIDES
**Address** 0x401580b8
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | override_rramcache_vddcut | | | | | | RSVD | | | override_rramcache_vddcut_val | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17:12 | R/W | override_rramcache_vddcut | If set, software now has direct control of cpu rram cache's head switches.<br>[17] data1 macro<br>[16] data0 macro<br>[15] tag3 macro<br>[14] tag2 macro<br>[13] tag1 macro<br>[12] tag0 macro | 6'd0 |
| 5:0 | R/W | override_rramcache_vddcut_val | If override is set, what is state of cpu rram cache's's head switches?<br>[5] data1 macro; [1] head switch open [0] head switch close<br>[4] data0 macro; [1] head switch open [0] head switch close<br>[3] tag3 macro; [1] head switch open [0] head switch close<br>[2] tag2 macro; [1] head switch open [0] head switch close<br>[1] tag1 macro; [1] head switch open [0] head switch close<br>[0] tag0 macro; [1] head switch open [0] head switch close | 6'd0 |

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**Name** RRAMCACHE_OVERRIDES2
**Address** 0x401580bc

**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_rramcache_clken | RSVD | override_rramcache_clken_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18** | **17 16 15 14 13 12** | **11 10 9 8 7 6** | **5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17:12 | R/W | override_rramcache_clken | If set, software now has direct control of cpu rram cache's clock enable signals.<br>[17] data1 macro<br>[16] data0 macro<br>[15] tag3 macro<br>[14] tag2 macro<br>[13] tag1 macro<br>[12] tag0 macro | 6'd0 |
| 5:0 | R/W | override_rramcache_clken_val | If override is set, what is state of cpu rram cache's clock enable signals?<br>[5] data1 macro<br>[4] data0 macro<br>[3] tag3 macro<br>[2] tag2 macro<br>[1] tag1 macro<br>[0] tag0 macro | 6'd0 |

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**Name** RRAMCACHE_OVERRIDES3
**Address** 0x401580c0
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | override | | override_ |
|---|---|---|---|---|

| Name | RSVD | override_rramcache_iso | RSVD | override_rramcache_iso_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18** | **17 16 15 14 13 12** | **11 10 9 8 7 6** | **5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17:12 | R/W | override_rramcache_iso | If set, software now has direct control of cpu rram cache's isolation signals.<br>[17] data1 macro<br>[16] data0 macro<br>[15] tag3 macro<br>[14] tag2 macro<br>[13] tag1 macro<br>[12] tag0 macro | 6'd0 |
| 5:0 | R/W | override_rramcache_iso_val | If override is set, what is state of cpu rram cache's isolation signals?<br>[5] data1 macro<br>[4] data0 macro<br>[3] tag3 macro<br>[2] tag2 macro<br>[1] tag1 macro<br>[0] tag0 macro | 6'd0 |

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**Name** RRAMCACHE_OVERRIDES4
**Address** 0x401580c4
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_rramcache_ls | RSVD | override_rramcache_ls_va |
|---|---|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 1 | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17:12 | R/W | override_rramcache_ls | If set, software now has direct control of cpu rram cache's LS signals.<br>[17] data1 macro<br>[16] data0 macro<br>[15] tag3 macro<br>[14] tag2 macro<br>[13] tag1 macro<br>[12] tag0 macro | 6'd0 |
| 5:0 | R/W | override_rramcache_ls_val | If override is set, what is state of cpu rram cache's LS signals?<br>[5] data1 macro<br>[4] data0 macro<br>[3] tag3 macro<br>[2] tag2 macro<br>[1] tag1 macro<br>[0] tag0 macro | 6'd0 |

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**Name** RRAMCACHE_OVERRIDES5
**Address** 0x401580c8
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_rramcache_ds | RSVD | override_rramcache_ds_val | RSVD |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18** | **17 16** | **15 14 13 12 11 10 9 8 7 6** | **5 4** | **3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17:16 | R/W | override_rramcache_ds | If set, software now has direct control of cpu rram cache's DS signals.<br>[17] data1 macro<br>[16] data0 macro | 2'd0 |
| 5:4 | R/W | override_rramcache_ds_val | If override is set, what is state of cpu rram cache's DS signals?<br>[5] data1 macro<br>[4] data0 macro | 2'd0 |

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**Name** RRAMCACHE_OVERRIDES6

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**Address** 0x401580cc
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_rramcache_sd | RSVD | override_rramcache_sd_val | RSVD |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18** | **17 16** | **15 14 13 12 11 10 9 8 7 6** | **5 4** | **3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17:16 | R/W | override_rramcache_sd | If set, software now has direct control of cpu rram cache's SD signals.<br>[17] data1 macro<br>[16] data0 macro | 2'd0 |
| 5:4 | R/W | override_rramcache_sd_val | If override is set, what is state of cpu rram cache's SD signals?<br>[5] data1 macro<br>[4] data0 macro | 2'd0 |

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**Name** RRAMCACHE_OVERRIDES7
**Address** 0x401580d0
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | block_rramcache_vddc |
|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | u t | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:0 | R/W | block_rramcache_vddcut | Block core's FSM from cutting the vdd supplies to the rram cache.<br>[5] data1 macro<br>[4] data0 macro<br>[3] tag3 macro<br>[2] tag2 macro<br>[1] tag1 macro<br>[0] tag0 macro | 6'd0 |

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**Name** RRAMCACHE_OVERRIDES8
**Address** 0x401580d4
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | | | | | | | | | | | | | | | | | | | | | | | | | | | b l o c k _ r r a m c a c h e _ s d | | R S V D | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:4 | R/W | block_rramcache_sd | Block core's FSM from setting the SD signal to the rram cache.<br>[5] data1 macro<br>[4] data0 macro | 2'd0 |

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**Name** QSPICACHE_OVERRIDES
**Address** 0x401580d8
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | o v e r r i d e _ q | o v e r r i d e _ q s p | o v e r r i d e _ q | o v e r r i d e _ q s p | o v e r r i d e _ | o v e r r i d e _ q s | o v e r r i d e | o v e r r i d e _ q | o v e r r i d e | o v e r r i d e _ q | o v e r r i d e | o v e r r i d e | o v e r r i d e _ q | b l o c k _ q s p | b l o c k _ q |

| Name | RSVD | spicache_vddcut | icache_vddcut_val | spicache_clken | icache_clken_val | qspicache_iso | picache_iso_val | _qspicache_ls | spicache_ls_val | _qspicache_ds | spicache_ds_val | _qspicache_sd | spicache_sd_val | icache_vddcut | spicache_sd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13 | R/W | override_qspicache_vddcut | if set, head switch on qspi cache is now software controllable | 1'b0 |
| 12 | R/W | override_qspicache_vddcut_val | if override is set, should head switch on qspi cache be open or close?<br>1=open<br>0=close | 1'b0 |
| 11 | R/W | override_qspicache_clken | if set, clock gating to qspi cache is now under software control | 1'b0 |
| 10 | R/W | override_qspicache_clken_val | if override is set, should the clock gate on the qspi cache be enabled?<br>1=clock running<br>0=clock gated off | 1'b0 |
| 9 | R/W | override_qspicache_iso | if set, software now has direct control of the qspi cache ISO pin | 1'b0 |
| 8 | R/W | override_qspicache_iso_val | if override is set, what value should be driven onto qspi cache ISO pin? | 1'b0 |
| 7 | R/W | override_qspicache_ls | if set, software now has direct control of the qspi cache LS pin | 1'b0 |
| 6 | R/W | override_qspicache_ls_val | if override is set, what value should be driven onto qspi cache LS pin? | 1'b0 |
| 5 | R/W | override_qspicache_ds | if set, software now has direct control of the qspi cache DS pin | 1'b0 |
| 4 | R/W | override_qspicache_ds_val | if override is set, what value should be driven onto qspi cache DS pin? | 1'b0 |
| 3 | R/W | override_qspicache_sd | if set, software now has direct control of the qspi cache SD pin | 1'b0 |
| 2 | R/W | override_qspicache_sd_val | if override is set, what value should be driven onto qspi cache SD pin? | 1'b0 |
| 1 | R/W | block_qspicache_vddcut | if set, this core's FSM cannot power off the qspi cache | 1'b0 |
| 0 | R/W | block_qspicache_sd | if set, this core's FSM cannot assert the qspi cache SD pin | 1'b0 |

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**Name** COUNTER_CONTROL
**Address** 0x401580dc
**Description** control signals for counters
**Reset** 0x00000000

| Name | RSVD | load_count_down_t | halt_real_time_co |
|---|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | i m e r | u n t e r |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | load_count_down_timer | loads on positive edge; hardware doesn't reset this bit | 1'b0 |
| 0 | R/W | halt_real_time_counter | stop the real time counter from advancing | 1'b0 |

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**Name** CURRENT_REAL_TIME
**Address** 0x401580e0
**Description** counter value
**Reset** 0x00000000

| Name | v a l | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | real time in clk_lpc clock cycles | 32'd0 |

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**Name** CURRENT_COUNT_DOWN_TIME
**Address** 0x401580e4
**Description** counter value
**Reset** 0x00000000

| Name | v a l | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | this value is on 32kHz domain; read twice and check value is the same; if it's not, reread | 32'd0 |

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**Name** INIT_COUNT_DOWN
**Address** 0x401580e8
**Description** counter value
**Reset** 0x00000fff

| Name | v a l | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | val | initial value for count down timer | 32'hfff |

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**Name** INST_PENDING
**Address** 0x401580ec
**Description** instructions from clk_mpc to clk_lpc still pending
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | wdog_timer_disable | cntdwn_timer_load | RSVD | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17 | R | wdog_timer_disable | disable/enable of watchdog timer is still pending | 1'b0 |
| 16 | R | cntdwn_timer_load | load of count down timer is still pending | 1'b0 |

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**Name** STATUS
**Address** 0x401580f0
**Description** status signals
**Reset** 0x00000000

| Name | RSVD | | | | | | ble_osc_on_triggered | wurx_triggered | brownout_triggered | endoflife_triggered | energy4TX_triggered | energy4CPU_triggered | pmu_lpcomp_triggered | pmu_timer_triggered | i2c_triggered | spi_triggered | shub_triggered | dbg_triggered | ksm_triggered | qdec_triggered | wurx1_triggered | wurx0_triggered | gpio_triggered | timer_triggered | RSVD | | | | | in_hibernate_state | in_cpu_active_state | in_active_state |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25 | R | ble_osc_on_triggered | wake triggered by ble_osc_on | 1'b0 |
| 24 | R | wurx_triggered | wake triggered by multiple events of wurx. | 1'b0 |
| 23 | R | brownout_triggered | wake triggered by pmu brownout | 1'b0 |
| 22 | R | endoflife_triggered | wake triggered by pmu endoflife | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 21 | R | energy4TX_triggered | wake triggered by pmu energy4TX | 1'b0 |
| 20 | R | energy4CPU_triggered | wake triggered by pmu energy4CPU | 1'b0 |
| 19 | R | pmu_lpcomp_triggered | wake triggered by pmu lpcomp threshold crossing | 1'b0 |
| 18 | R | pmu_timer_triggered | wake triggered by pmu timer expiring | 1'b0 |
| 17 | R | i2c_triggered | wake triggered by i2c slave activity | 1'b0 |
| 16 | R | spi_triggered | wake triggered by spi slave activity | 1'b0 |
| 15 | R | shub_triggered | wake triggered by sensor hub | 1'b0 |
| 14 | R | dbg_triggered | wake triggered by debugger | 1'b0 |
| 13 | R | ksm_triggered | wake triggered by ksm activity | 1'b0 |
| 12 | R | qdec_triggered | wake triggered by qdec activity | 1'b0 |
| 11 | R | wurx1_triggered | wake triggered by wurx1 | 1'b0 |
| 10 | R | wurx0_triggered | wake triggered by wurx0 | 1'b0 |
| 9 | R | gpio_triggered | wake triggered by gpio | 1'b0 |
| 8 | R | timer_triggered | wake triggered by count down timer | 1'b0 |
| 2 | R | in_hibernate_state | pseq is currently in hibernate state | 1'b0 |
| 1 | R | in_cpu_active_state | pseq is currently in cpu active state | 1'b0 |
| 0 | R | in_active_state | pseq is currently in active state | 1'b0 |

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**Name** PERSISTENT0
**Address** 0x401580f4
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT1
**Address** 0x401580f8
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT2
**Address** 0x401580fc
**Description** persistent scratchpad register
**Reset** 0x00000000

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| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT3
**Address** 0x40158100
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT4
**Address** 0x40158104
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT5
**Address** 0x40158108
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT6
**Address** 0x4015810c
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT7
**Address** 0x40158110
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** SENSOR_HUB_CONTROL
**Address** 0x40158114
**Description** control signals for sensor hub
**Reset** 0xf00000e2

| Name | shub_sram3_acc_en | shub_sram2_acc_en | shub_sram1_acc_en | shub_sram0_acc_en | RSVD | shub_hib_exit_en | shub_start_adv_ovrd | shub_start_adv_val | shub_adv_enable | shub_breakout_imm_en | shub_start_ovrd | shub_start_val | shub_clken_slow | shub_reset_slow | shub_vddcut | shub_iso | shub_clken | shub_running_ovrd | shub_running_val | shub_reset | enable_func |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27 26 25 24 23 22 21 20 19 18 17 16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | shub_sram3_acc_en | 1= allow shub access to sram3 | 1'b1 |

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| | | | | |
|---|---|---|---|---|
| 30 | R/W | shub_sram2_acc_en | 1= allow shub access to sram2 | 1'b1 |
| 29 | R/W | shub_sram1_acc_en | 1= allow shub access to sram1 | 1'b1 |
| 28 | R/W | shub_sram0_acc_en | 1= allow shub access to sram0 | 1'b1 |
| 15 | R/W | shub_hib_exit_en | 1= exit out of hib, 0=exit out of retain. | 1'b0 |
| 14 | R/W | shub_start_adv_ovrd | 1= override | 1'b0 |
| 13 | R/W | shub_start_adv_val | value fo shub_start_adv when override is set | 1'b0 |
| 12 | R/W | shub_adv_enable | 1= enable adv along with sensor hub. | 1'b0 |
| 11 | R/W | shub_breakout_imm_en | 1= breakout sensor asap when a valid out-of-HIB event is triggered. 0= wait for sensor hub wrap-up. | 1'b0 |
| 10 | R/W | shub_start_ovrd | 1= override | 1'b0 |
| 9 | R/W | shub_start_val | value fo shub_start when override is set | 1'b0 |
| 8 | R/W | shub_clken_slow | for 32kHz portion; sensor hub clock enable<br>0 = clock off<br>1 = clock running | 1'b0 |
| 7 | R/W | shub_reset_slow | for 32kHz portion; 1 = keep shub in reset state; 0 = release shub from reset | 1'b1 |
| 6 | R/W | shub_vddcut | sensor hub head switch control<br>0 = head switch is closed<br>1 = head switch is open | 1'b1 |
| 5 | R/W | shub_iso | sensor hub isolation; do not assert/deassert in same cycle as shub_clken<br>0 = not isolated<br>1 = isolated | 1'b1 |
| 4 | R/W | shub_clken | sensor hub clock enable<br>0 = clock gated off<br>1 = clock running | 1'b0 |
| 3 | R/W | shub_running_ovrd | 1= override | 1'b0 |
| 2 | R/W | shub_running_val | value fo shub_running when override is set | 1'b0 |
| 1 | R/W | shub_reset | 1= keep shub in reset state; 0= release shub from reset | 1'b1 |
| 0 | R/W | enable_func | 1= sensor hub can trigger pseq out of hibernation early due to a threshold crossing event in sensor hub.<br>0= sensor hub cannot trigger pseq out of hibernation.<br>This bit does not disable the trigger signal to start the sensor functionality at the onset of hibernation. | 1'b0 |

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**Name** KSMQDEC_CONTROL
**Address** 0x40158118
**Description** keyboard and mouse power domain controls
**Reset** 0x0000000e

| Name | RSVD | ksmqdec_vddcut | ksmqdec_iso | ksmqdec_first | ksmqdec_clken |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | ksmqdec_vddcut | head switch control<br>0 = head switch is closed<br>1 = head switch is open | 1'b1 |
| | | | isolation | |

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| | | | | |
|---|---|---|---|---|
| 2 | R/W | ksmqdec_iso | 0 = not isolated<br>1 = isolated | 1'b1 |
| 1 | R/W | ksmqdec_frst | forced reset<br>1 = reset<br>0 = not reset | 1'b1 |
| 0 | R/W | ksmqdec_clken | clock enable<br>0 = clock gated off<br>1 = clock running | 1'b0 |

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**Name** SPII2C_CONTROL
**Address** 0x4015811c
**Description** spi and i2c power domain controls
**Reset** 0x0000000e

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | i2c1_slv_id_sel | i2c0_slv_id_sel | spii2c_vddcut | spii2c_iso | spii2c_frst | spii2c_clken |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | i2c1_slv_id_sel | select secondary i2c device id for i2c1 | 1'b0 |
| 4 | R/W | i2c0_slv_id_sel | select secondary i2c device id for i2c0 | 1'b0 |
| 3 | R/W | spii2c_vddcut | head switch control<br>0 = head switch is closed<br>1 = head switch is open | 1'b1 |
| 2 | R/W | spii2c_iso | isolation<br>0 = not isolated<br>1 = isolated | 1'b1 |
| 1 | R/W | spii2c_frst | forced reset<br>1 = reset<br>0 = not reset | 1'b1 |
| 0 | R/W | spii2c_clken | clock enable<br>0 = clock gated off<br>1 = clock running | 1'b0 |

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**Name** FLASH_CONTROL
**Address** 0x40158120
**Description** low power control of external flash needs
**Reset** 0x100b9aba

| Name | RS | pd_b4_ | expm | expm_o | pd_op | opc | mo | rpd_has | power_c |
|---|---|---|---|---|---|---|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | VD | | sleep | _en | | | | pcode | | | | | | | code | | | | | | | | ode | | | de | | _clock | | ycle_en |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29 | R/W | pd_b4_sleep | 1=issue flash deep power down prior to low power mode<br>0=no command issue. | 1'b0 |
| 28 | R/W | expm_en | 1=exit performance mode before issuing pd_opcode<br>0=not in performance mode. | 1'b1 |
| 27:20 | R/W | expm_opcode | opcode that takes flash out of performance mode | 8'h00 |
| 19:12 | R/W | pd_opcode | opcode that puts external flash into deep power down mode | 8'hb9 |
| 11:4 | R/W | opcode | opcode that wakes up the external flash from deep power down mode | 8'hab |
| 3:2 | R/W | mode | defining interface bus width of opcode; 0,3=single 1=dual 2=quad<br>For Macronix, set it to 0 because opcode is always serialized;<br>For Micron, it depends on the pre-configured interface bus width on the flash device. | 2'h2 |
| 1 | R/W | rpd_has_clock | 1=flash requires opcode to wakeup<br>0=flash requires only CS to wakeup | 1'b1 |
| 0 | R/W | power_cycle_en | 1=trigger command to release flash from power down when waking up<br>0= no action | 1'b0 |

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**Name** FLASH_CONTROL2
**Address** 0x40158124
**Description** low power control of external flash needs
**Reset** 0x0002007f

| Name | RSVD | | | | | | | | | | | | | enpm_clr_en | expm_mode | RSVD | | | | | | | | pseq_state_match | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 18 | R/W | enpm_clr_en | 1=clears enable_performance_mode bit in qspi_ahb too if expm_opcode is sent to flash. | 1'b0 |
| 17:16 | R/W | expm_mode | defining interface bus width of performance exit command; 0,3=single 1=dual 2=quad<br>For Macronix, set it to 2 as it's the only interface bus width equipped with performance enhancement mode<br>For Micron, it depends on the pre-configured interface bus width on the flash device, likely quad. | 2'h2 |
| 7:0 | R/W | pseq_state_match | pseq state to trigger flash deep power down. | 8'h7f |

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**Name** PMU_STATUS
**Address** 0x40158128
**Description** pmu status information
**Reset** 0x00000000

| Name | RSVD | soc_off_wkup_det | RSVD | brownout | endoflife | energy4TX | energy4CPU |
|---|---|---|---|---|---|---|---|
| **Bits** | **31:8** | **7:5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:5 | R | soc_off_wkup_det | returned from SOC Off state | 3'b0 |
| 3 | R | brownout | brownout detected; go to low power state as quickly as possible | 1'b0 |
| 2 | R | endoflife | battery is detected to be at end of life; do not start flash writes or radio operation | 1'b0 |
| 1 | R | energy4TX | enough energy is present to do a transmit | 1'b0 |
| 0 | R | energy4CPU | enough energy is present to operation the CPU | 1'b0 |

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**Name** PMU_INTERRUPT
**Address** 0x4015812c
**Description** pmu interrupt source
**Reset** 0x00000000

| Name | RSVD | clear_fe | clear_re | intmask |
|---|---|---|---|---|
| **Bits** | **31:29** | **28:25** | **24:21** | **20:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 28 | R/W | clear_fe_brownout | clear the rising edge capture blocks | 5'b0 |
| 27 | | clear_fe_endOfLife | | |
| 26 | | clear_fe_energy4TX | | |
| 25 | | clear_fe_energy4CPU | | |
| 23 | R/W | clear_re_brownout | clear the falling edge capture blocks | 5'b0 |
| 22 | | clear_re_endOfLife | | |
| 21 | | clear_re_energy4TX | | |
| 20 | | clear_re_energy4CPU | | |
| 15 | | brownout_falling | 0 = pmu_status_energy4CPU level | |
| 14 | | brownout_rising | | |
| 13 | | brownout_low | | |

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|---|---|---|---|---|
| 12 | R/W | brownout_high | 1 = pmu_status_energy4CPU negated level<br>2 = pmu_status_energy4CPU rising edge<br>3 = pmu_status_energy4CPU falling edge<br>4 = pmu_status_energy4TX level<br>5 = pmu_status_energy4TX negated level<br>6 = pmu_status_energy4TX rising edge<br>7 = pmu_status_energy4TX falling edge<br>8 = pmu_status_endOfLife level<br>9 = pmu_status_endOfLife negated level<br>10 = pmu_status_endOfLife rising edge<br>11 = pmu_status_endOfLife falling edge<br>12 = pmu_status_brownout level<br>13 = pmu_status_brownout negated level<br>14 = pmu_status_brownout rising edge<br>15 = pmu_status_brownout falling edge | 20'b0 |
| 11 | | endOfLife_falling | | |
| 10 | | endOfLife_rising | | |
| 9 | | endOfLife_low | | |
| 8 | | endOfLife_high | | |
| 7 | | energy4TX_falling | | |
| 6 | | energy4TX_rising | | |
| 5 | | energy4TX_low | | |
| 4 | | energy4TX_high | | |
| 3 | | energy4CPU_falling | | |
| 2 | | energy4CPU_rising | | |
| 1 | | energy4CPU_low | | |
| 0 | | energy4CPU_high | | |

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**Name** SYSRAM_WRITE_BLOCK_CFG
**Address** 0x40158130
**Description** Contains register fields associated with sysram_write_block_cfg.
**Reset** 0x00000000

| Name | RSVD | | | | | en_wrblk | a_hi | | | | | | | | | | | | a_lo | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 26 | R/W | en_wrblk | enable the sysram write lock; can't be undone unless a porst_soc is issued by pmu | 1'b0 |
| 25:13 | R/W | a_hi | high address on word boundary | 13'b0 |
| 12:0 | R/W | a_lo | low address on word boundary; any address from high address to low address is no longer writable once enable is set | 13'b0 |

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**Name** SYSRAM_SSRS_CFG
**Address** 0x40158134
**Description** Contains register fields associated with sysram_ssrs_cfg.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | enable | addr | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15 | R/W | enable | when set, enables r/w access to sysram during ssrs operation | 1'b0 |
| 14:0 | R/W | addr | sysram starting address for ssrs operation | 15'd0 |

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**Name** INTERRUPT_STATUS
**Address** 0x40158138
**Description** interrupt status
**Reset** 0x00000000

| Name | RSVD | returned_from_lp_state |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | returned_from_lp_state | indicates system returned from low power state (as oppposed to first power on) | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x4015813c
**Description** interrupt mask
**Reset** 0x00000001

| Name | RSVD | intrpt_mask |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt_mask | if nth bit set, the nth interrupt source is allowed to propogate to core's interrupt | 1'b1 |

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**Name** RESET_INTERRUPT
**Address** 0x40158140
**Description** interrupt resets
**Reset** 0x00000000

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| Name | RSVD | intrpt_reset |
|---|---|---|
| Bits | 31:1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt_reset | not self clearing | 1'b0 |

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**Name** FPGA_CFG
**Address** 0x40158144
**Description** Contains register fields associated with fpga_cfg.
**Reset** 0x00000000

| Name | RSVD | cfg |
|---|---|---|
| Bits | 31:16 | 15:0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | cfg | (no description) | 16'b0 |

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**Name** PLL
**Address** 0x40158148
**Description** Contains register fields associated with pll.
**Reset** 0x00000000

| Name | RSVD | divaccess | divref | divfb | divout | div2out | cpcur | filtc | filtr | RSVD |
|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31:30 | 29 | 28:26 | 25:20 | 19:17 | 16 | 15:10 | 9:8 | 7:4 | 3:0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29 | R/W | divaccess | (no description) | 1'b0 |
| 28:26 | R/W | divref | (no description) | 3'b0 |
| 25:20 | R/W | divfb | (no description) | 6'b0 |
| 19:17 | R/W | divout | (no description) | 3'b0 |
| 16 | R/W | div2out | (no description) | 1'b0 |
| 15:10 | R/W | cpcur | (no description) | 6'b0 |
| 9:8 | R/W | filtc | (no description) | 2'b0 |

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| | | | | |
|---|---|---|---|---|
| 7:4 | R/W | filtr | (no description) | 4'b0 |

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**Name** RADIO_STATUS
**Address** 0x4015814c
**Description** Contains register fields associated with radio_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | xtal_stable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | xtal_stable | (no description) | 1'b0 |

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**Name** RRAM_READ_CONFIG_LO
**Address** 0x40158150
**Description** Contains register fields associated with rram_read_config_lo.
**Reset** 0x00000000

| Name | dout | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dout | (no description) | 32'b0 |

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**Name** RRAM_READ_CONFIG_HI
**Address** 0x40158154
**Description** Contains register fields associated with rram_read_config_hi.
**Reset** 0x00000000

| Name | dout | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dout | (no description) | 32'b0 |

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**Name** SENSOR_HUB_IMM_BREAKOUT_MASK

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**Address** 0x40158158
**Description** control signals for sensor hub
**Reset** 0x0000ffff

| Name | RSVD | watch_wurx | watch_brownout | watch_endoflife | watch_energy4TX | watch_energy4CPU | watch_pmu_timer | watch_pmu_lpcomp | watch_i2c | watch_spi | watch_shub | watch_ksm | watch_qdec | watch_wurx1 | watch_wurx0 | watch_cntdown | watch_gpio |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15 | R/W | watch_wurx | valid to immediately breakout on multiple wurx detects in certain amount of time | 1'b1 |
| 14 | R/W | watch_brownout | valid to immediately breakout on pmu asserting brownout | 1'b1 |
| 13 | R/W | watch_endoflife | valid to immediately breakout on pmu asserting endoflife | 1'b1 |
| 12 | R/W | watch_energy4TX | valid to immediately breakout on pmu asserting energy4TX | 1'b1 |
| 11 | R/W | watch_energy4CPU | valid to immediately breakout on pmu asserting energy4CPU | 1'b1 |
| 10 | R/W | watch_pmu_timer | valid to immediately breakout on pmu timer expiring | 1'b1 |
| 9 | R/W | watch_pmu_lpcomp | valid to immediately breakout on pmu lpcomp threshold crossing | 1'b1 |
| 8 | R/W | watch_i2c | valid to immediately breakout on i2c slave activity | 1'b1 |
| 7 | R/W | watch_spi | valid to immediately breakout on spi slave activity | 1'b1 |
| 6 | R/W | watch_shub | valid to immediately breakout on sensor hub trigger | 1'b1 |
| 5 | R/W | watch_ksm | valid to immediately breakout on keyboard activity | 1'b1 |
| 4 | R/W | watch_qdec | valid to immediately breakout on qdec activity | 1'b1 |
| 3 | R/W | watch_wurx1 | valid to immediately breakout on wurx1 detect | 1'b1 |
| 2 | R/W | watch_wurx0 | valid to immediately breakout on wurx0 detect | 1'b1 |
| 1 | R/W | watch_cntdown | valid to immediately breakout on count down timer expiring | 1'b1 |
| 0 | R/W | watch_gpio | valid to immediately breakout on gpio activity | 1'b1 |

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**Name** FSM_TUNINGA
**Address** 0x4015815c
**Description** Contains register fields associated with fsm_tuningA.
**Reset** 0x00008000

| Name | cnt9_lsb | cnt8 | cnt7 | cnt6 | cnt5 | cnt4 | cnt3 | cnt2 | cnt1 | cnt0 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27** | **26 25 24** | **23 22 21** | **20 19 18** | **17 16 15** | **14 13 12** | **11 10 9** | **8 7 6** | **5 4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:27 | R/W | cnt9_lsb | additional clk_lpc cycles to wait in states tied to cnt9 (lsb's); determines | 5'b0 |

| | | | | |
|---|---|---|---|---|
| | | | cpu_ldo wait to setttle time | |
| 26:24 | R/W | cnt8 | additional clk_lpc cycles to wait in states tied to cnt8 | 3'b0 |
| 23:21 | R/W | cnt7 | additional clk_lpc cycles to wait in states tied to cnt7 | 3'b0 |
| 20:18 | R/W | cnt6 | additional clk_lpc cycles to wait in states tied to cnt6 | 3'b0 |
| 17:15 | R/W | cnt5 | additional clk_lpc cycles to wait in states tied to cnt5 | 3'b1 |
| 14:12 | R/W | cnt4 | additional clk_lpc cycles to wait in states tied to cnt4 | 3'b0 |
| 11:9 | R/W | cnt3 | additional clk_lpc cycles to wait in states tied to cnt3 | 3'b0 |
| 8:6 | R/W | cnt2 | additional clk_lpc cycles to wait in states tied to cnt2 | 3'b0 |
| 5:3 | R/W | cnt1 | additional clk_lpc cycles to wait in states tied to cnt1 | 3'b0 |
| 2:0 | R/W | cnt0 | additional clk_lpc cycles to wait in states tied to cnt0 | 3'b0 |

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**Name** FSM_TUNINGB
**Address** 0x40158160
**Description** Contains register fields associated with fsm_tuningB.
**Reset** 0x00000000

| Name | cnt9_msb | rcnt9 | | | rcnt8 | | | rcnt7 | | | rcnt6 | | | rcnt5 | | | rcnt4 | | | rcnt3 | | | rcnt2 | | | rcnt1 | | | rcnt0 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | cnt9_msb | additional clk_lpc cycles to wait in states tied to cnt9 (msb's) | 2'b0 |
| 29:27 | R/W | rcnt9 | additional clk_lpc cycles to wait in states tied to rcnt9 | 3'b0 |
| 26:24 | R/W | rcnt8 | additional clk_lpc cycles to wait in states tied to rcnt8 | 3'b0 |
| 23:21 | R/W | rcnt7 | additional clk_lpc cycles to wait in states tied to rcnt7 | 3'b0 |
| 20:18 | R/W | rcnt6 | additional clk_lpc cycles to wait in states tied to rcnt6 | 3'b0 |
| 17:15 | R/W | rcnt5 | additional clk_lpc cycles to wait in states tied to rcnt5 | 3'b0 |
| 14:12 | R/W | rcnt4 | additional clk_lpc cycles to wait in states tied to rcnt4 | 3'b0 |
| 11:9 | R/W | rcnt3 | additional clk_lpc cycles to wait in states tied to rcnt3 | 3'b0 |
| 8:6 | R/W | rcnt2 | additional clk_lpc cycles to wait in states tied to rcnt2 | 3'b0 |
| 5:3 | R/W | rcnt1 | additional clk_lpc cycles to wait in states tied to rcnt1 | 3'b0 |
| 2:0 | R/W | rcnt0 | additional clk_lpc cycles to wait in states tied to rcnt0 | 3'b0 |

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**Name** WDOG_CONTROL
**Address** 0x40158164
**Description** control signals for 32KHz watchdog timer
**Reset** 0x00000001

| Name | RSVD | internal_sc | timer_val_s | disable |
|---|---|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | internal_scan | (no description) | 1'b0 |
| 1 | R/W | timer_val_sel | selects timer value to load<br>1= 1-sec 0= 100-msec in 32KHz cycles | 1'b0 |
| 0 | R/W | disable | 1= disables wdog timer | 1'b1 |

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**Name** WDOG_TIMER
**Address** 0x40158168
**Description** Read current 32KHz watchdog timer value
**Reset** 0x00000000

| Name | R S V D | | | | | | | | | | | | | | | | v a l | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R | val | this value is on 32kHz domain; read twice and check value is the same; if it's not, reread | 16'd0 |

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**Name** CORE_ID
**Address** 0x40158ffc
**Description** core's id
**Reset** 0x50534551

| Name | i d | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | PSEQ in ASCII | 32'h50534551 |

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# SWD_NONSECURE

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**Name** DAP_ADDR_CTRL
**Address** 0x40159000
**Description** Contains register fields associated with dap_addr_ctrl.
**Reset** 0x00000000

| Name | R S V D | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | d a p _ a d d r | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | dap_addr | (no description) | 2'b0 |

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---

**Name** DAP_WDATA_CTRL
**Address** 0x40159004
**Description** Contains register fields associated with dap_wdata_ctrl.
**Reset** 0x00000000

| Name | dap_wdata | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | dap_wdata | (no description) | 32'b0 |

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---

**Name** SWD_CLK_DIV_CTRL
**Address** 0x40159008
**Description** Contains register fields associated with swd_clk_div_ctrl.
**Reset** 0x00000003

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | swd_clk_div | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2:0 | R/W | swd_clk_div | clock divider = 2^(swd_clk_div) | 3'd3 |

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---

**Name** PATTERN_UPPER_CTRL
**Address** 0x4015900c
**Description** Contains register fields associated with pattern_upper_ctrl.
**Reset** 0xffffffff

| Name | patte | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | r<br>n | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | pattern | pattern to switch from jtag to swd , lsb is shifted out first | 32'hffffffff |

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---

**Name** PATTERN_LOWER_CTRL
**Address** 0x40159010
**Description** Contains register fields associated with pattern_lower_ctrl.
**Reset** 0xffffffff

| Name | p<br>a<br>t<br>t<br>e<br>r<br>n | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | pattern | pattern to switch from jtag to swd , lsb is shifted out first | 32'hffffffff |

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**Name** TAIL_DUR
**Address** 0x40159014
**Description** Contains register fields associated with tail_dur.
**Reset** 0x00000005

| Name | R<br>S<br>V<br>D | | | | | | | | | | | | | | | | | | | | | | | | | | | | c<br>o<br>u<br>n<br>t | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3:0 | R/W | count | exact number of extra cycles after a packet. 0 means no tail. must be at least 5 cycles after a DAP packet | 4'h5 |

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**Name** PATTERN_DUR
**Address** 0x40159018
**Description** Contains register fields associated with pattern_dur.
**Reset** 0x0000003f

| Name | R<br>S<br>V<br>D | | | | | | | | | | | | | | | | | | | | | | | | c<br>o<br>u<br>n<br>t | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| | | | number cycles to drive the pattern MSB first is count + 1. e.g. 64 cycles | |

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| | | | | |
|---|---|---|---|---|
| 7:0 | R/W | count | by default | 8'd63 |

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**Name** DAP_BRIDGE_STATUS
**Address** 0x4015901c
**Description** Contains register fields associated with dap_bridge_status.
**Reset** 0x00000000

| Name | RSVD | busy |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | busy | (no description) | 1'b0 |

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**Name** DAP_RESP_STATUS
**Address** 0x40159020
**Description** Contains register fields associated with dap_resp_status.
**Reset** 0x00000000

| Name | RSVD | par_err | dap_resp |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | par_err | 1= RDATA has parity error, 0= otherwise | 1'b0 |
| 2:0 | R | dap_resp | (no description) | 3'b0 |

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**Name** RW_CTRL
**Address** 0x40159024
**Description** Contains register fields associated with RW_ctrl.
**Reset** 0x00000000

| Name | RSVD | RnW |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | RnW | 1 for Read, 0 for Write | 1'b0 |

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**Name** APNDP_CTRL
**Address** 0x40159028

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**Description** Contains register fields associated with apndp_ctrl.
**Reset** 0x00000000

| Name | RSVD | APnDP |
|---|---|---|
| **Bits** | **31:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | APnDP | 0 for DPACC, 1 for APACC | 1'b0 |

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**Name** TRIGGER_CTRL
**Address** 0x4015902c
**Description** Contains register fields associated with trigger_ctrl.
**Reset** 0x00000000

| Name | RSVD | go_pattern | RSVD | go_packet |
|---|---|---|---|---|
| **Bits** | **31:3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | go_pattern | 1 to start resetting by sending a pattern of bits as prescribed by pattern_upper_ctrl and pattern_lower_ctrl.<br>The length of the pattern is set by pattern_dur. not self-resetting.<br>The reset pattern is to bring the target DAP to an inital state. | 1'b0 |
| 0 | R/W | go_packet | 1 to start a dap packet , not self-resetting.<br>This trigger does not start a DAP reset. See go_pattern to for DAP reset. | 1'b0 |

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**Name** DAP_RDATA
**Address** 0x40159030
**Description** Contains register fields associated with dap_rdata.
**Reset** 0x00000000

| Name | dap_rdata |
|---|---|
| **Bits** | **31:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dap_rdata | (no description) | 32'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x40159034
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000000

| Name | RSVD | passthru1 | passthru0 |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | passthru1 | 1=allow interrupt1 (par err on readback data) to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 (read done) to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x40159038
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | interrupt1 | interrupt0 |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | interrupt1 | saw a rising edge on par_err; independent of mask | 1'b0 |
| 0 | R | interrupt0 | saw a rising edge on done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x4015903c
**Description** not auto cleared
**Reset** 0x00000000

| Name | RSVD | set_inter | set_inter |
|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | rupt1 | rupt0 |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | set_interrupt1 | set interrupt 1 | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt 0 | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x40159040
**Description** not auto cleared
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | reset_interrupt1 | reset_interrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | reset_interrupt1 | reset interrupt 1 | 1'b0 |
| 0 | R/W | reset_interrupt0 | reset interrupt 0 | 1'b0 |

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**Name** DTOP_BYPASS_CTRL
**Address** 0x40159044
**Description** Contains register fields associated with dtop_bypass_ctrl.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | byp_rd | hw_own |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | byp_rd | 1 to bypass read modify write | 1'b0 |
| 0 | R/W | hw_own | 1 to indicate hw own and sw control is disabled | 1'b0 |

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**Name** DTOP_BYPASS_STAT
**Address** 0x40159048
**Description** Contains register fields associated with dtop_bypass_stat.
**Reset** 0x00000000

| Name | RSVD | mode |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | mode | 1 if fpga is in dtop_bypass mode (built with FPGA_DTOP_BYP defined) | 1'b0 |

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**Name** DTOP_BYPASS_GPO
**Address** 0x4015904c
**Description** Contains register fields associated with dtop_bypass_gpo.
**Reset** 0x00000000

| Name | RSVD | gpo |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | gpo | 1 selects ext clock as source in DTOP_BYPASS, 0 selects internal pll. | 16'b0 |

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**Name** DTOP_BYPASS_RIF_MODE_CNTL
**Address** 0x40159050
**Description** Contains register fields associated with dtop_bypass_rif_mode_cntl.
**Reset** 0x500040ac

| Name | addr |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | addr | rif mode control address. | 32'h500040ac |

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**Name** DTOP_BYPASS_RIF_CAL_CNTL
**Address** 0x40159054
**Description** Contains register fields associated with dtop_bypass_rif_cal_cntl.
**Reset** 0x500040b0

| Name | addr |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | addr | rif cal control address. | 32'h500040b0 |

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**Name** DTOP_BYPASS_RIF_RXDC7
**Address** 0x40159058
**Description** Contains register fields associated with dtop_bypass_rif_rxdc7.
**Reset** 0x50004090

| Name | addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | addr | rif rxdc7 address. | 32'h50004090 |

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**Name** CORE_ID
**Address** 0x40159ffc
**Description** Contains register fields associated with core_id.
**Reset** 0x53574420

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | SWD in ASCII | 32'h53574420 |

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# GADC_NONSECURE

**Name** CTRL
**Address** 0x4015a000
**Description** control signals
**Reset** 0x00000100

| Name | watch_channels | | | | | | | | | | | | | | | | averaging_amount | | | wait_amount | | | | | | | | clkdiv | | code | | warmup | mode | enable_dp |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | watch_channels | Which channels are active?<br>bit 0 = Not Used<br>bit 1 = Vbatt<br>bit 2 = Vstor<br>bit 3 = core supply<br>bit 4 = temperature<br>bit 5 = external port 1; differential<br>bit 6 = external port 0; differential<br>bit 7 = external port 1; single ended pos<br>bit 8 = external port 1; single ended neg<br>bit 9 = external port 0; single ended pos<br>bit 10 = external port 0; single ended neg<br>bit 11 = Vli-ion<br>bit 12-14 = input short<br>bit 15 = No-Connect | 16'h0 |
| 15:13 | R/W | averaging_amount | average 2^averaging_amount samples | 3'd0 |
| 12:6 | R/W | wait_amount | Specifies how many wait cycles in gadc clock cycles. | 7'd4 |
| 5:4 | R/W | clkdiv | gadc clock frequency<br>0 = 2.00MHz<br>1 = 1.00MHz<br>2 = 0.50MHz<br>3 = 0.25MHz | 2'd0 |
| 3 | R/W | code | Specifies ADC input data code bit representation.<br>0 = MSB of input gadc data is inverted<br>1 = MSB of input gadc data is not inverted Data in pedestal format is converted to 2's complement. | 1'b0 |
| 2 | R/W | warmup | Do 8 clocks cycle of warmup at beginning of round. No data sampled during this period. | 1'b0 |
| 1 | R/W | mode | What is the mode of operation?<br>0 = continuous<br>1 = one-shot | 1'b0 |
| 0 | R/W | enable_dp | Enables digital datapath. | 1'b0 |

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**Name** CTRL1
**Address** 0x4015a004
**Description** control signals
**Reset** 0x00000000

| Name | RSVD | ddr_en | ch_sel_ovrd_val | ch_sel_ovrd |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6** | **5** | **4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | ddr_en | for debug; brings out gadc_out as a 6b wide ddr bus | 1'b0 |
| 4:1 | R/W | ch_sel_ovrd_val | when override is set, what's the gadc channel to select. | 4'd0 |
| 0 | R/W | ch_sel_ovrd | when set overrides gadc channel select. | 1'b0 |

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**Name** CH1_DATAPATH_CONFIG
**Address** 0x4015a008
**Description** channel-1 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH2_DATAPATH_CONFIG
**Address** 0x4015a00c
**Description** channel-2 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH3_DATAPATH_CONFIG
**Address** 0x4015a010
**Description** channel-3 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH4_DATAPATH_CONFIG
**Address** 0x4015a014
**Description** channel-4 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH5_DATAPATH_CONFIG
**Address** 0x4015a018
**Description** channel-5 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH6_DATAPATH_CONFIG
**Address** 0x4015a01c
**Description** channel-6 offset and gain datpath parameters
**Reset** 0x00002800

| | off | | | | | | | | | | | | | gai | | | | | g | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH7_DATAPATH_CONFIG
**Address** 0x4015a020
**Description** channel-7 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH8_DATAPATH_CONFIG
**Address** 0x4015a024
**Description** channel-8 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH9_DATAPATH_CONFIG
**Address** 0x4015a028

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**Description** channel-9 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH10_DATAPATH_CONFIG
**Address** 0x4015a02c
**Description** channel-10 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH11_DATAPATH_CONFIG
**Address** 0x4015a030
**Description** channel-11 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |

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| | | | | |
|---|---|---|---|---|
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH12_DATAPATH_CONFIG
**Address** 0x4015a034
**Description** channel-12 offset and gain datpath parameters applied to unused channels 0,13,14 by defaults
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** GAIN_CONFIG0
**Address** 0x4015a038
**Description** channels 1-10 gain settings selection
**Reset** 0x00000000

| Name | RSVD | ch10_gain_sel | | | ch9_gain_sel | | | ch8_gain_sel | | | ch7_gain_sel | | | ch6_gain_sel | | | ch5_gain_sel | | | ch4_gain_sel | | | ch3_gain_sel | | | ch2_gain_sel | | | ch1_gain_sel | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29:27 | R/W | ch10_gain_sel | channel-10 gain settings | 3'd0 |
| 26:24 | R/W | ch9_gain_sel | channel-9 gain settings | 3'd0 |
| 23:21 | R/W | ch8_gain_sel | channel-8 gain settings | 3'd0 |
| 20:18 | R/W | ch7_gain_sel | channel-7 gain settings | 3'd0 |
| 17:15 | R/W | ch6_gain_sel | channel-6 gain settings | 3'd0 |
| 14:12 | R/W | ch5_gain_sel | channel-5 gain settings | 3'd0 |
| 11:9 | R/W | ch4_gain_sel | channel-4 gain settings | 3'd0 |
| 8:6 | R/W | ch3_gain_sel | channel-3 gain settings | 3'd0 |
| 5:3 | R/W | ch2_gain_sel | channel-2 gain settings | 3'd0 |
| 2:0 | R/W | ch1_gain_sel | channel-1 gain settings | 3'd0 |

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**Name** GAIN_CONFIG1
**Address** 0x4015a03c
**Description** channels 11-15 gain settings selection
**Reset** 0x00000000

| Name | RSVD | ch12_gain_sel | ch11_gain_sel |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6** | **5 4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:3 | R/W | ch12_gain_sel | channel-12 gain settings | 3'd0 |
| 2:0 | R/W | ch11_gain_sel | channel-11 gain settings | 3'd0 |

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**Name** DATAPATH_OUTPUT
**Address** 0x4015a040
**Description** a read of this register pops the data FIFO
**Reset** 0x00000000

| Name | empty | RSVD | data |
|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21** | **20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R | empty | if 1, ignore data sample because fifo has been drained | 1'b0 |
| 20:0 | R | data | {4b channel id, 12b processed ADC sample Q<10,1>, 5b exponent} | 21'd0 |

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**Name** INTERRUPTS
**Address** 0x4015a044
**Description** interrupt status
**Reset** 0x00000000

| Name | RSVD | intrpt2 | intrpt1 | intrpt0 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R | intrpt2 | one shot or one round of continuous mode has completed | 1'b0 |
| 1 | R | intrpt1 | fifo overran | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 0 | R | intrpt0 | hardware written to fifo | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x4015a048
**Description** interrupt masking
**Reset** 0x00000007

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | mask_intrpt2 | mask_intrpt1 | mask_intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | mask_intrpt2 | allow intrpt2 to propogate to core's single output interrupt | 1'b1 |
| 1 | R/W | mask_intrpt1 | allow intrpt1 to propogate to core's single output interrupt | 1'b1 |
| 0 | R/W | mask_intrpt0 | allow intrpt0 to propogate to core's single output interrupt | 1'b1 |

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**Name** INTERRUPT_CLEAR
**Address** 0x4015a04c
**Description** clear interrupts
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | clear_intrpt2 | clear_intrpt1 | clear_intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clear_intrpt2 | clear interrupt 2; not self reseting | 1'b0 |
| 1 | R/W | clear_intrpt1 | clear interrupt 1; not self reseting | 1'b0 |
| 0 | R/W | clear_intrpt0 | clear interrupt 0; not self reseting | 1'b0 |

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**Name** CORE_ID
**Address** 0x4015affc
**Description** core ID
**Reset** 0x47414443

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| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | GADC in ASCII | 32'h47414443 |

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# TRNG_NONSECURE

**Name** CONTROL
**Address** 0x4015b000
**Description** Contains register fields associated with control.
**Reset** 0x00010010

| Name | RSVD | | | | | | | | | | | | | | | | radio_warmup_cnt | | | | | | | | | | block_force_radio_into_trng_mode | launch_on_radio_up | radio_gain_override | lsb_source | adc_capture_edge | go |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16:6 | R/W | radio_warmup_cnt | how long to wait for radio to warmup in clk_mpc cycles | 11'h400 |
| 5 | R/W | block_force_radio_into_trng_mode | (no description) | 1'b0 |
| 4 | R/W | launch_on_radio_up | when radio is first coming back from power down mode, generate automatically (without software intervension) a new random number; this core will assert the radio gain override pin, wait 50usec, and then start the bit gathering process | 1'b1 |
| 3 | R/W | radio_gain_override | (no description) | 1'b0 |
| 2 | R/W | lsb_source | 0 = rif.rif_i[0]; 1 = radio.adc_i[0] | 1'b0 |
| 1 | R/W | adc_capture_edge | 0 = rising edge; 1 = falling edge | 1'b0 |

| | | | | |
|---|---|---|---|---|
| 0 | R/W | go | start the process of generating a true random number ; not self reseting | 1'b0 |

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**Name** STATUS
**Address** 0x4015b004
**Description** Contains register fields associated with status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | fsm_running | running |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | fsm_running | this status is asserted when the fsm gets out of ST_IDLE | 1'b0 |
| 0 | R | running | this status is asserted after RADIO warm-up | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x4015b008
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | trng_trouble | trng_ready |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | trng_trouble | (no description) | 1'b0 |
| 0 | R | trng_ready | (no description) | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x4015b00c
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000003

| | R | | | | | | | | | | | | | | | | | | | | | | | | | | | | intr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | SVD | intrpt_mask |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_mask | if nth bit set, the nth interrupt source is allowed to propogate to core's interrupt | 2'h3 |

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**Name** SET_INTERRUPT
**Address** 0x4015b010
**Description** Contains register fields associated with set_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_set |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_set | not self clearing | 2'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x4015b014
**Description** Contains register fields associated with reset_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_reset |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_reset | not self clearing | 2'b0 |

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**Name** TRNG

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**Address** 0x4015b018
**Description** Contains register fields associated with trng.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | the generated true random number | 32'b0 |

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**Name** CORE_ID
**Address** 0x4015bffc
**Description** Contains register fields associated with core_id.
**Reset** 0x54524e47

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | TRNG in ASCII | 32'h54524e47 |

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# RCOS_CAL_NONSECURE

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**Name** CAL_CTRL
**Address** 0x4015c000
**Description** Contains register fields associated with cal_ctrl.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | hw_cal_sel | sw_cal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | hw_cal_sel | 1=choose cal from HW; 0=choose cal from SW | 1'b0 |
| 0 | R/W | sw_cal | rising edge starts SW cal, falling edge terminates cal if cal is not done; best to leave it on once set. | 1'b0 |

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**Name** SLOW_CLK_SET
**Address** 0x4015c004
**Description** Contains register fields associated with slow_clk_set.
**Reset** 0x00000064

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| Name | RSVD | slow_clk_set |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12** | **11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 11:0 | R/W | slow_clk_set | set the number of cycles to run slow clock; counting stops when this number is reached or cal is pulled down. constraint: 1 <= slow_clk_set <= 63; if set 0, hardware will use 1. | 12'd100 |

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**Name** SLOW_CLK_CNT
**Address** 0x4015c008
**Description** Contains register fields associated with slow_clk_cnt.
**Reset** 0x00000000

| Name | RSVD | slow_clk_cnt |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12** | **11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 11:0 | R | slow_clk_cnt | number of cycles counted by slow clock. | 12'b0 |

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**Name** FAST_CLK_CNT
**Address** 0x4015c00c
**Description** Contains register fields associated with fast_clk_cnt.
**Reset** 0x00000000

| Name | RSVD | fast_clk_cnt |
|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 21:0 | R | fast_clk_cnt | number of cycles counted by fast clock. 10 bits more than slow_clk_cnt/set because fast clock is 1000 times faster. | 22'b0 |

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**Name** SOURCE_CLK_SEL
**Address** 0x4015c010
**Description** Contains register fields associated with source_clk_sel.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | fast_clk_sel | slow_clk_sel |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | fast_clk_sel | 0=fast clk source is PCLk; 1=fast clk source is clk_calfc | 1'b0 |
| 0 | R/W | slow_clk_sel | 0=slow clk source is clk_lpc;1=slow clk source is clk_calfc | 1'b0 |

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**Name** FORCE_RESET
**Address** 0x4015c014
**Description** Contains register fields associated with force_reset.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | force_reset |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | force_reset | 1=apply force reset to fast domain and slow domain.<br>0=release force reset. | 1'b0 |

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**Name** STAT
**Address** 0x4015c018
**Description** Contains register fields associated with stat.
**Reset** 0x00000000

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | fast_domain_reset_stat | slow_domain_reset_stat | high_precision_run | running |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | fast_domain_reset_stat | 1=fast_clk domain is still in reset; don't triger cal | 1'b0 |
| 2 | R | slow_domain_reset_stat | 1=slow_clk domain is still in reset; don't triger cal | 1'b0 |
| 1 | R | high_precision_run | 1=high-precisioned clock was used at the beginning of cal; 0=low precisioned clock | 1'b0 |
| 0 | R | running | (no description) | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x4015c01c
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | rcos_done |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | rcos_done | (no description) | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x4015c020
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000001

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | int |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | intrpt_mask |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt_mask | if nth bit set, the nth interrupt source is allowed to propogate to core's interrupt | 1'h1 |

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**Name** SET_INTERRUPT
**Address** 0x4015c024
**Description** Contains register fields associated with set_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_set |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt_set | not self clearing | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x4015c028
**Description** Contains register fields associated with reset_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_reset |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt_reset | not self clearing | 1'b0 |

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**Name** CORE_ID
**Address** 0x4015cffc
**Description** Contains register fields associated with core_id.
**Reset** 0x52434f53

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | RCOS in ASCII | 32'h52434f53 |

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# SHUB_NONSECURE

**Name** SETUP0
**Address** 0x4015d000
**Description** Contains register fields associated with setup0.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | shub_disable | shub_trig_mode | adv_mode | store_in_sram | breakout | go |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | shub_disable | 0=enable sensor hub. | 1'b0 |
| 4 | R/W | shub_trig_mode | 0=continuous mode 1=trigger mode. | 1'b0 |
| 3 | R/W | adv_mode | 1=sensor hub's cpu enabled. | 1'b0 |
| 2 | R/W | store_in_sram | 0=data gets stored in flash, 1=data gets stored in SRAM | 1'b0 |
| 1 | R/W | breakout | useful for testing | 1'b0 |
| 0 | R/W | go | not self reseting | 1'b0 |

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**Name** SETUP1
**Address** 0x4015d004
**Description** Contains register fields associated with setup1.
**Reset** 0x00003fff

| Name | loop_ti |
|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | m<br>e | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | loop_time | how long should each iteration (sensor read and flash write) be ? (800 = wr_delay*2 + 16bytes read + overhead). The multiplier of 2 is needed beause when there are mulitple data bytes straddling the page boundary, the shub issues the page program twice. The first one sends all bytes up to the at the page boundary, and the second sends the rest of the data bytes. | 32'h3fff |

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**Name** PORT0_SETUP
**Address** 0x4015d008
**Description** Contains register fields associated with port0_setup.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | trig_en | RSVD | | | | num_sensors | | | RSVD | | spi_n_i2c | port_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 11 | R/W | trig_en | 1= enable threshold comparison with measurement results. | 1'd0 |
| 6:4 | R/W | num_sensors | number of sensors connected to port = num_sensors + 1 | 3'd0 |
| 1 | R/W | spi_n_i2c | 1= use spi for port0, 0= use i2c | 1'b0 |
| 0 | R/W | port_en | (no description) | 1'b0 |

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**Name** PORT0_SENSOR0_SETUP
**Address** 0x4015d00c
**Description** Contains register fields associated with port0_sensor0_setup.
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | num_bytes | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR1_SETUP

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**Address** 0x4015d010
**Description** Contains register fields associated with port0_sensor1_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR2_SETUP
**Address** 0x4015d014
**Description** Contains register fields associated with port0_sensor2_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR3_SETUP
**Address** 0x4015d018
**Description** Contains register fields associated with port0_sensor3_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR4_SETUP
**Address** 0x4015d01c
**Description** Contains register fields associated with port0_sensor4_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR5_SETUP
**Address** 0x4015d020
**Description** Contains register fields associated with port0_sensor5_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR6_SETUP
**Address** 0x4015d024
**Description** Contains register fields associated with port0_sensor6_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR7_SETUP
**Address** 0x4015d028
**Description** Contains register fields associated with port0_sensor7_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 | 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SPI_ENG0_CTRL
**Address** 0x4015d02c
**Description** Contains register fields associated with port0_spi_eng0_ctrl.
**Reset** 0x00000000

| Name | cpol | cpha | RSVD | sensor_opcode |
|---|---|---|---|---|
| Bits | 31 | 30 | 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | cpol | spi clock polarity mode. 0 = posedge-sensitive, 1 = negedge | 1'd0 |
| 30 | R/W | cpha | spi clcok phase mode | 1'd0 |
| 7:0 | R/W | sensor_opcode | opcode used to communicate with sensor in spi mode | 8'd0 |

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**Name** PORT0_I2C_IDW_0
**Address** 0x4015d030
**Description** id(read) of sensor0 on port0 prior to pointer address write
**Reset** 0x0001fe00

| | | ext_ack | ext_ma | | e | e | e | e | ext_ |
|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_1
**Address** 0x4015d034
**Description** id(read) of sensor1 on port0 prior to pointer address write
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_2
**Address** 0x4015d038
**Description** id(read) of sensor2 on port0 prior to pointer address write
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_3
**Address** 0x4015d03c
**Description** id(read) of sensor3 on port0 prior to pointer address write
**Reset** 0x0001fe00

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | ext_ | ext_ | | | | | | | e |

| Name | RSVD | ack_value_to_drive | master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_4
**Address** 0x4015d040
**Description** id(read) of sensor4 on port0 prior to pointer address write
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_5
**Address** 0x4015d044
**Description** id(read) of sensor5 on port0 prior to pointer address write
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26** | **25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_6
**Address** 0x4015d048
**Description** id(read) of sensor6 on port0 prior to pointer address write
**Reset** 0x0001fe00

| | | e | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_7
**Address** 0x4015d04c
**Description** id(read) of sensor7 on port0 prior to pointer address write
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_0
**Address** 0x4015d050
**Description** pointer addr of sensor0 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 | 24 | 23 | 22 21 | 20 19 | 18 | 17 | 16 15 14 13 12 11 10 9 | 8 7 6 5 4 3 2 1 | 0 |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_1
**Address** 0x4015d054
**Description** pointer addr of sensor1 on port0

**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10** | **9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_2
**Address** 0x4015d058
**Description** pointer addr of sensor2 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_dri | ext_master_drives_a | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|

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| | | | | | | | | ve | ck | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_3
**Address** 0x4015d05c
**Description** pointer addr of sensor3 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_4

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**Address** 0x4015d060
**Description** pointer addr of sensor4 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10** | **9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_5
**Address** 0x4015d064
**Description** pointer addr of sensor5 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_ | ext_master_drives | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ov |
|---|---|---|---|---|---|---|---|---|---|---|

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | drive | _ack | | | | | | | | | | | | | | | | | | | | | rd |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_6
**Address** 0x4015d068
**Description** pointer addr of sensor6 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_7
**Address** 0x4015d06c
**Description** pointer addr of sensor7 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10** | **9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_0
**Address** 0x4015d070
**Description** id(read) of sensor0 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_ | ext_master_dr | ext_tai | ext_hea | ext_clk_ | ext_data | ext_data | ext_data | ext_data_ie |
|---|---|---|---|---|---|---|---|---|---|---|

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | to_drive | ives_ack | l | d | pu | _pu | _oe | | | | | | | _o | | | | | | | | _ovrd |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_1
**Address** 0x4015d074
**Description** id(read) of sensor1 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10** | **9 8 7 6 5 4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |

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| | | | | |
|---|---|---|---|---|
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_2
**Address** 0x4015d078
**Description** id(read) of sensor2 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:25** | **24** | **23** | **22:21** | **20:19** | **18** | **17** | **16:9** | **8:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_3
**Address** 0x4015d07c
**Description** id(read) of sensor3 on port0 prior to data read
**Reset** 0x0001fe00

| | RS | ext_ack_val | ext_master | ext_ | ext_ | ext_c | ext_d | ext_d | ext_d | ext_data |
|---|---|---|---|---|---|---|---|---|---|---|

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| Name | VD | ue_to_drive | _drives_ack | tail | head | lk_pu | ata_pu | ata_oe | ata_o | _ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_4
**Address** 0x4015d080
**Description** id(read) of sensor4 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_5
**Address** 0x4015d084
**Description** id(read) of sensor5 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_6
**Address** 0x4015d088
**Description** id(read) of sensor6 on port0 prior to data read
**Reset** 0x0001fe00

| | | ext_ack_ | ext_mas | e | e | ex | ex | ex | ex | ext_d |
|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | | | | | | | value_to_drive | ter_drives_ack | xt_tail | | xt_head | | t_clk_pu | t_data_pu | t_data_oe | | | | | | | t_data_o | | | | | | | | ata_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_7
**Address** 0x4015d08c
**Description** id(read) of sensor7 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSVD | | | | | | | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | | ext_head | | ext_clk_pu | ext_data_pu | ext_data_oe | | | | | | | ext_data_o | | | | | | | | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |

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| | | | | |
|---|---|---|---|---|
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_DAT_CTRL
**Address** 0x4015d090
**Description** slave(port0) data byte read (used in not the last byte)
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_DAT_LAST_CTRL
**Address** 0x4015d094
**Description** slave(port0) data byte read (used in the last byte)
**Reset** 0x0001fe00

| | | ext_a | ext_m | | | | | | | ex |
|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | | | | | | | ck_value_to_drive | aster_drives_ack | ext_tail | | ext_head | | ext_clk_pu | ext_data_pu | ext_data_oe | | | | | | | ext_data_o | | | | | | | | t_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_ALM0_QUAN_CTRL
**Address** 0x4015d098
**Description** alarm0 quantity's definition
**Reset** 0x05500000

| Name | quan_dat_is_signed | lsb_first | RSVD | | quan3_size | | quan2_size | | quan1_size | | quan0_size | | quan3_pos | | | | quan2_pos | | | | quan1_pos | | | | quan0_pos | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | quan_dat_is_signed | quantity data is a signed data | 1'b0 |
| 30 | R/W | lsb_first | 1= quantity data is big endian ; 0= small endian | 1'b0 |
| 27:26 | R/W | quan3_size | number of bytes making up the quantity data = programmed quan3_size + 1 | 2'd1 |

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| | | | | |
|---|---|---|---|---|
| 25:24 | R/W | quan2_size | (no description) | 2'd1 |
| 23:22 | R/W | quan1_size | (no description) | 2'd1 |
| 21:20 | R/W | quan0_size | (no description) | 2'd1 |
| 19:15 | R/W | quan3_pos | starting position of the quantity | 5'd0 |
| 14:10 | R/W | quan2_pos | (no description) | 5'd0 |
| 9:5 | R/W | quan1_pos | (no description) | 5'd0 |
| 4:0 | R/W | quan0_pos | (no description) | 5'd0 |

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**Name** PORT0_ALM1_QUAN_CTRL
**Address** 0x4015d09c
**Description** alarm1 quantity's definition
**Reset** 0x05500000

| Name | quan_dat_is_signed | lsb_first | RSVD | quan3_size | quan2_size | quan1_size | quan0_size | quan3_pos | quan2_pos | quan1_pos | quan0_pos |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29 28** | **27 26** | **25 24** | **23 22** | **21 20** | **19 18 17 16 15** | **14 13 12 11 10** | **9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | quan_dat_is_signed | quantity data is a signed data | 1'b0 |
| 30 | R/W | lsb_first | 1= quantity data is big endian ; 0= small endian | 1'b0 |
| 27:26 | R/W | quan3_size | number of bytes making up the quantity data = programmed quan3_size + 1 | 2'd1 |
| 25:24 | R/W | quan2_size | (no description) | 2'd1 |
| 23:22 | R/W | quan1_size | (no description) | 2'd1 |
| 21:20 | R/W | quan0_size | (no description) | 2'd1 |
| 19:15 | R/W | quan3_pos | starting position of the quantity | 5'd0 |
| 14:10 | R/W | quan2_pos | (no description) | 5'd0 |
| 9:5 | R/W | quan1_pos | (no description) | 5'd0 |
| 4:0 | R/W | quan0_pos | (no description) | 5'd0 |

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**Name** PORT0_ALM0_TRIG_CTRL
**Address** 0x4015d0a0
**Description** alarm0 trigger function: trigger = trig_func_en( out_of_range(quan3), out_of_range(quan2),out_of_range(quan1),out_of_range(quan0))
**Reset** 0x00000000

| | | |
|---|---|---|
| | R | trig_ |

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| Name | SVD | | | | | | | | | | | | | | | func_en | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | trig_func_en | (no description) | 16'b0 |

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**Name** PORT0_ALM1_TRIG_CTRL
**Address** 0x4015d0a4
**Description** alarm1 trigger function: trigger = trig_func_en( out_of_range(quan3), out_of_range(quan2),out_of_range(quan1),out_of_range(quan0))
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | trig_func_en | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | trig_func_en | (no description) | 16'b0 |

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**Name** PORT0_ALM0_THRHLD_MAX_0
**Address** 0x4015d0a8
**Description** threshold of sensor results
**Reset** 0xffffffff

| Name | byte_max_03 | | | | | | | | byte_max_02 | | | | | | | | byte_max_01 | | | | | | | | byte_max_00 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_03 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_02 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_01 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_00 | (no description) | 8'hff |

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**Name** PORT0_ALM0_THRHLD_MAX_1
**Address** 0x4015d0ac
**Description** threshold of sensor results
**Reset** 0xffffffff

| | byte_max_07 | byte_max_06 | byte_max_05 | byte_max_04 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_07 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_06 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_05 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_04 | (no description) | 8'hff |

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**Name** PORT0_ALM0_THRHLD_MAX_2
**Address** 0x4015d0b0
**Description** threshold of sensor results
**Reset** 0xffffffff

| | byte_max_11 | byte_max_10 | byte_max_09 | byte_max_08 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_11 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_10 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_09 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_08 | (no description) | 8'hff |

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**Name** PORT0_ALM0_THRHLD_MIN_0
**Address** 0x4015d0b4
**Description** threshold of sensor results
**Reset** 0x00000000

| | b | b | b | b |
|---|---|---|---|---|

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| Name | byte_min_03 | byte_min_02 | byte_min_01 | byte_min_00 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_03 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_02 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_01 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_00 | (no description) | 8'h00 |

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**Name** PORT0_ALM0_THRHLD_MIN_1
**Address** 0x4015d0b8
**Description** threshold of sensor results
**Reset** 0x00000000

| Name | byte_min_07 | byte_min_06 | byte_min_05 | byte_min_04 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_07 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_06 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_05 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_04 | (no description) | 8'h00 |

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**Name** PORT0_ALM0_THRHLD_MIN_2
**Address** 0x4015d0bc
**Description** threshold of sensor results
**Reset** 0x00000000

| Name | byte_min_11 | byte_min_10 | byte_min_09 | byte_min_08 |
|---|---|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_11 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_10 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_09 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_08 | (no description) | 8'h00 |

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**Name** PORT0_ALM1_THRHLD_MAX_0
**Address** 0x4015d0c0
**Description** threshold of sensor results
**Reset** 0xffffffff

| Name | byte_max_03 | byte_max_02 | byte_max_01 | byte_max_00 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_03 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_02 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_01 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_00 | (no description) | 8'hff |

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**Name** PORT0_ALM1_THRHLD_MAX_1
**Address** 0x4015d0c4
**Description** threshold of sensor results
**Reset** 0xffffffff

| Name | byte_max_07 | byte_max_06 | byte_max_05 | byte_max_04 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_07 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_06 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_05 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_04 | (no description) | 8'hff |

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**Name** PORT0_ALM1_THRHLD_MAX_2
**Address** 0x4015d0c8
**Description** threshold of sensor results
**Reset** 0xffffffff

| | byte_max_11 | | | | | | | | byte_max_10 | | | | | | | | byte_max_09 | | | | | | | | byte_max_08 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_11 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_10 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_09 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_08 | (no description) | 8'hff |

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**Name** PORT0_ALM1_THRHLD_MIN_0
**Address** 0x4015d0cc
**Description** threshold of sensor results
**Reset** 0x00000000

| | byte_min_03 | | | | | | | | byte_min_02 | | | | | | | | byte_min_01 | | | | | | | | byte_min_00 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_03 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_02 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_01 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_00 | (no description) | 8'h00 |

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**Name** PORT0_ALM1_THRHLD_MIN_1
**Address** 0x4015d0d0
**Description** threshold of sensor results
**Reset** 0x00000000

| | byte_ | byte_ | byte_ | byte_ |
|---|---|---|---|---|

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| Name | te_min_07 | | | | | | | | te_min_06 | | | | | | | | te_min_05 | | | | | | | | te_min_04 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_07 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_06 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_05 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_04 | (no description) | 8'h00 |

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**Name** PORT0_ALM1_THRHLD_MIN_2
**Address** 0x4015d0d4
**Description** threshold of sensor results
**Reset** 0x00000000

| Name | byte_min_11 | | | | | | | | byte_min_10 | | | | | | | | byte_min_09 | | | | | | | | byte_min_08 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_11 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_10 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_09 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_08 | (no description) | 8'h00 |

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**Name** PORT0_STATUS
**Address** 0x4015d0d8
**Description** Contains register fields associated with port0_status.
**Reset** 0x00000000

| Name | RSVD | triggere | RSVD | out_of_range |
|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | d | | | | | | | | | | | | | | | | | | | | _stat | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R | triggered | (no description) | 1'b0 |
| 3:0 | R | out_of_range_stat | (no description) | 4'd0 |

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**Name** PORT1_SETUP
**Address** 0x4015d0dc
**Description** Contains register fields associated with port1_setup.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | trig_en | RSVD | | | | | num_sensors | | | RSVD | | spi_n_i2c | port_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 11 | R/W | trig_en | 1= enable threshold comparison with measurement results. | 1'd0 |
| 6:4 | R/W | num_sensors | number of sensors connected to port = num_sensors + 1 | 3'd0 |
| 1 | R/W | spi_n_i2c | 1= use spi for port0, 0= use i2c | 1'b0 |
| 0 | R/W | port_en | (no description) | 1'b0 |

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**Name** PORT1_SENSOR0_SETUP
**Address** 0x4015d0e0
**Description** Contains register fields associated with port1_sensor0_setup.
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | num_bytes | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT1_SPI_ENG0_CTRL

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**Address** 0x4015d0e4
**Description** Contains register fields associated with port1_spi_eng0_ctrl.
**Reset** 0x00000000

| Name | cpol | cpha | RSVD | | | | | | | | | | | | | | | | | | | | | sensor_opcode | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | cpol | spi clock polarity mode. 0 = posedge-sensitive, 1 = negedge | 1'd0 |
| 30 | R/W | cpha | spi clcok phase mode | 1'd0 |
| 7:0 | R/W | sensor_opcode | opcode used to communicate with sensor in spi mode | 8'd0 |

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**Name** PORT1_I2C_IDW_0
**Address** 0x4015d0e8
**Description** id(read) of sensor0 on port0 prior to pointer address write
**Reset** 0x0001fe00

| Name | RSVD | | | | | | | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | | ext_head | | ext_clk_pu | ext_data_pu | ext_data_oe | | | | | | | | ext_data_o | | | | | | | | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT1_I2C_ADDR_0
**Address** 0x4015d0ec
**Description** pointer addr of sensor0 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT1_I2C_IDR_0
**Address** 0x4015d0f0
**Description** id(read) of sensor0 on port0 prior to data read
**Reset** 0x0001fe00

| | | ext_ack_v | ext_maste | ex | ex | ext | ext | ext | ext | ext_da |
|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | alue_to_drive | r_drives_ack | t_tail | t_head | _clk_pu | _data_pu | _data_oe | _data_o | ta_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT1_I2C_DAT_CTRL
**Address** 0x4015d0f4
**Description** slave(port0) data byte read (used in not the last byte)
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |

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| | | | | |
|---|---|---|---|---|
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT1_I2C_DAT_LAST_CTRL
**Address** 0x4015d0f8
**Description** slave(port0) data byte read (used in the last byte)
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10** | **9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT1_ALM0_QUAN_CTRL
**Address** 0x4015d0fc
**Description** alarm0 quantity's definition
**Reset** 0x05500000

| | quan_d | ls | | qu | qu | qu | qu | qu | qu | qu | qu |
|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | at_is_signed | lsb_first | RSVD | quan3_size | quan2_size | quan1_size | quan0_size | quan3_pos | quan2_pos | quan1_pos | quan0_pos |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29 28** | **27 26** | **25 24** | **23 22** | **21 20** | **19 18 17 16 15** | **14 13 12 11 10** | **9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | quan_dat_is_signed | quantity data is a signed data | 1'b0 |
| 30 | R/W | lsb_first | 1= quantity data is big endian ; 0= small endian | 1'b0 |
| 27:26 | R/W | quan3_size | number of bytes making up the quantity data = programmed quan3_size + 1 | 2'd1 |
| 25:24 | R/W | quan2_size | (no description) | 2'd1 |
| 23:22 | R/W | quan1_size | (no description) | 2'd1 |
| 21:20 | R/W | quan0_size | (no description) | 2'd1 |
| 19:15 | R/W | quan3_pos | starting position of the quantity | 5'd0 |
| 14:10 | R/W | quan2_pos | (no description) | 5'd0 |
| 9:5 | R/W | quan1_pos | (no description) | 5'd0 |
| 4:0 | R/W | quan0_pos | (no description) | 5'd0 |

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**Name** PORT1_ALM0_TRIG_CTRL
**Address** 0x4015d100
**Description** alarm0 trigger function: trigger = trig_func_en( out_of_range(quan3), out_of_range(quan2),out_of_range(quan1),out_of_range(quan0))
**Reset** 0x00000000

| Name | RSVD | trig_func_en |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | trig_func_en | (no description) | 16'b0 |

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**Name** PORT1_ALM0_THRHLD_MAX_0
**Address** 0x4015d104
**Description** threshold of sensor results
**Reset** 0xffffffff

| | by | by | by | by |
|---|---|---|---|---|

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| Name | byte_max_03 | byte_max_02 | byte_max_01 | byte_max_00 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_03 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_02 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_01 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_00 | (no description) | 8'hff |

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**Name** PORT1_ALM0_THRHLD_MAX_1
**Address** 0x4015d108
**Description** threshold of sensor results
**Reset** 0xffffffff

| Name | byte_max_07 | byte_max_06 | byte_max_05 | byte_max_04 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_07 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_06 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_05 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_04 | (no description) | 8'hff |

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**Name** PORT1_ALM0_THRHLD_MAX_2
**Address** 0x4015d10c
**Description** threshold of sensor results
**Reset** 0xffffffff

| Name | byte_max_11 | byte_max_10 | byte_max_09 | byte_max_08 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_11 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_10 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_09 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_08 | (no description) | 8'hff |

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**Name** PORT1_ALM0_THRHLD_MIN_0
**Address** 0x4015d110
**Description** threshold of sensor results
**Reset** 0x00000000

| Name | byte_min_03 | | | | | | | | byte_min_02 | | | | | | | | byte_min_01 | | | | | | | | byte_min_00 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_03 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_02 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_01 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_00 | (no description) | 8'h00 |

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**Name** PORT1_ALM0_THRHLD_MIN_1
**Address** 0x4015d114
**Description** threshold of sensor results
**Reset** 0x00000000

| Name | byte_min_07 | | | | | | | | byte_min_06 | | | | | | | | byte_min_05 | | | | | | | | byte_min_04 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_07 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_06 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_05 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_04 | (no description) | 8'h00 |

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**Name** PORT1_ALM0_THRHLD_MIN_2
**Address** 0x4015d118
**Description** threshold of sensor results
**Reset** 0x00000000

| Name | byte_min_11 | | | | | | | | byte_min_10 | | | | | | | | byte_min_09 | | | | | | | | byte_min_08 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_11 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_10 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_09 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_08 | (no description) | 8'h00 |

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**Name** PORT1_STATUS
**Address** 0x4015d11c
**Description** Contains register fields associated with port1_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | triggered | RSVD | | | | | | | | | | | | | | | | | | | | out_of_range_stat | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R | triggered | (no description) | 1'b0 |
| 3:0 | R | out_of_range_stat | (no description) | 4'd0 |

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**Name** FLASH_CTRL0
**Address** 0x4015d120
**Description** Contains register fields associated with flash_ctrl0.
**Reset** 0xabb90206

| | | | | o |
|---|---|---|---|---|

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| Name | opcode_rpd | opcode_pd | opcode_pp | opcode_wren |
|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | opcode_rpd | release_power_down (not needed for macronix) | 8'hab |
| 23:16 | R/W | opcode_pd | power down opcode | 8'hb9 |
| 15:8 | R/W | opcode_pp | page program opcode | 8'h02 |
| 7:0 | R/W | opcode_wren | write_enable opcode | 8'h06 |

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**Name** FLASH_CTRL1
**Address** 0x4015d124
**Description** Contains register fields associated with flash_ctrl1.
**Reset** 0x00000000

| Name | RSVD | wr_start_address |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R/W | wr_start_address | (no description) | 24'h0 |

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**Name** FLASH_CTRL2
**Address** 0x4015d128
**Description** Contains register fields associated with flash_ctrl2.
**Reset** 0x00000000

| Name | RSVD | wr_max_add |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 27:20 | R/W | pd_delay | delay cycles (pd_delay+1) after flash power down / release from power down | 8'd2 |
| 11:0 | R/W | pp_delay | delay cycles (pp_delay+1) after flash write. (320 cycles of 32k clock = 10ms pp time from Macronix) | 12'd320 |

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**Name** FLASH_STATUS0
**Address** 0x4015d134
**Description** Contains register fields associated with flash_status0.
**Reset** 0x00000000

| Name | RSVD | wr_saturated | wr_current_address |
|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 | 24 | 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R | wr_saturated | indicates overflow. | 1'b0 |
| 23:0 | R | wr_current_address | the next address to write to. Number of bytes in storage = wr_current_address - wr_start_address. | 24'h0 |

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**Name** FLASH_STATUS1
**Address** 0x4015d138
**Description** Contains register fields associated with flash_status1.
**Reset** 0x00000000

| Name | lost_write_cnt |
|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | lost_write_cnt | (no description) | 32'h0 |

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**Name** SHUB_STATUS0
**Address** 0x4015d13c
**Description** Contains register fields associated with shub_status0.
**Reset** 0x00000000

| Name | cpu_running | RSVD | state |
|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R | cpu_running | useful for testing | 1'b0 |
| 4:0 | R | state | useful for testing | 5'b0 |

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**Name** PC_CTRL
**Address** 0x4015d140
**Description** Contains register fields associated with pc_ctrl.
**Reset** 0x00000000

| Name | pc_init |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | pc_init | iniital pc value | 32'b0 |

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**Name** PC_CTRL2
**Address** 0x4015d144
**Description** Contains register fields associated with pc_ctrl2.
**Reset** 0x00000000

| Name | RSVD | pc_init_l |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | o<br>a<br>d |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | pc_init_load | iniital pc value load; not self-clearing | 1'b0 |

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**Name** REG7
**Address** 0x4015d148
**Description** Contains register fields associated with reg7.
**Reset** 0x00000000

| Name | r<br>7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | r7 | value of reg7 | 32'b0 |

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**Name** SHUB_SRAM_CLK_SEL
**Address** 0x4015d14c
**Description** clock selection between PCLK and clk_lpc as to drive SRAM.
**Reset** 0x00000000

| Name | R<br>S<br>V<br>D | | | | | | | | | | | | | | | | | | | | | | | | | | | | s<br>h<br>u<br>b<br>_<br>s<br>r<br>a<br>m<br>_<br>c<br>l<br>k<br>_<br>s<br>e<br>l<br>_<br>v<br>a<br>l | s<br>h<br>u<br>b<br>_<br>s<br>r<br>a<br>m<br>_<br>c<br>l<br>k<br>_<br>s<br>e<br>l<br>_<br>o<br>v<br>r<br>d |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | shub_sram_clk_sel_val | override val to shub_sram_clk mux. 0=PCLK ; 1=clk_lpc | 1'b0 |
| 0 | R/W | shub_sram_clk_sel_ovrd | override bit to shub_sram_clk mux | 1'b0 |

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**Name** REGFILE_SEL
**Address** 0x4015d150

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**Description** Contains register fields associated with regfile_sel.
**Reset** 0x00000007

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | sel | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3:0 | R/W | sel | select which reg in regfile to be connected to reg7 | 4'd7 |

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**Name** REGFILE_REGW
**Address** 0x4015d154
**Description** Contains register fields associated with regfile_regw.
**Reset** 0x00000000

| Name | regfile_wdata | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | regfile_wdata | write data to selected reg in regfile. The selection pointer is regfile_sel | 32'b0 |

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**Name** ID
**Address** 0x4015dffc
**Description** Contains register fields associated with id.
**Reset** 0x53485542

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | SHUB in ASCII | 32'h53485542 |

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# PDM0_NONSECURE

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**Name** CONTROL
**Address** 0x40160000
**Description** Contains register fields associated with control.

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**Reset** 0x000006f0

| Name | RSVD | | | | | | | | | | | | | | | stereo_mode | pdm_in_dbg | pcm_bit_width | | pcm_rate_sel | | | | pdm_clk_sel | | | | | pack_samples | swap_samples | capture_on_rising | enable_dp |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16 | R/W | stereo_mode | selects between stereo and mono mode of operation<br>1= stereo mode; act as datapath enable for other stereo datapath pair. When this bit is set along with enable_dp, it enables stereo datapath pair concurrently to avoid pdm data capture synchronization issues.<br>At that time, corresponding bits for the other datapath pair shouldn't be set.<br>0= mono mode; use enable_dp bit to enable each datapath. | 1'd0 |
| 15 | R/W | pdm_in_dbg | if 1, bypass pdm bitstream and send in dc value 1 | 1'd0 |
| 14:13 | R/W | pcm_bit_width | pcm output sample bit width<br>0= 16bit<br>1= 20bit<br>2/3= 24bit | 2'd0 |
| 12:9 | R/W | pcm_rate_sel | pcm output rate selection if msb bit3=1 it's in pcm_rate override mode<br>0= 6KHz<br>1= 8KHz<br>2= 12KHz<br>3= 16KHz<br>4= 24KHz<br>5= 32KHz<br>6= 48KHz<br>7= 64KHz | 4'd3 |
| 8:4 | R/W | pdm_clk_sel | pdm clock frequency = f_pclk/(pdm_clk_sel+1)<br>valid range: 3 to 31<br>possible rates when pcm_rate_sel[0]=1:<br>31= 500KHz<br>15= 1MHz<br>7= 2MHz<br>possible rates when pcm_rate_sel[0]=0:<br>23= 500KHz<br>11= 1MHz<br>5= 2MHz | 5'd15 |
| 3 | R/W | pack_samples | pack samples in ping-pong buffer<br>default: samples are stored unpacked<br>applies to 20/24 bit samples only | 1'b0 |
| 2 | R/W | swap_samples | swap packed samples in pingpong buffer location<br>default: former samples uses lsb bits and later samples uses msb bits<br>applies to 16bit samples only | 1'b0 |
| 1 | R/W | capture_on_rising | selects edge of pdm clock to capture pdm data<br>0= pdm data sampled on falling edge of pdm clock (i.e. pdm data transitions on rising edge of pdm clock)<br>1= pdm data sampled on rising edge of pdm clock (i.e. pdm data transitions on falling edge of pdm clock) | 1'b0 |
| 0 | R/W | enable_dp | enable the datapath | 1'b0 |

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**Name** FILTER_CONFIG
**Address** 0x40160004
**Description** Contains register fields associated with filter_config.
**Reset** 0x00062a50

| Name | RSVD | | | | | | downsample_ratio | | lpf_bypass_ovrd_val | lpf_bw | | | hbf_bypass_ovrd_val | hpf2_init_shift | | | | hpf1_init_shift | | | | hpf_init_enable | hpf_bypass | hpf_bw | | | | | hpf_ord | cic_ord | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25:24 | R/W | downsample_ratio | downsample override<br>used only during pcm_rate override mode:<br>downsampling by 1/(1+downsample_ratio) | 2'd0 |
| 23 | R/W | lpf_bypass_ovrd_val | lpf bypass override value<br>used only during pcm_rate override mode:<br>0= enable lpf<br>1= bypass lpf; output rate=input rate | 1'd0 |
| 22:20 | R/W | lpf_bw | lpf bandwidth override selection<br>used only during pcm_rate override mode:<br>if pcm_rate_sel[0]=1, bandwidth will be (KHz):<br>0= 4<br>1= 6<br>2= 8<br>3= 10<br>4= 12<br>5= 14<br>6/7= 15.5<br>if pcm_rate_sel[0]=0, bandwidth will be (KHz):<br>0= 3<br>1= 4.5<br>2= 6<br>3= 7.5<br>4= 9<br>5= 10<br>6/7= 11.5 | 3'd0 |
| 19 | R/W | hbf_bypass_ovrd_val | hbf bypass override value<br>used only during pcm_rate override mode:<br>0= enable hbf<br>1= bypass hbf; output rate=input rate | 1'd0 |
| 18:14 | R/W | hpf2_init_shift | hpf2 initialization shift amount, max=24<br>for values > max, operation is same as max | 5'd24 |
| 13:10 | R/W | hpf1_init_shift | hpf1 initialization shift amount, max=10<br>for values > max, operation is same as max | 4'd10 |
| 9 | R/W | hpf_init_enable | 1= enable hpf initialization | 1'b1 |
| 8 | R/W | hpf_bypass | 1= bypass hpf | 1'b0 |
| | | | hpf bandwidth selection | |

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| | | | | |
|---|---|---|---|---|
| 7:4 | R/W | hpf_bw | when hpf_ord=0, bandwidth will be (Hz):<br>0= 10, 1= 20, 2= 40, 3= 80, 4= 161, 5= 324, 6= 658, 7= 1359<br>when hpf_ord=1, bandwidth will be (Hz):<br>0= 9, 1= 19, 2= 37, 3= 75, 4= 151, 5= 304, 6= 616, 7= 1269<br>when hpf_ord=2, bandwidth will be (Hz):<br>0= 15, 1= 30, 2= 60, 3= 120, 4= 242, 5= 487, 6= 989, 7= 2036 | 4'd5 |
| 3:2 | R/W | hpf_ord | hpf order<br>0= 1st order<br>1= 2nd order<br>2/3= 3rd order | 2'd0 |
| 1:0 | R/W | cic_ord | cic order<br>0= 3rd order<br>1= 4th order<br>2/3= unused | 2'd0 |

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**Name** FILTER_CONFIG1
**Address** 0x40160008
**Description** Contains register fields associated with filter_config1.
**Reset** 0x000003f8

| Name | RSVD | interp_ph_idx_updt | interp_ph_idx1 | interp_ph_idx0 | interp_add_sample | interp_skp_sample | interp_bypass |
|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20** | **19** | **18 17 16 15 14 13 12 11** | **10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19 | R/W | interp_ph_idx_updt | interpolator phase indices update indicator for HW to capture, auto-clear | 1'b0 |
| 18:11 | R/W | interp_ph_idx1 | index-1 into interpolator coefficient table; for use only when add sample | 8'd0 |
| 10:3 | R/W | interp_ph_idx0 | index-0 into interpolator coefficient table | 8'd127 |
| 2 | R/W | interp_add_sample | add one sample to accomodate cfo frequency drift, auto-clear | 1'b0 |
| 1 | R/W | interp_skp_sample | skip one sample to accomodate cfo frequency drift, auto-clear | 1'b0 |
| 0 | R/W | interp_bypass | 1= bypass interpolator, also resets the fir filter taps | 1'b0 |

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**Name** GAIN_CONTROL_CONFIG
**Address** 0x4016000c
**Description** Contains register fields associated with gain_control_config.
**Reset** 0x00000040

| Name | RS | hbf_gain | hpf_gain | cic_gain | co |
|---|---|---|---|---|---|

| | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | in_sel | | | in_sel | | | in_sel | | | code | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:13 | R/W | hbf_gain_sel | allow gain-scaling at the output of hbf through bit shifting<br>zero-padding for left shift; rounding, dropping lsbs for right shift.<br>0= no scaling<br>1-4= up scaling; zero-padding +1 to +4 respectively<br>5-7= down scaling; dropping lsbs -1 to -3 respectively | 3'd0 |
| 12:10 | R/W | hpf_gain_sel | allow gain-scaling at the output of hpf through bit shifting<br>zero-padding for left shift; rounding, dropping lsbs for right shift.<br>0= no scaling<br>1-4= up scaling; zero-padding +1 to +4 respectively<br>5-7= down scaling; dropping lsbs -1 to -3 respectively | 3'd0 |
| 9:7 | R/W | cic_gain_sel | allow gain-scaling at the output of cic through bit shifting<br>zero-padding for left shift; rounding, dropping lsbs for right shift.<br>0= no scaling<br>1-4= up scaling; zero-padding +1 to +4 respectively<br>5-7= down scaling; dropping lsbs -1 to -3 respectively | 3'd0 |
| 6:0 | R/W | code | gain code for LUT | 7'd64 |

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**Name** INTERRUPTS
**Address** 0x40160018
**Description** Contains register fields associated with interrupts.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | intrpt9 | intrpt8 | intrpt7 | intrpt6 | intrpt5 | intrpt4 | intrpt3 | intrpt2 | intrpt1 | intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R | intrpt9 | error in buffer D; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 8 | R | intrpt8 | error in buffer C; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 7 | R | intrpt7 | ping pong buffer D is locked and needs to be emptied | 1'b0 |
| 6 | R | intrpt6 | ping pong buffer C is locked and needs to be emptied | 1'b0 |
| 5 | R | intrpt5 | error in buffer B; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 4 | R | intrpt4 | error in buffer A; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 3 | R | intrpt3 | ping pong buffer B is locked and needs to be emptied | 1'b0 |
| 2 | R | intrpt2 | ping pong buffer A is locked and needs to be emptied | 1'b0 |
| 1 | R | intrpt1 | hw wrote to ping pong buffer | 1'b0 |
| 0 | R | intrpt0 | interpolator done processing current pcm sample | 1'b0 |

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**Name** INTERRUPT_MASK

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**Address** 0x4016001c
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x000003ff

| Name | RSVD | mask_intrpt9 | mask_intrpt8 | mask_intrpt7 | mask_intrpt6 | mask_intrpt5 | mask_intrpt4 | mask_intrpt3 | mask_intrpt2 | mask_intrpt1 | mask_intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R/W | mask_intrpt9 | allow intrpt9 to propogate to core's single output interrupt | 1'b1 |
| 8 | R/W | mask_intrpt8 | allow intrpt8 to propogate to core's single output interrupt | 1'b1 |
| 7 | R/W | mask_intrpt7 | allow intrpt7 to propogate to core's single output interrupt | 1'b1 |
| 6 | R/W | mask_intrpt6 | allow intrpt6 to propogate to core's single output interrupt | 1'b1 |
| 5 | R/W | mask_intrpt5 | allow intrpt5 to propogate to core's single output interrupt | 1'b1 |
| 4 | R/W | mask_intrpt4 | allow intrpt4 to propogate to core's single output interrupt | 1'b1 |
| 3 | R/W | mask_intrpt3 | allow intrpt3 to propogate to core's single output interrupt | 1'b1 |
| 2 | R/W | mask_intrpt2 | allow intrpt2 to propogate to core's single output interrupt | 1'b1 |
| 1 | R/W | mask_intrpt1 | allow intrpt1 to propogate to core's single output interrupt | 1'b1 |
| 0 | R/W | mask_intrpt0 | allow intrpt0 to propogate to core's single output interrupt | 1'b1 |

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**Name** INTERRUPT_CLEAR
**Address** 0x40160020
**Description** Contains register fields associated with interrupt_clear.
**Reset** 0x00000000

| Name | RSVD | clear_intrpt9 | clear_intrpt8 | clear_intrpt7 | clear_intrpt6 | clear_intrpt5 | clear_intrpt4 | clear_intrpt3 | clear_intrpt2 | clear_intrpt1 | clear_intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R/W | clear_intrpt9 | clear interrupt 9; not self reseting | 1'b0 |
| 8 | R/W | clear_intrpt8 | clear interrupt 8; not self reseting | 1'b0 |
| 7 | R/W | clear_intrpt7 | clear interrupt 7; not self reseting | 1'b0 |
| 6 | R/W | clear_intrpt6 | clear interrupt 6; not self reseting | 1'b0 |
| 5 | R/W | clear_intrpt5 | clear interrupt 5; not self reseting | 1'b0 |
| 4 | R/W | clear_intrpt4 | clear interrupt 4; not self reseting | 1'b0 |
| 3 | R/W | clear_intrpt3 | clear interrupt 3; not self reseting | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 2 | R/W | clear_intrpt2 | clear interrupt 2; not self reseting | 1'b0 |
| 1 | R/W | clear_intrpt1 | clear interrupt 1; not self reseting | 1'b0 |
| 0 | R/W | clear_intrpt0 | clear interrupt 0; not self reseting | 1'b0 |

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**Name** BUFFER_ACCESS_MODE
**Address** 0x40160024
**Description** Contains register fields associated with buffer_access_mode.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | dma_mode |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | dma_mode | 1= pingpong buffer has fifo-like interface.<br>read address is auto-incremented from base address of buffer A by an internal counter, counting from 0 to 63 and wraps around.<br>buffer full interrupt is cleared by HW at the end of the last read of the buffer.<br>The auto-increment counter is cleared when fifo_mode is 0. | 1'b0 |

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**Name** AUTO_COUNTER
**Address** 0x40160028
**Description** Contains register fields associated with auto_counter.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | count | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:0 | R | count | current value of the auto increment counter. | 6'b0 |

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**Name** BUFFER_DEPTH
**Address** 0x4016002c
**Description** Contains register fields associated with buffer_depth.
**Reset** 0x00000010

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | depth | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 4:0 | R | depth | one buffer depth | 5'd16 |

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**Name** CORE_ID
**Address** 0x40160ff4
**Description** Contains register fields associated with core_id.
**Reset** 0x50444d20

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | 'PDM ' in ASCII | 32'h50444d20 |

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**Name** REV_HASH
**Address** 0x40160ff8
**Description** Contains register fields associated with rev_hash.
**Reset** 0x00000000

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | crc32 of this document; don't change value to anything other than 32'h00000000; scripts will fill it out in the verilog | 32'h00000000 |

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**Name** REV_KEY
**Address** 0x40160ffc
**Description** Contains register fields associated with rev_key.
**Reset** 0x52455620

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | REV in ASCII | 32'h52455620 |

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# PDM1_NONSECURE

**Name** CONTROL
**Address** 0x40161000
**Description** Contains register fields associated with control.
**Reset** 0x000006f0

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | p | | | | | | | | | | | | | | | c a p | |

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| Name | RSVD | | | | | | | | | | | | | | | stereo_mode | pdm_in_dbg | pcm_bit_width | | pcm_rate_sel | | | | pdm_clk_sel | | | | | pack_samples | swap_samples | capture_on_rising | enable_dp |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16 | R/W | stereo_mode | selects between stereo and mono mode of operation<br>1= stereo mode; act as datapath enable for other stereo datapath pair. When this bit is set along with enable_dp, it enables stereo datapath pair concurrently to avoid pdm data capture synchronization issues. At that time, corresponding bits for the other datapath pair shouldn't be set.<br>0= mono mode; use enable_dp bit to enable each datapath. | 1'd0 |
| 15 | R/W | pdm_in_dbg | if 1, bypass pdm bitstream and send in dc value 1 | 1'd0 |
| 14:13 | R/W | pcm_bit_width | pcm output sample bit width<br>0= 16bit<br>1= 20bit<br>2/3= 24bit | 2'd0 |
| 12:9 | R/W | pcm_rate_sel | pcm output rate selection if msb bit3=1 it's in pcm_rate override mode<br>0= 6KHz<br>1= 8KHz<br>2= 12KHz<br>3= 16KHz<br>4= 24KHz<br>5= 32KHz<br>6= 48KHz<br>7= 64KHz | 4'd3 |
| 8:4 | R/W | pdm_clk_sel | pdm clock frequency = f_pclk/(pdm_clk_sel+1)<br>valid range: 3 to 31<br>possible rates when pcm_rate_sel[0]=1:<br>31= 500KHz<br>15= 1MHz<br>7= 2MHz<br>possible rates when pcm_rate_sel[0]=0:<br>23= 500KHz<br>11= 1MHz<br>5= 2MHz | 5'd15 |
| 3 | R/W | pack_samples | pack samples in ping-pong buffer<br>default: samples are stored unpacked<br>applies to 20/24 bit samples only | 1'b0 |
| 2 | R/W | swap_samples | swap packed samples in pingpong buffer location<br>default: former samples uses lsb bits and later samples uses msb bits<br>applies to 16bit samples only | 1'b0 |
| 1 | R/W | capture_on_rising | selects edge of pdm clock to capture pdm data<br>0= pdm data sampled on falling edge of pdm clock (i.e. pdm data transitions on rising edge of pdm clock)<br>1= pdm data sampled on rising edge of pdm clock (i.e. pdm data transitions on falling edge of pdm clock) | 1'b0 |
| 0 | R/W | enable_dp | enable the datapath | 1'b0 |

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**Name** FILTER_CONFIG
**Address** 0x40161004

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**Description** Contains register fields associated with filter_config.
**Reset** 0x00062a50

| Name | RSVD | | | | | | downsample_ratio | | lpf_bypass_ovrd_val | lpf_bw | | | hbf_bypass_ovrd_val | hpf2_init_shift | | | | hpf1_init_shift | | | | hpf_init_enable | hpf_bypass | hpf_bw | | | | hpf_ord | cic_ord | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25:24 | R/W | downsample_ratio | downsample override<br>used only during pcm_rate override mode:<br>downsampling by 1/(1+downsample_ratio) | 2'd0 |
| 23 | R/W | lpf_bypass_ovrd_val | lpf bypass override value<br>used only during pcm_rate override mode:<br>0= enable lpf<br>1= bypass lpf; output rate=input rate | 1'd0 |
| 22:20 | R/W | lpf_bw | lpf bandwidth override selection<br>used only during pcm_rate override mode:<br>if pcm_rate_sel[0]=1, bandwidth will be (KHz):<br>0= 4<br>1= 6<br>2= 8<br>3= 10<br>4= 12<br>5= 14<br>6/7= 15.5<br>if pcm_rate_sel[0]=0, bandwidth will be (KHz):<br>0= 3<br>1= 4.5<br>2= 6<br>3= 7.5<br>4= 9<br>5= 10<br>6/7= 11.5 | 3'd0 |
| 19 | R/W | hbf_bypass_ovrd_val | hbf bypass override value<br>used only during pcm_rate override mode:<br>0= enable hbf<br>1= bypass hbf; output rate=input rate | 1'd0 |
| 18:14 | R/W | hpf2_init_shift | hpf2 initialization shift amount, max=24<br>for values > max, operation is same as max | 5'd24 |
| 13:10 | R/W | hpf1_init_shift | hpf1 initialization shift amount, max=10<br>for values > max, operation is same as max | 4'd10 |
| 9 | R/W | hpf_init_enable | 1= enable hpf initialization | 1'b1 |
| 8 | R/W | hpf_bypass | 1= bypass hpf | 1'b0 |
| 7:4 | R/W | hpf_bw | hpf bandwidth selection<br>when hpf_ord=0, bandwidth will be (Hz):<br>0= 10, 1= 20, 2= 40, 3= 80, 4= 161, 5= 324, 6= 658, 7= 1359<br>when hpf_ord=1, bandwidth will be (Hz):<br>0= 9, 1= 19, 2= 37, 3= 75, 4= 151, 5= 304, 6= 616, 7= 1269<br>when hpf_ord=2, bandwidth will be (Hz): | 4'd5 |

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| | | | | |
|---|---|---|---|---|
| | | | 0= 15, 1= 30, 2= 60, 3= 120, 4= 242, 5= 487, 6= 989, 7= 2036 | |
| 3:2 | R/W | hpf_ord | hpf order<br>0= 1st order<br>1= 2nd order<br>2/3= 3rd order | 2'd0 |
| 1:0 | R/W | cic_ord | cic order<br>0= 3rd order<br>1= 4th order<br>2/3= unused | 2'd0 |

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**Name** FILTER_CONFIG1
**Address** 0x40161008
**Description** Contains register fields associated with filter_config1.
**Reset** 0x000003f8

| Name | RSVD | interp_ph_idx_updt | interp_ph_idx1 | interp_ph_idx0 | interp_add_sample | interp_skp_sample | interp_bypass |
|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20** | **19** | **18 17 16 15 14 13 12 11** | **10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19 | R/W | interp_ph_idx_updt | interpolator phase indices update indicator for HW to capture, auto-clear | 1'b0 |
| 18:11 | R/W | interp_ph_idx1 | index-1 into interpolator coefficient table; for use only when add sample | 8'd0 |
| 10:3 | R/W | interp_ph_idx0 | index-0 into interpolator coefficient table | 8'd127 |
| 2 | R/W | interp_add_sample | add one sample to accomodate cfo frequency drift, auto-clear | 1'b0 |
| 1 | R/W | interp_skp_sample | skip one sample to accomodate cfo frequency drift, auto-clear | 1'b0 |
| 0 | R/W | interp_bypass | 1= bypass interpolator, also resets the fir filter taps | 1'b0 |

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**Name** GAIN_CONTROL_CONFIG
**Address** 0x4016100c
**Description** Contains register fields associated with gain_control_config.
**Reset** 0x00000040

| Name | RSVD | hbf_gain_se | hpf_gain_se | cic_gain_se | code |
|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | 1 | | | 1 | | | 1 | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:13 | R/W | hbf_gain_sel | allow gain-scaling at the output of hbf through bit shifting<br>zero-padding for left shift; rounding, dropping lsbs for right shift.<br>0= no scaling<br>1-4= up scaling; zero-padding +1 to +4 respectively<br>5-7= down scaling; dropping lsbs -1 to -3 respectively | 3'd0 |
| 12:10 | R/W | hpf_gain_sel | allow gain-scaling at the output of hpf through bit shifting<br>zero-padding for left shift; rounding, dropping lsbs for right shift.<br>0= no scaling<br>1-4= up scaling; zero-padding +1 to +4 respectively<br>5-7= down scaling; dropping lsbs -1 to -3 respectively | 3'd0 |
| 9:7 | R/W | cic_gain_sel | allow gain-scaling at the output of cic through bit shifting<br>zero-padding for left shift; rounding, dropping lsbs for right shift.<br>0= no scaling<br>1-4= up scaling; zero-padding +1 to +4 respectively<br>5-7= down scaling; dropping lsbs -1 to -3 respectively | 3'd0 |
| 6:0 | R/W | code | gain code for LUT | 7'd64 |

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**Name** INTERRUPTS
**Address** 0x40161018
**Description** Contains register fields associated with interrupts.
**Reset** 0x00000000

| Name | RSVD | intrpt9 | intrpt8 | intrpt7 | intrpt6 | intrpt5 | intrpt4 | intrpt3 | intrpt2 | intrpt1 | intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R | intrpt9 | error in buffer D; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 8 | R | intrpt8 | error in buffer C; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 7 | R | intrpt7 | ping pong buffer D is locked and needs to be emptied | 1'b0 |
| 6 | R | intrpt6 | ping pong buffer C is locked and needs to be emptied | 1'b0 |
| 5 | R | intrpt5 | error in buffer B; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 4 | R | intrpt4 | error in buffer A; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 3 | R | intrpt3 | ping pong buffer B is locked and needs to be emptied | 1'b0 |
| 2 | R | intrpt2 | ping pong buffer A is locked and needs to be emptied | 1'b0 |
| 1 | R | intrpt1 | hw wrote to ping pong buffer | 1'b0 |
| 0 | R | intrpt0 | interpolator done processing current pcm sample | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x4016101c
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x000003ff

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| Name | RSVD | mask_intrpt9 | mask_intrpt8 | mask_intrpt7 | mask_intrpt6 | mask_intrpt5 | mask_intrpt4 | mask_intrpt3 | mask_intrpt2 | mask_intrpt1 | mask_intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R/W | mask_intrpt9 | allow intrpt9 to propogate to core's single output interrupt | 1'b1 |
| 8 | R/W | mask_intrpt8 | allow intrpt8 to propogate to core's single output interrupt | 1'b1 |
| 7 | R/W | mask_intrpt7 | allow intrpt7 to propogate to core's single output interrupt | 1'b1 |
| 6 | R/W | mask_intrpt6 | allow intrpt6 to propogate to core's single output interrupt | 1'b1 |
| 5 | R/W | mask_intrpt5 | allow intrpt5 to propogate to core's single output interrupt | 1'b1 |
| 4 | R/W | mask_intrpt4 | allow intrpt4 to propogate to core's single output interrupt | 1'b1 |
| 3 | R/W | mask_intrpt3 | allow intrpt3 to propogate to core's single output interrupt | 1'b1 |
| 2 | R/W | mask_intrpt2 | allow intrpt2 to propogate to core's single output interrupt | 1'b1 |
| 1 | R/W | mask_intrpt1 | allow intrpt1 to propogate to core's single output interrupt | 1'b1 |
| 0 | R/W | mask_intrpt0 | allow intrpt0 to propogate to core's single output interrupt | 1'b1 |

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**Name** INTERRUPT_CLEAR
**Address** 0x40161020
**Description** Contains register fields associated with interrupt_clear.
**Reset** 0x00000000

| Name | RSVD | clear_intrpt9 | clear_intrpt8 | clear_intrpt7 | clear_intrpt6 | clear_intrpt5 | clear_intrpt4 | clear_intrpt3 | clear_intrpt2 | clear_intrpt1 | clear_intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R/W | clear_intrpt9 | clear interrupt 9; not self reseting | 1'b0 |
| 8 | R/W | clear_intrpt8 | clear interrupt 8; not self reseting | 1'b0 |
| 7 | R/W | clear_intrpt7 | clear interrupt 7; not self reseting | 1'b0 |
| 6 | R/W | clear_intrpt6 | clear interrupt 6; not self reseting | 1'b0 |
| 5 | R/W | clear_intrpt5 | clear interrupt 5; not self reseting | 1'b0 |
| 4 | R/W | clear_intrpt4 | clear interrupt 4; not self reseting | 1'b0 |
| 3 | R/W | clear_intrpt3 | clear interrupt 3; not self reseting | 1'b0 |
| 2 | R/W | clear_intrpt2 | clear interrupt 2; not self reseting | 1'b0 |
| 1 | R/W | clear_intrpt1 | clear interrupt 1; not self reseting | 1'b0 |
| 0 | R/W | clear_intrpt0 | clear interrupt 0; not self reseting | 1'b0 |

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**Name** BUFFER_ACCESS_MODE
**Address** 0x40161024
**Description** Contains register fields associated with buffer_access_mode.
**Reset** 0x00000000

| Name | RSVD | dma_mode |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | dma_mode | 1= pingpong buffer has fifo-like interface.<br>read address is auto-incremented from base address of buffer A by an internal counter, counting from 0 to 63 and wraps around.<br>buffer full interrupt is cleared by HW at the end of the last read of the buffer.<br>The auto-increment counter is cleared when fifo_mode is 0. | 1'b0 |

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**Name** AUTO_COUNTER
**Address** 0x40161028
**Description** Contains register fields associated with auto_counter.
**Reset** 0x00000000

| Name | RSVD | count |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6** | **5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:0 | R | count | current value of the auto increment counter. | 6'b0 |

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**Name** BUFFER_DEPTH
**Address** 0x4016102c
**Description** Contains register fields associated with buffer_depth.
**Reset** 0x00000010

| Name | RSVD | depth |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | depth | one buffer depth | 5'd16 |

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**Name** CORE_ID
**Address** 0x40161ff4
**Description** Contains register fields associated with core_id.
**Reset** 0x50444d20

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | 'PDM ' in ASCII | 32'h50444d20 |

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**Name** REV_HASH
**Address** 0x40161ff8
**Description** Contains register fields associated with rev_hash.
**Reset** 0x00000000

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | crc32 of this document; don't change value to anything other than 32'h00000000; scripts will fill it out in the verilog | 32'h00000000 |

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**Name** REV_KEY
**Address** 0x40161ffc
**Description** Contains register fields associated with rev_key.
**Reset** 0x52455620

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | REV in ASCII | 32'h52455620 |

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# WRPR2_NONSECURE

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**Name** APB0_CTRL
**Address** 0x40163000
**Description** Contains register fields associated with apb0_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_se | srese | clk_ena |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | l | t | ble |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB1_CTRL
**Address** 0x40163004
**Description** Contains register fields associated with apb1_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB2_CTRL
**Address** 0x40163008
**Description** Contains register fields associated with apb2_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB3_CTRL
**Address** 0x4016300c
**Description** Contains register fields associated with apb3_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB4_CTRL
**Address** 0x40163010
**Description** Contains register fields associated with apb4_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB5_CTRL
**Address** 0x40163014
**Description** Contains register fields associated with apb5_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enabl |
|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | e |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB6_CTRL
**Address** 0x40163018
**Description** Contains register fields associated with apb6_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB7_CTRL
**Address** 0x4016301c
**Description** Contains register fields associated with apb7_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB8_CTRL
**Address** 0x40163020

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**Description** Contains register fields associated with apb8_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB9_CTRL
**Address** 0x40163024
**Description** Contains register fields associated with apb9_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB10_CTRL
**Address** 0x40163028
**Description** Contains register fields associated with apb10_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB11_CTRL
**Address** 0x4016302c
**Description** Contains register fields associated with apb11_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB12_CTRL
**Address** 0x40163030
**Description** Contains register fields associated with apb12_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB13_CTRL
**Address** 0x40163034
**Description** Contains register fields associated with apb13_ctrl.
**Reset** 0x00000002

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| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB14_CTRL
**Address** 0x40163038
**Description** Contains register fields associated with apb14_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB15_CTRL
**Address** 0x4016303c
**Description** Contains register fields associated with apb15_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** CORE_ID
**Address** 0x40163ffc
**Description** Contains register fields associated with core_id.
**Reset** 0x57525052

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | WRPR in ASCII | 32'h57525052 |

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# I2S_NONSECURE

**Name** I2S_CTRL0
**Address** 0x40165000
**Description** Contains register fields associated with i2s_ctrl0.
**Reset** 0x06100900

| Name | RSVD | ws_nedge_st_tx | ws_nedge_st_rx | sck_nedge_sd_tx | sck_nedge_sd_rx | ws_init_tx | ws_init_rx | dma_en | sck_init_cnt | | | | | | | | RSVD | | snk_cnt | | | | src_cnt | | | | mstr_sckws_tx | mstr_sckws_rx | src_snk | | src_snk_en | | sck_nedge_ws_tx | sck_nedge_ws_rx |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 30 | R/W | ws_nedge_st_tx | 1 = tx starts on negative edge of ws | 1'b0 |
| 29 | R/W | ws_nedge_st_rx | 1 = rx starts on negative edge of ws | 1'b0 |
| 28 | R/W | sck_nedge_sd_tx | 1 = drive/latch tx sd on negative edge of sck, 0 = drive/latch tx sd on positive edge of sck | 1'b0 |
| 27 | R/W | sck_nedge_sd_rx | 1 = drive/latch rx sd on negative edge of sck, 0 = drive/latch rx sd on positive edge of sck | 1'b0 |
| 26 | R/W | ws_init_tx | 0 = word select is deasserted at the starting half of justieid mode 1 = word select is asserted at the starting half of justieid mode this applies to right or left justified mode, ws_init should be set to 1 for PCM mode | 1'b1 |
| 25 | R/W | ws_init_rx | 0 = word select is deasserted at the starting half of justieid mode 1 = word select is asserted at the starting half of justieid mode this applies to right or left justified mode, ws_init should be set to 1 for PCM mode | 1'b1 |
| 24 | R/W | dma_en | dma is enabled to access ping pong buffer | 1'b0 |

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|---|---|---|---|---|
| 23:16 | R/W | sck_init_cnt | initial number of sck before going to transmit or receive state | 8'd16 |
| 13:11 | R/W | snk_cnt | number of sinks for i2s | 3'b1 |
| 10:8 | R/W | src_cnt | number of sources for i2s | 3'b1 |
| 7 | R/W | mstr_sckws_tx | 0 = slave sck and ws, 1 = master sck and ws | 1'b0 |
| 6 | R/W | mstr_sckws_rx | 0 = slave sck and ws, 1 = master sck and ws | 1'b0 |
| 5:4 | R/W | src_snk | 0 = pcm (pdm2pcm processed data is transmitted), 1 = ahb (data in memory 0 is transmitted or data is received and stored in memory 0), 2 = ahb (data in memory 1 is transmitted or data is received and stored in memory 1) this is mainly used for rx to tx loop back simulation and eimulation | 2'b0 |
| 3:2 | R/W | src_snk_en | Enable tranceiver tx or rx or both After it's enabled, first data valid assertion initiates tx and first ws deassertion initiates rx<br>0 = i2s disabled<br>1 = source serial data<br>2 = sink serial data<br>3 = source serial data and sink serial data | 2'b0 |
| 1 | R/W | sck_nedge_ws_tx | 1 = drive/latch tx ws on negative edge of sck, 0 = drive/latch tx ws on positive edge of sck | 1'b0 |
| 0 | R/W | sck_nedge_ws_rx | 1 = drive/latch rx ws on negative edge of sck, 0 = drive/latch rx ws on positive edge of sck | 1'b0 |

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**Name** I2S_CTRL1_TX
**Address** 0x40165004
**Description** Contains register fields associated with i2s_ctrl1_tx.
**Reset** 0x00200010

| Name | sck2ws_rt | | | | | | | | | | | | | | | | ck2sck_rt | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | sck2ws_rt | serial clock to word select ratio - number of serial clock per word select period | 16'd32 |
| 15:0 | R/W | ck2sck_rt | clock to serial clock ratio - number of clocks per serial clock period | 16'd16 |

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**Name** I2S_CTRL1_RX
**Address** 0x40165008
**Description** Contains register fields associated with i2s_ctrl1_rx.
**Reset** 0x00200010

| Name | sck2ws_rt | | | | | | | | | | | | | | | | ck2sck_rt | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | sck2ws_rt | serial clock to word select ratio - number of serial clock per word select period | 16'd32 |
| 15:0 | R/W | ck2sck_rt | clock to serial clock ratio - number of clocks per serial clock period | 16'd16 |

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**Name** I2S_CTRL2_TX
**Address** 0x4016500c
**Description** Contains register fields associated with i2s_ctrl2_tx.
**Reset** 0x40001000

| Name | vld_extd_cnt | | | | | | | | RSVD | | pb_cnt | | | | | | RSVD | | sdw | | | | | | sd_offst | | | | wssd_md | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | vld_extd_cnt | cycle count to extend vld to ensure it is latched by slow backplane clock | 8'd64 |
| 21:16 | R/W | pb_cnt | number of zero padding bits - used in right justified mode. | 6'd0 |
| 13:8 | R/W | sdw | width of valid serial data - additioanl bits in a word beyond the width are either padded for tx or discarded for rx valid range is [1:32] | 6'd16 |
| 7:4 | R/W | sd_offst | number of cycles after word select edge to wait before driving serial data - valid range is [0:15] | 4'd0 |
| 3:0 | R/W | wssd_md | word select serial data mode - specify the behavior of word select and serial data 0 = PCM mode, 1 = left justfied I2S, 2 = right justified I2S | 4'd0 |

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**Name** I2S_CTRL2_RX
**Address** 0x40165010
**Description** Contains register fields associated with i2s_ctrl2_rx.
**Reset** 0x40001000

| Name | vld_extd_cnt | | | | | | | | RSVD | | pb_cnt | | | | | | RSVD | | sdw | | | | | | sd_offst | | | | wssd_md | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | vld_extd_cnt | cycle count to extend vld to ensure it is latched by slow backplane clock | 8'd64 |
| 21:16 | R/W | pb_cnt | number of zero padding bits - used in right justified mode. | 6'd0 |
| 13:8 | R/W | sdw | width of valid serial data - additioanl bits in a word beyond the width are either padded for tx or discarded for rx valid range is [1:32] | 6'd16 |

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| | | | | |
|---|---|---|---|---|
| 7:4 | R/W | sd_offst | number of cycles after word select edge to wait before driving serial data - valid range is [0:15] | 4'd0 |
| 3:0 | R/W | wssd_md | word select serial data mode - specify the behavior of word select and serial data 0 = PCM mode, 1 = left justfied I2S, 2 = right justified I2S | 4'd0 |

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**Name** I2S_CTRL3
**Address** 0x40165014
**Description** Contains register fields associated with i2s_ctrl3.
**Reset** 0x80007f04

| Name | intrp_byp | RSVD | intrp_use_msb_smpl | phs_idx_add | phs_idx | RSVD | use_msb_smpl | pck_smpl | swp_smpl |
|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25** | **24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | intrp_byp | bypass interpolator for cfo correction for tx | 1'b1 |
| 24 | R/W | intrp_use_msb_smpl | Use top 24 bit of 32 bit sample for tx interpolator | 1'b0 |
| 23:16 | R/W | phs_idx_add | index into the coefficient table for insert sample | 8'b0 |
| 15:8 | R/W | phs_idx | index into the coefficient table | 8'd127 |
| 2 | R/W | use_msb_smpl | use msb of the samples | 1'b1 |
| 1 | R/W | pck_smpl | pack samples in ping-pong buffer<br>default: samples are stored unpacked<br>applies to 20/24 bit samples only | 1'b0 |
| 0 | R/W | swp_smpl | swap packed samples in pingpong buffer location<br>default: former samples uses lsb bits and later samples uses msb bits<br>applies to 16bit samples only | 1'b0 |

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**Name** I2S_CTRL4
**Address** 0x40165018
**Description** Contains register fields associated with i2s_ctrl4.
**Reset** 0x80007f00

| Name | intrp_ | RSVD | intrp_use_m | phs_idx | phs_i | RSVD | use_msb_ | pck_sm | swp_sm |
|---|---|---|---|---|---|---|---|---|---|

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| | byp | | | | | | | sb_smpl | | | | | _add | | | | | | | dx | | | | | | | | | smpl | pl | pl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | intrp_byp | bypass interpolator for cfo correction for rx | 1'b1 |
| 24 | R/W | intrp_use_msb_smpl | Use top 24 bit of 32 bit sample for rx interpolator | 1'b0 |
| 23:16 | R/W | phs_idx_add | index into the coefficient table for insert sample | 8'b0 |
| 15:8 | R/W | phs_idx | index into the coefficient table | 8'd127 |
| 2 | R/W | use_msb_smpl | use msb of the samples | 1'b0 |
| 1 | R/W | pck_smpl | pack samples in ping-pong buffer<br>default: samples are stored unpacked<br>applies to 20/24 bit samples only | 1'b0 |
| 0 | R/W | swp_smpl | swap packed samples in pingpong buffer location<br>default: former samples uses lsb bits and later samples uses msb bits<br>applies to 16bit samples only | 1'b0 |

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**Name** I2S_CTRL5_TX
**Address** 0x4016501c
**Description** Contains register fields associated with i2s_ctrl5_tx.
**Reset** 0x00000000

| Name | RSVD | frq_drft_corr_en |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | frq_drft_corr_en | 0 = do not enable freq drift correction 1 = enable freq drift correction | 1'b0 |

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**Name** I2S_CTRL5_RX
**Address** 0x40165020
**Description** Contains register fields associated with i2s_ctrl5_rx.
**Reset** 0x00000000

| | frq_d |
|---|---|

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| Name | RSVD | frq_drft_corr_en |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | frq_drft_corr_en | 0 = do not enable freq drift correction 1 = enable freq drift correction | 1'b0 |

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**Name** I2S_INTRP_CTRL
**Address** 0x40165024
**Description** Contains register fields associated with i2s_intrp_ctrl.
**Reset** 0x00000000

| Name | RSVD | rx_updt_phs_idx | tx_updt_phs_idx | rx_add_smpl | rx_skp_smpl | tx_add_smpl | tx_skp_smpl |
|---|---|---|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | rx_updt_phs_idx | latch the phase index for rx interpolator | 1'b0 |
| 4 | R/W | tx_updt_phs_idx | latch the phase index for tx interpolator | 1'b0 |
| 3 | R/W | rx_add_smpl | insert one sample to accomodate frequecy drift | 1'b0 |
| 2 | R/W | rx_skp_smpl | skip one sample to accomodate frequecy drift | 1'b0 |
| 1 | R/W | tx_add_smpl | insert one sample to accomodate frequecy drift | 1'b0 |
| 0 | R/W | tx_skp_smpl | skip one sample to accomodate frequecy drift | 1'b0 |

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**Name** I2S_PP1_RDATA
**Address** 0x40165028
**Description** Contains register fields associated with i2s_pp1_rdata.
**Reset** 0x00000000

| Name | val |
|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:0 | R | val | read data at the top of ping pong set 1 buffer 3 | 32'b0 |

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**Name** I2S_PP0_WDATA
**Address** 0x4016502c
**Description** Contains register fields associated with i2s_pp0_wdata.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | W | val | write data at the bottom of ping pong set 0 buffer 3 | 32'b0 |

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**Name** I2S_PP1_CTRL
**Address** 0x40165030
**Description** Contains register fields associated with i2s_pp1_ctrl.
**Reset** 0x00000008

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | thrshld | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6:0 | R/W | thrshld | number of available words that are empty and can be filled | 7'd8 |

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**Name** I2S_PP0_CTRL
**Address** 0x40165034
**Description** Contains register fields associated with i2s_pp0_ctrl.
**Reset** 0x00000008

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | thrshld | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6:0 | R/W | thrshld | number of available words that are empty and can be filled | 7'd8 |

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**Name** I2S_IRQ0
**Address** 0x40165038
**Description** Contains register fields associated with i2s_irq0.

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**Reset** 0x00000000

| Name | pdm_ufof | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R | pdm_ufof | pdm underflow or overflow | 1'b0 |
| 13 | R | intrp_dout_vld | interpolator data valid | 1'b0 |
| 12 | R | pp_ep_thrshld | ping pong buffer empty threshold hit | 1'b0 |
| 11 | R | pp3_uf | ping pong set 0 buffer 3 under flow | 1'b0 |
| 10 | R | pp2_uf | ping pong set 0 buffer 2 under flow | 1'b0 |
| 9 | R | pp1_uf | ping pong set 0 buffer 1 under flow | 1'b0 |
| 8 | R | pp0_uf | ping pong set 0 buffer 0 under flow | 1'b0 |
| 7 | R | pp3_of | ping pong set 0 buffer 3 over flow | 1'b0 |
| 6 | R | pp2_of | ping pong set 0 buffer 2 over flow | 1'b0 |
| 5 | R | pp1_of | ping pong set 0 buffer 1 over flow | 1'b0 |
| 4 | R | pp0_of | ping pong set 0 buffer 0 over flow | 1'b0 |
| 3 | R | pp3_fl | ping pong set 0 buffer 3 full | 1'b0 |
| 2 | R | pp2_fl | ping pong set 0 buffer 2 full | 1'b0 |
| 1 | R | pp1_fl | ping pong set 0 buffer 1 full | 1'b0 |
| 0 | R | pp0_fl | ping pong set 0 buffer 0 full | 1'b0 |

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**Name** I2S_IRQM0
**Address** 0x4016503c
**Description** Contains register fields associated with i2s_irqm0.
**Reset** 0x00000fff

| Name | pdm_ufof | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | pdm_ufof | pdm underflow or overflow | 1'b0 |
| 13 | R/W | intrp_dout_vld | interpolator data valid mask | 1'b0 |

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|---|---|---|---|---|
| 12 | R/W | pp_ep_thrshld | ping pong buffer empty threshold hit mask | 1'b0 |
| 11 | R/W | pp3_uf | ping pong set 0 buffer 3 under flow mask | 1'b1 |
| 10 | R/W | pp2_uf | ping pong set 0 buffer 2 under flow mask | 1'b1 |
| 9 | R/W | pp1_uf | ping pong set 0 buffer 1 under flow mask | 1'b1 |
| 8 | R/W | pp0_uf | ping pong set 0 buffer 0 under flow mask | 1'b1 |
| 7 | R/W | pp3_of | ping pong set 0 buffer 3 over flow mask | 1'b1 |
| 6 | R/W | pp2_of | ping pong set 0 buffer 2 over flow mask | 1'b1 |
| 5 | R/W | pp1_of | ping pong set 0 buffer 1 over flow mask | 1'b1 |
| 4 | R/W | pp0_of | ping pong set 0 buffer 0 over flow mask | 1'b1 |
| 3 | R/W | pp3_fl | ping pong set 0 buffer 3 full mask | 1'b1 |
| 2 | R/W | pp2_fl | ping pong set 0 buffer 2 full mask | 1'b1 |
| 1 | R/W | pp1_fl | ping pong set 0 buffer 1 full mask | 1'b1 |
| 0 | R/W | pp0_fl | ping pong set 0 buffer 0 full mask | 1'b1 |

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**Name** I2S_IRQS0
**Address** 0x40165040
**Description** Contains register fields associated with i2s_irqs0.
**Reset** 0x00000000

| Name | pdm_ufof | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | pdm_ufof | pdm underflow or overflow | 1'b0 |
| 13 | R/W | intrp_dout_vld | interpolator data valid set | 1'b0 |
| 12 | R/W | pp_ep_thrshld | ping pong buffer empty threshold hit set | 1'b0 |
| 11 | R/W | pp3_uf | ping pong set 0 buffer 3 under flow set | 1'b0 |
| 10 | R/W | pp2_uf | ping pong set 0 buffer 2 under flow set | 1'b0 |
| 9 | R/W | pp1_uf | ping pong set 0 buffer 1 under flow set | 1'b0 |
| 8 | R/W | pp0_uf | ping pong set 0 buffer 0 under flow set | 1'b0 |
| 7 | R/W | pp3_of | ping pong set 0 buffer 3 over flow set | 1'b0 |
| 6 | R/W | pp2_of | ping pong set 0 buffer 2 over flow set | 1'b0 |
| 5 | R/W | pp1_of | ping pong set 0 buffer 1 over flow set | 1'b0 |
| 4 | R/W | pp0_of | ping pong set 0 buffer 0 over flow set | 1'b0 |
| 3 | R/W | pp3_fl | ping pong set 0 buffer 3 full set | 1'b0 |
| 2 | R/W | pp2_fl | ping pong set 0 buffer 2 full set | 1'b0 |
| 1 | R/W | pp1_fl | ping pong set 0 buffer 1 full set | 1'b0 |
| 0 | R/W | pp0_fl | ping pong set 0 buffer 0 full set | 1'b0 |

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**Name** I2S_IRQC0
**Address** 0x40165044

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**Description** Contains register fields associated with i2s_irqc0.
**Reset** 0x00000000

| Name | pdm_ufof | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | pdm_ufof | pdm underflow or overflow | 1'b0 |
| 13 | R/W | intrp_dout_vld | interpolator data valid clear | 1'b0 |
| 12 | R/W | pp_ep_thrshld | ping pong buffer empty threshold hit clear | 1'b0 |
| 11 | R/W | pp3_uf | ping pong set 0 buffer 3 under flow clear | 1'b0 |
| 10 | R/W | pp2_uf | ping pong set 0 buffer 2 under flow clear | 1'b0 |
| 9 | R/W | pp1_uf | ping pong set 0 buffer 1 under flow clear | 1'b0 |
| 8 | R/W | pp0_uf | ping pong set 0 buffer 0 under flow clear | 1'b0 |
| 7 | R/W | pp3_of | ping pong set 0 buffer 3 over flow clear | 1'b0 |
| 6 | R/W | pp2_of | ping pong set 0 buffer 2 over flow clear | 1'b0 |
| 5 | R/W | pp1_of | ping pong set 0 buffer 1 over flow clear | 1'b0 |
| 4 | R/W | pp0_of | ping pong set 0 buffer 0 over flow clear | 1'b0 |
| 3 | R/W | pp3_fl | ping pong set 0 buffer 3 full clear | 1'b0 |
| 2 | R/W | pp2_fl | ping pong set 0 buffer 2 full clear | 1'b0 |
| 1 | R/W | pp1_fl | ping pong set 0 buffer 1 full clear | 1'b0 |
| 0 | R/W | pp0_fl | ping pong set 0 buffer 0 full clear | 1'b0 |

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**Name** I2S_IRQ1
**Address** 0x40165048
**Description** Contains register fields associated with i2s_irq1.
**Reset** 0x00000000

| Name | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13 | R | intrp_dout_vld | interpolator data valid | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 12 | R | pp_ep_thrshld | ping pong buffer empty threshold hit | 1'b0 |
| 11 | R | pp3_uf | ping pong set 1 buffer 3 under flow | 1'b0 |
| 10 | R | pp2_uf | ping pong set 1 buffer 2 under flow | 1'b0 |
| 9 | R | pp1_uf | ping pong set 1 buffer 1 under flow | 1'b0 |
| 8 | R | pp0_uf | ping pong set 1 buffer 0 under flow | 1'b0 |
| 7 | R | pp3_of | ping pong set 1 buffer 3 over flow | 1'b0 |
| 6 | R | pp2_of | ping pong set 1 buffer 2 over flow | 1'b0 |
| 5 | R | pp1_of | ping pong set 1 buffer 1 over flow | 1'b0 |
| 4 | R | pp0_of | ping pong set 1 buffer 0 over flow | 1'b0 |
| 3 | R | pp3_fl | ping pong set 1 buffer 3 full | 1'b0 |
| 2 | R | pp2_fl | ping pong set 1 buffer 2 full | 1'b0 |
| 1 | R | pp1_fl | ping pong set 1 buffer 1 full | 1'b0 |
| 0 | R | pp0_fl | ping pong set 1 buffer 0 full | 1'b0 |

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**Name** I2S_IRQM1
**Address** 0x4016504c
**Description** Contains register fields associated with i2s_irqm1.
**Reset** 0x00000fff

| Name | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13 | R/W | intrp_dout_vld | interpolator data valid mask | 1'b0 |
| 12 | R/W | pp_ep_thrshld | ping pong buffer empty threshold hit mask | 1'b0 |
| 11 | R/W | pp3_uf | ping pong set 1 buffer 3 under flow mask | 1'b1 |
| 10 | R/W | pp2_uf | ping pong set 1 buffer 2 under flow mask | 1'b1 |
| 9 | R/W | pp1_uf | ping pong set 1 buffer 1 under flow mask | 1'b1 |
| 8 | R/W | pp0_uf | ping pong set 1 buffer 0 under flow mask | 1'b1 |
| 7 | R/W | pp3_of | ping pong set 1 buffer 3 over flow mask | 1'b1 |
| 6 | R/W | pp2_of | ping pong set 1 buffer 2 over flow mask | 1'b1 |
| 5 | R/W | pp1_of | ping pong set 1 buffer 1 over flow mask | 1'b1 |
| 4 | R/W | pp0_of | ping pong set 1 buffer 0 over flow mask | 1'b1 |
| 3 | R/W | pp3_fl | ping pong set 1 buffer 3 full mask | 1'b1 |
| 2 | R/W | pp2_fl | ping pong set 1 buffer 2 full mask | 1'b1 |
| 1 | R/W | pp1_fl | ping pong set 1 buffer 1 full mask | 1'b1 |
| 0 | R/W | pp0_fl | ping pong set 1 buffer 0 full mask | 1'b1 |

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**Name** I2S_IRQS1
**Address** 0x40165050
**Description** Contains register fields associated with i2s_irqs1.

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**Reset** 0x00000000

| Name | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13 | R/W | intrp_dout_vld | interpolator data valid set | 1'b0 |
| 12 | R/W | pp_ep_thrshld | ping pong buffer empty threshold hit set | 1'b0 |
| 11 | R/W | pp3_uf | ping pong set 1 buffer 3 under flow set | 1'b0 |
| 10 | R/W | pp2_uf | ping pong set 1 buffer 2 under flow set | 1'b0 |
| 9 | R/W | pp1_uf | ping pong set 1 buffer 1 under flow set | 1'b0 |
| 8 | R/W | pp0_uf | ping pong set 1 buffer 0 under flow set | 1'b0 |
| 7 | R/W | pp3_of | ping pong set 1 buffer 3 over flow set | 1'b0 |
| 6 | R/W | pp2_of | ping pong set 1 buffer 2 over flow set | 1'b0 |
| 5 | R/W | pp1_of | ping pong set 1 buffer 1 over flow set | 1'b0 |
| 4 | R/W | pp0_of | ping pong set 1 buffer 0 over flow set | 1'b0 |
| 3 | R/W | pp3_fl | ping pong set 1 buffer 3 full set | 1'b0 |
| 2 | R/W | pp2_fl | ping pong set 1 buffer 2 full set | 1'b0 |
| 1 | R/W | pp1_fl | ping pong set 1 buffer 1 full set | 1'b0 |
| 0 | R/W | pp0_fl | ping pong set 1 buffer 0 full set | 1'b0 |

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**Name** I2S_IRQC1
**Address** 0x40165054
**Description** Contains register fields associated with i2s_irqc1.
**Reset** 0x00000000

| Name | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13 | R/W | intrp_dout_vld | interpolator data valid clear | 1'b0 |
| 12 | R/W | pp_ep_thrshld | ping pong buffer empty threshold hit clear | 1'b0 |
| 11 | R/W | pp3_uf | ping pong set 1 buffer 3 under flow clear | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 10 | R/W | pp2_uf | ping pong set 1 buffer 2 under flow clear | 1'b0 |
| 9 | R/W | pp1_uf | ping pong set 1 buffer 1 under flow clear | 1'b0 |
| 8 | R/W | pp0_uf | ping pong set 1 buffer 0 under flow clear | 1'b0 |
| 7 | R/W | pp3_of | ping pong set 1 buffer 3 over flow clear | 1'b0 |
| 6 | R/W | pp2_of | ping pong set 1 buffer 2 over flow clear | 1'b0 |
| 5 | R/W | pp1_of | ping pong set 1 buffer 1 over flow clear | 1'b0 |
| 4 | R/W | pp0_of | ping pong set 1 buffer 0 over flow clear | 1'b0 |
| 3 | R/W | pp3_fl | ping pong set 1 buffer 3 full clear | 1'b0 |
| 2 | R/W | pp2_fl | ping pong set 1 buffer 2 full clear | 1'b0 |
| 1 | R/W | pp1_fl | ping pong set 1 buffer 1 full clear | 1'b0 |
| 0 | R/W | pp0_fl | ping pong set 1 buffer 0 full clear | 1'b0 |

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**Name** BUFFER_DEPTH
**Address** 0x40165058
**Description** Contains register fields associated with buffer_depth.
**Reset** 0x00000010

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | depth | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | depth | one buffer depth | 5'd16 |

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**Name** PP_ST
**Address** 0x4016505c
**Description** Contains register fields associated with pp_st.
**Reset** 0x00001010

| Name | RSVD | | | | | | | | | | | | | | | | | | pp1_ep_cnt | | | | | | RSVD | | pp0_ep_cnt | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13:8 | R | pp1_ep_cnt | number of empty slot in ping poing set 1. | 6'd16 |
| 5:0 | R | pp0_ep_cnt | number of empty slot in ping poing set 0. | 6'd16 |

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**Name** I2S_SPARE
**Address** 0x40165060
**Description** Contains register fields associated with i2s_spare.
**Reset** 0x00000000

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| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | val | spare bits | 32'b0 |

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**Name** I2S_DBG
**Address** 0x40165064
**Description** Expose debug signals on mdm_out
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | sel | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:0 | R/W | sel | 0123456 | 6'd0 |

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**Name** ID
**Address** 0x40165ffc
**Description** Contains register fields associated with id.
**Reset** 0x49325320

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | 'I2S' in ASCII | 32'h49325320 |

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# CLKRSTGEN_NONSECURE

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**Name** CLK_BP_CTRL
**Address** 0x40166000
**Description** Contains register fields associated with clk_bp_ctrl.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | cluster | slow_clust | double_clu | pll_div2_clu | pll_clust |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | _sel | | | er_sel | | | ster_sel | ster_sel | er_sel |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9:6 | R/W | cluster_sel | selects which cluster is to drive the backplane clock<br>0 = clk16x<br>1 = slow clocks<br>2 = doubler cluster<br>4 = pll cluster<br>8 = pll div2 cluster<br>always cycle through 0 so that only one bits changes per write | 4'b0 |
| 5:3 | R/W | slow_cluster_sel | selects which clock is output by this cluster<br>0 = 16MHz<br>1 = 8MHz<br>3 = 4MHz<br>2 = 2MHz<br>6 = 1MHz<br>7 = 500kHz<br>notice the gray encoding; only change one bit per write<br>always leave this cluster in 16MHz (0 setting) before leaving it | 3'b0 |
| 2 | R/W | doubler_cluster_sel | selects which clock is output by this cluster<br>0 = 16MHz<br>1 = 32MHz<br>always leave this cluster in 16MHz (0 setting) before leaving it | 1'b0 |
| 1 | R/W | pll_div2_cluster_sel | selects which clock is output by this cluster<br>0 = 16MHz<br>1 = (selected pll frequency)/2<br>always leave this cluster in 16MHz (0 setting) before leaving it | 1'b0 |
| 0 | R/W | pll_cluster_sel | selects which clock is output by this cluster<br>0 = 16MHz<br>1 = selected pll frequency<br>always leave this cluster in 16MHz (0 setting) before leaving it | 1'b0 |

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**Name** CLK_BP_CTRL_STAT
**Address** 0x40166004
**Description** status of clk_bp_ctrl.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cluster_sel_st | | | | slow_cluster_sel_ | | | doubler_cluster_se | pll_div2_cluster_se | pll_cluster_sel_ |

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | l_stat | l_stat | stat |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9:6 | R | cluster_sel_stat | status of cluster_sel | 4'b0 |
| 5:3 | R | slow_cluster_sel_stat | status of slow_cluster_sel | 3'b0 |
| 2 | R | doubler_cluster_sel_stat | status of doubler_cluster_sel | 1'b0 |
| 1 | R | pll_div2_cluster_sel_stat | status of pll_div2_cluster_sel | 1'b0 |
| 0 | R | pll_cluster_sel_stat | status of pll_cluster_sel | 1'b0 |

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**Name** CLK_AUD_CTRL
**Address** 0x40166008
**Description** Contains register fields associated with clk_aud_ctrl.
**Reset** 0x00000000

| Name | RSVD | i2s_clk_enable | i2s_sel | pdm_clk_enable | RSVD | pdm_sel |
|---|---|---|---|---|---|---|
| Bits | 31:10 | 9 | 8:5 | 4 | 3:1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R/W | i2s_clk_enable | enable i2s clock; make frequency selection (i2s_sel) only when disabled | 1'b0 |
| 8:5 | R/W | i2s_sel | selects the i2s clock source<br>0 = 16MHz<br>1 = 32MHz<br>2 = (pll frequency/1)<br>3 = (pll frequency/2)<br>4 = (pll frequency/3)<br>5 = (pll frequency/4)<br>6 = (pll frequency/5)<br>7 = (pll frequency/6)<br>8 = (pll frequency/7)<br>9 = (pll frequency/8) | 4'b0 |
| 4 | R/W | pdm_clk_enable | enable pdm clock; make frequency selection (pdm_sel) only when disabled | 1'b0 |
| 0 | R/W | pdm_sel | selects the pdm clock source<br>0 = i2s clock divided by 1<br>1 = i2s clock divided by 2 | 1'b0 |

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**Name** PLL_CTRL
**Address** 0x4016600c
**Description** Contains register fields associated with pll_ctrl.

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**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | pll_update | pll_enable | pll_freq_sel | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | pll_update | (no description) | 1'b0 |
| 4 | R/W | pll_enable | (no description) | 1'b0 |
| 3:0 | R/W | pll_freq_sel | 0 = 48 MHz<br>1 = 56 MHz / reserved / don't use<br>2 = 64 MHz<br>3 = 72 MHz / reserved / don't use<br>4 = 80 MHz<br>5 = 88 MHz / reserved / don't use<br>6 = 96 MHz | 4'b0 |

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**Name** USER_RESETS
**Address** 0x40166010
**Description** Contains register fields associated with user_resets.
**Reset** 0x00000000

| Name | use_arm_clkdivs | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | mdm_rif | ble_lowp | ble_intf | ble_core |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | use_arm_clkdivs | (no description) | 1'b0 |
| 3 | R/W | mdm_rif | forces a reset into both mdm and rif cores | 1'b0 |
| 2 | R/W | ble_lowp | (no description) | 1'b0 |
| 1 | R/W | ble_intf | (no description) | 1'b0 |
| 0 | R/W | ble_core | (no description) | 1'b0 |

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**Name** USER_CLK_DISABLES
**Address** 0x40166014

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**Description** Contains register fields associated with user_clk_disables.
**Reset** 0x00000020

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | cal | hyper | cpu_lpc | i2s | pdm | mdm_rif |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | cal | radio.calfc_out | 1'b1 |
| 4 | R/W | hyper | (no description) | 1'b0 |
| 3 | R/W | cpu_lpc | (no description) | 1'b0 |
| 2 | R/W | i2s | (no description) | 1'b0 |
| 1 | R/W | pdm | (no description) | 1'b0 |
| 0 | R/W | mdm_rif | turns off 8MHz and 16MHz clocks going to mdm and rif cores | 1'b0 |

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**Name** USER_CLK_GATE_FORCE_ON
**Address** 0x40166018
**Description** Contains register fields associated with user_clk_gate_force_on.
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | ble52_hgclk |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | ble52_hgclk | The hgclk port of ble52 is normally clock gated by the ble52 core itself. If you want this clock to always run (for instance, for debug) then assert this signal. | 1'b1 |

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---

**Name** USER_CLK_GATE_CTRL
**Address** 0x4016601c
**Description** Contains register fields associated with user_clk_gate_ctrl.
**Reset** 0x00001655

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | sticky_clk | ble_hclk | romc_hclk | sha_hclk | dma_hclk | apb2_hclk | apb1_hclk | apb0_hclk |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 14 | R/W | sticky_clk | 0=forced off; 1=forced on; clock enable signal to sticky bits. | 1'b0 |
| 13:12 | R/W | ble_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b1 |
| 11:10 | R/W | romc_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b1 |
| 9:8 | R/W | sha_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b10 |
| 7:6 | R/W | dma_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b1 |
| 5:4 | R/W | apb2_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b1 |
| 3:2 | R/W | apb1_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b1 |
| 1:0 | R/W | apb0_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b1 |

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**Name** CONFIGURATION
**Address** 0x40166020
**Description** Contains register fields associated with configuration.
**Reset** 0x00000000

| Name | RSVD | pll_ready | index |
|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 | 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8 | R | pll_ready | indicates when PLL output is no longer being gated and PLL clock is available | 1'b0 |
| 7:0 | R | index | which fpga configuration? | 8'b0 |

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**Name** CLKSYNC
**Address** 0x40166024
**Description** Contains register fields associated with clksync.
**Reset** 0x00000000

| Name | RSVD | clk16_src | div_val |
|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 | 4 | 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | clk16_src | selects the source of the 16 MHz clock domain<br>0 = crystal<br>1 = divided down pll | 1'b0 |
| 3:0 | R/W | div_val | how much to divide the pll to get 16 MHz? | 4'b0 |

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**Name** CORE_ID
**Address** 0x40166ffc
**Description** core id
**Reset** 0x434c4b52

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | CLKR in ASCII | 32'h434c4b52 |

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# AES_NONSECURE

**Name** ALERT_TEST
**Address** 0x40167000
**Description** control signals for all pwm subcores
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | recov_ctrl_update_err | fatal_fault |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | recov_ctrl_update_err | Write 1 to trigger one alert event of this kind | 1'b0 |
| 0 | R/W | fatal_fault | Write 1 to trigger one alert event of this kind | 1'b0 |

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**Name** KEY_SHARE0_0
**Address** 0x40167004
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

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| Name | key_share0_0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_0 | initial Key Share 0 | 32'b0 |

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**Name** KEY_SHARE0_1
**Address** 0x40167008
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share0_1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_1 | For KEY_SHARE01 | 32'b0 |

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**Name** KEY_SHARE0_2
**Address** 0x4016700c
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| | key_ |
|---|---|

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| Name | key_share0_2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_2 | For KEY_SHARE02 | 32'b0 |

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**Name** KEY_SHARE0_3
**Address** 0x40167010
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share0_3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_3 | For KEY_SHARE03 | 32'b0 |

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**Name** KEY_SHARE0_4
**Address** 0x40167014
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share |
|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | key_share0_4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_4 | For KEY_SHARE04 | 32'b0 |

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**Name** KEY_SHARE0_5
**Address** 0x40167018
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | key_share0_5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_5 | For KEY_SHARE05 | 32'b0 |

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**Name** KEY_SHARE0_6
**Address** 0x4016701c
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | key_share0_6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_6 | For KEY_SHARE06 | 32'b0 |

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**Name** KEY_SHARE0_7
**Address** 0x40167020
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share0_7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_7 | For KEY_SHARE07 | 32'b0 |

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**Name** KEY_SHARE1_0
**Address** 0x40167024
**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share1_0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_0 | initial Key Share 1 | 32'b0 |

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**Name** KEY_SHARE1_1
**Address** 0x40167028
**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share1_1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_1 | For KEY_SHARE11 | 32'b0 |

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**Name** KEY_SHARE1_2
**Address** 0x4016702c
**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share1_2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_2 | For KEY_SHARE12 | 32'b0 |

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**Name** KEY_SHARE1_3
**Address** 0x40167030
**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write

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random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

| Name | key_share1_3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_3 | For KEY_SHARE13 | 32'b0 |

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**Name** KEY_SHARE1_4

**Address** 0x40167034

**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

| Name | key_share1_4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_4 | For KEY_SHARE14 | 32'b0 |

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**Name** KEY_SHARE1_5

**Address** 0x40167038

**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

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| Name | key_share1_5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_5 | For KEY_SHARE15 | 32'b0 |

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---

**Name** KEY_SHARE1_6
**Address** 0x4016703c
**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share1_6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_6 | For KEY_SHARE16 | 32'b0 |

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**Name** KEY_SHARE1_7
**Address** 0x40167040
**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| | key_s |
|---|---|

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| Name | key_share1_7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_7 | For KEY_SHARE17 | 32'b0 |

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**Name** IV_0
**Address** 0x40167044
**Description** Initialization Vector Registers. The initialization vector (IV) or initial counter value must be written to these registers when starting a new message in CBC or CTR mode (see Control Register), respectively. In CBC and CTR modes, the AES unit does not start encryption/decryption with a partially updated IV. Each register has to be written at least once. The order in which the registers are written does not matter. If the AES unit is non-idle, writes to these registers are ignored. Whenever starting a new message, the corresponding IV value must be provided by the processor. Once started, the AES unit automatically updates the contents of these registers. In ECB mode, the IV registers are not used and do not need to be configured. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | iv_0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | iv_0 | Initialization Vector | 32'b0 |

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**Name** IV_1
**Address** 0x40167048
**Description** Initialization Vector Registers. The initialization vector (IV) or initial counter value must be written to these registers when starting a new message in CBC or CTR mode (see Control Register), respectively. In CBC and CTR modes, the AES unit does not start encryption/decryption with a partially updated IV. Each register has to be written at least once. The order in which the registers are written does not matter. If the AES unit is non-idle, writes to these registers are ignored. Whenever starting a new message, the corresponding IV value must be provided by the processor. Once started, the AES unit automatically updates the contents of these registers. In ECB mode, the IV registers are not used and do not need to be configured. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | iv_1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | iv_1 | For IV1 | 32'b0 |

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**Name** IV_2
**Address** 0x4016704c
**Description** Initialization Vector Registers. The initialization vector (IV) or initial counter value must be written to these registers when

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starting a new message in CBC or CTR mode (see Control Register), respectively. In CBC and CTR modes, the AES unit does not start encryption/decryption with a partially updated IV. Each register has to be written at least once. The order in which the registers are written does not matter. If the AES unit is non-idle, writes to these registers are ignored. Whenever starting a new message, the corresponding IV value must be provided by the processor. Once started, the AES unit automatically updates the contents of these registers. In ECB mode, the IV registers are not used and do not need to be configured. Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

| Name | iv_2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | iv_2 | For IV2 | 32'b0 |

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**Name** IV_3

**Address** 0x40167050

**Description** Initialization Vector Registers. The initialization vector (IV) or initial counter value must be written to these registers when starting a new message in CBC or CTR mode (see Control Register), respectively. In CBC and CTR modes, the AES unit does not start encryption/decryption with a partially updated IV. Each register has to be written at least once. The order in which the registers are written does not matter. If the AES unit is non-idle, writes to these registers are ignored. Whenever starting a new message, the corresponding IV value must be provided by the processor. Once started, the AES unit automatically updates the contents of these registers. In ECB mode, the IV registers are not used and do not need to be configured. Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

| Name | iv_3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | iv_3 | For IV3 | 32'b0 |

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**Name** DATA_IN_0

**Address** 0x40167054

**Description** Input Data Registers. If MANUAL_OPERATION=0 (see Control Register), the AES unit automatically starts encryption/decryption after all Input Data registers have been written. Each register has to be written at least once. The order in which the registers are written does not matter. Loaded into the internal State register upon starting encryption/decryption of the next block. After that, the processor can update the Input Data registers (See INPUT_READY field of Status Register). Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

| Name | data_in_0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data_in_0 | Input Data | 32'b0 |

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**Name** DATA_IN_1
**Address** 0x40167058
**Description** Input Data Registers. If MANUAL_OPERATION=0 (see Control Register), the AES unit automatically starts encryption/decryption after all Input Data registers have been written. Each register has to be written at least once. The order in which the registers are written does not matter. Loaded into the internal State register upon starting encryption/decryption of the next block. After that, the processor can update the Input Data registers (See INPUT_READY field of Status Register). Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | data_in_1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data_in_1 | For DATA_IN1 | 32'b0 |

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**Name** DATA_IN_2
**Address** 0x4016705c
**Description** Input Data Registers. If MANUAL_OPERATION=0 (see Control Register), the AES unit automatically starts encryption/decryption after all Input Data registers have been written. Each register has to be written at least once. The order in which the registers are written does not matter. Loaded into the internal State register upon starting encryption/decryption of the next block. After that, the processor can update the Input Data registers (See INPUT_READY field of Status Register). Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | data_in_2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data_in_2 | For DATA_IN2 | 32'b0 |

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**Name** DATA_IN_3
**Address** 0x40167060
**Description** Input Data Registers. If MANUAL_OPERATION=0 (see Control Register), the AES unit automatically starts encryption/decryption after all Input Data registers have been written. Each register has to be written at least once. The order in which the registers are written does not matter. Loaded into the internal State register upon starting encryption/decryption of the next block. After that, the processor can update the Input Data registers (See INPUT_READY field of Status Register). Upon reset, these registers are cleared with pseudo-random data.

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**Reset** 0x00000000

| Name | data_in_3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data_in_3 | For DATA_IN3 | 32'b0 |

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**Name** DATA_OUT_0
**Address** 0x40167064
**Description** Output Data Register. Holds the output data produced by the AES unit during the last encryption/decryption operation. If MANUAL_OPERATION=0 (see Control Register), the AES unit is stalled when the previous output data has not yet been read and is about to be overwritten. Each register has to be read at least once. The order in which the registers are read does not matter. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | data_out_0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data_out_0 | Output Data | 32'h0 |

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**Name** DATA_OUT_1
**Address** 0x40167068
**Description** Output Data Register. Holds the output data produced by the AES unit during the last encryption/decryption operation. If MANUAL_OPERATION=0 (see Control Register), the AES unit is stalled when the previous output data has not yet been read and is about to be overwritten. Each register has to be read at least once. The order in which the registers are read does not matter. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | data_out_1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data_out_1 | For DATA_OUT1 | 32'h0 |

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**Name** DATA_OUT_2
**Address** 0x4016706c
**Description** Output Data Register. Holds the output data produced by the AES unit during the last encryption/decryption operation. If MANUAL_OPERATION=0 (see Control Register), the AES unit is stalled when the previous output data has not yet been read and is about to be overwritten. Each register has to be read at least once. The order in which the registers are read does not matter. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | data_out_2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data_out_2 | For DATA_OUT2 | 32'h0 |

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**Name** DATA_OUT_3
**Address** 0x40167070
**Description** Output Data Register. Holds the output data produced by the AES unit during the last encryption/decryption operation. If MANUAL_OPERATION=0 (see Control Register), the AES unit is stalled when the previous output data has not yet been read and is about to be overwritten. Each register has to be read at least once. The order in which the registers are read does not matter. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | data_out_3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data_out_3 | For DATA_OUT3 | 32'h0 |

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**Name** CTRL_SHADOWED
**Address** 0x40167074
**Description** Control Register. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to this register are ignored. This register is shadowed, meaning two subsequent write operations are required to change its content. If the two write operations try to set a different value, a recoverable alert is triggered (See Status Register). A read operation clears the internal phase tracking: The next write operation is always considered a first write operation of an update sequence. Any

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write operation to this register will clear the status tracking required for automatic mode (See MANUAL_OPERATION field). A write to the Control Register is considered the start of a new message. Hence, software needs to provide new key, IV and input data afterwards.

**Reset** 0x000000c0

| Name | RSVD | | | | | | | | | | | | | | | | | | FORCE_ZERO_MASKS | MANUAL_OPERATION | SIDELOAD | KEY_LEN | | | MODE | | | | | OPERATION |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 12 | R/W | FORCE_ZERO_MASKS | Use masks generated by internal masking PRNG (0) or force all masks constantly to zero (1). Setting all masks to constant zero can be useful when performing SCA. To completely disable the masking, the second key share (KEY_SHARE1_0 - KEY_SHARE1_7) must be zero as well. Only applicable if both the Masking parameter and the SecAllowForcingMasks parameter are set to one. | 1'b0 |
| 11 | R/W | MANUAL_OPERATION | Controls whether the AES unit is operated in normal/automatic mode (0) or fully manual mode (1). In automatic mode (0), the AES unit automatically i) starts to encrypt/decrypt when it receives new input data, and ii) stalls during the last encryption/decryption cycle if the previous output data has not yet been read. This is the most efficient mode to operate in. Note that the corresponding status tracking is automatically cleared upon a write to the Control Register. In manual mode (1), the AES unit i) only starts to encrypt/decrypt after receiving a start trigger (see Trigger Register), and ii) overwrites previous output data irrespective of whether it has been read out or not. This mode is useful if software needs full control over the AES unit. | 1'b0 |
| 10 | R/W | SIDELOAD | Controls whether the AES unit uses the key provided by the key manager via key sideload interface (1) or<br>the key provided by software via Initial Key Registers KEY_SHARE1_0 - KEY_SHARE1_7 (0). | 1'b0 |
| 9:7 | R/W | KEY_LEN | 3-bit one-hot field to select AES key length.<br>Invalid input values, i.e., values with multiple bits set, value 3'b000, and value 3'b010 in case 192-bit keys are not supported (because disabled at compile time) are mapped to AES_256 (3'b100).<br>1 AES_128 3'b001 128-bit key length.<br>2 AES_192 3'b010 192-bit key length.<br>In case support for 192-bit keys has been disabled at compile time, setting this value results in configuring AES_256 (3'b100).<br>4 AES_256 3'b100 256-bit key length.<br>Invalid input values, i.e., values with multiple bits set, value 3'b000, and value 3'b010 in case 192-bit keys are not supported (because disabled at compile time) are mapped to AES_256. | 3'd1 |
| 6:1 | R/W | MODE | 6-bit one-hot field to select AES block cipher mode. Invalid input values, i.e., values with multiple bits set and value 6'b00_0000, are mapped to AES_NONE (6'b10_0000).<br>1 AES_ECB 6'b00_0001: Electronic Codebook (ECB) mode.<br>2 AES_CBC 6'b00_0010: Cipher Block Chaining (CBC) mode.<br>4 AES_CFB 6'b00_0100: Cipher Feedback (CFB) mode.<br>8 AES_OFB 6'b00_1000: Output Feedback (OFB) mode.<br>16 AES_CTR 6'b01_0000: Counter (CTR) mode.<br>32 AES_NONE 6'b10_0000: Invalid input values, i.e., value with multiple bits set and value 6'b00_0000, are mapped to AES_NONE. | 6'h20 |

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| | | | | |
|---|---|---|---|---|
| | | | Other values are reserved. | |
| 0 | R/W | OPERATION | Select encryption(0) or decryption(1) operation of AES unit. | 1'b0 |

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**Name** TRIGGER
**Address** 0x40167078
**Description** Trigger Register. Each bit is individually cleared to zero when executing the corresponding trigger. While executing any of the triggered operations, the AES unit will set the IDLE bit in the Status Register to zero. The processor must check the Status Register before triggering further actions. For example, writes to Initial Key and IV Registers are ignored while the AES unit is busy. Writes to the Input Data Registers are not ignored but the data will be cleared if a KEY_IV_DATA_IN_CLEAR operation is pending.
**Reset** 0x0000000e

| Name | RSVD | PRNG_RESEED | DATA_OUT_CLEAR | KEY_IV_DATA_IN_CLEAR | START |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | PRNG_RESEED | Keep continuing with the current internal state of the internal pseudo-random number generator<br>used for register clearing (0) or perform a reseed of the internal state from the connected entropy source (1). | 1'b1 |
| 2 | R/W | DATA_OUT_CLEAR | Keep current values in Output Data registers (0) or clear those registers with pseudo-random data (1). | 1'b1 |
| 1 | R/W | KEY_IV_DATA_IN_CLEAR | Keep current values in Initial Key, internal Full Key and Decryption Key registers,<br>IV registers and Input Data registers (0) or clear all those registers with pseudo-random data (1). | 1'b1 |
| 0 | R/W | START | Keep AES unit paused (0) or trigger the encryption/decryption of one data block (1).<br>This trigger is ignored if MANUAL_OPERATION=0 (see Control Register). | 1'b0 |

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**Name** STATUS
**Address** 0x4016707c
**Description** Status Register
**Reset** 0x00000000

| | | ALERT | | | | | |
|---|---|---|---|---|---|---|---|

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | ALERT_FATAL_FAULT | _RECOV_CTRL_UPDATE_ERR | INPUT_READY | OUTPUT_VALID | OUTPUT_LOST | STALL | IDLE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6 | R | ALERT_FATAL_FAULT | No fatal fault has occurred inside the AES unit (0).<br>A fatal fault has occurred and the AES unit needs to be reset (1).<br>Examples for fatal faults include i) storage errors in the Control Register,<br>ii) if any internal FSM enters an invalid state,<br>iii) if any sparsely encoded signal takes on an invalid value,<br>iv) errors in the internal round counter,<br>v) escalations triggered by the life cycle controller, and<br>vi) fatal integrity failures on the TL-UL bus. | 1'h0 |
| 5 | R | ALERT_RECOV_CTRL_UPDATE_ERR | An update error has not occurred (0) or has occurred (1) in the shadowed Control Register.<br>AES operation needs to be restarted by re-writing the Control Register. | 1'b0 |
| 4 | R | INPUT_READY | The AES unit is ready (1) or not ready (0) to receive new data input via the DATA_IN registers.<br>If the present values in the DATA_IN registers have not yet been loaded into the module this flag is 0 (not ready). | 1'b0 |
| 3 | R | OUTPUT_VALID | The AES unit has no valid output (0) or has valid output data (1). | 1'b0 |
| 2 | R | OUTPUT_LOST | All previous output data has been fully read by the processor (0)<br>or at least one previous output data block has been lost (1).<br>It has been overwritten by the AES unit before the processor could fully read it.<br>Once set to 1, this flag remains set until AES operation is restarted by re-writing the Control Register.<br>The primary use of this flag is for design verification.<br>This flag is not meaningful if MANUAL_OPERATION=0 (see Control Register). | 1'b0 |
| 1 | R | STALL | The AES unit is not stalled (0) or stalled (1)<br>because there is previous output data that must be read by the processor before the AES unit can overwrite this data.<br>This flag is not meaningful if MANUAL_OPERATION=1 (see Control Register). | 1'b0 |
| 0 | R | IDLE | The AES unit is idle (1) or busy (0).<br>This flag is 0 if one of the following operations is currently running:<br>i) encryption/decryption, ii) register clearing or iii) PRNG reseeding.<br>This flag is also 0 if an encryption/decryption is running but the AES unit is stalled. | 1'b0 |

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**Name** STATUS1
**Address** 0x40167ff4
**Description** Status Register

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**Reset** 0x00000000

| Name | key_len_err | RSVD | sideload_wptr |
|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R | key_len_err | indicates meta key size error. Asserted by HW when sideload_val is written 1 while the meta key size does not match operation key size. cleared by core reset. | 1'b0 |
| 2:0 | R | sideload_wptr | The current value of the sideload key write pointer. This pointer shows the next slot the new word will be stored. | 3'b0 |

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**Name** SIDELOAD_CTRL
**Address** 0x40167ff8
**Description** Contains register fields associated with sideload_ctrl.
**Reset** 0x00000002

| Name | RSVD | sideload_update_en | sideload_val |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | sideload_update_en | write 1 to enable sideload buffer push. | 1'b1 |
| 0 | R/W | sideload_val | write 1 to assert that sideload key is valid. Write 0 to invalidate.<br>Write 1 will also clear the sideload key write pointer.<br>The sideload key is composed of 8 words, which are stored in a fifo like buffer.<br>A write pointer points to the next slot in the buffer in which a new word will be stored. | 1'b0 |

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**Name** ID
**Address** 0x40167ffc

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**Description** Contains register fields associated with id.
**Reset** 0x00414553

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | AES in ASCII | 32'h414553 |

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# AT_DMA_NONSECURE

**Name** OPMODE
**Address** 0x40170000
**Description** Contains register fields associated with opmode.
**Reset** 0x00000000

| Name | go | stop | RSVD | const_tar_addr | dat_inv | const_trans |
|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** **28** **27** **26** **25** **24** **23** **22** **21** **20** **19** **18** **17** **16** **15** **14** **13** **12** **11** **10** **9** **8** **7** **6** **5** **4** **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | go | op code start when rising edge on 'go' is detected. HW clears after. | 1'b0 |
| 30 | R/W | stop | to stop pending transaction . Not self clearing. | 1'b0 |
| 2 | R/W | const_tar_addr | 1= write address remains constant at tar_addr, which is required to be word-aligned in this mode.<br>0= write address increments as defined by the size parameter. | 1'b0 |
| 1 | R/W | dat_inv | 0= non-inverted write ; 1= inverted write (write data are bit-wise inverted) | 1'b0 |
| 0 | R/W | const_trans | op code start when rising edge on 'go' is detected. | 1'b0 |

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**Name** CONST_WDATA
**Address** 0x40170004
**Description** Contains register fields associated with const_wdata.
**Reset** 0x00000000

| Name | const_wda |
|---|---|

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| | t a |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | const_wdata | constant value to be written to target when the operation is in const_trans mode. | 32'b0 |

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**Name** SRC_ADDR
**Address** 0x40170008
**Description** Contains register fields associated with src_addr.
**Reset** 0x00000000

| Name | s r c _ a d d r |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | src_addr | source address | 32'b0 |

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**Name** TAR_ADDR
**Address** 0x4017000c
**Description** Contains register fields associated with tar_addr.
**Reset** 0x00000000

| Name | t a r _ a d d r |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | tar_addr | target address | 32'b0 |

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**Name** SIZE
**Address** 0x40170010
**Description** Contains register fields associated with size.
**Reset** 0x00000000

| Name | R S V D | s i z e |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R/W | size | payload size in bytes. | 24'b0 |

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**Name** SRC_CTRL
**Address** 0x40170014
**Description** Contains register fields associated with src_ctrl.
**Reset** 0x00000004

| Name | RSVD | src_bus_size | src_type |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | src_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | src_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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**Name** TAR_CTRL
**Address** 0x40170018
**Description** Contains register fields associated with tar_ctrl.
**Reset** 0x00000004

| Name | RSVD | tar_bus_size | tar_type |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | tar_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | tar_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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**Name** FIFO_DPTH_ADDR
**Address** 0x4017001c
**Description** Contains register fields associated with fifo_dpth_addr.
**Reset** 0x00000000

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| Name | addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | addr | address of the external fifo depth.<br>Statemachine reads this to determine how many push/pull accesses to do per interrupt. | 32'b0 |

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**Name** FIFO_PORT_SEL
**Address** 0x40170020
**Description** Contains register fields associated with fifo_port_sel.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | tar_port_sel | | RSVD | | | | | | | | | | | | src_port_sel | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 18:16 | R/W | tar_port_sel | selecting the target fifo port;<br>5 = pwd fifo<br>4 = i2s fifo<br>3 = rsvd<br>2 = rsvd<br>1 = uart1 tx<br>0 = uart0 tx | 3'b0 |
| 2:0 | R/W | src_port_sel | selecting the source fifo port;<br>5 = rsvd<br>4 = i2s fifo<br>3 = pdm1 fifo<br>2 = pdm0 fifo<br>1 = uart1 rx<br>0 = uart0 rx | 3'd0 |

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**Name** SPI_PORT_SEL
**Address** 0x40170024
**Description** Contains register fields associated with spi_port_sel.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | spi_sel |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | spi_sel | 1=select spi1 for this channel, 0=select spi0 for this channel. | 1'b0 |

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**Name** ERR_STAT
**Address** 0x40170028
**Description** The status is cleared when the err interrupt is cleared
**Reset** 0x00000000

| Name | RSVD | addr_ovfl | bus_err | addr_ovlp | size_0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | addr_ovfl | 0= no error, 1= error due to size + src_addr/tar_addr > 32'hffff_ffff. | 1'b0 |
| 2 | R | bus_err | 0= no bus error, 1= a bus error occur during the transfer or at the launch of 'go'. | 1'b0 |
| 1 | R | addr_ovlp | 0= no overlap between src and tar addr range, 1= there is an overlap | 1'b0 |
| 0 | R | size_0 | 0= size is not 0 when go is launched, 1= size is 0 when go is launched. | 1'b0 |

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**Name** STATUS
**Address** 0x4017002c
**Description** Contains register fields associated with status.
**Reset** 0x00000000

| Name | RSVD | busy | running |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | busy | fsm is in some non-idle state including waiting for the bus or an interrupt. | 1'b0 |
| 0 | R | running | fsm is in an active state, not waiting for the bus or an interrupt. | 1'b0 |

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**Name** TOTAL_WRITE_REMAINDER
**Address** 0x40170030
**Description** Contains register fields associated with total_write_remainder.
**Reset** 0x00000000

| | | tota |
|---|---|---|

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| Name | RSVD | total_write_remainder |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R | total_write_remainder | number of bytes left to write at the time of stop | 24'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x40170034
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | dma_err | dma_done |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | dma_err | source of interrupt due to an error (chan0). | 1'b0 |
| 0 | R | dma_done | source of interrupt due to completion (chan0). | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x40170038
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000001

| Name | RSVD | interrupt_mask |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_mask | if nth bit set, the nth interrupt is enabled | 2'b01 |

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**Name** SET_INTERRUPT
**Address** 0x4017003c
**Description** Contains register fields associated with set_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_set |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_set | if nth bit set, the nth interrupt is set; HW clears all bits the next cycle. | 2'b00 |

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**Name** RESET_INTERRUPT
**Address** 0x40170040
**Description** Contains register fields associated with reset_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_reset |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_reset | if nth bit set, the nth interrupt is reset; HW clears all bits the next cycle. | 2'b0 |

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**Name** CFG_HNONSEC
**Address** 0x40170044
**Description** Contains register fields associated with cfg_hnonsec.
**Reset** 0x00000001

| Name | RS | cfg_hn |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | V D | | | | | | | | | | | | | | onsec |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | cfg_hnonsec | control HNONSEC output signal of this channel. This register is only Write-able by Secure program. | 1'b1 |

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**Name** CHAN1_OPMODE
**Address** 0x40170048
**Description** Contains register fields associated with chan1_opmode.
**Reset** 0x00000000

| Name | | go | stop | | | | | | | | | | | | | RSVD | | | | | | | | | | | | | | const_tar_addr | dat_inv | const_trans |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | go | op code start when rising edge on 'go' is detected. HW clears after. | 1'b0 |
| 30 | R/W | stop | to stop pending transaction . Not self clearing. | 1'b0 |
| 2 | R/W | const_tar_addr | 1= write address remains constant at tar_addr, which is required to be word-aligned in this mode.<br>0= write address increments as defined by the size parameter. | 1'b0 |
| 1 | R/W | dat_inv | 0= non-inverted write ; 1= inverted write (write data are bit-wise inverted) | 1'b0 |
| 0 | R/W | const_trans | op code start when rising edge on 'go' is detected. | 1'b0 |

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**Name** CHAN1_CONST_WDATA
**Address** 0x4017004c
**Description** Contains register fields associated with chan1_const_wdata.
**Reset** 0x00000000

| Name | const_wdata |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | const_wdata | constant value to be written to target when the operation is in const_trans mode. | 32'b0 |

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**Name** CHAN1_SRC_ADDR
**Address** 0x40170050
**Description** Contains register fields associated with chan1_src_addr.
**Reset** 0x00000000

| Name | src_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | src_addr | source address | 32'b0 |

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**Name** CHAN1_TAR_ADDR
**Address** 0x40170054
**Description** Contains register fields associated with chan1_tar_addr.
**Reset** 0x00000000

| Name | tar_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | tar_addr | target address | 32'b0 |

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**Name** CHAN1_SIZE
**Address** 0x40170058
**Description** Contains register fields associated with chan1_size.
**Reset** 0x00000000

| Name | RSVD | | | | | | | size | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R/W | size | payload size in bytes. | 24'b0 |

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**Name** CHAN1_SRC_CTRL
**Address** 0x4017005c
**Description** Contains register fields associated with chan1_src_ctrl.
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | src_bus_size | | | src_type | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | src_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | src_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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**Name** CHAN1_TAR_CTRL
**Address** 0x40170060
**Description** Contains register fields associated with chan1_tar_ctrl.
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | tar_bus_size | | | tar_type | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | tar_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | tar_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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**Name** CHAN1_FIFO_DPTH_ADDR
**Address** 0x40170064
**Description** Contains register fields associated with chan1_fifo_dpth_addr.
**Reset** 0x00000000

| Name | addr |
|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | addr | address of the external fifo depth.<br>Statemachine reads this to determine how many push/pull accesses to do per interrupt. | 32'b0 |

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**Name** CHAN1_FIFO_PORT_SEL
**Address** 0x40170068
**Description** Contains register fields associated with chan1_fifo_port_sel.
**Reset** 0x00000000

| Name | RSVD | tar_port_sel | RSVD | src_port_sel |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 | 18 17 16 | 15 14 13 12 11 10 9 8 7 6 5 4 3 | 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 18:16 | R/W | tar_port_sel | selecting the target fifo port;<br>5 = pwd fifo<br>4 = i2s fifo<br>3 = rsvd<br>2 = rsvd<br>1 = uart1 tx<br>0 = uart0 tx | 3'b0 |
| 2:0 | R/W | src_port_sel | selecting the source fifo port;<br>5 = rsvd<br>4 = i2s fifo<br>3 = pdm1 fifo<br>2 = pdm0 fifo<br>1 = uart1 rx<br>0 = uart0 rx | 3'd0 |

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**Name** CHAN1_SPI_PORT_SEL
**Address** 0x4017006c
**Description** Contains register fields associated with chan1_spi_port_sel.
**Reset** 0x00000000

| Name | RSVD | spi_sel |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | spi_sel | 1=select spi1 for this channel, 0=select spi0 for this channel. | 1'b0 |

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**Name** CHAN1_ERR_STAT
**Address** 0x40170070
**Description** The status is cleared when the err interrupt is cleared
**Reset** 0x00000000

| Name | RSVD | addr_ovfl | bus_err | addr_ovlp | size_0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | addr_ovfl | 0= no error, 1= error due to size + src_addr/tar_addr > 32'hffff_ffff. | 1'b0 |
| 2 | R | bus_err | 0= no bus error, 1= a bus error occur during the transfer or at the launch of 'go'. | 1'b0 |
| 1 | R | addr_ovlp | 0= no overlap between src and tar addr range, 1= there is an overlap | 1'b0 |
| 0 | R | size_0 | 0= size is not 0 when go is launched, 1= size is 0 when go is launched. | 1'b0 |

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**Name** CHAN1_STATUS
**Address** 0x40170074
**Description** Contains register fields associated with chan1_status.
**Reset** 0x00000000

| Name | RSVD | busy | running |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | busy | fsm is in some non-idle state including waiting for the bus or an interrupt. | 1'b0 |
| 0 | R | running | fsm is in an active state, not waiting for the bus or an interrupt. | 1'b0 |

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**Name** CHAN1_TOTAL_WRITE_REMAINDER
**Address** 0x40170078
**Description** Contains register fields associated with chan1_total_write_remainder.
**Reset** 0x00000000

| | | total_wri |
|---|---|---|

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| Name | RSVD | total_write_remainder |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R | total_write_remainder | number of bytes left to write at the time of stop | 24'b0 |

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**Name** CHAN1_INTERRUPT_STATUS
**Address** 0x4017007c
**Description** Contains register fields associated with chan1_interrupt_status.
**Reset** 0x00000000

| Name | RSVD | dma_err | dma_done |
|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | dma_err | source of interrupt due to an error (chan0). | 1'b0 |
| 0 | R | dma_done | source of interrupt due to completion (chan0). | 1'b0 |

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**Name** CHAN1_INTERRUPT_MASK
**Address** 0x40170080
**Description** Contains register fields associated with chan1_interrupt_mask.
**Reset** 0x00000001

| Name | RSVD | intrpt_mask |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_mask | if nth bit set, the nth interrupt is enabled | 2'b01 |

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**Name** CHAN1_SET_INTERRUPT
**Address** 0x40170084
**Description** Contains register fields associated with chan1_set_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_set |
|---|---|---|
| **Bits** | **31:2** | **1:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_set | if nth bit set, the nth interrupt is set; HW clears all bits the next cycle. | 2'b00 |

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**Name** CHAN1_RESET_INTERRUPT
**Address** 0x40170088
**Description** Contains register fields associated with chan1_reset_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_reset |
|---|---|---|
| **Bits** | **31:2** | **1:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_reset | if nth bit set, the nth interrupt is reset; HW clears all bits the next cycle. | 2'b0 |

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**Name** CHAN1_CFG_HNONSEC
**Address** 0x4017008c
**Description** Contains register fields associated with chan1_cfg_hnonsec.
**Reset** 0x00000001

| Name | RSVD | cfg_hnonse |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | c |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | cfg_hnonsec | control HNONSEC output signal of this channel. This register is only Write-able by Secure program. | 1'b1 |

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**Name** CHAN2_OPMODE
**Address** 0x40170090
**Description** Contains register fields associated with chan2_opmode.
**Reset** 0x00000000

| Name | go | stop | RSVD | const_tar_addr | dat_inv | const_trans |
|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | go | op code start when rising edge on 'go' is detected. HW clears after. | 1'b0 |
| 30 | R/W | stop | to stop pending transaction . Not self clearing. | 1'b0 |
| 2 | R/W | const_tar_addr | 1= write address remains constant at tar_addr, which is required to be word-aligned in this mode.<br>0= write address increments as defined by the size parameter. | 1'b0 |
| 1 | R/W | dat_inv | 0= non-inverted write ; 1= inverted write (write data are bit-wise inverted) | 1'b0 |
| 0 | R/W | const_trans | op code start when rising edge on 'go' is detected. | 1'b0 |

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**Name** CHAN2_CONST_WDATA
**Address** 0x40170094
**Description** Contains register fields associated with chan2_const_wdata.
**Reset** 0x00000000

| Name | const_wdata |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | const_wdata | constant value to be written to target when the operation is in const_trans mode. | 32'b0 |

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**Name** CHAN2_SRC_ADDR
**Address** 0x40170098
**Description** Contains register fields associated with chan2_src_addr.
**Reset** 0x00000000

| Name | src_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | src_addr | source address | 32'b0 |

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**Name** CHAN2_TAR_ADDR
**Address** 0x4017009c
**Description** Contains register fields associated with chan2_tar_addr.
**Reset** 0x00000000

| Name | tar_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | tar_addr | target address | 32'b0 |

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**Name** CHAN2_SIZE
**Address** 0x401700a0
**Description** Contains register fields associated with chan2_size.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | size | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R/W | size | payload size in bytes. | 24'b0 |

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**Name** CHAN2_SRC_CTRL

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**Address** 0x401700a4
**Description** Contains register fields associated with chan2_src_ctrl.
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | src_bus_size | | src_type | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | src_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | src_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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**Name** CHAN2_TAR_CTRL
**Address** 0x401700a8
**Description** Contains register fields associated with chan2_tar_ctrl.
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | tar_bus_size | | tar_type | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | tar_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | tar_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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**Name** CHAN2_FIFO_DPTH_ADDR
**Address** 0x401700ac
**Description** Contains register fields associated with chan2_fifo_dpth_addr.
**Reset** 0x00000000

| Name | addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:0 | R/W | addr | address of the external fifo depth.<br>Statemachine reads this to determine how many push/pull accesses to do per interrupt. | 32'b0 |

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**Name** CHAN2_FIFO_PORT_SEL
**Address** 0x401700b0
**Description** Contains register fields associated with chan2_fifo_port_sel.
**Reset** 0x00000000

| Name | RSVD | tar_port_sel | RSVD | src_port_sel |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19** | **18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 18:16 | R/W | tar_port_sel | selecting the target fifo port;<br>5 = pwd fifo<br>4 = i2s fifo<br>3 = rsvd<br>2 = rsvd<br>1 = uart1 tx<br>0 = uart0 tx | 3'b0 |
| 2:0 | R/W | src_port_sel | selecting the source fifo port;<br>5 = rsvd<br>4 = i2s fifo<br>3 = pdm1 fifo<br>2 = pdm0 fifo<br>1 = uart1 rx<br>0 = uart0 rx | 3'd0 |

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**Name** CHAN2_SPI_PORT_SEL
**Address** 0x401700b4
**Description** Contains register fields associated with chan2_spi_port_sel.
**Reset** 0x00000000

| Name | RSVD | spi_sel |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | spi_sel | 1=select spi1 for this channel, 0=select spi0 for this channel. | 1'b0 |

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**Name** CHAN2_ERR_STAT

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**Address** 0x401700b8
**Description** The status is cleared when the err interrupt is cleared
**Reset** 0x00000000

| Name | RSVD | addr_ovfl | bus_err | addr_ovlp | size_0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | addr_ovfl | 0= no error, 1= error due to size + src_addr/tar_addr > 32'hffff_ffff. | 1'b0 |
| 2 | R | bus_err | 0= no bus error, 1= a bus error occur during the transfer or at the launch of 'go'. | 1'b0 |
| 1 | R | addr_ovlp | 0= no overlap between src and tar addr range, 1= there is an overlap | 1'b0 |
| 0 | R | size_0 | 0= size is not 0 when go is launched, 1= size is 0 when go is launched. | 1'b0 |

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**Name** CHAN2_STATUS
**Address** 0x401700bc
**Description** Contains register fields associated with chan2_status.
**Reset** 0x00000000

| Name | RSVD | busy | running |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | busy | fsm is in some non-idle state including waiting for the bus or an interrupt. | 1'b0 |
| 0 | R | running | fsm is in an active state, not waiting for the bus or an interrupt. | 1'b0 |

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**Name** CHAN2_TOTAL_WRITE_REMAINDER
**Address** 0x401700c0
**Description** Contains register fields associated with chan2_total_write_remainder.
**Reset** 0x00000000

| Name | RSVD | total_write_r |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | e m a i n d e r |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R | total_write_remainder | number of bytes left to write at the time of stop | 24'b0 |

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**Name** CHAN2_INTERRUPT_STATUS
**Address** 0x401700c4
**Description** Contains register fields associated with chan2_interrupt_status.
**Reset** 0x00000000

| Name | R S V D | | | | | | | | | | | | | | | | | | | | | | | | | | | | d m a _ e r r | d m a _ d o n e |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | dma_err | source of interrupt due to an error (chan0). | 1'b0 |
| 0 | R | dma_done | source of interrupt due to completion (chan0). | 1'b0 |

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**Name** CHAN2_INTERRUPT_MASK
**Address** 0x401700c8
**Description** Contains register fields associated with chan2_interrupt_mask.
**Reset** 0x00000001

| Name | R S V D | | | | | | | | | | | | | | | | | | | | | | | | | | | | i n t r p t _ m a s k | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_mask | if nth bit set, the nth interrupt is enabled | 2'b01 |

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**Name** CHAN2_SET_INTERRUPT
**Address** 0x401700cc

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**Description** Contains register fields associated with chan2_set_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_set |
|---|---|---|
| **Bits** | **31:2** | **1:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_set | if nth bit set, the nth interrupt is set; HW clears all bits the next cycle. | 2'b00 |

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**Name** CHAN2_RESET_INTERRUPT
**Address** 0x401700d0
**Description** Contains register fields associated with chan2_reset_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_reset |
|---|---|---|
| **Bits** | **31:2** | **1:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_reset | if nth bit set, the nth interrupt is reset; HW clears all bits the next cycle. | 2'b0 |

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---

**Name** CHAN2_CFG_HNONSEC
**Address** 0x401700d4
**Description** Contains register fields associated with chan2_cfg_hnonsec.
**Reset** 0x00000001

| Name | RSVD | cfg_hnonsec |
|---|---|---|
| **Bits** | **31:1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | cfg_hnonsec | control HNONSEC output signal of this channel. This register is only Write-able by Secure program. | 1'b1 |

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**Name** CHAN3_OPMODE
**Address** 0x401700d8
**Description** Contains register fields associated with chan3_opmode.
**Reset** 0x00000000

| Name | go | stop | RSVD | const_tar_addr | dat_inv | const_trans |
|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29:3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | go | op code start when rising edge on 'go' is detected. HW clears after. | 1'b0 |
| 30 | R/W | stop | to stop pending transaction . Not self clearing. | 1'b0 |
| 2 | R/W | const_tar_addr | 1= write address remains constant at tar_addr, which is required to be word-aligned in this mode.<br>0= write address increments as defined by the size parameter. | 1'b0 |
| 1 | R/W | dat_inv | 0= non-inverted write ; 1= inverted write (write data are bit-wise inverted) | 1'b0 |
| 0 | R/W | const_trans | op code start when rising edge on 'go' is detected. | 1'b0 |

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**Name** CHAN3_CONST_WDATA
**Address** 0x401700dc
**Description** Contains register fields associated with chan3_const_wdata.
**Reset** 0x00000000

| Name | const_wdata |
|---|---|
| **Bits** | **31:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | const_wdata | constant value to be written to target when the operation is in const_trans mode. | 32'b0 |

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**Name** CHAN3_SRC_ADDR
**Address** 0x401700e0
**Description** Contains register fields associated with chan3_src_addr.
**Reset** 0x00000000

| Name | src_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | src_addr | source address | 32'b0 |

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---

**Name** CHAN3_TAR_ADDR
**Address** 0x401700e4
**Description** Contains register fields associated with chan3_tar_addr.
**Reset** 0x00000000

| Name | tar_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | tar_addr | target address | 32'b0 |

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---

**Name** CHAN3_SIZE
**Address** 0x401700e8
**Description** Contains register fields associated with chan3_size.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | size | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R/W | size | payload size in bytes. | 24'b0 |

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---

**Name** CHAN3_SRC_CTRL
**Address** 0x401700ec
**Description** Contains register fields associated with chan3_src_ctrl.
**Reset** 0x00000004

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| Name | RSVD | src_bus_size | src_type |
|---|---|---|---|
| **Bits** | **31:5** | **4:2** | **1:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | src_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | src_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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---

**Name** CHAN3_TAR_CTRL
**Address** 0x401700f0
**Description** Contains register fields associated with chan3_tar_ctrl.
**Reset** 0x00000004

| Name | RSVD | tar_bus_size | tar_type |
|---|---|---|---|
| **Bits** | **31:5** | **4:2** | **1:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | tar_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | tar_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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**Name** CHAN3_FIFO_DPTH_ADDR
**Address** 0x401700f4
**Description** Contains register fields associated with chan3_fifo_dpth_addr.
**Reset** 0x00000000

| Name | addr |
|---|---|
| **Bits** | **31:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | addr | address of the external fifo depth.<br>Statemachine reads this to determine how many push/pull accesses to do per interrupt. | 32'b0 |

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**Name** CHAN3_FIFO_PORT_SEL
**Address** 0x401700f8
**Description** Contains register fields associated with chan3_fifo_port_sel.
**Reset** 0x00000000

| Name | RSVD | tar_port_sel | RSVD | src_port_sel |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19** | **18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 18:16 | R/W | tar_port_sel | selecting the target fifo port;<br>5 = pwd fifo<br>4 = i2s fifo<br>3 = rsvd<br>2 = rsvd<br>1 = uart1 tx<br>0 = uart0 tx | 3'b0 |
| 2:0 | R/W | src_port_sel | selecting the source fifo port;<br>5 = rsvd<br>4 = i2s fifo<br>3 = pdm1 fifo<br>2 = pdm0 fifo<br>1 = uart1 rx<br>0 = uart0 rx | 3'd0 |

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**Name** CHAN3_SPI_PORT_SEL
**Address** 0x401700fc
**Description** Contains register fields associated with chan3_spi_port_sel.
**Reset** 0x00000000

| Name | RSVD | spi_sel |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | spi_sel | 1=select spi1 for this channel, 0=select spi0 for this channel. | 1'b0 |

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**Name** CHAN3_ERR_STAT
**Address** 0x40170100
**Description** The status is cleared when the err interrupt is cleared
**Reset** 0x00000000

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| Name | RSVD | addr_ovfl | bus_err | addr_ovlp | size_0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | addr_ovfl | 0= no error, 1= error due to size + src_addr/tar_addr > 32'hffff_ffff. | 1'b0 |
| 2 | R | bus_err | 0= no bus error, 1= a bus error occur during the transfer or at the launch of 'go'. | 1'b0 |
| 1 | R | addr_ovlp | 0= no overlap between src and tar addr range, 1= there is an overlap | 1'b0 |
| 0 | R | size_0 | 0= size is not 0 when go is launched, 1= size is 0 when go is launched. | 1'b0 |

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**Name** CHAN3_STATUS
**Address** 0x40170104
**Description** Contains register fields associated with chan3_status.
**Reset** 0x00000000

| Name | RSVD | busy | running |
|---|---|---|---|
| **Bits** | **31:2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | busy | fsm is in some non-idle state including waiting for the bus or an interrupt. | 1'b0 |
| 0 | R | running | fsm is in an active state, not waiting for the bus or an interrupt. | 1'b0 |

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**Name** CHAN3_TOTAL_WRITE_REMAINDER
**Address** 0x40170108
**Description** Contains register fields associated with chan3_total_write_remainder.
**Reset** 0x00000000

| Name | RSVD | total_write_remain |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | d<br>e<br>r |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R | total_write_remainder | number of bytes left to write at the time of stop | 24'b0 |

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**Name** CHAN3_INTERRUPT_STATUS
**Address** 0x4017010c
**Description** Contains register fields associated with chan3_interrupt_status.
**Reset** 0x00000000

| Name | R<br>S<br>V<br>D | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | d<br>m<br>a<br>_<br>e<br>r<br>r | d<br>m<br>a<br>_<br>d<br>o<br>n<br>e |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | dma_err | source of interrupt due to an error (chan0). | 1'b0 |
| 0 | R | dma_done | source of interrupt due to completion (chan0). | 1'b0 |

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**Name** CHAN3_INTERRUPT_MASK
**Address** 0x40170110
**Description** Contains register fields associated with chan3_interrupt_mask.
**Reset** 0x00000001

| Name | R<br>S<br>V<br>D | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | i<br>n<br>t<br>r<br>p<br>t<br>_<br>m<br>a<br>s<br>k | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_mask | if nth bit set, the nth interrupt is enabled | 2'b01 |

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**Name** CHAN3_SET_INTERRUPT
**Address** 0x40170114
**Description** Contains register fields associated with chan3_set_interrupt.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | i<br>n |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | intrpt_set |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_set | if nth bit set, the nth interrupt is set; HW clears all bits the next cycle. | 2'b00 |

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**Name** CHAN3_RESET_INTERRUPT
**Address** 0x40170118
**Description** Contains register fields associated with chan3_reset_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_reset |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_reset | if nth bit set, the nth interrupt is reset; HW clears all bits the next cycle. | 2'b0 |

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**Name** CHAN3_CFG_HNONSEC
**Address** 0x4017011c
**Description** Contains register fields associated with chan3_cfg_hnonsec.
**Reset** 0x00000001

| Name | RSVD | cfg_hnonsec |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | cfg_hnonsec | control HNONSEC output signal of this channel. This register is only Write-able by Secure program. | 1'b1 |

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**Name** ERR_INTERRUPT_DLY
**Address** 0x40170120
**Description** err interrupt delayed in cycles
**Reset** 0x0000000a

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | delay_amnt | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | delay_amnt | number of cycles to delay the error interrupt event.<br>Without this delay the interrupt may come up immediately after the go trigger leaving CPU no time to go into WFI. | 8'd10 |

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**Name** ARBITER_CTRL
**Address** 0x40170124
**Description** Contains register fields associated with arbiter_ctrl.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | priority_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | priority_en | 0= no priority 1=lower numbered channel has higher priority when the bus owernership is relinquished. | 1'd0 |

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**Name** ARBITER_GRANT_STAT
**Address** 0x40170128
**Description** Contains register fields associated with arbiter_grant_stat.
**Reset** 0x00000000

| Name | RSVD | chanx_bus_ |
|---|---|---|

| | | | | | | | | | | | | | | | | | | | | | | | | | | | granted | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3:0 | R | chanx_bus_granted | one-hot signal indicating channel with AHB bus ownership. | 4'd0 |

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**Name** DIAGCNTL
**Address** 0x4017012c
**Description** Contains register fields associated with diagcntl.
**Reset** 0x00000000

| Name | RSVD | | | | | | diag0 | | RSVD | | | | | | diag1 | | RSVD | | | | | | diag2 | | RSVD | | | | | | diag3 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25:24 | R/W | diag0 | selects a group of signal to drive dbg[31:24]. | 2'd0 |
| 17:16 | R/W | diag1 | selects a group of signal to drive dbg[23:16]. | 2'd0 |
| 9:8 | R/W | diag2 | selects a group of signal to drive dbg[15:8]. | 2'd0 |
| 1:0 | R/W | diag3 | selects a group of signal to drive dbg[7:0]. | 2'd0 |

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**Name** ID
**Address** 0x40170ffc
**Description** Contains register fields associated with id.
**Reset** 0x444d4120

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | 'DMA ' | 32'h444d4120 |

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# GPIO0_SECURE

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**Name** DATA
**Address** 0x50100000
**Description** Data value[15:0]
**Reset** 0x00000000

| Name | RSV | | | | | | | | | | | | | | | | Dat | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | D | | | | | | | | | | | | | | | a | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | Data | Read Sampled at pin.<br>Write To data output register.<br>Read back value goes through double flip-flop synchronization logic with a delay of two cycles. | 16'b0 |

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**Name** DATAOUT
**Address** 0x50100004
**Description** Data output Register value [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | Dataout | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | Dataout | Read Current value of data output register.<br>Write To data output register | 16'b0 |

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**Name** INENABLE_SET
**Address** 0x50100008
**Description** Input enable set [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | inenable_set | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inenable_set | Write 1 Sets the input enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the input signal is disabled.<br>Read back 1 Indicates the input signal is enabled. | 16'b0 |

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**Name** INENABLE_CLR
**Address** 0x5010000c
**Description** Input enable clear [15:0]
**Reset** 0x00000000

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| Name | RSVD | inenable_clr |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inenable_clr | Write 1 Clears the input enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the input signal is disabled.<br>Read back 1 Indicates the input signal is enabled. | 16'b0 |

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**Name** OUTENSET
**Address** 0x50100010
**Description** Output enable set [15:0]
**Reset** 0x00000000

| Name | RSVD | outenset |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | outenset | Write 1 Sets the output enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the signal direction as input.<br>Read back 1 Indicates the signal direction as output. | 16'b0 |

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**Name** OUTENCLR
**Address** 0x50100014
**Description** Output enable clear [15:0]
**Reset** 0x00000000

| Name | RSVD | outenclr |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

CONFIDENTIAL

| | | | | |
|---|---|---|---|---|
| 15:0 | R/W | outenclr | Write 1 Clears the output enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the signal direction as input.<br>Read back 1 Indicates the signal direction as output. | 16'b0 |

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**Name** ALTFUNCSET
**Address** 0x50100018
**Description** Alternative function set [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | altfuncset | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | altfuncset | Write 1 Sets the ALTFUNC bit.<br>Write 0 No effect.<br>Read back 0 For I/O.<br>Read back 1 For an alternate function. | 16'b0 |

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**Name** ALTFUNCCLR
**Address** 0x5010001c
**Description** Alternative function clear [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | altfunccclr | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | altfuncclr | Write 1 Clears the ALTFUNC bit.<br>Write 0 No effect.<br>Read back 0 For I/O.<br>Read back 1 For an alternate function. | 16'b0 |

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**Name** INTENSET
**Address** 0x50100020
**Description** Interrupt enable set [15:0]
**Reset** 0x00000000

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| Name | RSVD | intenset |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intenset | Write 1 Sets the interrupt enable bit.<br>Write 0 No effect.<br>Read back 0 Interrupt disabled.<br>Read back 1 Interrupt enabled. | 16'b0 |

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**Name** INTENCLR
**Address** 0x50100024
**Description** Interrupt enable clear [15:0]
**Reset** 0x00000000

| Name | RSVD | intenclr |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intenclr | Write 1 Clears the interrupt enable bit.<br>Write 0 No effect.<br>Read back 0 Interrupt disabled.<br>Read back 1 Interrupt enabled. | 16'b0 |

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**Name** INTTYPESET
**Address** 0x50100028
**Description** Interrupt type set [15:0]
**Reset** 0x00000000

| Name | RSVD | inttypeset |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inttypeset | Write 1 Sets the interrupt type bit.<br>Write 0 No effect.<br>Read back 0 For LOW or HIGH level.<br>Read back 1 For falling edge or rising edge. | 16'b0 |

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**Name** INTTYPECLR
**Address** 0x5010002c
**Description** Interrupt type clear [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | inttypeclr | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inttypeclr | Write 1 Clears the interrupt enable bit.<br>Write 0 No effect.<br>Read back 0 For LOW or HIGH level.<br>Read back 1 For falling edge or rising edge. | 16'b0 |

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**Name** INTPOLSET
**Address** 0x50100030
**Description** Polarity-level, edge IRQ configuration [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | intpolset | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intpolset | Write 1 Sets the interrupt polarity bit.<br>Write 0 No effect.<br>Read back 0 For LOW level or falling edge.<br>Read back 1 For HIGH level or rising edge. | 16'b0 |

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**Name** INTPOLCLR
**Address** 0x50100034
**Description** Interrupt type clear [15:0]
**Reset** 0x00000000

| Name | RSV | | | | | | | | | | | | | | | | intpol | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | D | | | | | | | | | | | | | | | | c l r | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intpolclr | Write 1 Clears the interrupt polarity bit.<br>Write 0 No effect.<br>Read back 0 For LOW level or falling edge.<br>Read back 1 For HIGH level or rising edge. | 16'b0 |

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**Name** INTSTATUS_CLEAR
**Address** 0x50100038
**Description** Write one to clear interrupt request:
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | R S V D | | | | | | | | | | | | | | | | i n t s t a t u s _ c l e a r | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intstatus_clear | Write 1 clears the interrupt request.<br>Write 0 No effect.<br>Read back IRQ status Register. | 16'b0 |

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**Name** PULLUP_ENABLE_SET
**Address** 0x5010005c
**Description** pullup enable set [15:0]
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | R S V D | | | | | | | | | | | | | | | | p u l l u p _ e n a b l e _ s e t | | | | | | | | | | | | | |

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pullup_enable_set | Write 1 Sets the pullup enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled up.<br>Read back 1 Indicates the pin is pulled up. | 16'b0 |

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**Name** PULLUP_ENABLE_CLR
**Address** 0x50100060
**Description** pullup enable clear [15:0]
**Reset** 0x00000000

| Name | RSVD | pullup_enable_clr |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pullup_enable_clr | Write 1 Clears the pullup enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled up.<br>Read back 1 Indicates the pin is pulled up. | 16'b0 |

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**Name** PULLDOWN_ENABLE_SET
**Address** 0x50100064
**Description** pulldown enable set [15:0]
**Reset** 0x00000000

| Name | RSVD | pulldown_enable_s |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | et |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pulldown_enable_set | Write 1 Sets the pulldown enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled down.<br>Read back 1 Indicates the pin is pulled down. | 16'b0 |

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**Name** PULLDOWN_ENABLE_CLR
**Address** 0x50100068
**Description** pulldown enable clear [15:0]
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | pulldown_enable_clr | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pulldown_enable_clr | Write 1 Clears the pulldown enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled down.<br>Read back 1 Indicates the pin is pulled down. | 16'b0 |

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**Name** MASKLOWBYTE
**Address** 0x50100400
**Description** Lower eight bits masked access. The register actually spans from 0x100 to 0x1FF
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | masklowbyte | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | masklowbyte | Data for lower byte access, whith bits[9:2] of address value used as enable mask for each bit. | 8'b0 |

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**Name** MASKHIGHBYTE
**Address** 0x50100800
**Description** Higher eight bits masked access. The register actually spans from 0x200 to 0x2FF
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | maskhighbyte | | | | | | | | RSVD | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R/W | maskhighbyte | Data for higher byte access, whith bits[9:2] of address value used as enable mask for each bit. | 8'b0 |

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**Name** PID4
**Address** 0x50100fd0
**Description** PID4
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | block_count | | | | jep106_c_code | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | block_count | Block count | 4'b0 |
| 3:0 | R | jep106_c_code | jep106_c_code | 4'd4 |

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**Name** PID5
**Address** 0x50100fd4
**Description** PID5
**Reset** 0x00000000

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid5 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid5 | Peripheral ID Register 5. | 8'b0 |

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**Name** PID6
**Address** 0x50100fd8
**Description** PID6
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid6 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid6 | Peripheral ID Register 6. | 8'b0 |

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**Name** PID7
**Address** 0x50100fdc
**Description** PID7
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid7 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid7 | Peripheral ID Register 7. | 8'b0 |

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**Name** PID0
**Address** 0x50100fe0
**Description** PID0
**Reset** 0x00000021

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | part_num | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 7:0 | R | part_num | Part number[7:0]. | 8'h21 |

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**Name** PID1
**Address** 0x50100fe4
**Description** PID1
**Reset** 0x000000b8

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | jep106_id_3_0 | | | | part_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | jep106_id_3_0 | jep106_id_3_0 | 4'hb |
| 3:0 | R | part_num | Part number[11:8]. | 4'h8 |

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**Name** PID2
**Address** 0x50100fe8
**Description** PID2
**Reset** 0x0000001b

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | rev | | | | jedec_used | jep106_id_6_4 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | rev | Revision | 4'h1 |
| 3 | R | jedec_used | jedec_used | 1'h1 |
| 2:0 | R | jep106_id_6_4 | jep106_id_6_4 | 3'h3 |

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**Name** PID3
**Address** 0x50100fec
**Description** PID3
**Reset** 0x00000000

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| Name | RSVD | ecorevnum | mod_num |
|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 | 7 6 5 4 | 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | ecorevnum | ECO revision number. | 4'h0 |
| 3:0 | R | mod_num | modification number. | 4'h0 |

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**Name** CID0
**Address** 0x50100ff0
**Description** CID0
**Reset** 0x0000000d

| Name | RSVD | cid0 |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid0 | Component ID Register 0 | 8'h0d |

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**Name** CID1
**Address** 0x50100ff4
**Description** CID1
**Reset** 0x000000f0

| Name | RSVD | cid1 |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid1 | Component ID Register 1 | 8'hf0 |

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**Name** CID2
**Address** 0x50100ff8
**Description** CID2
**Reset** 0x00000005

| Name | RSVD | cid2 |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

CONFIDENTIAL

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid2 | Component ID Register 2 | 8'h05 |

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**Name** CID3
**Address** 0x50100ffc
**Description** CID3
**Reset** 0x000000b1

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid3 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid3 | Component ID Register 3 | 8'hb1 |

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# GPIO1_SECURE

**Name** DATA
**Address** 0x50101000
**Description** Data value[15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | Data | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | Data | Read Sampled at pin.<br>Write To data output register.<br>Read back value goes through double flip-flop synchronization logic with a delay of two cycles. | 16'b0 |

Back to top

**Name** DATAOUT
**Address** 0x50101004
**Description** Data output Register value [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | Dataout | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| | | | Read Current value of data output register. | |

CONFIDENTIAL

| | | | | |
|---|---|---|---|---|
| 15:0 | R/W | Dataout | Write To data output register | 16'b0 |

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**Name** INENABLE_SET
**Address** 0x50101008
**Description** Input enable set [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | inenable_set | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inenable_set | Write 1 Sets the input enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the input signal is disabled.<br>Read back 1 Indicates the input signal is enabled. | 16'b0 |

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**Name** INENABLE_CLR
**Address** 0x5010100c
**Description** Input enable clear [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | inenable_clr | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inenable_clr | Write 1 Clears the input enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the input signal is disabled.<br>Read back 1 Indicates the input signal is enabled. | 16'b0 |

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**Name** OUTENSET
**Address** 0x50101010
**Description** Output enable set [15:0]
**Reset** 0x00000000

CONFIDENTIAL

| Name | RSVD | outenset |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | outenset | Write 1 Sets the output enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the signal direction as input.<br>Read back 1 Indicates the signal direction as output. | 16'b0 |

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**Name** OUTENCLR
**Address** 0x50101014
**Description** Output enable clear [15:0]
**Reset** 0x00000000

| Name | RSVD | outenclr |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | outenclr | Write 1 Clears the output enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the signal direction as input.<br>Read back 1 Indicates the signal direction as output. | 16'b0 |

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**Name** ALTFUNCSET
**Address** 0x50101018
**Description** Alternative function set [15:0]
**Reset** 0x00000000

| Name | RSVD | altfuncset |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | altfuncset | Write 1 Sets the ALTFUNC bit.<br>Write 0 No effect. | 16'b0 |

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| | | | | |
|---|---|---|---|---|
| | | | Read back 0 For I/O.<br>Read back 1 For an alternate function. | |

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**Name** ALTFUNCCLR
**Address** 0x5010101c
**Description** Alternative function clear [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | altfuncclr | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | altfuncclr | Write 1 Clears the ALTFUNC bit.<br>Write 0 No effect.<br>Read back 0 For I/O.<br>Read back 1 For an alternate function. | 16'b0 |

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**Name** INTENSET
**Address** 0x50101020
**Description** Interrupt enable set [15:0]
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | intenset | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intenset | Write 1 Sets the interrupt enable bit.<br>Write 0 No effect.<br>Read back 0 Interrupt disabled.<br>Read back 1 Interrupt enabled. | 16'b0 |

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**Name** INTENCLR
**Address** 0x50101024
**Description** Interrupt enable clear [15:0]
**Reset** 0x00000000

| | RS | | | | | | | | | | | | | | | inte | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

CONFIDENTIAL

| Name | RSVD | intenclr |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intenclr | Write 1 Clears the interrupt enable bit.<br>Write 0 No effect.<br>Read back 0 Interrupt disabled.<br>Read back 1 Interrupt enabled. | 16'b0 |

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**Name** INTTYPESET
**Address** 0x50101028
**Description** Interrupt type set [15:0]
**Reset** 0x00000000

| Name | RSVD | inttypeset |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inttypeset | Write 1 Sets the interrupt type bit.<br>Write 0 No effect.<br>Read back 0 For LOW or HIGH level.<br>Read back 1 For falling edge or rising edge. | 16'b0 |

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**Name** INTTYPECLR
**Address** 0x5010102c
**Description** Interrupt type clear [15:0]
**Reset** 0x00000000

| Name | RSVD | inttypeclr |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | inttypeclr | Write 1 Clears the interrupt enable bit.<br>Write 0 No effect.<br>Read back 0 For LOW or HIGH level.<br>Read back 1 For falling edge or rising edge. | 16'b0 |

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**Name** INTPOLSET
**Address** 0x50101030
**Description** Polarity-level, edge IRQ configuration [15:0]
**Reset** 0x00000000

| Name | RSVD | intpolset |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intpolset | Write 1 Sets the interrupt polarity bit.<br>Write 0 No effect.<br>Read back 0 For LOW level or falling edge.<br>Read back 1 For HIGH level or rising edge. | 16'b0 |

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**Name** INTPOLCLR
**Address** 0x50101034
**Description** Interrupt type clear [15:0]
**Reset** 0x00000000

| Name | RSVD | intpolclr |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intpolclr | Write 1 Clears the interrupt polarity bit.<br>Write 0 No effect.<br>Read back 0 For LOW level or falling edge.<br>Read back 1 For HIGH level or rising edge. | 16'b0 |

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**Name** INTSTATUS_CLEAR
**Address** 0x50101038
**Description** Write one to clear interrupt request:
**Reset** 0x00000000

| Name | RS | intstatu |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | V D | | | | | | | | | | | | s _ c l e a r | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | intstatus_clear | Write 1 clears the interrupt request.<br>Write 0 No effect.<br>Read back IRQ status Register. | 16'b0 |

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**Name** PULLUP_ENABLE_SET
**Address** 0x5010105c
**Description** pullup enable set [15:0]
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | | | | | | | | | | | | | | | | | R S V D | | | | | | | | | | | | p u l l u p _ e n a b l e _ s e t | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pullup_enable_set | Write 1 Sets the pullup enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled up.<br>Read back 1 Indicates the pin is pulled up. | 16'b0 |

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**Name** PULLUP_ENABLE_CLR
**Address** 0x50101060
**Description** pullup enable clear [15:0]
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | | | | | | | | | | | | | | | | | R S V D | | | | | | | | | | | | p u l l u p _ e n a b | |

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | l<br>e<br>_<br>c<br>l<br>r |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pullup_enable_clr | Write 1 Clears the pullup enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled up.<br>Read back 1 Indicates the pin is pulled up. | 16'b0 |

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**Name** PULLDOWN_ENABLE_SET
**Address** 0x50101064
**Description** pulldown enable set [15:0]
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | | | | | | | | | | | | | | | | | p<br>u<br>l<br>l<br>d<br>o<br>w<br>n<br>_<br>e<br>n<br>a<br>b<br>l<br>e<br>_<br>s<br>e<br>t | | | | | | | | | | | | | | | | |
| | R<br>S<br>V<br>D | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pulldown_enable_set | Write 1 Sets the pulldown enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled down.<br>Read back 1 Indicates the pin is pulled down. | 16'b0 |

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**Name** PULLDOWN_ENABLE_CLR
**Address** 0x50101068
**Description** pulldown enable clear [15:0]
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | p<br>u<br>l<br>l<br>d<br>o<br>w<br>n<br>_ | | | | | | | | | | | | | | |
| | R<br>S | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

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| Name | VD | | | | | | | | | | | | | | | | enable_clr | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | pulldown_enable_clr | Write 1 Clears the pulldown enable bit.<br>Write 0 No effect.<br>Read back 0 Indicates the pin is not pulled down.<br>Read back 1 Indicates the pin is pulled down. | 16'b0 |

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**Name** MASKLOWBYTE
**Address** 0x50101400
**Description** Lower eight bits masked access. The register actually spans from 0x100 to 0x1FF
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | masklowbyte | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | masklowbyte | Data for lower byte access, whith bits[9:2] of address value used as enable mask for each bit. | 8'b0 |

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**Name** MASKHIGHBYTE
**Address** 0x50101800
**Description** Higher eight bits masked access. The register actually spans from 0x200 to 0x2FF
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | maskhighbyte | | | | | | | | RSVD | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

CONFIDENTIAL

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R/W | maskhighbyte | Data for higher byte access, whith bits[9:2] of address value used as enable mask for each bit. | 8'b0 |

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**Name** PID4
**Address** 0x50101fd0
**Description** PID4
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | block_count | | | | jep106_c_code | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | block_count | Block count | 4'b0 |
| 3:0 | R | jep106_c_code | jep106_c_code | 4'd4 |

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**Name** PID5
**Address** 0x50101fd4
**Description** PID5
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | pid5 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid5 | Peripheral ID Register 5. | 8'b0 |

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**Name** PID6
**Address** 0x50101fd8
**Description** PID6
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | pid6 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| | | | | |

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| | | | | |
|---|---|---|---|---|
| 7:0 | R | pid6 | Peripheral ID Register 6. | 8'b0 |

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**Name** PID7
**Address** 0x50101fdc
**Description** PID7
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid7 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid7 | Peripheral ID Register 7. | 8'b0 |

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**Name** PID0
**Address** 0x50101fe0
**Description** PID0
**Reset** 0x00000021

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | part_num | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | part_num | Part number[7:0]. | 8'h21 |

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**Name** PID1
**Address** 0x50101fe4
**Description** PID1
**Reset** 0x000000b8

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | jep106_id_3_0 | | | | part_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 7:4 | R | jep106_id_3_0 | jep106_id_3_0 | 4'hb |
| 3:0 | R | part_num | Part number[11:8]. | 4'h8 |

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**Name** PID2
**Address** 0x50101fe8
**Description** PID2
**Reset** 0x0000001b

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | rev | | | | jedec_used | jep106_id_6_4 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | rev | Revision | 4'h1 |
| 3 | R | jedec_used | jedec_used | 1'h1 |
| 2:0 | R | jep106_id_6_4 | jep106_id_6_4 | 3'h3 |

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**Name** PID3
**Address** 0x50101fec
**Description** PID3
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | ecorevnum | | | | mod_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | ecorevnum | ECO revision number. | 4'h0 |
| 3:0 | R | mod_num | modification number. | 4'h0 |

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**Name** CID0
**Address** 0x50101ff0
**Description** CID0
**Reset** 0x0000000d

| | RS | ci |
|---|---|---|

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| Name | RSVD | cid0 |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid0 | Component ID Register 0 | 8'h0d |

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**Name** CID1
**Address** 0x50101ff4
**Description** CID1
**Reset** 0x000000f0

| Name | RSVD | cid1 |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid1 | Component ID Register 1 | 8'hf0 |

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**Name** CID2
**Address** 0x50101ff8
**Description** CID2
**Reset** 0x00000005

| Name | RSVD | cid2 |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid2 | Component ID Register 2 | 8'h05 |

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**Name** CID3
**Address** 0x50101ffc
**Description** CID3
**Reset** 0x000000b1

| Name | RSVD | cid3 |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid3 | Component ID Register 3 | 8'hb1 |

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# SHA2_SECURE

**Name** CONTROL
**Address** 0x50103000
**Description** Contains register fields associated with control.
**Reset** 0x00000000

| Name | RSVD | digest_swizzle | byte_swizzle | hmac_en | sha_en |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | digest_swizzle | digest word's byte swizzle | 1'b0 |
| 2 | R/W | byte_swizzle | message word byte swizzle | 1'b0 |
| 1 | R/W | hmac_en | HMAC datapath enable. If this bit is 1, HMAC operates when hash_start toggles | 1'b0 |
| 0 | R/W | sha_en | SHA256 enable. If 0, SHA engine won't initiate compression,<br>this is used to stop operation of the SHA engine until configuration has been done.<br>When the SHA engine is disabled the digest is cleared. | 1'b0 |

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**Name** SIDELOAD_CTRL
**Address** 0x50103004
**Description** Contains register fields associated with sideload_ctrl.
**Reset** 0x00000000

| Name | RSVD | sideload_update_en | sideload_ptr_clr |
|---|---|---|---|
| **Bits** | **31:2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | sideload_update_en | write 1 to enable sideload buffer push. | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 0 | R/W | sideload_ptr_clr | write 1 to clear the write pointer. Not Self-clearing. | 1'b0 |

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**Name** CMD
**Address** 0x50103008
**Description** Contains register fields associated with cmd.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | hash_process | hash_start |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | hash_process | If writes 1 into this field, SHA256 or HMAC calculates the digest or signing based on currently received message. | 1'b0 |
| 0 | R/W | hash_start | If writes 1 into this field, SHA256 or HMAC begins its operation. CPU should configure relative information first, such as message_length, secret_key. | 1'b0 |

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**Name** MESSAGE_LENGTH_LO
**Address** 0x5010300c
**Description** Contains register fields associated with message_length_lo.
**Reset** 0x00000000

| Name | length_lo | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | length_lo | Received Message Length calculated by the HMAC in bits [31:0] Message is byte granularity. lower 3bits [2:0] are ignored. | 32'b0 |

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**Name** MESSAGE_LENGTH_HI
**Address** 0x50103010
**Description** Contains register fields associated with message_length_hi.
**Reset** 0x00000000

| | len |
|---|---|

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| Name | length_hi | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | length_hi | Received Message Length calculated by the HMAC in bits [63:32]<br>Message is byte granularity. lower 3bits [2:0] are ignored. | 32'b0 |

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**Name** WIPE_V
**Address** 0x50103014
**Description** Contains register fields associated with wipe_v.
**Reset** 0x00000000

| Name | wipe_v | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | wipe_v | a write to this register triggers a wipe_secret action. | 32'b0 |

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**Name** FIFO_PUSH
**Address** 0x50103018
**Description** Contains register fields associated with fifo_push.
**Reset** 0x00000000

| Name | wdata | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | wdata | Message FIFO input port. Byte writes are supported | 32'b0 |

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**Name** DIGEST0
**Address** 0x5010301c
**Description** Contains register fields associated with digest0.
**Reset** 0x00000000

| Name | digest0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest0 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST1
**Address** 0x50103020
**Description** Contains register fields associated with digest1.
**Reset** 0x00000000

| Name | digest1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest1 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST2
**Address** 0x50103024
**Description** Contains register fields associated with digest2.
**Reset** 0x00000000

| Name | digest2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest2 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST3
**Address** 0x50103028
**Description** Contains register fields associated with digest3.
**Reset** 0x00000000

| Name | digest3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | t<br>3 |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest3 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST4
**Address** 0x5010302c
**Description** Contains register fields associated with digest4.
**Reset** 0x00000000

| Name | d<br>i<br>g<br>e<br>s<br>t<br>4 |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest4 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST5
**Address** 0x50103030
**Description** Contains register fields associated with digest5.
**Reset** 0x00000000

| Name | d<br>i<br>g<br>e<br>s<br>t<br>5 |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest5 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST6
**Address** 0x50103034
**Description** Contains register fields associated with digest6.
**Reset** 0x00000000

| | d<br>i<br>g |
|---|---|

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| Name | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | e<br>s<br>t<br>6 |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest6 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** DIGEST7
**Address** 0x50103038
**Description** Contains register fields associated with digest7.
**Reset** 0x00000000

| Name | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | d<br>i<br>g<br>e<br>s<br>t<br>7 | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | digest7 | Digest output. If HMAC is disabled, the register shows result of SHA256 Order of the digest is: digest[255:0] = {DIGEST0, DIGEST1, DIGEST2, ... , DIGEST7}; | 32'b0 |

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**Name** KEY_0
**Address** 0x5010303c
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | k<br>e<br>y<br>_<br>0 | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_0 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_1
**Address** 0x50103040
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | k<br>e<br>y | | |

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | _1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_1 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_2
**Address** 0x50103044
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | key_2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_2 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_3
**Address** 0x50103048
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | key_3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_3 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_4
**Address** 0x5010304c
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | key_4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_4 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_5
**Address** 0x50103050
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | key_5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_5 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_6
**Address** 0x50103054
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | key_6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_6 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** KEY_7
**Address** 0x50103058
**Description** SHA256 assumes secret key is hashed 256bit key. Order of the secret key is: key[255:0] = {KEY0, KEY1, KEY2, ... , KEY7};
**Reset** 0x00000000

| Name | key_7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_7 | 32-bit chunk of 256-bit Secret Key. | 32'b0 |

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**Name** STATUS
**Address** 0x5010305c
**Description** Contains register fields associated with status.

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**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | fifo_depth | | | | | reserved | | fifo_full | fifo_empty |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8:4 | R | fifo_depth | FIFO entry count | 5'b0 |
| 3:2 | R | reserved | (no description) | 2'b0 |
| 1 | R | fifo_full | FIFO full. Data written to the FIFO whilst it is full will cause back-pressure on the interconnect | 1'b0 |
| 0 | R | fifo_empty | FIFO empty | 1'b1 |

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**Name** INTERRUPT_STATUS
**Address** 0x50103060
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | hmac_err | pop_when_empty | push_when_full | hmac_done | chunk_done | fifo_full | fifo_empty |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6 | R | hmac_err | HMAC error occurred. ERR_CODE register shows which error occurred | 1'b0 |
| 5 | R | pop_when_empty | pop when Message FIFO is full condition | 1'b0 |
| 4 | R | push_when_full | push when Message FIFO empty condition | 1'b0 |
| 3 | R | hmac_done | HMAC-256 completes a message with key | 1'b0 |
| 2 | R | chunk_done | one chunk of message done. (512 bits of message and padded bits) | 1'b0 |
| 1 | R | fifo_full | Message FIFO full condition | 1'b0 |
| 0 | R | fifo_empty | Message FIFO empty condition | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x50103064
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | p | p | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | hmac_err | pop_when_empty | push_when_full | hmac_done | chunk_done | fifo_full | fifo_empty |
|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6 | R/W | hmac_err | interrupt mask bit | 1'b0 |
| 5 | R/W | pop_when_empty | interrupt mask bit | 1'b0 |
| 4 | R/W | push_when_full | interrupt mask bit | 1'b0 |
| 3 | R/W | hmac_done | interrupt mask bit | 1'b0 |
| 2 | R/W | chunk_done | interrupt mask bit | 1'b0 |
| 1 | R/W | fifo_full | interrupt mask bit | 1'b0 |
| 0 | R/W | fifo_empty | interrupt mask bit | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x50103068
**Description** Contains register fields associated with set_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_set |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7** | **6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6:0 | R/W | intrpt_set | if nth bit set, the nth interrupt is set; | 7'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x5010306c
**Description** Contains register fields associated with reset_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_re |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | s<br>e<br>t | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6:0 | R/W | intrpt_reset | if nth bit set, the nth interrupt is reset; | 7'b0 |

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**Name** ERR_CODE
**Address** 0x50103070
**Description** Contains register fields associated with err_code.
**Reset** 0x00000000

| Name | e<br>r<br>r<br>_<br>c<br>o<br>d<br>e | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | err_code | NoError = 32'h 0000_0000,<br>SwPushMsgWhenShaDisabled = 32'h 0000_0001,<br>SwHashStartWhenShaDisabled = 32'h 0000_0002,<br>SwUpdateSecretKeyInProcess = 32'h 0000_0003,<br>SwHashStartWhenActive = 32'h 0000_0004,<br>SwPushMsgWhenDisallowed = 32'h 0000_0005 | 32'b0 |

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**Name** ID
**Address** 0x50103ffc
**Description** Contains register fields associated with id.
**Reset** 0x53484132

| Name | i<br>d | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | 'SHA2' | 32'h53484132 |

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# WRPR0_SECURE

**Name** APB0_CTRL
**Address** 0x50143000
**Description** Contains register fields associated with apb0_ctrl.
**Reset** 0x00000002

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | c | s | c<br>l<br>k |

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| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB1_CTRL
**Address** 0x50143004
**Description** Contains register fields associated with apb1_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB2_CTRL
**Address** 0x50143008
**Description** Contains register fields associated with apb2_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB3_CTRL
**Address** 0x5014300c
**Description** Contains register fields associated with apb3_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB4_CTRL
**Address** 0x50143010
**Description** Contains register fields associated with apb4_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB5_CTRL
**Address** 0x50143014
**Description** Contains register fields associated with apb5_ctrl.
**Reset** 0x00000002

| | RS | clk | sre | clk_e |
|---|---|---|---|---|

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| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB6_CTRL
**Address** 0x50143018
**Description** Contains register fields associated with apb6_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB7_CTRL
**Address** 0x5014301c
**Description** Contains register fields associated with apb7_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB8_CTRL
**Address** 0x50143020
**Description** Contains register fields associated with apb8_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB9_CTRL
**Address** 0x50143024
**Description** Contains register fields associated with apb9_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB10_CTRL
**Address** 0x50143028
**Description** Contains register fields associated with apb10_ctrl.
**Reset** 0x00000002

| Name | RSV | clk_ | sres | clk_en |
|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | D | | | | | | | | | | | | | | | | | | | | | | | | | | | s e l | e t | a b l e |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB11_CTRL
**Address** 0x5014302c
**Description** Contains register fields associated with apb11_ctrl.
**Reset** 0x00000002

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | R S V D | | | | | | | | | | | | | | | | | | | | | | | | | | | c l k _ s e l | s r e s e t | c l k _ e n a b l e |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB12_CTRL
**Address** 0x50143030
**Description** Contains register fields associated with apb12_ctrl.
**Reset** 0x00000002

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | R S V D | | | | | | | | | | | | | | | | | | | | | | | | | | | c l k _ s e l | s r e s e t | c l k _ e n a b l e |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB13_CTRL
**Address** 0x50143034
**Description** Contains register fields associated with apb13_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB14_CTRL
**Address** 0x50143038
**Description** Contains register fields associated with apb14_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB15_CTRL
**Address** 0x5014303c
**Description** Contains register fields associated with apb15_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_se | sreset | clk_enab |
|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | l | l e |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** CLK_HPC_PIN_OUT_CTRL
**Address** 0x50143040
**Description** Contains register fields associated with clk_hpc_pin_out_ctrl.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_hpc_sel | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | clk_hpc_sel | (no description) | 2'b0 |

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**Name** CLK_MPC_PIN_OUT_CTRL
**Address** 0x50143044
**Description** Contains register fields associated with clk_mpc_pin_out_ctrl.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_mpc_sel | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | clk_mpc_sel | (no description) | 2'b0 |

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**Name** CLK_LPC_PIN_OUT_CTRL
**Address** 0x50143048
**Description** Contains register fields associated with clk_lpc_pin_out_ctrl.

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**Reset** 0x00000000

| Name | RSVD | clk_lpc_sel |
|---|---|---|
| **Bits** | **31:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | clk_lpc_sel | (no description) | 1'b0 |

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**Name** DBG_CTRL
**Address** 0x5014304c
**Description** Contains register fields associated with dbg_ctrl.
**Reset** 0x0006c000

| Name | RSVD | dbg_cg | dbg_skip_6x6_holes | trace_is_ddr | trigger_soc_rst | sse200_mstexp0_clk_enable | sse200_mstexp1_clk_enable | sse200_exp_nmi | sse200_expclkreq_in | sse200_sysfclkqactive | sse200_exp_mpc_g_sel | sse200_dbgen_sel | sse200_nideen_sel | sse200_certdisableext | em_block_addr | em_block_wdata | dbg_enable | dbg_shft | dbg_sel |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6:5** | **4:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 22 | R/W | dbg_cg | (no description) | 1'b0 |
| 21 | R/W | dbg_skip_6x6_holes | (no description) | 1'b0 |
| 20 | R/W | trace_is_ddr | (no description) | 1'b0 |
| 19 | R/W | trigger_soc_rst | (no description) | 1'b0 |
| 18 | R/W | sse200_mstexp0_clk_enable | (no description) | 1'b1 |
| 17 | R/W | sse200_mstexp1_clk_enable | (no description) | 1'b1 |
| 16 | R/W | sse200_exp_nmi | (no description) | 1'b0 |
| 15 | R/W | sse200_expclkreq_in | (no description) | 1'b1 |
| 14 | R/W | sse200_sysfclkqactive | (no description) | 1'b1 |
| 13 | R/W | sse200_exp_mpc_g_sel | (no description) | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 12 | R/W | sse200_dbgen_sel | (no description) | 1'b0 |
| 11 | R/W | sse200_niden_sel | (no description) | 1'b0 |
| 10 | R/W | sse200_certdisableext | (no description) | 1'b0 |
| 9 | R/W | em_block_addr | not intended to be used; helps with cpf isolation issues | 1'b0 |
| 8 | R/W | em_block_wdata | not intended to be used; helps with cpf isolation issues | 1'b0 |
| 7 | R/W | dbg_enable | (no description) | 1'd0 |
| 6:4 | R/W | dbg_shft | (no description) | 3'd0 |
| 3:0 | R/W | dbg_sel | (no description) | 4'd0 |

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**Name** PSEL_A
**Address** 0x50143050
**Description** Contains register fields associated with psel_A.
**Reset** 0x00000201

| Name | p3_sel | | | | | | | | p2_sel | | | | | | | | p1_sel | | | | | | | | p0_sel | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p3_sel | (no description) | 8'b0 |
| 23:16 | R/W | p2_sel | (no description) | 8'b0 |
| 15:8 | R/W | p1_sel | by default P1 is SWDIO | 8'd2 |
| 7:0 | R/W | p0_sel | by default P0 is SWDCLK | 8'd1 |

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**Name** PSEL_B
**Address** 0x50143054
**Description** Contains register fields associated with psel_B.
**Reset** 0x00000000

| Name | p7_sel | | | | | | | | p6_sel | | | | | | | | p5_sel | | | | | | | | p4_sel | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p7_sel | (no description) | 8'b0 |
| 23:16 | R/W | p6_sel | (no description) | 8'b0 |
| 15:8 | R/W | p5_sel | (no description) | 8'b0 |
| 7:0 | R/W | p4_sel | (no description) | 8'b0 |

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**Name** PSEL_C
**Address** 0x50143058
**Description** Contains register fields associated with psel_C.
**Reset** 0x00000000

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| Name | p11_sel | | | | | | | | p10_sel | | | | | | | | p9_sel | | | | | | | | p8_sel | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p11_sel | (no description) | 8'b0 |
| 23:16 | R/W | p10_sel | (no description) | 8'b0 |
| 15:8 | R/W | p9_sel | (no description) | 8'b0 |
| 7:0 | R/W | p8_sel | (no description) | 8'b0 |

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**Name** PSEL_D
**Address** 0x5014305c
**Description** Contains register fields associated with psel_D.
**Reset** 0x00000000

| Name | p15_sel | | | | | | | | p14_sel | | | | | | | | p13_sel | | | | | | | | p12_sel | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p15_sel | (no description) | 8'b0 |
| 23:16 | R/W | p14_sel | (no description) | 8'b0 |
| 15:8 | R/W | p13_sel | (no description) | 8'b0 |
| 7:0 | R/W | p12_sel | (no description) | 8'b0 |

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**Name** PSEL_E
**Address** 0x50143060
**Description** Contains register fields associated with psel_E.
**Reset** 0x00000000

| Name | p19_sel | | | | | | | | p18_sel | | | | | | | | p17_sel | | | | | | | | p16_sel | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p19_sel | (no description) | 8'b0 |
| 23:16 | R/W | p18_sel | (no description) | 8'b0 |
| 15:8 | R/W | p17_sel | (no description) | 8'b0 |
| 7:0 | R/W | p16_sel | (no description) | 8'b0 |

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**Name** PSEL_F
**Address** 0x50143064
**Description** Contains register fields associated with psel_F.
**Reset** 0x00000000

| Name | p23_sel | | | | | | | | p22_sel | | | | | | | | p21_sel | | | | | | | | p20_sel | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p23_sel | (no description) | 8'b0 |
| 23:16 | R/W | p22_sel | (no description) | 8'b0 |
| 15:8 | R/W | p21_sel | (no description) | 8'b0 |
| 7:0 | R/W | p20_sel | (no description) | 8'b0 |

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**Name** PSEL_G
**Address** 0x50143068
**Description** Contains register fields associated with psel_G.
**Reset** 0x00000000

| Name | p27_sel | | | | | | | | p26_sel | | | | | | | | p25_sel | | | | | | | | p24_sel | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | p27_sel | (no description) | 8'b0 |
| 23:16 | R/W | p26_sel | (no description) | 8'b0 |
| 15:8 | R/W | p25_sel | (no description) | 8'b0 |
| 7:0 | R/W | p24_sel | (no description) | 8'b0 |

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**Name** PSEL_H
**Address** 0x5014306c
**Description** Contains register fields associated with psel_H.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | p30_sel | | | | | | | | p29_sel | | | | | | | | p28_sel | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 23:16 | R/W | p30_sel | (no description) | 8'b0 |
| 15:8 | R/W | p29_sel | (no description) | 8'b0 |
| 7:0 | R/W | p28_sel | (no description) | 8'b0 |

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**Name** PSEL_I
**Address** 0x50143070
**Description** Contains register fields associated with psel_I.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | jtag_interface | RSVD | | | | | | | | | | | | atb_enable | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16 | R/W | jtag_interface | (no description) | 1'b0 |
| 4:0 | R/W | atb_enable | (no description) | 5'b0 |

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**Name** PUPD_OVRD
**Address** 0x50143074
**Description** pin 0-31 pull up/down override control
**Reset** 0x00000000

| Name | RSVD | pin | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 30:0 | R/W | pin | when set, overrides pull-up/pull-down from periphery | 31'd0 |

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**Name** PUPD_OVRD_VAL
**Address** 0x50143078
**Description** pin 0-31 pull up/down override value
**Reset** 0x00000000

| Name | RSVD | pin | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 30:0 | R/W | pin | pull-up/pull_down override value<br>when pupd_ovrd is set:<br>1= pull-up override<br>0= pull-down override | 31'd0 |

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**Name** REMAP
**Address** 0x5014307c
**Description** Contains register fields associated with remap.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | qspi_drop |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | qspi_drop | drops the at_ahb_qspi address start from 0x10000000 to 0x0. No change in range. | 1'b0 |

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**Name** AHB_INTERPOSERS
**Address** 0x50143080
**Description** Contains register fields associated with ahb_interposers.
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | qspi_intp_low_ceiling | qspi_intp_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | qspi_intp_low_ceiling | (no description) | 1'b0 |
| 0 | R/W | qspi_intp_enable | (no description) | 1'b1 |

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**Name** SCRATCHPAD_A
**Address** 0x50143084
**Description** Contains register fields associated with scratchpad_A.
**Reset** 0x00000000

| Name | misc | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | misc | mostly likely will be used to inject random seed in sim | 32'b0 |

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---

**Name** SCRATCHPAD_B
**Address** 0x50143088
**Description** Contains register fields associated with scratchpad_B.
**Reset** 0x00000000

| Name | misc | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | misc | mostly likely will be used to inject random seed in sim | 32'b0 |

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---

**Name** SCRATCHPAD_C
**Address** 0x5014308c
**Description** Contains register fields associated with scratchpad_C.
**Reset** 0x00000000

| Name | misc | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | misc | mostly likely will be used to inject random seed in sim | 32'b0 |

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---

**Name** SCRATCHPAD_D
**Address** 0x50143090
**Description** Contains register fields associated with scratchpad_D.
**Reset** 0x00000000

| Name | misc | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | misc | mostly likely will be used to inject random seed in sim | 32'b0 |

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---

**Name** RRAM_WRITE_PROTECTION0
**Address** 0x50143094
**Description** Contains register fields associated with rram_write_protection0.
**Reset** 0xffffffff

| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | if a bit is set, invalidates any writes to corresponding to 2kB block | 32'hffffffff |

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---

**Name** RRAM_WRITE_PROTECTION1
**Address** 0x50143098
**Description** Contains register fields associated with rram_write_protection1.
**Reset** 0xffffffff

| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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---

**Name** RRAM_WRITE_PROTECTION2
**Address** 0x5014309c
**Description** Contains register fields associated with rram_write_protection2.
**Reset** 0xffffffff

| | disa |
|---|---|

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| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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---

**Name** RRAM_WRITE_PROTECTION3
**Address** 0x501430a0
**Description** Contains register fields associated with rram_write_protection3.
**Reset** 0xffffffff

| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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---

**Name** RRAM_WRITE_PROTECTION4
**Address** 0x501430a4
**Description** Contains register fields associated with rram_write_protection4.
**Reset** 0xffffffff

| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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**Name** RRAM_WRITE_PROTECTION5
**Address** 0x501430a8
**Description** Contains register fields associated with rram_write_protection5.
**Reset** 0xffffffff

| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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**Name** RRAM_WRITE_PROTECTION6
**Address** 0x501430ac
**Description** Contains register fields associated with rram_write_protection6.
**Reset** 0xffffffff

| Name | disable_bits | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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**Name** RRAM_WRITE_PROTECTION7
**Address** 0x501430b0
**Description** Contains register fields associated with rram_write_protection7.
**Reset** 0xffffffff

| Name | disable_bi |
|---|---|

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| | t<br>s | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | disable_bits | (no description) | 32'hffffffff |

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**Name** RRAM_STICKY_WRITE_PROTECTION0
**Address** 0x501430b4
**Description** Contains register fields associated with rram_sticky_write_protection0.
**Reset** 0x00000000

| Name | s<br>e<br>t | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | if a bit is set, invalidates any writes to the corresponding 2kB block, can only revert with cpu reset | 32'h0 |

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---

**Name** RRAM_STICKY_WRITE_PROTECTION1
**Address** 0x501430b8
**Description** Contains register fields associated with rram_sticky_write_protection1.
**Reset** 0x00000000

| Name | s<br>e<br>t | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION2
**Address** 0x501430bc
**Description** Contains register fields associated with rram_sticky_write_protection2.
**Reset** 0x00000000

| Name | s<br>e<br>t | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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---

**Name** RRAM_STICKY_WRITE_PROTECTION3
**Address** 0x501430c0
**Description** Contains register fields associated with rram_sticky_write_protection3.
**Reset** 0x00000000

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| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION4
**Address** 0x501430c4
**Description** Contains register fields associated with rram_sticky_write_protection4.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION5
**Address** 0x501430c8
**Description** Contains register fields associated with rram_sticky_write_protection5.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION6
**Address** 0x501430cc
**Description** Contains register fields associated with rram_sticky_write_protection6.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION7
**Address** 0x501430d0

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**Description** Contains register fields associated with rram_sticky_write_protection7.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS0
**Address** 0x501430d4
**Description** Contains register fields associated with rram_sticky_write_protection_status0.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | if read back value is 1, then the sticky bit has been set | 32'h0 |

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---

**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS1
**Address** 0x501430d8
**Description** Contains register fields associated with rram_sticky_write_protection_status1.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS2
**Address** 0x501430dc
**Description** Contains register fields associated with rram_sticky_write_protection_status2.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS3
**Address** 0x501430e0
**Description** Contains register fields associated with rram_sticky_write_protection_status3.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS4
**Address** 0x501430e4
**Description** Contains register fields associated with rram_sticky_write_protection_status4.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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---

**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS5
**Address** 0x501430e8
**Description** Contains register fields associated with rram_sticky_write_protection_status5.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS6
**Address** 0x501430ec
**Description** Contains register fields associated with rram_sticky_write_protection_status6.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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**Name** RRAM_STICKY_WRITE_PROTECTION_STATUS7
**Address** 0x501430f0
**Description** Contains register fields associated with rram_sticky_write_protection_status7.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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**Name** RRAM_STICKY_READ_PROTECTION0
**Address** 0x501430f4
**Description** Contains register fields associated with rram_sticky_read_protection0.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | if a bit is set, invalidates any reads to the corresponding 2kB block, can only revert with cpu reset | 32'h0 |

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**Name** RRAM_STICKY_READ_PROTECTION_STATUS0
**Address** 0x501430f8
**Description** Contains register fields associated with rram_sticky_read_protection_status0.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | if read back value is 1, then the sticky bit has been set | 32'h0 |

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**Name** PROT_BITS_SET0
**Address** 0x501430fc
**Description** Contains register fields associated with prot_bits_set0.
**Reset** 0x00000000

| | rom_rea |
|---|---|

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| Name | rom_read_locks_set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | rom_read_locks_set | These bits go on to set sticky bits that lock up rom read. Each bit maps to one 2kB sector. Not self clearing | 32'h00000000 |

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**Name** PROT_BITS_SET1
**Address** 0x50143100
**Description** Not self-clearing
**Reset** 0x00000000

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | suppress_otp_writes | suppress_rom_patch | apb_shub_lock_set | apb_qspi_lock_set | uart1_lock_set | uart0_lock_set |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | suppress_otp_writes | sets sticky bit preventing OTP writes | 1'b0 |
| 4 | R/W | suppress_rom_patch | sets sticky bit preventing ROM patches from being applied | 1'b0 |
| 3 | R/W | apb_shub_lock_set | sets sticky bit disabling the update of write address range. | 1'b0 |
| 2 | R/W | apb_qspi_lock_set | sets sticky bit disabling APB cores accessing qspi. | 1'b0 |
| 1 | R/W | uart1_lock_set | sets sticky bit disabling uart1_tx. | 1'b0 |
| 0 | R/W | uart0_lock_set | sets sticky bit disabling uart0_tx. | 1'b0 |

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**Name** PROT_BITS_STAT0
**Address** 0x50143104
**Description** Contains register fields associated with prot_bits_stat0.
**Reset** 0x00000000

| | rom |
|---|---|

| Name | rom_read_locks_stat | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | rom_read_locks_stat | sticky bits that lock up rom read. Each bit maps to one 2kB sector. Not self clearing | 32'h00000000 |

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**Name** PROT_BITS_STAT1
**Address** 0x50143108
**Description** Not self-clearing
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | suppress_otp_writes_stat | suppress_rom_patch_stat | apb_shub_lock_stat | apb_qspi_lock_stat | uart1_lock_stat | uart0_lock_stat |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R | suppress_otp_writes_stat | sticky bit preventing OTP writes | 1'b0 |
| 4 | R | suppress_rom_patch_stat | sticky bit preventing ROM patches from being applied | 1'b0 |
| 3 | R | apb_shub_lock_stat | sticky bit disabling shub write address range. | 1'b0 |
| 2 | R | apb_qspi_lock_stat | sticky bit disabling APB cores accessing qspi. | 1'b0 |
| 1 | R | uart1_lock_stat | sticky bit disabling uart1_tx. | 1'b0 |
| 0 | R | uart0_lock_stat | sticky bit disabling uart0_tx. | 1'b0 |

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**Name** QSPI_STICKY_WRITE_PROTECTION0
**Address** 0x5014310c
**Description** Contains register fields associated with qspi_sticky_write_protection0.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | if a bit is set, invalidates any writes to the corresponding 2kB block, can only revert with cpu reset | 32'h0 |

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**Name** QSPI_STICKY_WRITE_PROTECTION1
**Address** 0x50143110
**Description** Contains register fields associated with qspi_sticky_write_protection1.
**Reset** 0x00000000

| Name | set | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | set | (no description) | 32'h0 |

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**Name** QSPI_STICKY_WRITE_PROTECTION_STATUS0
**Address** 0x50143114
**Description** Contains register fields associated with qspi_sticky_write_protection_status0.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | if read back value is 1, then the sticky bit has been set | 32'h0 |

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**Name** QSPI_STICKY_WRITE_PROTECTION_STATUS1
**Address** 0x50143118
**Description** Contains register fields associated with qspi_sticky_write_protection_status1.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | (no description) | 32'h0 |

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**Name** PDSN
**Address** 0x5014311c
**Description** pin drive strength
**Reset** 0x00000000

| Name | RSVD | is_high | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 30:0 | R/W | is_high | 0=high drive stength/lower impedance<br>1=low drive strength/higher impedance | 31'd0 |

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**Name** CALCOUNTRC
**Address** 0x50143120
**Description** Contains register fields associated with calCountRC.
**Reset** 0x00000000

| Name | RSVD | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 22:0 | R | val | (no description) | 23'd0 |

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**Name** SECURE_DEBUG_CTRL
**Address** 0x50143124
**Description** Contains register fields associated with secure_debug_ctrl.
**Reset** 0x00000000

| Name | RSVD | sticky_disabled | soft_disabled |
|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 1 | R/W | sticky_disabled | software writes 1 during bootload time to disable SWD interface.<br>Read value reflects the state of the secure debug FSM. | 1'd0 |
| 0 | R/W | soft_disabled | if sticky_disabled is 0, SW can write 0 to enable SWD interface.<br>Read value reflects the state of the secure debug FSM. | 1'd0 |

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**Name** SECURE_DEBUG_STAT
**Address** 0x50143128
**Description** Contains register fields associated with secure_debug_stat.
**Reset** 0x00000000

| Name | RSVD | sec_dbg_fsm |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2:0 | R | sec_dbg_fsm | state of secure debug FSM. | 3'd0 |

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**Name** CHIPID1
**Address** 0x50143ff0
**Description** Contains register fields associated with chipid1.
**Reset** 0x50617269

| Name | char0 | char1 | char2 | char3 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R | char0 | P | 8'h50 |
| 23:16 | R | char1 | a | 8'h61 |
| 15:8 | R | char2 | r | 8'h72 |
| 7:0 | R | char3 | i | 8'h69 |

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**Name** CHIPID2
**Address** 0x50143ff4
**Description** Contains register fields associated with chipid2.
**Reset** 0x73000000

| Name | char4 | char5 | char6 | char7 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R | char4 | s | 8'h73 |
| 23:16 | R | char5 | nul | 8'h0 |
| 15:8 | R | char6 | nul | 8'h0 |
| 7:0 | R | char7 | nul | 8'h0 |

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**Name** CHIPREV
**Address** 0x50143ff8
**Description** Contains register fields associated with chiprev.
**Reset** 0x00000100

| Name | RSVD | | | | | | | | | | | | | | | running_off_32khz_xtal | major | | | | | | | minor | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16 | R | running_off_32khz_xtal | (no description) | 1'b0 |
| 15:8 | R | major | (no description) | 8'd1 |
| 7:0 | R | minor | (no description) | 8'd0 |

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**Name** CORE_ID
**Address** 0x50143ffc
**Description** Contains register fields associated with core_id.
**Reset** 0x57525052

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | WRPR in ASCII | 32'h57525052 |

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# UART0_SECURE

**Name** DATA
**Address** 0x50144000
**Description** Contains register fields associated with DATA.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | Data | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | Data | Read Received data.<br>Write Transmit data. | 8'b0 |

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**Name** STATE
**Address** 0x50144004
**Description** Contains register fields associated with STATE.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | tx_locked_stat | tx_idle | stop_bit_err | nUARTCTS | nUARTRTS | rx_overrun | tx_overrun | rx_buf_full | tx_buf_full |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8 | R/W | tx_locked_stat | 1=(Sticky) indicates whether tx is locked. Once locked, it remains locked until a system reset. (Read only) // (Atmosic) | 1'b0 |
| 7 | R/W | tx_idle | 1=tx state is idle (Read only) // (Atmosic) | 1'b0 |
| 6 | R/W | stop_bit_err | 1=stop_bit err (Read only) // (Atmosic) | 1'b0 |
| 5 | R/W | nUARTCTS | current status of nUARTCTS (Read only) // (Atmosic) | 1'b0 |
| 4 | R/W | nUARTRTS | current status of nUARTRTS (Read only) // (Atmosic) | 1'b0 |
| 3 | R/W | rx_overrun | RX buffer overrun, write 1 to clear. | 1'b0 |
| 2 | R/W | tx_overrun | TX buffer overrun, write 1 to clear. | 1'b0 |
| 1 | R/W | rx_buf_full | 1=RX buffer is not empty, 0= RX buffer is empty read-only. | 1'b0 |
| 0 | R/W | tx_buf_full | TX buffer full, read-only. | 1'b0 |

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**Name** CTRL
**Address** 0x50144008
**Description** Contains register fields associated with CTRL.
**Reset** 0x00000000

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| Name | RSVD | dma_mode_rx | dma_mode_tx | hs_testmode | rx_overrun_int_en | tx_overrun_int_en | RxIntEn | TxIntEn | RxEn | TxEn |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8 | R/W | dma_mode_rx | 1= dma can drain the rx fifo and the rx_intr is cleared when empty | 1'b0 |
| 7 | R/W | dma_mode_tx | 1= dma can fill the tx fifo and the tx_intr is cleared when full or done. | 1'b0 |
| 6 | R/W | hs_testmode | High-speed test mode for TX only. | 1'b0 |
| 5 | R/W | rx_overrun_int_en | RX overrun interrupt enable. | 1'b0 |
| 4 | R/W | tx_overrun_int_en | TX overrun interrupt enable. | 1'b0 |
| 3 | R/W | RxIntEn | RX interrupt enable. | 1'b0 |
| 2 | R/W | TxIntEn | TX interrupt enable. | 1'b0 |
| 1 | R/W | RxEn | RX enable. | 1'b0 |
| 0 | R/W | TxEn | TX enable. | 1'b0 |

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**Name** INTSTATUS_CLEAR
**Address** 0x5014400c
**Description** Contains register fields associated with INTSTATUS_CLEAR.
**Reset** 0x00000000

| Name | RSVD | rx_overflow_intr | tx_overflow_intr | rxintr | txintr |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | rx_overflow_intr | RX overrun interrupt. Write 1 to clear. | 1'b0 |
| 2 | R/W | tx_overflow_intr | TX overrun interrupt. Write 1 to clear. | 1'b0 |
| 1 | R/W | rxintr | RX interrupt. Write 1 to clear. | 1'b0 |
| 0 | R/W | txintr | TX interrupt. Write 1 to clear. | 1'b0 |

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**Name** BAUDDIV
**Address** 0x50144010
**Description** Contains register fields associated with BAUDDIV.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | baud_div | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:0 | R/W | baud_div | Baud rate divider. The minimum number is 8. | 20'b0 |

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**Name** RX_LWM
**Address** 0x50144014
**Description** Contains register fields associated with RX_LWM.
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | Rx_LWM | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | Rx_LWM | low water mark of RX FIFO; a trigger is issued when (Atmosic)<br>RX FIFO has waited long enough to have received these many items, starting from<br>the first item.<br>Constraint: ( 1<= RX_LWM <= 14) | 5'd1 |

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**Name** TX_LWM
**Address** 0x50144018
**Description** Contains register fields associated with TX_LWM.
**Reset** 0x00000010

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | Tx_LWM | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | Tx_LWM | low water mark of TX FIF0; a trigger is issued when (Atmosic)<br>tX FIFO is drained to these many unoccupied slots.<br>Constraint (1<= TX_LWM <= 16) | 5'd16 |

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**Name** RX_FIFO_SPACES
**Address** 0x5014401c
**Description** Contains register fields associated with RX_FIFO_SPACES.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_fifo_spaces | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | rx_fifo_spaces | number of unoccupied spaces in RX_FIFO (Atmosic) | 5'b0 |

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**Name** TX_FIFO_SPACES
**Address** 0x50144020
**Description** Contains register fields associated with TX_FIFO_SPACES.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | tx_fifo_spaces | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | tx_fifo_spaces | number of unoccupied spaces in TX_FIFO (Atmosic) | 5'b0 |

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**Name** HW_FLOW_OVRD
**Address** 0x50144024
**Description** Contains register fields associated with HW_FLOW_OVRD.
**Reset** 0x0000002a

| | R | | | | | | | | | | | | | | | | | | | | | | | | | | | RTS_T | nCT | nCT | nRT | nRT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Name | SVD | | | | | | | | | | | | | | | | | | | | | | | | | RIG_LVL | | | S_OVRD | S_VAL | S_OVRD | S_VAL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6:4 | R/W | RTS_TRIG_LVL | threshold of rx fifo to de-assert nRTS. The value corresponds to the number of open spaces in RX fifo.<br>minimum programmable value is 2. | 3'd2 |
| 3 | R/W | nCTS_OVRD | 0= nUARTCTS override override disabled; 1(default)= nUARTCTS override enabled (Atmosic) | 1'b1 |
| 2 | R/W | nCTS_VAL | 0(default) = nCTS is asserted to enable TX; 1= nCTS is deasserted | 1'b0 |
| 1 | R/W | nRTS_OVRD | 0= nUARTRTS override override disabled; 1(default)= nUARTRTS override enabled ;(Atmosic) | 1'b1 |
| 0 | R/W | nRTS_VAL | 0(default) = nRTS is asserted to enable external TX; 1= nRTS is deasserted | 1'b0 |

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**Name** BAUD_RATE_DETECT
**Address** 0x50144028
**Description** Contains register fields associated with BAUD_RATE_DETECT.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | brd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | brd | 1=set by SW, HW treat the next frame/s as baud rate detection byte, cleared by HW once baud rate is detected. 0=normal mode | 1'b0 |

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**Name** BAUD_DIV_CEIL
**Address** 0x5014402c
**Description** Contains register fields associated with BAUD_DIV_CEIL.
**Reset** 0x000fffff

| Name | RSVD | | | | | | | | | | | | baud_div_ceil | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 19:0 | R/W | bauddiv_ceil | sets max reference for baud rate in auto detection. Auto detection restarts if start bit is not detected within this window. | 20'hfffff |

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**Name** BRD_STAT
**Address** 0x50144030
**Description** Contains register fields associated with BRD_STAT.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | start_bit_timeout | on_going |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | start_bit_timeout | Previous synch bit too long. The calibration restarts after timeout. | 1'b0 |
| 0 | R | on_going | auto detection is on going | 1'b0 |

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**Name** INTMASK
**Address** 0x50144034
**Description** Contains register fields associated with INTMASK.
**Reset** 0x0000000f

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_overflow_int_en | tx_overflow_int_en | rxint_en | txint_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | rx_overflow_int_en | mask enable for RX overflow interrupt. | 1'b1 |
| 2 | R/W | tx_overflow_int_en | mask enable for TX overflow interrupt. | 1'b1 |

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| | | | | |
|---|---|---|---|---|
| 1 | R/W | rxint_en | mask enable for RX interrupt. | 1'b1 |
| 0 | R/W | txint_en | mask enable for TX interrupt. | 1'b1 |

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**Name** RX_FIFO_ENTRIES
**Address** 0x50144038
**Description** Contains register fields associated with RX_FIFO_ENTRIES.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_fifo_entries | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | rx_fifo_entries | number of occupied spaces in RX_FIFO. DMA reads this value to determine how many reads to do in each round. | 5'b0 |

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**Name** INTSTATUS_SET
**Address** 0x5014403c
**Description** Contains register fields associated with INTSTATUS_SET.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_overflow_intr_set | tx_overflow_intr_set | rxintr_set | txintr_set |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | rx_overflow_intr_set | RX overrun interrupt. Write 1 to set. (self-clearing) | 1'b0 |
| 2 | R/W | tx_overflow_intr_set | TX overrun interrupt. Write 1 to set. (self-clearing) | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 1 | R/W | rxintr_set | RX interrupt. Write 1 to set. (self-clearing) | 1'b0 |
| 0 | R/W | txintr_set | TX interrupt. Write 1 to set. (self-clearing) | 1'b0 |

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**Name** PID4
**Address** 0x50144fd0
**Description** Contains register fields associated with PID4.
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | block_count | | | | jep106_c_code | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | block_count | Block count | 4'b0 |
| 3:0 | R | jep106_c_code | jep106_c_code | 4'd4 |

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**Name** PID5
**Address** 0x50144fd4
**Description** Contains register fields associated with PID5.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid5 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid5 | Peripheral ID Register 5. | 8'b0 |

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**Name** PID6
**Address** 0x50144fd8
**Description** Contains register fields associated with PID6.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid6 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid6 | Peripheral ID Register 6. | 8'b0 |

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**Name** PID7
**Address** 0x50144fdc
**Description** Contains register fields associated with PID7.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid7 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid7 | Peripheral ID Register 7. | 8'b0 |

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**Name** PID0
**Address** 0x50144fe0
**Description** Contains register fields associated with PID0.
**Reset** 0x00000021

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | part_num | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | part_num | Part number[7:0]. | 8'h21 |

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**Name** PID1
**Address** 0x50144fe4
**Description** Contains register fields associated with PID1.
**Reset** 0x000000b8

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | jep106_id_3_0 | | | | part_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | jep106_id_3_0 | jep106_id_3_0 | 4'hb |

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| | | | | |
|---|---|---|---|---|
| 3:0 | R | part_num | Part number[11:8]. | 4'h8 |

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**Name** PID2
**Address** 0x50144fe8
**Description** Contains register fields associated with PID2.
**Reset** 0x0000001b

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | rev | | | | jedec_used | jep106_id_6_4 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | rev | Revision | 4'h1 |
| 3 | R | jedec_used | jedec_used | 1'h1 |
| 2:0 | R | jep106_id_6_4 | jep106_id_6_4 | 3'h3 |

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**Name** PID3
**Address** 0x50144fec
**Description** Contains register fields associated with PID3.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | ecorevnum | | | | mod_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | ecorevnum | ECO revision number. | 4'h0 |
| 3:0 | R | mod_num | modification number. | 4'h0 |

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**Name** CID0
**Address** 0x50144ff0
**Description** Contains register fields associated with CID0.
**Reset** 0x0000000d

| Name | RSV | | | | | | | | | | | | | | | | | | | | | | | | | | | | cid | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | D | | | | | | | | | | | | | | | | | | | | | | 0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid0 | Component ID Register 0 | 8'h0d |

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**Name** CID1
**Address** 0x50144ff4
**Description** Contains register fields associated with CID1.
**Reset** 0x000000f0

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid1 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid1 | Component ID Register 1 | 8'hf0 |

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**Name** CID2
**Address** 0x50144ff8
**Description** Contains register fields associated with CID2.
**Reset** 0x00000005

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid2 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid2 | Component ID Register 2 | 8'h05 |

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**Name** CID3
**Address** 0x50144ffc
**Description** Contains register fields associated with CID3.
**Reset** 0x000000b1

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid3 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid3 | Component ID Register 3 | 8'hb1 |

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# UART1_SECURE

CONFIDENTIAL

**Name** DATA
**Address** 0x50145000
**Description** Contains register fields associated with DATA.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | Data | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | Data | Read Received data.<br>Write Transmit data. | 8'b0 |

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**Name** STATE
**Address** 0x50145004
**Description** Contains register fields associated with STATE.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | tx_locked_stat | tx_idle | stop_bit_err | nUARTCTS | nUARTRTS | rx_overrun | tx_overrun | rx_buf_full | tx_buf_full |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8 | R/W | tx_locked_stat | 1=(Sticky) indicates whether tx is locked. Once locked, it remains locked until a system reset. (Read only) // (Atmosic) | 1'b0 |
| 7 | R/W | tx_idle | 1=tx state is idle (Read only) // (Atmosic) | 1'b0 |
| 6 | R/W | stop_bit_err | 1=stop_bit err (Read only) // (Atmosic) | 1'b0 |
| 5 | R/W | nUARTCTS | current status of nUARTCTS (Read only) // (Atmosic) | 1'b0 |
| 4 | R/W | nUARTRTS | current status of nUARTRTS (Read only) // (Atmosic) | 1'b0 |
| 3 | R/W | rx_overrun | RX buffer overrun, write 1 to clear. | 1'b0 |
| 2 | R/W | tx_overrun | TX buffer overrun, write 1 to clear. | 1'b0 |
| 1 | R/W | rx_buf_full | 1=RX buffer is not empty, 0= RX buffer is empty read-only. | 1'b0 |
| 0 | R/W | tx_buf_full | TX buffer full, read-only. | 1'b0 |

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**Name** CTRL
**Address** 0x50145008
**Description** Contains register fields associated with CTRL.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_ | tx_ | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Name | RSVD | dma_mode_rx | dma_mode_tx | hs_testmode | overrun_int_en | overrun_int_en | RxIntEn | TxIntEn | RxEn | TxEn |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8 | R/W | dma_mode_rx | 1= dma can drain the rx fifo and the rx_intr is cleared when empty | 1'b0 |
| 7 | R/W | dma_mode_tx | 1= dma can fill the tx fifo and the tx_intr is cleared when full or done. | 1'b0 |
| 6 | R/W | hs_testmode | High-speed test mode for TX only. | 1'b0 |
| 5 | R/W | rx_overrun_int_en | RX overrun interrupt enable. | 1'b0 |
| 4 | R/W | tx_overrun_int_en | TX overrun interrupt enable. | 1'b0 |
| 3 | R/W | RxIntEn | RX interrupt enable. | 1'b0 |
| 2 | R/W | TxIntEn | TX interrupt enable. | 1'b0 |
| 1 | R/W | RxEn | RX enable. | 1'b0 |
| 0 | R/W | TxEn | TX enable. | 1'b0 |

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**Name** INTSTATUS_CLEAR
**Address** 0x5014500c
**Description** Contains register fields associated with INTSTATUS_CLEAR.
**Reset** 0x00000000

| Name | RSVD | rx_overflow_intr | tx_overflow_intr | rxintr | txintr |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | rx_overflow_intr | RX overrun interrupt. Write 1 to clear. | 1'b0 |
| 2 | R/W | tx_overflow_intr | TX overrun interrupt. Write 1 to clear. | 1'b0 |
| 1 | R/W | rxintr | RX interrupt. Write 1 to clear. | 1'b0 |
| 0 | R/W | txintr | TX interrupt. Write 1 to clear. | 1'b0 |

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**Name** BAUDDIV
**Address** 0x50145010

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**Description** Contains register fields associated with BAUDDIV.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | baud_div | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:0 | R/W | baud_div | Baud rate divider. The minimum number is 8. | 20'b0 |

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**Name** RX_LWM
**Address** 0x50145014
**Description** Contains register fields associated with RX_LWM.
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | Rx_LWM | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | Rx_LWM | low water mark of RX FIFO; a trigger is issued when (Atmosic)<br>RX FIFO has waited long enough to have received these many items, starting from<br>the first item.<br>Constraint: ( 1<= RX_LWM <= 14) | 5'd1 |

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**Name** TX_LWM
**Address** 0x50145018
**Description** Contains register fields associated with TX_LWM.
**Reset** 0x00000010

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | Tx_LWM | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | Tx_LWM | low water mark of TX FIF0; a trigger is issued when (Atmosic)<br>tX FIFO is drained to these many unoccupied slots.<br>Constraint (1<= TX_LWM <= 16) | 5'd16 |

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**Name** RX_FIFO_SPACES

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**Address** 0x5014501c
**Description** Contains register fields associated with RX_FIFO_SPACES.
**Reset** 0x00000000

| Name | RSVD | rx_fifo_spaces |
|---|---|---|
| **Bits** | **31:5** | **4:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | rx_fifo_spaces | number of unoccupied spaces in RX_FIFO (Atmosic) | 5'b0 |

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**Name** TX_FIFO_SPACES
**Address** 0x50145020
**Description** Contains register fields associated with TX_FIFO_SPACES.
**Reset** 0x00000000

| Name | RSVD | tx_fifo_spaces |
|---|---|---|
| **Bits** | **31:5** | **4:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | tx_fifo_spaces | number of unoccupied spaces in TX_FIFO (Atmosic) | 5'b0 |

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**Name** HW_FLOW_OVRD
**Address** 0x50145024
**Description** Contains register fields associated with HW_FLOW_OVRD.
**Reset** 0x0000002a

| Name | RSV | RTS_TRI | nCTS_ | nCTS_ | nRTS_ | nRTS_ |
|---|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | D | | | | | | | | | | | | | | | | | | | | | | | | G_LVL | | | OVRD | VAL | OVRD | VAL |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6:4 | R/W | RTS_TRIG_LVL | threshold of rx fifo to de-assert nRTS. The value corresponds to the number of open spaces in RX fifo.<br>minimum programmable value is 2. | 3'd2 |
| 3 | R/W | nCTS_OVRD | 0= nUARTCTS override override disabled; 1(default)= nUARTCTS override enabled (Atmosic) | 1'b1 |
| 2 | R/W | nCTS_VAL | 0(default) = nCTS is asserted to enable TX; 1= nCTS is deasserted | 1'b0 |
| 1 | R/W | nRTS_OVRD | 0= nUARTRTS override override disabled; 1(default)= nUARTRTS override enabled ;(Atmosic) | 1'b1 |
| 0 | R/W | nRTS_VAL | 0(default) = nRTS is asserted to enable external TX; 1= nRTS is deasserted | 1'b0 |

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**Name** BAUD_RATE_DETECT
**Address** 0x50145028
**Description** Contains register fields associated with BAUD_RATE_DETECT.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | brd |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | brd | 1=set by SW, HW treat the next frame/s as baud rate detection byte, cleared by HW once baud rate is detected. 0=normal mode | 1'b0 |

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**Name** BAUD_DIV_CEIL
**Address** 0x5014502c
**Description** Contains register fields associated with BAUD_DIV_CEIL.
**Reset** 0x000fffff

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | bauddiv_ceil | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:0 | R/W | bauddiv_ceil | sets max reference for baud rate in auto detection. Auto detection restarts if start bit is not detected within this window. | 20'hfffff |

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**Name** BRD_STAT
**Address** 0x50145030
**Description** Contains register fields associated with BRD_STAT.
**Reset** 0x00000000

| Name | RSVD | start_bit_timeout | on_going |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | start_bit_timeout | Previous synch bit too long. The calibration restarts after timeout. | 1'b0 |
| 0 | R | on_going | auto detection is on going | 1'b0 |

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**Name** INTMASK
**Address** 0x50145034
**Description** Contains register fields associated with INTMASK.
**Reset** 0x0000000f

| Name | RSVD | rx_overflow_int_en | tx_overflow_int_en | rxint_en | txint_en |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | rx_overflow_int_en | mask enable for RX overflow interrupt. | 1'b1 |
| 2 | R/W | tx_overflow_int_en | mask enable for TX overflow interrupt. | 1'b1 |
| 1 | R/W | rxint_en | mask enable for RX interrupt. | 1'b1 |
| 0 | R/W | txint_en | mask enable for TX interrupt. | 1'b1 |

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**Name** RX_FIFO_ENTRIES
**Address** 0x50145038
**Description** Contains register fields associated with RX_FIFO_ENTRIES.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_fifo_entries | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | rx_fifo_entries | number of occupied spaces in RX_FIFO. DMA reads this value to determine how many reads to do in each round. | 5'b0 |

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**Name** INTSTATUS_SET
**Address** 0x5014503c
**Description** Contains register fields associated with INTSTATUS_SET.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | rx_overflow_intr_set | tx_overflow_intr_set | rxintr_set | txintr_set |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | rx_overflow_intr_set | RX overrun interrupt. Write 1 to set. (self-clearing) | 1'b0 |
| 2 | R/W | tx_overflow_intr_set | TX overrun interrupt. Write 1 to set. (self-clearing) | 1'b0 |
| 1 | R/W | rxintr_set | RX interrupt. Write 1 to set. (self-clearing) | 1'b0 |
| 0 | R/W | txintr_set | TX interrupt. Write 1 to set. (self-clearing) | 1'b0 |

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**Name** PID4
**Address** 0x50145fd0
**Description** Contains register fields associated with PID4.
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | block_count | | | | jep106_c_code | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | block_count | Block count | 4'b0 |
| 3:0 | R | jep106_c_code | jep106_c_code | 4'd4 |

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**Name** PID5
**Address** 0x50145fd4
**Description** Contains register fields associated with PID5.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid5 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid5 | Peripheral ID Register 5. | 8'b0 |

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**Name** PID6
**Address** 0x50145fd8
**Description** Contains register fields associated with PID6.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid6 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid6 | Peripheral ID Register 6. | 8'b0 |

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**Name** PID7
**Address** 0x50145fdc
**Description** Contains register fields associated with PID7.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pid7 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | pid7 | Peripheral ID Register 7. | 8'b0 |

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**Name** PID0
**Address** 0x50145fe0
**Description** Contains register fields associated with PID0.
**Reset** 0x00000021

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | part_num | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | part_num | Part number[7:0]. | 8'h21 |

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**Name** PID1
**Address** 0x50145fe4
**Description** Contains register fields associated with PID1.
**Reset** 0x000000b8

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | jep106_id_3_0 | | | | part_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | jep106_id_3_0 | jep106_id_3_0 | 4'hb |
| 3:0 | R | part_num | Part number[11:8]. | 4'h8 |

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**Name** PID2
**Address** 0x50145fe8
**Description** Contains register fields associated with PID2.
**Reset** 0x0000001b

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | rev | | | | jedec_used | jep106_id_6_4 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | rev | Revision | 4'h1 |
| 3 | R | jedec_used | jedec_used | 1'h1 |
| 2:0 | R | jep106_id_6_4 | jep106_id_6_4 | 3'h3 |

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**Name** PID3
**Address** 0x50145fec
**Description** Contains register fields associated with PID3.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | ecorevnum | | | | mod_num | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:4 | R | ecorevnum | ECO revision number. | 4'h0 |
| 3:0 | R | mod_num | modification number. | 4'h0 |

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**Name** CID0
**Address** 0x50145ff0
**Description** Contains register fields associated with CID0.
**Reset** 0x0000000d

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid0 | Component ID Register 0 | 8'h0d |

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**Name** CID1
**Address** 0x50145ff4
**Description** Contains register fields associated with CID1.
**Reset** 0x000000f0

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid1 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid1 | Component ID Register 1 | 8'hf0 |

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**Name** CID2
**Address** 0x50145ff8
**Description** Contains register fields associated with CID2.
**Reset** 0x00000005

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid2 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid2 | Component ID Register 2 | 8'h05 |

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**Name** CID3
**Address** 0x50145ffc
**Description** Contains register fields associated with CID3.
**Reset** 0x000000b1

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cid3 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cid3 | Component ID Register 3 | 8'hb1 |

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# PWM_SECURE

**Name** PWM_CTRL
**Address** 0x50146000

**Description** control signals for all pwm channels
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | alt_ok_to_run | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | alt_ok_to_run | alternative mechanism to start pwm channels; useful if application requires all used pwm channels start together<br>[0] = start pwm0 channel<br>[1] = start pwm1 channel<br>[2] = start pwm2 channel<br>[3] = start pwm3 channel<br>[4] = start pwm4 channel<br>[5] = start pwm5 channel<br>[6] = start pwm6 channel<br>[7] = start pwm7 channel | 8'd0 |

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**Name** PWM0_CTRL
**Address** 0x50146004
**Description** control signals for pwm0 channel
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | idle_polarity | invert | mode | | ok_to_run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run. | 2'd0 |

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| | | | | |
|---|---|---|---|---|
| | | | 11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM0_DUR
**Address** 0x50146008
**Description** durations config for pwm0 channel
**Reset** 0x00000000

| Name | hi_dur | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM0_CFG
**Address** 0x5014600c
**Description** configuration for pwm0 channel
**Reset** 0x00000000

| Name | update | RSVD | | | | | | | | | | | | | | | | frame_count | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** PWM1_CTRL
**Address** 0x50146010
**Description** control signals for pwm1 channel
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | i | | |

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| Name | RSVD | idle_polarity | invert | mode | ok_to_run |
|---|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run.<br>11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | 2'd0 |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM1_DUR
**Address** 0x50146014
**Description** durations config for pwm1 channel
**Reset** 0x00000000

| Name | hi_dur | lo_dur |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM1_CFG
**Address** 0x50146018
**Description** configuration for pwm1 channel
**Reset** 0x00000000

| u | fra |
|---|---|

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| Name | pdate | RSVD | | | | | | | | | | | | | | | | me_count | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** PWM2_CTRL
**Address** 0x5014601c
**Description** control signals for pwm2 channel
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | idle_polarity | invert | mode | | ok_to_run | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run.<br>11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | 2'd0 |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM2_DUR
**Address** 0x50146020
**Description** durations config for pwm2 channel
**Reset** 0x00000000

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| Name | hi_dur | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM2_CFG
**Address** 0x50146024
**Description** configuration for pwm2 channel
**Reset** 0x00000000

| Name | update | RSVD | | | | | | | | | | | | | | | | frame_count | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** PWM3_CTRL
**Address** 0x50146028
**Description** control signals for pwm3 channel
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | idle_polarity | invert | mode | ok_to_run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run.<br>11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | 2'd0 |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM3_DUR
**Address** 0x5014602c
**Description** durations config for pwm3 channel
**Reset** 0x00000000

| Name | hi_dur | | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM3_CFG
**Address** 0x50146030
**Description** configuration for pwm3 channel
**Reset** 0x00000000

| Name | update | RSVD | | | | | | | | | | | | | | | | frame_count | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** PWM4_CTRL
**Address** 0x50146034
**Description** control signals for pwm4 channel
**Reset** 0x00000000

| Name | RSVD | idle_polarity | invert | mode | ok_to_run |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4** | **3** | **2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run.<br>11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | 2'd0 |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM4_DUR
**Address** 0x50146038
**Description** durations config for pwm4 channel
**Reset** 0x00000000

| Name | hi_dur | lo_dur |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM4_CFG
**Address** 0x5014603c
**Description** configuration for pwm4 channel
**Reset** 0x00000000

| Name | update | RSVD | | | | | | | | | | | | | | | | frame_count | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** PWM5_CTRL
**Address** 0x50146040
**Description** control signals for pwm5 channel
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | idle_polarity | invert | mode | | ok_to_run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run.<br>11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | 2'd0 |

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| | | | | |
|---|---|---|---|---|
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM5_DUR
**Address** 0x50146044
**Description** durations config for pwm5 channel
**Reset** 0x00000000

| Name | hi_dur | | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM5_CFG
**Address** 0x50146048
**Description** configuration for pwm5 channel
**Reset** 0x00000000

| Name | update | RSVD | | | | | | | | | | | | | | | | frame_count | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** PWM6_CTRL
**Address** 0x5014604c
**Description** control signals for pwm6 channel
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | idle_ | i | | ok_ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | polarity | invert | mode | | to_run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | idle_polarity | polarity in idle state | 1'b0 |
| 3 | R/W | invert | invert pwm output. | 1'b0 |
| 2:1 | R/W | mode | mode of operations:<br>00 = CONTINUOUS_MODE: each frame is repeated indefinately. hi_dur and lo_dur parameters are sampled once per frame.<br>01 = COUNTER_MODE: a group of frames is run once after ok_to_run is set. hi_dur and lo_dur parameters are sampled once at the start of the group.<br>10 = IR_MODE: a group of frames is run and repeated indefinitely after ok_to_run is set. hi_dur, lo_dur and frame_count paramters are sampled once before each group is run.<br>11 = IR_FIFO_MODE: the same IR_MODE but the group parameters are pulled from the fifo. The mode is reserved if the fifo is not assigned to the channel. | 2'd0 |
| 0 | R/W | ok_to_run | start pwm channel with its rising edge. Its de-assertion stops the channel at the end of the next frame. | 1'b0 |

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**Name** PWM6_DUR
**Address** 0x50146050
**Description** durations config for pwm6 channel
**Reset** 0x00000000

| Name | hi_dur | | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM6_CFG
**Address** 0x50146054
**Description** configuration for pwm6 channel
**Reset** 0x00000000

| Name | upda | RSV | frame_c |
|---|---|---|---|

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| | te | D | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | out | | | | | | | | | | | | | | | | | | |

| | u r | | | | | | | | | | | | | | | u r | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** PWM7_CFG
**Address** 0x50146060
**Description** configuration for pwm7 channel
**Reset** 0x00000000

| Name | update | RSVD | frame_count |
|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | update | this bit synchronize channel parameters (lo/hi dur + frame count) update, when set it triggers HW capture and it auto-cleares afterwards during CONTINUOUS and IR_FIFO modes, this bit is don't care. | 1'b0 |
| 13:0 | R/W | frame_count | number of frames in the group = frame_count + 1 | 14'd0 |

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**Name** FSM_STATE
**Address** 0x50146064
**Description** fsm state
**Reset** 0x00000000

| Name | RSVD | ch7 | ch6 | ch5 | ch4 | ch3 | ch2 | ch1 | ch0 |
|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14** | **13 12** | **11 10** | **9 8** | **7 6** | **5 4** | **3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:14 | R | ch7 | current state of pwm7 channel. | 2'd0 |
| 13:12 | R | ch6 | current state of pwm6 channel. | 2'd0 |
| 11:10 | R | ch5 | current state of pwm5 channel. | 2'd0 |
| 9:8 | R | ch4 | current state of pwm4 channel. | 2'd0 |
| 7:6 | R | ch3 | current state of pwm3 channel. | 2'd0 |
| 5:4 | R | ch2 | current state of pwm2 channel. | 2'd0 |
| 3:2 | R | ch1 | current state of pwm1 channel. | 2'd0 |
| 1:0 | R | ch0 | current state of pwm0 channel. | 2'd0 |

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**Name** FIFO_CFG
**Address** 0x50146068
**Description** fifo configuration
**Reset** 0x0000000f

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | ch_stop | flush | RSVD | | | | lwm | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7 | R/W | ch_stop | when set, stops channel running in IR_FIFO_MODE after last fifo command is done.<br>otherwise, the last command keeps repeating. | 1'b0 |
| 6 | R/W | flush | Signal's rising edge clears out fifo. Not self-clearing | 1'b0 |
| 4:0 | R/W | lwm | low water mark (lwm) of FIF0; a trigger is issued when FIFO has been drained to these many open slots.<br>Constraint (1<= LWM <= 16)<br>triggers when the number of open slots changes from (lwm - 1) to (lwm).<br>lwm = 15 means an interrupt is issued as the number of open slots changes from 14 to 15.<br>In other words, the number of entries in the fifo changes from 2 to 1. | 5'd15 |

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**Name** FIFO_CARRIER1_DUR
**Address** 0x5014606c
**Description** carrier1 frequency durations for IR_FIFO_MODE
**Reset** 0x00000000

| Name | hi_dur | | | | | | | | | | | | | | | | lo_dur | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** FIFO_CARRIER2_DUR
**Address** 0x50146070
**Description** carrier2 frequency durations for IR_FIFO_MODE
**Reset** 0x00000000

| Name | hi_dur | lo_dur |
|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | hi_dur | duration of high portion of period in clk_mpc clock cycles = hi_dur. (hi_dur=1/lo_dur=0 not supported) | 16'd0 |
| 15:0 | R/W | lo_dur | duration of low portion of period in clk_mpc clock cycles = lo_dur. (lo_dur=1/hi_dur=0 not supported) | 16'd0 |

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**Name** FIFO_STAT
**Address** 0x50146074
**Description** fifo status
**Reset** 0x00000000

| Name | RSVD | empty | full |
|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | empty | fifo is empty. | 1'b0 |
| 0 | R | full | fifo is full. APB is stalled if a write is requested. | 1'b0 |

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**Name** FIFO_STAT1
**Address** 0x50146078
**Description** fifo status
**Reset** 0x00000010

| Name | RSVD | num_open_slots |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | num_open_slots | open slots in FIFO | 5'd16 |

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**Name** FIFO_SEL_ERR_SOURCE
**Address** 0x5014607c
**Description** HW sets and SW write 0 to clear and 1 to set these bits. Set takes precedence.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | ch | ch | ch | ch | ch | ch | ch | ch |

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | 7_fifo_sel_err | 6_fifo_sel_err | 5_fifo_sel_err | 4_fifo_sel_err | 3_fifo_sel_err | 2_fifo_sel_err | 1_fifo_sel_err | 0_fifo_sel_err |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7 | R/W | ch7_fifo_sel_err | pwm7 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 6 | R/W | ch6_fifo_sel_err | pwm6 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 5 | R/W | ch5_fifo_sel_err | pwm5 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 4 | R/W | ch4_fifo_sel_err | pwm4 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 3 | R/W | ch3_fifo_sel_err | pwm3 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 2 | R/W | ch2_fifo_sel_err | pwm2 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 1 | R/W | ch1_fifo_sel_err | pwm1 channel caused fifo_sel_err_intrpt; | 1'b0 |
| 0 | R/W | ch0_fifo_sel_err | pwm0 channel caused fifo_sel_err_intrpt; | 1'b0 |

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**Name** INTERRUPTS
**Address** 0x50146080
**Description** Contains register fields associated with interrupts.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | fifo_cmd_done_intrpt | fifo_sel_err_intrpt | fifo_lwm_hit_intrpt | fifo_overflow_intrpt |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | fifo_cmd_done_intrpt | HW sets when a channel is in IR_FIFO_MODE and last cmmand in fifo is done. | 1'b0 |
| 2 | R | fifo_sel_err_intrpt | HW sets when a channel is in IR_FIFO_MODE but not given the FIFO; FIFO being used by another channel.<br>With the conflict condition as described, an error assertion happens when channel is in idle and ok_to_run is issued.<br>Consequently, the internal state registers of the channel revert to default | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| | | | values. | |
| 1 | R | fifo_lwm_hit_intrpt | HW sets when lwm condition is hit. | 1'b0 |
| 0 | R | fifo_ovrflow_intrpt | HW sets when a write to fifo happens when fifo is full. | 1'b0 |

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**Name** INTERRUPTS_MASK
**Address** 0x50146084
**Description** Contains register fields associated with interrupts_mask.
**Reset** 0x00000003

| Name | RSVD | mask_intrpt3 | mask_intrpt2 | mask_intrpt1 | mask_intrpt0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | mask_intrpt3 | allows fifo_cmd_done_intrpt to propagate to channel's combined interrupt. | 1'b0 |
| 2 | R/W | mask_intrpt2 | allows fifo_sel_err_intrpt to propagate to channel's combined interrupt. | 1'b0 |
| 1 | R/W | mask_intrpt1 | allows lwm_hit_intrpt to propogate to channel's combined interrupt. | 1'b1 |
| 0 | R/W | mask_intrpt0 | allows fifo_ovrflow_intrpt to propogate to channel's combined interrupt. | 1'b1 |

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**Name** INTERRUPTS_CLEAR
**Address** 0x50146088
**Description** Contains register fields associated with interrupts_clear.
**Reset** 0x00000000

| Name | RSVD | clear_intrpt3 | clear_intrpt2 | clear_intrpt1 | clear_intrpt0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | clear_intrpt3 | clears fifo_cmd_done_intrpt; not auto cleared. | 1'b0 |
| 2 | R/W | clear_intrpt2 | clears fifo_sel_error_intrpt; not auto cleared. | 1'b0 |
| 1 | R/W | clear_intrpt1 | clears lwm_hit_intrpt; not auto cleared. | 1'b0 |
| 0 | R/W | clear_intrpt0 | clears fifo_overflow_intrpt; not auto cleared. | 1'b0 |

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**Name** INTERRUPTS_SET
**Address** 0x5014608c
**Description** Contains register fields associated with interrupts_set.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | set_intrpt3 | set_intrpt2 | set_intrpt1 | set_intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | set_intrpt3 | sets fifo_cmd_done_intrpt; not auto cleared. | 1'b0 |
| 2 | R/W | set_intrpt2 | sets fifo_sel_error_intrpt; not auto cleared. | 1'b0 |
| 1 | R/W | set_intrpt1 | sets lwm_hit_intrpt; not auto cleared. | 1'b0 |
| 0 | R/W | set_intrpt0 | sets fifo_overflow_intrpt; not auto cleared. | 1'b0 |

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**Name** FIFO_DATA
**Address** 0x50146ff0
**Description** write data to be stored in fifo
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | wdata | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | wdata | [15] selects carrier frequency<br>0: use carrier frequency defined by fifo_carrier1_dur<br>1: use carrier frequency defined by fifo_carrier2_dur<br>[14] selects between mark and space IR regions<br>0: output lo for the space region in IR.<br>space region frame: hi_dur=0, lo_dur= hi+lo dur of selected carrier frequency<br>max lo_dur is limited to 16bits.<br>1: output value defined by hi/lo dur for the mark region in IR<br>[13:0]: frame_count | 16'd0 |

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**Name** CORE_ID
**Address** 0x50146ff4
**Description** Contains register fields associated with core_id.
**Reset** 0x50574d20

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | PWM in ASCII | 32'h50574d20 |

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**Name** REV_HASH
**Address** 0x50146ff8
**Description** Contains register fields associated with rev_hash.
**Reset** 0x00000000

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | crc32 of this document; don't change value to anything other than 32'h00000000; scripts will fill it out in the verilog | 32'h00000000 |

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**Name** REV_KEY
**Address** 0x50146ffc
**Description** Contains register fields associated with rev_key.
**Reset** 0x52455620

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | REV in ASCII | 32'h52455620 |

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# SPI0_SECURE

**Name** CTRL
**Address** 0x50147000
**Description** SPI Control Register
**Reset** 0x00010301

| Name | dma_mode | RSVD | | | | | | | | | | spi3wire_en | s_wr_opcode | | | | | | | | s_rd_opcode | | | | | | | | cpol | cpha | mstr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | dma_mode | 1 = dma controls spi core | 1'b0 |
| 19 | R/W | spi3wire_en | 1=enable 3-wire spi transactions, mosi pad becomes bi-directional<br>0=default 4-wire spi transactions | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 18:11 | R/W | s_wr_opcode | slave write opcode | 8'h20 |
| 10:3 | R/W | s_rd_opcode | slave read opcode | 8'h60 |
| 2 | R/W | cpol | spi clock polarity 0=spi clock low in idle; cs high 1=spi clock high in idle; cs high | 1'b0 |
| 1 | R/W | cpha | spi clock phase 0=leading edge sample, trailing edge toggle 1=leading edge toggle, trailing edge sample | 1'b0 |
| 0 | R/W | mstr | 1=master interface<br>0=slave interface | 1'b1 |

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**Name** TRANSACTION_SETUP
**Address** 0x50147004
**Description** transaction setup
**Reset** 0x00000000

| Name | csn_bubble | invert_mosi | dummy_cycles | | | | csn_stays_low | loopback | opcode | | | | | | clkdiv | | | | | | | | rwb | num_data_bytes | | | | start |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | csn_bubble | introduce a delay between spi_csn falling and first bit on mosi | 1'b0 |
| 30 | R/W | invert_mosi | invert mosi; keep static during entire spi transaction | 1'b0 |
| 29:26 | R/W | dummy_cycles | number of extra shift cycles to help with turn-around | 4'd0 |
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 24 | R/W | loopback | if 1, mosi is internally connected to miso | 1'b0 |
| 23:16 | R/W | opcode | opcode | 8'd0 |
| 15:6 | R/W | clkdiv | freq_sclk = f_pclk/(2*(clkdiv+1)) | 10'd0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; not auto cleared | 1'b0 |

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**Name** TRANSACTION_SETUP_DMA
**Address** 0x50147008
**Description** transaction setup by DMA
**Reset** 0x00000000

| Name | RSVD | csn_stays | RSVD | rwb | num_data_ | star |
|---|---|---|---|---|---|---|

CONFIDENTIAL

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | _low | | | | | | | | | | | | | | | | | | | bytes | | | | t |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; auto cleared | 1'b0 |

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**Name** TRANSACTION_STATUS
**Address** 0x5014700c
**Description** transaction status
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | opcode_status | | | | | | | | RSVD | | | | | | running | done |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R | opcode_status | master or slave opcode status | 8'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | done | 1=indicate master or slave read/write transaction is done | 1'b0 |

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**Name** DATA_BYTES_LOWER
**Address** 0x50147010
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores read/write address
**Reset** 0x00000000

| Name | byte3 | | | | | | | | byte2 | | | | | | | | byte1 | | | | | | | | byte0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte3 | data byte3 | 8'b0 |
| 23:16 | R/W | byte2 | data byte2 | 8'b0 |
| 15:8 | R/W | byte1 | data byte1 | 8'b0 |

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| | | | | |
|---|---|---|---|---|
| 7:0 | R/W | byte0 | data byte0 | 8'b0 |

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**Name** DATA_BYTES_UPPER
**Address** 0x50147014
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores write data; preload read data
**Reset** 0x00000000

| Name | byte7 | | | | | | | | byte6 | | | | | | | | byte5 | | | | | | | | byte4 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte7 | data byte7 | 8'b0 |
| 23:16 | R/W | byte6 | data byte6 | 8'b0 |
| 15:8 | R/W | byte5 | data byte5 | 8'b0 |
| 7:0 | R/W | byte4 | data byte4 | 8'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x50147018
**Description** interrupt mask
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | passthru3 | passthru2 | passthru1 | passthru0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | passthru3 | 1=allow interrupt3 to be OR'ed into core interrupt | 1'b0 |
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x5014701c
**Description** interrupt status
**Reset** 0x00000000

| Name | RSV | | | | | | | | | | | | | | | | | | | | | | | | | | | | interr | interr | interr | interr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | D | | | | | | | | | | | | | | | | | | | | | | | | | | | | upt3 | upt2 | upt1 | upt0 |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | interrupt3 | from slave, error read-data invalid; independent of mask | 1'b0 |
| 2 | R | interrupt2 | from slave, error unknown command; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, transaction done capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, transaction done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x50147020
**Description** interrupt set
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | set_interrupt3 | set_interrupt2 | set_interrupt1 | set_interrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | set_interrupt3 | set interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | set_interrupt2 | set interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | set_interrupt1 | set interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt0; not auto cleared | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x50147024
**Description** interrupt clear
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | reset_interrupt | reset_interrupt | reset_interrupt | reset_interrupt |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | t3 | t2 | t1 | t0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | reset_interrupt3 | clear interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | reset_interrupt2 | clear interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | reset_interrupt1 | clear interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | reset_interrupt0 | clear interrupt0; not auto cleared | 1'b0 |

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**Name** CORE_ID
**Address** 0x50147ffc
**Description** core ID
**Reset** 0x53504920

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | SPI in ASCII | 32'h53504920 |

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# KSM_SECURE

**Name** CTRL0
**Address** 0x5014a000
**Description** Contains register fields associated with ctrl0.
**Reset** 0x00000800

| Name | RSVD | | | | | | | | | | | | | | | | | | | ksm_not_empty_to_interrupt | consecscan | RSVD | | | | | | | flush | pop | loopback | go |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| | | | | |

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| | | | | |
|---|---|---|---|---|
| 12 | R/W | ksm_not_empty_to_intrpt | in addition to fifo_write, /not empty/ signal will also feed into setting intrpt[0] | 1'b0 |
| 11:8 | R/W | consecscan | how many consecutive scans are needed before key press or key release event? (0 is INVALID) | 4'd8 |
| 3 | R/W | flush | flushes the key packet fifo | 1'b0 |
| 2 | R/W | pop | read from keyboard event fifo; not self reseting | 1'b0 |
| 1 | R/W | loopback | enable lpback_ksi (see below) | 1'b0 |
| 0 | R/W | go | start the keyboard scanning; will scan while go is high; not self reseting | 1'b0 |

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**Name** TIME_PARAM0
**Address** 0x5014a004
**Description** Contains register fields associated with time_param0.
**Reset** 0x00100010

| Name | t1 | | | | | | | | | | | | | | | | t2 | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | t1 | from col falling to sampling first row | 16'd16 |
| 15:0 | R/W | t2 | from sampling last row to col rising | 16'd16 |

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**Name** TIME_PARAM1
**Address** 0x5014a008
**Description** Contains register fields associated with time_param1.
**Reset** 0x00080040

| Name | t3 | | | | | | | | | | | | | | | | t4 | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | t3 | gap between adjacent row samplings | 16'd8 |
| 15:0 | R/W | t4 | gap between col rising and next col falling | 16'd64 |

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**Name** MATRIX_SIZE
**Address** 0x5014a00c
**Description** Contains register fields associated with matrix_size.
**Reset** 0x00001307

| Name | RSVD | | | | | | | | | | | | | | | | | | | num_row | | | | | RSVD | | | num_col | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 12:8 | R/W | num_row | the actual number of rows is num_row + 1 | 5'd19 |
| 4:0 | R/W | num_col | the actual number of cols is num_col + 1 | 5'd7 |

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**Name** KEYBOARD_PACKET
**Address** 0x5014a040
**Description** Contains register fields associated with keyboard_packet.
**Reset** 0x00000000

| Name | RSVD | empty | pressed_released_n | col | row |
|---|---|---|---|---|---|
| **Bits** | **31:12** | **11** | **10** | **9:5** | **4:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 11 | R | empty | if 1, ignore this pack et | 1'd0 |
| 10 | R | pressed_released_n | 1=pressed 0=released | 1'b0 |
| 9:5 | R | col | the column of the pressed or released key; 0-indexed | 5'd0 |
| 4:0 | R | row | the row of the pressed or released key; 0-indexed | 5'd0 |

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**Name** LOW_POWER
**Address** 0x5014a044
**Description** Contains register fields associated with low_power.
**Reset** 0x00000000

| Name | RSVD | key |
|---|---|---|
| **Bits** | **31:11** | **10:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 10:0 | R | key | this is the key press or release that woke the device | 11'd0 |

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**Name** LPBACK_KSI
**Address** 0x5014a058
**Description** Contains register fields associated with lpback_ksi.
**Reset** 0x000fffff

| | RS | lpbac |
|---|---|---|

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| Name | VD | | | | | | | | | | | lpback_ksi | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:0 | R/W | lpback_ksi | the values to be driven on each of the ksi lines (while in loopback) | 20'hfffff |

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**Name** STATUS
**Address** 0x5014a080
**Description** Contains register fields associated with status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | command_taken |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | command_taken | fsm in on 32kHz clock; it takes two 32kHz cycles for the command to be picked up | 1'b0 |

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**Name** INTERRUPTS
**Address** 0x5014a0c0
**Description** Contains register fields associated with interrupts.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | intrpt3 | intrpt2 | intrpt1 | intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | intrpt3 | flush read is done | 1'b0 |
| 2 | R | intrpt2 | pop read is done | 1'b0 |
| 1 | R | intrpt1 | keyboard event fifo overran | 1'b0 |
| 0 | R | intrpt0 | hardware wrote packet to keyboard event fifo | 1'b0 |

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**Name** INTERRUPT_MASK

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**Address** 0x5014a0c4
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x0000000f

| Name | RSVD | mask_intrpt3 | mask_intrpt2 | mask_intrpt1 | mask_intrpt0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | mask_intrpt3 | allow intrpt3 to propogate to core's single output interrupt | 1'b1 |
| 2 | R/W | mask_intrpt2 | allow intrpt2 to propogate to core's single output interrupt | 1'b1 |
| 1 | R/W | mask_intrpt1 | allow intrpt1 to propogate to core's single output interrupt | 1'b1 |
| 0 | R/W | mask_intrpt0 | allow intrpt0 to propogate to core's single output interrupt | 1'b1 |

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**Name** INTERRUPT_CLEAR
**Address** 0x5014a0c8
**Description** Contains register fields associated with interrupt_clear.
**Reset** 0x00000000

| Name | RSVD | clear_intrpt3 | clear_intrpt2 | clear_intrpt1 | clear_intrpt0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | clear_intrpt3 | clear interrupt 3; not self reseting | 1'b0 |
| 2 | R/W | clear_intrpt2 | clear interrupt 2; not self reseting | 1'b0 |
| 1 | R/W | clear_intrpt1 | clear interrupt 1; not self reseting | 1'b0 |
| 0 | R/W | clear_intrpt0 | clear interrupt 0; not self reseting | 1'b0 |

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**Name** DEBUG
**Address** 0x5014aff4
**Description** Contains register fields associated with debug.
**Reset** 0x00000000

| | fi | k |
|---|---|---|

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| Name | fo | | | | | | | | | | | | | | | b | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R | fifo | {overrun, underrun, full, empty, 4'd0, rdaddr(unsigned int; 4b), wraddr(unsigned int; 4b)} | 16'b0 |
| 15:0 | R | kb | {4'b0, debounce_available(array of booleans; 12b)} | 16'b0 |

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**Name** DEBUG2
**Address** 0x5014aff8
**Description** Contains register fields associated with debug2.
**Reset** 0x00000000

| Name | kp | | | | | | | | | | | | | | | kr | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R | kp | how many keys have been pressed? | 16'b0 |
| 15:0 | R | kr | how many keys have been released? | 16'b0 |

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**Name** CORE_ID
**Address** 0x5014affc
**Description** Contains register fields associated with core_id.
**Reset** 0x4b534d20

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | KSM in ASCII | 32'h4b534d20 |

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# SPI2_SECURE

**Name** CTRL
**Address** 0x5014b000
**Description** SPI Control Register
**Reset** 0x00010301

| Name | dma_mode | RSVD | spi3wire_e | s_wr_opcod | s_rd_opcod | cpol | cpha | mstr |
|---|---|---|---|---|---|---|---|---|

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| | | | | | | | | | | | | | n | | | e | | | | | | e | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | dma_mode | 1 = dma controls spi core | 1'b0 |
| 19 | R/W | spi3wire_en | 1=enable 3-wire spi transactions, mosi pad becomes bi-directional<br>0=default 4-wire spi transactions | 1'b0 |
| 18:11 | R/W | s_wr_opcode | slave write opcode | 8'h20 |
| 10:3 | R/W | s_rd_opcode | slave read opcode | 8'h60 |
| 2 | R/W | cpol | spi clock polarity 0=spi clock low in idle; cs high 1=spi clock high in idle; cs high | 1'b0 |
| 1 | R/W | cpha | spi clock phase 0=leading edge sample, trailing edge toggle 1=leading edge toggle, trailing edge sample | 1'b0 |
| 0 | R/W | mstr | 1=master interface<br>0=slave interface | 1'b1 |

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**Name** TRANSACTION_SETUP
**Address** 0x5014b004
**Description** transaction setup
**Reset** 0x00000000

| Name | csn_bubble | invert_mosi | dummy_cycles | csn_stays_low | loopback | opcode | clkdiv | rwb | num_data_bytes | start |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29 28 27 26** | **25** | **24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6** | **5** | **4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | csn_bubble | introduce a delay between spi_csn falling and first bit on mosi | 1'b0 |
| 30 | R/W | invert_mosi | invert mosi; keep static during entire spi transaction | 1'b0 |
| 29:26 | R/W | dummy_cycles | number of extra shift cycles to help with turn-around | 4'd0 |
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 24 | R/W | loopback | if 1, mosi is internally connected to miso | 1'b0 |
| 23:16 | R/W | opcode | opcode | 8'd0 |
| 15:6 | R/W | clkdiv | freq_sclk = f_pclk/(2*(clkdiv+1)) | 10'd0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; not auto cleared | 1'b0 |

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**Name** TRANSACTION_SETUP_DMA
**Address** 0x5014b008
**Description** transaction setup by DMA
**Reset** 0x00000000

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| Name | RSVD | csn_stays_low | RSVD | rwb | num_data_bytes | start |
|---|---|---|---|---|---|---|
| **Bits** | **31:26** | **25** | **24:6** | **5** | **4:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; auto cleared | 1'b0 |

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**Name** TRANSACTION_STATUS
**Address** 0x5014b00c
**Description** transaction status
**Reset** 0x00000000

| Name | RSVD | opcode_status | RSVD | running | done |
|---|---|---|---|---|---|
| **Bits** | **31:16** | **15:8** | **7:2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R | opcode_status | master or slave opcode status | 8'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | done | 1=indicate master or slave read/write transaction is done | 1'b0 |

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**Name** DATA_BYTES_LOWER
**Address** 0x5014b010
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores read/write address
**Reset** 0x00000000

| Name | byte | byte | byte | byte |
|---|---|---|---|---|

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| | 3 | | | | | | | | 2 | | | | | | | | 1 | | | | | | | | 0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte3 | data byte3 | 8'b0 |
| 23:16 | R/W | byte2 | data byte2 | 8'b0 |
| 15:8 | R/W | byte1 | data byte1 | 8'b0 |
| 7:0 | R/W | byte0 | data byte0 | 8'b0 |

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**Name** DATA_BYTES_UPPER
**Address** 0x5014b014
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores write data; preload read data
**Reset** 0x00000000

| | b y t e 7 | | | | | | | | b y t e 6 | | | | | | | | b y t e 5 | | | | | | | | b y t e 4 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte7 | data byte7 | 8'b0 |
| 23:16 | R/W | byte6 | data byte6 | 8'b0 |
| 15:8 | R/W | byte5 | data byte5 | 8'b0 |
| 7:0 | R/W | byte4 | data byte4 | 8'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x5014b018
**Description** interrupt mask
**Reset** 0x00000000

| | R S V D | | | | | | | | | | | | | | | | | | | | | | | | | | | | p a s s t h r u 3 | p a s s t h r u 2 | p a s s t h r u 1 | p a s s t h r u 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | passthru3 | 1=allow interrupt3 to be OR'ed into core interrupt | 1'b0 |
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x5014b01c
**Description** interrupt status

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**Reset** 0x00000000

| Name | RSVD | interrupt3 | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | interrupt3 | from slave, error read-data invalid; independent of mask | 1'b0 |
| 2 | R | interrupt2 | from slave, error unknown command; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, transaction done capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, transaction done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x5014b020
**Description** interrupt set
**Reset** 0x00000000

| Name | RSVD | set_interrupt3 | set_interrupt2 | set_interrupt1 | set_interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | set_interrupt3 | set interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | set_interrupt2 | set interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | set_interrupt1 | set interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt0; not auto cleared | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x5014b024
**Description** interrupt clear
**Reset** 0x00000000

| | | reset_ | reset_ | reset_ | reset_ |
|---|---|---|---|---|---|

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| Name | RSVD | interrupt3 | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | reset_interrupt3 | clear interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | reset_interrupt2 | clear interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | reset_interrupt1 | clear interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | reset_interrupt0 | clear interrupt0; not auto cleared | 1'b0 |

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**Name** CORE_ID
**Address** 0x5014bffc
**Description** core ID
**Reset** 0x53504920

| Name | id |
|---|---|
| **Bits** | **31:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | SPI in ASCII | 32'h53504920 |

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# QDEC_SECURE

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**Name** SETUP_FOR_AXIS_0
**Address** 0x5014c000
**Description** Contains register fields associated with setup_for_axis_0.
**Reset** 0x00000030

| Name | RSVD | purge_err_post_read | purge_accum_post_rea | reverse_polarity | qsig_a_loopback_val | qsig_b_loopback_val | enable_loopback |
|---|---|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | d | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | purge_err_post_read | (no description) | 1'b1 |
| 4 | R/W | purge_accum_post_read | (no description) | 1'b1 |
| 3 | R/W | reverse_polarity | (no description) | 1'b0 |
| 2 | R/W | qsig_a_loopback_val | (no description) | 1'b0 |
| 1 | R/W | qsig_b_loopback_val | (no description) | 1'b0 |
| 0 | R/W | enable_loopback | (no description) | 1'b0 |

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**Name** ACCUM_PURGE_STATUS_FOR_AXIS_0
**Address** 0x5014c004
**Description** Contains register fields associated with accum_purge_status_for_axis_0.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | accum_purge_sts |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | accum_purge_sts | (no description) | 1'b0 |

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**Name** ACCUMULATOR_VALUE_FOR_AXIS_0
**Address** 0x5014c008
**Description** Contains register fields associated with accumulator_value_for_axis_0.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | accum_val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | accum_val | (no description) | 32'b0 |

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**Name** ERR_PURGE_STATUS_FOR_AXIS_0
**Address** 0x5014c00c
**Description** Contains register fields associated with err_purge_status_for_axis_0.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | err_purge_sts |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | err_purge_sts | (no description) | 1'b0 |

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**Name** ERROR_COUNT_FOR_AXIS_0
**Address** 0x5014c010
**Description** Contains register fields associated with error_count_for_axis_0.
**Reset** 0x00000000

| Name | err_cnt | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | err_cnt | (no description) | 32'b0 |

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**Name** SETUP_FOR_AXIS_1
**Address** 0x5014c014
**Description** Contains register fields associated with setup_for_axis_1.
**Reset** 0x00000030

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | purge_err_pos | purge_accum_pol | reverse_pola | qsig_a_loopba | qsig_b_loopba | enable_loop |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | t_read | st_read | rity | ck_val | ck_val | back |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | purge_err_post_read | (no description) | 1'b1 |
| 4 | R/W | purge_accum_post_read | (no description) | 1'b1 |
| 3 | R/W | reverse_polarity | (no description) | 1'b0 |
| 2 | R/W | qsig_a_loopback_val | (no description) | 1'b0 |
| 1 | R/W | qsig_b_loopback_val | (no description) | 1'b0 |
| 0 | R/W | enable_loopback | (no description) | 1'b0 |

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**Name** ACCUM_PURGE_STATUS_FOR_AXIS_1
**Address** 0x5014c018
**Description** Contains register fields associated with accum_purge_status_for_axis_1.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | accum_purge_sts |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | accum_purge_sts | (no description) | 1'b0 |

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---

**Name** ACCUMULATOR_VALUE_FOR_AXIS_1
**Address** 0x5014c01c
**Description** Contains register fields associated with accumulator_value_for_axis_1.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | accum_val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | accum_val | (no description) | 32'b0 |

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**Name** ERR_PURGE_STATUS_FOR_AXIS_1
**Address** 0x5014c020
**Description** Contains register fields associated with err_purge_status_for_axis_1.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | err_purge_sts |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | err_purge_sts | (no description) | 1'b0 |

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**Name** ERROR_COUNT_FOR_AXIS_1
**Address** 0x5014c024
**Description** Contains register fields associated with error_count_for_axis_1.
**Reset** 0x00000000

| Name | err_cnt | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | err_cnt | (no description) | 32'b0 |

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**Name** SETUP_FOR_AXIS_2
**Address** 0x5014c028
**Description** Contains register fields associated with setup_for_axis_2.
**Reset** 0x00000030

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | purge_e | purge_ac | revers | qsig_a_ | qsig_b_ | enabl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Name | RSVD | err_post_read | cum_post_read | e_polarity | loopback_val | loopback_val | e_loopback |
|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | purge_err_post_read | (no description) | 1'b1 |
| 4 | R/W | purge_accum_post_read | (no description) | 1'b1 |
| 3 | R/W | reverse_polarity | (no description) | 1'b0 |
| 2 | R/W | qsig_a_loopback_val | (no description) | 1'b0 |
| 1 | R/W | qsig_b_loopback_val | (no description) | 1'b0 |
| 0 | R/W | enable_loopback | (no description) | 1'b0 |

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**Name** ACCUM_PURGE_STATUS_FOR_AXIS_2
**Address** 0x5014c02c
**Description** Contains register fields associated with accum_purge_status_for_axis_2.
**Reset** 0x00000000

| Name | RSVD | accum_purge_sts |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | accum_purge_sts | (no description) | 1'b0 |

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**Name** ACCUMULATOR_VALUE_FOR_AXIS_2
**Address** 0x5014c030
**Description** Contains register fields associated with accumulator_value_for_axis_2.
**Reset** 0x00000000

| Name | accum |
|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | _val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | accum_val | (no description) | 32'b0 |

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**Name** ERR_PURGE_STATUS_FOR_AXIS_2
**Address** 0x5014c034
**Description** Contains register fields associated with err_purge_status_for_axis_2.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | err_purge_sts |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | err_purge_sts | (no description) | 1'b0 |

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**Name** ERROR_COUNT_FOR_AXIS_2
**Address** 0x5014c038
**Description** Contains register fields associated with error_count_for_axis_2.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | err_cnt | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | err_cnt | (no description) | 32'b0 |

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**Name** CORE_ID
**Address** 0x5014cffc
**Description** Contains register fields associated with core_id.
**Reset** 0x51444543

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | i | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

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| Name | d | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | QDEC in ASCII | 32'h51444543 |

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# SPI1_SECURE

**Name** CTRL
**Address** 0x5014d000
**Description** SPI Control Register
**Reset** 0x00010301

| Name | dma_mode | RSVD | spi3wire_en | s_wr_opcode | s_rd_opcode | cpol | cpha | mstr |
|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30:20** | **19** | **18:11** | **10:3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | dma_mode | 1 = dma controls spi core | 1'b0 |
| 19 | R/W | spi3wire_en | 1=enable 3-wire spi transactions, mosi pad becomes bi-directional<br>0=default 4-wire spi transactions | 1'b0 |
| 18:11 | R/W | s_wr_opcode | slave write opcode | 8'h20 |
| 10:3 | R/W | s_rd_opcode | slave read opcode | 8'h60 |
| 2 | R/W | cpol | spi clock polarity 0=spi clock low in idle; cs high 1=spi clock high in idle; cs high | 1'b0 |
| 1 | R/W | cpha | spi clock phase 0=leading edge sample, trailing edge toggle 1=leading edge toggle, trailing edge sample | 1'b0 |
| 0 | R/W | mstr | 1=master interface<br>0=slave interface | 1'b1 |

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**Name** TRANSACTION_SETUP
**Address** 0x5014d004
**Description** transaction setup
**Reset** 0x00000000

| Name | csn_bubb | invert_m | dummy_cyc | csn_stays | loopbac | opcode | clkdiv | rwb | num_data_b | start |
|---|---|---|---|---|---|---|---|---|---|---|

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| | le | osi | les | _low | k | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | csn_bubble | introduce a delay between spi_csn falling and first bit on mosi | 1'b0 |
| 30 | R/W | invert_mosi | invert mosi; keep static during entire spi transaction | 1'b0 |
| 29:26 | R/W | dummy_cycles | number of extra shift cycles to help with turn-around | 4'd0 |
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 24 | R/W | loopback | if 1, mosi is internally connected to miso | 1'b0 |
| 23:16 | R/W | opcode | opcode | 8'd0 |
| 15:6 | R/W | clkdiv | freq_sclk = f_pclk/(2*(clkdiv+1)) | 10'd0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; not auto cleared | 1'b0 |

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**Name** TRANSACTION_SETUP_DMA
**Address** 0x5014d008
**Description** transaction setup by DMA
**Reset** 0x00000000

| Name | RSVD | csn_stays_low | RSVD | rwb | num_data_bytes | start |
|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26** | **25** | **24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6** | **5** | **4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; auto cleared | 1'b0 |

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**Name** TRANSACTION_STATUS
**Address** 0x5014d00c
**Description** transaction status
**Reset** 0x00000000

| | R | opcode | R | run | d |
|---|---|---|---|---|---|

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| Name | SVD | _status | SVD | ning | one |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R | opcode_status | master or slave opcode status | 8'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | done | 1=indicate master or slave read/write transaction is done | 1'b0 |

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**Name** DATA_BYTES_LOWER
**Address** 0x5014d010
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores read/write address
**Reset** 0x00000000

| Name | byte3 | byte2 | byte1 | byte0 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte3 | data byte3 | 8'b0 |
| 23:16 | R/W | byte2 | data byte2 | 8'b0 |
| 15:8 | R/W | byte1 | data byte1 | 8'b0 |
| 7:0 | R/W | byte0 | data byte0 | 8'b0 |

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**Name** DATA_BYTES_UPPER
**Address** 0x5014d014
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores write data; preload read data
**Reset** 0x00000000

| Name | byte7 | byte6 | byte5 | byte4 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte7 | data byte7 | 8'b0 |
| 23:16 | R/W | byte6 | data byte6 | 8'b0 |
| 15:8 | R/W | byte5 | data byte5 | 8'b0 |
| 7:0 | R/W | byte4 | data byte4 | 8'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x5014d018
**Description** interrupt mask
**Reset** 0x00000000

| Name | RSVD | passthru3 | passthru2 | passthru1 | passthru0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | passthru3 | 1=allow interrupt3 to be OR'ed into core interrupt | 1'b0 |
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x5014d01c
**Description** interrupt status
**Reset** 0x00000000

| Name | RSVD | interrupt3 | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | interrupt3 | from slave, error read-data invalid; independent of mask | 1'b0 |
| 2 | R | interrupt2 | from slave, error unknown command; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, transaction done capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, transaction done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x5014d020
**Description** interrupt set
**Reset** 0x00000000

| | RS | set_int | set_int | set_int | set_int |
|---|---|---|---|---|---|

CONFIDENTIAL

| Name | VD | | |

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# SLWTIMER_SECURE

**Name** CONTROL
**Address** 0x5014e000
**Description** control
**Reset** 0x00000000

| Name | RSVD | auto_reload | RSVD | reset_thres2 | reset_thres1 | reset_thres0 | reset_expired | load | clear |
|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:9** | **8** | **7:6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8 | R/W | auto_reload | auto reload slow timer with initial value after expiring | 1'b0 |
| 5 | R/W | reset_thres2 | reset thres2 interrupt; not self clearing | 1'b0 |
| 4 | R/W | reset_thres1 | reset thres1 interrupt; not self clearing | 1'b0 |
| 3 | R/W | reset_thres0 | reset thres0 interrupt; not self clearing | 1'b0 |
| 2 | R/W | reset_expired | reset expired interrupt; not self clearing | 1'b0 |
| 1 | R/W | load | load slow timer | 1'b0 |
| 0 | R/W | clear | clear slow timer | 1'b0 |

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**Name** INIT_LOW
**Address** 0x5014e004
**Description** initial count value
**Reset** 0x00000000

| Name | init_low |
|---|---|
| **Bits** | **31:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | init_low | lower 32b of the 40b initial count value | 32'b0 |

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**Name** INIT_HIGH
**Address** 0x5014e008
**Description** initial count value
**Reset** 0x00000000

CONFIDENTIAL

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | init_high | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | init_high | upper 8b of the 40b initial count value | 8'b0 |

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**Name** CNT_LOW
**Address** 0x5014e00c
**Description** current count value
**Reset** 0x00000000

| Name | cnt_low | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | cnt_low | lower 32b of the 40b current count value | 32'b0 |

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**Name** CNT_HIGH
**Address** 0x5014e010
**Description** current count value
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | cnt_high | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | cnt_high | upper 8b of the 40b current count value | 8'b0 |

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**Name** THRES0_LOW
**Address** 0x5014e014
**Description** threshold 0 value
**Reset** 0x00000000

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| Name | thres0_low | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | thres0_low | lower 32b of the 40b threshold 0 value; when counter decrements from one above to this value, an optional interrupt triggers | 32'b0 |

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**Name** THRES0_HIGH
**Address** 0x5014e018
**Description** threshold 0 value
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | thres0_high | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | thres0_high | upper 8b of the 40b theshold 0 value | 8'b0 |

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**Name** THRES1_LOW
**Address** 0x5014e01c
**Description** threshold 1 value
**Reset** 0x00000000

| Name | thres1_low | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | thres1_low | lower 32b of the 40b threshold 1 value; when counter decrements from one above to this value, an optional interrupt triggers | 32'b0 |

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**Name** THRES1_HIGH
**Address** 0x5014e020
**Description** threshold 1 value
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | thres1_high | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | thres1_high | upper 8b of the 40b theshold 1 value | 8'b0 |

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**Name** THRES2_LOW
**Address** 0x5014e024
**Description** threshold 2 value
**Reset** 0x00000000

| Name | thres2_low | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | thres2_low | lower 32b of the 40b threshold 2 value; when counter decrements from one above to this value, an optional interrupt triggers | 32'b0 |

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**Name** THRES2_HIGH
**Address** 0x5014e028
**Description** threshold 2 value
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | thres2_hig | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | h | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | thres2_high | upper 8b of the 40b theshold 2 value | 8'b0 |

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**Name** STATUS
**Address** 0x5014e02c
**Description** status information
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | op_running |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | op_running | the operation (either load or clear) is still running; handoff to 32kHz domain takes time | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x5014e030
**Description** interrupt status
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | hit_thres2 | hit_thres1 | hit_thres0 | expired |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | hit_thres2 | threshold 2 interrupt triggered | 1'b0 |
| 2 | R | hit_thres1 | threshold 1 interrupt triggered | 1'b0 |
| 1 | R | hit_thres0 | threshold 0 interrupt triggered | 1'b0 |
| 0 | R | expired | counter transitioned from 1 to 0 | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x5014e034
**Description** interrupt mask
**Reset** 0x0000000f

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| Name | RSVD | passthru_hit_thres2 | passthru_hit_thres1 | passthru_hit_thres0 | passthru_expired |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | passthru_hit_thres2 | 1=allow hit_thres2 to be OR'd into core interrupt | 1'b1 |
| 2 | R/W | passthru_hit_thres1 | 1=allow hit_thres1 to be OR'd into core interrupt | 1'b1 |
| 1 | R/W | passthru_hit_thres0 | 1=allow hit_thres0 to be OR'd into core interrupt | 1'b1 |
| 0 | R/W | passthru_expired | 1=allow counter expiring to be OR'd into core interrupt | 1'b1 |

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**Name** CORE_ID
**Address** 0x5014effc
**Description** core ID
**Reset** 0x53544d52

| Name | id |
|---|---|
| **Bits** | **31:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | 'STMR' in hex | 32'h53544d52 |

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# QSPI_SECURE

**Name** TRANSACTION_SETUP
**Address** 0x5014f000
**Description** Contains register fields associated with transaction_setup.
**Reset** 0x22000000

| | | remote | remote_ah | | | | | d | | d | | d | | d | s |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | | remote_spi_has_control | remote_ahb_qspi_has_control | RSVD | | csn_val | clk_val | RSVD | | dout_3_ctrl | | RSVD | | dout_2_ctrl | | RSVD | | dout_1_ctrl | | RSVD | | dout_0_ctrl | | sample_din | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29 | R/W | remote_spi_has_control | (no description) | 1'b1 |
| 28 | R/W | remote_ahb_qspi_has_control | (no description) | 1'b0 |
| 25 | R/W | csn_val | (no description) | 1'b1 |
| 24 | R/W | clk_val | (no description) | 1'b0 |
| 21:20 | R/W | dout_3_ctrl | {oe val, drive val} | 2'h0 |
| 17:16 | R/W | dout_2_ctrl | {oe val, drive val} | 2'h0 |
| 13:12 | R/W | dout_1_ctrl | {oe val, drive val} | 2'h0 |
| 9:8 | R/W | dout_0_ctrl | {oe val, drive val} | 2'h0 |
| 7:0 | R/W | sample_din | (no description) | 8'h0 |

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**Name** READ_DATA
**Address** 0x5014f004
**Description** Contains register fields associated with read_data.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | rdata | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R | rdata | (no description) | 8'b0 |

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**Name** OVERRIDE_DIN
**Address** 0x5014f008
**Description** Contains register fields associated with override_din.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | over | din_ov |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | override_din | din_override_val |
|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 | 4 | 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | override_din | (no description) | 1'b0 |
| 3:0 | R/W | din_override_val | (no description) | 4'b0 |

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**Name** MODE
**Address** 0x5014f00c
**Description** Contains register fields associated with mode.
**Reset** 0x00000002

| Name | RSVD | is_quad | is_dual |
|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | is_quad | high priority than is_dual | 1'b1 |
| 0 | R/W | is_dual | for debug | 1'b0 |

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**Name** REMOTE_AHB_SETUP
**Address** 0x5014f010
**Description** Contains register fields associated with remote_ahb_setup.
**Reset** 0x02030000

| Name | RSVD | slip_half_cycle | hyper | skew_csn_act_wen | serialize_pp_addre | invalidate_entire_c | enable_fine_clock_ga | enable_cache | enable_clocks | wdata_wordswap | wdata_halfword_sw | wdata_byteswap | quad_overhead | rdata_byteswap | clkdivsel | opcode | RSVD | is_opcode | mode | dummy_cycles |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | | | | | s s | a c h e | t i n g | | | | a p | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 30 | R/W | slip_half_cycle | (no description) | 1'b0 |
| 29 | R/W | hyper | core runs at 32MHz instead of 16MHz | 1'b0 |
| 28 | R/W | skew_csn_act_wen | in a write sequence, controller stalls the start of the sequence; set if div-by-1 | 1'b0 |
| 27 | R/W | serialize_pp_address | (no description) | 1'b0 |
| 26 | R/W | invalidate_entire_cache | (no description) | 1'b0 |
| 25 | R/W | enable_fine_clock_gating | (no description) | 1'b1 |
| 24 | R/W | enable_cache | (no description) | 1'b0 |
| 23 | R/W | enable_clocks | for the core clocks not the qspi interface clock | 1'b0 |
| 22 | R/W | wdata_word_swap | (no description) | 1'b0 |
| 21 | R/W | wdata_halfword_swap | (no description) | 1'b0 |
| 20 | R/W | wdata_byte_swap | (no description) | 1'b0 |
| 19 | R/W | quad_overhead | adds overhead cycles during quad I/O read mode<br>1 = opcode is fully serial (8-cycles), adds two cycles for performance enhance; (macronix, gigadev, winbond)<br>0 = opcode is parallelized (2-cycles); no performance enhance cycles needed (micron) | 1'b0 |
| 18 | R/W | rdata_byte_swap | does a byte swap before passing rdata to backplane | 1'b0 |
| 17:16 | R/W | clkdivsel | 0=1x 1=2x 2=4x 3=8x | 2'd3 |
| 15:8 | R/W | opcode | (no description) | 8'h0 |
| 6 | R/W | is_opcode | (no description) | 1'b0 |
| 5:4 | R/W | mode | 0,3=single 1=dual 2=quad | 2'd0 |
| 3:0 | R/W | dummy_cycles | (no description) | 4'd0 |

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**Name** REMOTE_AHB_SETUP_2
**Address** 0x5014f014
**Description** Contains register fields associated with remote_ahb_setup_2.
**Reset** 0x20050206

| Name | o p c o d e _ s e | | | | | | | | o p c o d e _ w i p | | | | | | | | o p c o d e _ p p | | | | | | | | o p c o d e _ w e | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | opcode_se | (no description) | 8'h20 |
| 23:16 | R/W | opcode_wip | (no description) | 8'h05 |
| 15:8 | R/W | opcode_pp | (no description) | 8'h02 |
| 7:0 | R/W | opcode_we | (no description) | 8'h06 |

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**Name** REMOTE_AHB_SETUP_3
**Address** 0x5014f018
**Description** Contains register fields associated with remote_ahb_setup_3.
**Reset** 0x09968000

| Name | check_wle | wip_bit | | | wip_polarity | wle_bit | | | wle_polarity | enable_performance_mode | opcode_performance_mode | | | | | | | RSVD | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | check_wle | (no description) | 1'b0 |
| 30:28 | R/W | wip_bit | (no description) | 3'd0 |
| 27 | R/W | wip_polarity | (no description) | 1'b1 |
| 26:24 | R/W | wle_bit | (no description) | 3'd1 |
| 23 | R/W | wle_polarity | (no description) | 1'b1 |
| 22 | R/W | enable_performance_mode | enable performace enhance mode for supported flash | 1'b0 |
| 21:14 | R/W | opcode_performance_mode | performance enhance mode opcode<br>macronix = 8'h5A<br>gigadev = 8'hAx<br>winbond = 8'h20 | 8'h5A |

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**Name** REMOTE_AHB_SETUP_4
**Address** 0x5014f01c
**Description** Contains register fields associated with remote_ahb_setup_4.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | invert_addr | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 23:0 | R/W | invert_addr | (no description) | 24'd0 |

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**Name** REMOTE_AHB_SETUP_5
**Address** 0x5014f020
**Description** Contains register fields associated with remote_ahb_setup_5.
**Reset** 0x02008200

| Name | RSVD | stall_we2pp | stall_wle | pp_stall_wip |
|---|---|---|---|---|
| **Bits** | **31 30** | **29 28 27 26 25 24 23 22 21 20** | **19 18 17 16 15 14 13 12 11 10** | **9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29:20 | R/W | stall_we2pp | wen or wle to pp depending on check_wle setting | 10'd32 |
| 19:10 | R/W | stall_wle | (no description) | 10'd32 |
| 9:0 | R/W | pp_stall_wip | (no description) | 10'd512 |

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**Name** REMOTE_AHB_SETUP_6
**Address** 0x5014f024
**Description** Contains register fields associated with remote_ahb_setup_6.
**Reset** 0x00001fff

| Name | RSVD | se_stall_wip |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | se_stall_wip | (no description) | 16'd8191 |

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**Name** REMOTE_AHB_SETUP_7
**Address** 0x5014f028
**Description** Contains register fields associated with remote_ahb_setup_7.
**Reset** 0x35000000

| | | | rem |
|---|---|---|---|

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| Name | RSVD | remap_legacy | remap_msb_loc_sub2 | remap_table |
|---|---|---|---|---|
| **Bits** | **31 30** | **29** | **28 27 26 25 24** | **23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29 | R/W | remap_legacy | (no description) | 1'b1 |
| 28:24 | R/W | remap_msb_loc_sub2 | (no description) | 5'd21 |
| 23:0 | R/W | remap_table | (no description) | 24'b0 |

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**Name** REMOTE_AHB_WLE_CNT
**Address** 0x5014f02c
**Description** Contains register fields associated with remote_ahb_wle_cnt.
**Reset** 0x00000000

| Name | RSVD | wle_cnt |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10** | **9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9:0 | R | wle_cnt | number of wle checks before it is true. | 10'd0 |

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**Name** REMOTE_AHB_WIP_CNT
**Address** 0x5014f030
**Description** Contains register fields associated with remote_ahb_wip_cnt.
**Reset** 0x00000000

| Name | RSVD | wip_cnt |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10** | **9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9:0 | R | wip_cnt | number of wip checks before it is over. | 10'd0 |

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**Name** REMOTE_AHB_DBG0
**Address** 0x5014f034
**Description** Contains register fields associated with remote_ahb_dbg0.
**Reset** 0x00000000

| Name | dbg0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dbg0 | (no description) | 32'd0 |

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**Name** REMOTE_AHB_DBG1
**Address** 0x5014f038
**Description** Contains register fields associated with remote_ahb_dbg1.
**Reset** 0x00000000

| Name | dbg1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dbg1 | (no description) | 32'd0 |

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**Name** REMOTE_AHB_DBG2
**Address** 0x5014f03c
**Description** Contains register fields associated with remote_ahb_dbg2.
**Reset** 0x00000000

| Name | dbg2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dbg2 | (no description) | 32'd0 |

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**Name** CORE_ID
**Address** 0x5014fffc
**Description** Contains register fields associated with core_id.
**Reset** 0x51535049

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | QSPI in ASCII | 32'h51535049 |

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# I2C0_SECURE

**Name** TRANSACTION_SETUP
**Address** 0x50150000
**Description** Contains register fields associated with transaction_setup.
**Reset** 0x00200000

| Name | RSVD | mstr | go | RSVD | ack_value_to_drive | master_drives_ack | RSVD | tail | RSVD | head |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22** | **21** | **20** | **19 18** | **17** | **16** | **15 14 13 12 11 10 9 8 7 6** | **5 4** | **3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 21 | R/W | mstr | - 1=master interface<br>- 0=slave interface | 1'b1 |
| 20 | R/W | go | transaction lauches on rising edge; not self reseting | 1'b0 |
| 17 | R/W | ack_value_to_drive | (no description) | 1'd0 |
| 16 | R/W | master_drives_ack | (no description) | 1'd0 |
| 5:4 | R/W | tail | - 0=stop<br>- 1=stall<br>- 2=restart | 2'd0 |
| 1:0 | R/W | head | - 0=start<br>- 1=stall | 2'd0 |

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**Name** CLOCK_CONTROL
**Address** 0x50150004
**Description** Contains register fields associated with clock_control.
**Reset** 0x00000007

| Name | RSVD | clk_pu | clkdiv |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11** | **10** | **9 8 7 6 5 4 3 2 1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 10 | R/W | clk_pu | scl pin pullup enable | 1'b0 |
| 9:0 | R/W | clkdiv | freq_sclk = f_pclk/(4*(clkdiv+1)) | 10'd7 |

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**Name** OUTGOING_DATA
**Address** 0x50150008
**Description** Contains register fields associated with outgoing_data.
**Reset** 0x0000ff00

| Name | RSVD | | | | | | | | | | | | | | data_pu | data_oe | | | | | | | | data_o | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16 | R/W | data_pu | sda pin pullup enable | 1'b0 |
| 15:8 | R/W | data_oe | (no description) | 8'hff |
| 7:0 | R/W | data_o | can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator | 8'b0 |

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**Name** OUTGOING_DATA1
**Address** 0x5015000c
**Description** stores outgoing read data
**Reset** 0x00000000

| Name | data1_o | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data1_o | slave: stores outgoing read data | 32'd0 |

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**Name** IO_CTRL
**Address** 0x50150010
**Description** Contains register fields associated with io_ctrl.
**Reset** 0x00000000

| Name | RSVD | data_ie_o |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | vrd |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | data_ie_ovrd | - 1= ie is high<br>- 0= ie is ~oe | 1'b0 |

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**Name** INCOMING_DATA
**Address** 0x50150014
**Description** stores incoming write data
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | data_i | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data_i | master: only lsb byte used slave: all 4 bytes can be used | 32'd0 |

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**Name** INCOMING_DATA1
**Address** 0x50150018
**Description** stores read/write address
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | data1_i | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data1_i | slave: stores read/write address | 32'd0 |

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**Name** TRANSACTION_STATUS
**Address** 0x5015001c
**Description** Contains register fields associated with transaction_status.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | s_rwb | runnin | ack_val |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | g | ue |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R | s_rwb | slave read/write status bit<br>- 0=write<br>- 1=read | 1'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | ack_value | useless if mast_drives_ack is set | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x50150020
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | passthru2 | passthru1 | passthru0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x50150024
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R | interrupt2 | from slave, rising edge on read data valid error; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, rising edge on done transaction capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, rising edge on done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x50150028
**Description** not auto cleared
**Reset** 0x00000000

| Name | RSVD | set_interrupt2 | set_interrupt1 | set_interrupt0 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | set_interrupt2 | set interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | set_interrupt1 | set interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt0; not auto cleared | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x5015002c
**Description** not auto cleared
**Reset** 0x00000000

| Name | RSVD | reset_interrupt2 | reset_interrupt1 | reset_interrupt0 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | reset_interrupt2 | clear interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | reset_interrupt1 | clear interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | reset_interrupt0 | clear interrupt0; not auto cleared | 1'b0 |

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**Name** ID
**Address** 0x50150ffc
**Description** Contains register fields associated with id.
**Reset** 0x49324320

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| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | I2C in ASCII | 32'h49324320 |

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# I2C1_SECURE

**Name** TRANSACTION_SETUP
**Address** 0x50151000
**Description** Contains register fields associated with transaction_setup.
**Reset** 0x00200000

| Name | RSVD | | | | | | | | | | mstr | go | RSVD | | ack_value_to_drive | master_drives_ack | RSVD | | | | | | | | | | tail | | RSVD | | head | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 21 | R/W | mstr | - 1=master interface<br>- 0=slave interface | 1'b1 |
| 20 | R/W | go | transaction lauches on rising edge; not self reseting | 1'b0 |
| 17 | R/W | ack_value_to_drive | (no description) | 1'd0 |
| 16 | R/W | master_drives_ack | (no description) | 1'd0 |
| 5:4 | R/W | tail | - 0=stop<br>- 1=stall<br>- 2=restart | 2'd0 |
| 1:0 | R/W | head | - 0=start<br>- 1=stall | 2'd0 |

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**Name** CLOCK_CONTROL
**Address** 0x50151004
**Description** Contains register fields associated with clock_control.
**Reset** 0x00000007

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | c | c |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | clk_pu | clkdiv |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11** | **10** | **9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 10 | R/W | clk_pu | scl pin pullup enable | 1'b0 |
| 9:0 | R/W | clkdiv | freq_sclk = f_pclk/(4*(clkdiv+1)) | 10'd7 |

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**Name** OUTGOING_DATA
**Address** 0x50151008
**Description** Contains register fields associated with outgoing_data.
**Reset** 0x0000ff00

| Name | RSVD | data_pu | data_oe | data_o |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17** | **16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16 | R/W | data_pu | sda pin pullup enable | 1'b0 |
| 15:8 | R/W | data_oe | (no description) | 8'hff |
| 7:0 | R/W | data_o | can be device id, address within device, or write data; if it's device id, the LSB is the R/Wb indicator | 8'b0 |

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**Name** OUTGOING_DATA1
**Address** 0x5015100c
**Description** stores outgoing read data
**Reset** 0x00000000

| Name | data1_o |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data1_o | slave: stores outgoing read data | 32'd0 |

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**Name** IO_CTRL
**Address** 0x50151010
**Description** Contains register fields associated with io_ctrl.
**Reset** 0x00000000

| | | d |
|---|---|---|

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| Name | RSVD | data_ie_ovrd |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | data_ie_ovrd | - 1= ie is high<br>- 0= ie is ~oe | 1'b0 |

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---

**Name** INCOMING_DATA
**Address** 0x50151014
**Description** stores incoming write data
**Reset** 0x00000000

| Name | data_i |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data_i | master: only lsb byte used slave: all 4 bytes can be used | 32'd0 |

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---

**Name** INCOMING_DATA1
**Address** 0x50151018
**Description** stores read/write address
**Reset** 0x00000000

| Name | data1_i |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data1_i | slave: stores read/write address | 32'd0 |

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**Name** TRANSACTION_STATUS
**Address** 0x5015101c
**Description** Contains register fields associated with transaction_status.
**Reset** 0x00000000

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | s_rwb | running | ack_value |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R | s_rwb | slave read/write status bit<br>- 0=write<br>- 1=read | 1'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | ack_value | useless if mast_drives_ack is set | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x50151020
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | passthru2 | passthru1 | passthru0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x50151024
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R | interrupt2 | from slave, rising edge on read data valid error; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, rising edge on done transaction capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, rising edge on done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x50151028
**Description** not auto cleared
**Reset** 0x00000000

| Name | RSVD | set_interrupt2 | set_interrupt1 | set_interrupt0 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | set_interrupt2 | set interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | set_interrupt1 | set interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt0; not auto cleared | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x5015102c
**Description** not auto cleared
**Reset** 0x00000000

| Name | RSVD | reset_interrupt2 | reset_interrupt1 | reset_interrupt0 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | reset_interrupt2 | clear interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | reset_interrupt1 | clear interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | reset_interrupt0 | clear interrupt0; not auto cleared | 1'b0 |

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**Name** ID
**Address** 0x50151ffc
**Description** Contains register fields associated with id.
**Reset** 0x49324320

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | I2C in ASCII | 32'h49324320 |

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# NVM_SECURE

**Name** OPMODE
**Address** 0x50152000
**Description** Contains register fields associated with opmode.
**Reset** 0x00000000

| Name | go | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | program | read |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | go | opcode start when rising edge on 'go' is detected. | 1'b0 |
| 1 | R/W | program | program opcode | 1'b0 |
| 0 | R/W | read | read opcode | 1'b0 |

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**Name** CTRL
**Address** 0x50152004
**Description** Contains register fields associated with ctrl.
**Reset** 0x00000100

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | num_pgm_bit | | | | | | | | burst_mode | ser_dout_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8:2 | R/W | num_pgm_bit | In single mode, specifies bit location to program; actual bit location is (programmed number - 1)<br>ex. to program bit 0; num_pgm_bit=1, to program bit 63; num_pgm_bit=64<br>In burst mode, program bits location from 0 up to (programmed number - 1). | 7'd64 |
| 1 | R/W | burst_mode | burst or single bit programming<br>1 = serial burst program bits starting at bit 0 up to specified bit number in num_pgm_bit register using specified input data from data0/data1 registers<br>0 = single program bit specified in num_pgm_bit register programming sets efuse bit to 1 | 1'b0 |
| 0 | R/W | ser_dout_en | enable serial data out and allow soft repair during read | 1'b0 |

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**Name** DATA0
**Address** 0x50152008
**Description** Contains register fields associated with data0.
**Reset** 0x00000000

| Name | data0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data0 | otp data word-0,<br>can be used for program or read data based on target operation | 32'd0 |

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**Name** DATA1
**Address** 0x5015200c
**Description** Contains register fields associated with data1.
**Reset** 0x00000000

| Name | data1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data1 | otp data word-1,<br>can be used for program or read data based on target operation | 32'd0 |

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**Name** STATUS
**Address** 0x50152010
**Description** Contains register fields associated with status.
**Reset** 0x00000000

| | r | | pro | re |
|---|---|---|---|---|

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| Name | running | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | program_done | read_done |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R | running | read or program running | 1'b0 |
| 1 | R | program_done | programming done | 1'b0 |
| 0 | R | read_done | read done | 1'b0 |

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**Name** T_SP_VQ
**Address** 0x50152014
**Description** VDDQ to CSB setup time into program mode
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 10 nsec;<br>each lsb accounts for 16usec, max when all bits programmed is around 500usec<br>default 16usec based on 16MHz system clock | 5'd1 |

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**Name** T_HP_VQ
**Address** 0x50152018
**Description** VDDQ to CSB hold time out of program mode
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 10 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_SP_PG
**Address** 0x5015201c

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**Description** PGM to CSB setup time into program mode
**Reset** 0x00000000

| Name | RSVD | cycles |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 5 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_HP_PG
**Address** 0x50152020
**Description** PGM to CSB hold time into program mode
**Reset** 0x00000000

| Name | RSVD | cycles |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 5 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_HP_CK
**Address** 0x50152024
**Description** SCLK to CSB hold time into program mode
**Reset** 0x00000001

| Name | RSVD | cycles |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 31 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd1 |

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**Name** T_S_PGM
**Address** 0x50152028
**Description** PGM to SCLK setup time to program bits

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**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:0 | R/W | cycles | want >= 8 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 6'd0 |

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**Name** T_H_PGM
**Address** 0x5015202c
**Description** PGM to SCLK hold time to program bits
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:0 | R/W | cycles | want >= 8 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 6'd0 |

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**Name** T_CKHP
**Address** 0x50152030
**Description** SCLK high period for program
**Reset** 0x000000a0

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | cycles | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 10:0 | R/W | cycles | want >= 9 usec and <= 11 usec; default for 10 usec<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 11'd160 |

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**Name** T_CKLP
**Address** 0x50152034
**Description** SCLK low period for program
**Reset** 0x00000000

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 10 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_SPS_CK
**Address** 0x50152038
**Description** CSB to SCLK hold time out of program mode
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 31 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd1 |

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**Name** T_SR_CK
**Address** 0x5015203c
**Description** SCLK to CSB setup time into read mode
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 6 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_HR_CK
**Address** 0x50152040
**Description** SCLK to CSB hold time into read mode
**Reset** 0x0000000e

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | cycles | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9:0 | R/W | cycles | want >= 940 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 10'd14 |

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**Name** T_CKLR
**Address** 0x50152044
**Description** SCLK low period for read
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 10 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_CKHR
**Address** 0x50152048
**Description** SCLK high period for read
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 10 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** T_SRS_CK
**Address** 0x5015204c
**Description** CSB to SCLK hold time out of read mode
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | c |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | cycles | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | cycles | want >= 20 nsec;<br>actual delay is (programmed number + 1), default based on 16MHz system clock | 5'd0 |

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**Name** INTERRUPT_STATUS
**Address** 0x50152050
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | intrpt0 | saw a rising edge on read or program done; independent of mask | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x50152054
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt0 | 1=allow intrpt0 to be OR'ed into core interrupt | 1'b0 |

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---

**Name** INTERRUPT_SET
**Address** 0x50152058
**Description** not auto cleared
**Reset** 0x00000000

| Name | RS | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | intr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | V<br>D | | | | | | | | | | | | | | | | | | | | | | | | | | | | | p<br>t<br>0 |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt0 | (no description) | 1'b0 |

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**Name** INTERRUPT_RESET
**Address** 0x5015205c
**Description** not auto cleared
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | R<br>S<br>V<br>D | | | | | | | | | | | | | | | | | | | | | | | | | | | | | i<br>n<br>t<br>r<br>p<br>t<br>0 |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt0 | (no description) | 1'b0 |

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**Name** CORE_ID
**Address** 0x50152ffc
**Description** CORE ID
**Reset** 0x4e564d20

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | i<br>d | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | NVM in ASCII | 32'h4e564d20 |

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# WRPR1_SECURE

**Name** APB0_CTRL
**Address** 0x50153000
**Description** Contains register fields associated with apb0_ctrl.
**Reset** 0x00000002

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | R<br>S<br>V<br>D | | | | | | | | | | | | | | | | | | | | | | | | | | | c<br>l<br>k<br>_<br>s<br>e | s<br>r<br>e<br>s<br>e<br>t | c<br>l<br>k<br>_<br>e<br>n<br>a<br>b |

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | l | | le |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB1_CTRL
**Address** 0x50153004
**Description** Contains register fields associated with apb1_ctrl.
**Reset** 0x00000002

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB2_CTRL
**Address** 0x50153008
**Description** Contains register fields associated with apb2_ctrl.
**Reset** 0x00000002

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB3_CTRL

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**Address** 0x5015300c
**Description** Contains register fields associated with apb3_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB4_CTRL
**Address** 0x50153010
**Description** Contains register fields associated with apb4_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB5_CTRL
**Address** 0x50153014
**Description** Contains register fields associated with apb5_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB6_CTRL
**Address** 0x50153018
**Description** Contains register fields associated with apb6_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB7_CTRL
**Address** 0x5015301c
**Description** Contains register fields associated with apb7_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB8_CTRL
**Address** 0x50153020
**Description** Contains register fields associated with apb8_ctrl.

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**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB9_CTRL
**Address** 0x50153024
**Description** Contains register fields associated with apb9_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB10_CTRL
**Address** 0x50153028
**Description** Contains register fields associated with apb10_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB11_CTRL
**Address** 0x5015302c
**Description** Contains register fields associated with apb11_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB12_CTRL
**Address** 0x50153030
**Description** Contains register fields associated with apb12_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB13_CTRL
**Address** 0x50153034
**Description** Contains register fields associated with apb13_ctrl.
**Reset** 0x00000002

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB14_CTRL
**Address** 0x50153038
**Description** Contains register fields associated with apb14_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB15_CTRL
**Address** 0x5015303c
**Description** Contains register fields associated with apb15_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** CORE_ID
**Address** 0x50153ffc
**Description** Contains register fields associated with core_id.
**Reset** 0x57525052

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | WRPR in ASCII | 32'h57525052 |

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# RADIO_SECURE

**Name** CTRL
**Address** 0x50155000
**Description** SPI Control Register
**Reset** 0x00010301

| Name | dma_mode | RSVD | | | | | | | | | | | spi3wire_en | s_wr_opcode | | | | | | | | s_rd_opcode | | | | | | | | cpol | cpha | mstr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | dma_mode | 1 = dma controls spi core | 1'b0 |
| 19 | R/W | spi3wire_en | 1=enable 3-wire spi transactions, mosi pad becomes bi-directional<br>0=default 4-wire spi transactions | 1'b0 |
| 18:11 | R/W | s_wr_opcode | slave write opcode | 8'h20 |
| 10:3 | R/W | s_rd_opcode | slave read opcode | 8'h60 |
| 2 | R/W | cpol | spi clock polarity 0=spi clock low in idle; cs high 1=spi clock high in idle; cs high | 1'b0 |
| 1 | R/W | cpha | spi clock phase 0=leading edge sample, trailing edge toggle 1=leading edge toggle, trailing edge sample | 1'b0 |
| 0 | R/W | mstr | 1=master interface<br>0=slave interface | 1'b1 |

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**Name** TRANSACTION_SETUP
**Address** 0x50155004

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**Description** transaction setup
**Reset** 0x00000000

| Name | csn_bubble | invert_mosi | dummy_cycles | | | | csn_stays_low | loopback | opcode | | | | | | clkdiv | | | | | | | | rwb | num_data_bytes | | | | start |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | csn_bubble | introduce a delay between spi_csn falling and first bit on mosi | 1'b0 |
| 30 | R/W | invert_mosi | invert mosi; keep static during entire spi transaction | 1'b0 |
| 29:26 | R/W | dummy_cycles | number of extra shift cycles to help with turn-around | 4'd0 |
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 24 | R/W | loopback | if 1, mosi is internally connected to miso | 1'b0 |
| 23:16 | R/W | opcode | opcode | 8'd0 |
| 15:6 | R/W | clkdiv | freq_sclk = f_pclk/(2*(clkdiv+1)) | 10'd0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; not auto cleared | 1'b0 |

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**Name** TRANSACTION_SETUP_DMA
**Address** 0x50155008
**Description** transaction setup by DMA
**Reset** 0x00000000

| Name | RSVD | | | | | csn_stays_low | RSVD | | | | | | | | | | | | | | | | rwb | num_data_bytes | | | | start |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; auto cleared | 1'b0 |

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**Name** TRANSACTION_STATUS
**Address** 0x5015500c
**Description** transaction status
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | opcode_status | | | | | | | | RSVD | | | | | | running | done |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R | opcode_status | master or slave opcode status | 8'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | done | 1=indicate master or slave read/write transaction is done | 1'b0 |

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**Name** DATA_BYTES_LOWER
**Address** 0x50155010
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores read/write address
**Reset** 0x00000000

| Name | byte3 | | | | | | | | byte2 | | | | | | | | byte1 | | | | | | | | byte0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte3 | data byte3 | 8'b0 |
| 23:16 | R/W | byte2 | data byte2 | 8'b0 |
| 15:8 | R/W | byte1 | data byte1 | 8'b0 |
| 7:0 | R/W | byte0 | data byte0 | 8'b0 |

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**Name** DATA_BYTES_UPPER
**Address** 0x50155014
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores write data; preload read data
**Reset** 0x00000000

| Name | byte7 | | | | | | | | byte6 | | | | | | | | byte5 | | | | | | | | byte4 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte7 | data byte7 | 8'b0 |
| 23:16 | R/W | byte6 | data byte6 | 8'b0 |
| 15:8 | R/W | byte5 | data byte5 | 8'b0 |
| 7:0 | R/W | byte4 | data byte4 | 8'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x50155018
**Description** interrupt mask
**Reset** 0x00000000

| Name | RSVD | passthru3 | passthru2 | passthru1 | passthru0 |
|---|---|---|---|---|---|
| **Bits** | **31 … 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | passthru3 | 1=allow interrupt3 to be OR'ed into core interrupt | 1'b0 |
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x5015501c
**Description** interrupt status
**Reset** 0x00000000

| Name | RSVD | interrupt3 | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31 … 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | interrupt3 | from slave, error read-data invalid; independent of mask | 1'b0 |
| 2 | R | interrupt2 | from slave, error unknown command; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, transaction done capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, transaction done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT

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**Address** 0x50155020
**Description** interrupt set
**Reset** 0x00000000

| Name | RSVD | set_interrupt3 | set_interrupt2 | set_interrupt1 | set_interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31 … 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | set_interrupt3 | set interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | set_interrupt2 | set interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | set_interrupt1 | set interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt0; not auto cleared | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x50155024
**Description** interrupt clear
**Reset** 0x00000000

| Name | RSVD | reset_interrupt3 | reset_interrupt2 | reset_interrupt1 | reset_interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31 … 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | reset_interrupt3 | clear interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | reset_interrupt2 | clear interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | reset_interrupt1 | clear interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | reset_interrupt0 | clear interrupt0; not auto cleared | 1'b0 |

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**Name** CORE_ID
**Address** 0x50155ffc
**Description** core ID

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**Reset** 0x53504920

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | SPI in ASCII | 32'h53504920 |

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# PMU_SECURE

**Name** CTRL
**Address** 0x50157000
**Description** SPI Control Register
**Reset** 0x00010301

| Name | dma_mode | RSVD | | | | | | | | | | spi3wire_en | s_wr_opcode | | | | | | | | s_rd_opcode | | | | | | | | cpol | cpha | mstr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | dma_mode | 1 = dma controls spi core | 1'b0 |
| 19 | R/W | spi3wire_en | 1=enable 3-wire spi transactions, mosi pad becomes bi-directional<br>0=default 4-wire spi transactions | 1'b0 |
| 18:11 | R/W | s_wr_opcode | slave write opcode | 8'h20 |
| 10:3 | R/W | s_rd_opcode | slave read opcode | 8'h60 |
| 2 | R/W | cpol | spi clock polarity 0=spi clock low in idle; cs high 1=spi clock high in idle; cs high | 1'b0 |
| 1 | R/W | cpha | spi clock phase 0=leading edge sample, trailing edge toggle 1=leading edge toggle, trailing edge sample | 1'b0 |
| 0 | R/W | mstr | 1=master interface<br>0=slave interface | 1'b1 |

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**Name** TRANSACTION_SETUP
**Address** 0x50157004
**Description** transaction setup
**Reset** 0x00000000

| Name | csn_b | invert | dummy_ | csn_sta | loop | opc | clk | rw | num_dat | sta |
|---|---|---|---|---|---|---|---|---|---|---|

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| | csn_bubble | invert_mosi | dummy_cycles | csn_stays_low | loopback | opcode | clkdiv | rwb | num_data_bytes | start |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29 28 27 26** | **25** | **24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6** | **5** | **4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | csn_bubble | introduce a delay between spi_csn falling and first bit on mosi | 1'b0 |
| 30 | R/W | invert_mosi | invert mosi; keep static during entire spi transaction | 1'b0 |
| 29:26 | R/W | dummy_cycles | number of extra shift cycles to help with turn-around | 4'd0 |
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 24 | R/W | loopback | if 1, mosi is internally connected to miso | 1'b0 |
| 23:16 | R/W | opcode | opcode | 8'd0 |
| 15:6 | R/W | clkdiv | freq_sclk = f_pclk/(2*(clkdiv+1)) | 10'd0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; not auto cleared | 1'b0 |

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**Name** TRANSACTION_SETUP_DMA
**Address** 0x50157008
**Description** transaction setup by DMA
**Reset** 0x00000000

| Name | RSVD | csn_stays_low | RSVD | rwb | num_data_bytes | start |
|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26** | **25** | **24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6** | **5** | **4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25 | R/W | csn_stays_low | at end of transaction, leave csn low | 1'b0 |
| 5 | R/W | rwb | 1=read 0=write; note it's possibly redundant with opcode | 1'b0 |
| 4:1 | R/W | num_data_bytes | Master data; don't add 1 for opcode | 4'd0 |
| 0 | R/W | start | Master FSM needs to see low to high transition to start; auto cleared | 1'b0 |

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**Name** TRANSACTION_STATUS
**Address** 0x5015700c
**Description** transaction status
**Reset** 0x00000000

| | | opc | | |
|---|---|---|---|---|

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| Name | RSVD | opcode_status | RSVD | running | done |
|---|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:8 | R | opcode_status | master or slave opcode status | 8'b0 |
| 1 | R | running | 1=indicate master or slave running based on which interface selected by 'mstr' bit | 1'b0 |
| 0 | R | done | 1=indicate master or slave read/write transaction is done | 1'b0 |

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**Name** DATA_BYTES_LOWER
**Address** 0x50157010
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores read/write address
**Reset** 0x00000000

| Name | byte3 | byte2 | byte1 | byte0 |
|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte3 | data byte3 | 8'b0 |
| 23:16 | R/W | byte2 | data byte2 | 8'b0 |
| 15:8 | R/W | byte1 | data byte1 | 8'b0 |
| 7:0 | R/W | byte0 | data byte0 | 8'b0 |

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**Name** DATA_BYTES_UPPER
**Address** 0x50157014
**Description** master: preload if write; postread if read; readback data will change if num_data_bytes changes post done slave: stores write data; preload read data
**Reset** 0x00000000

| Name | byte7 | byte6 | byte5 | byte4 |
|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte7 | data byte7 | 8'b0 |
| 23:16 | R/W | byte6 | data byte6 | 8'b0 |
| 15:8 | R/W | byte5 | data byte5 | 8'b0 |
| 7:0 | R/W | byte4 | data byte4 | 8'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x50157018
**Description** interrupt mask
**Reset** 0x00000000

| Name | RSVD | passthru3 | passthru2 | passthru1 | passthru0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | passthru3 | 1=allow interrupt3 to be OR'ed into core interrupt | 1'b0 |
| 2 | R/W | passthru2 | 1=allow interrupt2 to be OR'ed into core interrupt | 1'b0 |
| 1 | R/W | passthru1 | 1=allow interrupt1 to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x5015701c
**Description** interrupt status
**Reset** 0x00000000

| Name | RSVD | interrupt3 | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | interrupt3 | from slave, error read-data invalid; independent of mask | 1'b0 |
| 2 | R | interrupt2 | from slave, error unknown command; independent of mask | 1'b0 |
| 1 | R | interrupt1 | from slave, transaction done capture; independent of mask | 1'b0 |
| 0 | R | interrupt0 | from master, transaction done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x50157020
**Description** interrupt set
**Reset** 0x00000000

| | | set_i | set_i | set_i | set_i |
|---|---|---|---|---|---|

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| Name | RSVD | interrupt3 | interrupt2 | interrupt1 | interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | set_interrupt3 | set interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | set_interrupt2 | set interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | set_interrupt1 | set interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt0; not auto cleared | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x50157024
**Description** interrupt clear
**Reset** 0x00000000

| Name | RSVD | reset_interrupt3 | reset_interrupt2 | reset_interrupt1 | reset_interrupt0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | reset_interrupt3 | clear interrupt3; not auto cleared | 1'b0 |
| 2 | R/W | reset_interrupt2 | clear interrupt2; not auto cleared | 1'b0 |
| 1 | R/W | reset_interrupt1 | clear interrupt1; not auto cleared | 1'b0 |
| 0 | R/W | reset_interrupt0 | clear interrupt0; not auto cleared | 1'b0 |

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**Name** CORE_ID
**Address** 0x50157ffc
**Description** core ID
**Reset** 0x53504920

| Name | id |
|---|---|
| **Bits** | **31:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | SPI in ASCII | 32'h53504920 |

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# PSEQ_SECURE

**Name** CTRL0
**Address** 0x50158000
**Description** signals to prepare for and to request low power state
**Reset** 0x00280000

| Name | req_shub_adv | spi_latch_open | i2c_latch_open | pinpu_latch_open | req_soc_off | gpio_latch_open | ksm_latch_open | pinsel_latch_open | RSVD | wurx_det1_sel | wurx_det0_sel | block_dbg_wake | RSVD | pwm_latch_open | RSVD | req_hibernate_state | req_retain_all_state | RSVD | manage_xtal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17:7** | **6** | **5:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | req_shub_adv | If set, requests SOC to go to do CPU-less adv. | 1'b0 |
| 30 | R/W | spi_latch_open | If set, opens all latches on spi0 slave control signals. | 1'b0 |
| 29 | R/W | i2c_latch_open | If set, opens all latches on i2c 0/1 sck/sda pu signals. | 1'b0 |
| 28 | R/W | pinpu_latch_open | If set, opens all latches on all pin pu signals. | 1'b0 |
| 27 | R/W | req_soc_off | If set, requests SOC to go SOC Off state. | 1'b0 |
| 26 | R/W | gpio_latch_open | If set, opens all latches on gpio control signals. | 1'b0 |
| 25 | R/W | ksm_latch_open | If set, opens all latches on ksm control signals. | 1'b0 |
| 24 | R/W | pinsel_latch_open | If set, opens all latches all pin select signals. | 1'b0 |
| 22:21 | R/W | wurx_det1_sel | Is wurx1 detect level or edge?<br>0=level<br>1=rising<br>2,3=falling | 2'd1 |
| 20:19 | R/W | wurx_det0_sel | Is wurx0 detect level or edge?<br>0=level<br>1=rising<br>2,3=falling | 2'd1 |
| 18 | R/W | block_dbg_wake | If set, debugger cannot wake chip from low power state. | 1'b0 |
| 6 | R/W | pwm_latch_open | If set, opens all latches on pwm output prior to entering low power to hold pwm. | 1'b0 |
| 3 | R/W | req_hibernate_state | If set, requests SOC to go to Hibernate low power state. | 1'b0 |
| 2 | R/W | req_retain_all_state | If set, requests SOC to go to Retain All low power state. | 1'b0 |
| 0 | R/W | manage_xtal | If set, permits SOC to power down clk_mpc crystal circuit during low power mode. | 1'b0 |

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**Name** CTRL1

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**Address** 0x50158004
**Description** signals to prepare for and to request low power state
**Reset** 0x00000055

| Name | RSVD | reset_pmu_spi_regs | reset_radio_spi_regs | reset_wurx_spi_regs | brownout_sel | endoflife_sel | energy4TX_sel | energy4CPU_sel |
|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:11** | **10** | **9** | **8** | **7:6** | **5:4** | **3:2** | **1:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 10 | R/W | reset_pmu_spi_regs | (no description) | 1'b0 |
| 9 | R/W | reset_radio_spi_regs | (no description) | 1'b0 |
| 8 | R/W | reset_wurx_spi_regs | (no description) | 1'b0 |
| 7:6 | R/W | brownout_sel | Is brownout level or edge?<br>0=level<br>1=rising<br>2=falling<br>3=negate level | 2'd1 |
| 5:4 | R/W | endoflife_sel | Is endoflife level or edge?<br>0=level<br>1=rising<br>2=falling<br>3=negate level | 2'd1 |
| 3:2 | R/W | energy4TX_sel | Is energy4TX level or edge?<br>0=level<br>1=rising<br>2=falling<br>3=negate level | 2'd1 |
| 1:0 | R/W | energy4CPU_sel | Is energy4CPU level or edge?<br>0=level<br>1=rising<br>2=falling<br>3=negate level | 2'd1 |

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**Name** RETAIN_ALL_WAKE_MASK
**Address** 0x50158008
**Description** signals to select which events can generate a wake event while in Retain All
**Reset** 0x00000000

| | qualif | quali | | wat | wa | wat | wat | wat | wat | wat | w | w | wa | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Name | qualify_on_energy4CPU | qualify_on_energy4TX | RSVD | watch_ble_osc_on | watch_wurx | watch_brownout | watch_endoflife | watch_energy4TX | watch_energy4CPU | watch_pmu_timer | watch_pmu_lpcomp | watch_i2c | watch_spi | watch_shub | watch_ksm | watch_qdec | watch_wurx1 | watch_wurx0 | watch_cntdown | watch_gpio |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29:16** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | qualify_on_energy4CPU | (no description) | 1'b0 |
| 30 | R/W | qualify_on_energy4TX | (no description) | 1'b0 |
| 16 | R/W | watch_ble_osc_on | (no description) | 1'b0 |
| 15 | R/W | watch_wurx | valid to wake on multiple wurx detects in certain amount of time | 1'b0 |
| 14 | R/W | watch_brownout | valid to wake on pmu asserting brownout | 1'b0 |
| 13 | R/W | watch_endoflife | valid to wake on pmu asserting endoflife | 1'b0 |
| 12 | R/W | watch_energy4TX | valid to wake on pmu asserting energy4TX | 1'b0 |
| 11 | R/W | watch_energy4CPU | valid to wake on pmu asserting energy4CPU | 1'b0 |
| 10 | R/W | watch_pmu_timer | valid to wake on pmu timer expiring | 1'b0 |
| 9 | R/W | watch_pmu_lpcomp | valid to wake on pmu lpcomp threshold crossing | 1'b0 |
| 8 | R/W | watch_i2c | valid to wake on i2c slave activity | 1'b0 |
| 7 | R/W | watch_spi | valid to wake on spi slave activity | 1'b0 |
| 6 | R/W | watch_shub | valid to wake on sensor hub trigger | 1'b0 |
| 5 | R/W | watch_ksm | valid to wake on keyboard activity | 1'b0 |
| 4 | R/W | watch_qdec | valid to wake on qdec activity | 1'b0 |
| 3 | R/W | watch_wurx1 | valid to wake on wurx1 detect | 1'b0 |
| 2 | R/W | watch_wurx0 | valid to wake on wurx0 detect | 1'b0 |
| 1 | R/W | watch_cntdown | valid to wake on count down timer expiring | 1'b0 |
| 0 | R/W | watch_gpio | valid to wake on gpio activity | 1'b0 |

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**Name** HIB_WAKE_MASK
**Address** 0x5015800c
**Description** signals to select which events can generate a wake event while in Hiberbate
**Reset** 0x00000000

| Name | qualify_on_energ | qualify_on_ener | RSVD | watch_ble_osc | watch_wurx | watch_brownou | watch_endofli | watch_energy4 | watch_energy4 | watch_pmu_tim | watch_pmu_lpc | watch_i2c | watch_spi | watch_shub | watch_ksm | watch_qdec | watch_wurx1 | watch_wurx0 | watch_cntdow | watch_gpio |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| | y4CPU | gy4TX | | | | | | | | | | | | | | _on | | ut | fe | TX | CPU | er | omp | | | | | | | | n | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | qualify_on_energy4CPU | (no description) | 1'b0 |
| 30 | R/W | qualify_on_energy4TX | (no description) | 1'b0 |
| 16 | R/W | watch_ble_osc_on | (no description) | 1'b0 |
| 15 | R/W | watch_wurx | valid to wake on multiple wurx detects in certain amount of time | 1'b0 |
| 14 | R/W | watch_brownout | valid to wake on pmu asserting brownout | 1'b0 |
| 13 | R/W | watch_endoflife | valid to wake on pmu asserting endoflife | 1'b0 |
| 12 | R/W | watch_energy4TX | valid to wake on pmu asserting energy4TX | 1'b0 |
| 11 | R/W | watch_energy4CPU | valid to wake on pmu asserting energy4CPU | 1'b0 |
| 10 | R/W | watch_pmu_timer | valid to wake on pmu timer expiring | 1'b0 |
| 9 | R/W | watch_pmu_lpcomp | valid to wake on pmu lpcomp threshold crossing | 1'b0 |
| 8 | R/W | watch_i2c | valid to wake on i2c slave activity | 1'b0 |
| 7 | R/W | watch_spi | valid to wake on spi slave activity | 1'b0 |
| 6 | R/W | watch_shub | valid to wake on sensor hub trigger | 1'b0 |
| 5 | R/W | watch_ksm | valid to wake on keyboard activity | 1'b0 |
| 4 | R/W | watch_qdec | valid to wake on qdec activity | 1'b0 |
| 3 | R/W | watch_wurx1 | valid to wake on wurx1 detect | 1'b0 |
| 2 | R/W | watch_wurx0 | valid to wake on wurx0 detect | 1'b0 |
| 1 | R/W | watch_cntdown | valid to wake on count down timer expiring | 1'b0 |
| 0 | R/W | watch_gpio | valid to wake on gpio activity | 1'b0 |

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**Name** GPIO_WAKE_MASK
**Address** 0x50158018
**Description** mask function to specify which gpio's can generate a gpio event
**Reset** 0x00000000

| Name | upper | | | | | | | | | | | | | | | | lower | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | upper | upper 16 gpio's | 16'h0 |
| 15:0 | R/W | lower | lower 16 gpio's | 16'h0 |

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**Name** GPIO_WAKE_TYPE
**Address** 0x5015801c
**Description** specifies how a gpio generates an event
**Reset** 0x00000000

| Name | upper | lower |
|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | upper | (per bit) 1 = edge sensitive , 0 = level | 16'h0 |
| 15:0 | R/W | lower | (per bit) 1 = edge sensitive , 0 = level | 16'h0 |

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**Name** GPIO_WAKE_POL
**Address** 0x50158020
**Description** specifies how a gpio generates an event
**Reset** 0xffffffff

| Name | upper | | | | | | | | | | | | | | | | lower | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | upper | (per bit) 1 = level high/rising edge , 0 = level low/falling edge | 16'hffff |
| 15:0 | R/W | lower | (per bit) 1 = level high/rising edge , 0 = level low/falling edge | 16'hffff |

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**Name** GPIO_WAKE_BOTH_EDGES
**Address** 0x50158024
**Description** specifies how a gpio generates an event
**Reset** 0x00000000

| Name | upper | | | | | | | | | | | | | | | | lower | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | upper | (per bit) 1 = either edge will wake; higher precedence than gpio_wake_type and gpio_wake_pol | 16'h0 |
| 15:0 | R/W | lower | (per bit) 1 = either edge will wake; higher precedence than gpio_wake_type and gpio_wake_pol | 16'h0 |

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**Name** GPIO_WAKE_STATUS
**Address** 0x50158028
**Description** specifies which gpio caused the wake
**Reset** 0x00000000

| Name | upper | | | | | | | | | | | | | | | | lower | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:16 | R | upper | (no description) | 16'h0 |
| 15:0 | R | lower | (no description) | 16'h0 |

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**Name** WURX_CONFIG
**Address** 0x5015802c
**Description** wurx configuration
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | wurx_cutvdd_b | wurx_rstb | wurx_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | wurx_cutvdd_b | if low, removes power from wurx | 1'b0 |
| 1 | R/W | wurx_rstb | if low, holds wurx in reset | 1'b0 |
| 0 | R/W | wurx_en | if low, disables wurx datapath | 1'b0 |

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**Name** WURX_CONFIG1
**Address** 0x50158030
**Description** wurx configuration
**Reset** 0x00000022

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | num_wurx_event | | | | wurx_src_sel | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:2 | R/W | num_wurx_event | Number of multiple wurx detect events to watch | 4'd8 |
| 1:0 | R/W | wurx_src_sel | which multiple wurx detect source to watch<br>0=wurx0<br>1=wurx1<br>2,3=wurx0 or wurx1 | 2'd2 |

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**Name** WURX_CONFIG2
**Address** 0x50158034
**Description** wurx configuration
**Reset** 0x80008000

| Name | num_wurx_time1_cyc | | | | | | | | | | | | | | | num_wurx_time0_cyc | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:17 | R/W | num_wurx_time1_cyc | Max time window between two consecutive wurx detect events in 32KHz cycles, default 500msec window | 15'd16384 |
| 16:0 | R/W | num_wurx_time0_cyc | Full time window during multiple wurx detect events watching in 32KHz cycles, default 1sec window | 17'd32768 |

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**Name** GADC_CONFIG
**Address** 0x50158038
**Description** gadc configuration
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | gadc_cutvdd_b | gadc_rstb | gadc_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | gadc_cutvdd_b | if low, removes power from gadc | 1'b0 |
| 1 | R/W | gadc_rstb | if low, holds gadc in reset | 1'b0 |
| 0 | R/W | gadc_en | if low, disables gadc datapath | 1'b0 |

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**Name** XTAL_BITS0
**Address** 0x5015803c
**Description** control signals for clk_mpc crystal circuit

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**Reset** 0x000b6766

| Name | RSVD | xobufp | xobufn | xocapout | xocapin | RSVD |
|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21** | **20 19** | **18 17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 20:19 | R/W | xobufp | (no description) | 2'b01 |
| 18:17 | R/W | xobufn | (no description) | 2'b01 |
| 16:9 | R/W | xocapout | 16MHz crystal tuning cap on XTALO_16M pin, typically set to same value as xocapin | 8'hb3 |
| 8:1 | R/W | xocapin | 16MHz crystal tuning cap on XTALI_16M pin, typically set to same value as xocapout | 8'hb3 |

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**Name** XTAL_BITS1
**Address** 0x50158040
**Description** control signals for clk_mpc crystal circuit
**Reset** 0x00164000

| Name | RSVD | extclk | clkmpc_en | clkhpc_en | doublerdcc_en | rcoscfreq | RSVD |
|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22** | **21** | **20** | **19** | **18** | **17 16 15 14** | **13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 21 | R/W | extclk | Set to 1 if supplying external clock source on XTALI_16M pin | 1'b0 |
| 20 | R/W | clkmpc_en | enable root clk_mpc buffer; chip will not be operational if cleared | 1'b1 |
| 19 | R/W | clkhpc_en | enable root clk_hpc buffer | 1'b0 |
| 18 | R/W | doublerdcc_en | enable duty cycle correction for clock doubler | 1'b1 |
| 17:14 | R/W | rcoscfreq | 16MHz RC oscillator frequency tuning, 0 - slowest, 15 - fastest | 4'h9 |

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**Name** OVERRIDES
**Address** 0x50158044
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | forc | force | force |
|---|---|---|---|---|

| Name | RSVD | force_osc_on_req | force_precision_val | force_precision_req |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | force_osc_on_req | force that clk_mpc oscillator request always be on | 1'b0 |
| 1 | R/W | force_precision_val | when type of precision for clk_mpc is forced, what should it be?<br>1=high/xtal<br>0=low/osc | 1'b0 |
| 0 | R/W | force_precision_req | force the type of precision when clk_mpc is requested | 1'b0 |

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**Name** OVERRIDES2
**Address** 0x50158048
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | block_efuse_vddcut | block_rram_vddcut | block_sysrom_vddcut |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | block_efuse_vddcut | if set, this core's FSM cannot power down the e-fuse macros | 1'b0 |
| 1 | R/W | block_rram_vddcut | if set, this core's FSM cannot power down the rram macros | 1'b0 |
| 0 | R/W | block_sysrom_vddcut | if set, this core's FSM cannot power down the ROM macros | 1'b0 |

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**Name** OVERRIDES3
**Address** 0x5015804c
**Description** override portion of pmu control signals
**Reset** 0x00000000

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| Name | RSVD | override_ram_iso | override_ram_iso_val | override_efuse_vddcut | override_efuse_vddcut_val | override_efuse_iso | override_efuse_iso_val | override_rram_vddcut | override_rram_vddcut_val | override_rram_rst | override_rram_rst_val | override_rram_poc | override_rram_poc_val | override_rram_nap | override_rram_nap_val | override_rram_iso | override_rram_iso_val | override_rram_clken | override_rram_clken_val | override_sysrom_ls | override_sysrom_ls_val | override_sysrom_iso | override_sysrom_iso_val | override_sysrom_clken | override_sysrom_clken_val |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23 | R/W | override_ram_iso | if set, software now has direct control of the pd_mem ISO pins | 1'b0 |
| 22 | R/W | override_ram_iso_val | if override is set, what value should be driven onto pd_mem ISO pins? | 1'b0 |
| 21 | R/W | override_efuse_vddcut | if set, head switch on e-fuse is now software controllable | 1'b0 |
| 20 | R/W | override_efuse_vddcut_val | if override is set, should head switch on e-fuse be open or close?<br>1=open<br>0=close | 1'b0 |
| 19 | R/W | override_efuse_iso | if set, software now has direct control of the e-fuse ISO pin | 1'b0 |
| 18 | R/W | override_efuse_iso_val | if override is set, what value should be driven onto e-fuse ISO pin? | 1'b0 |
| 17 | R/W | override_rram_vddcut | if set, head switch on romc (rom,rram) is now software controllable | 1'b0 |
| 16 | R/W | override_rram_vddcut_val | if override is set, should head switch on romc be open or close?<br>1=open<br>0=close | 1'b0 |
| 15 | R/W | override_rram_rst | if set, software now has direct control of the rram RST pin | 1'b0 |
| 14 | R/W | override_rram_rst_val | if override is set, what value should be driven onto rram RST pin? | 1'b0 |
| 13 | R/W | override_rram_poc | if set, software now has direct control of the rram POC_H pin | 1'b0 |
| 12 | R/W | override_rram_poc_val | if override is set, what value should be driven onto rram POC_H pin? | 1'b0 |
| 11 | R/W | override_rram_nap | if set, software now has direct control of the rram NAP pin | 1'b0 |
| 10 | R/W | override_rram_nap_val | if override is set, what value should be driven onto rram NAP pin? | 1'b0 |
| 9 | R/W | override_rram_iso | if set, software now has direct control of the rram ISO pin | 1'b0 |
| 8 | R/W | override_rram_iso_val | if override is set, what value should be driven onto rram ISO pin? | 1'b0 |
| 7 | R/W | override_rram_clken | if set, software now has direct control of the rram clock enable pin | 1'b0 |
| 6 | R/W | override_rram_clken_val | if override is set, what value should be driven onto rram clock enable pin? | 1'b0 |
| 5 | R/W | override_sysrom_ls | if set, software now has direct control of the rom LS pin | 1'b0 |
| 4 | R/W | override_sysrom_ls_val | if override is set, what value should be driven onto rom LS pin? | 1'b0 |
| 3 | R/W | override_sysrom_iso | if set, software now has direct control of the rom ISO pin | 1'b0 |
| 2 | R/W | override_sysrom_iso_val | if override is set, what value should be driven onto rom ISO pin? | 1'b0 |
| 1 | R/W | override_sysrom_clken | if set, software now has direct control of the rom clock enable pin | 1'b0 |
| 0 | R/W | override_sysrom_clken_val | if override is set, what value should be driven onto rom clock enable pin? | 1'b0 |

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**Name** OVERRIDES4
**Address** 0x50158050

**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | override_pnr_iso1 | override_pnr_iso1_val | override_pnr_save | override_pnr_save_val | override_pnr_restore | override_pnr_restore_val | override_pnr_sleep | override_pnr_sleep_val | override_pnr_kill | override_pnr_kill_val | override_pnr_iso | override_pnr_iso_val | override_pnr_frst | override_pnr_frst_val | override_pnr_clken | override_pnr_clken_val | override_ble_nrestore | override_ble_nrestore_val | override_ble_sleep | override_ble_sleep_val | override_ble_kill | override_ble_kill_val | override_ble_iso | override_ble_iso_val | override_ble_frst | override_ble_frst_val | override_ble_clken | override_ble_clken_val |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29 | R/W | override_pnr_iso1 | if set, software now has direct control of the iso1 signal going to the cpu domain; for scan_outs only | 1'b0 |
| 28 | R/W | override_pnr_iso1_val | if override is set, what value should be driven onto the cpu domain's iso1 signal? | 1'b0 |
| 27 | R/W | override_pnr_save | if set, software now has direct control of the SAVE signal going to the cpu domain | 1'b0 |
| 26 | R/W | override_pnr_save_val | if override is set, what value should be driven onto the cpu domain's SAVE signal? | 1'b0 |
| 25 | R/W | override_pnr_restore | if set, software now has direct control of the RESTORE signal going to the cpu domain | 1'b0 |
| 24 | R/W | override_pnr_restore_val | if override is set, what value should be driven onto the cpu domain's RESTORE signal? | 1'b0 |
| 23 | R/W | override_pnr_sleep | if set, software now has direct control of the sleep signal going to the cpu domain | 1'b0 |
| 22 | R/W | override_pnr_sleep_val | if override is set, what value should be driven onto the cpu domain's sleep signal? | 1'b0 |
| 21 | R/W | override_pnr_kill | if set, software now has direct control of the kill signal going to the cpu domain | 1'b0 |
| 20 | R/W | override_pnr_kill_val | if override is set, what value should be driven onto the cpu domain's kill signal? | 1'b0 |
| 19 | R/W | override_pnr_iso | if set, software now has direct control of the iso signal going to the cpu domain | 1'b0 |
| 18 | R/W | override_pnr_iso_val | if override is set, what value should be driven onto the cpu domain's iso signal? | 1'b0 |
| 17 | R/W | override_pnr_frst | if set, software now has direct control of the forced reset signal going to the cpu domain | 1'b0 |
| 16 | R/W | override_pnr_frst_val | if override is set, what value should be driven onto the cpu domain's forced reset signal? | 1'b0 |
| 15 | R/W | override_pnr_clken | if set, software now has direct control of the clock enable signal going to the cpu domain | 1'b0 |
| 14 | R/W | override_pnr_clken_val | if override is set, what value should be driven onto the cpu domain's clock enable signal? | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 13 | R/W | override_ble_save | if set, software now has direct control of the SAVE signal going to the ble domain | 1'b0 |
| 12 | R/W | override_ble_save_val | if override is set, what value should be driven onto the ble domain's SAVE signal? | 1'b0 |
| 11 | R/W | override_ble_nrestore | if set, software now has direct control of the NRESTORE signal going to the ble domain | 1'b0 |
| 10 | R/W | override_ble_nrestore_val | if override is set, what value should be driven onto the ble domain's NRESTORE signal? | 1'b0 |
| 9 | R/W | override_ble_sleep | if set, software now has direct control of the sleep signal going to the ble domain | 1'b0 |
| 8 | R/W | override_ble_sleep_val | if override is set, what value should be driven onto the ble domain's sleep signal? | 1'b0 |
| 7 | R/W | override_ble_kill | if set, software now has direct control of the kill signal going to the ble domain | 1'b0 |
| 6 | R/W | override_ble_kill_val | if override is set, what value should be driven onto the ble domain's kill signal? | 1'b0 |
| 5 | R/W | override_ble_iso | if set, software now has direct control of the iso signal going to the ble domain | 1'b0 |
| 4 | R/W | override_ble_iso_val | if override is set, what value should be driven onto the ble domain's iso signal? | 1'b0 |
| 3 | R/W | override_ble_frst | if set, software now has direct control of the forced reset signal going to the ble domain | 1'b0 |
| 2 | R/W | override_ble_frst_val | if override is set, what value should be driven onto the ble domain's forced reset signal? | 1'b0 |
| 1 | R/W | override_ble_clken | if set, software now has direct control of the clock enable signal going to the ble domain | 1'b0 |
| 0 | R/W | override_ble_clken_val | if override is set, what value should be driven onto the ble domain's clock enable signal? | 1'b0 |

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**Name** OVERRIDES5
**Address** 0x50158054
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | override_xtal | override_xtal_val | override_retv | override_retv_val | override_radio_en_o | override_radio_en_o_val | override_cutvdd_b | override_cutvdd_b_val | override_radio_frst_b | override_radio_frst_b_val | override_wreq | override_wreq_val | override_radio_iso | override_radio_iso_val | override_ble_frstdp | override_ble_frstdp_val | override_pnr_frstdp | override_pnr_frstdp_val | RSVD |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | override_xtal | If set, software now has direct control of clk_mpc's crystal circuit enable. | 1'b0 |
| 30 | R/W | override_xtal_val | If override is set, what value is clk_mpc's cystal circuit enable?<br>0=off<br>1=on | 1'b0 |
| 29 | R/W | override_retv | If set, software now has direct control of the memory retention voltage. | 1'b0 |
| 28 | R/W | override_retv_val | If override is set, what value is the memory retention voltage?<br>0=nominal<br>1=ultra low | 1'b0 |
| 27 | R/W | override_radio_en_o | If set, software now has direct control of radio enable. | 1'b0 |
| 26 | R/W | override_radio_en_o_val | If override is set, what value is the radio enable?<br>0=off<br>1=on | 1'b0 |
| 25 | R/W | override_cutvdd_b | If set, software now has direct control of radio head switches. | 1'b0 |
| 24 | R/W | override_cutvdd_b_val | If override is set, what value is the radio enable?<br>0=head switches are open<br>1=head switches are closed | 1'b0 |
| 23 | R/W | override_radio_frst_b | If set, software now has direct control of radio forced reset . | 1'b0 |
| 22 | R/W | override_radio_frst_b_val | If override is set, what value is the radio forced reset?<br>0=forced reset is enabled<br>1=forced reset is disabled | 1'b0 |
| 21 | R/W | override_wreq | If set, software now has direct control of wake up request. | 1'b0 |
| 20 | R/W | override_wreq_val | If override is set, what value is value of wake up request? Useful for debug/test of pseq interrupt.<br>0=no wake<br>1=wake | 1'b0 |
| 19 | R/W | override_radio_iso | If set, software now has direct control of radio isolation signal . | 1'b0 |
| 18 | R/W | override_radio_iso_val | If override is set, what value is the radio isolation signal? | 1'b0 |
| 17 | R/W | override_ble_frstdp | If set, software now has direct control of ble domain's datapath forced reset.<br>Not all state is retained even in Retain All state. | 1'b0 |
| 16 | R/W | override_ble_frstdp_val | If override is set, what value is the ble domain's datapath forced reset. | 1'b0 |
| 15 | R/W | override_pnr_frstdp | If set, software now has direct control of cpu domain's datapath forced reset.<br>Not all state is retained even in Retain All state. | 1'b0 |
| 14 | R/W | override_pnr_frstdp_val | If override is set, what value is the cpu domain's datapath forced reset. | 1'b0 |

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**Name** SYSRAM_OVERRIDES
**Address** 0x50158058
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_sysram_vdd | RSVD | override_sysram_vdd |
|---|---|---|---|---|

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | d d c u t | | | | | | | | | | | | | c u t _ v a l | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:12 | R/W | override_sysram_vddcut | If set, software now has direct control of sysram's head switches.<br>[23:20] reserved<br>[19] 16kB macro at high address space<br>[18] 16kB macro<br>[17] 16kB macro<br>[16] 16kB macro<br>[15] 16kB macro<br>[14] 16kB macro<br>[13] 16kB macro<br>[12] 16kB macro at lowest address space | 8'd0 |
| 7:0 | R/W | override_sysram_vddcut_val | If override is set, what is state of sysram's head switches?<br>[11:8] reserved<br>[7] 16kB macro at high address space; [1] head switch open [0] head switch close<br>[6] 16kB macro; [1] head switch open [0] head switch close<br>[5] 16kB macro; [1] head switch open [0] head switch close<br>[4] 16kB macro; [1] head switch open [0] head switch close<br>[3] 16kB macro; [1] head switch open [0] head switch close<br>[2] 16kB macro; [1] head switch open [0] head switch close<br>[1] 16kB macro; [1] head switch open [0] head switch close<br>[0] 16kB macro at lowest address space; [1] head switch open [0] head switch close | 8'd0 |

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**Name** SYSRAM_OVERRIDES2
**Address** 0x5015805c
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | R S V D | | | | | | | | | | | | o v e r r i d e _ s y s r a m _ c l k e n | | | | | | | | R S V D | | | | o v e r r i d e _ s y s r a m _ c l k e n _ v a l | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:12 | R/W | override_sysram_clken | If set, software now has direct control of sysram's clock enable signals.<br>[23:20] reserved<br>[19] 16kB macro at high address space<br>[18] 16kB macro<br>[17] 16kB macro<br>[16] 16kB macro<br>[15] 16kB macro<br>[14] 16kB macro<br>[13] 16kB macro<br>[12] 16kB macro at lowest address space | 8'd0 |
| 7:0 | R/W | override_sysram_clken_val | If override is set, what is state of sysram's clock enable signals?<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** SYSRAM_OVERRIDES3
**Address** 0x50158060
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_sysram_iso | RSVD | override_sysram_iso_val |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 | 19 18 17 16 15 14 13 12 | 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:12 | R/W | override_sysram_iso | If set, software now has direct control of sysram's isolation signals.<br>[23:20] reserved<br>[19] 16kB macro at high address space<br>[18] 16kB macro<br>[17] 16kB macro<br>[16] 16kB macro<br>[15] 16kB macro<br>[14] 16kB macro<br>[13] 16kB macro<br>[12] 16kB macro at lowest address space | 8'd0 |

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| | | | | |
|---|---|---|---|---|
| 7:0 | R/W | override_sysram_iso_val | If override is set, what is state of sysram's isolation signals?<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** SYSRAM_OVERRIDES4
**Address** 0x50158064
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_sysram_ls | RSVD | override_sysram_ls_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20** | **19 18 17 16 15 14 13 12** | **11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:12 | R/W | override_sysram_ls | If set, software now has direct control of sysram's LS signals.<br>[23:20] reserved<br>[19] 16kB macro at high address space<br>[18] 16kB macro<br>[17] 16kB macro<br>[16] 16kB macro<br>[15] 16kB macro<br>[14] 16kB macro<br>[13] 16kB macro<br>[12] 16kB macro at lowest address space | 8'd0 |
| 7:0 | R/W | override_sysram_ls_val | If override is set, what is state of sysram's LS signals?<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** SYSRAM_OVERRIDES5
**Address** 0x50158068
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_sysram_ds | RSVD | override_sysram_ds_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20** | **19 18 17 16 15 14 13 12** | **11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:12 | R/W | override_sysram_ds | If set, software now has direct control of sysram's DS signals.<br>[23:20] reserved<br>[19] 16kB macro at high address space<br>[18] 16kB macro<br>[17] 16kB macro<br>[16] 16kB macro<br>[15] 16kB macro<br>[14] 16kB macro<br>[13] 16kB macro<br>[12] 16kB macro at lowest address space | 8'd0 |
| 7:0 | R/W | override_sysram_ds_val | If override is set, what is state of sysram's DS signals?<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** SYSRAM_OVERRIDES6
**Address** 0x5015806c
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | override_sysram_ds | | override_sysram_ds_val |
|---|---|---|---|---|

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| Name | RSVD | override_sysram_sd | RSVD | override_sysram_sd_val |
|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19:12 | R/W | override_sysram_sd | If set, software now has direct control of sysram's SD signals.<br>[23:20] reserved<br>[19] 16kB macro at high address space<br>[18] 16kB macro<br>[17] 16kB macro<br>[16] 16kB macro<br>[15] 16kB macro<br>[14] 16kB macro<br>[13] 16kB macro<br>[12] 16kB macro at lowest address space | 8'd0 |
| 7:0 | R/W | override_sysram_sd_val | If override is set, what is state of sysram's SD signals?<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** SYSRAM_OVERRIDES7
**Address** 0x50158070
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | block_sysram_vddcut |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | block_sysram_vddcut | Block core's FSM from cutting the vdd supplies to the sysram.<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** SYSRAM_OVERRIDES8
**Address** 0x50158074
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | block_sysram_sd | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | block_sysram_sd | Block core's FSM from setting the SD signal to the sysram.<br>[11:8] reserved<br>[7] 16kB macro at high address space<br>[6] 16kB macro<br>[5] 16kB macro<br>[4] 16kB macro<br>[3] 16kB macro<br>[2] 16kB macro<br>[1] 16kB macro<br>[0] 16kB macro at lowest address space | 8'd0 |

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**Name** EMRAM_OVERRIDES
**Address** 0x50158078
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | override_e |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Name | RSVD | override_emram_vddcut | RSVD | override_emram_vddcut_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13 12** | **11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13:12 | R/W | override_emram_vddcut | If set, software now has direct control of exchange memory vddcut signals. | 2'd0 |
| 1:0 | R/W | override_emram_vddcut_val | If override is set, what are the states of exchange memory vddcut signals? | 2'd0 |

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**Name** EMRAM_OVERRIDES2
**Address** 0x5015807c
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_emram_clken | RSVD | override_emram_clken_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13 12** | **11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13:12 | R/W | override_emram_clken | If set, software now has direct control of exchange memory clock enable signals. | 2'd0 |
| 1:0 | R/W | override_emram_clken_val | If override is set, what are the states of exchange memory clock enable signals? | 2'd0 |

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**Name** EMRAM_OVERRIDES3

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**Address** 0x50158080
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_emram_iso | RSVD | override_emram_iso_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13 12** | **11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13:12 | R/W | override_emram_iso | If set, software now has direct control of exchange memory isolation signals. | 2'd0 |
| 1:0 | R/W | override_emram_iso_val | If override is set, what are the states of exchange memory isolation signals? | 2'd0 |

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**Name** EMRAM_OVERRIDES4
**Address** 0x50158084
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_emram_ls | RSVD | override_emram_ls_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13 12** | **11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 13:12 | R/W | override_emram_ls | If set, software now has direct control of exchange memory LS signals. | 2'd0 |
| 1:0 | R/W | override_emram_ls_val | If override is set, what are the states of exchange memory LS signals? | 2'd0 |

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**Name** EMRAM_OVERRIDES5
**Address** 0x50158088
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_emram_ds | RSVD | override_emram_ds_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13 12** | **11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13:12 | R/W | override_emram_ds | If set, software now has direct control of exchange memory DS signals. | 2'd0 |
| 1:0 | R/W | override_emram_ds_val | If override is set, what are the states of exchange memory DS signals? | 2'd0 |

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**Name** EMRAM_OVERRIDES6
**Address** 0x5015808c
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_emram_sd | RSVD | override_emram_sd_val |
|---|---|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13:12 | R/W | override_emram_sd | If set, software now has direct control of exchange memory SD signals. | 2'd0 |
| 1:0 | R/W | override_emram_sd_val | If override is set, what are the states of exchange memory SD signals? | 2'd0 |

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**Name** EMRAM_OVERRIDES7
**Address** 0x50158090
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | block_emram_vddcut |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | block_emram_vddcut | If set, core's FSM cannot open the exchange memory head switches. | 2'd0 |

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**Name** EMRAM_OVERRIDES8
**Address** 0x50158094
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | block_emram_sd |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | block_emram_sd | If set, core's FSM cannot set the exchange memory SD signals. | 2'd0 |

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**Name** ICACHE_OVERRIDES
**Address** 0x50158098
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | override_icache_vddcut | | | | RSVD | | | | | | | | override_icache_vddcut_val | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:12 | R/W | override_icache_vddcut | If set, software now has direct control of cpu icache's head switches.<br>[15] data1 macro<br>[14] data0 macro<br>[13] tag1 macro<br>[12] tag0 macro | 4'd0 |
| 3:0 | R/W | override_icache_vddcut_val | If override is set, what is state of cpu icache's head switches?<br>[3] data1 macro; [1] head switch open [0] head switch close<br>[2] data0 macro; [1] head switch open [0] head switch close<br>[1] tag1 macro; [1] head switch open [0] head switch close<br>[0] tag0 macro; [1] head switch open [0] head switch close | 4'd0 |

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**Name** ICACHE_OVERRIDES2
**Address** 0x5015809c
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | | | override_ic |
|---|---|---|---|---|

| Name | RSVD | | | | | | | | | | | | | | | | icache_clken | | | | RSVD | | | | | | | | acache_clken_val | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:12 | R/W | override_icache_clken | If set, software now has direct control of cpu icache's clock enable signals.<br>[15] data1 macro<br>[14] data0 macro<br>[13] tag1 macro<br>[12] tag0 macro | 4'd0 |
| 3:0 | R/W | override_icache_clken_val | If override is set, what is state of cpu icache's clock enable signals?<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** ICACHE_OVERRIDES3
**Address** 0x501580a0
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | override_icache_iso | | | | RSVD | | | | | | | | override_icache_iso_val | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:12 | R/W | override_icache_iso | If set, software now has direct control of cpu icache's isolation signals.<br>[15] data1 macro<br>[14] data0 macro<br>[13] tag1 macro | 4'd0 |

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| | | | | |
|---|---|---|---|---|
| | | | [12] tag0 macro | |
| 3:0 | R/W | override_icache_iso_val | If override is set, what is state of cpu icache's isolation signals?<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** ICACHE_OVERRIDES4
**Address** 0x501580a4
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | override_icache_ls | | | | RSVD | | | | | | override_icache_ls_val | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:12 | R/W | override_icache_ls | If set, software now has direct control of cpu icache's LS signals.<br>[15] data1 macro<br>[14] data0 macro<br>[13] tag1 macro<br>[12] tag0 macro | 4'd0 |
| 3:0 | R/W | override_icache_ls_val | If override is set, what is state of cpu icache's LS signals?<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** ICACHE_OVERRIDES5
**Address** 0x501580a8
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | | | override_i | | | override_icache_ls_val |
|---|---|---|---|---|---|---|---|

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| Name | RSVD | override_icache_ds | RSVD | override_icache_ds_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12** | **11 10 9 8 7 6 5 4** | **3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:12 | R/W | override_icache_ds | If set, software now has direct control of cpu icache's DS signals.<br>[15] data1 macro<br>[14] data0 macro<br>[13] tag1 macro<br>[12] tag0 macro | 4'd0 |
| 3:0 | R/W | override_icache_ds_val | If override is set, what is state of cpu icache's DS signals?<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** ICACHE_OVERRIDES6
**Address** 0x501580ac
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_icache_sd | RSVD | override_icache_sd_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12** | **11 10 9 8 7 6 5 4** | **3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:12 | R/W | override_icache_sd | If set, software now has direct control of cpu icache's SD signals.<br>[15] data1 macro<br>[14] data0 macro<br>[13] tag1 macro<br>[12] tag0 macro | 4'd0 |

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| | | | | |
|---|---|---|---|---|
| 3:0 | R/W | override_icache_sd_val | If override is set, what is state of cpu icache's SD signals?<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** ICACHE_OVERRIDES7
**Address** 0x501580b0
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | block_icache_vddcut | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3:0 | R/W | block_icache_vddcut | Block core's FSM from cutting the vdd supplies to the icache.<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** ICACHE_OVERRIDES8
**Address** 0x501580b4
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | block_icache_sd | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3:0 | R/W | block_icache_sd | Block core's FSM from setting the SD signal to the icache.<br>[3] data1 macro<br>[2] data0 macro<br>[1] tag1 macro<br>[0] tag0 macro | 4'd0 |

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**Name** RRAMCACHE_OVERRIDES
**Address** 0x501580b8
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_rramcache_vddcut | RSVD | override_rramcache_vddcut_val |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 | 17 16 15 14 13 12 | 11 10 9 8 7 6 | 5 4 3 2 1 0 |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17:12 | R/W | override_rramcache_vddcut | If set, software now has direct control of cpu rram cache's head switches.<br>[17] data1 macro<br>[16] data0 macro<br>[15] tag3 macro<br>[14] tag2 macro<br>[13] tag1 macro<br>[12] tag0 macro | 6'd0 |
| 5:0 | R/W | override_rramcache_vddcut_val | If override is set, what is state of cpu rram cache's's head switches?<br>[5] data1 macro; [1] head switch open [0] head switch close<br>[4] data0 macro; [1] head switch open [0] head switch close<br>[3] tag3 macro; [1] head switch open [0] head switch close<br>[2] tag2 macro; [1] head switch open [0] head switch close<br>[1] tag1 macro; [1] head switch open [0] head switch close<br>[0] tag0 macro; [1] head switch open [0] head switch close | 6'd0 |

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**Name** RRAMCACHE_OVERRIDES2

**Address** 0x501580bc
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_rramcache_clken | RSVD | override_rramcache_clken_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18** | **17 16 15 14 13 12** | **11 10 9 8 7 6** | **5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17:12 | R/W | override_rramcache_clken | If set, software now has direct control of cpu rram cache's clock enable signals.<br>[17] data1 macro<br>[16] data0 macro<br>[15] tag3 macro<br>[14] tag2 macro<br>[13] tag1 macro<br>[12] tag0 macro | 6'd0 |
| 5:0 | R/W | override_rramcache_clken_val | If override is set, what is state of cpu rram cache's clock enable signals?<br>[5] data1 macro<br>[4] data0 macro<br>[3] tag3 macro<br>[2] tag2 macro<br>[1] tag1 macro<br>[0] tag0 macro | 6'd0 |

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**Name** RRAMCACHE_OVERRIDES3
**Address** 0x501580c0
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | override | | override |
|---|---|---|---|---|

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| Name | RSVD | override_rramcache_iso | RSVD | override_rramcache_iso_val |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18** | **17 16 15 14 13 12** | **11 10 9 8 7 6** | **5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17:12 | R/W | override_rramcache_iso | If set, software now has direct control of cpu rram cache's isolation signals.<br>[17] data1 macro<br>[16] data0 macro<br>[15] tag3 macro<br>[14] tag2 macro<br>[13] tag1 macro<br>[12] tag0 macro | 6'd0 |
| 5:0 | R/W | override_rramcache_iso_val | If override is set, what is state of cpu rram cache's isolation signals?<br>[5] data1 macro<br>[4] data0 macro<br>[3] tag3 macro<br>[2] tag2 macro<br>[1] tag1 macro<br>[0] tag0 macro | 6'd0 |

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**Name** RRAMCACHE_OVERRIDES4
**Address** 0x501580c4
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_rramcache_ls | RSVD | override_rramcache_ls_v |
|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | a l | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17:12 | R/W | override_rramcache_ls | If set, software now has direct control of cpu rram cache's LS signals.<br>[17] data1 macro<br>[16] data0 macro<br>[15] tag3 macro<br>[14] tag2 macro<br>[13] tag1 macro<br>[12] tag0 macro | 6'd0 |
| 5:0 | R/W | override_rramcache_ls_val | If override is set, what is state of cpu rram cache's LS signals?<br>[5] data1 macro<br>[4] data0 macro<br>[3] tag3 macro<br>[2] tag2 macro<br>[1] tag1 macro<br>[0] tag0 macro | 6'd0 |

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**Name** RRAMCACHE_OVERRIDES5
**Address** 0x501580c8
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_rramcache_ds | RSVD | override_rramcache_ds_val | RSVD |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18** | **17 16** | **15 14 13 12 11 10 9 8 7 6** | **5 4** | **3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17:16 | R/W | override_rramcache_ds | If set, software now has direct control of cpu rram cache's DS signals.<br>[17] data1 macro<br>[16] data0 macro | 2'd0 |
| 5:4 | R/W | override_rramcache_ds_val | If override is set, what is state of cpu rram cache's DS signals?<br>[5] data1 macro<br>[4] data0 macro | 2'd0 |

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**Name** RRAMCACHE_OVERRIDES6

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**Address** 0x501580cc
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | override_rramcache_sd | RSVD | override_rramcache_sd_val | RSVD |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18** | **17 16** | **15 14 13 12 11 10 9 8 7 6** | **5 4** | **3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17:16 | R/W | override_rramcache_sd | If set, software now has direct control of cpu rram cache's SD signals.<br>[17] data1 macro<br>[16] data0 macro | 2'd0 |
| 5:4 | R/W | override_rramcache_sd_val | If override is set, what is state of cpu rram cache's SD signals?<br>[5] data1 macro<br>[4] data0 macro | 2'd0 |

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**Name** RRAMCACHE_OVERRIDES7
**Address** 0x501580d0
**Description** override portion of pmu control signals
**Reset** 0x00000000

| Name | RSVD | block_rramcache_vddc |
|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | u t | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:0 | R/W | block_rramcache_vddcut | Block core's FSM from cutting the vdd supplies to the rram cache.<br>[5] data1 macro<br>[4] data0 macro<br>[3] tag3 macro<br>[2] tag2 macro<br>[1] tag1 macro<br>[0] tag0 macro | 6'd0 |

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**Name** RRAMCACHE_OVERRIDES8
**Address** 0x501580d4
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | R S V D | | | | | | | | | | | | | | | | | | | | | | | | | | b l o c k _ r r a m c a c h e _ s d | | R S V D | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:4 | R/W | block_rramcache_sd | Block core's FSM from setting the SD signal to the rram cache.<br>[5] data1 macro<br>[4] data0 macro | 2'd0 |

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**Name** QSPICACHE_OVERRIDES
**Address** 0x501580d8
**Description** override portion of pmu control signals
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| Name | RSVD | override_qspicache_vddcut | override_qspicache_vddcut_val | override_qspicache_clken | override_qspicache_clken_val | override_qspicache_iso | override_qspicache_iso_val | override_qspicache_ls | override_qspicache_ls_val | override_qspicache_ds | override_qspicache_ds_val | override_qspicache_sd | override_qspicache_sd_val | block_qspicache_vddcut | block_qspicache_sd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13 | R/W | override_qspicache_vddcut | if set, head switch on qspi cache is now software controllable | 1'b0 |
| 12 | R/W | override_qspicache_vddcut_val | if override is set, should head switch on qspi cache be open or close?<br>1=open<br>0=close | 1'b0 |
| 11 | R/W | override_qspicache_clken | if set, clock gating to qspi cache is now under software control | 1'b0 |
| 10 | R/W | override_qspicache_clken_val | if override is set, should the clock gate on the qspi cache be enabled?<br>1=clock running<br>0=clock gated off | 1'b0 |
| 9 | R/W | override_qspicache_iso | if set, software now has direct control of the qspi cache ISO pin | 1'b0 |
| 8 | R/W | override_qspicache_iso_val | if override is set, what value should be driven onto qspi cache ISO pin? | 1'b0 |
| 7 | R/W | override_qspicache_ls | if set, software now has direct control of the qspi cache LS pin | 1'b0 |
| 6 | R/W | override_qspicache_ls_val | if override is set, what value should be driven onto qspi cache LS pin? | 1'b0 |
| 5 | R/W | override_qspicache_ds | if set, software now has direct control of the qspi cache DS pin | 1'b0 |
| 4 | R/W | override_qspicache_ds_val | if override is set, what value should be driven onto qspi cache DS pin? | 1'b0 |
| 3 | R/W | override_qspicache_sd | if set, software now has direct control of the qspi cache SD pin | 1'b0 |
| 2 | R/W | override_qspicache_sd_val | if override is set, what value should be driven onto qspi cache SD pin? | 1'b0 |
| 1 | R/W | block_qspicache_vddcut | if set, this core's FSM cannot power off the qspi cache | 1'b0 |
| 0 | R/W | block_qspicache_sd | if set, this core's FSM cannot assert the qspi cache SD pin | 1'b0 |

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**Name** COUNTER_CONTROL
**Address** 0x501580dc
**Description** control signals for counters
**Reset** 0x00000000

| Name | RSVD | load_count_down_halt_real_time_c |
|---|---|---|
| **Bits** | **31:2** | **1** | **0** |

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | timer | counter |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | load_count_down_timer | loads on positive edge; hardware doesn't reset this bit | 1'b0 |
| 0 | R/W | halt_real_time_counter | stop the real time counter from advancing | 1'b0 |

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**Name** CURRENT_REAL_TIME
**Address** 0x501580e0
**Description** counter value
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | real time in clk_lpc clock cycles | 32'd0 |

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**Name** CURRENT_COUNT_DOWN_TIME
**Address** 0x501580e4
**Description** counter value
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | this value is on 32kHz domain; read twice and check value is the same; if it's not, reread | 32'd0 |

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**Name** INIT_COUNT_DOWN
**Address** 0x501580e8
**Description** counter value
**Reset** 0x00000fff

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | val | initial value for count down timer | 32'hfff |

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**Name** INST_PENDING
**Address** 0x501580ec
**Description** instructions from clk_mpc to clk_lpc still pending
**Reset** 0x00000000

| Name | RSVD | wdog_timer_disable | cntdwn_timer_load | RSVD |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18** | **17** | **16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 17 | R | wdog_timer_disable | disable/enable of watchdog timer is still pending | 1'b0 |
| 16 | R | cntdwn_timer_load | load of count down timer is still pending | 1'b0 |

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**Name** STATUS
**Address** 0x501580f0
**Description** status signals
**Reset** 0x00000000

| Name | RSVD | ble_osc_on_triggered | wurx_triggered | brownout_triggered | endoflife_triggered | energy4TX_triggered | energy4CPU_triggered | pmu_lpcomp_triggered | pmu_timer_triggered | i2c_triggered | spi_triggered | shub_triggered | dbg_triggered | ksm_triggered | qdec_triggered | wurx1_triggered | wurx0_triggered | gpio_triggered | timer_triggered | RSVD | in_hibernate_state | in_cpu_active_state | in_active_state |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25 | R | ble_osc_on_triggered | wake triggered by ble_osc_on | 1'b0 |
| 24 | R | wurx_triggered | wake triggered by multiple events of wurx. | 1'b0 |
| 23 | R | brownout_triggered | wake triggered by pmu brownout | 1'b0 |
| 22 | R | endoflife_triggered | wake triggered by pmu endoflife | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 21 | R | energy4TX_triggered | wake triggered by pmu energy4TX | 1'b0 |
| 20 | R | energy4CPU_triggered | wake triggered by pmu energy4CPU | 1'b0 |
| 19 | R | pmu_lpcomp_triggered | wake triggered by pmu lpcomp threshold crossing | 1'b0 |
| 18 | R | pmu_timer_triggered | wake triggered by pmu timer expiring | 1'b0 |
| 17 | R | i2c_triggered | wake triggered by i2c slave activity | 1'b0 |
| 16 | R | spi_triggered | wake triggered by spi slave activity | 1'b0 |
| 15 | R | shub_triggered | wake triggered by sensor hub | 1'b0 |
| 14 | R | dbg_triggered | wake triggered by debugger | 1'b0 |
| 13 | R | ksm_triggered | wake triggered by ksm activity | 1'b0 |
| 12 | R | qdec_triggered | wake triggered by qdec activity | 1'b0 |
| 11 | R | wurx1_triggered | wake triggered by wurx1 | 1'b0 |
| 10 | R | wurx0_triggered | wake triggered by wurx0 | 1'b0 |
| 9 | R | gpio_triggered | wake triggered by gpio | 1'b0 |
| 8 | R | timer_triggered | wake triggered by count down timer | 1'b0 |
| 2 | R | in_hibernate_state | pseq is currently in hibernate state | 1'b0 |
| 1 | R | in_cpu_active_state | pseq is currently in cpu active state | 1'b0 |
| 0 | R | in_active_state | pseq is currently in active state | 1'b0 |

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**Name** PERSISTENT0
**Address** 0x501580f4
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT1
**Address** 0x501580f8
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT2
**Address** 0x501580fc
**Description** persistent scratchpad register
**Reset** 0x00000000

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| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT3
**Address** 0x50158100
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT4
**Address** 0x50158104
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT5
**Address** 0x50158108
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT6
**Address** 0x5015810c
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** PERSISTENT7
**Address** 0x50158110
**Description** persistent scratchpad register
**Reset** 0x00000000

| Name | data | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data | holds persistent state; persists through hibernation; might be of use to sw | 32'h0 |

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**Name** SENSOR_HUB_CONTROL
**Address** 0x50158114
**Description** control signals for sensor hub
**Reset** 0xf00000e2

| Name | shub_sram3_acc_en | shub_sram2_acc_en | shub_sram1_acc_en | shub_sram0_acc_en | RSVD | shub_hib_exit_en | shub_start_adv_ovrd | shub_start_adv_val | shub_adv_enable | shub_breakout_imm_en | shub_start_ovrd | shub_start_val | shub_clken_slow | shub_reset_slow | shub_vddcut | shub_iso | shub_clken | shub_running_ovrd | shub_running_val | shub_reset | enable_func |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27 26 25 24 23 22 21 20 19 18 17 16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | shub_sram3_acc_en | 1= allow shub access to sram3 | 1'b1 |

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| | | | | |
|---|---|---|---|---|
| 30 | R/W | shub_sram2_acc_en | 1= allow shub access to sram2 | 1'b1 |
| 29 | R/W | shub_sram1_acc_en | 1= allow shub access to sram1 | 1'b1 |
| 28 | R/W | shub_sram0_acc_en | 1= allow shub access to sram0 | 1'b1 |
| 15 | R/W | shub_hib_exit_en | 1= exit out of hib, 0=exit out of retain. | 1'b0 |
| 14 | R/W | shub_start_adv_ovrd | 1= override | 1'b0 |
| 13 | R/W | shub_start_adv_val | value fo shub_start_adv when override is set | 1'b0 |
| 12 | R/W | shub_adv_enable | 1= enable adv along with sensor hub. | 1'b0 |
| 11 | R/W | shub_breakout_imm_en | 1= breakout sensor asap when a valid out-of-HIB event is triggered. 0= wait for sensor hub wrap-up. | 1'b0 |
| 10 | R/W | shub_start_ovrd | 1= override | 1'b0 |
| 9 | R/W | shub_start_val | value fo shub_start when override is set | 1'b0 |
| 8 | R/W | shub_clken_slow | for 32kHz portion; sensor hub clock enable<br>0 = clock off<br>1 = clock running | 1'b0 |
| 7 | R/W | shub_reset_slow | for 32kHz portion; 1 = keep shub in reset state; 0 = release shub from reset | 1'b1 |
| 6 | R/W | shub_vddcut | sensor hub head switch control<br>0 = head switch is closed<br>1 = head switch is open | 1'b1 |
| 5 | R/W | shub_iso | sensor hub isolation; do not assert/deassert in same cycle as shub_clken<br>0 = not isolated<br>1 = isolated | 1'b1 |
| 4 | R/W | shub_clken | sensor hub clock enable<br>0 = clock gated off<br>1 = clock running | 1'b0 |
| 3 | R/W | shub_running_ovrd | 1= override | 1'b0 |
| 2 | R/W | shub_running_val | value fo shub_running when override is set | 1'b0 |
| 1 | R/W | shub_reset | 1= keep shub in reset state; 0= release shub from reset | 1'b1 |
| 0 | R/W | enable_func | 1= sensor hub can trigger pseq out of hibernation early due to a threshold crossing event in sensor hub.<br>0= sensor hub cannot trigger pseq out of hibernation.<br>This bit does not disable the trigger signal to start the sensor functionality at the onset of hibernation. | 1'b0 |

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**Name** KSMQDEC_CONTROL
**Address** 0x50158118
**Description** keyboard and mouse power domain controls
**Reset** 0x0000000e

| Name | RSVD | ksmqdec_vddcut | ksmqdec_iso | ksmqdec_first | ksmqdec_clken |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | ksmqdec_vddcut | head switch control<br>0 = head switch is closed<br>1 = head switch is open | 1'b1 |
| | | | isolation | |

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| | | | | |
|---|---|---|---|---|
| 2 | R/W | ksmqdec_iso | 0 = not isolated<br>1 = isolated | 1'b1 |
| 1 | R/W | ksmqdec_frst | forced reset<br>1 = reset<br>0 = not reset | 1'b1 |
| 0 | R/W | ksmqdec_clken | clock enable<br>0 = clock gated off<br>1 = clock running | 1'b0 |

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**Name** SPII2C_CONTROL
**Address** 0x5015811c
**Description** spi and i2c power domain controls
**Reset** 0x0000000e

| Name | RSVD | i2c1_slv_id_sel | i2c0_slv_id_sel | spii2c_vddcut | spii2c_iso | spii2c_frst | spii2c_clken |
|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | i2c1_slv_id_sel | select secondary i2c device id for i2c1 | 1'b0 |
| 4 | R/W | i2c0_slv_id_sel | select secondary i2c device id for i2c0 | 1'b0 |
| 3 | R/W | spii2c_vddcut | head switch control<br>0 = head switch is closed<br>1 = head switch is open | 1'b1 |
| 2 | R/W | spii2c_iso | isolation<br>0 = not isolated<br>1 = isolated | 1'b1 |
| 1 | R/W | spii2c_frst | forced reset<br>1 = reset<br>0 = not reset | 1'b1 |
| 0 | R/W | spii2c_clken | clock enable<br>0 = clock gated off<br>1 = clock running | 1'b0 |

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**Name** FLASH_CONTROL
**Address** 0x50158120
**Description** low power control of external flash needs
**Reset** 0x100b9aba

| | RS | pd_b4 | exp | expm_ | pd_o | opc | mo | rpd_ha | power_c |
|---|---|---|---|---|---|---|---|---|---|

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| Name | VD | _sleep | m_en | opcode | pcode | ode | de | s_clock | ycle_en |
|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30** | **29** | **28** | **27 26 25 24** | **23 22 21 20 19 18** | **17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29 | R/W | pd_b4_sleep | 1=issue flash deep power down prior to low power mode<br>0=no command issue. | 1'b0 |
| 28 | R/W | expm_en | 1=exit performance mode before issuing pd_opcode<br>0=not in performance mode. | 1'b1 |
| 27:20 | R/W | expm_opcode | opcode that takes flash out of performance mode | 8'h00 |
| 19:12 | R/W | pd_opcode | opcode that puts external flash into deep power down mode | 8'hb9 |
| 11:4 | R/W | opcode | opcode that wakes up the external flash from deep power down mode | 8'hab |
| 3:2 | R/W | mode | defining interface bus width of opcode; 0,3=single 1=dual 2=quad<br>For Macronix, set it to 0 because opcode is always serialized;<br>For Micron, it depends on the pre-configured interface bus width on the flash device. | 2'h2 |
| 1 | R/W | rpd_has_clock | 1=flash requires opcode to wakeup<br>0=flash requires only CS to wakeup | 1'b1 |
| 0 | R/W | power_cycle_en | 1=trigger command to release flash from power down when waking up<br>0= no action | 1'b0 |

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**Name** FLASH_CONTROL2
**Address** 0x50158124
**Description** low power control of external flash needs
**Reset** 0x0002007f

| Name | RSVD | enpm_clr_en | expm_mode | RSVD | pseq_state_match |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19** | **18** | **17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 18 | R/W | enpm_clr_en | 1=clears enable_performance_mode bit in qspi_ahb too if expm_opcode is sent to flash. | 1'b0 |
| 17:16 | R/W | expm_mode | defining interface bus width of performance exit command; 0,3=single 1=dual 2=quad<br>For Macronix, set it to 2 as it's the only interface bus width equipped with performance enhancement mode<br>For Micron, it depends on the pre-configured interface bus width on the flash device, likely quad. | 2'h2 |
| 7:0 | R/W | pseq_state_match | pseq state to trigger flash deep power down. | 8'h7f |

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**Name** PMU_STATUS
**Address** 0x50158128
**Description** pmu status information
**Reset** 0x00000000

| Name | RSVD | soc_off_wkup_det | RSVD | brownout | endoflife | energy4TX | energy4CPU |
|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8** | **7 6 5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:5 | R | soc_off_wkup_det | returned from SOC Off state | 3'b0 |
| 3 | R | brownout | brownout detected; go to low power state as quickly as possible | 1'b0 |
| 2 | R | endoflife | battery is detected to be at end of life; do not start flash writes or radio operation | 1'b0 |
| 1 | R | energy4TX | enough energy is present to do a transmit | 1'b0 |
| 0 | R | energy4CPU | enough energy is present to operation the CPU | 1'b0 |

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**Name** PMU_INTERRUPT
**Address** 0x5015812c
**Description** pmu interrupt source
**Reset** 0x00000000

| Name | RSVD | clear_fe | clear_re | intmask |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8** | **7 6 5 4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 28 | R/W | clear_fe_brownout | clear the rising edge capture blocks | 5'b0 |
| 27 | | clear_fe_endOfLife | | |
| 26 | | clear_fe_energy4TX | | |
| 25 | | clear_fe_energy4CPU | | |
| 23 | R/W | clear_re_brownout | clear the falling edge capture blocks | 5'b0 |
| 22 | | clear_re_endOfLife | | |
| 21 | | clear_re_energy4TX | | |
| 20 | | clear_re_energy4CPU | | |
| 15 | | brownout_falling | | |

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| | | | | |
|---|---|---|---|---|
| 14 | R/W | brownout_rising | 0 = pmu_status_energy4CPU level<br>1 = pmu_status_energy4CPU negated level<br>2 = pmu_status_energy4CPU rising edge<br>3 = pmu_status_energy4CPU falling edge<br>4 = pmu_status_energy4TX level<br>5 = pmu_status_energy4TX negated level<br>6 = pmu_status_energy4TX rising edge<br>7 = pmu_status_energy4TX falling edge<br>8 = pmu_status_endOfLife level<br>9 = pmu_status_endOfLife negated level<br>10 = pmu_status_endOfLife rising edge<br>11 = pmu_status_endOfLife falling edge<br>12 = pmu_status_brownout level<br>13 = pmu_status_brownout negated level<br>14 = pmu_status_brownout rising edge<br>15 = pmu_status_brownout falling edge | 20'b0 |
| 13 | | brownout_low | | |
| 12 | | brownout_high | | |
| 11 | | endOfLife_falling | | |
| 10 | | endOfLife_rising | | |
| 9 | | endOfLife_low | | |
| 8 | | endOfLife_high | | |
| 7 | | energy4TX_falling | | |
| 6 | | energy4TX_rising | | |
| 5 | | energy4TX_low | | |
| 4 | | energy4TX_high | | |
| 3 | | energy4CPU_falling | | |
| 2 | | energy4CPU_rising | | |
| 1 | | energy4CPU_low | | |
| 0 | | energy4CPU_high | | |

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**Name** SYSRAM_WRITE_BLOCK_CFG
**Address** 0x50158130
**Description** Contains register fields associated with sysram_write_block_cfg.
**Reset** 0x00000000

| Name | RSVD | | | | | en_wrblk | a_hi | | | | | | | | | | | | a_lo | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 26 | R/W | en_wrblk | enable the sysram write lock; can't be undone unless a porst_soc is issued by pmu | 1'b0 |
| 25:13 | R/W | a_hi | high address on word boundary | 13'b0 |
| 12:0 | R/W | a_lo | low address on word boundary; any address from high address to low address is no longer writable once enable is set | 13'b0 |

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**Name** SYSRAM_SSRS_CFG
**Address** 0x50158134
**Description** Contains register fields associated with sysram_ssrs_cfg.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | enable | addr | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15 | R/W | enable | when set, enables r/w access to sysram during ssrs operation | 1'b0 |
| 14:0 | R/W | addr | sysram starting address for ssrs operation | 15'd0 |

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**Name** INTERRUPT_STATUS
**Address** 0x50158138
**Description** interrupt status
**Reset** 0x00000000

| Name | RSVD | returned_from_lp_state |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | returned_from_lp_state | indicates system returned from low power state (as oppposed to first power on) | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x5015813c
**Description** interrupt mask
**Reset** 0x00000001

| Name | RSVD | intrpt_mask |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt_mask | if nth bit set, the nth interrupt source is allowed to propogate to core's interrupt | 1'b1 |

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**Name** RESET_INTERRUPT
**Address** 0x50158140

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**Description** interrupt resets
**Reset** 0x00000000

| Name | RSVD | intrpt_reset |
|---|---|---|
| **Bits** | **31:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt_reset | not self clearing | 1'b0 |

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**Name** FPGA_CFG
**Address** 0x50158144
**Description** Contains register fields associated with fpga_cfg.
**Reset** 0x00000000

| Name | RSVD | cfg |
|---|---|---|
| **Bits** | **31:16** | **15:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | cfg | (no description) | 16'b0 |

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**Name** PLL
**Address** 0x50158148
**Description** Contains register fields associated with pll.
**Reset** 0x00000000

| Name | RSVD | divaccess | divref | divfb | divout | div2out | cpcur | filtc | filtr | RSVD |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:30** | **29** | **28:26** | **25:20** | **19:17** | **16** | **15:10** | **9:8** | **7:4** | **3:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29 | R/W | divaccess | (no description) | 1'b0 |
| 28:26 | R/W | divref | (no description) | 3'b0 |
| 25:20 | R/W | divfb | (no description) | 6'b0 |
| 19:17 | R/W | divout | (no description) | 3'b0 |
| 16 | R/W | div2out | (no description) | 1'b0 |
| 15:10 | R/W | cpcur | (no description) | 6'b0 |

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| | | | | |
|---|---|---|---|---|
| 9:8 | R/W | filtc | (no description) | 2'b0 |
| 7:4 | R/W | filtr | (no description) | 4'b0 |

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**Name** RADIO_STATUS
**Address** 0x5015814c
**Description** Contains register fields associated with radio_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | xtal_stable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | xtal_stable | (no description) | 1'b0 |

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**Name** RRAM_READ_CONFIG_LO
**Address** 0x50158150
**Description** Contains register fields associated with rram_read_config_lo.
**Reset** 0x00000000

| Name | dout | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dout | (no description) | 32'b0 |

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**Name** RRAM_READ_CONFIG_HI
**Address** 0x50158154
**Description** Contains register fields associated with rram_read_config_hi.
**Reset** 0x00000000

| Name | dout | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dout | (no description) | 32'b0 |

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**Name** SENSOR_HUB_IMM_BREAKOUT_MASK
**Address** 0x50158158
**Description** control signals for sensor hub
**Reset** 0x0000ffff

| Name | RSVD | watch_wurx | watch_brownout | watch_endoflife | watch_energy4TX | watch_energy4CPU | watch_pmu_timer | watch_pmu_lpcomp | watch_i2c | watch_spi | watch_shub | watch_ksm | watch_qdec | watch_wurx1 | watch_wurx0 | watch_cntdown | watch_gpio |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15 | R/W | watch_wurx | valid to immediately breakout on multiple wurx detects in certain amount of time | 1'b1 |
| 14 | R/W | watch_brownout | valid to immediately breakout on pmu asserting brownout | 1'b1 |
| 13 | R/W | watch_endoflife | valid to immediately breakout on pmu asserting endoflife | 1'b1 |
| 12 | R/W | watch_energy4TX | valid to immediately breakout on pmu asserting energy4TX | 1'b1 |
| 11 | R/W | watch_energy4CPU | valid to immediately breakout on pmu asserting energy4CPU | 1'b1 |
| 10 | R/W | watch_pmu_timer | valid to immediately breakout on pmu timer expiring | 1'b1 |
| 9 | R/W | watch_pmu_lpcomp | valid to immediately breakout on pmu lpcomp threshold crossing | 1'b1 |
| 8 | R/W | watch_i2c | valid to immediately breakout on i2c slave activity | 1'b1 |
| 7 | R/W | watch_spi | valid to immediately breakout on spi slave activity | 1'b1 |
| 6 | R/W | watch_shub | valid to immediately breakout on sensor hub trigger | 1'b1 |
| 5 | R/W | watch_ksm | valid to immediately breakout on keyboard activity | 1'b1 |
| 4 | R/W | watch_qdec | valid to immediately breakout on qdec activity | 1'b1 |
| 3 | R/W | watch_wurx1 | valid to immediately breakout on wurx1 detect | 1'b1 |
| 2 | R/W | watch_wurx0 | valid to immediately breakout on wurx0 detect | 1'b1 |
| 1 | R/W | watch_cntdown | valid to immediately breakout on count down timer expiring | 1'b1 |
| 0 | R/W | watch_gpio | valid to immediately breakout on gpio activity | 1'b1 |

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**Name** FSM_TUNINGA
**Address** 0x5015815c
**Description** Contains register fields associated with fsm_tuningA.
**Reset** 0x00008000

| Name | cnt9_lsb | cnt8 | cnt7 | cnt6 | cnt5 | cnt4 | cnt3 | cnt2 | cnt1 | cnt0 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27** | **26 25 24 23 22 21** | **20 19 18** | **17 16** | **15 14 13 12** | **11 10** | **9 8** | **7 6** | **5 4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 31:27 | R/W | cnt9_lsb | additional clk_lpc cycles to wait in states tied to cnt9 (lsb's); determines cpu_ldo wait to setttle time | 5'b0 |
| 26:24 | R/W | cnt8 | additional clk_lpc cycles to wait in states tied to cnt8 | 3'b0 |
| 23:21 | R/W | cnt7 | additional clk_lpc cycles to wait in states tied to cnt7 | 3'b0 |
| 20:18 | R/W | cnt6 | additional clk_lpc cycles to wait in states tied to cnt6 | 3'b0 |
| 17:15 | R/W | cnt5 | additional clk_lpc cycles to wait in states tied to cnt5 | 3'b1 |
| 14:12 | R/W | cnt4 | additional clk_lpc cycles to wait in states tied to cnt4 | 3'b0 |
| 11:9 | R/W | cnt3 | additional clk_lpc cycles to wait in states tied to cnt3 | 3'b0 |
| 8:6 | R/W | cnt2 | additional clk_lpc cycles to wait in states tied to cnt2 | 3'b0 |
| 5:3 | R/W | cnt1 | additional clk_lpc cycles to wait in states tied to cnt1 | 3'b0 |
| 2:0 | R/W | cnt0 | additional clk_lpc cycles to wait in states tied to cnt0 | 3'b0 |

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**Name** FSM_TUNINGB
**Address** 0x50158160
**Description** Contains register fields associated with fsm_tuningB.
**Reset** 0x00000000

| Name | cnt9_msb | rcnt9 | rcnt8 | rcnt7 | rcnt6 | rcnt5 | rcnt4 | rcnt3 | rcnt2 | rcnt1 | rcnt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30** | **29 28 27** | **26 25 24** | **23 22 21** | **20 19 18** | **17 16 15** | **14 13 12** | **11 10 9** | **8 7 6** | **5 4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:30 | R/W | cnt9_msb | additional clk_lpc cycles to wait in states tied to cnt9 (msb's) | 2'b0 |
| 29:27 | R/W | rcnt9 | additional clk_lpc cycles to wait in states tied to rcnt9 | 3'b0 |
| 26:24 | R/W | rcnt8 | additional clk_lpc cycles to wait in states tied to rcnt8 | 3'b0 |
| 23:21 | R/W | rcnt7 | additional clk_lpc cycles to wait in states tied to rcnt7 | 3'b0 |
| 20:18 | R/W | rcnt6 | additional clk_lpc cycles to wait in states tied to rcnt6 | 3'b0 |
| 17:15 | R/W | rcnt5 | additional clk_lpc cycles to wait in states tied to rcnt5 | 3'b0 |
| 14:12 | R/W | rcnt4 | additional clk_lpc cycles to wait in states tied to rcnt4 | 3'b0 |
| 11:9 | R/W | rcnt3 | additional clk_lpc cycles to wait in states tied to rcnt3 | 3'b0 |
| 8:6 | R/W | rcnt2 | additional clk_lpc cycles to wait in states tied to rcnt2 | 3'b0 |
| 5:3 | R/W | rcnt1 | additional clk_lpc cycles to wait in states tied to rcnt1 | 3'b0 |
| 2:0 | R/W | rcnt0 | additional clk_lpc cycles to wait in states tied to rcnt0 | 3'b0 |

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**Name** WDOG_CONTROL
**Address** 0x50158164
**Description** control signals for 32KHz watchdog timer
**Reset** 0x00000001

| Name | RSVD | internal_s | timer_val_ | disable |
|---|---|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | can | sel | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | internal_scan | (no description) | 1'b0 |
| 1 | R/W | timer_val_sel | selects timer value to load<br>1= 1-sec 0= 100-msec in 32KHz cycles | 1'b0 |
| 0 | R/W | disable | 1= disables wdog timer | 1'b1 |

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**Name** WDOG_TIMER
**Address** 0x50158168
**Description** Read current 32KHz watchdog timer value
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | val | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R | val | this value is on 32kHz domain; read twice and check value is the same; if it's not, reread | 16'd0 |

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**Name** CORE_ID
**Address** 0x50158ffc
**Description** core's id
**Reset** 0x50534551

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | PSEQ in ASCII | 32'h50534551 |

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# SWD_SECURE

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**Name** DAP_ADDR_CTRL
**Address** 0x50159000
**Description** Contains register fields associated with dap_addr_ctrl.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | dap_add | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | dap_addr | (no description) | 2'b0 |

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**Name** DAP_WDATA_CTRL
**Address** 0x50159004
**Description** Contains register fields associated with dap_wdata_ctrl.
**Reset** 0x00000000

| Name | dap_wdata | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | dap_wdata | (no description) | 32'b0 |

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**Name** SWD_CLK_DIV_CTRL
**Address** 0x50159008
**Description** Contains register fields associated with swd_clk_div_ctrl.
**Reset** 0x00000003

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | swd_clk_div | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2:0 | R/W | swd_clk_div | clock divider = 2^(swd_clk_div) | 3'd3 |

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**Name** PATTERN_UPPER_CTRL
**Address** 0x5015900c
**Description** Contains register fields associated with pattern_upper_ctrl.
**Reset** 0xffffffff

| Name | patt | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | e r n |
|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | pattern | pattern to switch from jtag to swd , lsb is shifted out first | 32'hffffffff |

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**Name** PATTERN_LOWER_CTRL
**Address** 0x50159010
**Description** Contains register fields associated with pattern_lower_ctrl.
**Reset** 0xffffffff

| Name | p a t t e r n |
|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | pattern | pattern to switch from jtag to swd , lsb is shifted out first | 32'hffffffff |

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**Name** TAIL_DUR
**Address** 0x50159014
**Description** Contains register fields associated with tail_dur.
**Reset** 0x00000005

| Name | R S V D | c o u n t |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3:0 | R/W | count | exact number of extra cycles after a packet. 0 means no tail. must be at least 5 cycles after a DAP packet | 4'h5 |

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**Name** PATTERN_DUR
**Address** 0x50159018
**Description** Contains register fields associated with pattern_dur.
**Reset** 0x0000003f

| Name | R S V D | c o u n t |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 7:0 | R/W | count | number cycles to drive the pattern MSB first is count + 1. e.g. 64 cycles by default | 8'd63 |

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**Name** DAP_BRIDGE_STATUS
**Address** 0x5015901c
**Description** Contains register fields associated with dap_bridge_status.
**Reset** 0x00000000

| Name | RSVD | busy |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | busy | (no description) | 1'b0 |

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**Name** DAP_RESP_STATUS
**Address** 0x50159020
**Description** Contains register fields associated with dap_resp_status.
**Reset** 0x00000000

| Name | RSVD | par_err | dap_resp |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | par_err | 1= RDATA has parity error, 0= otherwise | 1'b0 |
| 2:0 | R | dap_resp | (no description) | 3'b0 |

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**Name** RW_CTRL
**Address** 0x50159024
**Description** Contains register fields associated with RW_ctrl.
**Reset** 0x00000000

| Name | RSVD | RnW |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | RnW | 1 for Read, 0 for Write | 1'b0 |

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**Name** APNDP_CTRL
**Address** 0x50159028

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**Description** Contains register fields associated with apndp_ctrl.
**Reset** 0x00000000

| Name | RSVD | APnDP |
|---|---|---|
| **Bits** | **31:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | APnDP | 0 for DPACC, 1 for APACC | 1'b0 |

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**Name** TRIGGER_CTRL
**Address** 0x5015902c
**Description** Contains register fields associated with trigger_ctrl.
**Reset** 0x00000000

| Name | RSVD | go_pattern | RSVD | go_packet |
|---|---|---|---|---|
| **Bits** | **31:3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | go_pattern | 1 to start resetting by sending a pattern of bits as prescribed by pattern_upper_ctrl and pattern_lower_ctrl.<br>The length of the pattern is set by pattern_dur. not self-resetting.<br>The reset pattern is to bring the target DAP to an inital state. | 1'b0 |
| 0 | R/W | go_packet | 1 to start a dap packet , not self-resetting.<br>This trigger does not start a DAP reset. See go_pattern to for DAP reset. | 1'b0 |

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**Name** DAP_RDATA
**Address** 0x50159030
**Description** Contains register fields associated with dap_rdata.
**Reset** 0x00000000

| Name | dap_rdata |
|---|---|
| **Bits** | **31:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | dap_rdata | (no description) | 32'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x50159034
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000000

| Name | RSVD | passthru1 | passthru0 |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | passthru1 | 1=allow interrupt1 (par err on readback data) to be OR'ed into core interrupt | 1'b0 |
| 0 | R/W | passthru0 | 1=allow interrupt0 (read done) to be OR'ed into core interrupt | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x50159038
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | interrupt1 | interrupt0 |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | interrupt1 | saw a rising edge on par_err; independent of mask | 1'b0 |
| 0 | R | interrupt0 | saw a rising edge on done; independent of mask | 1'b0 |

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**Name** SET_INTERRUPT
**Address** 0x5015903c
**Description** not auto cleared
**Reset** 0x00000000

| Name | RSVD | set_inter | set_inter |
|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | rupt1 | rupt0 |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | set_interrupt1 | set interrupt 1 | 1'b0 |
| 0 | R/W | set_interrupt0 | set interrupt 0 | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x50159040
**Description** not auto cleared
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | reset_interrupt1 | reset_interrupt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | reset_interrupt1 | reset interrupt 1 | 1'b0 |
| 0 | R/W | reset_interrupt0 | reset interrupt 0 | 1'b0 |

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**Name** DTOP_BYPASS_CTRL
**Address** 0x50159044
**Description** Contains register fields associated with dtop_bypass_ctrl.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | byp_rd | hw_own |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | byp_rd | 1 to bypass read modify write | 1'b0 |
| 0 | R/W | hw_own | 1 to indicate hw own and sw control is disabled | 1'b0 |

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**Name** DTOP_BYPASS_STAT
**Address** 0x50159048
**Description** Contains register fields associated with dtop_bypass_stat.
**Reset** 0x00000000

| Name | RSVD | mode |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | mode | 1 if fpga is in dtop_bypass mode (built with FPGA_DTOP_BYP defined) | 1'b0 |

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**Name** DTOP_BYPASS_GPO
**Address** 0x5015904c
**Description** Contains register fields associated with dtop_bypass_gpo.
**Reset** 0x00000000

| Name | RSVD | gpo |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | gpo | 1 selects ext clock as source in DTOP_BYPASS, 0 selects internal pll. | 16'b0 |

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**Name** DTOP_BYPASS_RIF_MODE_CNTL
**Address** 0x50159050
**Description** Contains register fields associated with dtop_bypass_rif_mode_cntl.
**Reset** 0x500040ac

| Name | addr |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | addr | rif mode control address. | 32'h500040ac |

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**Name** DTOP_BYPASS_RIF_CAL_CNTL
**Address** 0x50159054
**Description** Contains register fields associated with dtop_bypass_rif_cal_cntl.
**Reset** 0x500040b0

| Name | addr |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | addr | rif cal control address. | 32'h500040b0 |

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**Name** DTOP_BYPASS_RIF_RXDC7
**Address** 0x50159058
**Description** Contains register fields associated with dtop_bypass_rif_rxdc7.
**Reset** 0x50004090

| Name | addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | addr | rif rxdc7 address. | 32'h50004090 |

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**Name** CORE_ID
**Address** 0x50159ffc
**Description** Contains register fields associated with core_id.
**Reset** 0x53574420

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | SWD in ASCII | 32'h53574420 |

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# GADC_SECURE

**Name** CTRL
**Address** 0x5015a000
**Description** control signals
**Reset** 0x00000100

| Name | watch_channels | | | | | | | | | | | | | | | | averaging_amount | | | wait_amount | | | | | | | | clkdiv | | code | warmup | mode | enable_dp |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | watch_channels | Which channels are active?<br>bit 0 = Not Used<br>bit 1 = Vbatt<br>bit 2 = Vstor<br>bit 3 = core supply<br>bit 4 = temperature<br>bit 5 = external port 1; differential<br>bit 6 = external port 0; differential<br>bit 7 = external port 1; single ended pos<br>bit 8 = external port 1; single ended neg<br>bit 9 = external port 0; single ended pos<br>bit 10 = external port 0; single ended neg<br>bit 11 = Vli-ion<br>bit 12-14 = input short<br>bit 15 = No-Connect | 16'h0 |
| 15:13 | R/W | averaging_amount | average 2^averaging_amount samples | 3'd0 |
| 12:6 | R/W | wait_amount | Specifies how many wait cycles in gadc clock cycles. | 7'd4 |
| 5:4 | R/W | clkdiv | gadc clock frequency<br>0 = 2.00MHz<br>1 = 1.00MHz<br>2 = 0.50MHz<br>3 = 0.25MHz | 2'd0 |
| 3 | R/W | code | Specifies ADC input data code bit representation.<br>0 = MSB of input gadc data is inverted<br>1 = MSB of input gadc data is not inverted Data in pedestal format is converted to 2's complement. | 1'b0 |
| 2 | R/W | warmup | Do 8 clocks cycle of warmup at beginning of round. No data sampled during this period. | 1'b0 |
| 1 | R/W | mode | What is the mode of operation?<br>0 = continuous<br>1 = one-shot | 1'b0 |
| 0 | R/W | enable_dp | Enables digital datapath. | 1'b0 |

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**Name** CTRL1
**Address** 0x5015a004
**Description** control signals
**Reset** 0x00000000

| Name | RSVD | ddr_en | ch_sel_ovrd_val | ch_sel_ovrd |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 | 5 | 4 3 2 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | ddr_en | for debug; brings out gadc_out as a 6b wide ddr bus | 1'b0 |
| 4:1 | R/W | ch_sel_ovrd_val | when override is set, what's the gadc channel to select. | 4'd0 |
| 0 | R/W | ch_sel_ovrd | when set overrides gadc channel select. | 1'b0 |

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**Name** CH1_DATAPATH_CONFIG
**Address** 0x5015a008
**Description** channel-1 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH2_DATAPATH_CONFIG
**Address** 0x5015a00c
**Description** channel-2 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH3_DATAPATH_CONFIG
**Address** 0x5015a010
**Description** channel-3 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH4_DATAPATH_CONFIG
**Address** 0x5015a014
**Description** channel-4 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH5_DATAPATH_CONFIG
**Address** 0x5015a018
**Description** channel-5 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH6_DATAPATH_CONFIG
**Address** 0x5015a01c
**Description** channel-6 offset and gain datpath parameters
**Reset** 0x00002800

| | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH7_DATAPATH_CONFIG
**Address** 0x5015a020
**Description** channel-7 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH8_DATAPATH_CONFIG
**Address** 0x5015a024
**Description** channel-8 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH9_DATAPATH_CONFIG
**Address** 0x5015a028

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**Description** channel-9 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH10_DATAPATH_CONFIG
**Address** 0x5015a02c
**Description** channel-10 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH11_DATAPATH_CONFIG
**Address** 0x5015a030
**Description** channel-11 offset and gain datpath parameters
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |

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| | | | | |
|---|---|---|---|---|
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** CH12_DATAPATH_CONFIG
**Address** 0x5015a034
**Description** channel-12 offset and gain datpath parameters applied to unused channels 0,13,14 by defaults
**Reset** 0x00002800

| Name | offset | | | | | | | | | | | | | | gain_exp | | | | | gain | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:18 | R/W | offset | Q<9,4> offset | 14'h0000 |
| 17:13 | R/W | gain_exp | Q<4,0> gain exponent | 5'h01 |
| 12:0 | R/W | gain | Q<0,12> gain number | 13'h0800 |

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**Name** GAIN_CONFIG0
**Address** 0x5015a038
**Description** channels 1-10 gain settings selection
**Reset** 0x00000000

| Name | RSVD | | ch10_gain_sel | | | ch9_gain_sel | | | ch8_gain_sel | | | ch7_gain_sel | | | ch6_gain_sel | | | ch5_gain_sel | | | ch4_gain_sel | | | ch3_gain_sel | | | ch2_gain_sel | | | ch1_gain_sel | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29:27 | R/W | ch10_gain_sel | channel-10 gain settings | 3'd0 |
| 26:24 | R/W | ch9_gain_sel | channel-9 gain settings | 3'd0 |
| 23:21 | R/W | ch8_gain_sel | channel-8 gain settings | 3'd0 |
| 20:18 | R/W | ch7_gain_sel | channel-7 gain settings | 3'd0 |
| 17:15 | R/W | ch6_gain_sel | channel-6 gain settings | 3'd0 |
| 14:12 | R/W | ch5_gain_sel | channel-5 gain settings | 3'd0 |
| 11:9 | R/W | ch4_gain_sel | channel-4 gain settings | 3'd0 |
| 8:6 | R/W | ch3_gain_sel | channel-3 gain settings | 3'd0 |
| 5:3 | R/W | ch2_gain_sel | channel-2 gain settings | 3'd0 |
| 2:0 | R/W | ch1_gain_sel | channel-1 gain settings | 3'd0 |

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**Name** GAIN_CONFIG1
**Address** 0x5015a03c
**Description** channels 11-15 gain settings selection
**Reset** 0x00000000

| Name | RSVD | ch12_gain_sel | ch11_gain_sel |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6** | **5 4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:3 | R/W | ch12_gain_sel | channel-12 gain settings | 3'd0 |
| 2:0 | R/W | ch11_gain_sel | channel-11 gain settings | 3'd0 |

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**Name** DATAPATH_OUTPUT
**Address** 0x5015a040
**Description** a read of this register pops the data FIFO
**Reset** 0x00000000

| Name | empty | RSVD | data |
|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21** | **20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R | empty | if 1, ignore data sample because fifo has been drained | 1'b0 |
| 20:0 | R | data | {4b channel id, 12b processed ADC sample Q<10,1>, 5b exponent} | 21'd0 |

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**Name** INTERRUPTS
**Address** 0x5015a044
**Description** interrupt status
**Reset** 0x00000000

| Name | RSVD | intrpt2 | intrpt1 | intrpt0 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R | intrpt2 | one shot or one round of continuous mode has completed | 1'b0 |
| 1 | R | intrpt1 | fifo overran | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 0 | R | intrpt0 | hardware written to fifo | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x5015a048
**Description** interrupt masking
**Reset** 0x00000007

| Name | RSVD | mask_intrpt2 | mask_intrpt1 | mask_intrpt0 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | mask_intrpt2 | allow intrpt2 to propogate to core's single output interrupt | 1'b1 |
| 1 | R/W | mask_intrpt1 | allow intrpt1 to propogate to core's single output interrupt | 1'b1 |
| 0 | R/W | mask_intrpt0 | allow intrpt0 to propogate to core's single output interrupt | 1'b1 |

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**Name** INTERRUPT_CLEAR
**Address** 0x5015a04c
**Description** clear interrupts
**Reset** 0x00000000

| Name | RSVD | clear_intrpt2 | clear_intrpt1 | clear_intrpt0 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clear_intrpt2 | clear interrupt 2; not self reseting | 1'b0 |
| 1 | R/W | clear_intrpt1 | clear interrupt 1; not self reseting | 1'b0 |
| 0 | R/W | clear_intrpt0 | clear interrupt 0; not self reseting | 1'b0 |

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**Name** CORE_ID
**Address** 0x5015affc
**Description** core ID
**Reset** 0x47414443

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| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | GADC in ASCII | 32'h47414443 |

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# TRNG_SECURE

**Name** CONTROL
**Address** 0x5015b000
**Description** Contains register fields associated with control.
**Reset** 0x00010010

| Name | RSVD | | | | | | | | | | | | | | | radio_warmup_cnt | | | | | | | | | | block_force_radio_into_trng_mode | launch_on_radio_up | radio_gain_override | lsb_source | adc_capture_edge | go |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16:6 | R/W | radio_warmup_cnt | how long to wait for radio to warmup in clk_mpc cycles | 11'h400 |
| 5 | R/W | block_force_radio_into_trng_mode | (no description) | 1'b0 |
| 4 | R/W | launch_on_radio_up | when radio is first coming back from power down mode, generate automatically (without software intervension) a new random number; this core will assert the radio gain override pin, wait 50usec, and then start the bit gathering process | 1'b1 |
| 3 | R/W | radio_gain_override | (no description) | 1'b0 |
| 2 | R/W | lsb_source | 0 = rif.rif_i[0]; 1 = radio.adc_i[0] | 1'b0 |
| 1 | R/W | adc_capture_edge | 0 = rising edge; 1 = falling edge | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 0 | R/W | go | start the process of generating a true random number ; not self reseting | 1'b0 |

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**Name** STATUS
**Address** 0x5015b004
**Description** Contains register fields associated with status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | fsm_running | running |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | fsm_running | this status is asserted when the fsm gets out of ST_IDLE | 1'b0 |
| 0 | R | running | this status is asserted after RADIO warm-up | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x5015b008
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | trng_trouble | trng_ready |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | trng_trouble | (no description) | 1'b0 |
| 0 | R | trng_ready | (no description) | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x5015b00c
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000003

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | intr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | intrpt_mask |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_mask | if nth bit set, the nth interrupt source is allowed to propogate to core's interrupt | 2'h3 |

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**Name** SET_INTERRUPT
**Address** 0x5015b010
**Description** Contains register fields associated with set_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_set |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_set | not self clearing | 2'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x5015b014
**Description** Contains register fields associated with reset_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_reset |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_reset | not self clearing | 2'b0 |

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**Name** TRNG

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**Address** 0x5015b018
**Description** Contains register fields associated with trng.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | the generated true random number | 32'b0 |

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**Name** CORE_ID
**Address** 0x5015bffc
**Description** Contains register fields associated with core_id.
**Reset** 0x54524e47

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | TRNG in ASCII | 32'h54524e47 |

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# RCOS_CAL_SECURE

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**Name** CAL_CTRL
**Address** 0x5015c000
**Description** Contains register fields associated with cal_ctrl.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | hw_cal_sel | sw_cal |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | hw_cal_sel | 1=choose cal from HW; 0=choose cal from SW | 1'b0 |
| 0 | R/W | sw_cal | rising edge starts SW cal, falling edge terminates cal if cal is not done; best to leave it on once set. | 1'b0 |

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**Name** SLOW_CLK_SET
**Address** 0x5015c004
**Description** Contains register fields associated with slow_clk_set.
**Reset** 0x00000064

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| Name | RSVD | slow_clk_set |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12** | **11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 11:0 | R/W | slow_clk_set | set the number of cycles to run slow clock; counting stops when this number is reached or cal is pulled down. constraint: 1 <= slow_clk_set <= 63; if set 0, hardware will use 1. | 12'd100 |

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**Name** SLOW_CLK_CNT
**Address** 0x5015c008
**Description** Contains register fields associated with slow_clk_cnt.
**Reset** 0x00000000

| Name | RSVD | slow_clk_cnt |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12** | **11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 11:0 | R | slow_clk_cnt | number of cycles counted by slow clock. | 12'b0 |

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**Name** FAST_CLK_CNT
**Address** 0x5015c00c
**Description** Contains register fields associated with fast_clk_cnt.
**Reset** 0x00000000

| Name | RSVD | fast_clk_cnt |
|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 21:0 | R | fast_clk_cnt | number of cycles counted by fast clock. 10 bits more than slow_clk_cnt/set because fast clock is 1000 times faster. | 22'b0 |

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**Name** SOURCE_CLK_SEL
**Address** 0x5015c010
**Description** Contains register fields associated with source_clk_sel.
**Reset** 0x00000000

| Name | RSVD | fast_clk_sel | slow_clk_sel |
|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | fast_clk_sel | 0=fast clk source is PCLk; 1=fast clk source is clk_calfc | 1'b0 |
| 0 | R/W | slow_clk_sel | 0=slow clk source is clk_lpc;1=slow clk source is clk_calfc | 1'b0 |

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**Name** FORCE_RESET
**Address** 0x5015c014
**Description** Contains register fields associated with force_reset.
**Reset** 0x00000000

| Name | RSVD | force_reset |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | force_reset | 1=apply force reset to fast domain and slow domain.<br>0=release force reset. | 1'b0 |

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**Name** STAT
**Address** 0x5015c018
**Description** Contains register fields associated with stat.
**Reset** 0x00000000

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | fast_domain_reset_stat | slow_domain_reset_stat | high_precision_run | running |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | fast_domain_reset_stat | 1=fast_clk domain is still in reset; don't triger cal | 1'b0 |
| 2 | R | slow_domain_reset_stat | 1=slow_clk domain is still in reset; don't triger cal | 1'b0 |
| 1 | R | high_precision_run | 1=high-precisioned clock was used at the beginning of cal; 0=low precisioned clock | 1'b0 |
| 0 | R | running | (no description) | 1'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x5015c01c
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | rcos_done |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R | rcos_done | (no description) | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x5015c020
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000001

| | | int |
|---|---|---|

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| Name | RSVD | intrpt_mask |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt_mask | if nth bit set, the nth interrupt source is allowed to propogate to core's interrupt | 1'h1 |

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**Name** SET_INTERRUPT
**Address** 0x5015c024
**Description** Contains register fields associated with set_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_set |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt_set | not self clearing | 1'b0 |

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**Name** RESET_INTERRUPT
**Address** 0x5015c028
**Description** Contains register fields associated with reset_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_reset |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | intrpt_reset | not self clearing | 1'b0 |

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**Name** CORE_ID
**Address** 0x5015cffc
**Description** Contains register fields associated with core_id.
**Reset** 0x52434f53

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | RCOS in ASCII | 32'h52434f53 |

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# SHUB_SECURE

**Name** SETUP0
**Address** 0x5015d000
**Description** Contains register fields associated with setup0.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | shub_disable | shub_trig_mode | adv_mode | store_in_sram | breakout | go |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | shub_disable | 0=enable sensor hub. | 1'b0 |
| 4 | R/W | shub_trig_mode | 0=continuous mode 1=trigger mode. | 1'b0 |
| 3 | R/W | adv_mode | 1=sensor hub's cpu enabled. | 1'b0 |
| 2 | R/W | store_in_sram | 0=data gets stored in flash, 1=data gets stored in SRAM | 1'b0 |
| 1 | R/W | breakout | useful for testing | 1'b0 |
| 0 | R/W | go | not self reseting | 1'b0 |

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**Name** SETUP1
**Address** 0x5015d004
**Description** Contains register fields associated with setup1.
**Reset** 0x00003fff

| Name | loop_ti |
|---|---|

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| | m<br>e | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | loop_time | how long should each iteration (sensor read and flash write) be ? (800 = wr_delay*2 + 16bytes read + overhead). The multiplier of 2 is needed beause when there are mulitple data bytes straddling the page boundary, the shub issues the page program twice. The first one sends all bytes up to the at the page boundary, and the second sends the rest of the data bytes. | 32'h3fff |

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**Name** PORT0_SETUP
**Address** 0x5015d008
**Description** Contains register fields associated with port0_setup.
**Reset** 0x00000000

| Name | RSVD | trig_en | RSVD | num_sensors | RSVD | spi_n_i2c | port_en |
|---|---|---|---|---|---|---|---|
| **Bits** | **31:12** | **11** | **10:9** | **8:6** | **5:4** | **3:2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 11 | R/W | trig_en | 1= enable threshold comparison with measurement results. | 1'd0 |
| 6:4 | R/W | num_sensors | number of sensors connected to port = num_sensors + 1 | 3'd0 |
| 1 | R/W | spi_n_i2c | 1= use spi for port0, 0= use i2c | 1'b0 |
| 0 | R/W | port_en | (no description) | 1'b0 |

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**Name** PORT0_SENSOR0_SETUP
**Address** 0x5015d00c
**Description** Contains register fields associated with port0_sensor0_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31:5** | **4:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR1_SETUP

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**Address** 0x5015d010
**Description** Contains register fields associated with port0_sensor1_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR2_SETUP
**Address** 0x5015d014
**Description** Contains register fields associated with port0_sensor2_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR3_SETUP
**Address** 0x5015d018
**Description** Contains register fields associated with port0_sensor3_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR4_SETUP
**Address** 0x5015d01c
**Description** Contains register fields associated with port0_sensor4_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31:5** | **4:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR5_SETUP
**Address** 0x5015d020
**Description** Contains register fields associated with port0_sensor5_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31:5** | **4:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR6_SETUP
**Address** 0x5015d024
**Description** Contains register fields associated with port0_sensor6_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31:5** | **4:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SENSOR7_SETUP
**Address** 0x5015d028
**Description** Contains register fields associated with port0_sensor7_setup.
**Reset** 0x00000001

| Name | RSVD | num_bytes |
|---|---|---|
| **Bits** | **31:5** | **4:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT0_SPI_ENG0_CTRL
**Address** 0x5015d02c
**Description** Contains register fields associated with port0_spi_eng0_ctrl.
**Reset** 0x00000000

| Name | cpol | cpha | RSVD | sensor_opcode |
|---|---|---|---|---|
| **Bits** | **31** | **30** | **29:8** | **7:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | cpol | spi clock polarity mode. 0 = posedge-sensitive, 1 = negedge | 1'd0 |
| 30 | R/W | cpha | spi clcok phase mode | 1'd0 |
| 7:0 | R/W | sensor_opcode | opcode used to communicate with sensor in spi mode | 8'd0 |

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**Name** PORT0_I2C_IDW_0
**Address** 0x5015d030
**Description** id(read) of sensor0 on port0 prior to pointer address write
**Reset** 0x0001fe00

| | | ext_ack | ext_ma | | e | ex | ex | e | ext_ |
|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:18** | **17** | **16** | **15:14** | **13** | **12** | **11:10** | **9:8** | **7:0** |

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| Name | RSVD | | | | | | | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | | ext_head | | ext_clk_pu | ext_data_pu | ext_data_oe | | | | | | | | ext_data_o | | | | | | | | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_1
**Address** 0x5015d034
**Description** id(read) of sensor1 on port0 prior to pointer address write
**Reset** 0x0001fe00

| Name | RSVD | | | | | | | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | | ext_head | | ext_clk_pu | ext_data_pu | ext_data_oe | | | | | | | | ext_data_o | | | | | | | | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_2
**Address** 0x5015d038
**Description** id(read) of sensor2 on port0 prior to pointer address write
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:25** | **24** | **23** | **22:21** | **20:19** | **18** | **17** | **16:9** | **8:1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_3
**Address** 0x5015d03c
**Description** id(read) of sensor3 on port0 prior to pointer address write
**Reset** 0x0001fe00

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | ext_ | ext_ | | | | | | | e |

| Name | RSVD | ack_value_to_drive | master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_4
**Address** 0x5015d040
**Description** id(read) of sensor4 on port0 prior to pointer address write
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_5
**Address** 0x5015d044
**Description** id(read) of sensor5 on port0 prior to pointer address write
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26** | **25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_6
**Address** 0x5015d048
**Description** id(read) of sensor6 on port0 prior to pointer address write
**Reset** 0x0001fe00

| | | e | e | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDW_7
**Address** 0x5015d04c
**Description** id(read) of sensor7 on port0 prior to pointer address write
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_0
**Address** 0x5015d050
**Description** pointer addr of sensor0 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 | 24 | 23 | 22 21 | 20 19 | 18 | 17 | 16 15 14 13 12 11 10 9 | 8 7 6 5 4 3 2 1 | 0 |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_1
**Address** 0x5015d054
**Description** pointer addr of sensor1 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_2
**Address** 0x5015d058
**Description** pointer addr of sensor2 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_driv | ext_master_drives_ac | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | e | k | | | | | | | | | | | | | | | | | | | | | | | |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_3
**Address** 0x5015d05c
**Description** pointer addr of sensor3 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 | 24 | 23 | 22 21 | 20 19 | 18 | 17 | 16 15 14 13 12 11 10 | 9 8 7 6 5 4 3 | 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_4

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**Address** 0x5015d060
**Description** pointer addr of sensor4 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_5
**Address** 0x5015d064
**Description** pointer addr of sensor5 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_d | ext_master_drives | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovr |
|---|---|---|---|---|---|---|---|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | rive | _ack | | | | | | | | | | | | | | | | | | | | | d |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_6
**Address** 0x5015d068
**Description** pointer addr of sensor6 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:25** | **24** | **23** | **22:21** | **20:19** | **18** | **17** | **16:9** | **8:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_ADDR_7
**Address** 0x5015d06c
**Description** pointer addr of sensor7 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:30** | **29** | **28** | **27:26** | **25:24** | **23:22** | **21:20** | **19:18** | **17** | **16:9** | **8:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_0
**Address** 0x5015d070
**Description** id(read) of sensor0 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to | ext_master_driv | ext_tail | ext_head | ext_clk_pu | ext_data_p | ext_data_o | ext_data_o | ext_data_ie_o |
|---|---|---|---|---|---|---|---|---|---|---|

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| | | | | | | | | _drive | es_ack | | | | | u | e | | | | | | | | | | | vrd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_1
**Address** 0x5015d074
**Description** id(read) of sensor1 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_2
**Address** 0x5015d078
**Description** id(read) of sensor2 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:25** | **24** | **23** | **22:21** | **20:19** | **18** | **17** | **16:9** | **8:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_3
**Address** 0x5015d07c
**Description** id(read) of sensor3 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value | ext_master_dr | ext_ta | ext_he | ext_clk | ext_data | ext_data | ext_data_i |
|---|---|---|---|---|---|---|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | | | | | | | | ext_data_o | | | | | | | ext_data_ie_ovrd |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_4
**Address** 0x5015d080
**Description** id(read) of sensor4 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSVD | | | | | | | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | | | | | | | | ext_data_o | | | | | | | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |

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| | | | | |
|---|---|---|---|---|
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_5
**Address** 0x5015d084
**Description** id(read) of sensor5 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_6
**Address** 0x5015d088
**Description** id(read) of sensor6 on port0 prior to data read
**Reset** 0x0001fe00

| | RS | ext_ack_val | ext_master | ext_ | ext_ | ext_c | ext_d | ext_d | ext_d | ext_data |
|---|---|---|---|---|---|---|---|---|---|---|

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| Name | VD | ue_to_drive | _drives_ack | tail | head | lk_pu | ata_pu | ata_oe | ata_o | _ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_IDR_7
**Address** 0x5015d08c
**Description** id(read) of sensor7 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_DAT_CTRL
**Address** 0x5015d090
**Description** slave(port0) data byte read (used in not the last byte)
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_I2C_DAT_LAST_CTRL
**Address** 0x5015d094
**Description** slave(port0) data byte read (used in the last byte)
**Reset** 0x0001fe00

| | | ext_ack_v | ext_mast | ex | ex | ext | ext | ext | ext | ext_da |
|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | | | | | | | alue_to_drive | er_drives_ack | t_tail | | t_head | | _clk_pu | _data_pu | _data_oe | | | | | | | _data_o | | | | | | | ta_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT0_ALM0_QUAN_CTRL
**Address** 0x5015d098
**Description** alarm0 quantity's definition
**Reset** 0x05500000

| Name | quan_dat_is_signed | lsb_first | RSVD | | quan3_size | | quan2_size | | quan1_size | | quan0_size | | | quan3_pos | | | | quan2_pos | | | | quan1_pos | | | | quan0_pos | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | quan_dat_is_signed | quantity data is a signed data | 1'b0 |
| 30 | R/W | lsb_first | 1= quantity data is big endian ; 0= small endian | 1'b0 |
| 27:26 | R/W | quan3_size | number of bytes making up the quantity data = programmed quan3_size + 1 | 2'd1 |
| 25:24 | R/W | quan2_size | (no description) | 2'd1 |
| 23:22 | R/W | quan1_size | (no description) | 2'd1 |
| 21:20 | R/W | quan0_size | (no description) | 2'd1 |

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| | | | | |
|---|---|---|---|---|
| 19:15 | R/W | quan3_pos | starting position of the quantity | 5'd0 |
| 14:10 | R/W | quan2_pos | (no description) | 5'd0 |
| 9:5 | R/W | quan1_pos | (no description) | 5'd0 |
| 4:0 | R/W | quan0_pos | (no description) | 5'd0 |

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**Name** PORT0_ALM1_QUAN_CTRL
**Address** 0x5015d09c
**Description** alarm1 quantity's definition
**Reset** 0x05500000

| Name | quan_dat_is_signed | lsb_first | RSVD | quan3_size | quan2_size | quan1_size | quan0_size | quan3_pos | quan2_pos | quan1_pos | quan0_pos |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29 28** | **27 26** | **25 24** | **23 22** | **21 20** | **19 18 17 16 15** | **14 13 12 11 10** | **9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | quan_dat_is_signed | quantity data is a signed data | 1'b0 |
| 30 | R/W | lsb_first | 1= quantity data is big endian ; 0= small endian | 1'b0 |
| 27:26 | R/W | quan3_size | number of bytes making up the quantity data = programmed quan3_size + 1 | 2'd1 |
| 25:24 | R/W | quan2_size | (no description) | 2'd1 |
| 23:22 | R/W | quan1_size | (no description) | 2'd1 |
| 21:20 | R/W | quan0_size | (no description) | 2'd1 |
| 19:15 | R/W | quan3_pos | starting position of the quantity | 5'd0 |
| 14:10 | R/W | quan2_pos | (no description) | 5'd0 |
| 9:5 | R/W | quan1_pos | (no description) | 5'd0 |
| 4:0 | R/W | quan0_pos | (no description) | 5'd0 |

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**Name** PORT0_ALM0_TRIG_CTRL
**Address** 0x5015d0a0
**Description** alarm0 trigger function: trigger = trig_func_en( out_of_range(quan3), out_of_range(quan2),out_of_range(quan1),out_of_range(quan0))
**Reset** 0x00000000

| Name | RSVD | trig_func |
|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | _<br>e<br>n |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | trig_func_en | (no description) | 16'b0 |

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**Name** PORT0_ALM1_TRIG_CTRL
**Address** 0x5015d0a4
**Description** alarm1 trigger function: trigger = trig_func_en( out_of_range(quan3), out_of_range(quan2),out_of_range(quan1),out_of_range(quan0))
**Reset** 0x00000000

| Name | R<br>S<br>V<br>D | | | | | | | | | | | | | | | | t<br>r<br>i<br>g<br>_<br>f<br>u<br>n<br>c<br>_<br>e<br>n | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | trig_func_en | (no description) | 16'b0 |

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**Name** PORT0_ALM0_THRHLD_MAX_0
**Address** 0x5015d0a8
**Description** threshold of sensor results
**Reset** 0xffffffff

| Name | b<br>y<br>t<br>e<br>_<br>m<br>a<br>x<br>_<br>0<br>3 | | | | | | | | b<br>y<br>t<br>e<br>_<br>m<br>a<br>x<br>_<br>0<br>2 | | | | | | | | b<br>y<br>t<br>e<br>_<br>m<br>a<br>x<br>_<br>0<br>1 | | | | | | | | b<br>y<br>t<br>e<br>_<br>m<br>a<br>x<br>_<br>0<br>0 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_03 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_02 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_01 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_00 | (no description) | 8'hff |

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**Name** PORT0_ALM0_THRHLD_MAX_1
**Address** 0x5015d0ac
**Description** threshold of sensor results
**Reset** 0xffffffff

| | byte_max_07 | | | | | | | | byte_max_06 | | | | | | | | byte_max_05 | | | | | | | | byte_max_04 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_07 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_06 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_05 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_04 | (no description) | 8'hff |

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**Name** PORT0_ALM0_THRHLD_MAX_2
**Address** 0x5015d0b0
**Description** threshold of sensor results
**Reset** 0xffffffff

| | byte_max_11 | | | | | | | | byte_max_10 | | | | | | | | byte_max_09 | | | | | | | | byte_max_08 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_11 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_10 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_09 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_08 | (no description) | 8'hff |

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**Name** PORT0_ALM0_THRHLD_MIN_0
**Address** 0x5015d0b4
**Description** threshold of sensor results
**Reset** 0x00000000

| | byte | | | | byte | | | | byte | | | | byte | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | byte_min_03 | byte_min_02 | byte_min_01 | byte_min_00 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_03 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_02 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_01 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_00 | (no description) | 8'h00 |

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**Name** PORT0_ALM0_THRHLD_MIN_1
**Address** 0x5015d0b8
**Description** threshold of sensor results
**Reset** 0x00000000

| Name | byte_min_07 | byte_min_06 | byte_min_05 | byte_min_04 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_07 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_06 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_05 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_04 | (no description) | 8'h00 |

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**Name** PORT0_ALM0_THRHLD_MIN_2
**Address** 0x5015d0bc
**Description** threshold of sensor results
**Reset** 0x00000000

| Name | byte_min_11 | byte_min_10 | byte_min_09 | byte_min_08 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_11 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_10 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_09 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_08 | (no description) | 8'h00 |

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**Name** PORT0_ALM1_THRHLD_MAX_0
**Address** 0x5015d0c0
**Description** threshold of sensor results
**Reset** 0xffffffff

| Name | byte_max_03 | | | | | | | | byte_max_02 | | | | | | | | byte_max_01 | | | | | | | | byte_max_00 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_03 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_02 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_01 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_00 | (no description) | 8'hff |

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**Name** PORT0_ALM1_THRHLD_MAX_1
**Address** 0x5015d0c4
**Description** threshold of sensor results
**Reset** 0xffffffff

| Name | byte_max_07 | | | | | | | | byte_max_06 | | | | | | | | byte_max_05 | | | | | | | | byte_max_04 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_07 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_06 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_05 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_04 | (no description) | 8'hff |

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**Name** PORT0_ALM1_THRHLD_MAX_2
**Address** 0x5015d0c8
**Description** threshold of sensor results
**Reset** 0xffffffff

| | byte_max_11 | | | | | | | | byte_max_10 | | | | | | | | byte_max_09 | | | | | | | | byte_max_08 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_11 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_10 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_09 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_08 | (no description) | 8'hff |

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**Name** PORT0_ALM1_THRHLD_MIN_0
**Address** 0x5015d0cc
**Description** threshold of sensor results
**Reset** 0x00000000

| | byte_min_03 | | | | | | | | byte_min_02 | | | | | | | | byte_min_01 | | | | | | | | byte_min_00 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_03 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_02 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_01 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_00 | (no description) | 8'h00 |

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**Name** PORT0_ALM1_THRHLD_MIN_1
**Address** 0x5015d0d0
**Description** threshold of sensor results
**Reset** 0x00000000

| | byte | byte | byte | byte |
|---|---|---|---|---|

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| Name | byte_min_07 | byte_min_06 | byte_min_05 | byte_min_04 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_07 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_06 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_05 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_04 | (no description) | 8'h00 |

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**Name** PORT0_ALM1_THRHLD_MIN_2
**Address** 0x5015d0d4
**Description** threshold of sensor results
**Reset** 0x00000000

| Name | byte_min_11 | byte_min_10 | byte_min_09 | byte_min_08 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_11 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_10 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_09 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_08 | (no description) | 8'h00 |

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**Name** PORT0_STATUS
**Address** 0x5015d0d8
**Description** Contains register fields associated with port0_status.
**Reset** 0x00000000

| Name | RSVD | triggered | RSVD | out_of_range_s |
|---|---|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | t a t | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R | triggered | (no description) | 1'b0 |
| 3:0 | R | out_of_range_stat | (no description) | 4'd0 |

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**Name** PORT1_SETUP
**Address** 0x5015d0dc
**Description** Contains register fields associated with port1_setup.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | trig_en | RSVD | | | | | num_sensors | | | spi_n_i2c | port_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 11 | R/W | trig_en | 1= enable threshold comparison with measurement results. | 1'd0 |
| 6:4 | R/W | num_sensors | number of sensors connected to port = num_sensors + 1 | 3'd0 |
| 1 | R/W | spi_n_i2c | 1= use spi for port0, 0= use i2c | 1'b0 |
| 0 | R/W | port_en | (no description) | 1'b0 |

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**Name** PORT1_SENSOR0_SETUP
**Address** 0x5015d0e0
**Description** Contains register fields associated with port1_sensor0_setup.
**Reset** 0x00000001

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | num_bytes | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R/W | num_bytes | total number of bytes in sensor data = num_bytes+1 | 5'd1 |

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**Name** PORT1_SPI_ENG0_CTRL
**Address** 0x5015d0e4
**Description** Contains register fields associated with port1_spi_eng0_ctrl.

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**Reset** 0x00000000

| Name | cpol | cpha | RSVD | sensor_opcode |
|---|---|---|---|---|
| **Bits** | **31** | **30** | **29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | cpol | spi clock polarity mode. 0 = posedge-sensitive, 1 = negedge | 1'd0 |
| 30 | R/W | cpha | spi clcok phase mode | 1'd0 |
| 7:0 | R/W | sensor_opcode | opcode used to communicate with sensor in spi mode | 8'd0 |

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**Name** PORT1_I2C_IDW_0
**Address** 0x5015d0e8
**Description** id(read) of sensor0 on port0 prior to pointer address write
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |

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| | | | | |
|---|---|---|---|---|
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT1_I2C_ADDR_0
**Address** 0x5015d0ec
**Description** pointer addr of sensor0 on port0
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT1_I2C_IDR_0
**Address** 0x5015d0f0
**Description** id(read) of sensor0 on port0 prior to data read
**Reset** 0x0001fe00

| Name | RSV | ext_ack_valu | ext_master_ | ext_t | ext_h | ext_cl | ext_da | ext_da | ext_da | ext_data_ |
|---|---|---|---|---|---|---|---|---|---|---|

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| | D | | | | | | | e_to_drive | drives_ack | ail | ead | k_pu | ta_pu | ta_oe | ta_o | ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT1_I2C_DAT_CTRL
**Address** 0x5015d0f4
**Description** slave(port0) data byte read (used in not the last byte)
**Reset** 0x0001fe00

| Name | RSVD | | | | | | | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT1_I2C_DAT_LAST_CTRL
**Address** 0x5015d0f8
**Description** slave(port0) data byte read (used in the last byte)
**Reset** 0x0001fe00

| Name | RSVD | ext_ack_value_to_drive | ext_master_drives_ack | ext_tail | ext_head | ext_clk_pu | ext_data_pu | ext_data_oe | ext_data_o | ext_data_ie_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23** | **22 21** | **20 19** | **18** | **17** | **16 15 14 13 12 11 10 9** | **8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R/W | ext_ack_value_to_drive | (no description) | 1'b0 |
| 23 | R/W | ext_master_drives_ack | (no description) | 1'b0 |
| 22:21 | R/W | ext_tail | (no description) | 2'h0 |
| 20:19 | R/W | ext_head | (no description) | 2'h0 |
| 18 | R/W | ext_clk_pu | (no description) | 1'b0 |
| 17 | R/W | ext_data_pu | (no description) | 1'b0 |
| 16:9 | R/W | ext_data_oe | (no description) | 8'hff |
| 8:1 | R/W | ext_data_o | (no description) | 8'h0 |
| 0 | R/W | ext_data_ie_ovrd | (no description) | 1'b0 |

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**Name** PORT1_ALM0_QUAN_CTRL
**Address** 0x5015d0fc
**Description** alarm0 quantity's definition
**Reset** 0x05500000

| Name | quan_dat_lsb_f | RS | quan3 | quan2 | quan1 | quan0 | quan3 | quan2 | quan1 | quan0 |
|---|---|---|---|---|---|---|---|---|---|---|

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| | is_signed | lsb_first | VD | quan3_size | quan2_size | quan1_size | quan0_size | quan3_pos | quan2_pos | quan1_pos | quan0_pos |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29 28 27** | **26 25** | **24 23** | **22 21** | **20 19** | **18 17 16 15** | **14 13 12 11 10** | **9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | quan_dat_is_signed | quantity data is a signed data | 1'b0 |
| 30 | R/W | lsb_first | 1= quantity data is big endian ; 0= small endian | 1'b0 |
| 27:26 | R/W | quan3_size | number of bytes making up the quantity data = programmed quan3_size + 1 | 2'd1 |
| 25:24 | R/W | quan2_size | (no description) | 2'd1 |
| 23:22 | R/W | quan1_size | (no description) | 2'd1 |
| 21:20 | R/W | quan0_size | (no description) | 2'd1 |
| 19:15 | R/W | quan3_pos | starting position of the quantity | 5'd0 |
| 14:10 | R/W | quan2_pos | (no description) | 5'd0 |
| 9:5 | R/W | quan1_pos | (no description) | 5'd0 |
| 4:0 | R/W | quan0_pos | (no description) | 5'd0 |

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**Name** PORT1_ALM0_TRIG_CTRL
**Address** 0x5015d100
**Description** alarm0 trigger function: trigger = trig_func_en( out_of_range(quan3), out_of_range(quan2),out_of_range(quan1),out_of_range(quan0))
**Reset** 0x00000000

| Name | RSVD | trig_func_en |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:0 | R/W | trig_func_en | (no description) | 16'b0 |

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**Name** PORT1_ALM0_THRHLD_MAX_0
**Address** 0x5015d104
**Description** threshold of sensor results
**Reset** 0xffffffff

| | byte_ | byte_ | byte_ | byte_ |
|---|---|---|---|---|

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| Name | byte_max_03 | byte_max_02 | byte_max_01 | byte_max_00 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_03 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_02 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_01 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_00 | (no description) | 8'hff |

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**Name** PORT1_ALM0_THRHLD_MAX_1
**Address** 0x5015d108
**Description** threshold of sensor results
**Reset** 0xffffffff

| Name | byte_max_07 | byte_max_06 | byte_max_05 | byte_max_04 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_07 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_06 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_05 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_04 | (no description) | 8'hff |

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**Name** PORT1_ALM0_THRHLD_MAX_2
**Address** 0x5015d10c
**Description** threshold of sensor results
**Reset** 0xffffffff

| Name | byte_max_11 | byte_max_10 | byte_max_09 | byte_max_08 |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 | 23 22 21 20 19 18 17 16 | 15 14 13 12 11 10 9 8 | 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:24 | R/W | byte_max_11 | (no description) | 8'hff |
| 23:16 | R/W | byte_max_10 | (no description) | 8'hff |
| 15:8 | R/W | byte_max_09 | (no description) | 8'hff |
| 7:0 | R/W | byte_max_08 | (no description) | 8'hff |

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**Name** PORT1_ALM0_THRHLD_MIN_0
**Address** 0x5015d110
**Description** threshold of sensor results
**Reset** 0x00000000

| Name | byte_min_03 | | | | | | | | byte_min_02 | | | | | | | | byte_min_01 | | | | | | | | byte_min_00 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_03 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_02 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_01 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_00 | (no description) | 8'h00 |

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**Name** PORT1_ALM0_THRHLD_MIN_1
**Address** 0x5015d114
**Description** threshold of sensor results
**Reset** 0x00000000

| Name | byte_min_07 | | | | | | | | byte_min_06 | | | | | | | | byte_min_05 | | | | | | | | byte_min_04 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_07 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_06 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_05 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_04 | (no description) | 8'h00 |

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**Name** PORT1_ALM0_THRHLD_MIN_2

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**Address** 0x5015d118
**Description** threshold of sensor results
**Reset** 0x00000000

| Name | byte_min_11 | byte_min_10 | byte_min_09 | byte_min_08 |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | byte_min_11 | (no description) | 8'h00 |
| 23:16 | R/W | byte_min_10 | (no description) | 8'h00 |
| 15:8 | R/W | byte_min_09 | (no description) | 8'h00 |
| 7:0 | R/W | byte_min_08 | (no description) | 8'h00 |

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**Name** PORT1_STATUS
**Address** 0x5015d11c
**Description** Contains register fields associated with port1_status.
**Reset** 0x00000000

| Name | RSVD | triggered | RSVD | out_of_range_stat |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25** | **24** | **23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R | triggered | (no description) | 1'b0 |
| 3:0 | R | out_of_range_stat | (no description) | 4'd0 |

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**Name** FLASH_CTRL0
**Address** 0x5015d120
**Description** Contains register fields associated with flash_ctrl0.
**Reset** 0xabb90206

| | op | o | o | op |
|---|---|---|---|---|

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| Name | opcode_rpd | | | | | | | | opcode_pd | | | | | | | | opcode_pp | | | | | | | | opcode_wren | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | opcode_rpd | release_power_down (not needed for macronix) | 8'hab |
| 23:16 | R/W | opcode_pd | power down opcode | 8'hb9 |
| 15:8 | R/W | opcode_pp | page program opcode | 8'h02 |
| 7:0 | R/W | opcode_wren | write_enable opcode | 8'h06 |

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**Name** FLASH_CTRL1
**Address** 0x5015d124
**Description** Contains register fields associated with flash_ctrl1.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | wr_start_address | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R/W | wr_start_address | (no description) | 24'h0 |

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**Name** FLASH_CTRL2
**Address** 0x5015d128
**Description** Contains register fields associated with flash_ctrl2.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | wr_max_addr | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | e s s |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R/W | wr_max_address | (no description) | 24'h0 |

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**Name** FLASH_CTRL3
**Address** 0x5015d12c
**Description** Contains register fields associated with flash_ctrl3.
**Reset** 0x00000082

| Name | R S V D | | | | | | | | | | | | | | | | | | | | | | | | rpd_has_clock | serialize_pp_address | clear_lost_write_cnt | clear_flash_overflow | qspi_mode | | serialize_opcode | power_cycle_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7 | R/W | rpd_has_clock | for Macronix flash, rpd has no clock | 1'b1 |
| 6 | R/W | serialize_pp_address | (no description) | 1'b0 |
| 5 | R/W | clear_lost_write_cnt | (no description) | 1'b0 |
| 4 | R/W | clear_flash_overflow | (no description) | 1'b0 |
| 3:2 | R/W | qspi_mode | 0,1,3 = single spi mode; 2 = qspi mode | 2'b0 |
| 1 | R/W | serialize_opcode | 1= opcode is serial even in quad mode. | 1'b1 |
| 0 | R/W | power_cycle_en | 1= cycle flash into and out of deep power down mode. | 1'h0 |

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**Name** FLASH_CTRL4
**Address** 0x5015d130
**Description** Contains register fields associated with flash_ctrl4.
**Reset** 0x00200140

| Name | R S V D | | | | pd_delay | | | | | | | | R S V D | | | | | | | | pp_delay | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 27:20 | R/W | pd_delay | delay cycles (pd_delay+1) after flash power down / release from power down | 8'd2 |
| 11:0 | R/W | pp_delay | delay cycles (pp_delay+1) after flash write. (320 cycles of 32k clock = 10ms pp time from Macronix) | 12'd320 |

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**Name** FLASH_STATUS0
**Address** 0x5015d134
**Description** Contains register fields associated with flash_status0.
**Reset** 0x00000000

| Name | RSVD | | | | | | | wr_saturated | wr_current_address | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 24 | R | wr_saturated | indicates overflow. | 1'b0 |
| 23:0 | R | wr_current_address | the next address to write to. Number of bytes in storage = wr_current_address - wr_start_address. | 24'h0 |

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**Name** FLASH_STATUS1
**Address** 0x5015d138
**Description** Contains register fields associated with flash_status1.
**Reset** 0x00000000

| Name | lost_write_cnt | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:0 | R | lost_write_cnt | (no description) | 32'h0 |

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**Name** SHUB_STATUS0
**Address** 0x5015d13c
**Description** Contains register fields associated with shub_status0.
**Reset** 0x00000000

| Name | cpu_running | RSVD | state |
|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5** | **4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R | cpu_running | useful for testing | 1'b0 |
| 4:0 | R | state | useful for testing | 5'b0 |

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**Name** PC_CTRL
**Address** 0x5015d140
**Description** Contains register fields associated with pc_ctrl.
**Reset** 0x00000000

| Name | pc_init |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | pc_init | iniital pc value | 32'b0 |

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**Name** PC_CTRL2
**Address** 0x5015d144
**Description** Contains register fields associated with pc_ctrl2.
**Reset** 0x00000000

| Name | RSVD | pc_init_lo |
|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | pc_init_load | iniital pc value load; not self-clearing | 1'b0 |

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**Name** REG7
**Address** 0x5015d148
**Description** Contains register fields associated with reg7.
**Reset** 0x00000000

| Name | r 7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | r7 | value of reg7 | 32'b0 |

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**Name** SHUB_SRAM_CLK_SEL
**Address** 0x5015d14c
**Description** clock selection between PCLK and clk_lpc as to drive SRAM.
**Reset** 0x00000000

| Name | R S V D | | | | | | | | | | | | | | | | | | | | | | | | | | | | shub_sram_clk_sel_val | shub_sram_clk_sel_ovrd |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | shub_sram_clk_sel_val | override val to shub_sram_clk mux. 0=PCLK ; 1=clk_lpc | 1'b0 |
| 0 | R/W | shub_sram_clk_sel_ovrd | override bit to shub_sram_clk mux | 1'b0 |

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**Name** REGFILE_SEL
**Address** 0x5015d150
**Description** Contains register fields associated with regfile_sel.

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**Reset** 0x00000007

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | sel | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3:0 | R/W | sel | select which reg in regfile to be connected to reg7 | 4'd7 |

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**Name** REGFILE_REGW
**Address** 0x5015d154
**Description** Contains register fields associated with regfile_regw.
**Reset** 0x00000000

| Name | regfile_wdata | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | regfile_wdata | write data to selected reg in regfile. The selection pointer is regfile_sel | 32'b0 |

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**Name** ID
**Address** 0x5015dffc
**Description** Contains register fields associated with id.
**Reset** 0x53485542

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | SHUB in ASCII | 32'h53485542 |

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# PDM0_SECURE

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**Name** CONTROL
**Address** 0x50160000
**Description** Contains register fields associated with control.
**Reset** 0x000006f0

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| Name | RSVD | stereo_mode | pdm_in_dbg | pcm_bit_width | pcm_rate_sel | pdm_clk_sel | pack_samples | swap_samples | capture_on_rising | enable_dp |
|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:17** | **16** | **15** | **14:13** | **12:9** | **8:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16 | R/W | stereo_mode | selects between stereo and mono mode of operation<br>1= stereo mode; act as datapath enable for other stereo datapath pair. When this bit is set along with enable_dp, it enables stereo datapath pair concurrently to avoid pdm data capture synchronization issues. At that time, corresponding bits for the other datapath pair shouldn't be set.<br>0= mono mode; use enable_dp bit to enable each datapath. | 1'd0 |
| 15 | R/W | pdm_in_dbg | if 1, bypass pdm bitstream and send in dc value 1 | 1'd0 |
| 14:13 | R/W | pcm_bit_width | pcm output sample bit width<br>0= 16bit<br>1= 20bit<br>2/3= 24bit | 2'd0 |
| 12:9 | R/W | pcm_rate_sel | pcm output rate selection if msb bit3=1 it's in pcm_rate override mode<br>0= 6KHz<br>1= 8KHz<br>2= 12KHz<br>3= 16KHz<br>4= 24KHz<br>5= 32KHz<br>6= 48KHz<br>7= 64KHz | 4'd3 |
| 8:4 | R/W | pdm_clk_sel | pdm clock frequency = f_pclk/(pdm_clk_sel+1)<br>valid range: 3 to 31<br>possible rates when pcm_rate_sel[0]=1:<br>31= 500KHz<br>15= 1MHz<br>7= 2MHz<br>possible rates when pcm_rate_sel[0]=0:<br>23= 500KHz<br>11= 1MHz<br>5= 2MHz | 5'd15 |
| 3 | R/W | pack_samples | pack samples in ping-pong buffer<br>default: samples are stored unpacked<br>applies to 20/24 bit samples only | 1'b0 |
| 2 | R/W | swap_samples | swap packed samples in pingpong buffer location<br>default: former samples uses lsb bits and later samples uses msb bits<br>applies to 16bit samples only | 1'b0 |
| 1 | R/W | capture_on_rising | selects edge of pdm clock to capture pdm data<br>0= pdm data sampled on falling edge of pdm clock (i.e. pdm data transitions on rising edge of pdm clock)<br>1= pdm data sampled on rising edge of pdm clock (i.e. pdm data transitions on falling edge of pdm clock) | 1'b0 |
| 0 | R/W | enable_dp | enable the datapath | 1'b0 |

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**Name** FILTER_CONFIG
**Address** 0x50160004
**Description** Contains register fields associated with filter_config.
**Reset** 0x00062a50

| Name | RSVD | | | | | | downsample_ratio | | lpf_bypass_ovrd_val | lpf_bw | | | hbf_bypass_ovrd_val | hpf2_init_shift | | | | hpf1_init_shift | | | | hpf_init_enable | hpf_bypass | hpf_bw | | | | hpf_ord | | cic_ord | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25:24 | R/W | downsample_ratio | downsample override<br>used only during pcm_rate override mode:<br>downsampling by 1/(1+downsample_ratio) | 2'd0 |
| 23 | R/W | lpf_bypass_ovrd_val | lpf bypass override value<br>used only during pcm_rate override mode:<br>0= enable lpf<br>1= bypass lpf; output rate=input rate | 1'd0 |
| 22:20 | R/W | lpf_bw | lpf bandwidth override selection<br>used only during pcm_rate override mode:<br>if pcm_rate_sel[0]=1, bandwidth will be (KHz):<br>0= 4<br>1= 6<br>2= 8<br>3= 10<br>4= 12<br>5= 14<br>6/7= 15.5<br>if pcm_rate_sel[0]=0, bandwidth will be (KHz):<br>0= 3<br>1= 4.5<br>2= 6<br>3= 7.5<br>4= 9<br>5= 10<br>6/7= 11.5 | 3'd0 |
| 19 | R/W | hbf_bypass_ovrd_val | hbf bypass override value<br>used only during pcm_rate override mode:<br>0= enable hbf<br>1= bypass hbf; output rate=input rate | 1'd0 |
| 18:14 | R/W | hpf2_init_shift | hpf2 initialization shift amount, max=24<br>for values > max, operation is same as max | 5'd24 |
| 13:10 | R/W | hpf1_init_shift | hpf1 initialization shift amount, max=10<br>for values > max, operation is same as max | 4'd10 |
| 9 | R/W | hpf_init_enable | 1= enable hpf initialization | 1'b1 |
| 8 | R/W | hpf_bypass | 1= bypass hpf | 1'b0 |
| | | | hpf bandwidth selection<br>when hpf_ord=0, bandwidth will be (Hz): | |

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|---|---|---|---|---|
| 7:4 | R/W | hpf_bw | 0= 10, 1= 20, 2= 40, 3= 80, 4= 161, 5= 324, 6= 658, 7= 1359<br>when hpf_ord=1, bandwidth will be (Hz):<br>0= 9, 1= 19, 2= 37, 3= 75, 4= 151, 5= 304, 6= 616, 7= 1269<br>when hpf_ord=2, bandwidth will be (Hz):<br>0= 15, 1= 30, 2= 60, 3= 120, 4= 242, 5= 487, 6= 989, 7= 2036 | 4'd5 |
| 3:2 | R/W | hpf_ord | hpf order<br>0= 1st order<br>1= 2nd order<br>2/3= 3rd order | 2'd0 |
| 1:0 | R/W | cic_ord | cic order<br>0= 3rd order<br>1= 4th order<br>2/3= unused | 2'd0 |

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**Name** FILTER_CONFIG1
**Address** 0x50160008
**Description** Contains register fields associated with filter_config1.
**Reset** 0x000003f8

| Name | RSVD | interp_ph_idx_updt | interp_ph_idx1 | interp_ph_idx0 | interp_add_sample | interp_skp_sample | interp_bypass |
|---|---|---|---|---|---|---|---|
| **Bits** | **31:20** | **19** | **18:11** | **10:3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19 | R/W | interp_ph_idx_updt | interpolator phase indices update indicator for HW to capture, auto-clear | 1'b0 |
| 18:11 | R/W | interp_ph_idx1 | index-1 into interpolator coefficient table; for use only when add sample | 8'd0 |
| 10:3 | R/W | interp_ph_idx0 | index-0 into interpolator coefficient table | 8'd127 |
| 2 | R/W | interp_add_sample | add one sample to accomodate cfo frequency drift, auto-clear | 1'b0 |
| 1 | R/W | interp_skp_sample | skip one sample to accomodate cfo frequency drift, auto-clear | 1'b0 |
| 0 | R/W | interp_bypass | 1= bypass interpolator, also resets the fir filter taps | 1'b0 |

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**Name** GAIN_CONTROL_CONFIG
**Address** 0x5016000c
**Description** Contains register fields associated with gain_control_config.
**Reset** 0x00000040

| Name | RSV | hbf_gain | hpf_gain | cic_gain | cod |
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| | D | | | | | | | | | | | | | | | n_sel | | | n_sel | | | n_sel | | | e | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:13 | R/W | hbf_gain_sel | allow gain-scaling at the output of hbf through bit shifting<br>zero-padding for left shift; rounding, dropping lsbs for right shift.<br>0= no scaling<br>1-4= up scaling; zero-padding +1 to +4 respectively<br>5-7= down scaling; dropping lsbs -1 to -3 respectively | 3'd0 |
| 12:10 | R/W | hpf_gain_sel | allow gain-scaling at the output of hpf through bit shifting<br>zero-padding for left shift; rounding, dropping lsbs for right shift.<br>0= no scaling<br>1-4= up scaling; zero-padding +1 to +4 respectively<br>5-7= down scaling; dropping lsbs -1 to -3 respectively | 3'd0 |
| 9:7 | R/W | cic_gain_sel | allow gain-scaling at the output of cic through bit shifting<br>zero-padding for left shift; rounding, dropping lsbs for right shift.<br>0= no scaling<br>1-4= up scaling; zero-padding +1 to +4 respectively<br>5-7= down scaling; dropping lsbs -1 to -3 respectively | 3'd0 |
| 6:0 | R/W | code | gain code for LUT | 7'd64 |

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**Name** INTERRUPTS
**Address** 0x50160018
**Description** Contains register fields associated with interrupts.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | intrpt9 | intrpt8 | intrpt7 | intrpt6 | intrpt5 | intrpt4 | intrpt3 | intrpt2 | intrpt1 | intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R | intrpt9 | error in buffer D; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 8 | R | intrpt8 | error in buffer C; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 7 | R | intrpt7 | ping pong buffer D is locked and needs to be emptied | 1'b0 |
| 6 | R | intrpt6 | ping pong buffer C is locked and needs to be emptied | 1'b0 |
| 5 | R | intrpt5 | error in buffer B; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 4 | R | intrpt4 | error in buffer A; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 3 | R | intrpt3 | ping pong buffer B is locked and needs to be emptied | 1'b0 |
| 2 | R | intrpt2 | ping pong buffer A is locked and needs to be emptied | 1'b0 |
| 1 | R | intrpt1 | hw wrote to ping pong buffer | 1'b0 |
| 0 | R | intrpt0 | interpolator done processing current pcm sample | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x5016001c

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**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x000003ff

| Name | RSVD | mask_intrpt9 | mask_intrpt8 | mask_intrpt7 | mask_intrpt6 | mask_intrpt5 | mask_intrpt4 | mask_intrpt3 | mask_intrpt2 | mask_intrpt1 | mask_intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R/W | mask_intrpt9 | allow intrpt9 to propogate to core's single output interrupt | 1'b1 |
| 8 | R/W | mask_intrpt8 | allow intrpt8 to propogate to core's single output interrupt | 1'b1 |
| 7 | R/W | mask_intrpt7 | allow intrpt7 to propogate to core's single output interrupt | 1'b1 |
| 6 | R/W | mask_intrpt6 | allow intrpt6 to propogate to core's single output interrupt | 1'b1 |
| 5 | R/W | mask_intrpt5 | allow intrpt5 to propogate to core's single output interrupt | 1'b1 |
| 4 | R/W | mask_intrpt4 | allow intrpt4 to propogate to core's single output interrupt | 1'b1 |
| 3 | R/W | mask_intrpt3 | allow intrpt3 to propogate to core's single output interrupt | 1'b1 |
| 2 | R/W | mask_intrpt2 | allow intrpt2 to propogate to core's single output interrupt | 1'b1 |
| 1 | R/W | mask_intrpt1 | allow intrpt1 to propogate to core's single output interrupt | 1'b1 |
| 0 | R/W | mask_intrpt0 | allow intrpt0 to propogate to core's single output interrupt | 1'b1 |

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**Name** INTERRUPT_CLEAR
**Address** 0x50160020
**Description** Contains register fields associated with interrupt_clear.
**Reset** 0x00000000

| Name | RSVD | clear_intrpt9 | clear_intrpt8 | clear_intrpt7 | clear_intrpt6 | clear_intrpt5 | clear_intrpt4 | clear_intrpt3 | clear_intrpt2 | clear_intrpt1 | clear_intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R/W | clear_intrpt9 | clear interrupt 9; not self reseting | 1'b0 |
| 8 | R/W | clear_intrpt8 | clear interrupt 8; not self reseting | 1'b0 |
| 7 | R/W | clear_intrpt7 | clear interrupt 7; not self reseting | 1'b0 |
| 6 | R/W | clear_intrpt6 | clear interrupt 6; not self reseting | 1'b0 |
| 5 | R/W | clear_intrpt5 | clear interrupt 5; not self reseting | 1'b0 |
| 4 | R/W | clear_intrpt4 | clear interrupt 4; not self reseting | 1'b0 |
| 3 | R/W | clear_intrpt3 | clear interrupt 3; not self reseting | 1'b0 |
| 2 | R/W | clear_intrpt2 | clear interrupt 2; not self reseting | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 1 | R/W | clear_intrpt1 | clear interrupt 1; not self reseting | 1'b0 |
| 0 | R/W | clear_intrpt0 | clear interrupt 0; not self reseting | 1'b0 |

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**Name** BUFFER_ACCESS_MODE
**Address** 0x50160024
**Description** Contains register fields associated with buffer_access_mode.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | dma_mode |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | dma_mode | 1= pingpong buffer has fifo-like interface.<br>read address is auto-incremented from base address of buffer A by an internal counter, counting from 0 to 63 and wraps around.<br>buffer full interrupt is cleared by HW at the end of the last read of the buffer.<br>The auto-increment counter is cleared when fifo_mode is 0. | 1'b0 |

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**Name** AUTO_COUNTER
**Address** 0x50160028
**Description** Contains register fields associated with auto_counter.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | count | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:0 | R | count | current value of the auto increment counter. | 6'b0 |

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**Name** BUFFER_DEPTH
**Address** 0x5016002c
**Description** Contains register fields associated with buffer_depth.
**Reset** 0x00000010

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | depth | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | depth | one buffer depth | 5'd16 |

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**Name** CORE_ID
**Address** 0x50160ff4
**Description** Contains register fields associated with core_id.
**Reset** 0x50444d20

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | 'PDM ' in ASCII | 32'h50444d20 |

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**Name** REV_HASH
**Address** 0x50160ff8
**Description** Contains register fields associated with rev_hash.
**Reset** 0x00000000

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | crc32 of this document; don't change value to anything other than 32'h00000000; scripts will fill it out in the verilog | 32'h00000000 |

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**Name** REV_KEY
**Address** 0x50160ffc
**Description** Contains register fields associated with rev_key.
**Reset** 0x52455620

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | REV in ASCII | 32'h52455620 |

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# PDM1_SECURE

**Name** CONTROL
**Address** 0x50161000
**Description** Contains register fields associated with control.
**Reset** 0x000006f0

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | s | p | pc | | pc | | p | pa | sw | capt | |

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| Name | RSVD | | | | | | | | | | | | | | | stereo_mode | pdm_in_dbg | pcm_bit_width | | pcm_rate_sel | | | | pdm_clk_sel | | | | | ck_samples | ap_samples | ure_on_rising | enable_dp |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 16 | R/W | stereo_mode | selects between stereo and mono mode of operation<br>1= stereo mode; act as datapath enable for other stereo datapath pair. When this bit is set along with enable_dp, it enables stereo datapath pair concurrently to avoid pdm data capture synchronization issues. At that time, corresponding bits for the other datapath pair shouldn't be set.<br>0= mono mode; use enable_dp bit to enable each datapath. | 1'd0 |
| 15 | R/W | pdm_in_dbg | if 1, bypass pdm bitstream and send in dc value 1 | 1'd0 |
| 14:13 | R/W | pcm_bit_width | pcm output sample bit width<br>0= 16bit<br>1= 20bit<br>2/3= 24bit | 2'd0 |
| 12:9 | R/W | pcm_rate_sel | pcm output rate selection if msb bit3=1 it's in pcm_rate override mode<br>0= 6KHz<br>1= 8KHz<br>2= 12KHz<br>3= 16KHz<br>4= 24KHz<br>5= 32KHz<br>6= 48KHz<br>7= 64KHz | 4'd3 |
| 8:4 | R/W | pdm_clk_sel | pdm clock frequency = f_pclk/(pdm_clk_sel+1)<br>valid range: 3 to 31<br>possible rates when pcm_rate_sel[0]=1:<br>31= 500KHz<br>15= 1MHz<br>7= 2MHz<br>possible rates when pcm_rate_sel[0]=0:<br>23= 500KHz<br>11= 1MHz<br>5= 2MHz | 5'd15 |
| 3 | R/W | pack_samples | pack samples in ping-pong buffer<br>default: samples are stored unpacked<br>applies to 20/24 bit samples only | 1'b0 |
| 2 | R/W | swap_samples | swap packed samples in pingpong buffer location<br>default: former samples uses lsb bits and later samples uses msb bits<br>applies to 16bit samples only | 1'b0 |
| 1 | R/W | capture_on_rising | selects edge of pdm clock to capture pdm data<br>0= pdm data sampled on falling edge of pdm clock (i.e. pdm data transitions on rising edge of pdm clock)<br>1= pdm data sampled on rising edge of pdm clock (i.e. pdm data transitions on falling edge of pdm clock) | 1'b0 |
| 0 | R/W | enable_dp | enable the datapath | 1'b0 |

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**Name** FILTER_CONFIG
**Address** 0x50161004
**Description** Contains register fields associated with filter_config.

**Reset** 0x00062a50

| Name | RSVD | | | | | | downsample_ratio | | lpf_bypass_ovrd_val | lpf_bw | | | hbf_bypass_ovrd_val | hpf2_init_shift | | | | hpf1_init_shift | | | | hpf_init_enable | hpf_bypass | hpf_bw | | | | hpf_ord | cic_ord | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25:24 | R/W | downsample_ratio | downsample override<br>used only during pcm_rate override mode:<br>downsampling by 1/(1+downsample_ratio) | 2'd0 |
| 23 | R/W | lpf_bypass_ovrd_val | lpf bypass override value<br>used only during pcm_rate override mode:<br>0= enable lpf<br>1= bypass lpf; output rate=input rate | 1'd0 |
| 22:20 | R/W | lpf_bw | lpf bandwidth override selection<br>used only during pcm_rate override mode:<br>if pcm_rate_sel[0]=1, bandwidth will be (KHz):<br>0= 4<br>1= 6<br>2= 8<br>3= 10<br>4= 12<br>5= 14<br>6/7= 15.5<br>if pcm_rate_sel[0]=0, bandwidth will be (KHz):<br>0= 3<br>1= 4.5<br>2= 6<br>3= 7.5<br>4= 9<br>5= 10<br>6/7= 11.5 | 3'd0 |
| 19 | R/W | hbf_bypass_ovrd_val | hbf bypass override value<br>used only during pcm_rate override mode:<br>0= enable hbf<br>1= bypass hbf; output rate=input rate | 1'd0 |
| 18:14 | R/W | hpf2_init_shift | hpf2 initialization shift amount, max=24<br>for values > max, operation is same as max | 5'd24 |
| 13:10 | R/W | hpf1_init_shift | hpf1 initialization shift amount, max=10<br>for values > max, operation is same as max | 4'd10 |
| 9 | R/W | hpf_init_enable | 1= enable hpf initialization | 1'b1 |
| 8 | R/W | hpf_bypass | 1= bypass hpf | 1'b0 |
| 7:4 | R/W | hpf_bw | hpf bandwidth selection<br>when hpf_ord=0, bandwidth will be (Hz):<br>0= 10, 1= 20, 2= 40, 3= 80, 4= 161, 5= 324, 6= 658, 7= 1359<br>when hpf_ord=1, bandwidth will be (Hz):<br>0= 9, 1= 19, 2= 37, 3= 75, 4= 151, 5= 304, 6= 616, 7= 1269<br>when hpf_ord=2, bandwidth will be (Hz):<br>0= 15, 1= 30, 2= 60, 3= 120, 4= 242, 5= 487, 6= 989, 7= 2036 | 4'd5 |

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| | | | | |
|---|---|---|---|---|
| 3:2 | R/W | hpf_ord | hpf order<br>0= 1st order<br>1= 2nd order<br>2/3= 3rd order | 2'd0 |
| 1:0 | R/W | cic_ord | cic order<br>0= 3rd order<br>1= 4th order<br>2/3= unused | 2'd0 |

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**Name** FILTER_CONFIG1
**Address** 0x50161008
**Description** Contains register fields associated with filter_config1.
**Reset** 0x000003f8

| Name | RSVD | interp_ph_idx_updt | interp_ph_idx1 | interp_ph_idx0 | interp_add_sample | interp_skp_sample | interp_bypass |
|---|---|---|---|---|---|---|---|
| **Bits** | **31:20** | **19** | **18:11** | **10:3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 19 | R/W | interp_ph_idx_updt | interpolator phase indices update indicator for HW to capture, auto-clear | 1'b0 |
| 18:11 | R/W | interp_ph_idx1 | index-1 into interpolator coefficient table; for use only when add sample | 8'd0 |
| 10:3 | R/W | interp_ph_idx0 | index-0 into interpolator coefficient table | 8'd127 |
| 2 | R/W | interp_add_sample | add one sample to accomodate cfo frequency drift, auto-clear | 1'b0 |
| 1 | R/W | interp_skp_sample | skip one sample to accomodate cfo frequency drift, auto-clear | 1'b0 |
| 0 | R/W | interp_bypass | 1= bypass interpolator, also resets the fir filter taps | 1'b0 |

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**Name** GAIN_CONTROL_CONFIG
**Address** 0x5016100c
**Description** Contains register fields associated with gain_control_config.
**Reset** 0x00000040

| Name | RSVD | hbf_gain_sel | hpf_gain_sel | cic_gain_sel | code |
|---|---|---|---|---|---|
| **Bits** | **31:8** | **7:6** | **5:4** | **3:2** | **1:0** |

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 15:13 | R/W | hbf_gain_sel | allow gain-scaling at the output of hbf through bit shifting<br>zero-padding for left shift; rounding, dropping lsbs for right shift.<br>0= no scaling<br>1-4= up scaling; zero-padding +1 to +4 respectively<br>5-7= down scaling; dropping lsbs -1 to -3 respectively | 3'd0 |
| 12:10 | R/W | hpf_gain_sel | allow gain-scaling at the output of hpf through bit shifting<br>zero-padding for left shift; rounding, dropping lsbs for right shift.<br>0= no scaling<br>1-4= up scaling; zero-padding +1 to +4 respectively<br>5-7= down scaling; dropping lsbs -1 to -3 respectively | 3'd0 |
| 9:7 | R/W | cic_gain_sel | allow gain-scaling at the output of cic through bit shifting<br>zero-padding for left shift; rounding, dropping lsbs for right shift.<br>0= no scaling<br>1-4= up scaling; zero-padding +1 to +4 respectively<br>5-7= down scaling; dropping lsbs -1 to -3 respectively | 3'd0 |
| 6:0 | R/W | code | gain code for LUT | 7'd64 |

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**Name** INTERRUPTS
**Address** 0x50161018
**Description** Contains register fields associated with interrupts.
**Reset** 0x00000000

| Name | RSVD | intrpt9 | intrpt8 | intrpt7 | intrpt6 | intrpt5 | intrpt4 | intrpt3 | intrpt2 | intrpt1 | intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R | intrpt9 | error in buffer D; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 8 | R | intrpt8 | error in buffer C; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 7 | R | intrpt7 | ping pong buffer D is locked and needs to be emptied | 1'b0 |
| 6 | R | intrpt6 | ping pong buffer C is locked and needs to be emptied | 1'b0 |
| 5 | R | intrpt5 | error in buffer B; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 4 | R | intrpt4 | error in buffer A; hw wants to write to buffer but lock hasn't been released by sw | 1'b0 |
| 3 | R | intrpt3 | ping pong buffer B is locked and needs to be emptied | 1'b0 |
| 2 | R | intrpt2 | ping pong buffer A is locked and needs to be emptied | 1'b0 |
| 1 | R | intrpt1 | hw wrote to ping pong buffer | 1'b0 |
| 0 | R | intrpt0 | interpolator done processing current pcm sample | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x5016101c
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x000003ff

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ma | ma | ma | ma | ma | ma | ma | ma | ma | ma |

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| Name | RSVD | sk_intrpt9 | sk_intrpt8 | sk_intrpt7 | sk_intrpt6 | sk_intrpt5 | sk_intrpt4 | sk_intrpt3 | sk_intrpt2 | sk_intrpt1 | sk_intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R/W | mask_intrpt9 | allow intrpt9 to propogate to core's single output interrupt | 1'b1 |
| 8 | R/W | mask_intrpt8 | allow intrpt8 to propogate to core's single output interrupt | 1'b1 |
| 7 | R/W | mask_intrpt7 | allow intrpt7 to propogate to core's single output interrupt | 1'b1 |
| 6 | R/W | mask_intrpt6 | allow intrpt6 to propogate to core's single output interrupt | 1'b1 |
| 5 | R/W | mask_intrpt5 | allow intrpt5 to propogate to core's single output interrupt | 1'b1 |
| 4 | R/W | mask_intrpt4 | allow intrpt4 to propogate to core's single output interrupt | 1'b1 |
| 3 | R/W | mask_intrpt3 | allow intrpt3 to propogate to core's single output interrupt | 1'b1 |
| 2 | R/W | mask_intrpt2 | allow intrpt2 to propogate to core's single output interrupt | 1'b1 |
| 1 | R/W | mask_intrpt1 | allow intrpt1 to propogate to core's single output interrupt | 1'b1 |
| 0 | R/W | mask_intrpt0 | allow intrpt0 to propogate to core's single output interrupt | 1'b1 |

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**Name** INTERRUPT_CLEAR
**Address** 0x50161020
**Description** Contains register fields associated with interrupt_clear.
**Reset** 0x00000000

| Name | RSVD | clear_intrpt9 | clear_intrpt8 | clear_intrpt7 | clear_intrpt6 | clear_intrpt5 | clear_intrpt4 | clear_intrpt3 | clear_intrpt2 | clear_intrpt1 | clear_intrpt0 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R/W | clear_intrpt9 | clear interrupt 9; not self reseting | 1'b0 |
| 8 | R/W | clear_intrpt8 | clear interrupt 8; not self reseting | 1'b0 |
| 7 | R/W | clear_intrpt7 | clear interrupt 7; not self reseting | 1'b0 |
| 6 | R/W | clear_intrpt6 | clear interrupt 6; not self reseting | 1'b0 |
| 5 | R/W | clear_intrpt5 | clear interrupt 5; not self reseting | 1'b0 |
| 4 | R/W | clear_intrpt4 | clear interrupt 4; not self reseting | 1'b0 |
| 3 | R/W | clear_intrpt3 | clear interrupt 3; not self reseting | 1'b0 |
| 2 | R/W | clear_intrpt2 | clear interrupt 2; not self reseting | 1'b0 |
| 1 | R/W | clear_intrpt1 | clear interrupt 1; not self reseting | 1'b0 |
| 0 | R/W | clear_intrpt0 | clear interrupt 0; not self reseting | 1'b0 |

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**Name** BUFFER_ACCESS_MODE
**Address** 0x50161024
**Description** Contains register fields associated with buffer_access_mode.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | dma_mode |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | dma_mode | 1= pingpong buffer has fifo-like interface.<br>read address is auto-incremented from base address of buffer A by an internal counter, counting from 0 to 63 and wraps around.<br>buffer full interrupt is cleared by HW at the end of the last read of the buffer.<br>The auto-increment counter is cleared when fifo_mode is 0. | 1'b0 |

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**Name** AUTO_COUNTER
**Address** 0x50161028
**Description** Contains register fields associated with auto_counter.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | count | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:0 | R | count | current value of the auto increment counter. | 6'b0 |

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**Name** BUFFER_DEPTH
**Address** 0x5016102c
**Description** Contains register fields associated with buffer_depth.
**Reset** 0x00000010

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | depth | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | depth | one buffer depth | 5'd16 |

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**Name** CORE_ID
**Address** 0x50161ff4

CONFIDENTIAL

**Description** Contains register fields associated with core_id.
**Reset** 0x50444d20

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | 'PDM ' in ASCII | 32'h50444d20 |

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**Name** REV_HASH
**Address** 0x50161ff8
**Description** Contains register fields associated with rev_hash.
**Reset** 0x00000000

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | crc32 of this document; don't change value to anything other than 32'h00000000; scripts will fill it out in the verilog | 32'h00000000 |

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---

**Name** REV_KEY
**Address** 0x50161ffc
**Description** Contains register fields associated with rev_key.
**Reset** 0x52455620

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | REV in ASCII | 32'h52455620 |

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# WRPR2_SECURE

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**Name** APB0_CTRL
**Address** 0x50163000
**Description** Contains register fields associated with apb0_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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---

**Name** APB1_CTRL
**Address** 0x50163004
**Description** Contains register fields associated with apb1_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB2_CTRL
**Address** 0x50163008
**Description** Contains register fields associated with apb2_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 | 2 | 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB3_CTRL
**Address** 0x5016300c
**Description** Contains register fields associated with apb3_ctrl.

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**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB4_CTRL
**Address** 0x50163010
**Description** Contains register fields associated with apb4_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB5_CTRL
**Address** 0x50163014
**Description** Contains register fields associated with apb5_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB6_CTRL
**Address** 0x50163018
**Description** Contains register fields associated with apb6_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB7_CTRL
**Address** 0x5016301c
**Description** Contains register fields associated with apb7_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB8_CTRL
**Address** 0x50163020
**Description** Contains register fields associated with apb8_ctrl.
**Reset** 0x00000002

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| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB9_CTRL
**Address** 0x50163024
**Description** Contains register fields associated with apb9_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB10_CTRL
**Address** 0x50163028
**Description** Contains register fields associated with apb10_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB11_CTRL
**Address** 0x5016302c
**Description** Contains register fields associated with apb11_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB12_CTRL
**Address** 0x50163030
**Description** Contains register fields associated with apb12_ctrl.
**Reset** 0x00000002

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | clk_sel | sreset | clk_enable |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB13_CTRL
**Address** 0x50163034
**Description** Contains register fields associated with apb13_ctrl.
**Reset** 0x00000002

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | c |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB14_CTRL
**Address** 0x50163038
**Description** Contains register fields associated with apb14_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** APB15_CTRL
**Address** 0x5016303c
**Description** Contains register fields associated with apb15_ctrl.
**Reset** 0x00000002

| Name | RSVD | clk_sel | sreset | clk_enable |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 2 | R/W | clk_sel | [0]=PCLK [1]=PCLK_ALT | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 1 | R/W | sreset | (no description) | 1'b1 |
| 0 | R/W | clk_enable | (no description) | 1'b0 |

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**Name** CORE_ID
**Address** 0x50163ffc
**Description** Contains register fields associated with core_id.
**Reset** 0x57525052

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | WRPR in ASCII | 32'h57525052 |

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# I2S_SECURE

**Name** I2S_CTRL0
**Address** 0x50165000
**Description** Contains register fields associated with i2s_ctrl0.
**Reset** 0x06100900

| Name | RSVD | ws_nedge_st_tx | ws_nedge_st_rx | sck_nedge_sd_tx | sck_nedge_sd_rx | ws_init_tx | ws_init_rx | dma_en | sck_init_cnt | | | | | | | | RSVD | snk_cnt | | | src_cnt | | | | mstr_sckws_tx | mstr_sckws_rx | src_snk | | src_snk_en | | sck_nedge_ws_tx | sck_nedge_ws_rx |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 30 | R/W | ws_nedge_st_tx | 1 = tx starts on negative edge of ws | 1'b0 |
| 29 | R/W | ws_nedge_st_rx | 1 = rx starts on negative edge of ws | 1'b0 |
| 28 | R/W | sck_nedge_sd_tx | 1 = drive/latch tx sd on negative edge of sck, 0 = drive/latch tx sd on positive edge of sck | 1'b0 |
| 27 | R/W | sck_nedge_sd_rx | 1 = drive/latch rx sd on negative edge of sck, 0 = drive/latch rx sd on positive edge of sck | 1'b0 |
| 26 | R/W | ws_init_tx | 0 = word select is deasserted at the starting half of justieid mode 1 = word select is asserted at the starting half of justieid mode this applies to right or left justified mode, ws_init should be set to 1 for PCM mode | 1'b1 |
| 25 | R/W | ws_init_rx | 0 = word select is deasserted at the starting half of justieid mode 1 = word select is asserted at the starting half of justieid mode this applies to right or left justified mode, ws_init should be set to 1 for PCM mode | 1'b1 |
| 24 | R/W | dma_en | dma is enabled to access ping pong buffer | 1'b0 |
| 23:16 | R/W | sck_init_cnt | initial number of sck before going to transmit or receive state | 8'd16 |
| 13:11 | R/W | snk_cnt | number of sinks for i2s | 3'b1 |

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| | | | | |
|---|---|---|---|---|
| 10:8 | R/W | src_cnt | number of sources for i2s | 3'b1 |
| 7 | R/W | mstr_sckws_tx | 0 = slave sck and ws, 1 = master sck and ws | 1'b0 |
| 6 | R/W | mstr_sckws_rx | 0 = slave sck and ws, 1 = master sck and ws | 1'b0 |
| 5:4 | R/W | src_snk | 0 = pcm (pdm2pcm processed data is transmitted), 1 = ahb (data in memory 0 is transmitted or data is received and stored in memory 0), 2 = ahb (data in memory 1 is transmitted or data is received and stored in memory 1) this is mainly used for rx to tx loop back simulation and eimulation | 2'b0 |
| 3:2 | R/W | src_snk_en | Enable tranceiver tx or rx or both After it's enabled, first data valid assertion initiates tx and first ws deassertion initiates rx<br>0 = i2s disabled<br>1 = source serial data<br>2 = sink serial data<br>3 = source serial data and sink serial data | 2'b0 |
| 1 | R/W | sck_nedge_ws_tx | 1 = drive/latch tx ws on negative edge of sck, 0 = drive/latch tx ws on positive edge of sck | 1'b0 |
| 0 | R/W | sck_nedge_ws_rx | 1 = drive/latch rx ws on negative edge of sck, 0 = drive/latch rx ws on positive edge of sck | 1'b0 |

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**Name** I2S_CTRL1_TX
**Address** 0x50165004
**Description** Contains register fields associated with i2s_ctrl1_tx.
**Reset** 0x00200010

| Name | sck2ws_rt | | | | | | | | | | | | | | | | ck2sck_rt | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:16 | R/W | sck2ws_rt | serial clock to word select ratio - number of serial clock per word select period | 16'd32 |
| 15:0 | R/W | ck2sck_rt | clock to serial clock ratio - number of clocks per serial clock period | 16'd16 |

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**Name** I2S_CTRL1_RX
**Address** 0x50165008
**Description** Contains register fields associated with i2s_ctrl1_rx.
**Reset** 0x00200010

| Name | sck2ws_rt | | | | | | | | | | | | | | | | ck2sck_rt | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| | | | serial clock to word select ratio - number of serial clock per word select | |

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| | | | | |
|---|---|---|---|---|
| 31:16 | R/W | sck2ws_rt | period | 16'd32 |
| 15:0 | R/W | ck2sck_rt | clock to serial clock ratio - number of clocks per serial clock period | 16'd16 |

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**Name** I2S_CTRL2_TX
**Address** 0x5016500c
**Description** Contains register fields associated with i2s_ctrl2_tx.
**Reset** 0x40001000

| Name | vld_extd_cnt | | | | | | | | RSVD | | pb_cnt | | | | | | RSVD | | sdw | | | | | | sd_offst | | | | wssd_md | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | vld_extd_cnt | cycle count to extend vld to ensure it is latched by slow backplane clock | 8'd64 |
| 21:16 | R/W | pb_cnt | number of zero padding bits - used in right justified mode. | 6'd0 |
| 13:8 | R/W | sdw | width of valid serial data - additioanl bits in a word beyond the width are either padded for tx or discarded for rx valid range is [1:32] | 6'd16 |
| 7:4 | R/W | sd_offst | number of cycles after word select edge to wait before driving serial data - valid range is [0:15] | 4'd0 |
| 3:0 | R/W | wssd_md | word select serial data mode - specify the behavior of word select and serial data 0 = PCM mode, 1 = left justfied I2S, 2 = right justified I2S | 4'd0 |

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**Name** I2S_CTRL2_RX
**Address** 0x50165010
**Description** Contains register fields associated with i2s_ctrl2_rx.
**Reset** 0x40001000

| Name | vld_extd_cnt | | | | | | | | RSVD | | pb_cnt | | | | | | RSVD | | sdw | | | | | | sd_offst | | | | wssd_md | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:24 | R/W | vld_extd_cnt | cycle count to extend vld to ensure it is latched by slow backplane clock | 8'd64 |
| 21:16 | R/W | pb_cnt | number of zero padding bits - used in right justified mode. | 6'd0 |
| 13:8 | R/W | sdw | width of valid serial data - additioanl bits in a word beyond the width are either padded for tx or discarded for rx valid range is [1:32] | 6'd16 |
| | | | number of cycles after word select edge to wait before driving serial data - | |

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|---|---|---|---|---|
| 7:4 | R/W | sd_offst | valid range is [0:15] | 4'd0 |
| 3:0 | R/W | wssd_md | word select serial data mode - specify the behavior of word select and serial data 0 = PCM mode, 1 = left justfied I2S, 2 = right justified I2S | 4'd0 |

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**Name** I2S_CTRL3
**Address** 0x50165014
**Description** Contains register fields associated with i2s_ctrl3.
**Reset** 0x80007f04

| Name | intrp_byp | RSVD | intrp_use_msb_smpl | phs_idx_add | phs_idx | RSVD | use_msb_smpl | pck_smpl | swp_smpl |
|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25** | **24** | **23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8** | **7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | intrp_byp | bypass interpolator for cfo correction for tx | 1'b1 |
| 24 | R/W | intrp_use_msb_smpl | Use top 24 bit of 32 bit sample for tx interpolator | 1'b0 |
| 23:16 | R/W | phs_idx_add | index into the coefficient table for insert sample | 8'b0 |
| 15:8 | R/W | phs_idx | index into the coefficient table | 8'd127 |
| 2 | R/W | use_msb_smpl | use msb of the samples | 1'b1 |
| 1 | R/W | pck_smpl | pack samples in ping-pong buffer<br>default: samples are stored unpacked<br>applies to 20/24 bit samples only | 1'b0 |
| 0 | R/W | swp_smpl | swap packed samples in pingpong buffer location<br>default: former samples uses lsb bits and later samples uses msb bits<br>applies to 16bit samples only | 1'b0 |

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**Name** I2S_CTRL4
**Address** 0x50165018
**Description** Contains register fields associated with i2s_ctrl4.
**Reset** 0x80007f00

| Name | intrp_by | RSVD | intrp_use_ms | phs_idx_a | phs_idx | RSVD | use_msb_s | pck_smp | swp_smp |
|---|---|---|---|---|---|---|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | p | | | | | | | b_smpl | | | | dd | | | | | | | | | | | | | | | mpl | l | l |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | intrp_byp | bypass interpolator for cfo correction for rx | 1'b1 |
| 24 | R/W | intrp_use_msb_smpl | Use top 24 bit of 32 bit sample for rx interpolator | 1'b0 |
| 23:16 | R/W | phs_idx_add | index into the coefficient table for insert sample | 8'b0 |
| 15:8 | R/W | phs_idx | index into the coefficient table | 8'd127 |
| 2 | R/W | use_msb_smpl | use msb of the samples | 1'b0 |
| 1 | R/W | pck_smpl | pack samples in ping-pong buffer<br>default: samples are stored unpacked<br>applies to 20/24 bit samples only | 1'b0 |
| 0 | R/W | swp_smpl | swap packed samples in pingpong buffer location<br>default: former samples uses lsb bits and later samples uses msb bits<br>applies to 16bit samples only | 1'b0 |

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**Name** I2S_CTRL5_TX
**Address** 0x5016501c
**Description** Contains register fields associated with i2s_ctrl5_tx.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | frq_drft_corr_en |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | frq_drft_corr_en | 0 = do not enable freq drift correction 1 = enable freq drift correction | 1'b0 |

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**Name** I2S_CTRL5_RX
**Address** 0x50165020
**Description** Contains register fields associated with i2s_ctrl5_rx.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | frq_dr |

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| Name | RSVD | frq_drft_corr_en |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 | 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | frq_drft_corr_en | 0 = do not enable freq drift correction 1 = enable freq drift correction | 1'b0 |

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**Name** I2S_INTRP_CTRL
**Address** 0x50165024
**Description** Contains register fields associated with i2s_intrp_ctrl.
**Reset** 0x00000000

| Name | RSVD | rx_updt_phs_idx | tx_updt_phs_idx | rx_add_smpl | rx_skp_smpl | tx_add_smpl | tx_skp_smpl |
|---|---|---|---|---|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 | 5 | 4 | 3 | 2 | 1 | 0 |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | rx_updt_phs_idx | latch the phase index for rx interpolator | 1'b0 |
| 4 | R/W | tx_updt_phs_idx | latch the phase index for tx interpolator | 1'b0 |
| 3 | R/W | rx_add_smpl | insert one sample to accomodate frequecy drift | 1'b0 |
| 2 | R/W | rx_skp_smpl | skip one sample to accomodate frequecy drift | 1'b0 |
| 1 | R/W | tx_add_smpl | insert one sample to accomodate frequecy drift | 1'b0 |
| 0 | R/W | tx_skp_smpl | skip one sample to accomodate frequecy drift | 1'b0 |

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**Name** I2S_PP1_RDATA
**Address** 0x50165028
**Description** Contains register fields associated with i2s_pp1_rdata.
**Reset** 0x00000000

| Name | val |
|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | val | read data at the top of ping pong set 1 buffer 3 | 32'b0 |

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**Name** I2S_PP0_WDATA
**Address** 0x5016502c
**Description** Contains register fields associated with i2s_pp0_wdata.
**Reset** 0x00000000

| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | W | val | write data at the bottom of ping pong set 0 buffer 3 | 32'b0 |

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**Name** I2S_PP1_CTRL
**Address** 0x50165030
**Description** Contains register fields associated with i2s_pp1_ctrl.
**Reset** 0x00000008

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | thrshld | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6:0 | R/W | thrshld | number of available words that are empty and can be filled | 7'd8 |

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**Name** I2S_PP0_CTRL
**Address** 0x50165034
**Description** Contains register fields associated with i2s_pp0_ctrl.
**Reset** 0x00000008

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | thrshld | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6:0 | R/W | thrshld | number of available words that are empty and can be filled | 7'd8 |

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**Name** I2S_IRQ0
**Address** 0x50165038
**Description** Contains register fields associated with i2s_irq0.
**Reset** 0x00000000

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| Name | pdm_ufof | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R | pdm_ufof | pdm underflow or overflow | 1'b0 |
| 13 | R | intrp_dout_vld | interpolator data valid | 1'b0 |
| 12 | R | pp_ep_thrshld | ping pong buffer empty threshold hit | 1'b0 |
| 11 | R | pp3_uf | ping pong set 0 buffer 3 under flow | 1'b0 |
| 10 | R | pp2_uf | ping pong set 0 buffer 2 under flow | 1'b0 |
| 9 | R | pp1_uf | ping pong set 0 buffer 1 under flow | 1'b0 |
| 8 | R | pp0_uf | ping pong set 0 buffer 0 under flow | 1'b0 |
| 7 | R | pp3_of | ping pong set 0 buffer 3 over flow | 1'b0 |
| 6 | R | pp2_of | ping pong set 0 buffer 2 over flow | 1'b0 |
| 5 | R | pp1_of | ping pong set 0 buffer 1 over flow | 1'b0 |
| 4 | R | pp0_of | ping pong set 0 buffer 0 over flow | 1'b0 |
| 3 | R | pp3_fl | ping pong set 0 buffer 3 full | 1'b0 |
| 2 | R | pp2_fl | ping pong set 0 buffer 2 full | 1'b0 |
| 1 | R | pp1_fl | ping pong set 0 buffer 1 full | 1'b0 |
| 0 | R | pp0_fl | ping pong set 0 buffer 0 full | 1'b0 |

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**Name** I2S_IRQM0
**Address** 0x5016503c
**Description** Contains register fields associated with i2s_irqm0.
**Reset** 0x00000fff

| Name | pdm_ufof | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | pdm_ufof | pdm underflow or overflow | 1'b0 |
| 13 | R/W | intrp_dout_vld | interpolator data valid mask | 1'b0 |
| 12 | R/W | pp_ep_thrshld | ping pong buffer empty threshold hit mask | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 11 | R/W | pp3_uf | ping pong set 0 buffer 3 under flow mask | 1'b1 |
| 10 | R/W | pp2_uf | ping pong set 0 buffer 2 under flow mask | 1'b1 |
| 9 | R/W | pp1_uf | ping pong set 0 buffer 1 under flow mask | 1'b1 |
| 8 | R/W | pp0_uf | ping pong set 0 buffer 0 under flow mask | 1'b1 |
| 7 | R/W | pp3_of | ping pong set 0 buffer 3 over flow mask | 1'b1 |
| 6 | R/W | pp2_of | ping pong set 0 buffer 2 over flow mask | 1'b1 |
| 5 | R/W | pp1_of | ping pong set 0 buffer 1 over flow mask | 1'b1 |
| 4 | R/W | pp0_of | ping pong set 0 buffer 0 over flow mask | 1'b1 |
| 3 | R/W | pp3_fl | ping pong set 0 buffer 3 full mask | 1'b1 |
| 2 | R/W | pp2_fl | ping pong set 0 buffer 2 full mask | 1'b1 |
| 1 | R/W | pp1_fl | ping pong set 0 buffer 1 full mask | 1'b1 |
| 0 | R/W | pp0_fl | ping pong set 0 buffer 0 full mask | 1'b1 |

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**Name** I2S_IRQS0
**Address** 0x50165040
**Description** Contains register fields associated with i2s_irqs0.
**Reset** 0x00000000

| Name | pdm_ufof | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | pdm_ufof | pdm underflow or overflow | 1'b0 |
| 13 | R/W | intrp_dout_vld | interpolator data valid set | 1'b0 |
| 12 | R/W | pp_ep_thrshld | ping pong buffer empty threshold hit set | 1'b0 |
| 11 | R/W | pp3_uf | ping pong set 0 buffer 3 under flow set | 1'b0 |
| 10 | R/W | pp2_uf | ping pong set 0 buffer 2 under flow set | 1'b0 |
| 9 | R/W | pp1_uf | ping pong set 0 buffer 1 under flow set | 1'b0 |
| 8 | R/W | pp0_uf | ping pong set 0 buffer 0 under flow set | 1'b0 |
| 7 | R/W | pp3_of | ping pong set 0 buffer 3 over flow set | 1'b0 |
| 6 | R/W | pp2_of | ping pong set 0 buffer 2 over flow set | 1'b0 |
| 5 | R/W | pp1_of | ping pong set 0 buffer 1 over flow set | 1'b0 |
| 4 | R/W | pp0_of | ping pong set 0 buffer 0 over flow set | 1'b0 |
| 3 | R/W | pp3_fl | ping pong set 0 buffer 3 full set | 1'b0 |
| 2 | R/W | pp2_fl | ping pong set 0 buffer 2 full set | 1'b0 |
| 1 | R/W | pp1_fl | ping pong set 0 buffer 1 full set | 1'b0 |
| 0 | R/W | pp0_fl | ping pong set 0 buffer 0 full set | 1'b0 |

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**Name** I2S_IRQC0
**Address** 0x50165044
**Description** Contains register fields associated with i2s_irqc0.

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**Reset** 0x00000000

| Name | pdm_ufof | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | pdm_ufof | pdm underflow or overflow | 1'b0 |
| 13 | R/W | intrp_dout_vld | interpolator data valid clear | 1'b0 |
| 12 | R/W | pp_ep_thrshld | ping pong buffer empty threshold hit clear | 1'b0 |
| 11 | R/W | pp3_uf | ping pong set 0 buffer 3 under flow clear | 1'b0 |
| 10 | R/W | pp2_uf | ping pong set 0 buffer 2 under flow clear | 1'b0 |
| 9 | R/W | pp1_uf | ping pong set 0 buffer 1 under flow clear | 1'b0 |
| 8 | R/W | pp0_uf | ping pong set 0 buffer 0 under flow clear | 1'b0 |
| 7 | R/W | pp3_of | ping pong set 0 buffer 3 over flow clear | 1'b0 |
| 6 | R/W | pp2_of | ping pong set 0 buffer 2 over flow clear | 1'b0 |
| 5 | R/W | pp1_of | ping pong set 0 buffer 1 over flow clear | 1'b0 |
| 4 | R/W | pp0_of | ping pong set 0 buffer 0 over flow clear | 1'b0 |
| 3 | R/W | pp3_fl | ping pong set 0 buffer 3 full clear | 1'b0 |
| 2 | R/W | pp2_fl | ping pong set 0 buffer 2 full clear | 1'b0 |
| 1 | R/W | pp1_fl | ping pong set 0 buffer 1 full clear | 1'b0 |
| 0 | R/W | pp0_fl | ping pong set 0 buffer 0 full clear | 1'b0 |

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**Name** I2S_IRQ1
**Address** 0x50165048
**Description** Contains register fields associated with i2s_irq1.
**Reset** 0x00000000

| Name | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13 | R | intrp_dout_vld | interpolator data valid | 1'b0 |
| 12 | R | pp_ep_thrshld | ping pong buffer empty threshold hit | 1'b0 |

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| | | | | |
|---|---|---|---|---|
| 11 | R | pp3_uf | ping pong set 1 buffer 3 under flow | 1'b0 |
| 10 | R | pp2_uf | ping pong set 1 buffer 2 under flow | 1'b0 |
| 9 | R | pp1_uf | ping pong set 1 buffer 1 under flow | 1'b0 |
| 8 | R | pp0_uf | ping pong set 1 buffer 0 under flow | 1'b0 |
| 7 | R | pp3_of | ping pong set 1 buffer 3 over flow | 1'b0 |
| 6 | R | pp2_of | ping pong set 1 buffer 2 over flow | 1'b0 |
| 5 | R | pp1_of | ping pong set 1 buffer 1 over flow | 1'b0 |
| 4 | R | pp0_of | ping pong set 1 buffer 0 over flow | 1'b0 |
| 3 | R | pp3_fl | ping pong set 1 buffer 3 full | 1'b0 |
| 2 | R | pp2_fl | ping pong set 1 buffer 2 full | 1'b0 |
| 1 | R | pp1_fl | ping pong set 1 buffer 1 full | 1'b0 |
| 0 | R | pp0_fl | ping pong set 1 buffer 0 full | 1'b0 |

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**Name** I2S_IRQM1
**Address** 0x5016504c
**Description** Contains register fields associated with i2s_irqm1.
**Reset** 0x00000fff

| Name | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13 | R/W | intrp_dout_vld | interpolator data valid mask | 1'b0 |
| 12 | R/W | pp_ep_thrshld | ping pong buffer empty threshold hit mask | 1'b0 |
| 11 | R/W | pp3_uf | ping pong set 1 buffer 3 under flow mask | 1'b1 |
| 10 | R/W | pp2_uf | ping pong set 1 buffer 2 under flow mask | 1'b1 |
| 9 | R/W | pp1_uf | ping pong set 1 buffer 1 under flow mask | 1'b1 |
| 8 | R/W | pp0_uf | ping pong set 1 buffer 0 under flow mask | 1'b1 |
| 7 | R/W | pp3_of | ping pong set 1 buffer 3 over flow mask | 1'b1 |
| 6 | R/W | pp2_of | ping pong set 1 buffer 2 over flow mask | 1'b1 |
| 5 | R/W | pp1_of | ping pong set 1 buffer 1 over flow mask | 1'b1 |
| 4 | R/W | pp0_of | ping pong set 1 buffer 0 over flow mask | 1'b1 |
| 3 | R/W | pp3_fl | ping pong set 1 buffer 3 full mask | 1'b1 |
| 2 | R/W | pp2_fl | ping pong set 1 buffer 2 full mask | 1'b1 |
| 1 | R/W | pp1_fl | ping pong set 1 buffer 1 full mask | 1'b1 |
| 0 | R/W | pp0_fl | ping pong set 1 buffer 0 full mask | 1'b1 |

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**Name** I2S_IRQS1
**Address** 0x50165050
**Description** Contains register fields associated with i2s_irqs1.
**Reset** 0x00000000

| Name | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13 | R/W | intrp_dout_vld | interpolator data valid set | 1'b0 |
| 12 | R/W | pp_ep_thrshld | ping pong buffer empty threshold hit set | 1'b0 |
| 11 | R/W | pp3_uf | ping pong set 1 buffer 3 under flow set | 1'b0 |
| 10 | R/W | pp2_uf | ping pong set 1 buffer 2 under flow set | 1'b0 |
| 9 | R/W | pp1_uf | ping pong set 1 buffer 1 under flow set | 1'b0 |
| 8 | R/W | pp0_uf | ping pong set 1 buffer 0 under flow set | 1'b0 |
| 7 | R/W | pp3_of | ping pong set 1 buffer 3 over flow set | 1'b0 |
| 6 | R/W | pp2_of | ping pong set 1 buffer 2 over flow set | 1'b0 |
| 5 | R/W | pp1_of | ping pong set 1 buffer 1 over flow set | 1'b0 |
| 4 | R/W | pp0_of | ping pong set 1 buffer 0 over flow set | 1'b0 |
| 3 | R/W | pp3_fl | ping pong set 1 buffer 3 full set | 1'b0 |
| 2 | R/W | pp2_fl | ping pong set 1 buffer 2 full set | 1'b0 |
| 1 | R/W | pp1_fl | ping pong set 1 buffer 1 full set | 1'b0 |
| 0 | R/W | pp0_fl | ping pong set 1 buffer 0 full set | 1'b0 |

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**Name** I2S_IRQC1
**Address** 0x50165054
**Description** Contains register fields associated with i2s_irqc1.
**Reset** 0x00000000

| Name | RSVD | intrp_dout_vld | pp_ep_thrshld | pp3_uf | pp2_uf | pp1_uf | pp0_uf | pp3_of | pp2_of | pp1_of | pp0_of | pp3_fl | pp2_fl | pp1_fl | pp0_fl |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13 | R/W | intrp_dout_vld | interpolator data valid clear | 1'b0 |
| 12 | R/W | pp_ep_thrshld | ping pong buffer empty threshold hit clear | 1'b0 |
| 11 | R/W | pp3_uf | ping pong set 1 buffer 3 under flow clear | 1'b0 |
| 10 | R/W | pp2_uf | ping pong set 1 buffer 2 under flow clear | 1'b0 |

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|---|---|---|---|---|
| 9 | R/W | pp1_uf | ping pong set 1 buffer 1 under flow clear | 1'b0 |
| 8 | R/W | pp0_uf | ping pong set 1 buffer 0 under flow clear | 1'b0 |
| 7 | R/W | pp3_of | ping pong set 1 buffer 3 over flow clear | 1'b0 |
| 6 | R/W | pp2_of | ping pong set 1 buffer 2 over flow clear | 1'b0 |
| 5 | R/W | pp1_of | ping pong set 1 buffer 1 over flow clear | 1'b0 |
| 4 | R/W | pp0_of | ping pong set 1 buffer 0 over flow clear | 1'b0 |
| 3 | R/W | pp3_fl | ping pong set 1 buffer 3 full clear | 1'b0 |
| 2 | R/W | pp2_fl | ping pong set 1 buffer 2 full clear | 1'b0 |
| 1 | R/W | pp1_fl | ping pong set 1 buffer 1 full clear | 1'b0 |
| 0 | R/W | pp0_fl | ping pong set 1 buffer 0 full clear | 1'b0 |

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**Name** BUFFER_DEPTH
**Address** 0x50165058
**Description** Contains register fields associated with buffer_depth.
**Reset** 0x00000010

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | depth | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:0 | R | depth | one buffer depth | 5'd16 |

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**Name** PP_ST
**Address** 0x5016505c
**Description** Contains register fields associated with pp_st.
**Reset** 0x00001010

| Name | RSVD | | | | | | | | | | | | | | | | | | pp1_ep_cnt | | | | | | RSVD | | pp0_ep_cnt | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13:8 | R | pp1_ep_cnt | number of empty slot in ping poing set 1. | 6'd16 |
| 5:0 | R | pp0_ep_cnt | number of empty slot in ping poing set 0. | 6'd16 |

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**Name** I2S_SPARE
**Address** 0x50165060
**Description** Contains register fields associated with i2s_spare.
**Reset** 0x00000000

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| Name | val | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | val | spare bits | 32'b0 |

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**Name** I2S_DBG
**Address** 0x50165064
**Description** Expose debug signals on mdm_out
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | sel | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5:0 | R/W | sel | 0123456 | 6'd0 |

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**Name** ID
**Address** 0x50165ffc
**Description** Contains register fields associated with id.
**Reset** 0x49325320

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | 'I2S' in ASCII | 32'h49325320 |

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# CLKRSTGEN_SECURE

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**Name** CLK_BP_CTRL
**Address** 0x50166000
**Description** Contains register fields associated with clk_bp_ctrl.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | cluster_ | | slow_clust | | | doubler_clus | | pll_div2_clu | | pll_cluste |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | sel | | | | | ter_sel | ster_sel | r_sel |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9:6 | R/W | cluster_sel | selects which cluster is to drive the backplane clock<br>0 = clk16x<br>1 = slow clocks<br>2 = doubler cluster<br>4 = pll cluster<br>8 = pll div2 cluster<br>always cycle through 0 so that only one bits changes per write | 4'b0 |
| 5:3 | R/W | slow_cluster_sel | selects which clock is output by this cluster<br>0 = 16MHz<br>1 = 8MHz<br>3 = 4MHz<br>2 = 2MHz<br>6 = 1MHz<br>7 = 500kHz<br>notice the gray encoding; only change one bit per write<br>always leave this cluster in 16MHz (0 setting) before leaving it | 3'b0 |
| 2 | R/W | doubler_cluster_sel | selects which clock is output by this cluster<br>0 = 16MHz<br>1 = 32MHz<br>always leave this cluster in 16MHz (0 setting) before leaving it | 1'b0 |
| 1 | R/W | pll_div2_cluster_sel | selects which clock is output by this cluster<br>0 = 16MHz<br>1 = (selected pll frequency)/2<br>always leave this cluster in 16MHz (0 setting) before leaving it | 1'b0 |
| 0 | R/W | pll_cluster_sel | selects which clock is output by this cluster<br>0 = 16MHz<br>1 = selected pll frequency<br>always leave this cluster in 16MHz (0 setting) before leaving it | 1'b0 |

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**Name** CLK_BP_CTRL_STAT
**Address** 0x50166004
**Description** status of clk_bp_ctrl.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | cluster_sel_st | | | | slow_cluster_sel | | | doubler_cluster_se | pll_div2_cluster_s | pll_cluster_sel_ |

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | a t | | _ s t a t | | l _ s t a t | e l _ s t a t | s t a t |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9:6 | R | cluster_sel_stat | status of cluster_sel | 4'b0 |
| 5:3 | R | slow_cluster_sel_stat | status of slow_cluster_sel | 3'b0 |
| 2 | R | doubler_cluster_sel_stat | status of doubler_cluster_sel | 1'b0 |
| 1 | R | pll_div2_cluster_sel_stat | status of pll_div2_cluster_sel | 1'b0 |
| 0 | R | pll_cluster_sel_stat | status of pll_cluster_sel | 1'b0 |

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**Name** CLK_AUD_CTRL
**Address** 0x50166008
**Description** Contains register fields associated with clk_aud_ctrl.
**Reset** 0x00000000

| Name | | | | | | | | | | | | | | | | | | | | | | i 2 s _ c l k _ e n a b l e | i 2 s _ s e l | | | | p d m _ c l k _ e n a b l e | | | R S V D | | p d m _ s e l |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | R S V D | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9 | R/W | i2s_clk_enable | enable i2s clock; make frequency selection (i2s_sel) only when disabled | 1'b0 |
| 8:5 | R/W | i2s_sel | selects the i2s clock source<br>0 = 16MHz<br>1 = 32MHz<br>2 = (pll frequency/1)<br>3 = (pll frequency/2)<br>4 = (pll frequency/3)<br>5 = (pll frequency/4)<br>6 = (pll frequency/5)<br>7 = (pll frequency/6)<br>8 = (pll frequency/7)<br>9 = (pll frequency/8) | 4'b0 |
| 4 | R/W | pdm_clk_enable | enable pdm clock; make frequency selection (pdm_sel) only when disabled | 1'b0 |
| 0 | R/W | pdm_sel | selects the pdm clock source<br>0 = i2s clock divided by 1<br>1 = i2s clock divided by 2 | 1'b0 |

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**Name** PLL_CTRL
**Address** 0x5016600c
**Description** Contains register fields associated with pll_ctrl.

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**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | pll_update | pll_enable | pll_freq_sel | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | pll_update | (no description) | 1'b0 |
| 4 | R/W | pll_enable | (no description) | 1'b0 |
| 3:0 | R/W | pll_freq_sel | 0 = 48 MHz<br>1 = 56 MHz / reserved / don't use<br>2 = 64 MHz<br>3 = 72 MHz / reserved / don't use<br>4 = 80 MHz<br>5 = 88 MHz / reserved / don't use<br>6 = 96 MHz | 4'b0 |

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**Name** USER_RESETS
**Address** 0x50166010
**Description** Contains register fields associated with user_resets.
**Reset** 0x00000000

| Name | use_arm_clkdivs | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | mdm_rif | ble_lowp | ble_intf | ble_core |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | use_arm_clkdivs | (no description) | 1'b0 |
| 3 | R/W | mdm_rif | forces a reset into both mdm and rif cores | 1'b0 |
| 2 | R/W | ble_lowp | (no description) | 1'b0 |
| 1 | R/W | ble_intf | (no description) | 1'b0 |
| 0 | R/W | ble_core | (no description) | 1'b0 |

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**Name** USER_CLK_DISABLES

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**Address** 0x50166014
**Description** Contains register fields associated with user_clk_disables.
**Reset** 0x00000020

| Name | RSVD | cal | hyper | cpu_lpc | i2s | pdm | mdm_rif |
|---|---|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 5 | R/W | cal | radio.calfc_out | 1'b1 |
| 4 | R/W | hyper | (no description) | 1'b0 |
| 3 | R/W | cpu_lpc | (no description) | 1'b0 |
| 2 | R/W | i2s | (no description) | 1'b0 |
| 1 | R/W | pdm | (no description) | 1'b0 |
| 0 | R/W | mdm_rif | turns off 8MHz and 16MHz clocks going to mdm and rif cores | 1'b0 |

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**Name** USER_CLK_GATE_FORCE_ON
**Address** 0x50166018
**Description** Contains register fields associated with user_clk_gate_force_on.
**Reset** 0x00000001

| Name | RSVD | ble52_hgclk |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | ble52_hgclk | The hgclk port of ble52 is normally clock gated by the ble52 core itself. If you want this clock to always run (for instance, for debug) then assert this signal. | 1'b1 |

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**Name** USER_CLK_GATE_CTRL
**Address** 0x5016601c
**Description** Contains register fields associated with user_clk_gate_ctrl.
**Reset** 0x00001655

| Name | RSVD | sticky_cl | ble_hclk | romc_hcl | sha_hclk | dma_hclk | apb2_hcl | apb1_hcl | apb0_hcl |
|---|---|---|---|---|---|---|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | k | | k | | | k | k | k | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 14 | R/W | sticky_clk | 0=forced off; 1=forced on; clock enable signal to sticky bits. | 1'b0 |
| 13:12 | R/W | ble_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b1 |
| 11:10 | R/W | romc_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b1 |
| 9:8 | R/W | sha_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b10 |
| 7:6 | R/W | dma_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b1 |
| 5:4 | R/W | apb2_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b1 |
| 3:2 | R/W | apb1_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b1 |
| 1:0 | R/W | apb0_hclk | 0,3=normal; 1=forced on; 2=forced off | 2'b1 |

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**Name** CONFIGURATION
**Address** 0x50166020
**Description** Contains register fields associated with configuration.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | pll_ready | index | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 8 | R | pll_ready | indicates when PLL output is no longer being gated and PLL clock is available | 1'b0 |
| 7:0 | R | index | which fpga configuration? | 8'b0 |

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**Name** CLKSYNC
**Address** 0x50166024
**Description** Contains register fields associated with clksync.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | clk16_src | div_val | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4 | R/W | clk16_src | selects the source of the 16 MHz clock domain<br>0 = crystal<br>1 = divided down pll | 1'b0 |
| 3:0 | R/W | div_val | how much to divide the pll to get 16 MHz? | 4'b0 |

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**Name** CORE_ID
**Address** 0x50166ffc
**Description** core id
**Reset** 0x434c4b52

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | CLKR in ASCII | 32'h434c4b52 |

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# AES_SECURE

**Name** ALERT_TEST
**Address** 0x50167000
**Description** control signals for all pwm subcores
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | recov_ctrl_update_err | fatal_fault |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | recov_ctrl_update_err | Write 1 to trigger one alert event of this kind | 1'b0 |
| 0 | R/W | fatal_fault | Write 1 to trigger one alert event of this kind | 1'b0 |

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**Name** KEY_SHARE0_0
**Address** 0x50167004
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with

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pseudo-random data.

**Reset** 0x00000000

| Name | key_share0_0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_0 | initial Key Share 0 | 32'b0 |

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**Name** KEY_SHARE0_1

**Address** 0x50167008

**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

| Name | key_share0_1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_1 | For KEY_SHARE01 | 32'b0 |

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**Name** KEY_SHARE0_2

**Address** 0x5016700c

**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

| | key |
|---|---|

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| Name | key_share0_2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_2 | For KEY_SHARE02 | 32'b0 |

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**Name** KEY_SHARE0_3
**Address** 0x50167010
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share0_3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_3 | For KEY_SHARE03 | 32'b0 |

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**Name** KEY_SHARE0_4
**Address** 0x50167014
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_sha | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| | key_share0_4 |
|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_4 | For KEY_SHARE04 | 32'b0 |

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**Name** KEY_SHARE0_5
**Address** 0x50167018
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share0_5 |
|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_5 | For KEY_SHARE05 | 32'b0 |

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**Name** KEY_SHARE0_6
**Address** 0x5016701c
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share0_6 |
|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_6 | For KEY_SHARE06 | 32'b0 |

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**Name** KEY_SHARE0_7
**Address** 0x50167020
**Description** Initial Key Registers Share 0. The actual initial key corresponds to Initial Key Registers Share 0 XORed with Initial Key Registers Share 1. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share0_7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share0_7 | For KEY_SHARE07 | 32'b0 |

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**Name** KEY_SHARE1_0
**Address** 0x50167024
**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share1_0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_0 | initial Key Share 1 | 32'b0 |

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**Name** KEY_SHARE1_1
**Address** 0x50167028
**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share1_1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_1 | For KEY_SHARE11 | 32'b0 |

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**Name** KEY_SHARE1_2
**Address** 0x5016702c
**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | key_share1_2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_2 | For KEY_SHARE12 | 32'b0 |

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**Name** KEY_SHARE1_3
**Address** 0x50167030

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**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

| Name | key_share1_3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_3 | For KEY_SHARE13 | 32'b0 |

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**Name** KEY_SHARE1_4

**Address** 0x50167034

**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

| Name | key_share1_4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_4 | For KEY_SHARE14 | 32'b0 |

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**Name** KEY_SHARE1_5

**Address** 0x50167038

**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with

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pseudo-random data.

**Reset** 0x00000000

| Name | key_share1_5 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_5 | For KEY_SHARE15 | 32'b0 |

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**Name** KEY_SHARE1_6

**Address** 0x5016703c

**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

| Name | key_share1_6 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_6 | For KEY_SHARE16 | 32'b0 |

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**Name** KEY_SHARE1_7

**Address** 0x50167040

**Description** Initial Key Registers Share 1. The actual initial key corresponds to Initial Key Registers Share 1 XORed with Initial Key Registers Share 0. Loaded into the internal Full Key register upon starting encryption/decryption of the next block. All key registers (Share 0 and Share 1) must be written at least once when the key is changed, regardless of key length (write random data for unused bits). The order in which the registers are updated does not matter. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to these registers are ignored. Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

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| Name | key_share1_7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | key_share1_7 | For KEY_SHARE17 | 32'b0 |

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**Name** IV_0
**Address** 0x50167044
**Description** Initialization Vector Registers. The initialization vector (IV) or initial counter value must be written to these registers when starting a new message in CBC or CTR mode (see Control Register), respectively. In CBC and CTR modes, the AES unit does not start encryption/decryption with a partially updated IV. Each register has to be written at least once. The order in which the registers are written does not matter. If the AES unit is non-idle, writes to these registers are ignored. Whenever starting a new message, the corresponding IV value must be provided by the processor. Once started, the AES unit automatically updates the contents of these registers. In ECB mode, the IV registers are not used and do not need to be configured. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | iv_0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | iv_0 | Initialization Vector | 32'b0 |

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**Name** IV_1
**Address** 0x50167048
**Description** Initialization Vector Registers. The initialization vector (IV) or initial counter value must be written to these registers when starting a new message in CBC or CTR mode (see Control Register), respectively. In CBC and CTR modes, the AES unit does not start encryption/decryption with a partially updated IV. Each register has to be written at least once. The order in which the registers are written does not matter. If the AES unit is non-idle, writes to these registers are ignored. Whenever starting a new message, the corresponding IV value must be provided by the processor. Once started, the AES unit automatically updates the contents of these registers. In ECB mode, the IV registers are not used and do not need to be configured. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | iv_1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | iv_1 | For IV1 | 32'b0 |

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**Name** IV_2
**Address** 0x5016704c
**Description** Initialization Vector Registers. The initialization vector (IV) or initial counter value must be written to these registers when starting a new message in CBC or CTR mode (see Control Register), respectively. In CBC and CTR modes, the AES unit does not start encryption/decryption with a partially updated IV. Each register has to be written at least once. The order in which the registers are written does not matter. If the AES unit is non-idle, writes to these registers are ignored. Whenever starting a new message, the corresponding IV value must be provided by the processor. Once started, the AES unit automatically updates the contents of these registers. In ECB mode, the IV registers are not used and do not need to be configured. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | iv_2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | iv_2 | For IV2 | 32'b0 |

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**Name** IV_3
**Address** 0x50167050
**Description** Initialization Vector Registers. The initialization vector (IV) or initial counter value must be written to these registers when starting a new message in CBC or CTR mode (see Control Register), respectively. In CBC and CTR modes, the AES unit does not start encryption/decryption with a partially updated IV. Each register has to be written at least once. The order in which the registers are written does not matter. If the AES unit is non-idle, writes to these registers are ignored. Whenever starting a new message, the corresponding IV value must be provided by the processor. Once started, the AES unit automatically updates the contents of these registers. In ECB mode, the IV registers are not used and do not need to be configured. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | iv_3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | iv_3 | For IV3 | 32'b0 |

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**Name** DATA_IN_0
**Address** 0x50167054
**Description** Input Data Registers. If MANUAL_OPERATION=0 (see Control Register), the AES unit automatically starts encryption/decryption after all Input Data registers have been written. Each register has to be written at least once. The order in which the registers are written does not matter. Loaded into the internal State register upon starting encryption/decryption of the next block. After that, the processor can update the Input Data registers (See INPUT_READY field of Status Register). Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | data_in_0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data_in_0 | Input Data | 32'b0 |

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**Name** DATA_IN_1
**Address** 0x50167058
**Description** Input Data Registers. If MANUAL_OPERATION=0 (see Control Register), the AES unit automatically starts encryption/decryption after all Input Data registers have been written. Each register has to be written at least once. The order in which the registers are written does not matter. Loaded into the internal State register upon starting encryption/decryption of the next block. After that, the processor can update the Input Data registers (See INPUT_READY field of Status Register). Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | data_in_1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data_in_1 | For DATA_IN1 | 32'b0 |

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**Name** DATA_IN_2
**Address** 0x5016705c
**Description** Input Data Registers. If MANUAL_OPERATION=0 (see Control Register), the AES unit automatically starts encryption/decryption after all Input Data registers have been written. Each register has to be written at least once. The order in which the registers are written does not matter. Loaded into the internal State register upon starting encryption/decryption of the next block. After that, the processor can update the Input Data registers (See INPUT_READY field of Status Register). Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | data_in_2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data_in_2 | For DATA_IN2 | 32'b0 |

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**Name** DATA_IN_3
**Address** 0x50167060
**Description** Input Data Registers. If MANUAL_OPERATION=0 (see Control Register), the AES unit automatically starts encryption/decryption after all Input Data registers have been written. Each register has to be written at least once. The order

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in which the registers are written does not matter. Loaded into the internal State register upon starting encryption/decryption of the next block. After that, the processor can update the Input Data registers (See INPUT_READY field of Status Register). Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

| Name | data_in_3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | data_in_3 | For DATA_IN3 | 32'b0 |

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**Name** DATA_OUT_0

**Address** 0x50167064

**Description** Output Data Register. Holds the output data produced by the AES unit during the last encryption/decryption operation. If MANUAL_OPERATION=0 (see Control Register), the AES unit is stalled when the previous output data has not yet been read and is about to be overwritten. Each register has to be read at least once. The order in which the registers are read does not matter. Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

| Name | data_out_0 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data_out_0 | Output Data | 32'h0 |

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**Name** DATA_OUT_1

**Address** 0x50167068

**Description** Output Data Register. Holds the output data produced by the AES unit during the last encryption/decryption operation. If MANUAL_OPERATION=0 (see Control Register), the AES unit is stalled when the previous output data has not yet been read and is about to be overwritten. Each register has to be read at least once. The order in which the registers are read does not matter. Upon reset, these registers are cleared with pseudo-random data.

**Reset** 0x00000000

| Name | data_out_ |
|---|---|

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| | 1 |
|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data_out_1 | For DATA_OUT1 | 32'h0 |

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**Name** DATA_OUT_2
**Address** 0x5016706c
**Description** Output Data Register. Holds the output data produced by the AES unit during the last encryption/decryption operation. If MANUAL_OPERATION=0 (see Control Register), the AES unit is stalled when the previous output data has not yet been read and is about to be overwritten. Each register has to be read at least once. The order in which the registers are read does not matter. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | data_out_2 |
|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data_out_2 | For DATA_OUT2 | 32'h0 |

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**Name** DATA_OUT_3
**Address** 0x50167070
**Description** Output Data Register. Holds the output data produced by the AES unit during the last encryption/decryption operation. If MANUAL_OPERATION=0 (see Control Register), the AES unit is stalled when the previous output data has not yet been read and is about to be overwritten. Each register has to be read at least once. The order in which the registers are read does not matter. Upon reset, these registers are cleared with pseudo-random data.
**Reset** 0x00000000

| Name | data_out_3 |
|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | data_out_3 | For DATA_OUT3 | 32'h0 |

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**Name** CTRL_SHADOWED
**Address** 0x50167074
**Description** Control Register. Can only be updated when the AES unit is idle. If the AES unit is non-idle, writes to this register are

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ignored. This register is shadowed, meaning two subsequent write operations are required to change its content. If the two write operations try to set a different value, a recoverable alert is triggered (See Status Register). A read operation clears the internal phase tracking: The next write operation is always considered a first write operation of an update sequence. Any write operation to this register will clear the status tracking required for automatic mode (See MANUAL_OPERATION field). A write to the Control Register is considered the start of a new message. Hence, software needs to provide new key, IV and input data afterwards.

**Reset** 0x000000c0

| Name | RSVD | | | | | | | | | | | | | | | | | | | | FORCE_ZERO_MASKS | MANUAL_OPERATION | SIDELOAD | KEY_LEN | | | MODE | | | | | OPERATION |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 12 | R/W | FORCE_ZERO_MASKS | Use masks generated by internal masking PRNG (0) or force all masks constantly to zero (1). Setting all masks to constant zero can be useful when performing SCA. To completely disable the masking, the second key share (KEY_SHARE1_0 - KEY_SHARE1_7) must be zero as well. Only applicable if both the Masking parameter and the SecAllowForcingMasks parameter are set to one. | 1'b0 |
| 11 | R/W | MANUAL_OPERATION | Controls whether the AES unit is operated in normal/automatic mode (0) or fully manual mode (1). In automatic mode (0), the AES unit automatically i) starts to encrypt/decrypt when it receives new input data, and ii) stalls during the last encryption/decryption cycle if the previous output data has not yet been read. This is the most efficient mode to operate in. Note that the corresponding status tracking is automatically cleared upon a write to the Control Register. In manual mode (1), the AES unit i) only starts to encrypt/decrypt after receiving a start trigger (see Trigger Register), and ii) overwrites previous output data irrespective of whether it has been read out or not. This mode is useful if software needs full control over the AES unit. | 1'b0 |
| 10 | R/W | SIDELOAD | Controls whether the AES unit uses the key provided by the key manager via key sideload interface (1) or<br>the key provided by software via Initial Key Registers KEY_SHARE1_0 - KEY_SHARE1_7 (0). | 1'b0 |
| 9:7 | R/W | KEY_LEN | 3-bit one-hot field to select AES key length.<br>Invalid input values, i.e., values with multiple bits set, value 3'b000, and value 3'b010 in case 192-bit keys are not supported (because disabled at compile time) are mapped to AES_256 (3'b100).<br>1 AES_128 3'b001 128-bit key length.<br>2 AES_192 3'b010 192-bit key length.<br>In case support for 192-bit keys has been disabled at compile time, setting this value results in configuring AES_256 (3'b100).<br>4 AES_256 3'b100 256-bit key length.<br>Invalid input values, i.e., values with multiple bits set, value 3'b000, and value 3'b010 in case 192-bit keys are not supported (because disabled at compile time) are mapped to AES_256. | 3'd1 |
| 6:1 | R/W | MODE | 6-bit one-hot field to select AES block cipher mode. Invalid input values, i.e., values with multiple bits set and value 6'b00_0000, are mapped to AES_NONE (6'b10_0000).<br>1 AES_ECB 6'b00_0001: Electronic Codebook (ECB) mode.<br>2 AES_CBC 6'b00_0010: Cipher Block Chaining (CBC) mode.<br>4 AES_CFB 6'b00_0100: Cipher Feedback (CFB) mode. | 6'h20 |

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| | | | | |
|---|---|---|---|---|
| | | | 8 AES_OFB 6'b00_1000: Output Feedback (OFB) mode.<br>16 AES_CTR 6'b01_0000: Counter (CTR) mode.<br>32 AES_NONE 6'b10_0000: Invalid input values, i.e., value with multiple bits set and value 6'b00_0000, are mapped to AES_NONE.<br>Other values are reserved. | |
| 0 | R/W | OPERATION | Select encryption(0) or decryption(1) operation of AES unit. | 1'b0 |

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**Name** TRIGGER
**Address** 0x50167078
**Description** Trigger Register. Each bit is individually cleared to zero when executing the corresponding trigger. While executing any of the triggered operations, the AES unit will set the IDLE bit in the Status Register to zero. The processor must check the Status Register before triggering further actions. For example, writes to Initial Key and IV Registers are ignored while the AES unit is busy. Writes to the Input Data Registers are not ignored but the data will be cleared if a KEY_IV_DATA_IN_CLEAR operation is pending.
**Reset** 0x0000000e

| Name | RSVD | PRNG_RESEED | DATA_OUT_CLEAR | KEY_IV_DATA_IN_CLEAR | START |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R/W | PRNG_RESEED | Keep continuing with the current internal state of the internal pseudo-random number generator<br>used for register clearing (0) or perform a reseed of the internal state from the connected entropy source (1). | 1'b1 |
| 2 | R/W | DATA_OUT_CLEAR | Keep current values in Output Data registers (0) or clear those registers with pseudo-random data (1). | 1'b1 |
| 1 | R/W | KEY_IV_DATA_IN_CLEAR | Keep current values in Initial Key, internal Full Key and Decryption Key registers,<br>IV registers and Input Data registers (0) or clear all those registers with pseudo-random data (1). | 1'b1 |
| 0 | R/W | START | Keep AES unit paused (0) or trigger the encryption/decryption of one data block (1).<br>This trigger is ignored if MANUAL_OPERATION=0 (see Control Register). | 1'b0 |

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**Name** STATUS
**Address** 0x5016707c
**Description** Status Register
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | A | |

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| Name | RSVD | ALERT_FATAL_FAULT | ALERT_RECOV_CTRL_UPDATE_ERR | INPUT_READY | OUTPUT_VALID | OUTPUT_LOST | STALL | IDLE |
|---|---|---|---|---|---|---|---|---|
| **Bits** | **31:7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 6 | R | ALERT_FATAL_FAULT | No fatal fault has occurred inside the AES unit (0).<br>A fatal fault has occurred and the AES unit needs to be reset (1).<br>Examples for fatal faults include i) storage errors in the Control Register,<br>ii) if any internal FSM enters an invalid state,<br>iii) if any sparsely encoded signal takes on an invalid value,<br>iv) errors in the internal round counter,<br>v) escalations triggered by the life cycle controller, and<br>vi) fatal integrity failures on the TL-UL bus. | 1'h0 |
| 5 | R | ALERT_RECOV_CTRL_UPDATE_ERR | An update error has not occurred (0) or has occurred (1) in the shadowed Control Register.<br>AES operation needs to be restarted by re-writing the Control Register. | 1'b0 |
| 4 | R | INPUT_READY | The AES unit is ready (1) or not ready (0) to receive new data input via the DATA_IN registers.<br>If the present values in the DATA_IN registers have not yet been loaded into the module this flag is 0 (not ready). | 1'b0 |
| 3 | R | OUTPUT_VALID | The AES unit has no valid output (0) or has valid output data (1). | 1'b0 |
| 2 | R | OUTPUT_LOST | All previous output data has been fully read by the processor (0)<br>or at least one previous output data block has been lost (1).<br>It has been overwritten by the AES unit before the processor could fully read it.<br>Once set to 1, this flag remains set until AES operation is restarted by re-writing the Control Register.<br>The primary use of this flag is for design verification.<br>This flag is not meaningful if MANUAL_OPERATION=0 (see Control Register). | 1'b0 |
| 1 | R | STALL | The AES unit is not stalled (0) or stalled (1)<br>because there is previous output data that must be read by the processor before the AES unit can overwrite this data.<br>This flag is not meaningful if MANUAL_OPERATION=1 (see Control Register). | 1'b0 |
| 0 | R | IDLE | The AES unit is idle (1) or busy (0).<br>This flag is 0 if one of the following operations is currently running:<br>i) encryption/decryption, ii) register clearing or iii) PRNG reseeding.<br>This flag is also 0 if an encryption/decryption is running but the AES unit is stalled. | 1'b0 |

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**Name** STATUS1
**Address** 0x50167ff4
**Description** Status Register
**Reset** 0x00000000

| Name | key_len_err | RSVD | sideload_wptr |
|---|---|---|---|
| **Bits** | **31** | **30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R | key_len_err | indicates meta key size error. Asserted by HW when sideload_val is written 1 while the meta key size does not match operation key size. cleared by core reset. | 1'b0 |
| 2:0 | R | sideload_wptr | The current value of the sideload key write pointer. This pointer shows the next slot the new word will be stored. | 3'b0 |

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**Name** SIDELOAD_CTRL
**Address** 0x50167ff8
**Description** Contains register fields associated with sideload_ctrl.
**Reset** 0x00000002

| Name | RSVD | sideload_update_en | sideload_val |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R/W | sideload_update_en | write 1 to enable sideload buffer push. | 1'b1 |
| 0 | R/W | sideload_val | write 1 to assert that sideload key is valid. Write 0 to invalidate.<br>Write 1 will also clear the sideload key write pointer.<br>The sideload key is composed of 8 words, which are stored in a fifo like buffer.<br>A write pointer points to the next slot in the buffer in which a new word will be stored. | 1'b0 |

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**Name** ID
**Address** 0x50167ffc
**Description** Contains register fields associated with id.
**Reset** 0x00414553

| Name | id | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | AES in ASCII | 32'h414553 |

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# AT_DMA_SECURE

**Name** OPMODE
**Address** 0x50170000
**Description** Contains register fields associated with opmode.
**Reset** 0x00000000

| Name | go | stop | RSVD | const_tar_addr | dat_inv | const_trans |
|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** **28** **27** **26** **25** **24** **23** **22** **21** **20** **19** **18** **17** **16** **15** **14** **13** **12** **11** **10** **9** **8** **7** **6** **5** **4** **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | go | op code start when rising edge on 'go' is detected. HW clears after. | 1'b0 |
| 30 | R/W | stop | to stop pending transaction . Not self clearing. | 1'b0 |
| 2 | R/W | const_tar_addr | 1= write address remains constant at tar_addr, which is required to be word-aligned in this mode.<br>0= write address increments as defined by the size parameter. | 1'b0 |
| 1 | R/W | dat_inv | 0= non-inverted write ; 1= inverted write (write data are bit-wise inverted) | 1'b0 |
| 0 | R/W | const_trans | op code start when rising edge on 'go' is detected. | 1'b0 |

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**Name** CONST_WDATA
**Address** 0x50170004
**Description** Contains register fields associated with const_wdata.
**Reset** 0x00000000

| | const |
|---|---|

CONFIDENTIAL

| Name | const_wdata | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | const_wdata | constant value to be written to target when the operation is in const_trans mode. | 32'b0 |

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**Name** SRC_ADDR
**Address** 0x50170008
**Description** Contains register fields associated with src_addr.
**Reset** 0x00000000

| Name | src_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | src_addr | source address | 32'b0 |

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**Name** TAR_ADDR
**Address** 0x5017000c
**Description** Contains register fields associated with tar_addr.
**Reset** 0x00000000

| Name | tar_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | tar_addr | target address | 32'b0 |

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**Name** SIZE
**Address** 0x50170010
**Description** Contains register fields associated with size.
**Reset** 0x00000000

| | RS | si |
|---|---|---|

CONFIDENTIAL

| Name | V D | z e |
|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R/W | size | payload size in bytes. | 24'b0 |

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**Name** SRC_CTRL
**Address** 0x50170014
**Description** Contains register fields associated with src_ctrl.
**Reset** 0x00000004

| Name | RSVD | src_bus_size | src_type |
|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | src_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | src_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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**Name** TAR_CTRL
**Address** 0x50170018
**Description** Contains register fields associated with tar_ctrl.
**Reset** 0x00000004

| Name | RSVD | tar_bus_size | tar_type |
|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | tar_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | tar_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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**Name** FIFO_DPTH_ADDR
**Address** 0x5017001c

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**Description** Contains register fields associated with fifo_dpth_addr.
**Reset** 0x00000000

| Name | addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | addr | address of the external fifo depth.<br>Statemachine reads this to determine how many push/pull accesses to do per interrupt. | 32'b0 |

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**Name** FIFO_PORT_SEL
**Address** 0x50170020
**Description** Contains register fields associated with fifo_port_sel.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | tar_port_sel | | | RSVD | | | | | | | | | | | | src_port_sel | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 18:16 | R/W | tar_port_sel | selecting the target fifo port;<br>5 = pwd fifo<br>4 = i2s fifo<br>3 = rsvd<br>2 = rsvd<br>1 = uart1 tx<br>0 = uart0 tx | 3'b0 |
| 2:0 | R/W | src_port_sel | selecting the source fifo port;<br>5 = rsvd<br>4 = i2s fifo<br>3 = pdm1 fifo<br>2 = pdm0 fifo<br>1 = uart1 rx<br>0 = uart0 rx | 3'd0 |

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**Name** SPI_PORT_SEL
**Address** 0x50170024
**Description** Contains register fields associated with spi_port_sel.
**Reset** 0x00000000

| Name | RSVD | spi_s |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | el |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | spi_sel | 1=select spi1 for this channel, 0=select spi0 for this channel. | 1'b0 |

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**Name** ERR_STAT
**Address** 0x50170028
**Description** The status is cleared when the err interrupt is cleared
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | addr_ovfl | bus_err | addr_ovlp | size_0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | addr_ovfl | 0= no error, 1= error due to size + src_addr/tar_addr > 32'hffff_ffff. | 1'b0 |
| 2 | R | bus_err | 0= no bus error, 1= a bus error occur during the transfer or at the launch of 'go'. | 1'b0 |
| 1 | R | addr_ovlp | 0= no overlap between src and tar addr range, 1= there is an overlap | 1'b0 |
| 0 | R | size_0 | 0= size is not 0 when go is launched, 1= size is 0 when go is launched. | 1'b0 |

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**Name** STATUS
**Address** 0x5017002c
**Description** Contains register fields associated with status.
**Reset** 0x00000000

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| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | busy | running |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | busy | fsm is in some non-idle state including waiting for the bus or an interrupt. | 1'b0 |
| 0 | R | running | fsm is in an active state, not waiting for the bus or an interrupt. | 1'b0 |

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**Name** TOTAL_WRITE_REMAINDER
**Address** 0x50170030
**Description** Contains register fields associated with total_write_remainder.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | t |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Name | RSVD | total_write_remainder |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R | total_write_remainder | number of bytes left to write at the time of stop | 24'b0 |

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**Name** INTERRUPT_STATUS
**Address** 0x50170034
**Description** Contains register fields associated with interrupt_status.
**Reset** 0x00000000

| Name | RSVD | dma_err | dma_done |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | dma_err | source of interrupt due to an error (chan0). | 1'b0 |
| 0 | R | dma_done | source of interrupt due to completion (chan0). | 1'b0 |

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**Name** INTERRUPT_MASK
**Address** 0x50170038
**Description** Contains register fields associated with interrupt_mask.
**Reset** 0x00000001

| Name | RSVD | intrpt_mask |
|---|---|---|

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| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_mask | if nth bit set, the nth interrupt is enabled | 2'b01 |

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**Name** SET_INTERRUPT
**Address** 0x5017003c
**Description** Contains register fields associated with set_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_set |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_set | if nth bit set, the nth interrupt is set; HW clears all bits the next cycle. | 2'b00 |

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**Name** RESET_INTERRUPT
**Address** 0x50170040
**Description** Contains register fields associated with reset_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_reset |
|---|---|---|
| Bits | 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 | 1 0 |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_reset | if nth bit set, the nth interrupt is reset; HW clears all bits the next cycle. | 2'b0 |

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**Name** CFG_HNONSEC
**Address** 0x50170044
**Description** Contains register fields associated with cfg_hnonsec.
**Reset** 0x00000001

| | | cfg |
|---|---|---|

CONFIDENTIAL

| Name | RSVD | | |

| | a | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | const_wdata | constant value to be written to target when the operation is in const_trans mode. | 32'b0 |

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**Name** CHAN1_SRC_ADDR
**Address** 0x50170050
**Description** Contains register fields associated with chan1_src_addr.
**Reset** 0x00000000

| Name | src_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | src_addr | source address | 32'b0 |

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**Name** CHAN1_TAR_ADDR
**Address** 0x50170054
**Description** Contains register fields associated with chan1_tar_addr.
**Reset** 0x00000000

| Name | tar_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | tar_addr | target address | 32'b0 |

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**Name** CHAN1_SIZE
**Address** 0x50170058
**Description** Contains register fields associated with chan1_size.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | size | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R/W | size | payload size in bytes. | 24'b0 |

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**Name** CHAN1_SRC_CTRL
**Address** 0x5017005c
**Description** Contains register fields associated with chan1_src_ctrl.
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | src_bus_size | | src_type | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | src_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | src_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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**Name** CHAN1_TAR_CTRL
**Address** 0x50170060
**Description** Contains register fields associated with chan1_tar_ctrl.
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | tar_bus_size | | tar_type | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | tar_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | tar_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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**Name** CHAN1_FIFO_DPTH_ADDR
**Address** 0x50170064
**Description** Contains register fields associated with chan1_fifo_dpth_addr.
**Reset** 0x00000000

| | a |
|---|---|

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| Name | addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | addr | address of the external fifo depth.<br>Statemachine reads this to determine how many push/pull accesses to do per interrupt. | 32'b0 |

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**Name** CHAN1_FIFO_PORT_SEL
**Address** 0x50170068
**Description** Contains register fields associated with chan1_fifo_port_sel.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | tar_port_sel | | RSVD | | | | | | | | | | | | src_port_sel | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 18:16 | R/W | tar_port_sel | selecting the target fifo port;<br>5 = pwd fifo<br>4 = i2s fifo<br>3 = rsvd<br>2 = rsvd<br>1 = uart1 tx<br>0 = uart0 tx | 3'b0 |
| 2:0 | R/W | src_port_sel | selecting the source fifo port;<br>5 = rsvd<br>4 = i2s fifo<br>3 = pdm1 fifo<br>2 = pdm0 fifo<br>1 = uart1 rx<br>0 = uart0 rx | 3'd0 |

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**Name** CHAN1_SPI_PORT_SEL
**Address** 0x5017006c
**Description** Contains register fields associated with chan1_spi_port_sel.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | spi_sel |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | spi_sel | 1=select spi1 for this channel, 0=select spi0 for this channel. | 1'b0 |

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**Name** CHAN1_ERR_STAT
**Address** 0x50170070
**Description** The status is cleared when the err interrupt is cleared
**Reset** 0x00000000

| Name | RSVD | addr_ovfl | bus_err | addr_ovlp | size_0 |
|---|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | addr_ovfl | 0= no error, 1= error due to size + src_addr/tar_addr > 32'hffff_ffff. | 1'b0 |
| 2 | R | bus_err | 0= no bus error, 1= a bus error occur during the transfer or at the launch of 'go'. | 1'b0 |
| 1 | R | addr_ovlp | 0= no overlap between src and tar addr range, 1= there is an overlap | 1'b0 |
| 0 | R | size_0 | 0= size is not 0 when go is launched, 1= size is 0 when go is launched. | 1'b0 |

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**Name** CHAN1_STATUS
**Address** 0x50170074
**Description** Contains register fields associated with chan1_status.
**Reset** 0x00000000

| Name | RSVD | busy | running |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | busy | fsm is in some non-idle state including waiting for the bus or an interrupt. | 1'b0 |
| 0 | R | running | fsm is in an active state, not waiting for the bus or an interrupt. | 1'b0 |

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**Name** CHAN1_TOTAL_WRITE_REMAINDER
**Address** 0x50170078
**Description** Contains register fields associated with chan1_total_write_remainder.
**Reset** 0x00000000

| | | total_ |
|---|---|---|

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| Name | RSVD | total_write_remainder |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24** | **23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R | total_write_remainder | number of bytes left to write at the time of stop | 24'b0 |

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**Name** CHAN1_INTERRUPT_STATUS
**Address** 0x5017007c
**Description** Contains register fields associated with chan1_interrupt_status.
**Reset** 0x00000000

| Name | RSVD | dma_err | dma_done |
|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | dma_err | source of interrupt due to an error (chan0). | 1'b0 |
| 0 | R | dma_done | source of interrupt due to completion (chan0). | 1'b0 |

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**Name** CHAN1_INTERRUPT_MASK
**Address** 0x50170080
**Description** Contains register fields associated with chan1_interrupt_mask.
**Reset** 0x00000001

| Name | RSVD | intrpt_mask |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_mask | if nth bit set, the nth interrupt is enabled | 2'b01 |

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**Name** CHAN1_SET_INTERRUPT
**Address** 0x50170084
**Description** Contains register fields associated with chan1_set_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_set |
|---|---|---|
| **Bits** | **31:2** | **1:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_set | if nth bit set, the nth interrupt is set; HW clears all bits the next cycle. | 2'b00 |

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**Name** CHAN1_RESET_INTERRUPT
**Address** 0x50170088
**Description** Contains register fields associated with chan1_reset_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_reset |
|---|---|---|
| **Bits** | **31:2** | **1:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_reset | if nth bit set, the nth interrupt is reset; HW clears all bits the next cycle. | 2'b0 |

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**Name** CHAN1_CFG_HNONSEC
**Address** 0x5017008c
**Description** Contains register fields associated with chan1_cfg_hnonsec.
**Reset** 0x00000001

| Name | RSVD | cfg_hnon |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | sec |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | cfg_hnonsec | control HNONSEC output signal of this channel. This register is only Write-able by Secure program. | 1'b1 |

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**Name** CHAN2_OPMODE
**Address** 0x50170090
**Description** Contains register fields associated with chan2_opmode.
**Reset** 0x00000000

| Name | go | stop | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | const_tar_addr | dat_inv | const_trans |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | go | op code start when rising edge on 'go' is detected. HW clears after. | 1'b0 |
| 30 | R/W | stop | to stop pending transaction . Not self clearing. | 1'b0 |
| 2 | R/W | const_tar_addr | 1= write address remains constant at tar_addr, which is required to be word-aligned in this mode.<br>0= write address increments as defined by the size parameter. | 1'b0 |
| 1 | R/W | dat_inv | 0= non-inverted write ; 1= inverted write (write data are bit-wise inverted) | 1'b0 |
| 0 | R/W | const_trans | op code start when rising edge on 'go' is detected. | 1'b0 |

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**Name** CHAN2_CONST_WDATA
**Address** 0x50170094
**Description** Contains register fields associated with chan2_const_wdata.
**Reset** 0x00000000

| Name | const_wdata | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|

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| | | | | |
|---|---|---|---|---|
| 31:0 | R/W | const_wdata | constant value to be written to target when the operation is in const_trans mode. | 32'b0 |

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**Name** CHAN2_SRC_ADDR
**Address** 0x50170098
**Description** Contains register fields associated with chan2_src_addr.
**Reset** 0x00000000

| Name | src_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | src_addr | source address | 32'b0 |

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**Name** CHAN2_TAR_ADDR
**Address** 0x5017009c
**Description** Contains register fields associated with chan2_tar_addr.
**Reset** 0x00000000

| Name | tar_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | tar_addr | target address | 32'b0 |

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**Name** CHAN2_SIZE
**Address** 0x501700a0
**Description** Contains register fields associated with chan2_size.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | size | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R/W | size | payload size in bytes. | 24'b0 |

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**Name** CHAN2_SRC_CTRL
**Address** 0x501700a4
**Description** Contains register fields associated with chan2_src_ctrl.
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | src_bus_size | | src_type | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | src_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | src_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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**Name** CHAN2_TAR_CTRL
**Address** 0x501700a8
**Description** Contains register fields associated with chan2_tar_ctrl.
**Reset** 0x00000004

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | tar_bus_size | | tar_type | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | tar_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | tar_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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**Name** CHAN2_FIFO_DPTH_ADDR
**Address** 0x501700ac
**Description** Contains register fields associated with chan2_fifo_dpth_addr.
**Reset** 0x00000000

| Name | addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | addr | address of the external fifo depth.<br>Statemachine reads this to determine how many push/pull accesses to do per interrupt. | 32'b0 |

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**Name** CHAN2_FIFO_PORT_SEL
**Address** 0x501700b0
**Description** Contains register fields associated with chan2_fifo_port_sel.
**Reset** 0x00000000

| Name | RSVD | tar_port_sel | RSVD | src_port_sel |
|---|---|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19** | **18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3** | **2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 18:16 | R/W | tar_port_sel | selecting the target fifo port;<br>5 = pwd fifo<br>4 = i2s fifo<br>3 = rsvd<br>2 = rsvd<br>1 = uart1 tx<br>0 = uart0 tx | 3'b0 |
| 2:0 | R/W | src_port_sel | selecting the source fifo port;<br>5 = rsvd<br>4 = i2s fifo<br>3 = pdm1 fifo<br>2 = pdm0 fifo<br>1 = uart1 rx<br>0 = uart0 rx | 3'd0 |

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**Name** CHAN2_SPI_PORT_SEL
**Address** 0x501700b4
**Description** Contains register fields associated with chan2_spi_port_sel.
**Reset** 0x00000000

| Name | RSVD | spi_sel |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16** | **15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | spi_sel | 1=select spi1 for this channel, 0=select spi0 for this channel. | 1'b0 |

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**Name** CHAN2_ERR_STAT
**Address** 0x501700b8
**Description** The status is cleared when the err interrupt is cleared
**Reset** 0x00000000

| Name | RSVD | addr_ovfl | bus_err | addr_ovlp | size_0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | addr_ovfl | 0= no error, 1= error due to size + src_addr/tar_addr > 32'hffff_ffff. | 1'b0 |
| 2 | R | bus_err | 0= no bus error, 1= a bus error occur during the transfer or at the launch of 'go'. | 1'b0 |
| 1 | R | addr_ovlp | 0= no overlap between src and tar addr range, 1= there is an overlap | 1'b0 |
| 0 | R | size_0 | 0= size is not 0 when go is launched, 1= size is 0 when go is launched. | 1'b0 |

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**Name** CHAN2_STATUS
**Address** 0x501700bc
**Description** Contains register fields associated with chan2_status.
**Reset** 0x00000000

| Name | RSVD | busy | running |
|---|---|---|---|
| **Bits** | **31:2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | busy | fsm is in some non-idle state including waiting for the bus or an interrupt. | 1'b0 |
| 0 | R | running | fsm is in an active state, not waiting for the bus or an interrupt. | 1'b0 |

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**Name** CHAN2_TOTAL_WRITE_REMAINDER
**Address** 0x501700c0
**Description** Contains register fields associated with chan2_total_write_remainder.
**Reset** 0x00000000

| Name | RSV | total_write_ |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | RSVD | | | | | | | | | | | | | | | | | | | | total_write_remainder | | | | | | | | | | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R | total_write_remainder | number of bytes left to write at the time of stop | 24'b0 |

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**Name** CHAN2_INTERRUPT_STATUS
**Address** 0x501700c4
**Description** Contains register fields associated with chan2_interrupt_status.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | dma_err | dma_done |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | dma_err | source of interrupt due to an error (chan0). | 1'b0 |
| 0 | R | dma_done | source of interrupt due to completion (chan0). | 1'b0 |

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**Name** CHAN2_INTERRUPT_MASK
**Address** 0x501700c8
**Description** Contains register fields associated with chan2_interrupt_mask.
**Reset** 0x00000001

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | intrpt_mask | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_mask | if nth bit set, the nth interrupt is enabled | 2'b01 |

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**Name** CHAN2_SET_INTERRUPT

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**Address** 0x501700cc
**Description** Contains register fields associated with chan2_set_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_set |
|---|---|---|
| **Bits** | **31:2** | **1:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_set | if nth bit set, the nth interrupt is set; HW clears all bits the next cycle. | 2'b00 |

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**Name** CHAN2_RESET_INTERRUPT
**Address** 0x501700d0
**Description** Contains register fields associated with chan2_reset_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_reset |
|---|---|---|
| **Bits** | **31:2** | **1:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_reset | if nth bit set, the nth interrupt is reset; HW clears all bits the next cycle. | 2'b0 |

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**Name** CHAN2_CFG_HNONSEC
**Address** 0x501700d4
**Description** Contains register fields associated with chan2_cfg_hnonsec.
**Reset** 0x00000001

| Name | RSVD | cfg_hnonsec |
|---|---|---|
| **Bits** | **31:1** | **0** |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | cfg_hnonsec | control HNONSEC output signal of this channel. This register is only Write-able by Secure program. | 1'b1 |

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**Name** CHAN3_OPMODE
**Address** 0x501700d8
**Description** Contains register fields associated with chan3_opmode.
**Reset** 0x00000000

| Name | go | stop | RSVD | const_tar_addr | dat_inv | const_trans |
|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | go | op code start when rising edge on 'go' is detected. HW clears after. | 1'b0 |
| 30 | R/W | stop | to stop pending transaction . Not self clearing. | 1'b0 |
| 2 | R/W | const_tar_addr | 1= write address remains constant at tar_addr, which is required to be word-aligned in this mode.<br>0= write address increments as defined by the size parameter. | 1'b0 |
| 1 | R/W | dat_inv | 0= non-inverted write ; 1= inverted write (write data are bit-wise inverted) | 1'b0 |
| 0 | R/W | const_trans | op code start when rising edge on 'go' is detected. | 1'b0 |

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**Name** CHAN3_CONST_WDATA
**Address** 0x501700dc
**Description** Contains register fields associated with chan3_const_wdata.
**Reset** 0x00000000

| Name | const_wdata |
|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | const_wdata | constant value to be written to target when the operation is in const_trans mode. | 32'b0 |

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**Name** CHAN3_SRC_ADDR
**Address** 0x501700e0
**Description** Contains register fields associated with chan3_src_addr.
**Reset** 0x00000000

| Name | src_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | src_addr | source address | 32'b0 |

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**Name** CHAN3_TAR_ADDR
**Address** 0x501700e4
**Description** Contains register fields associated with chan3_tar_addr.
**Reset** 0x00000000

| Name | tar_addr | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | tar_addr | target address | 32'b0 |

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**Name** CHAN3_SIZE
**Address** 0x501700e8
**Description** Contains register fields associated with chan3_size.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | size | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R/W | size | payload size in bytes. | 24'b0 |

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**Name** CHAN3_SRC_CTRL
**Address** 0x501700ec
**Description** Contains register fields associated with chan3_src_ctrl.
**Reset** 0x00000004

| Name | RSVD | src_bus_size | src_type |
|---|---|---|---|
| **Bits** | **31:5** | **4:2** | **1:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | src_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | src_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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---

**Name** CHAN3_TAR_CTRL
**Address** 0x501700f0
**Description** Contains register fields associated with chan3_tar_ctrl.
**Reset** 0x00000004

| Name | RSVD | tar_bus_size | tar_type |
|---|---|---|---|
| **Bits** | **31:5** | **4:2** | **1:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 4:2 | R/W | tar_bus_size | the width of fifo. one byte to 4 bytes. | 3'd1 |
| 1:0 | R/W | tar_type | 0=mem, 1=fifo, 2=periph master, 3=periph slave | 2'b0 |

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---

**Name** CHAN3_FIFO_DPTH_ADDR
**Address** 0x501700f4
**Description** Contains register fields associated with chan3_fifo_dpth_addr.
**Reset** 0x00000000

| Name | addr |
|---|---|
| **Bits** | **31:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | addr | address of the external fifo depth.<br>Statemachine reads this to determine how many push/pull accesses to do per interrupt. | 32'b0 |

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**Name** CHAN3_FIFO_PORT_SEL
**Address** 0x501700f8
**Description** Contains register fields associated with chan3_fifo_port_sel.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | tar_port_sel | | | RSVD | | | | | | | | | | | | | src_port_sel | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 18:16 | R/W | tar_port_sel | selecting the target fifo port;<br>5 = pwd fifo<br>4 = i2s fifo<br>3 = rsvd<br>2 = rsvd<br>1 = uart1 tx<br>0 = uart0 tx | 3'b0 |
| 2:0 | R/W | src_port_sel | selecting the source fifo port;<br>5 = rsvd<br>4 = i2s fifo<br>3 = pdm1 fifo<br>2 = pdm0 fifo<br>1 = uart1 rx<br>0 = uart0 rx | 3'd0 |

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**Name** CHAN3_SPI_PORT_SEL
**Address** 0x501700fc
**Description** Contains register fields associated with chan3_spi_port_sel.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | spi_sel |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | spi_sel | 1=select spi1 for this channel, 0=select spi0 for this channel. | 1'b0 |

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**Name** CHAN3_ERR_STAT
**Address** 0x50170100
**Description** The status is cleared when the err interrupt is cleared
**Reset** 0x00000000

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| Name | RSVD | addr_ovfl | bus_err | addr_ovlp | size_0 |
|---|---|---|---|---|---|
| **Bits** | **31:4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3 | R | addr_ovfl | 0= no error, 1= error due to size + src_addr/tar_addr > 32'hffff_ffff. | 1'b0 |
| 2 | R | bus_err | 0= no bus error, 1= a bus error occur during the transfer or at the launch of 'go'. | 1'b0 |
| 1 | R | addr_ovlp | 0= no overlap between src and tar addr range, 1= there is an overlap | 1'b0 |
| 0 | R | size_0 | 0= size is not 0 when go is launched, 1= size is 0 when go is launched. | 1'b0 |

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**Name** CHAN3_STATUS
**Address** 0x50170104
**Description** Contains register fields associated with chan3_status.
**Reset** 0x00000000

| Name | RSVD | busy | running |
|---|---|---|---|
| **Bits** | **31:2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | busy | fsm is in some non-idle state including waiting for the bus or an interrupt. | 1'b0 |
| 0 | R | running | fsm is in an active state, not waiting for the bus or an interrupt. | 1'b0 |

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**Name** CHAN3_TOTAL_WRITE_REMAINDER
**Address** 0x50170108
**Description** Contains register fields associated with chan3_total_write_remainder.
**Reset** 0x00000000

| Name | RSVD | total_write_remain |
|---|---|---|

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| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | d<br>e<br>r |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:0 | R | total_write_remainder | number of bytes left to write at the time of stop | 24'b0 |

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**Name** CHAN3_INTERRUPT_STATUS
**Address** 0x5017010c
**Description** Contains register fields associated with chan3_interrupt_status.
**Reset** 0x00000000

| Name | R<br>S<br>V<br>D | | | | | | | | | | | | | | | | | | | | | | | | | | | | | d<br>m<br>a<br>_<br>e<br>r<br>r | d<br>m<br>a<br>_<br>d<br>o<br>n<br>e |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1 | R | dma_err | source of interrupt due to an error (chan0). | 1'b0 |
| 0 | R | dma_done | source of interrupt due to completion (chan0). | 1'b0 |

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**Name** CHAN3_INTERRUPT_MASK
**Address** 0x50170110
**Description** Contains register fields associated with chan3_interrupt_mask.
**Reset** 0x00000001

| Name | R<br>S<br>V<br>D | | | | | | | | | | | | | | | | | | | | | | | | | | | | | i<br>n<br>t<br>r<br>p<br>t<br>_<br>m<br>a<br>s<br>k | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_mask | if nth bit set, the nth interrupt is enabled | 2'b01 |

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**Name** CHAN3_SET_INTERRUPT
**Address** 0x50170114
**Description** Contains register fields associated with chan3_set_interrupt.
**Reset** 0x00000000

| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | i<br>n |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

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| Name | RSVD | intrpt_set |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_set | if nth bit set, the nth interrupt is set; HW clears all bits the next cycle. | 2'b00 |

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**Name** CHAN3_RESET_INTERRUPT
**Address** 0x50170118
**Description** Contains register fields associated with chan3_reset_interrupt.
**Reset** 0x00000000

| Name | RSVD | intrpt_reset |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2** | **1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 1:0 | R/W | intrpt_reset | if nth bit set, the nth interrupt is reset; HW clears all bits the next cycle. | 2'b0 |

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**Name** CHAN3_CFG_HNONSEC
**Address** 0x5017011c
**Description** Contains register fields associated with chan3_cfg_hnonsec.
**Reset** 0x00000001

| Name | RSVD | cfg_hnonsec |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | cfg_hnonsec | control HNONSEC output signal of this channel. This register is only Write-able by Secure program. | 1'b1 |

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**Name** ERR_INTERRUPT_DLY
**Address** 0x50170120
**Description** err interrupt delayed in cycles
**Reset** 0x0000000a

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | delay_amnt | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 7:0 | R/W | delay_amnt | number of cycles to delay the error interrupt event.<br>Without this delay the interrupt may come up immediately after the go trigger leaving CPU no time to go into WFI. | 8'd10 |

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**Name** ARBITER_CTRL
**Address** 0x50170124
**Description** Contains register fields associated with arbiter_ctrl.
**Reset** 0x00000000

| Name | RSVD | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | priority_en |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | priority_en | 0= no priority 1=lower numbered channel has higher priority when the bus owernership is relinquished. | 1'd0 |

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**Name** ARBITER_GRANT_STAT
**Address** 0x50170128
**Description** Contains register fields associated with arbiter_grant_stat.
**Reset** 0x00000000

| Name | RSV | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | chanx_bus_ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | D | | | | | | | | | | | | | | | | | | | | | | | | | | | g r a n t e d | |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 3:0 | R | chanx_bus_granted | one-hot signal indicating channel with AHB bus ownership. | 4'd0 |

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**Name** DIAGCNTL
**Address** 0x5017012c
**Description** Contains register fields associated with diagcntl.
**Reset** 0x00000000

| Name | R S V D | | | | | | d i a g 0 | | R S V D | | | | | | d i a g 1 | | R S V D | | | | | | d i a g 2 | | R S V D | | | | d i a g 3 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 25:24 | R/W | diag0 | selects a group of signal to drive dbg[31:24]. | 2'd0 |
| 17:16 | R/W | diag1 | selects a group of signal to drive dbg[23:16]. | 2'd0 |
| 9:8 | R/W | diag2 | selects a group of signal to drive dbg[15:8]. | 2'd0 |
| 1:0 | R/W | diag3 | selects a group of signal to drive dbg[7:0]. | 2'd0 |

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**Name** ID
**Address** 0x50170ffc
**Description** Contains register fields associated with id.
**Reset** 0x444d4120

| Name | i d | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | 'DMA ' | 32'h444d4120 |

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# PMU_TOP

**Name** PMU0
**Address** 0xe1090000
**Description** timer_target_lsb - written by software
**Reset** 0x00000001

| | t i |
|---|---|

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| Name | timer_target_lsb | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | timer_target_lsb | written by software, target value to wake up from in soc-off mode<br>pmu starts from 0 and counts 32KHz clock cycles until it reaches this value to come out of soc-off mode if how2wkup reg is set to 1xx<br>31 lsb bits | 32'd1 |

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**Name** PMU1
**Address** 0xe1090004
**Description** software controlled registers
**Reset** 0x00000400

| Name | RSVD | | | | | doRCoscCal | use16MforPSEQ | RSVD | | | | | | | | | | | | | | lpcomp_in_sel | how2wkup | timer_target_msb | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 27 | R/W | doRCoscCal | set to 1 before entering low power state to do HW measurement in that state, clear after next wake up and read calCountRC value | 1'd0 |
| 26 | R/W | use16MforPSEQ | keep 16MHz on and use it as the clock source for 32KHz timers in PMU | 1'd0 |
| 11 | R/W | lpcomp_in_sel | 0: P3, 1: P4 | 1'd0 |
| 10 | R/W | how2wkup_timer | written by software - xx1: pin, x1x: lpcomp, 1xx: timer, write back to 0 after determining wake up cause to clear, write desired bits to 1 before entering low-power state, wait at least two 32KHz cycles between clearing and setting | 3'd4 |
| 9 | | how2wkup_lpcomp | | |
| 8 | | how2wkup_pin | | |
| 7:0 | R/W | timer_target_msb | written by software, target value to wake up from in soc-off mode<br>pmu starts from 0 and counts 32KHz clock cycles until it reaches this value to come out of soc-off mode if how2wkup reg is set to 1xx<br>8 msb bits | 8'd0 |

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**Name** PMU1A

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**Address** 0xe1090008
**Description** software controlled xtal related
**Reset** 0x00002040

| Name | RSVD | | | | | | | | | | | | | RCoscCalWait | | | | | | | | | | | RCoscCalPeriods | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 18:11 | R/W | RCoscCalWait | how long to wait for supply settling after entering low power state to start counting, ~1ms resolution | 8'd4 |
| 10:0 | R/W | RCoscCalPeriods | how many RCosc periods to run calibration | 11'd64 |

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**Name** PMU11
**Address** 0xe1090034
**Description** atb and clocks
**Reset** 0x00009800

| Name | en_sw_hsw_b_late_ovr_val | en_sw_hsw_b_late_ovr | en_mem_hsw_b_late_ovr_val | en_mem_hsw_b_late_ovr | RSVD | | | | | | | | | | | | lpcomp_ref | | | | lpcomp_ref_fall | | | | RSVD | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | en_sw_hsw_b_late_ovr_val | override en_sw_hsw_b_late val: Turn OFF switcher late hsw | 1'b0 |
| 30 | R/W | en_sw_hsw_b_late_ovr | override en_sw_hsw_b_late | 1'b0 |
| 29 | R/W | en_mem_hsw_b_late_ovr_val | override for en_mem_hsw_b_late val: Turn OFF memory late hsw | 1'b0 |

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|---|---|---|---|---|
| 28 | R/W | en_mem_hsw_b_late_ovr | override for en_mem_hsw_b_late | 1'b0 |
| 15:12 | R/W | lpcomp_ref | rising edge reference select for lpcomp<br>25*[66,62,58,54,50,46,42,38,34,30,26,22,18,14,10,6] for i=[15:0] | 4'd9 |
| 11:8 | R/W | lpcomp_ref_fall | falling edge reference select for lpcomp hysteresis, should be set lower than lpcomp_ref_rise<br>25*[65,61,57,53,49,45,41,37,33,29,25,21,17,13,9,5] for i=[15:0] | 4'd8 |

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**Name** PMU15
**Address** 0xe1090054
**Description** more pmu controls
**Reset** 0x3e13d4de

| Name | swRes_ctrsamecurr | swResVbat_ctrsamecurr | swResVbat_ctrcu | | | | | ctr_prtbypcurr_ovr_val | ctr_prtbypcurr_ovr | en_pmurectleakcu | en_shuntrf | en_compshreg | zerompptref_ctrcu | | | | decrectst_enhy | ctr_usevsto4avddreq | ctr_pmuadcref | ctr_refshreg | ctr_zerompptref | ctr_refvstoreq_lowR | ctr_refvstoreq | | ctr_refgenfastcurr | ctr_refvstoreq | | ctr_refstInsel | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | swRes_ctrsamecurr | control slow current in swRes comparator used in cold start<br>0: decrease the slow current<br>1: keep the slow current equal to the fast | 1'd0 |
| 30 | R/W | swResVbat_ctrsamecurr | control slow current in swResVbat comparator<br>0: decrease the slow current<br>1: keep the slow current equal to the fast | 1'd0 |
| 29:25 | R/W | swResVbat_ctrcu | control the swResVbat comparator current<br>ctr_cuLow with <1:0>, ctr_cuMid with <3:2>, ctr_cuMidx2 with <4> | 5'd31 |
| 24 | R/W | ctr_prtbypcurr_ovr_val | unused | 1'd0 |
| 23 | R/W | ctr_prtbypcurr_ovr | unused | 1'd0 |
| 22 | R/W | en_pmurectleakcu | add an imbalance current into the RF harvesting rectifiers to make sure they latch when OFF<br>to help prevent any stress issue in the rectifier core devices | 1'd0 |
| 21 | R/W | en_shuntrf | Shunt the RFHARV input to ground to check the maximum shunting ability of shunt regulator | 1'd0 |
| 20 | R/W | en_compshreg | Enable the shunt regulator on RFHARV input | 1'd1 |
| 19:15 | R/W | zerompptref_ctrcu | control the fast current of zerompptref comparator | 5'd7 |
| 14 | R/W | decrectst_enhy | enable hysteresis in the decrectstage_prtVharv comparator<br>1: When used to protect Vharv at ~3V during solar harvesting<br>0: During RF mppt that we want to regulate efficiently Vharv till the number of rectifier stages change | 1'd1 |
| 13 | R/W | ctr_usevsto4avddreq | Use the Vstore request comparator to sense avdd and generate avdd requests<br>It takes over the avdd request comparator thanks to its NMOS+PMOS full | 1'd0 |

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|---|---|---|---|---|
| | | | scale input. | |
| 12:11 | R/W | ctr_pmuadcref | Control pmuadcref, 25*(23+i-1) for i=3:1, vrefnat600 for i=0 | 2'd2 |
| 10:9 | R/W | ctr_refshreg | Control Vrefshreg used by the rectifier shunt regulator,25*(36+i*4), i=3:0 | 2'd2 |
| 8:7 | R/W | ctr_zerompptref | control Vrefzerompptref used by the zerompptref comparator<br>25*[9,8,6,4], i=3:0 | 2'd1 |
| 6 | R/W | ctr_refvstoreq_lowR | Selects different vstore comparator reference<br>1: reference generated by the lowR resistor in the pmuparef<br>0: reference generated by refgen | 1'd1 |
| 5:3 | R/W | ctr_refvstoreq | control VrefvstoReq= 25*(16+i*8), i=7:0 | 3'd3 |
| 2 | R/W | ctr_refgenfastcurr | control current of fast refgen | 1'd1 |
| 1:0 | R/W | ctr_reffstInsel | fast refgen feedback tap select. For reference=600mV<br>3: 625mV reflad<25><br>2: 600mV reflad<24><br>1: 575mV reflad<23><br>0: 550mV reflad<22> | 2'd2 |

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**Name** PMU16
**Address** 0xe1090058
**Description** more config bits
**Reset** 0x4e808039

| Name | chargeBatt | pmuadc_byrefbuf | pmuadc_enrefbuf | ctr_refcalpmuadc | ctr_refcalpmuadc | ctr_refcalpmuadc | ctr_refcalpmuadc | ctr_ref_vbatreg_ovr_val | ctr_ref_vbatreg_ovr | ovr_ReqHarvb_incrects t_ovr_val | ovr_ReqHarvb_incrects t_ovr | ovr_ReqCanMustHarvb_zerompptovr_val | ovr_ReqCanMustHarvb_zerompptovr | allowChrgInActive | decrectst_ctrcuMid | increctst_ctrcuMid1 | increctst_ctrcuMid0 | ovr_decrectstage_prtVharv_ovr_val | ovr_decrectstage_prtVharv_ovr | storeIsSup | fsm_lp_harv_reg | swreg_fsmTouse_ovr_val | swreg_fsmTouse_ovr | pRef | pRef | pRef | pRef | pRef |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31 | R/W | chargeBatt | Enable the VSTO->VBAT mode. Also unmasks the pmuadc indicator to set buck, boost for VSTO->VBAT. | 1'd0 |
| 30 | R/W | pmuadc_byrefbuf | bypass the pmuadc reference buffer | 1'd1 |

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|---|---|---|---|---|
| 29 | R/W | pmuadc_enrefbuf | enable the pmuadc reference buffer | 1'd0 |
| 28:25 | R/W | ctr_refcalpmuadc | calibrate the output of the pmuadc reference buffer | 4'd7 |
| 24:23 | R/W | ctr_ref_vbatreg_ovr_val | override value for VBATLI to VBATT LDO output level, 3.0V to 3.3V in 0.1V steps | 2'd1 |
| 22 | R/W | ctr_ref_vbatreg_ovr | override ctr_ref_vbatreg level | 1'd0 |
| 21 | R/W | ovr_ReqHarvb_increctst_ovr_val | override ovr_ReqCanMustHarvb_increctst val<br>0: When the chpuhswoff_increctstb is low, it forces can/mustharv requests to zero<br>1: chpuhswoff_increctstb has no effect on an/mustharv requests | 1'd0 |
| 20 | R/W | ovr_ReqHarvb_increcst_ovr | override ovr_ReqCanMustHarvb_increctst | 1'd0 |
| 19 | R/W | ovr_ReqCanMustHarvb_zeromppt_ovr_val | override ovr_ReqCanMustHarvb_zeromppt val<br>0: When the zerompptref is high, it forces can/mustharv requests to zero<br>1: zerompptref has no effect on an/mustharv requests | 1'd0 |
| 18 | R/W | ovr_ReqCanMustHarvb_zeromppt_ovr | override ovr_ReqCanMustHarvb_zeromppt | 1'd0 |
| 17 | R/W | allowChrgInActive | If set, the Vstore->Vbat charging can take place also in active state | 1'd0 |
| 16 | R/W | decrectst_ctrcuMid | Mid current setting of decrectstage_prtVharv comparator | 1'd0 |
| 15 | R/W | increctst_ctrcuMid1 | Mid current setting of chpuhsw_increctstage comparator when churge pump is enabled<br>Need faster comparator response for Vharv protection. | 1'd1 |
| 14 | R/W | increctst_ctrcuMid0 | Mid current setting of chpuhsw_increctstage comparator when churge pump is disabled. | 1'd0 |
| 13 | R/W | ovr_decrectstage_prtVharv_ovr_val | override ovr_decrectstage_prtVharv val:<br>0: the decrectstage_prtVharv forces can/mustharv to request<br>Important to be applied right after the number of stages is decreased during analog mppt<br>1: the decrectstage_prtVharv has no effect on can/mustharv | 1'd0 |
| 12 | R/W | ovr_decrectstage_prtVharv_ovr | override ovr_decrectstage_prtVharv. The ovr_decrectstage_prtVharv is controlled by the analog mppt algorithm. | 1'd0 |
| 11 | R/W | storeIsSup | When set the Vstore is a supply used ie for the PA. | 1'd0 |
| 10:9 | R/W | fsm_lp_harv_reg | selects idle state FSM if no harvsting, 0: FSM is tiny, 1: FSM is simple | 2'd0 |
| 8:7 | R/W | swreg_fsmToUse_ovr_val | override swreg_fsmToUse val: can force the FSM to use | 2'd0 |
| 6 | R/W | swreg_fsmToUse_ovr | override swreg_fsmToUse | 1'd0 |
| 5:0 | R/W | paRef | control the pa supply ref | 6'd57 |

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**Name** CHRG_CTRL
**Address** 0xe109005c
**Description** Charger control bits
**Reset** 0x00000000

| Name | RSVD | | | | | ctr_chrgpmosbyp | bypvstovbatli | RSVD | | en_VstoVbat | ctr_chrgcurr | | | | | | | en_chrg_curr | chrg_timeout | | | | | | | | | | chrg_timer_start | disable_chrg_ovr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 26 | R/W | ctr_chrgpmosbyp | Short VSTORE to VBATT and bypass charger current source | 1'b0 |

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|---|---|---|---|---|
| 25 | R/W | bypvstovbatli | Short VSTORE to VBATLI, should only be enabled if VBATLI is within OV and UV ranges | 1'b0 |
| 22 | R/W | en_VstoVbat | Enable VSTORE->VBATT charger current source path | 1'b0 |
| 21:14 | R/W | ctr_chrgcurr | Charge current source value, approx 60uA to 15mA in 60uA steps for VSTORE->VBATT path | 8'd0 |
| 13 | R/W | en_chrg_curr | Enable charger current source from VSTORE->VBATT | 1'b0 |
| 12:2 | R/W | chrg_timeout | Charger timeout countdown timer value in seconds, should be set longer than any battery measurement/charger settings update rate | 11'd0 |
| 1 | R/W | chrg_timer_start | Set high for at least 50us and then clear to restart timeout counter<br>This will start charging and make the configuration settings below take effect<br>SW needs to toggle this bit before charger timeout countdown timer expires to keep charging going | 1'b0 |
| 0 | R/W | disable_chrg_ovr | force disable all charging controls | 1'b0 |

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**Name** CHRG_CTRL2
**Address** 0xe1090060
**Description** More charger control bits
**Reset** 0x00000060

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| Name | RSVD | | | | | | | | | | | | | | | | | unused | | | | | | en_clk_cp | ctr_VrefstoGoodToChrg | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 13:8 | R/W | unused | unused, pmu_spares[5:0] | 6'd0 |
| 7 | R/W | en_clk_cp | Enable clock to drive external charge pump, this clock needs to be separately brought out to a GPIO through pin mux setting<br>Clock only actually runs if VSTORE is above ctr_VrefstoGoodToChrg level when this bit is enabled | 1'b0 |
| 6:0 | R/W | ctr_VrefstoGoodToChrg | VSTORE level above which to enable external charge pump clock in pmuadc units, = round[(vstore voltage - 600mv)/25mv]<br>There will be another 100mV rising hysteresis included in the comparison<br>Default setting for VSTORE = 3.0V with 100mV hysteresis = charge pump turns on at 3.1V and shuts off at 3.0V | 7'd96 |

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**Name** PMU_RB_MPPT
**Address** 0xe1090078
**Description** readback registers for customers
**Reset** 0x00000000

| Name | RSVD | ctr_mustharvfa | vstoreTooHigh_latch | mpptEventCount | mpptPeriodCount |
|---|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 23:20 | R | ctr_mustharvfa | readback vharv | 4'd0 |
| 19 | R | vstoreTooHigh_latch | readback vstoreToohigh latched value | 1'd0 |
| 18:9 | R | mpptEventCount | readback mpptEventCount value | 10'd0 |
| 8:0 | R | mpptPeriodCount | readback mpptPeriodCount value | 9'd0 |

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**Name** PMU_RB4
**Address** 0xe1090080
**Description** external OTP readback registers
**Reset** 0x00000000

| Name | RSVD | otp_spare | otp_harv_cfg | otp_reserved | otp_vddio_range | otp_batt_cfg | otp_supply_cfg | otp_disable_xtal32K |
|---|---|---|---|---|---|---|---|---|

| Bits | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 28 | R | otp_spare | (no description) | 1'd0 |
| 27:18 | R | otp_harv_cfg | {otp_vstore_max[2:0],otp_vstore_good[1:0],otp_vharv_start[1:0],otp_disable_rfharv,otp_mppt_type[1:0]} | 10'd0 |
| 17 | R | otp_reserved | (no description) | 1'd0 |
| 16 | R | otp_vddio_range | (no description) | 1'd0 |
| 15:5 | R | otp_batt_cfg | {otp_vbatt_brownout[4:0],otp_vbatt_good[2:0],otp_batt_type[1:0],otp_vbatt_level} | 11'd0 |
| 4:1 | R | otp_supply_cfg | {otp_disable_5v,otp_disable_vddgen,otp_disable_vddIOgen,otp_noind} | 4'd0 |

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|---|---|---|---|---|
| 0 | R | otp_disable_xtal32K | (no description) | 1'd0 |

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**Name** PMU_RB5
**Address** 0xe1090084
**Description** external soc watchdog reset readback status register
**Reset** 0x00000000

| Name | RSVD | soc_wdog_reset |
|---|---|---|
| **Bits** | **31:1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | soc_wdog_reset | HW writes sets status when soc wdog reset is asserted. SW writes clears status only. | 1'd0 |

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**Name** CORE_ID
**Address** 0xe1090088
**Description** Core ID
**Reset** 0x504d5520

| Name | id |
|---|---|
| **Bits** | **31:0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R | id | PMU in ASCII | 32'h504D5520 |

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# PMU_WURX

**Name** WURX_GENERAL
**Address** 0xe10a0000
**Description** Wake-up RX overrides and bias
**Reset** 0x00000000

| Name | RSVD | wurx_sel |
|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | e<br>c<br>t |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 0 | R/W | wurx_select | which algorithm to use: 0: algorithm 0, 1: algorithm 1 | 1'd0 |

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**Name** WURX0_0
**Address** 0xe10a0008
**Description** Wake-up RX unique ID, algorithm 0
**Reset** 0xaaaaaaaa

| Name | u<br>i<br>d | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | uid | User ID, upper 16 bits for group ID, lower 16 bits for user ID | 32'hAAAAAAAA |

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**Name** WURX0_1
**Address** 0xe10a000c
**Description** Wake-up RX global ID, algorithm 0
**Reset** 0xffffffff

| Name | g<br>i<br>d | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | gid | Global ID | 32'hFFFFFFFF |

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**Name** WURX0_2
**Address** 0xe10a0010
**Description** Wake-up RX settings, algorithm 0
**Reset** 0x0c00001f

| Name | R<br>S<br>V<br>D | t<br>h<br>r<br>_<br>l<br>e<br>n<br>g<br>t<br>h | c<br>o<br>r<br>r<br>_<br>l<br>e<br>n<br>g<br>t<br>h | R<br>S<br>V<br>D | e<br>r<br>r<br>_<br>t<br>h<br>r<br>e<br>s<br>h<br>o<br>l<br>d |
|---|---|---|---|---|---|
| **Bits** | **31** | **30 29 28 27** | **26 25 24 23 22 21** | **20 19 18 17 16 15 14 13 12 11 10 9 8 7 6** | **5 4 3 2 1 0** |

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|---|---|---|---|---|
| | | | | |

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| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 30:27 | R/W | thr_length | how many sybmols to adjust threshold on, thr_length = 1: 2 symbols | 4'd1 |
| 26:21 | R/W | corr_length | number of symbols in one ID | 6'd32 |
| 5:0 | R/W | err_threshold | number of bits of error between RX ID and detected ID | 6'd31 |

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**Name** WURX0_3
**Address** 0xe10a0014
**Description** Wake-up RX settings, algorithm 0
**Reset** 0x20100c04

| Name | RSVD | clk_freq | sym1_threshold | | | | | | | | | energy_toohigh | | | | | | | | | energy_toolow | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29 | R/W | clk_freq | 0 - 16KHz, 1- 32KHz | 1'b1 |
| 28:19 | R/W | sym1_threshold | number of counts above this means symbol 1 is detected | 10'd2 |
| 18:9 | R/W | energy_toohigh | received energy more than this --> increase threshold | 10'd6 |
| 8:0 | R/W | energy_toolow | received energy less than this --> decrease threshold | 9'd4 |

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**Name** WURX0_4
**Address** 0xe10a0018
**Description** Wake-up RX settings, algorithm 0
**Reset** 0x0000000f

| Name | RSVD | timer_max | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9:0 | R/W | timer_max | number of cycles in one symbol period - 1 | 10'd15 |

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**Name** WURX1_0
**Address** 0xe10a001c
**Description** Wake-up RX unique ID, algorithm 0
**Reset** 0xaaaaaaaa

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| Name | uid | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | uid | User ID, upper 16 bits for group ID, lower 16 bits for user ID | 32'hAAAAAAAA |

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**Name** WURX1_1
**Address** 0xe10a0020
**Description** Wake-up RX global ID, algorithm 0
**Reset** 0xffffffff

| Name | gid | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:0 | R/W | gid | Global ID | 32'hFFFFFFFF |

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**Name** WURX1_2
**Address** 0xe10a0024
**Description** Wake-up RX settings, algorithm 0
**Reset** 0x0c00001f

| Name | RSVD | thr_length | corr_length | RSVD | err_threshold | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 30:27 | R/W | thr_length | how many sybmols to adjust threshold on, thr_length = 1: 2 symbols | 4'd1 |
| 26:21 | R/W | corr_length | number of symbols in one ID | 6'd32 |
| 5:0 | R/W | err_threshold | number of bits of error between RX ID and detected ID | 6'd31 |

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**Name** WURX1_3
**Address** 0xe10a0028
**Description** Wake-up RX settings, algorithm 0
**Reset** 0x20100c04

| | | | sy | en | e |
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| Name | RSVD | clk_freq | sym1_threshold | energy_toohigh | energy_toolow |
|---|---|---|---|---|---|
| **Bits** | **31 30** | **29** | **28 27 26 25 24 23 22 21 20 19** | **18 17 16 15 14 13 12 11 10** | **9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 29 | R/W | clk_freq | 0 - 16KHz, 1- 32KHz | 1'b1 |
| 28:19 | R/W | sym1_threshold | number of counts above this means symbol 1 is detected | 10'd2 |
| 18:9 | R/W | energy_toohigh | received energy more than this --> increase threshold | 10'd6 |
| 8:0 | R/W | energy_toolow | received energy less than this --> decrease threshold | 9'd4 |

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**Name** WURX1_4
**Address** 0xe10a002c
**Description** Wake-up RX settings, algorithm 0
**Reset** 0x0000000f

| Name | RSVD | timer_max |
|---|---|---|
| **Bits** | **31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10** | **9 8 7 6 5 4 3 2 1 0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 9:0 | R/W | timer_max | number of cycles in one symbol period - 1 | 10'd15 |

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**Name** WURX_SPARE
**Address** 0xe10a003c
**Description** Wake-up RX spare bits
**Reset** 0x000045a1

| Name | spares | settle2_reg | settle1_reg | bypassAmp | delRst | longRst | wurx_dig_in_se |
|---|---|---|---|---|---|---|---|

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 1 |
| **Bits** | **31** | **30** | **29** | **28** | **27** | **26** | **25** | **24** | **23** | **22** | **21** | **20** | **19** | **18** | **17** | **16** | **15** | **14** | **13** | **12** | **11** | **10** | **9** | **8** | **7** | **6** | **5** | **4** | **3** | **2** | **1** | **0** |

| Bits | SW | Name | Description | Reset |
|---|---|---|---|---|
| 31:15 | R/W | spares | wurx local register spare bits | 17'd0 |
| 14:10 | R/W | settle2_reg | wait time after power on with load and bias set to default | 5'd17 |
| 9:5 | R/W | settle1_reg | wait time after power on for bias at max load | 5'd13 |
| 4 | R/W | bypassAmp | do not use amp to bias source follower | 1'd0 |
| 3:2 | R/W | delRst | adjusts output reset time | 2'd0 |
| 1 | R/W | longRst | resets the output while the clock is low (same as sydney) | 1'd0 |
| 0 | R/W | wurx_dig_in_sel | select between wurx analog and rectifier output | 1'd1 |

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# Revision History

| Date | Version | Description |
|---|---|---|
| March 10, 2023 | 0.53 | Changed Rev. B0 to Rev. B, updated control registers of PDM0 and PDM1. |
| January 3, 2023 | 0.52 | Updated PDSN register description. |
| August 30, 2022 | 0.51 | Updated title, highlighted features supported by ATM3330e, ATM3330. |
| August 29, 2022 | 0.50 | Initial version created. |

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